

TABLE 23-7 Zone Classification versus Minimum Extent of Preventive Measures against Danger of Ignition

Explosion hazard zone	Avoid ignition sources which can occur:
20/0	During normal operation, during frequent malfunction, and during rare malfunction
21/1	During normal operation and during frequent malfunction
22/2	During normal operation

Hazardous places are classified in terms of zones (divisions) on the basis of the frequency and duration of the occurrence of an explosive atmosphere (IEC 60079-10:200X, "Classification of Hazardous Areas"; European Standard EN 50281-3:200X, "Classification of Areas Where Combustible Dusts Are or May Be Present").

Fundamentally all 13 kinds of ignition sources mentioned and also described in detail in EN 1127-1 must be considered. In EN 1127-1, the descriptions refer to both the ignition mechanisms of the different kinds of ignition source and the necessary scope of protection (material- and zone-dependent). The ignition sources should be classified according to the likelihood of their occurrence in the following manner:

1. Sources of ignition which can occur *continuously or frequently*
2. Sources of ignition which can occur in *rare* situations
3. Sources of ignition which can occur in *very rare* situations

In terms of the equipment, protective systems, and components used, this classification must be considered equivalent to

1. Sources of ignition which can occur during *normal operation*
2. Sources of ignition which can occur solely as a result of *malfunctions*
3. Sources of ignition which can occur solely as a result of *rare malfunctions*

Ignition sources at devices, protective systems, and components must be avoided, depending on zones (Table 23-7); i.e., the zone determines the minimum extent of preventive measures against danger of ignition.

Flammable gas or vapor-air mixtures Due to their low minimum ignition energies (<1 mJ), avoidance of effective ignition sources in flammable gas or vapor-air mixtures is in principle possible only in exceptional cases. For *hot surfaces* a maximum permissible surface temperature $T_{\max}$ must be specified, with the help of the minimum ignition temperature (MIT) of flammable gases, vapors, or liquids, so that the temperature of all surfaces is not exceeded (Table 23-8).

Combustible dust-air mixtures For every installation a check has to be made to determine which ignition source may become effective and whether it can be prevented with a sufficient degree of safety. With more sensitive products and complex installations, it becomes more and more difficult to exclude ignition sources with ample safety.

With dusts the avoidance of effective ignition sources depends on the ignition sensitivity, i.e., on the temperature-related minimum ignition energy (MIE). Because a low MIE means that both the number of effective ignition sources rises and the probability of the release of an ignition of a dust-air mixture rises. Safe handling of particularly or extremely ignition-sensitive dusts requires in general increased use of preventive measures in comparison to normally ignitable sensitive dusts.

It is therefore usual to combine the evaluation of the ignition sensitivity (MIE) of the dusts with the type and extent of protective measures in accordance with Table 23-9.

TABLE 23-8 Measures against Ignition of Flammable Gas, Vapors, and Liquids by Hot Surfaces

Zone	Protection by limiting the hot surface temperature (EN 1127-1)
0	$T_{\max} \leq 0.8 \text{ MIT}$
1	$T_{\max} \leq 0.8 \text{ MIT}$
2	$T_{\max} \leq \text{MIT}$

TABLE 23-9 Ignition Sensitivity of Dusts as a Function of Protective Measures

Ignition sensitivity of dusts	Protective measures
Normally sensitive (MIE $\geq 10 \text{ mJ}$)	Avoidance of effective ignition sources.
Easily sensitive (3 mJ $\leq$ MIE $< 10 \text{ mJ}$)	Expert decision or Avoidance of effective ignition sources and additional protective measures.
Extremely sensitive (MIE $< 3 \text{ mJ}$)	Avoidance of effective ignition sources and additional protective measures.

Protective measures: Inerting or constructional explosion protection (e.g., containment).

For hot surfaces a maximum permissible surface temperature $T_{\max}$ must be specified, with the help of the minimum ignition temperature of a dust cloud MIT_C and the minimum ignition temperature of a dust layer MIT_L (usually 5-mm dust layer = glow temperature), so that the temperature of all surfaces is not exceeded. Independent of the zone the $T_{\max}$ for dust clouds is $\frac{2}{3}$ MIT_C and for dust layers MIT_L – 75 K.

Mechanically generated sparks and resultant hot surfaces together are regarded as one of the more important causes of ignition in industrial practice. The hot surfaces show considerably better incendiarity in comparison with the short-lived, mechanically generated sparks. Neither ignition source appears in industrial practice from the normal metallic materials of construction rubbing against each other or against stone if the relative circumferential speeds v_c are less than or equal to 1 m/s and the power requirement is no more than 4 kW (Table 23-10). This is not valid for cerium-iron, titanium, and zirconium.

Regarding *electrostatic ignition sources*, see the subsection "Static Electricity," later. In addition, see CENELEC CLC/TR 50404, "Electrostatics—Code of Practice for the Avoidance of Hazards due to Static Electricity," June 2003.

Explosion Protection through Design Measures Design measures which restrict the effects of an explosion to a safe level are always necessary when the goal of avoiding explosions cannot be achieved—or at least not with sufficient reliability—through the use of preventive explosion protection. This ensures that people are not injured and further that the protected equipment is usually ready for operation a short time after an explosion. In applying design measures, the possibility of an explosion is not prevented. Therefore, all exposed equipment has to be built to be explosion-pressure-resistant in order to withstand the anticipated explosion pressure. The anticipated explosion pressure may be the maximum explosion overpressure or the maximum reduced explosion overpressure. In addition, *any propagation of an explosion to other parts or process areas has to be prevented*. Depending on the anticipated explosion pressure, a distinction is made between the following explosion-pressure-resistant designs:

- Those capable of withstanding the maximum explosion overpressure
- Those capable of withstanding an explosion overpressure reduced by explosion suppression or explosion venting

The strength of the protected vessels or apparatus may be either explosion-pressure-resistant or explosion pressure shock-resistant.

Containment Containment is understood to mean the possibility of designing vessels and equipment for the full *maximum explosion overpressure*, which is generally from $P_{\max} = 7$ to 10 bar. The explosion-resistant vessel can then be designed as explosion-pressure-resistant or

TABLE 23-10 Influence of Relative Circumferential Speed v_c on Danger of Ignition for Combustible Dusts

$*v_c \leq 1 \text{ m/s}$	There is no danger of ignition.
$v_c > 1-10 \text{ m/s}$	Every case has to be judged separately, considering the product and material-specific characteristics.
$v_c > 10 \text{ m/s}$	In every case there is danger of ignition.

* In addition, low power requirements $W \leq 4 \text{ kW}$.

TABLE 23-11 Explosion Venting System Design Parameters

Explosion hazard definition	Venting system definition
Volume of vessel (free volume)	Type of venting device
Shape of vessel (cubic or elongated vessel)	Detection method for triggering a shutdown
Length-to-diameter ratio of vessel	Static activation overpressure P_{stat} of venting device
Strength of vessel	Venting capability of venting device
Type of dust cloud distribution (ISO method/ pneumatic-loading method)	Location of venting device on the vessel
Dust explosibility characteristics:	Position of equipment to be protected in the building
maximum explosion overpressure P_{max}	Length and shape of relief pipe if existent
maximum explosion constant K_{max}	Recoil force during venting
toxicity of the product	Duration of recoil force
Maximum flame length	Total transferred impulse
Pressure outside the vent areas	

explosion pressure shock-resistant. This protective measure is generally employed when small vessel volumes need to be protected, such as small filter units, fluidized-bed dryers, cyclones, rotary vales, or mill housings.

One has to consider that all connected devices must also withstand the maximum explosion overpressure. The NFPA 69 Standard, *Explosion Prevention System*, 1997; European Standard prEN 14460, *Explosion Resistant Equipment*, 2005; and Kirby and Siwek, "Preventing Failures of Equipment Subject to Explosions," *Chemical Engineering*, June 23, 1986, provide excellent guidance on the practice of containment.

Explosion venting The concept of explosion venting encompasses all measures used to open the originally closed vessels and equipment either briefly or permanently in a nonhazardous direction following an explosion. *Explosion venting is inadmissible when the escape of toxic or corrosive, irritating, carcinogenic, harmful-to-fruit, or genetically damaging substances is anticipated.* In contrast to containment, explosions in a vented vessel are characterized by the maximum reduced explosion overpressure $P_{red,max}$ instead of the maximum explosion overpressure P_{max} and by the maximum reduced rate of pressure rise $(dP/dt)_{red,max}$ instead of the maximum rate of pressure rise $(dP/dt)_{max}$.

By this method in general, the expected inherent maximum explosion overpressure on the order of $P_{max} = 7$ to 10 bar will be reduced to a value of $P_{red,max} < 2$ bar. In this case, the static activation overpressure of the venting device is $P_{stat} \leq 0.1$ bar. The resulting $P_{red,max}$ may not exceed the design pressure of the equipment. The explosion as such is not prevented; only the dangerous consequences are limited. However, subsequent fires must be expected.

Among other things, one prerequisite to calculate the pressure relief openings needed on the apparatus is knowledge about the explosion threat definition and venting system hardware definition. The various factors are summarized in Table 23-11.

The inertia, the opening behavior of a bursting disk or of the movable cover of an explosion device, and its arrangement (horizontal, vertical) can affect the venting efficiency and may result in a higher maximum reduced explosion overpressure inside the protected vessel (Fig. 23-13). This venting efficiency is mainly dependent upon the specific mass of the venting device.

If the specific mass of a venting device is $< 0.5 \text{ kg/m}^2$, then it has a venting efficiency of $EF = 1$ and is called inertia-free and does not impede the venting process (European Standard prEN 14797-2006, "Explosion Venting Devices"). For such explosion venting devices EF testing is therefore not required. Explosion venting devices with venting elements with a specific mass $> 0.5 \text{ kg/m}^2$ can influence the venting process by their opening and release behavior. Experiments have shown that explosion venting devices with a specific mass $> 0.5 \text{ kg/m}^2$ and $\leq 10 \text{ kg/m}^2$ can be considered as inertia-free, which means having a venting efficiency $EF = 1$ provided that

$$\frac{A}{V} < 0.07 \quad (23-11)$$

The equations are only valid for

- Vessel volumes of $0.1 \text{ m}^3 \leq V \leq 10,000 \text{ m}^3$
- Static activation overpressure of venting device $P_{stat} \leq 0.1$ bar

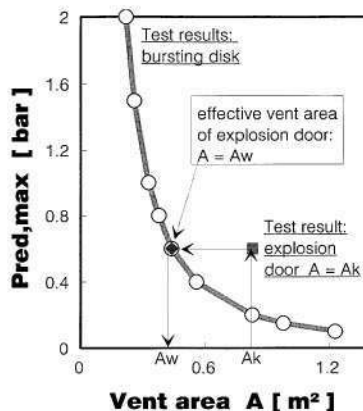


FIG. 23-13 Definition of the venting capability EF of an explosion door in comparison with a plastic foil rupture disk.

Venting Capability :

$$EF = \frac{A_w \cdot 100}{A_k} = \%$$

- Vessel strength ($= P_{red,max}$) of $0.1 \text{ bar} < P \leq 2 \text{ bar}$
- $P_{red,max} > P_{stat}$

For all other conditions EF has to be determined by tests (Fig. 23-13). EF and therefore the effective vent area A_w of a non-inertia-free explosion device are smaller than the venting efficiency of an inertia-free vent device (specific mass $< 0.5 \text{ kg/m}^2$) with the same vent area. Therefore, such devices need testing to determine the mechanical strength before actual use, and the EF or the pressure rise, respectively, has to be chosen relative to the $P_{red,max}$ of the rupture disk of the same area.

When explosion doors that close the vent area after the explosion are in use, the cooling of the hot gases of combustion may create a vacuum in the vessel, resulting in its deformation. To prevent this from happening, vacuum breakers have to be provided.

Sizing of vent areas The empirical equation (23-12) can be used to calculate the required vent area for flammable gas or solvent vapor explosions. The equation is valid for flammable gas-air mixtures which have been ignited in a quiescent state (nonturbulent) with an ignition source of $E = 10 \text{ J}$.

$$A = [(0.1265 \log K_{max} - 0.0567) P_{red,max}^{-0.5817} + 0.1754 P_{red,max}^{-0.5722} (P_{stat} - 0.1)] \times V^{2/3} + \frac{[AK_{max}(L/D - 2)^3]}{750} \quad (23-12)$$

The equation is valid for

- Vessel volumes $0.1 \text{ m}^3 \leq V \leq 1000 \text{ m}^3$
- Vessel length-to-diameter ratio $1 \leq L/D \leq 5$
- Static activation overpressure of rupture disk $0.1 \text{ bar} \leq P_{stat} \leq 0.5 \text{ bar}$
- Maximum reduced explosion overpressure $0.1 \text{ bar} \leq P_{red,max} \leq 2 \text{ bar}$
- $P_{red,max} > P_{stat} + 0.05 \text{ bar}$

- Maximum explosion constant $50 \text{ bar} \cdot \text{m/s} \leq K_{\max} \leq 550 \text{ bar} \cdot \text{m/s}$
- Gas-air mixtures ignited at zero turbulence
- Venting efficiency $EF = 1$

If it is necessary to locate equipment with explosion vents inside buildings, vent ducts should be used to direct vented material from the equipment to the outdoors. Vent ducts will significantly increase the pressure development in the equipment during venting. They require at least the same cross section as the vent area and the same design pressure as the protected equipment.

The use of vent ducts results in an increase in $P_{\text{red,max}}$. The maximum reduced explosion overpressure $P'_{\text{red,max}}$ caused by the downstream vent duct can be calculated with Eqs. (23-13) and (23-14).

Length of vent duct = $0 \text{ m} < L_D \leq 3 \text{ m}$:

$$P'_{\text{red,max}} = 1.24 P_{\text{red,max}}^{0.8614} \quad (23-13)$$

Length of vent duct = $3 \text{ m} < L_D \leq 6 \text{ m}$:

$$P'_{\text{red,max}} = 2.48 P_{\text{red,max}}^{0.5165} \quad (23-14)$$

where $P'_{\text{red,max}}$ = maximum explosion overpressure with vent duct, bar, and $P_{\text{red,max}}$ = maximum explosion overpressure without vent duct, bar.

The following empirical equations, Eqs. (23-15) to (23-17), allow the calculation of the size of a vent area A for combustible dust-air-mixture explosions.

For $P_{\text{red,max}} < 1.5 \text{ bar}$

$$A = B \left(1 + C \log \frac{L}{D} \right) \text{ m}^2 \quad (23-15)$$

For $P_{\text{red,max}} \geq 1.5 \text{ bar}$

$$A = B \text{ m}^2$$

with

$$B = [3.264 \cdot 10^{-5} \cdot P_{\max} K_{\max} P_{\text{red,max}}^{-0.569} + 0.27(P_{\text{stat}} - 0.1)P_{\text{red,max}}^{-0.5}] V^{0.753} \quad (23-16)$$

$$C = -4.305 \log P_{\text{red,max}} + 0.758 \quad (23-17)$$

The equations are valid for

- Vessel volumes $0.1 \text{ m}^3 \leq V \leq 10,000 \text{ m}^3$
- Static activation overpressure of venting device $0.1 \text{ bar} \leq P_{\text{stat}} \leq 1 \text{ bar}$
- Maximum reduced explosion overpressure $0.1 \text{ bar} < P_{\text{red,max}} \leq 2 \text{ bar}$
- $P_{\text{red,max}} > P_{\text{stat}}$
- Maximum explosion overpressure $5 \text{ bar} \leq P_{\max} \leq 10 \text{ bar}$ for a maximum explosion constant $10 \text{ bar} \cdot \text{m/s} \leq K_{\max} \leq 300 \text{ bar} \cdot \text{m/s}$
- Maximum explosion overpressure $5 \text{ bar} \leq P_{\max} \leq 12 \text{ bar}$ for a maximum explosion constant $300 \text{ bar} \cdot \text{m/s} \leq K_{\max} \leq 800 \text{ bar} \cdot \text{m/s}$
- Length-to-diameter ratio of the vessel $L/D \leq 20$
- L/D limited in that the maximum vent area shall not be greater than the cross-sectional area of the equipment
- Venting efficiency $EF = 1$

The required area for pressure venting increases with increased length (height) to diameter ratio of the vessel, in comparison with the area requirement for $L/D = 1$ vessel. For low $P_{\text{red,max}}$, the required effective vent area will be markedly influenced by the ratio L/D . Such influence diminishes with increasing reduced explosion overpressure and ceases at $P_{\text{red,max}} = 1.5 \text{ bar}$ as per experimental results. However, with $P_{\text{red,max}} \geq 1.5 \text{ bar}$, no influence of the L/D ratio of the vessel can be noticed.

The influence of the vent duct upon the pressure increase is most pronounced when the flame propagation from the secondary explosion in the vent duct reaches the velocity of sound. This is valid for vent ducts of

$$L_D = L_{DS} = 4.564 P_{\text{red,max}}^{-0.37} \text{ m} \quad (23-18)$$

Vent ducts with a length of $L_D > L_{DS}$ have no additional effect upon the pressure increase. Therefore, L_{DS} will be the maximum vent duct length that has to be considered. The above-mentioned equation (23-18) is not valid for metal dusts.

Independent of the location of the vent duct, the maximum reduced explosion overpressure $P'_{\text{red,max}}$ caused by the downstream vent duct can be calculated for vessels having an L/D ratio of 1 with Eq. (23-19).

Vessel ratio $L/D = 1$ (longitudinal und transversal):

$$P'_{\text{red,max}} = P_{\text{red,max}} [1 + 17.3 L_D (AV^{-0.753})^{1.6}] \quad (23-19)$$

where $P'_{\text{red,max}}$ = maximum explosion overpressure with vent duct, bar

$P_{\text{red,max}}$ = maximum explosion overpressure without vent duct, bar

A = vent area, m^2 , without vent duct

V = volume of protected vessel, m^3

L_D = length of vent duct, m

The equation is valid for

- Vessel volumes $0.1 \text{ m}^3 \leq V \leq 10,000 \text{ m}^3$
- Static activation overpressure of the venting device $0.1 \text{ bar} \leq P_{\text{stat}} \leq 1 \text{ bar}$
- Maximum reduced explosion overpressure $0.1 \text{ bar} < P_{\text{red,max}} \leq 2 \text{ bar}$, with $P_{\text{red,max}} > P_{\text{stat}}$
- Maximum explosion overpressure $5 \text{ bar} \leq P_{\max} \leq 12 \text{ bar}$ and a maximum explosion constant $10 \text{ bar} \cdot \text{m/s} \leq K_{\max} \leq 800 \text{ bar} \cdot \text{m/s}$
- Vessel $L/D = 1$
- Vent duct length $L_D \leq L_{DS}$

Experimental studies have proved that the influence of vent duct with longitudinal arrangement—located on the roof—decreases markedly with increased vessel length-to-diameter ratio. The increase of the maximum explosion overpressure is at its maximum if vessel ratio $L/D = 1$.

Hazard due to flame and pressure in the surroundings The maximum outside range of a flame L_F originating from a vessel increases with increased volume of the vented vessel.

Pressure and blast effects external to a vent arise from pressures generated by the vented explosion inside the plant and the explosion area outside the vent.

Explosion suppression During a suppression of an explosion, not products, residues from combustion, residues from gases, or flames can escape from the protected vessel, because an explosion suppression system reduces the effects of these explosions to a harmless level, by restricting the action of the flames during the initial phase of the explosion. This prevents the installation in question from being destroyed and people standing in the area of the installation from being injured. A further benefit of explosion suppression systems is that they can be deployed for combustible products with toxic properties and can be used irrespective of the equipment location.

An explosion can generally be considered suppressed if the expected maximum explosion pressure $P_{\max}$ at the optimum concentration of the combustible product (7 to 10 bar)—assuming the explosion suppression system has an activation overpressure P_a of 0.1 bar—is reduced to a maximum reduced explosion overpressure $P_{\text{red,max}} \leq 1 \text{ bar}$. This means that a vessel safeguarded in this way needs to be designed so that it is secured against explosions of up to 1 bar (equivalent to $P_{\text{red,max}}$). The activation overpressure P_a is that pressure at which an explosion suppression system will be activated.

To initiate an explosion suppression system, a detector is used to sense either an overpressure generated by, or a flame of, an incipient explosion. It is important to locate the detector in a position that ensures sufficient time for the suppression system to sense and activate the devices to extinguish the explosion.

Optical detectors shall be used in more open configurations where pressure buildup due to the incipient explosion is limited. Optical detectors shall not be used where high dust concentrations limit the reliability of the suppression system. Both uv and ir detectors are available for optical detection. The use of daylight-sensitive sensors shall be avoided to avoid spurious activation. The sensor shall be mounted such that the angle of vision allows it to cover all the protected hazard area. The performance of an optical detector will also be affected by any obstacles within its vision, and this shall be overcome by the introduction of more detectors. Optical detectors shall be fitted with air shields to keep the optical lens clean.

TABLE 23-12 Suppression System Design Parameters

Explosion hazard definition	Suppression system hardware definition
Volume of vessel (free volume V)	Type of explosion suppressant and its suppression efficiency
Shape of vessel (area and aspect ratio)	Type of HRD suppressors: number and free volume of HRD suppressors and the outlet diameter and valve opening time
Type of dust cloud distribution (ISO method/pneumatic-loading method)	Suppressant charge and propelling agent pressure
Dust explosibility characteristics:	Fittings: elbow and/or stub pipe and type of nozzle
Maximum explosion overpressure $P_{\max}$	Type of explosion detector(s): dynamic or threshold pressure, UV or IR radiation, effective system activation overpressure P_a
Maximum explosion constant $K_{\max}$	Hardware deployment: location of HRD suppressor(s) on vessel
Minimum ignition temperature MIT	

Pressure detection shall be used for closed enclosure applications. Threshold detectors provide an electric signal when a preset overpressure is exceeded. Dynamic detectors provide an electric signal to the control and indicating equipment (CIE). Typically they have both rate-of-rise and pressure threshold triggering points that can be configured specifically to the application conditions. Although this type of detector minimizes spurious activation of the isolation system (due to pressure fluctuations other than explosion pressure rise), care shall be taken to set up such detectors to meet appropriate detection response criteria for the particular application and protected enclosure geometry.

The suppressants deployed in suppression systems are water and dry and liquid chemicals. Apart from the effectiveness of the suppressant used, the compatibility of the suppressant with the process shall be considered. A suppressant is regarded as being very effective when an increase of the activation pressure P_a of the explosion suppression system leads to a small increase in the maximum reduced explosion overpressure $P_{\text{red,max}}$. The application of a suppressant is dependent upon how effective it is at suppressing an explosion. Testing shall be used to determine the effectiveness and performance of the suppressant, thus quantifying the applicability of the suppressant. The following parameters shall be considered when selecting a suppressant:

- Any adverse reaction with the process products
 - The toxicity levels of the suppressant relating to occupational exposure limits
 - The temperature stability of the suppressant
- In addition, the following properties shall be taken into account where necessary:

- Will the suppressant have to be food-compatible?
- Will the suppressant cause the onset of corrosion?
- Is the suppressant environmentally friendly?
- Can the suppressant be easily removed from the process?

Suppression system design parameters fall into the two categories of explosion hazard definition and suppression system hardware definition. The various influences are summarized in Table 23-12.

Comparison of explosion protection design measures In Table 23-13, comparison is made of the explosion protection design measures of containment, explosion venting and explosion suppression. In addition all three design measures are in combination with explosion isolation. Regarding the effectiveness of the different explosion design measures, all three techniques are equal if the design of these measures is performed properly and the design measures are inspected by a competent person at least once a year or more often depending on the process and/or environmental conditions.

Explosion Isolation For all equipment systems protected by design safety measures it is also necessary to prevent the propagation of an explosion from these protected vessels into operating areas or equipment connected via interconnecting pipeline. Such an approach is referred to as *explosion isolation*.

To prevent an explosion occurring in, e.g., a constructional protected installation from spreading through a pipeline ($\ell > 6$ m) to part of the installation fitted with preventive explosion protection, explosion isolation measures (see Fig. 23-14) must be implemented. As explosions are generally propagated by *flames* and not by pressure waves, it is especially important to detect, extinguish, or block this flame front at an early stage, i.e., to isolate or disengage the explosion. If there is no explosion isolation, then the flame issuing from the equipment, e.g., from the equipment protected through design (equipment part 1), through the connecting pipeline comes into contact with a highly turbulent recompressed mixture in the equipment with preventive protection (equipment part 2). The mixture will ignite in an instant and explode; a large increase in the rate of combustion reaction and, naturally, in the reduced explosion overpressure is the result. The equipment in question may be destroyed.

The isolation can be done with very different systems, which have in common that they become effective only by an explosion.

Since the action of the isolation systems requires the physical effects of an explosion, in the selection of a suitable system consideration must be given to process engineering and machine boundary

TABLE 23-13 Comparison of Explosion Protection Design Measures

	Containment with isolation	Explosion venting with isolation	Explosion suppression with isolation
Pressure resistance P	7–10 bar	Without relief pipe, up to 2 bar With relief pipe, up to 4 bar	St 1 up to 0.5 bar St 2+3 up to 1.0 bar
Location	Independent	Dependent	Independent
Limits of application	Products which decompose spontaneously	Toxic products and products which decompose spontaneously	Products which decompose spontaneously, metal dust hazard
Environmentally friendly	Yes	No (flame, pressure, and product)	Yes
Loss of material*	+++++	+++++	++
Maintenance requirements†	+++	++++	+++++

*The loss of material by using containment and explosion venting is always much greater than that by using explosion suppression.

†To ensure the reliability of explosion protection devices, regular servicing and maintenance are required. The nature and time intervals of these activities depend on technical specifications and on the plant situation. Normally, after commissioning of the plant, inspections are carried out in comparatively short intervals, e.g., every month. Positive experience may subsequently provide for longer service intervals (every three months). It is recommended to contract service and maintenance to reliable, specialized companies.

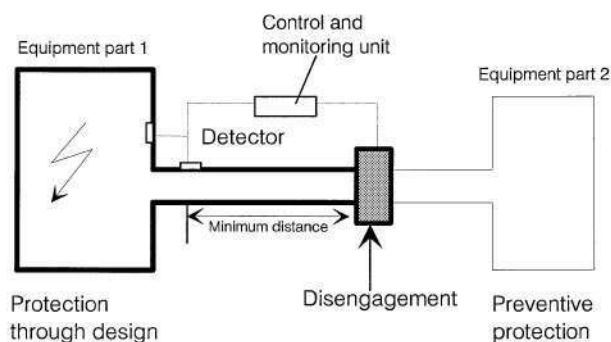


FIG. 23-14 Principle of the constructional measure explosion isolation.

conditions, particularly since the function and operability of these systems are not generally unrestricted.

Dust-carrying pipelines are different from gas-carrying pipelines— isolation devices can only be used which do not lose their function by the presence of the dust.

Today it is usual to divide the assigned isolation system in accordance with its mode of operation in *passive* isolation systems and *active* isolation systems. The passive isolation systems work without additional control units; i.e., the function (release) is determined by the physical effects of the explosion. Active isolation systems, however, are dependent on additional control and/or release mechanisms, without which they are nonfunctioning. Table 23-14 summarizes the different isolation systems.

Proper installation is dependent on the existing explosion protection measures and on the corresponding isolation system to be used. This guarantees operability. In addition the application and installation remarks of the manufacturer as well as the limits of application in accordance with the Type-Examination Certificate are to be obeyed in every detail. In Table 23-15 the most frequently used isolation systems for explosion protection measures are summarized.

An optical flame sensor installed at the beginning of the pipeline is the most suitable device for such an isolation system, since the propagating flame from the explosion has to be detected and extinguished. Pressure detectors alone are, in principle, not suited to the case on hand because there is no distinct separation between the pressure and flame fronts for explosion in pipelines. Optical ir sensors that have a relatively low sensitivity to daylight are normally chosen and have proved themselves amply in industrial practice. Therefore, daylight into the pipe in the vicinity of the sensor must be avoided. It is necessary to flush the optical lens with gas (e.g., nitrogen, air) to keep it dust-free.

TABLE 23-14 Different Isolation Devices

Type of isolation	Suitable for:
Rotary air lock (active)	Dusts
Extinguishing barrier (active)	Gas, dusts, and hybrid mixtures
Explosion protection valve (passive or active)	Gas, dusts, and hybrid mixtures
Explosion diverter (passive)	Gas, dusts, and hybrid mixtures
Double slide valve (active)	Gas, dusts, and hybrid mixtures
Product layer as a barrier (active)	Dusts and solvent humid products
Screw conveyer (passive)	Dusts and solvent humid products
Extinguishing barrier or explosion protection valve in combination with explosion diverter (active)	Gas, dusts, and hybrid mixtures
Flame arresters	Gas (see subsection "Flame Arresters," later)

TABLE 23-15 Most Frequently Used Isolation Systems as Related to the Installed Construction Explosion Protection Measure

Isolation system	Containment	Explosion venting	Explosion suppression
Double slide valve system	✓	✓	✓
Explosion diverter	✓	✓	✓
Explosion protection valve system	✓	✓	✓
Extinguishing barrier	—	✓	✓
Extinguishing barrier in combination with explosion diverter	—	✓	✓
Explosion protection valve in combination with explosion diverter	✓	✓	✓
Product layer as a barrier	—	✓	✓
Screw conveyer	✓	✓	✓
Rotary valve	✓	✓	✓

In difficult situations, it is recommended to install both types of sensors—pressure detector in the vessel and flame detector in the pipe—and they must be switched in an OR-logic to activate the isolation device (see Fig. 23-14).

The pressure detector in the vessel provides the earliest detection of an explosion in the interconnected vessel, whereas the flame detector in the pipeline provides assured detection, even for lazy flames, as they propagate down the pipeline toward the extinguishing barrier.

For the explosion protection valves always two installation distances are to be indicated, namely, the *minimum* and *maximum* installation distances. The *minimum installation distance* ensures that the given isolation system can react in time to prevent an explosion propagation beyond the installation place. The *maximum installation distance* ensures that a detonation in the piping cannot be formed up to the isolation system and/or that excessive pressure loads are avoided at the isolation system.

With extinguishing barriers the minimum and the maximum installation distances and the *extinguishing distance* must be considered after the extinguishing barrier. This *extinguishing distance* is specified as the minimum length after the extinguishing barrier to guarantee proper function of that extinguishing barrier. Only after this extinguishing distance may the piping connect to other equipment.

The minimum installation distance between the detector which activates the isolation system and the isolation system itself depends essentially on

- Type of detectors (pressure/flame)
- Volume of the vessel to be protected
- Maximum explosion overpressure $P_{\max}$ or maximum reduced explosion overpressure $P_{\text{red,max}}$ in the protected vessel
- Diameter of the pipeline
- Reaction time of the isolation system
- Maximum explosion overpressure $P_{\max}$ of the fuel
- Maximum explosion constant $K_{\max}$ of the fuel
- Minimum explosion constant $K_{\min}$ of the fuel (typically $\frac{1}{3} K_{\max}$, but minimum 50 m-bar/s)
- Forward air velocity in the pipeline
- Minimum ignition temperature of a dust cloud (only for extinguishing barriers)
- Type of suppressant (only for extinguishing barriers)

Finally, it must be pointed out that all explosion protection devices or systems used in practice may be used only when their pressure rating, flameproof, and functional testing have been proved by competent bodies and their test results including the limits of application are documented in a type test certificate.

Static Electricity

Nomenclature

C	Capacitance
C/kg	Unit of charge density
C/m ²	Unit surface charge density
F	Farad
J	Unit of energy (joules)
K _e	Relative dielectric constant, dimensionless
kV/m	Unit of electric field intensity
m	Meter
MIE	Minimum ignition energy, mJ
mJ	Millijoule
Ω ²	Resistivity value, ohms per square, usually used for fabrics and films
pS	Unit of conductivity (picosiemens)
pS/m	Unit of electrical conductivity of liquid
RH	Relative humidity, %
S	Siemen, formerly mho
T	Time, s
V	Electric potential, V
V/m	Unit of electrical field intensity

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Introduction Spark ignition hazards must be considered whenever static charges may accumulate in an environment that contains a flammable gas, liquid, or dust. The need for electrical bonding and grounding of conductive process equipment in hazardous (classified) locations is widely recognized. Less well understood are the ignition hazards associated with static charges on poorly conductive, flammable liquids, solids, and powders. Static charges, generated on these materials by normal handling and processing, cannot be conducted to ground quickly, and may cause hazardous charge accumulations. The electric fields associated with these charges may stress the surrounding air sufficiently to cause breakdown by some type of electrical discharge.

Electrical discharges from poorly conductive materials take several forms, each differing in its ability to ignite flammable mixtures. It is not possible to calculate the incendivity of most of these discharges, because of their varying time and spatial distributions. Several engineering rules of thumb for estimating the relative hazard of these discharges are discussed below.

An analysis of static ignition hazard should start with data on the ignition sensitivity of the particular flammable material at its most flammable concentration in air, i.e., its MIE. This is especially important for dusts. It is prudent to determine this value on fines of the specific dust of interest, rather than to rely on published data. *Hybrid mixtures*, i.e., mixtures of dust and vapor for which vapor concentrations may be below their lower flammable limit, can be ignited by smaller discharge energies than might be expected.

The key to safe operation is to provide an adequate means of charge dissipation from charged materials to ground. This requires mobility of charges in or on the charged material *plus* electrical continuity from the material to ground.

Definitions

Bonding A method of providing electrical continuity between two or more conductive objects to prevent electrical sparking between them.

Charge relaxation time The time required for a charge in a liquid or on a solid material to dissipate to 36.8 percent of its initial value when the material is grounded.

Electrical discharge A current flow that occurs when the electrical field strength exceeds the dielectric breakdown value of a medium such as air.

Flammable mixture A mixture of a gas, vapor, mist, or dust in air which is within its flammable range.

Grounding A special form of bonding in which a conductive object is connected to (earth) ground.

Incendive discharge Any electrical discharge that has sufficient energy to ignite a specified flammable mixture.

Minimum ignition energy The smallest amount of spark energy that has been found capable of igniting a specified flammable mixture in a standard test.

Static-dissipative (antistatic) material One with an electrical resistivity that is low enough to make it incapable of accumulating hazardous concentrations of static charges when grounded.

Electrostatic Charging The primary cause of electrostatic charging is contact electrification, which takes place when two different materials are brought into contact and separated. Other causes include induction charging, the formation of sprays, and impingement of charged mist or particles on an ungrounded conductor.

Contact electrification involves the contact and separation of solid-solid, solid-liquid, or liquid-liquid surfaces. Pure gases do not cause charging unless they carry droplets or dust particles.

Efforts to quantify the magnitude and polarity of contact charging have had limited success, because minute variations in the types and concentration of contaminants exert a large influence on charge separation. Even like solid-solid surfaces can produce significant charge separation. The charge density on separated surfaces is usually very nonuniform. Each surface may contain both + and – charges, with more of one polarity than of the other. After separation, the charges dissipate slowly or rapidly, depending upon the electrical resistivity of the material and the presence of a path to ground.

Contact electrification at liquid-liquid and liquid-solid surfaces is attributed to the adsorption of ions of one polarity by one surface. Ions of opposite polarity form a diffuse layer near the interface. If the diffuse layer is carried along by moving liquid, or in a pipeline, the flowing charges (called a *streaming current*) may create a sparking hazard downstream. One protective measure is to keep the charged liquid in a closed, grounded system (a *relaxation chamber*) long enough to allow for safe dissipation of the charges.

The magnitude of the streaming current in any given situation is not readily calculated. Equations, derived experimentally, for some liquids (Bustin and Dukek, *Electrostatic Hazards in the Petroleum Industry*, Research Studies Press, Letchworth, England, 1983) show that flow velocity and filters have the greatest influence on pipeline charging. Streaming currents can usually be limited to safe levels by limiting velocities to less than 1 m/s.

Charge induction takes place when a conducting object is exposed to electric fields from other objects. Examples include the induction charging of a human body by charged clothing, the charging of a conductive liquid in a charged plastic container, and the charging of the conductive coating on one-side-metallized film by static charges on the uncoated surface.

Although charge induction can take place whether or not the conductive object is grounded, a sparking hazard is present only if the conductor is not grounded. This phenomenon can convert a relatively innocuous charge buildup on a nonconductor to a serious sparking hazard by raising the potential of the conductor aboveground (Owens, "Spark Ignition Hazards Caused by Charge Induction," *Plant/Operations Progress* **7**, no. 1, pp. 37–39, 1988).

Droplets, formed by **spray nozzles**, tend to be highly charged, even if the conductivity of the liquid is high. Because there is no path to ground from the droplets, their charges can accumulate on an ungrounded conductor to cause sparking. If flammable vapor is present, as in some tank cleaning operations, it is essential that the spray nozzle and the tank be bonded or separately grounded. Other precautions include the use of a nonflammable cleaning solvent or the use of an inert gas.

Although charged **mists** are unable to cause ignition of flammable vapor by self-generated sparking, it is important that the mist not impinge on an ungrounded conductor.

Charge Dissipation It is an experimental fact that charged objects exert a force on other charged objects. This behavior is explained by the presence of an *electric field*, i.e., electric lines of force, each of which emanates from a + charge and terminates on a - charge. The *magnitude* of the field is defined as the force on a unit test charge, placed at the point of interest. The *direction* of the field is the direction of the force on a + test charge placed at that point.

Static charge generation causes an ignition hazard only if the accumulated charges create an electric field that is sufficient to produce an electrical discharge in a flammable atmosphere. In most processes, this means that the electric field intensity at some location must reach the dielectric breakdown strength of air (nominally 3×10^6 V/m). The objective of static control measures is to ensure that electric field intensities cannot reach this value.

Bonding and grounding are the primary means of dissipating charges from **conductive objects**. Bonding clamps should be of the single-point type, which bites through oxide or enamel coatings to make contact with the bare metal. Owing to the sturdy construction of bonding clamps and cables, their initial resistance is less than 1 Ω . It is good practice to visually inspect the condition of bonding cables and clamps during each use, and to measure the resistance of temporary bonding cables at least annually, to confirm that it is less than, say, 25 Ω .

Charge-dissipative materials allow static charges to dissipate without causing hazardous accumulations. Charge dissipation normally takes place by conduction along the material to ground. The charge-dissipating behavior of such materials is measured at a controlled temperature and relative humidity, in terms of Ω^2 (ohms per square) of *electrical surface resistivity*. The maximum safe resistivity depends, in part, upon the rate of charge generation, but is typically in the range of 10^8 to 10^{11} Ω^2 for fabrics and films [ASTM Standard Test Method D257-99 (2005), "DC Resistance or Conductance of Insulating Materials"].

An alternate test for charge-dissipating performance is the *charge decay* test, in which the time of charge decay is measured after a potential of 5 kV has been applied to the specimen (Federal Test Method Standard 101C, Method 4046.1). For many purposes, a charge decay time of 0.5 s to 500 V, measured at the RH in end use, indicates acceptable performance.

The electrical surface resistivity and the charge decay time of most materials vary substantially with RH. It is important that test specimens be conditioned and tested at the lowest RH expected during use. Items that are acceptable at 50 percent RH may not be safe at 20 percent RH.

Some fabrics contain a small percentage of conductive or antistatic fibers or staple, which limit charge accumulation by *air ionization*. These static-dissipative fabrics do not depend upon electrical conduction of static charges, and may not pass the resistivity or the charge decay test. Their performance is not humidity-dependent. Antistatic performance is determined by measuring the charge transferred in electrical discharges from the charged fabric, and by the ability of these discharges to ignite flammable mixtures having a known MIE.

The rate of dissipation of charges in **liquid**, assuming that its conductivity and dielectric properties are constant, can be expressed as

$$T = 8.85K_r/C \quad (23-20)$$

where T = time required for charge density to dissipate to 36.8% of its initial value, s

K_r = relative dielectric constant of liquid, dimensionless

C = electrical conductivity of liquid, pS/m

Flammable liquids are considered particularly static-prone if their electrical conductivity is within the range of 0.1 to 10 pS/m. If no particulates or immiscible liquids are present, these liquids are considered safe when their conductivity has been raised to 50 pS/m or higher. Blending operations or other two-phase mixing may cause such a high rate of charging that a conductivity of at least 1000 pS/m is needed for safe charge dissipation (British Standard 5958, part 1, "Control of Undesirable Static Electricity," para. 8, 1991).

Electrostatic Discharges An electrostatic discharge takes place when a gas- or a vapor-air mixture is stressed electrically to its break-

down value. Depending upon the specific circumstances, the breakdown usually appears as one of four types of discharges, which vary greatly in origin, appearance, duration, and incendivity.

Spark discharges are most common between solid conductors, although one electrode may be a conductive liquid or the human body. They appear as a narrow, luminous channel and carry a large peak current for a few microseconds or less. Sparks are the only form of discharge for which a maximum energy can be calculated, by using the expression

$$J = 0.5CV^2 \quad (23-21)$$

where J = total energy dissipated, J

C = capacitance of charged system, F

V = initial potential difference between electrodes, V

Incident investigations often require that estimates be made of the possible spark energy from an ungrounded conductor. If the discharge path contains significant resistance, some of the stored energy is dissipated in the resistance, thereby lowering the energy in the spark gap.

A **corona** is generated when a highly nonuniform electric field of sufficient strength terminates on a conductor that has a small radius of curvature, i.e., a point, wire, or knife edge. The luminous (breakdown) region is confined to a small volume near the corona electrode. Because of their small peak currents and long duration, corona discharges do not have sufficient energy to ignite most flammable materials found in industry, i.e., materials having an MIE greater than 0.2 mJ. For this reason, nonpowered devices that employ corona discharges for static neutralization can be used safely in most hazardous (classified) locations. Corona discharges *can* ignite hydrogen-air and oxygen-enriched gas mixtures.

Brush discharges take place between conductors and charged nonconductors, where the radius of curvature of the conductor is too large for corona generation. The name refers to the brushlike appearance of the discharge, which spreads from the conductor to discrete areas on the nonconductor. The brush discharge may have a hot "stem" near the conductor, which may cause ignition by raising the temperature of the flammable mixture to its *autoignition* value.

Brush discharges from - charged nonconductors have been found more incendive than those from + charged nonconductors. Brush discharges may ignite flammable mixtures that have an MIE of less than 4 mJ. This limitation arises because charges from a small area on the nonconductor are able to participate in the discharge. Most dust-air mixtures cannot be ignited by brush discharges, because their MIE exceeds 4 mJ.

Surface charge densities cannot exceed the theoretical value of 2.7×10^{-5} C/m² set by air breakdown, and are normally less than 1.5×10^{-5} C/m².

Propagating brush discharges are much less common than brush discharges. They may occur when a nonconductive film or plastic layer acquires a double layer of charges, i.e., + charges on one surface and - charges on the opposite surface. With the electric fields within the film, surface charge densities can be large, because they are not limited by air breakdown.

The double layer can be formed by contact (triboelectric) charging on one surface of the nonconductor while the opposite surface is in contact with a conductor, e.g., a nonconductive coating on a metal chute or a plastic-lined, metal pipe for powders. A less frequent cause is contact charging of one surface of a nonconductor while air ions collect on the opposite surface.

Investigations by Glor ("Discharges and Hazards Associated with the Handling of Powders," *Electrostatics* 1987, Inst. Phys. Conf. Ser. no. 85, pp. 207-216, 1987) and others conclude that propagating brush discharges require surface charge densities above 2.7×10^{-4} C/m². In addition, the dielectric breakdown voltage of the insulating layer must exceed 4 kV for a thickness of 10 μ m, or 8 kV for a thickness of 200 μ m.

If a conductor approaches the charged surface, the electric field will produce air ionization at the surface, which creates a semiconductive layer, thereby allowing charges from a large area to participate

in a single discharge. Because these discharges can have energies of 1 J or more, they are very hazardous in a flammable environment. They may also cause severe shocks to operators who reach into a nonconductive container that is receiving charged powder, pellets, or fibers.

Causes of Hazardous Discharges with Liquids Self-generated discharges in vapor-air mixtures can be ignited by static discharges from highly charged liquids. Such liquids are said to "carry their own match." Typical causes of such charging for poorly conductive ($< 50 \text{ pS/m}$) liquids include

1. High-velocity flow
2. Free-fall/splashing
3. Filtering
4. Spraying
5. Agitation with air
6. Blending with powder
7. Settling of an immiscible component, e.g., water in gasoline
8. Liquid sampling from pressurized lines, using ungrounded or nonconductive containers

Conductive liquids in nonconductive containers may cause sparking if the outside of the container is charged by rubbing.

External causes of incendive static discharges include

1. Sparks from ungrounded conductors or persons
2. Brush discharges from flexible intermediate bulk containers (FIBCs), plastic bags, stretch wrap, or other plastic film
3. Propagating brush discharges from metal-backed, plastic film or linings

Powders Contact charging of powders occurs whenever particles move relative to one another or to a third surface. Significant charging is most often generated by pneumatic transfer. Maximum charge densities (C/kg) on airborne powders increase as particle size decreases, because of larger surface/mass ratios. Dry fines can be expected to charge more highly than those containing moisture. While suspended in air, charged powder poses an ignition risk only if nonconductive piping is used in the conveying lines, or if conductive piping is not bonded. The collected powder may accumulate so much charge per unit volume that the associated electric fields cause breakdown of the surrounding air in the form of a corona or a brush discharge. For receiving containers larger than about 1 m^3 , bulking or cone discharges may be present. These discharges typically have energies of less than 10 mJ, but the value is sometimes higher. The ignition hazard from bulking discharges can be minimized by, e.g., using a rotary valve or bag filter to prebulk small volumes of charged powder prior to its collection in a large receiver.

Personnel and Clothing Sparks from ungrounded persons pose a serious ignition hazard in flammable gas-air, vapor-air, and some dust-air mixtures, because the body is a conductor and can store energies as high as 40 mJ. Induction of static charges on a person's ungrounded body by charged clothing is a common cause of personnel electrification. Even at the threshold of shock sensation, the stored energy is about 1 mJ.

It is essential that persons be grounded in hazardous (classified) locations. For most chemical operations, the resistance from skin to ground should not exceed $100 \text{ M}\Omega$. A lower allowable resistance may be specified for locations where the presence of primary explosives, hydrogen-air mixtures, oxygen-enriched mixtures, or certain solid-state devices requires faster charge dissipation.

The combination of conductive flooring and conductive (ESD) footwear is the preferred method of grounding. Untreated concrete flooring with conductive footwear is usually adequate, but the resistance to ground should be measured. Where this method is impractical, personnel-grounding devices are available.

In most chemical plants, grounded persons can wear any type of clothing safely. For the unusually sensitive environments noted above, charge-dissipative or conductive clothing should be worn, and personnel should be grounded. Removal of outer garments in a flammable location can cause hazardous discharges and should be avoided. Although most gloves used in the chemical industry show a resistance of less than $100 \text{ M}\Omega$ from the wearer's palm to a handheld electrode, this value should be verified.

CHEMICAL REACTIVITY

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Introduction Chemical reactivity is the tendency of substances to undergo chemical change. A chemical reactivity hazard is a situation with the potential for an uncontrolled chemical reaction that can result directly or indirectly in serious harm to people, property, or the environment. A chemical reaction can get out of control whenever the reaction environment is not able to safely absorb the energy and products released by the reaction. The possibility of such situations should be anticipated not only in the reaction step of chemical processes but also in storage, mixing, physical processing, purification, waste treatment, environmental control systems, and any other areas where reactive materials are handled or reactive interactions are possible.

The main business of most chemical companies is to manufacture products by means of controlled chemical reactions. The reactivity that makes chemicals useful can also make them hazardous. Chemical reactions are usually carried out without mishap, but sometimes they get out of control because of problems such as the wrong or contaminated raw material being used, changed operating conditions, unanticipated time delays, failed equipment, incompatible materials of construction, or loss of temperature control. Such mishaps can be worse if the chemistry under both normal and abnormal conditions is not fully understood. Therefore, it is essential that chemical process designers and operators understand the nature of the reactive materials and chemistry involved and what it takes to control intended reactions and avoid unintended reactions throughout the entire life cycle of a process facility.

Life-Cycle Considerations

Considering Chemical Reactivity during Process Development Decisions made at the early development stages of a process facility, including conceptual and research phases, will in large part determine the nature and magnitude of the chemical reactivity hazards that will need to be contained and controlled throughout the entire life cycle of the facility. For this reason, chemical reactivity hazards should be considered from the outset of process development, including creative thinking regarding feasible alternatives to the use of reactive materials or the employment of highly energetic reactive systems. What may seem reasonable to the research chemist—handling materials in very small quantities—will have vast implications to the design and ongoing operation of a full-scale facility that must safely control the intended chemical reactions and avoid unintended reactions throughout the entire facility lifetime.

Mosley et al. describe a chemistry hazard and operability (CHA-ZOP) analysis approach, similar to a HAZOP study but applied at the early development stages of a new process.

Many companies designate a particular person or position as the "owner" of the process chemistry; this responsibility is likely to change as the life cycle progresses from development to design, construction, and operation. Data on the hazardous properties of the chemical reactions to be employed and the materials to be handled should begin to be assembled into a formal documentation package. Screening tests (described later in this section) may also need to be performed early in the development process to identify consequences of abnormal reactions and of deviations such as exceeding the normal reaction temperature. This documentation package will then form part of the information base upon which safeguards can be developed to control chemical reactivity hazards.

Considering Inherently Safer Approaches Specific to Reactivity Hazards The basic concepts of inherently safer plants, and the general strategies for making a facility inherently safer, are detailed in the later subsection on Inherently Safer and More User-Friendly Design. Strategies that focus on chemical reactivity hazards, and steps to conduct a review of these strategies, are highlighted in that section.

Instead of choosing to receive and store a highly reactive raw material, it may be possible to use a less hazardous material that is one step farther along in the formulation or synthesis chain. Alternatively, a decision may be made to generate the material on demand and eliminate all or most storage and handling of the material. Many reactive materials can be handled in dilute solutions, dissolved in less hazardous solvents, or otherwise handled under inherently safer conditions. (For some reactive materials such as benzoyl peroxide, handling as a dilute paste or solution is essential to the safe handling of the material.)

Inherently safer facilities with respect to chemical reactivity hazards must focus on the magnitude of stored chemical energy, the kinetics of how fast the energy could be released, and the possible reaction products that may themselves have hazardous properties such as toxicity or flammability.

With respect to kinetics, a slower reaction might be considered at first glance to be the inherently safer option as compared to a rapid reaction. This may indeed be the case, if the energy and products of the slower reaction can always be dissipated safely without causing harm or loss. However, this is often not the case, for two important reasons. First, regardless of the speed of the reaction, the same potential chemical energy is still thermodynamically present if only the kinetics is changed, and may be available under abnormal conditions such as an external fire or the introduction of a catalytic contaminant. Second, a slower reaction may allow unreacted material to accumulate. Hence faster reactions are generally more desirable, as discussed in the general reaction considerations below.

Finally, with respect to reaction products, a chemical reaction that does not generate hazardous reaction products or by-products is inherently safer than one that does. Thought must be given not only to hazardous reaction products, however. The generation of any kind of noncondensable gases can cause a vessel rupture due to internal overpressurization, if not adequately vented or relieved.

The following is a typical agenda for an inherent safety review at the concept or development stage of a new facility involving reactivity hazards (Johnson et al. 2003):

1. Review what is known of the chemical reactivity hazards (as well as other hazards) that will need to be contained and controlled in the proposed process. This existing level of knowledge might come from past experience, suppliers, literature reviews, incident reports, etc.
2. Based on the level of knowledge of chemical reactivity hazards, determine if additional screening of reactivity hazards is necessary. Having reactive functional groups might indicate the need to perform literature searches, access databases, or run differential scanning calorimetry.
3. Discuss possible process alternatives and their relative hazards, including discussions on such topics as alternative solvents and possible incompatibilities to avoid.
4. Brainstorm and discuss possible ways to reduce the hazards.
5. Obtain consensus on significant unknowns that will need to be addressed.

6. Document the review, including attendees, scope, approach, and decisions.

7. Assign follow-up items, with responsibilities, goal completion dates, and a closure mechanism such as reconvening after a designated number of weeks.

Scale-up Considerations A key consideration when scaling up a reactive process, such as from a pilot plant to a full-scale facility, is to ensure adequate heat removal for normal or abnormal exothermic reactions. Heat generation is proportional to *volume* (mass) in a reactive system, whereas heat removal is only proportional to *area* (surface area) at best. Even though the reaction temperature can be easily controlled in the laboratory, this does not mean that it can be adequately controlled in a plant-scale reactor. Increasing the size of a reactor, or of another process or storage vessel where, e.g., polymerization or slow degradation can occur, without adequately considering heat transfer can have disastrous effects. The careful design of the agitation or recirculation system is likewise important when scaling up, and the combined effects on the design of the emergency relief system must be taken into account.

Scale-up can also have a significant effect on the basic process control system and safety systems in a reactive process. In particular, a larger process will likely require more temperature sensors at different locations in the process to be able to rapidly detect the onset of out-of-control situations. Consideration should be given to the impact of higher-temperature gradients in plant-scale equipment compared to a laboratory or pilot plant reactor (Hendershot 2002).

Designing Processes for Control of Intended Chemical Reactions

General Considerations The following should be taken into account whenever designing or operating a chemical process that involves intended chemical reactions (Hendershot 2002). CCPS (1999) also details many key issues and process safety practices to consider that are oriented toward the design and operation of batch reaction systems.

- *Know the heat of reaction for the intended and other potential chemical reactions.*
- *Calculate the maximum adiabatic temperature for the reaction mixture.*
- *Determine the stability of all individual components of the reaction mixture at the maximum adiabatic reaction temperature.* This might be done through literature searching, supplier contacts, or experimentation.
- *Understand the stability of the reaction mixture at the maximum adiabatic reaction temperature.* Are there any chemical reactions, other than the intended reaction, that can occur at the maximum adiabatic reaction temperature? Consider possible decomposition reactions, particularly those which generate gaseous products.
- *Determine the heat addition and heat removal capabilities of the reactor.* Don't forget to consider the reactor agitator as a source of energy—about 2550 Btu/(h·hp).
- *Identify potential reaction contaminants.* In particular, consider possible contaminants, which are ubiquitous in a plant environment, such as air, water, rust, oil, and grease. Think about possible catalytic effects of trace metal ions such as sodium, calcium, and others commonly present in process water.
- *Consider the impact of possible deviations from intended reactant charges and operating conditions.* For example, is a double charge of one of the reactants a possible deviation, and, if so, what is the impact?
- *Identify all heat sources connected to the reaction vessel and determine their maximum temperature.*
- *Determine the minimum temperature to which the reactor cooling sources could cool the reaction mixture.*
- *Understand the rate of all chemical reactions.* Thermal hazard calorimetry testing can provide useful kinetic data.
- *Consider possible vapor-phase reactions.* These might include combustion reactions, other vapor-phase reactions such as the reaction of organic vapors with a chlorine atmosphere, and vapor-phase decomposition of materials such as ethylene oxide or organic peroxide.

- *Understand the hazards of the products of both intended and unintended reactions.*
- *Rapid reactions are desirable.* In general, you want chemical reactions to occur immediately when the reactants come into contact.
- *Avoid batch processes in which all the potential chemical energy is present in the system at the start of the reaction step.*
- *Avoid using control of reaction mixture temperature as a means for limiting the reaction rate.*
- *Avoid feeding a material to a reactor at a higher temperature than the boiling point of the reactor contents.* This can cause rapid boiling of the reactor contents and vapor generation.

Exothermic Reactions and “Runaway Reactions” The term *runaway reaction* is often improperly used to refer to any uncontrolled chemical reaction. As properly used, it refers to loss of control of a kinetically limited, exothermic reaction that proceeds at a stable, controlled rate under normal conditions and that includes adequate removal of the heat of reaction (Fig. 23-15). When the situation changes such that the heat of reaction is not adequately removed, the excess heat increases the temperature of the reaction mass, which in turn increases the reaction rate and thus the rate of heat release as an exponential function of reaction temperature. If not limited by some means such as (1) the limiting reactant being exhausted, (2) a solvent removing the heat of reaction by boiling off, or (3) quenching or inhibiting the reaction, this “bootstrap” situation can result in an exponential temperature rise that can reach as high as hundreds of degrees Celsius per minute. The resulting temperature increase, generation of gaseous reaction products, and/or boiloff of evaporated liquid can easily exceed a pressure and/or thermal limit of the containment system, if not adequately relieved. The elevated temperatures may also initiate a secondary or side reaction that is even more rapid or energetic.

This runaway situation can be understood by comparing Fig. 23-15 with Fig. 23-16, which has two new lines added, for two possible upset conditions in a process with a cooling coil or other heat exchanger being used to absorb the heat of an exothermic reaction. The temperature of the cooling medium might increase (shift from line 1 to line 2), or the heat-transfer coefficient might decrease, such as by heat exchanger fouling (shift from line 1 to line 3). When one of these shifts gets past point T_{NR} (temperature of no return), the heat removal line no longer crosses the heat generation line, and stable operation is no longer possible. The heat of reaction causes the system temperature to increase, which further increases the rate of heat generation, which further increases the system temperature, etc.

Many possible abnormal situations can initiate a runaway reaction. These include

- Loss of flow of cooling medium to/from the reactor
- An increase in the temperature of the cooling medium

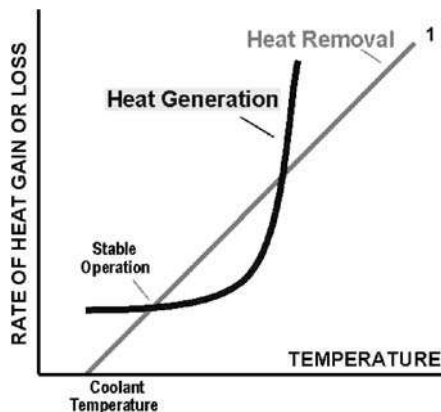


FIG. 23-15 For stable operation, all heat generated by an exothermic reaction is transferred to the surroundings, by whatever means (conduction, evaporation, etc.).

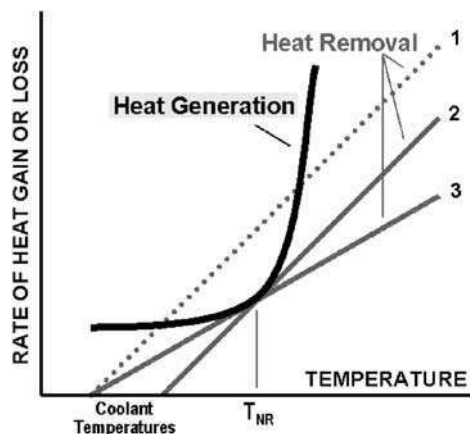


FIG. 23-16 For an exothermic reaction system with heat removal, e.g., to a vessel jacket and cooling coil, the limit of stable operation is reached as the reaction temperature increases to T_{NR} (temperature of no return), beyond which the rate of heat generation, which increases exponentially with increasing temperature, exceeds the capability of the system to remove the heat of reaction (see text).

- A general increase in the temperature of the storage or process configuration, such as due to an extreme ambient condition or loss of refrigeration
- Abnormal heat addition to the reactive material or mixture, such as by an external fire or the injection of steam to a vessel jacket or directly into the material or mixture
- Intentional heating of a vessel containing thermally sensitive material, due to lack of recognition of the runaway hazard or other reason
- Gradual fouling of the heat exchange surfaces to the point that maximum coolant flow is no longer sufficient to remove the heat of reaction
- Loss of agitation or circulation of the reactant mass or other reduction in the heat-transfer coefficient or contact with the heat exchange surface
- Insulation of the system, resulting in less heat dissipation
- Addition of a contaminant or excess catalyst which would increase the reaction rate
- Excess or too rapid addition of a limiting reactant
- Blockage of a vapor line or other means of increasing the system pressure
- Loss of a moderating diluent or solvent
- Inadequate inhibitor concentration in a storage container, or inadequate mixing of the inhibitor (including due to freezing of the material)
- Transfer of the reactive material or mixture to a location not capable of removing the heat of reaction

As can be seen from the above list, runaway reactions do not occur by a single mechanism. They can take place not only in reactors but also in raw material and product storage containers and vessels, purification systems, and anywhere else exothermic reactive systems and self-reacting materials (as described below) are involved.

Historical perspective An analysis of thermal runaways in the United Kingdom (Barton and Nolan, "Incidents in the Chemical Industry due to Thermal Runaway Chemical Reactions," *Hazards X: Process Safety in Fine and Specialty Chemical Plants*, IChem 115: 3–18) indicated that such incidents occur because of the following general causes:

- Inadequate understanding of the process chemistry and thermochemistry
- Inadequate design for heat removal
- Inadequate control systems and safety systems
- Inadequate operational procedures, including training

Semibatch reactions The inherently safer way to operate exothermic reaction processes is to determine a temperature at which the reaction occurs very rapidly (Hendershot 2002). The reactor can be

operated at this temperature, while feeding at least one of the reactants gradually to limit the potential energy contained in the reactor. This type of gradual addition process is often called *semibatch*. A physical limit to the possible rate of addition of the limiting reactant is desirable—e.g., a metering pump, flow limited by using a small feed line, or a restriction orifice. Ideally, the limiting reactant should react immediately, or very quickly, when it is charged. The reactant feed can be stopped if necessary, if there is any kind of a failure (e.g., loss of cooling, power failure, loss of agitation), and the reactor will contain little or no potential chemical energy from unreacted material. Some means to confirm actual reaction of the limiting reagent is also desirable. A direct measurement is best, but indirect methods such as monitoring of the demand for cooling from an exothermic batch reactor can also be effective.

Design of Emergency Relief and Effluent Treatment Systems Containment systems are only rarely designed with sufficient pressure and temperature rating to fully contain a runaway reaction. For this reason, overpressure protection is of obvious critical importance as a last line of defense against loss events that can result from runaway reactions. The latter sections in this chapter on Pressure Relief Systems and on Emergency Relief Device Effluent Collection and Handling address design basis selection, relief calculations, and effluent treatment system configurations for reactive system overpressure protection.

Endothermic Reactions An endothermic reaction process is generally easier to bring to a safe state if an out-of-control situation is detected. Discontinuing the heat input is usually the primary line of defense to stop the operation. In this regard, the endothermic reaction is inherently safer than an exothermic reaction.

The following should especially be taken into account:

- The final product of an endothermic chemical reaction has a greater energy content than the starting materials. For this reason, materials with net positive heats of formation are often termed *endothermic compounds*. (Most explosives, e.g., are endothermic compounds.) This energy content can potentially be released in an uncontrolled manner if sufficient energy is again added to the material, such as by heating it to a decomposition temperature.
- Likewise, if control is lost of an endothermic reaction process, such as by a heating control valve opening too far or by a steam leak directly into the reaction mass, a degradation reaction or other secondary or side reaction may be initiated that can be exothermic and can lead to a thermal runaway.
- Some endothermic compounds can gradually degrade, decompose, become more concentrated, or become sensitized over time.

Designing Facilities for Avoidance of Unintended Reactions

General Considerations The following general design and operational considerations for avoiding unintended chemical reactions are summarized from a CCPS Safety Alert (2001):

- Train all personnel to be aware of reactivity hazards and incompatibilities and to know maximum storage temperatures and quantities.
- Design storage and handling equipment with all compatible materials of construction.
- Avoid heating coils, space heaters, and all other heat sources for thermally sensitive materials.
- Avoid confinement when possible; otherwise, provide adequate emergency relief protection.
- Avoid the possibility of pumping a liquid reactive material against a closed or plugged line.
- Locate storage areas away from operating areas in secured and monitored locations.
- Monitor material and building temperatures where feasible with high-temperature alarms.
- Clearly label and identify all reactive materials and what must be avoided (e.g., heat, water).
- Positively segregate and separate incompatible materials, using dedicated equipment if possible.
- Use dedicated fittings and connections to avoid unloading a material to the wrong tank.
- Rotate inventories for materials that can degrade or react over time.

- Pay close attention to housekeeping and fire prevention around storage/handling areas.

Identifying Potential Reactions The U.S. Chemical Safety and Hazard Investigation Board's Hazard Investigation "Improving Reactive Hazard Management" (2002) highlighted the importance of identifying chemical reactivity hazards as a result of an examination of 167 previous reactive incidents. CCPS has published a preliminary screening methodology for identifying where reactive hazards are likely to exist (Johnson et al., 2003). The flowchart for the preliminary screening methodology is shown later in the Hazard Analysis subsection.

The following paragraphs break down the types of reactive materials and reactive interactions that an engineer may need to address in the design of a chemical process or other facility such as a warehouse where reactive materials are handled (Johnson and Lodol, 2003; Johnson et al., 2003). These can be considered to be in three larger categories:

- Self-reactive substances (polymerizing, decomposing, rearranging)
- Substances that are reactive with ubiquitous substances such as air (spontaneously combustible/pyrophoric, peroxide-forming), water (water-reactive), or ordinary combustibles (oxidizers)
- Incompatible materials

Polymerizing, Decomposing, and Rearranging Substances Most of these substances are stable under normal conditions or with an added inhibitor, but can energetically self-react with the input of thermal, mechanical, or other form of energy sufficient to overcome its activation energy barrier (see Sec. 4, Reaction Kinetics, Reactor Design, and Thermodynamics). The rate of self-reaction can vary from imperceptibly slow to violently explosive, and is likely to accelerate if the reaction is exothermic or self-catalytic.

The tendency of a material such as acrylic acid or styrene to *polymerize* is usually recognized, and the material safety data sheet should be checked and the supplier can be contacted as to whether hazardous polymerization might be expected. A less energetic means of self-reaction is by molecular *rearrangement* such as by isomerizing, tautomerizing, disproportionating, or condensing.

The *decomposition* of some materials into smaller, more stable molecules can be initiated by mechanical shock alone, and they are known as *shock-sensitive*. Many commercially important chemicals are *thermally sensitive* and decompose with the addition of heat. For storage situations, the critical temperature at which the thermal energy is sufficient to start an uncontrolled reaction in a particular storage configuration for a specified time is known as the *self-accelerating decomposition temperature* (SADT), as described in NFPA 49.

Decomposing materials are sometimes referred to as *unstable*, and generally they have a positive heat of formation such that energy will be released when the decomposition reaction occurs. Self-reactive materials can often be recognized by the presence of certain chemical structures that tend to confer reactivity. These include

- Carbon-carbon double bonds not in benzene rings (e.g., ethylene, styrene)
- Carbon-carbon triple bonds (e.g., acetylene)
- Nitrogen-containing compounds (NO₂ groups, adjacent N atoms, etc.)
- Oxygen-oxygen bonds (peroxides, hydroperoxides, ozonides)
- Ring compounds with only three or four atoms (e.g., ethylene oxide)
- Metal- and halogen-containing complexes (metal fulminates, halites, halates, etc.).

A more complete list is given by CCPS (1995), and specific compounds can be investigated in resources such as Urben (1999).

General considerations for avoiding unintended reactions with self-reacting substances include knowing the mechanisms and boundaries of what will initiate a self-reaction, maintaining diluents or inhibitors to extend the boundaries where feasible and avoiding the mechanisms (such as shock and overtemperature) that would initiate the self-reaction, and having reliable controls and last-resort safety systems in place to detect and deal with an incipient out-of-control condition.

Specific design considerations for a few substances including acrylic acid, styrene, organic peroxides, ethylene oxide, and 1,3-butadiene are given in CCPS (1995) on the basis of an industry-practice survey. Detailed information for other substances is distributed by industry

user groups. These include methacrylic acid and methacrylate esters (www.mpausa.org) and ethylene oxide (www.ethyleneoxide.com).

Spontaneously Combustible and Pyrophoric Substances *Spontaneously combustible* substances will readily react with the oxygen in the atmosphere, igniting and burning even without an ignition source. Ignition may be immediate, or may result from a self-heating process that may take minutes or hours (hence, some spontaneously combustible substances are known as *self-heating* materials).

Pyrophoric materials ignite spontaneously on short exposure to air under ordinary ambient conditions. Some materials that are considered pyrophoric require a minimum relative humidity in the atmosphere for spontaneous ignition to occur. The potential of pyrophoric materials to exhibit this behavior is usually well known due to the extreme care required for their safe handling.

Pyrophoric and other spontaneously combustible substances will generally be identified as such on their product literature, material safety data sheets (MSDSs), or International Chemical Safety Cards (ICSCs). If transported, these substances should be identified as DOT/UN Hazard Class 4.2 materials for shipping purposes and labeled as spontaneously combustible. For pyrophoric substances, the NFPA 704 diamond for container or vessel labeling has a red (top) quadrant with a rating of 4, indicating the highest severity of flammability hazard (NFPA 704, 2001). Note that pyrophoric materials often exhibit one or more other reactivity hazards as well, such as water reactivity.

A scenario that has resulted in many fires and explosions in petroleum refineries involves iron sulfide. An impure, pyrophoric sulfide is formed when streams containing hydrogen sulfide or other volatile sulfur compounds are processed in ferrous equipment. Oxidation of moist iron sulfide is highly exothermic. Opening sulfide-containing equipment without adequate purging can result in rapid self-heating and ignition of the iron sulfide, which can then ignite other residual flammable gases or liquids in the equipment.

Many scenarios involving spontaneous combustion involve a combination of materials exposed to sufficient air, often in an insulating situation that prevents heat from a slow oxidation reaction from dissipating, which results in a self-heating situation.

Lists of pyrophoric materials that include less common chemicals, including metals, can be found in volume 2 of *Bretherick's Handbook of Reactive Chemical Hazards* (Urben, 1999). Other spontaneously combustible substances are tabulated by their proper shipping names and UN/NA numbers in the U.S. Dept. of Transportation regulation 49 CFR 172.101.

Possible causes of uncontrolled reactions associated with pyrophoric and other spontaneously combustible materials are listed in Johnson et al. (2003).

Peroxide Formers *Peroxide formers* will react with the oxygen in the atmosphere to form unstable peroxides, which in turn might explosively decompose if concentrated. Peroxide formation, or peroxidation, usually happens slowly over time, when a peroxide-forming liquid is stored with limited access to air.

Substances that are peroxide formers will often have an inhibitor or stabilizer added to prevent peroxidation. They are often not easily identifiable as peroxide formers by using MSDSs or ICSCs. Rather, they are frequently identified by another characteristic, such as flammability, for storage and shipping purposes. Examples of peroxide formers include 1,3-butadiene, 1,1-dichloroethylene, isopropyl ether, and alkali metals. Johnson et al. (2003) tabulate other chemical structures susceptible to peroxide formation.

The total exclusion of air from vessels and equipment containing peroxide formers, and the establishment and observing of strict shelf life limitations, are basic strategies for managing peroxide-forming hazards.

Water-Reactive Substances *Water-reactive substances* will chemically react with water, particularly at normal ambient conditions. For fire protection purposes, a material is considered water-reactive if a gas or at least 30 cal/g (126 kJ/kg) of heat is generated when it is mixed with water (NFPA 704, 2001), using a two-drop mixing calorimeter.

Water reactivity can be hazardous by one or more of several mechanisms. The heat of reaction can cause thermal burns, ignite combustible materials, or initiate other chemical reactions. Flammable,

corrosive or toxic gases are often formed as reaction products. The violence of some reactions may disperse hazardous materials. Even slow reactions can generate sufficient heat and off-gases to overpressurize and rupture a closed container.

Substances that are water-reactive will nearly always be identified as such on their MSDSs or ICSCs. They may be identified as DOT/UN Hazard Class 4.3 materials for shipping purposes and labeled as dangerous when wet. However, some water-reactive materials are classified otherwise. Acetic anhydride is designated Class 8; it may also be identified as a combustible liquid.

The total exclusion of water from vessels and equipment containing water-reactive substances, and the maintenance of the primary containment integrity over time, are the obvious design and operational considerations when handling water-reactive substances. Drying of equipment prior to start-up and careful design of provisions for cleaning and purging of equipment are also essential.

Oxidizers and Organic Peroxides An oxidizer is any material that readily yields oxygen or other oxidizing gas, or that readily reacts to promote or initiate combustion of combustible materials (NFPA 430, 2000). Thus, most oxidizers can be thought of as being reactive with ordinary combustible liquids or solids, which are commonly used as process, packaging, general use, or structural materials. They can also react with many other reducing substances.

Oxidizers will nearly always be identified as such on their MSDSs or ICSCs. They may be identified as DOT/UN Hazard Class 5.1 materials for shipping purposes and labeled as oxidizers. However, some oxidizers are classified otherwise.

Volume 2 of *Bretherick's Handbook of Reactive Chemical Hazards* (Urban, 1999) lists many structures and individual chemical compounds having oxidizing properties. NFPA 432 can be consulted for typical organic peroxide formulations. Note, however, that some organic peroxide formulations burn with even less intensity than ordinary combustibles and present no chemical reactivity hazard.

NFPA 430 contains safety provisions for the storage of liquid and solid oxidizers. NFPA 432 contains safety provisions for the storage of organic peroxide formulations.

Incompatible Materials In this context, *incompatible* refers to two materials not able to contact each other without undesired consequences. ASTM E 1212 gives a method for preparing a binary compatibility chart for identifying incompatibilities. The NOAA Chemical Reactivity Worksheet uses a group compatibility method to predict the results of mixing any binary combination of the 6080 chemicals in the CAMEO database, including many common mixtures and solutions. Materials to be considered include not only raw materials and products but also by-products, waste products, cleaning solutions, normal and possible abnormal materials of construction, possible contaminants and degradation products, material that could be left in the process from a previous batch or cleanout, and materials in interconnected piping, heat-transfer systems, waste collection systems, or collocated storage.

The essence of the ASTM E 1212 approach is to determine incompatibility scenarios that could foreseeably occur by examining all possible binary combinations. It may be necessary to review a process by using a systematic method such as a process hazard analysis (PHA) to identify all incompatibility scenarios that have a significant likelihood of occurrence and severity of consequences. The same review can then be used to evaluate whether adequate safeguards exist or whether further risk reduction is warranted.

Where the consequences of combining two or more materials under given conditions of temperature, confinement, etc., are unknown and cannot be predicted with certainty, testing may need to be performed to screen for potential incompatibilities. Two common test methods used for this purpose are differential scanning calorimetry and mixing cell calorimetry (described later in this section).

Design considerations to avoid contact of incompatible materials include total exclusion of an incompatible substance from the facility; quality control and sampling of incoming materials; approval procedures for bringing new chemicals and materials of construction on-site; dedicated fittings and unloading spots; vessel, piping, and container labeling; dedicated or segregated storage; segregated diking, drainage, and vent systems; quality control of raw materials and of

materials of construction (both initial construction and ongoing maintenance and modifications); sealless pumps, double tube sheets, and other means of excluding seal fluid, heat-transfer fluid, and other utility substances; positive isolation of interconnections by physical disconnects, blinding, or double block and bleed valves; avoidance of manifolds with flexible connections; and use of compatible purge gases, cleaning solutions, heat-transfer fluids, insulation, fire-extinguishing and suppression agents whenever possible; and removal of unused materials from the site. These design considerations will always need to be accompanied by procedure training, hazard awareness, and operating discipline for them to be effective on an ongoing basis.

Designing Mitigation Systems to Handle Uncontrolled Reactions (From CCPS, *Guidelines for Safe Storage and Handling of Reactive Materials*, 1995, Chap. 5.) Last-resort safety systems are intended to be used in many reactive chemical storage and handling operations as last-ditch efforts to avert a loss event such as an explosion or a hazardous material release, if the operation exceeds safe operating limits and it is not possible to regain control by using the operation's normal control mechanisms.

Inhibitor Injection Inhibitor injection systems are primarily used with polymerizing materials such as vinyl acetate. If the material begins to self-react in an uncontrolled manner, then injection of a polymerization inhibitor can interfere with the reaction before sufficient pressure and temperature have built up to cause a release from the storage/handling containment. The type of inhibitor needed will depend on the nature of the polymerization reaction; e.g., a free-radical scavenger may be used as an inhibitor for a material that reacts by free-radical polymerization. The inhibitor is often the same inhibitor used for normal storage stability requirements, but injected in a much larger quantity. If a different inhibitor is used that is designed to quickly kill the reaction, it is generally called a *short-stop* system.

Inhibitor injection systems need to be carefully designed and maintained to provide a highly reliable last-resort safety system. Since the inhibitor injection system is on standby and may not be used for months, attention must be paid to how the system components can be functionally and effectively tested on a periodic basis, such as once a month, without excessive disruption of normal operations. CCPS' *Guidelines for Engineering Design for Process Safety* (CCPS-AICHE, New York, 1993, pp. 273-275) discusses testing of continuous-process safety systems. This functional testing is important not only for the checking of adequate inhibitor supply and properly functioning delivery system, but also as the means of detecting an out-of-control situation and actuating the inhibitor injection system. Such systems, as well as other last-resort safety systems, are likely to be considered *safety instrumented systems* (SISs); and they should be selected, designed, and maintained accordingly (see the later section on SISs).

Quench Systems Quench systems are used for essentially all types of reactive chemicals. A quench system involves the addition of flooding quantities of water or other quenching medium to the reactive material; the quenching medium might be a subcooled material such as liquid nitrogen or dry ice in special applications.

The means by which a quench system works depends on the nature of the reactive material; e.g., for water-reactive materials, a quench system will destroy the material in a last-resort situation and generally form less-hazardous products, and will at the same time absorb some of the heat of reaction. Most quench systems are designed to both cool down and dilute a material that may be reacting uncontrollably; the quenching medium may also actually interfere with the chemical reaction or deactivate a catalyst.

Dump Systems For an inhibitor injection or quench system, the inhibitor or quenching medium is transferred from an external supply to the reactive material; in a dump system, the reactive material is transferred from the storage/handling facility to a safer location that is the same size or, more commonly, larger than the normal capacity of the facility. This allows depressurizing and deinventory of the reacting mass from the facility in an out-of-control situation, such as an incipient runaway reaction.

Depressuring Systems A last-resort depressurizing system can be added to a reactive system to vent off excessive pressure buildup in a tank vessel in a controlled manner before reaching the relief valve or

rupture disk set pressure. Such a depressurizing system typically consists of a remotely actuated vent valve connected to the vapor space of the vessel, with the venting discharge directed to a scrubber or other treatment system of adequate capacity. The system can be designed to be actuated either manually, by a control room or field operator, or by detection of high pressure and/or high temperature in the vessel.

Reactive Hazard Reviews and Process Hazard Analyses Reactive hazards should be evaluated using reviews on all new processes and on all existing processes on a periodic basis. Reviews should include

1. Review of process chemistry, including reactions, side reactions, heat of reaction, potential pressure buildup, and characteristics of intermediate streams
2. Review of reactive chemicals test data for evidence of flammability characteristics, exotherms, shock sensitivity, and other evidence of instability
3. Review of planned operation of process, especially the possibility of upsets, modes of failure, unexpected delays, redundancy of equipment and instrumentation, critical instruments and controls, and worst-credible-case scenarios

These reviews can be either in addition to or combined with periodic process hazard analyses (PHAs) by using methods such as what-if analysis and HAZOP studies. The latter should consciously focus on identifying scenarios in which intended reactions could get out of control and unintended reactions could be initiated. One means of accomplishing this as part of a HAZOP study has been to include *chemical reaction* as one of the parameters to be investigated for each study node. Johnson and Unwin (2003) describe other PHA-related approaches for studying chemical reactivity hazards.

Worst-Case Thinking At every point in the operation, the process designer should conceive of the worst possible combination of circumstances that could realistically exist, such as loss of cooling water, power failure, wrong combination or amount of reactants, wrong valve position, plugged lines, instrument failure, loss of compressed air, air leakage, loss of agitation, deadheaded pumps, and raw material impurities. An engineering evaluation should then be made of the worst-case consequences, with the goal that the plant will be safe even if the worst case occurs. The previous discussion of calculating the maximum adiabatic temperature rise, then considering what might happen if it is realized, is an example of this type of analysis. A hazard and operability (HAZOP) study could be used to help identify abnormal situations and worst-case consequences.

Reactivity Testing Many of the data needed for the design of facilities with reactivity hazards involve the determination of thermal stability and of

1. The temperature at which an exothermic reaction starts
2. The rate of reaction as a function of temperature
3. Heat generated per unit mass of material

In many cases, data on the increase of pressure during a reaction are also required, especially for vent sizing, and on the composition of the product gases.

The term *onset temperature* T_{onset} is used in two contexts:

1. In a testing context, it refers to the first detection of exothermic activity on the thermogram. The differential scanning calorimeter (DSC) has a scan rate of 10°C/min, whereas the accelerating rate calorimeter (ARC®) has a sensitivity of 0.02°C/min. Consequently, the temperature at which thermal activity is detected by the DSC can be as much as 50°C different from ARC data.

2. The second context is the process reactor. There is a potential for a runaway if the net heat gain of the system exceeds its total heat loss capability. A self-heating rate of 3°C/day is not unusual for a monomer storage tank in the early stages of a runaway. This corresponds to 0.00208°C/min, which is 10 percent of the ARC's detection limit.

Sources of Reactivity Data Several important sources of reactivity data are described in the following paragraphs.

Calculations Potential energy that can be released by a chemical system can often be predicted by thermodynamic calculations. If there is little energy, the reaction still may be hazardous if gaseous products are produced. Kinetic data are usually not available in this way. Thermodynamic calculations should be backed up by actual tests.

Differential Scanning Calorimetry Sample and inert reference materials are heated in such a way that the temperatures are always equal. Onset-of-reaction temperatures reported by the DSC are higher than the true onset temperatures, so the test is mainly a screening test.

Differential Thermal Analysis (DTA) A sample and inert reference material are heated at a controlled rate in a single heating block. This test is basically qualitative and can be used for identifying exothermic reactions. Like the DSC, it is also a screening test. Reported temperatures are not reliable enough to be able to make quantitative conclusions. If an exothermic reaction is observed, it is advisable to conduct tests in the ARC.

Mixing Cell Calorimetry (MCC) The MCC provides information regarding the instantaneous temperature rise resulting from the mixing of two compounds. Together, DSC and MCC provide a reliable overview of the thermal events that may occur in a process.

Accelerating Rate Calorimetry (ARC) This equipment determines the self-heating rate of a chemical under near-adiabatic conditions. It usually gives a conservative estimate of the conditions for, and consequences of, a runaway reaction. Pressure and rate data from the ARC may sometimes be used for pressure vessel emergency relief design. Activation energy, heat of reaction, and approximate reaction order can usually be determined. For multiphase reactions, agitation can be provided. Nonstirred ARC runs may give answers that do not adequately duplicate plant results when there are reactants that may settle out or that require mixing for the reaction to be carried out (DeHaven and Dietsche, "Catalyst Explosion: A Case History," *Plant/Oper. Prog.*, April 1990).

Vent Sizing Package (VSP2™) The VSP is an extension of ARC technology. The VSP2 is a bench-scale apparatus for characterizing runaway chemical reactions. It makes possible the sizing of pressure relief systems with less engineering expertise than is required with the ARC or other methods.

Advanced Reactive System Screening Tool (ARSST™) The ARSST measures sample temperature and pressure within a sample containment vessel. The ARSST determines the potential for runaway reactions and measures the rate of temperature and pressure rise (for gassy reactions) to allow determinations of the energy and gas release rates. This information can be combined with simplified methods to assess reactor safety system relief vent requirements.

Shock Sensitivity Shock-sensitive materials react exothermically when subjected to a pressure pulse. Materials that do not show an exotherm on a DSC or DTA are presumed not to be shock-sensitive. Testing methods include

- **Drop weight test** A weight is dropped on a sample in a metal cup. The test measures the susceptibility of a chemical to decompose explosively when subjected to impact. This test should be applied to any materials known or suspected to contain unstable atomic groupings.
- **Confinement cap test** Detonability of a material is determined by using a blasting cap.
- **Adiabatic compression test** High pressure is applied rapidly to a liquid in a U-shaped metal tube. Bubbles of hot compressed gas are driven into the liquid and may cause explosive decomposition of the liquid. This test is intended to simulate water hammer and sloshing effects in transportation, such as humping of railway tank cars. It is very severe and gives worst-case results.

Obtaining test data for designing a facility with significant reactivity hazards requires familiarity with a range of test equipment and a significant amount of experience in the interpretation of test results.

TOXICITY

Introduction Many natural and artificial substances are toxic to humans (and animals). Liquids and solids can be ingested, or exposure can be through the skin, eyes, or other external passages to the body. Where these substances are gaseous or volatile, toxic effects can result from inhalation. As a result of accidents and tests, it has been discovered that some of these substances are more toxic than others. Quantification of the degree of hazard has become important in devising appropriate measures for containing these substances.

Several chemical companies have established toxicology laboratories to develop quantitative information concerning the toxicity of raw materials, products, by-products, and waste materials. They include Dow, Du Pont, Eastman Kodak, and Union Carbide (Fawcett and Wood, *Safety and Accident Prevention in Chemical Operations*, 2d ed., pp. 262 and 281, 1982). Also, the Chemical Industry Institute of Toxicology was established in 1975 to provide toxicological hazards services, and The Netherlands Organization of Applied Scientific Research [TNO] has performed similar toxicological research in the Netherlands (The Institution of Chemical Engineers, *Chlorine Toxicity Monograph*, p. 34, 1989).

The present "Process Safety Management" standard of the Occupational Safety and Health Act requires "toxicity information" and "a qualitative evaluation of a range of the possible safety and health effects of failure of controls on employees in the workplace" [U.S. Department of Labor, *Occupational Safety and Health Standards*, 29 CFR 1910.119(d)(1)(i), (e)(3)(vii), and (f)(1)(iii)(A), 1992]. Similarly, the "Risk Management Programs for Chemical Accidental Release Prevention" standard of the Environmental Protection Agency's Clean Air Act Amendments [U.S. Environmental Protection Agency, *Risk Management Programs for Chemical Accidental Release Prevention*, 40 CFR 68.15(b)(3), 15(c), 24(c)(7), and 26(b)(1), 1993] requires toxicity information, a qualitative evaluation of a range of the possible safety and health effects of failure of controls on public health and the environment, and analysis of the off-site consequences of the worst-case release scenario and the other more likely significant accidental release scenarios.

To perform safety and health evaluations, quantitative knowledge of the effects of exposure to toxic materials would be needed. Some of the available data for inhalation toxicity (quantitative), skin-absorption toxicity (qualitative), and ingestion (quantitative) of hazardous materials are presented in Table I (American Industrial Hygiene Association, *Emergency Response Planning Guidelines and Workplace Environmental Exposure Level*, No. AEAH05-559, 2005; National Institute for Occupational Safety and Health, *Pocket Guide to Chemical Hazards*, 1994; American Conference of Governmental Industrial Hygienists, *Threshold Limit Values for Chemical Substances and Physical Agents*, 2001; National Institute for Occupational Safety and Health, *Registry of Toxic Effects of Chemical Substances*, 1983). To facilitate use of these data, several types of graphical and "probit" equation methods are available (Griffiths, "The Use of Probit Expressions in the Assessment of Acute Population Impact of Toxic Releases," *Journal of Loss Prevention in the Process Industries*, 4: 49, January 1991; Prugh, "Quantitative Evaluation of Inhalation-Toxicity Hazards," 29th Annual Loss Prevention Symposium, 1995).

The scope of this section is limited to dangerous and life-threatening exposures of the public to toxic materials (primarily gases and vapors) and non-life-threatening exposures of employees to toxic materials. Data concerning life-threatening concentrations and doses of many toxic gases, vapors, and liquids are available (National Institute for Occupational Safety and Health, *Registry of Toxic Effects of Chemical Substances*, 1983).

Inhalation Toxicity: The Haber Equation In 1924, Fritz Haber reported on his analysis of the results of animal inhalation tests on chemical warfare agents. He discovered that the product of gas or vapor concentration and duration of exposure was nearly constant for a given physiological effect. This relationship has been termed the *Haber law* (Haber, *Fünf Vorträge aus den Jahren 1920–1923*, Springer-Verlag, 1924; Fleming et al., *Modern Occupational Medicine*, p. 78, 1960):

$$Ct = K = D \quad (23-22)$$

When the concentration C is expressed in parts per million (ppm) and the duration of exposure t is expressed in minutes, the values of the constant K and the dose D are in units of ppm-minutes.

It is now recognized that Haber's law does not apply for long exposures to low concentrations. Apparently, there are metabolic processes in the human body (and in animals) that can (for many toxic materials) result in biotransformation or detoxification, elimination, or excretion of toxic materials, or can repair damaged cells or tissues (Elkins, *The Chemistry of Industrial Toxicology*, 2d ed., p. 242, 1959; U.S. Federal

Emergency Management Agency, *Handbook of Chemical Hazard Analysis Procedures*, p. 6-7, 1989). It is likely that the absorption process functions in proportion to the square root of the duration of exposure (Perry, *Chemical Engineers' Handbook*, 4th ed., p. 14-13 and Figs. 14-7, 14-9, and 14-21, 1963).

Dosage Equation The Haber law apparently applies to short exposures (less than 30 min) (The Institution of Chemical Engineers, *Chlorine Toxicity Monograph*, Table 5, Rat Group Codes U, N, E, X, and Z, and Mouse Group Codes N and R, 1989), but does not apply for long exposures to toxic vapors and gases. Eisenberg and others (Eisenberg et al., *Vulnerability Model—A Simulation System for Assessing Damage Resulting from Marine Spills*, U.S. Coast Guard Report CG-D-136-75, pp. 77, 83–89, and 257–267, 1975); also, 3d International Symposium on Loss Prevention in the Process Industries, p. 15/1158, and Proceedings, p. 190, 1980) attempted to modify the dose equation to fit the data over a useful range of interest, between 5 min and 2 h (Lees, *Loss Prevention in the Process Industries*, pp. 206–209, 527, 594, 599, 651–653, and 661, 1980). They found that the following equation could be used:

$$C^n t = K \quad (23-23)$$

Eisenberg found that a value of 2.75 for n was appropriate for the chlorine and ammonia data which were available.

In the 20 years since Eisenberg's report, many inhalation toxicity tests have been conducted, and many of the earlier data have been reexamined, with the result that values for n ranging from 0.6 to 4.9 have been applied to the above dose equation. It appears (Griffiths, "The Use of Probit Expressions in the Assessment of Acute Population Impact of Toxic Releases," *Journal of Loss Prevention in the Process Industries*, 4, p. 49, 1991) that the value of n may be related to the degree of breathing rate stimulation (high value of n) or repression (low values of n), and the value of n apparently increases with increasing exposure times (decreasing slope of ordinate C versus abscissa t). A value of 1.0 is frequently used by investigators if there are few data.

Probit Equation The probit equation has been used in an attempt to quantitatively correlate hazardous material concentration, duration of exposure, and probability of effect/injury, for several types of exposures. The objective of such use is to transform the typical sigmoidal (S-shaped) relationship between cause and effect to a straight-line relationship (Mann, *Lees' Loss Prevention in the Process Industries*, 3d ed., p. 9/68, 2005).

Probit equations have the following form (Mann, *Lees' Loss Prevention in the Process Industries*, 3d ed., Table 9.29, 2005):

$$Y = k_1 + k_2 \ln V \quad (23-24)$$

where Y = probit value

V = value of "intensity of causative factor which harms the vulnerable resource"

k_1 = constant that is intercept of Y versus V line (where value of V is 1.0 and $\ln V$ is 0).

k_2 = constant that is slope of Y versus $\ln V$ line.

The following table can be used to convert from probit values to probability percentages (Mann, *Lees' Loss Prevention in the Process Industries*, 3d ed., Table 9.29, 2005):

Percentage	Probit value P
0.001 (1 in 10^5)	0.73
0.01 (1 in 10^4)	1.28
0.1 (1 in 10^3)	1.90
1 [1%] (1 in 100)	2.67
3 [3%] (3 in 100)	3.12
10 [10%] (1 of 10)	3.72
30 [30%] (3 of 10)	4.48
50 [50%] (5 of 10)	5.00

For the inhalation hazards of toxic vapors and gases, the function V has the form

$$V = C^n t \quad (23-25)$$

where C = concentration by volume, ppm

t = duration of exposure, min

n = exponent that expresses difference between dosage and dose

The term *dosage* typically refers to an environmental hazard and is the product of the concentration of toxic gas or vapor at a particular point and the duration of the hazardous environment at that point. Thus, the *dosage* can be expressed as an average concentration multiplied by an average duration, or

$$D = Ct \quad \text{ppm} \cdot \text{min} \quad (23-26)$$

Dose typically refers to the amount of toxic material actually retained and is sometimes referred to as the *toxic load*. Thus, the dose can be expressed as the product of a concentration term and a duration-of-exposure term, by either of the following relationships:

$$TL = V = C^n t \quad \text{ppm}^n \cdot \text{min} \quad (23-27)$$

$$TL = V = Ct^{1/n} \quad \text{ppm} \cdot \text{min}^{1/n} \quad (23-28)$$

Ingestion Toxicity Data are available for the *acute* (single-dose) ingestion/oral toxicity of many toxic materials (National Institute for Occupational Safety and Health, *Registry of Toxic Effects of Chemical Substances*, 1983; Lewis, *Sax's Dangerous Properties of Industrial Materials*, 9th ed., 1996). However, very few data are available for prolonged ingestion or periodic doses of toxic materials. It is likely that metabolic processes would operate to increase the total *burden* required for toxic effects for such chronic exposures, except for some materials (such as mercury and lead) which apparently can accumulate in the body.

The primary route for ingestion of toxic materials (especially dusts, mists, and vapors) is by the swallowing of mucus and saliva that has absorbed these materials during breathing. Cilia in the nose and esophagus (windpipe) sweep foreign materials that have been embedded or absorbed by these fluids toward the pharynx, where the contaminated fluid is swallowed (Guyton, *Textbook of Medical Physiology*, 3d ed., pp. 555, 556, 880, and 894, 1966).

Skin-Contact Toxicity Data for acute (short-term) exposures of the skin to corrosive and toxic liquids, solids, and gases are extremely limited, particularly where the consequences are severe or fatal injury, and the available data may not be useful, from an engineering standpoint. For example, the skin toxicity of hydrogen peroxide to rats is stated as 4060 mg/kg, but the skin area and duration of exposure are not stated. Thus, it is not possible (with the available data) to estimate the relationship among percent of body surface exposed to a corrosive material, the concentration of the corrosive material, the duration of exposure (before removal of the corrosive material), and the severity of the effect.

Somewhat in contrast, there are considerable data concerning the relatively long-term effects of exposure to toxic materials, where there are irritation, tumorigenic, reproductive, or mutation consequences (National Institute for Occupational Safety and Health, *Registry of Toxic Effects of Chemical Substances*, 1983). In the

absence of better skin-contact data, it might be appropriate to use parenteral or subcutaneous injection data for a worst-case exposure (e.g., through a cut in the skin). However, use of intravenous or intraperitoneal data might overstate the skin exposure toxicity of a material.

The U.S. Department of Transportation and others have developed guidance for the corrosivity of chemical substances (U.S. Department of Transportation, *Shippers—General Requirements for Shipments and Packagings*, 49 CFR 173.136, *Definitions*, and 49 CFR 173.137, *Assignment of Packing Group*, 1998); American Society for Testing and Materials, *Practice for Laboratory Immersion Corrosion Testing of Metals*, G-31, 2002; Organization for Economic Cooperation and Development, *Guideline for Testing of Chemicals—Acute Dermal Irritation/Corrosion*, no. 404, 1992). The following definitions apply for Class 8 corrosive materials:

Packing Group 1—Great Danger: Full thickness destruction of human skin (exposure time, 3 min or less; observation time, 60 min).

Packing Group 2—Medium Danger: Full thickness destruction of human skin (exposure time, 3 to 60 min; observation time, 14 days).

Packing Group 3—Minor Danger: Full thickness destruction of human skin (exposure time, 1 to 4 h; observation time, 14 days).

Examples of assignments to packing groups are shown in the table at the bottom of the page.

Another effect of skin-contact toxicity is dermatitis. This can be caused by "physical" agents, such as detergents and solvents that remove the natural oils from the skin and thereby render the skin susceptible to materials that ordinarily do not affect the skin (National Safety Council, *Fundamentals of Industrial Hygiene*, 3d ed., p. 23, 1988). Dermatitis also can be caused by desiccants and water-reactive chemicals that remove moisture from the skin, generating heat and causing burns. Other causes are oxidizers; protein precipitants; allergic or anaphylactic proteins; friction, pressure, and trauma; thermal and electromagnetic radiation; biological agents; and plant poisons. Dermatitis can be prevented or controlled by containment of skin-contact hazards and use of tools to avoid contact (engineering controls) or by the wearing of protective clothing, including gloves and eye and face protection, and good personal hygiene, including hand and face washing (administrative controls) (National Safety Council, *Fundamentals of Industrial Hygiene*, 3d ed., pp. 106, 108, 467, 469, and 471, 1988).

Compilation of Data Table 23-16 presents inhalation toxicity data for the following criteria:

The emergency response planning guidelines (ERPG) concentrations for the following types of effects, for 1-h exposures (American Industrial Hygiene Association, *Emergency Response Planning Guidelines and Workplace Environmental Exposure Level*, no. AEAH05-559, 2005):

- ERPG-1 Mild, transient health effects, without objectionable odor
- ERPG-2 No irreversible or action-impairing effects
- ERPG-3 No life-threatening effects

The immediately dangerous to life and health (IDLH) concentrations (National Institute for Occupational Safety and Health, *Pocket Guide to Chemical Hazards*, 1994), for 30-min exposures.

Values for workplace environmental exposure levels (WEELs) for many materials not listed in Table 23-16 can be obtained from the American Industrial Hygiene Association, at www.aiha.org.

The threshold limit values (TLVs) or time-weighted averages (TWAs) for 8-h exposures of workers (American Conference of Governmental Industrial Hygienists, *Threshold Limit Values for Chemical Substances and Physical Agents*, 2001).

Group	Acids	Alcohols	Oxides	Hydroxides	Halogens	Anhydrides
1	Hydrofluoric acid; selenic acid	Alkyl phenols			Fluorine; chlorine; bromine	
2	Hydrochloric acid; acetic acid; nitric acid	Pentol	Phosphorus pentoxide	Potassium hydroxide; sodium hydroxide		Acetic anhydride
3	Phosphoric acid		Calcium oxide	Ammonium hydroxide		Propionic anhydride

TABLE 23-16 ERPG Values and Other Toxicity Values for Toxic Materials

For materials that are listed in the USEPA 40 CFR 68.130 list and in the NJTCPA Group A and B lists (as of January 1, 2005; refer to current issues for changes).

Definitions:

ERPG-1 The maximum airborne concentration below which it is believed nearly all individuals could be exposed for up to 1 h without experiencing other than mild, transient adverse health effects or without perceiving a clearly defined objectionable odor.

ERPG-2 The maximum airborne concentration below which it is believed nearly all individuals could be exposed for up to 1 h without experiencing or developing irreversible or other serious health effects or symptoms that could impair an individual's ability to take protective action.

ERPG-3 The maximum airborne concentration below which it is believed nearly all individuals could be exposed for up to 1 h without experiencing or developing life-threatening health effects.

IDLH Immediately dangerous to life and health, for 1-h exposures. Where no IDLH data are available, the 50% lethal concentration is shown, as LC50:

ppm/time.

WEEL-8 Workplace environmental exposure level, for 8-h time-weighted average (TWA).

WEEL-C Workplace environmental exposure level, as a ceiling (not to be exceeded) value.

TLV-TWA Threshold limit value, time-weighted average for 8-h exposures, with ceiling concentrations shown as C, and with skin absorption hazard as S. The OSHA permissible exposure limit (PEL) is the lower of the TWA or the ceiling limit.

Oral LD50 data are recorded where they are available (— indicates a toxicity listing but no oral toxicity data). Where the material is a gas at normal temperatures and pressure (25°C and 1 atm), the atmospheric-pressure boiling point is given.

Material	CAS no.	ERPG-1	ERPG-2	ERPG-3	IDLH	WEEL-8	WEEL-C	TLV-TWA	Oral, mg/kg
Acetaldehyde	75-07-0	10 ppm	200 ppm	1000 ppm	Ca (2000 ppm)	—	—	0.25 ppm C	1930
Acrolein	107-02-8	0.1 ppm	0.5 ppm	3 ppm	2 ppm	—	—	0.1 ppm C	7
Acrylonitrile	107-13-1	10 ppm	35 ppm	75 ppm	LC50: 576/4 h	—	—	2 ppm	27
Ammonia	7664-41-7	25 ppm	150 ppm	750 ppm	300 ppm	—	—	25 ppm	350
Arsine	7784-42-1	—	0.5 ppm	1.5 ppm	Ca (3 ppm)	—	—	0.05 ppm	-62.5°C
Boron trifluoride	7637-07-2	2 mg/m ³	30 mg/m ³	100 mg/m ³	25 ppm	—	—	1 ppm C	-99.8°C
Bromine	7726-95-6	0.1 ppm	0.5 ppm	5 ppm	3 ppm	—	—	0.1 ppm	14
Carbon disulfide	75-15-0	1 ppm	50 ppm	500 ppm	500 ppm	—	—	10 ppm	2125
Carbon monoxide	630-08-0	200 ppm	350 ppm	500 ppm	1200 ppm	—	—	25 ppm	-191.5°C
Chlorine	7782-50-5	1 ppm	3 ppm	20 ppm	10 ppm	—	—	0.5 ppm	-34.1°C
Chlorine dioxide	10049-04-4	—	0.5 ppm	3 ppm	5 ppm	—	—	0.1 ppm	10.9°C
Chloroform	67-66-3	—	50 ppm	5000 ppm	Ca (500 ppm)	—	—	10 ppm	36
Chloropicrin	76-06-2	0.1 ppm	0.3 ppm	1.5 ppm	2 ppm	—	—	0.1 ppm	250
Chloroprene	126-99-8	—	—	—	Ca (300 ppm)	—	—	10 ppm	260
Dichlorosilane	4109-96-0	—	—	—	LC50: 215/	—	—	—	—
Diethylamine	109-89-7	—	—	—	200 ppm	—	—	5 ppm	540
Dimethylamine	124-40-3	0.6 ppm	100 ppm	350 ppm	500 ppm	—	—	5 ppm	240
Dimethylhydrazine	57-14-7	—	—	—	Ca (15 ppm)	—	—	0.01 ppm	122
Epichlorohydrin	106-89-8	2 ppm	20 ppm	100 ppm	Ca (75 ppm)	—	—	0.5 ppm	90
Ethylamine	75-04-7	—	—	—	600 ppm	—	—	5 ppm	400
Ethylenediamine	107-15-3	—	—	—	1000 ppm	—	—	10 ppm	470
Ethyleneimine	151-56-4	—	—	—	Ca (100 ppm)	—	—	0.5 ppm	15
Ethylene oxide	75-21-8	NA	50 ppm	500 ppm	Ca (800 ppm)	—	—	1 ppm	72
Ethylmercaptan	75-08-1	—	—	—	500 ppm	—	—	0.5 ppm	1960
Fluorine	7782-41-4	0.5 ppm	5 ppm	20 ppm	25 ppm	—	—	1 ppm	-188.3°C
Formaldehyde	50-00-0	1 ppm	10 ppm	25 ppm	Ca (20 ppm)	—	—	0.3 ppm C	42
Furan	110-00-9	—	—	—	LC50: 43/1 h	Minimize	—	—	—
Hydrazine	302-01-2	0.5 ppm	5 ppm	30 ppm	Ca (50 ppm)	—	—	0.01 ppm	49
Hydrogen bromide	10035-10-6	—	—	—	30 ppm	—	—	3 ppm C	-66.7°C
Hydrogen chloride	7647-01-0	3 ppm	20 ppm	150 ppm	50 ppm	—	—	5 ppm C	-85.0°C
Hydrogen cyanide	74-90-8	NA	10 ppm	25 ppm	50 ppm	—	—	4.7 ppm C	2
Hydrogen fluoride	7664-39-3	2 ppm	20 ppm	50 ppm	30 ppm	—	—	3 ppm C	-19.5°C
Hydrogen selenide	7783-07-5	NA	0.2 ppm	2 ppm	1 ppm	—	—	0.05 ppm	-41°C
Hydrogen sulfide	7783-06-4	0.1 ppm	30 ppm	100 ppm	100 ppm	—	—	10 ppm	-60°C
Iron pentacarbonyl	13463-40-6	—	—	—	LC50: 870/10 min	—	—	0.1 ppm	12
Isopropylamine	75-31-0	—	—	—	750 ppm	—	—	5 ppm	820
Ketene	143-50-0	—	—	—	5 ppm	—	—	0.5 ppm	1300
Methacrylonitrile	126-98-7	—	—	—	LC50: 36/4 h	—	—	1 ppm	15
Methylamine	74-89-5	10 ppm	100 ppm	500 ppm	100 ppm	—	—	5 ppm	-6.3°C
Methyl bromide	74-83-9	NA	50 ppm	200 ppm	Ca (250 ppm)	—	—	1 ppm	214
Methyl chloride	74-87-3	NA	400 ppm	1000 ppm	Ca (2000 ppm)	—	—	50 ppm	-24°C
Methylhydrazine	60-34-4	—	—	—	Ca (20 ppm)	—	—	0.01 ppm	22
Methyl iodide	74-88-4	25 ppm	50 ppm	125 ppm	Ca (100 ppm)	—	—	2 ppm	150
Methyl isocyanate	624-83-9	0.025 ppm	0.25 ppm	1.5 ppm	3 ppm	—	—	0.02 ppm	69
Methyl mercaptan	74-93-1	0.005 ppm	25 ppm	100 ppm	150 ppm	—	—	0.5 ppm	6.0°C
Methyl chlorosilane	75-79-6	0.5 ppm	3 ppm	15 ppm	LC50: 29/2 h	1 ppm	1 ppm	—	1000
Methylvinylketone	78-94-4	—	—	—	LC50: 2.4/4 h	—	—	0.2 ppm	31
Nickel carbonyl	13463-39-3	—	—	—	Ca (2 ppm)	—	—	0.05 ppm	43°C
Nitric acid (white fuming)	7697-37-2	1 ppm	6 ppm	78 ppm	25 ppm	—	—	2 ppm	430
Nitric oxide	10102-43-9	—	—	—	LC50: 315/15 min	—	—	25 ppm	-151.8°C
Nitrogen dioxide	10102-44-0	1 ppm	15 ppm	30 ppm	20 ppm	—	—	3 ppm	20.8°C
Oleum	8014-95-7	2 mg/m ³	10 mg/m ³	30 mg/m ³	LC50: 347/1 h	—	—	—	—
Ozone	10028-15-6	—	—	—	5 ppm	—	—	0.05 ppm	-112°C
Perchloryl fluoride	7616-94-6	—	—	—	100 ppm	—	—	3 ppm	-47°C
Phosgene	75-44-5	NA	0.2 ppm	1 ppm	2 ppm	—	—	0.1 ppm	7.6°C
Phosphine	7803-51-2	NA	0.5 ppm	5 ppm	50 ppm	—	—	0.3 ppm	-87°C
Phosphorus oxychloride	10025-87-3	—	—	—	LC50: 48/4 h	—	—	0.1 ppm	380
Phosphorus trichloride	7719-12-2	0.5 ppm	3 ppm	15 ppm	25 ppm	—	—	0.2 ppm	550
Propylene oxide	75-56-9	50 ppm	250 ppm	750 ppm	Ca (400 ppm)	—	—	2 ppm	380
Stibine	7803-52-3	ID	0.5 ppm	1.5 ppm	5 ppm	—	—	0.1 ppm	-18°C

TABLE 23-16 ERPG Values and Other Toxicity Values for Toxic Materials (Concluded)

Material	CAS no.	ERPG-1	ERPG-2	ERPG-3	IDLH	WEEL-8	WEEL-C	TLV-TWA	Oral, mg/kg
Sulfur dioxide	7446-09-5	0.3 ppm	3 ppm	15 ppm	100 ppm	—	—	2 ppm	−10°C
Sulfuric acid	7664-93-9	2 mg/m ³	10 mg/m ³	30 mg/m ³	15 mg/m ³	—	—	0.25 ppm	2140
Sulfur trioxide	7446-11-9	2 mg/m ³	10 mg/m ³	30 mg/m ³	LC50: >9/6 h	—	—	—	—
Tetranitromethane	509-14-8	—	—	—	4 ppm	—	—	0.005 ppm	130
Thionyl chloride	7719-09-7	0.2 ppm	2 ppm	10 ppm	LC50: 500/1 h	—	—	1 ppm	—
Titanium tetrachloride	7550-45-0	5 mg/m ³	20 mg/m ³	100 mg/m ³	LC50: 13/2 h	0.5 mg/m ³	—	—	—
Toluene di-isocyanate	584-84-9	0.01 ppm	0.15 ppm	0.6 ppm	Ca (2.5 ppm)	—	—	0.005 ppm	5800
Trimethylamine	75-50-3	—	—	—	LC50: 3500/4 h	1 ppm	—	5 ppm	2.9°C
Vinyl acetate	108-05-4	5 ppm	75 ppm	500 ppm	LC50: 1550/4 h	—	—	10 ppm	1613
Vinyl trichlorosilane	75-94-5	—	—	—	LC50: 500/4 h	1 ppm	1 ppm	—	1.280

The 50 percent lethal doses of ingested toxic materials that could cause fatal injury (National Institute for Occupational Safety and Health, *Registry of Toxic Effects of Chemical Substances*, 1983).

Where data for the above categories could not be found in the available literature, but the material was listed in the USEPA or NJTCPA standards, the LC50 value (op. cit.) was entered in the IDLH column.

Additional data concerning relatively long-term exposures of the public to toxic chemicals are presented in Table 23-17.

Safeguards against Toxicity Hazards Certainly the best protection against toxicity hazards is complete containment of hazardous materials within processing equipment.

Where complete containment is impractical, exhaust ventilation (preferably to a scrubber) can limit or eliminate exposure to toxic materials. The exhaust ventilation rate (velocity or volumetric rate) may be calculable for volatile liquids from spill size and vapor pressure (U.S. Environmental Protection Agency, *Risk Management Program Guidance for Offsite Consequence Analysis*, Appendix D, Equation D-1, 1999), but tests to determine concentrations in air usually would be needed for dusty processes and fugitive releases of gases.

If containment and exhaust ventilation are not considered adequate, cartridge respirators or self-contained breathing apparatus can provide protection against inhalation (and, in some cases, ingestion) of toxic materials. In 1994, the Occupational Safety and Health Standards were amended to require that "the employer shall assess the workplace to determine if hazards are present, or are likely to be present, which necessitate the use of personal protective equipment (PPE). If such hazards are present, or likely to be present, the employer shall select, and have each affected employee use, the types of PPE that will protect the affected employee from the hazards identified in the hazard assessment" [U.S. Department of Labor, *Occupational Safety and Health Standards*, 29 CFR 1910.132(d), 1999]. This hazard assessment would aid in determining the type of breathing protection that is appropriate for the toxicity hazard. Guidelines for appropriate use of breathing protection are given in this Standard (U.S. Department of Labor, *Occupational Safety and Health Standards*, 29 CFR 1910.134 and Appendices A, B, and C to 1910.134, and Appendix B to Subpart I, 1999). OSHA has not yet provided official Assigned Protection Factors, in Table I of this Standard, but manufacturers do provide these factors. As examples, a full-face cartridge respirator typically has a protection factor of 50 × PEL, and a self-contained breathing apparatus typically has a protection factor of 10,000 × PEL.

Conclusion Toxicity data are available for many thousands of solid, liquid, and gaseous chemicals and other materials. The data for inhalation toxicity provide guidance for concentration and duration limits, for protection of the public, chemical plant employees, and emergency response personnel. Similar data for ingestion and skin contact with toxic materials are not as readily available. Investigation into toxic effects is continuing, so that toxic materials can be handled safely.

OTHER HAZARDS

Hazards of Vacuum

Introduction Storage tanks and many other equipment items often have a relatively low resistance to the damage that can be caused

by internal vacuum. The low vacuum rating for such equipment and the amount of damage that can result are often surprising and potentially costly lessons learned by plant engineers and operators.

Equipment Limitations A robust internal pressure rating for a piece of equipment is no guarantee that it will withstand an appreciable vacuum. Industry loss experience includes failures of vessels with design pressure ratings in excess of 25 psig (Sanders, "Victims of Vacuum," *Proceedings of the 27th Annual Loss Prevention Symposium*, AIChE, 1993). Low-pressure storage tanks are particularly fragile. For example, an atmospheric fixed-roof storage tank may only withstand a vacuum of 2.5 mbar (0.036 psi or 1 in water) (British Petroleum, *Hazards of Trapped Pressure and Vacuum*, 2003).

Jacketed vessels can be particularly vulnerable to internal vacuum, since the operating pressure of the heat-transfer medium in the jacket adds to the differential pressure that would otherwise exist between the atmosphere and the vessel interior.

While many pressure vessels may withstand a significant vacuum, design calculations are required to confirm this. Unless specifically rated for vacuum service, equipment should be assumed to be subject to damage by vacuum. When equipment is procured, consideration should be given to including the vacuum rating in the pressure vessel calculations and code stamp. In many instances, the additional cost of doing so will be an insignificant fraction of the total procurement cost for the vessel (see Protective Measures for Equipment).

Consequences of Vacuum Damage Vessels, tank trucks, or railcars can be dimpled by partial collapse or, more significantly, crushed like used drink cans. Fortunately, equipment damaged by underpressurization does not fail explosively, as might occur with overpressurized equipment. Nevertheless, loss of containment of equipment contents is a real risk, due to damage to the vessel or to the piping connected to the vessel. Significant releases of toxic, flammable, or otherwise hazardous materials can result, with severe consequences.

Alternatively, vacuum within equipment could lead to ingress of air into inerted or fuel-rich systems, posing a fire or explosion hazard within the equipment.

The potential for "knock-on" effects resulting from equipment damage should be considered.

Common Causes of Equipment Underpressurization Equipment can be exposed to excessive vacuum due to an unanticipated mechanism creating a vacuum and/or the failure or inadequate design of protective systems provided to mitigate the hazard.

A common scenario involves the pumping, draining, or siphoning of liquid from a tank that has no, or an inadequate, venting capacity and thus cannot allow the entry of air at a rate sufficient to backfill behind the dropping liquid level.

Similarly, vacuums can be created when a blower, fan, compressor, or jet ejector removes gases from equipment. The magnitude of the vacuum attainable will be governed by the performance characteristics of the device. Other mechanisms for generating a vacuum, which have been demonstrated by industry experience, include the following.

- **Condensation of vapors or cooling of hot gases.** For example, steam is commonly used to clean vessels and, less frequently, to create an inert atmosphere inside of equipment. Steam condensing inside a closed vessel can create a significant vacuum, and vessels (e.g., railcars) have collapsed when all vessel inlets were

TABLE 23-17 Emergency and Continuous Exposure Guidance Levels for Selected Airborne Contaminants

Committee on Toxicology National Research Council

Vol.	Material (concentrations in ppm, except for mg/m ³)	Emergency exposure limits		Continuous exposure limit
		60-min	24-h	
I	Acetone	8,500	1,000	200
1	Acrolein	0.05	0.01	0.01
4	Aluminum oxide	15*	N/L	N/L
IV	Ammonia	10	N/L	N/L
1	Arsine	1.0	0.1	0.01
3	Bromotrifluoromethane	25,000	N/L	100
1	Carbon disulfide	200	100	50
VI	Carbon monoxide	25	N/L	N/L
4	Carbon monoxide	400	50	20
VIII	Chlorine	0.5	N/L	N/L
2	Chlorine	3	0.5	0.1
2	Chlorine trifluoride	1	N/L	N/L
1	Chloroform	100	30	1
2	Dichlorodifluoromethane	10,000	1,000	100
2	Dichlorofluoromethane	100	3	1
2	Dichlorotetrafluoroethane	10,000	1,000	100
V	Dimethyl hydrazine	15	N/L	N/L
5	Dimethyl hydrazine	0.24	0.01	N/L
2	Ethanolamine	50	3	0.5
4	Ethylene glycol	40	20	4
1	Fluorine	15	10	7.5
V	Hydrazine	5	N/L	N/L
5	Hydrazine	0.12	0.005	N/L
II	Hydrogen chloride	2	N/L	N/L
III	Hydrogen fluoride	4	N/L	N/L
4	Hydrogen sulfide	N/L	10	1
2	Isopropyl alcohol	400	200	1
1	Mercury vapor	N/L	0.2*	0.01*
1	Methane	N/L	5,000	5,000
4	Methanol	200	10	N/L
V	Methyl hydrazine	15	N/L	N/L
5	Methyl hydrazine	0.24	0.01	N/L
4	Nitrogen dioxide	1	0.04	N/L
1	Nitrogen oxides	2	N/L	N/L
4	Nitrous oxide	10,000	N/L	N/L
1	Ozone	1	0.1	0.02
2	Phosgene	0.2	0.02	0.01
1	Sulfuric acid	5*	2*	1*
2	Trichlorofluoromethane	1,500	500	100
2	Trichlorotrifluoroethane	1,500	500	100
2	Sodium hydroxide	2*	N/L	N/L
2	Sulfur dioxide	10	5	1
2	Vinylidene chloride	N/L	10	0.15
2	Xylene	200	100	50

N/L = not listed; no guidance is given.

*Concentration in milligrams per cubic meter.

closed immediately after steam cleaning. The rapid addition of cool liquid to a vessel containing a hot, volatile liquid can markedly reduce the vapor pressure of the liquid. The sudden cooling of a storage tank by a thunderstorm can create a vacuum when gases in the vessel head space cool and/or vapors of volatile liquids condense. The American Petroleum Institute (API, *Venting Atmospheric and Low-Pressure Storage Tanks*, Standard 2000, Washington, 1998) provides guidance for in-breathing requirements as a function of tank capacity to protect against this latter scenario.

- *Absorption of a gas in a liquid.* Vessels have collapsed when ammonia vapor from the head space dissolved in water within the vessel (Lees, *Loss Prevention in the Process Industries*, 2d ed., Butterworths, London, 1996). A similar potential should be considered for HCl and water.
- *Chemical reactions that remove gases from the head space.* The corrosion of the interior of a steel vessel, especially if the vessel is newly fabricated or has been chemically cleaned, can consume and

remove a significant quantity of the oxygen from the vessel atmosphere. Other chemical reactions (e.g., ammonia reacting with hydrogen chloride to form ammonium chloride) can reduce the amount of gas or vapor in the vessel.

Prudent design requires that equipment be protected from credible underpressurization scenarios. Equipment damage can result when such protections are omitted, improperly sized, incorrectly designed or installed, or inadequately maintained. Common failures include the following.

- *Failure to consider appropriate challenges when determining the required relief capacity* (e.g., maximum rates of liquid withdrawal or cooling of vessel contents). Credible contingencies (e.g., thunderstorm cooling a vessel during steam-out) should be considered.
- *Inadequate capacity, or failure, of vessel blanketing systems.* Inert gas supplies are often piped to vessels to maintain a reduced-oxygen atmosphere during liquid withdrawal. Coincident high demand for inert elsewhere, closure of a valve, or depletion of the supply could result in the failure to prevent a vacuum. A common means of initially inerting a vessel is to fill the vessel with liquid, then drain the liquid while allowing the blanketing system to backfill the head space with inert gas. Unless the blanketing system is sized to accommodate the drainage rate (which may exceed the normal process demand), there is a risk of collapsing the vessel.
- *Operating errors.* Many vessel collapses have resulted from closing or failing to open a valve in a vent line. For this reason, valves in vacuum relief lines should be avoided, and they may be prohibited by some design codes.
- *Maintenance errors.* One common error is the failure to remove an isolation blind in a vent line when returning a vessel to service. Even a thin sheet of plastic placed over an open nozzle may be sufficient to allow a vessel-damaging vacuum to be produced (BP, *Hazards of Trapped Pressure and Vacuum*, 2003).
- *Inappropriate modifications.* In one incident, a hose was connected to a vent line that was provided for both pressure and vacuum protection. The hose was submerged into a drum of liquid in an attempt to scrub vapors emitted from the vent. Only a few inches of submergence were required to ensure that the vent was effectively blocked the next time a vacuum was pulled on the vessel (Lees, *Loss Prevention in the Process Industries*, 2d ed., Butterworths, London, 1996).
- *Failure of vacuum control loop.* Control failures can either initiate events (e.g., increase the speed of an exhaustor) or disable protections (e.g., reduce the rate of supply of inert gas to a vessel).
- *Plugging of vent lines or devices.* Process materials can migrate into and occlude vent systems when they polymerize, crystallize, condense, or solidify. Monomers requiring an inhibitor to prevent polymerization can evaporate from a tank and then condense in the vent line, free of the inhibitor. Waxes and other high-melting-point materials can solidify upon cooling in the vent system, dusts can accumulate, and water vapor can condense to form liquid seals in low points of vent lines or freeze in the winter. Such scenarios are a particular problem in cases where flame arrestors, screens, and other devices introduce small apertures in the vent flow path. Plugging of vent lines by animal or insect nests is not uncommon.
- *Inadequate or incorrect maintenance.* Mechanical devices such as vacuum breakers and flame arrestors require routine maintenance attention to ensure that they provide their intended protective function. Incorrect maintenance (e.g., changing the vacuum breaker set pressure) could defeat the intended protection.

Lees (*Loss Prevention in the Process Industries*, 2d ed., Butterworths, London, 1996), BP (*Hazards of Trapped Pressure and Vacuum*, 2003), and Kletz (*What Went Wrong?—Case Histories of Process Plant Disaster*, Gulf Publishing Company, 1989) include additional case histories providing valuable lessons about how equipment failures and human errors can combine to inflict vacuum damage.

Protective Measures for Equipment If equipment is subject to experiencing a vacuum, the inherently safer alternative would be to design the equipment to withstand a full vacuum. While this may not be

economically feasible for large storage tanks, the incremental cost for smaller vessels may not be prohibitive, particularly when traded off against the capital and continued operating and maintenance costs of some alternatives (e.g., protective instrumentation systems). The incremental fabrication cost of providing a suitable vacuum rating can be less than 10 percent for vessels of up to 3000-gal nominal capacity and having a 15-psig pressure rating (Wintner, "Check the Vacuum Rating of Your Tanks," *Chemical Engineering*, pp. 157–159, February 1991).

Careful process hazards analysis may show that a particular vessel need not be designed to withstand a full vacuum (e.g., if the maximum attainable vacuum is limited to the performance characteristics of an exhaustor). Whatever the vacuum rating, rated vessels must be periodically inspected to ensure that internal or external corrosion has not diminished the vessel strength.

Reliable protections against excessive vacuum should be provided whenever equipment cannot withstand the vacuums that can credibly be achieved. In some low-risk situations, protections may consist of administrative controls implemented by adequately trained personnel. Where the risk of damage is higher or where design standards or codes require, engineered protections should be implemented.

Where process, safety, and environmental considerations permit, vacuum protection may be provided by properly sized ever-open vents. Alternatively, active protective devices and systems are required. Vacuum breaker valves designed to open and admit air at a predetermined vacuum in the vessel are commonly used on storage tanks, but may not be suitable for some applications involving flammable liquids. Inert gas blanketing systems may be used if adequate capacity and reliability can be ensured. Where the source of the vacuum can be deenergized or isolated, suitably reliable safety instrumented systems (e.g., interlocks) can be provided.

API (*Venting Atmospheric and Low-Pressure Storage Tanks*, Standard 2000, Washington, 1998) provides guidance for vacuum protection of low-pressure storage tanks. Where vacuum relief devices are provided, they should communicate directly with the vapor space in the vessel and should be installed so that they cannot be sealed off by the liquid contents in the vessel. Valves should be avoided in the inlets or outlets of vacuum relief devices unless the valves are reliably car-sealed or locked open, or excess relief capacity is provided (e.g., via multiple-way valves).

Hazards of Inerts

Introduction The use of inert gases to displace oxygen from equipment atmospheres in order to prevent combustion and, perhaps, consequent explosions has been described in the subsection "Flammability." Other applications for inerting exist, including preventing (1) corrosion or other deterioration of out-of-service equipment, (2) degradation of oxygen-sensitive products, or (3) exothermic reactions with air- or water-reactive materials. While the risk of personnel asphyxiation in an oxygen-deficient environment is the most frequently recognized concern, other hazards such as toxicity, temperature and pressure extremes, and chemical incompatibilities also need to be considered.

Sources of Inerts The most commonly used inert gases are N₂ and CO₂, but other gases and vapors such as argon (Ar), helium (He), steam, and exhaust gases from combustion devices are also used. The choice of the most appropriate inert for a given application must be based upon factors such as cost, availability, reliability of supply, effectiveness, and compatibility with process streams (Cumliff, "Avoiding Explosions by Means of Inerting Systems," IChemE Symposium Series no. 148, 2001; Grossel and Zalosh, *Guidelines for Safe Handling of Powders and Bulk Solids*, CCPS-AIChE, 2004).

Traditionally, inerts have been obtained from sources such as high-pressure gas cylinders or tube trailers or through evaporation of cryogenic liquids from bulk tanks. Other sources of inerts include (NFPA 69, *Standard on Explosion Prevention Systems*, National Fire Protection Association, 2002; FM Global, Loss Prevention Data Sheet 7-59, *Inerting and Purging of Tanks, Process Vessels, and Equipment*, 2000)

- On-site cryogenic air separation plants
- Gas generators burning or catalytically oxidizing a hydrocarbon to produce an oxygen-deficient product gas
- Nitrogen produced by the air oxidation of ammonia

- Nitrogen produced by removal of oxygen from air using pressure swing adsorption (PSA) or membrane separation units

Inert gas streams generated on site should be carefully monitored to ensure detection of an excessively high O₂ concentration in the product gas in the event of equipment failure or operational upset (e.g., due to a too high air-to-fuel ratio in a combustion generator or the failure of a membrane in a membrane separator). Consideration should be given to monitoring other indicators of problems in the inert generator (e.g., monitoring for low differential pressure across the membrane as an indication of the failure of a membrane separator).

Asphyxiation and Toxicity Hazards An asphyxiant is a chemical (either a gas or a vapor) that can cause death or unconsciousness by suffocation (BP, *Hazards of Nitrogen and Catalyst Handling*, 2003). A simple asphyxiant is a chemical, such as N₂, He, or Ar, whose effects are caused by the displacement of O₂ in air, reducing the O₂ concentration below its normal value of approximately 21 vol %. The physiological effects of oxygen concentration reduction by simple asphyxiants are illustrated in Table 23-18 (BP, *Hazards of Nitrogen and Catalyst Handling*, 2003).

The physiological processes leading to death from hypoxia (i.e., insufficient supply of oxygen to the body tissues) are described by Air Products (Air Products, *Dangers of Oxygen-Deficient Atmospheres*, Safetygram 17, 1998). At very low oxygen concentrations, loss of consciousness occurs within about 10 s of the first breath, followed by death within 2 to 4 min. A person exposed to an oxygen-deficient environment may not recognize the warning signs and may not be able to reason or take protective action before unconsciousness occurs. Victims removed from an O₂-deficient atmosphere require resuscitation through the administration of O₂ to prevent death [U.S. Chemical Safety and Hazard Investigation Board (CSB), *Hazards of Nitrogen Asphyxiation*, Safety Bulletin no. 2003-10-B, 2003].

Physical exertion increases oxygen demand and may result in oxygen deficiency symptoms at higher oxygen concentrations (CGA, *Oxygen-Deficient Atmospheres*, Publication SB-2, 2001), and individuals in poor health may be less tolerant of reduced oxygen concentrations. The guidance in Table 23-18 assumes a sea-level location and should be applied cautiously for facilities at significant altitudes; however, OSHA's Respiratory Protection Standard accepts 19.5 vol % as a safe O₂ concentration up to an altitude of 8000 ft (OSHA, 29 CFR 1910.134, Respiratory Protection Standard, 1998).

In its safety bulletin on the hazards of nitrogen asphyxiation, CSB identified 80 nitrogen asphyxiation deaths and 50 injuries occurring in 85 incidents between 1992 and 2002 (CSB, *Hazards of Nitrogen Asphyxiation*, Safety Bulletin no. 2003-10-B, 2003).

A chemical asphyxiant works by interfering with the body's ability to absorb or transport O₂ to the tissues. A relevant example of a chemical asphyxiant is CO, which can be present in inert gas streams produced

TABLE 23-18 Physiological Effects of Reduced O₂ Atmospheres

O ₂ (vol %)	Effects
23.5	Maximum "safe level" (23 vol % is often the high-level alarm of most O ₂ detectors)
21	Typical O ₂ concentration in air
19.5	Minimum "safe level" (19 vol % is often the low-level alarm of most O ₂ detectors)
15–19	First sign of hypoxia. Decreased ability to work strenuously. May induce early symptoms in persons with coronary, pulmonary, or circulatory problems.
12–14	Respiration increases with exertion; pulse up; impaired muscular coordination, perception, and judgment
10–12	Respiration further increases in rate and depth, poor judgment, lips blue
8–10	Mental failure, fainting, unconsciousness, ashen face, blueness of lips, nausea, vomiting, inability to move freely
6–8	6 min, 50% probability of death; 8 min, 100% probability of death
4–6	Coma in 40 s, convulsions, respiration ceases, death

TABLE 23-19 Physiological Effects of Exposure to CO₂

CO ₂ (vol %)	Effects
1	Slight increase in breathing rate.
2	Breathing rate increases to 50% above normal. Prolonged exposure can cause headache and tiredness.
3	Breathing increases to twice the normal rate and becomes labored. Weak narcotic effect. Impaired hearing, headache, increase in blood pressure and pulse rate.
4-5	Breathing increases to approximately four times the normal rate, symptoms of intoxication become evident, and slight choking may be felt.
5-10	Characteristic sharp odor noticeable. Very labored breathing, headache, visual impairment, and ringing in the ears. Judgment may be impaired, followed within minutes by loss of consciousness.
50-100	Unconsciousness occurs more rapidly above 10 vol % level. Prolonged exposure to high concentrations may eventually result in death from asphyxiation.

by combustion. Exposure to CO concentrations of approximately 1000 and 13,000 ppm can cause, respectively, loss of consciousness after 1 h and unconsciousness and danger of death after 1 to 3 min (Meidl, *Explosive and Toxic Hazardous Materials*, Table 28, p. 293, Glencoe Press, 1970).

Note that CO₂ acts as neither a simple asphyxiant (like N₂) nor a chemical asphyxiant (like CO). The normal concentration of CO₂ in air is approximately 300 ppm (0.03 vol %). Table 23-19 (Air Products, *Carbon Dioxide*, Safetygram 18, 1998) illustrates that exposure to air diluted by 5 vol % CO₂ (yielding an oxygen concentration of 21 × 0.95, or approximately 20 vol %) prompts physiological effects that are more severe than those inferred from Table 23-18 for dilution by the same amount of nitrogen.

Injuries and fatalities from asphyxiation are often associated with personnel entry into inerted equipment or enclosures. Guidance on safe procedures for confined space access are provided by OSHA (OSHA, 29 CFR 1910.146, *Confined Space Entry Standard*, 2000), the American National Standards Institute (ANSI, Z117.1, *Safety Requirements for Confined Spaces*, 2003), Hodson (Hodson, "Safe Entry into Confined Spaces," *Handbook of Chemical Health and Safety*, American Chemical Society, 2001), and BP (BP, *Hazards of Nitrogen and Catalyst Handling*, 2003). OSHA has established 19.5 vol % as the minimum safe oxygen concentration for confined space entry without supplemental oxygen supply (see Table 23-18). Note that OSHA imposes a safe upper limit on O₂ concentration of 23.5 vol % to protect against the enhanced flammability hazards associated with O₂-enriched atmospheres.

Physical Hazards A variety of physical hazards are presented by the various inerts in common usage.

High temperature The high-temperature off-gases from combustion-based sources of inerts typically must be quenched before use. Water scrubbing, in addition to reducing the temperature, can remove soot and sulfur compounds (which could react with moisture to form corrosive acids) present in the off-gas. The humidity of the resultant gas stream may make it unsuitable for inerting applications where moisture cannot be tolerated.

Use of steam as an inert requires that equipment be maintained at an elevated temperature to limit condensation that would lower the inert concentration. FM Global (FM Global, Loss Prevention Data Sheet 7-59, *Inerting and Purging of Tanks, Process Vessels, and Equipment*, 2000) recommends a minimum temperature of 160°F. The Compressed Gas Association (CGA, *Safe Handling of Compressed Gases in Containers*, Publication P-1, 2000) cautions against the use of steam in (1) systems where brittle materials (such as cast iron) may be stressed by thermal expansion, (2) systems with close clearances where high temperatures may cause permanent warping or maladjustment, and (3) systems where pipe coatings or plastic materials may be damaged by high temperatures. Protection for personnel to prevent thermal burns from equipment may be required.

In addition, some equipment or equipment supports may not have the strength to support a significant load of condensate, and provisions must be made for removal of condensate from the inerted equipment.

Low temperature The atmospheric boiling points for N₂, CO₂, He, and Ar are -196, -79, -269, and -186°C, respectively. The potential for cryogenic burns must be addressed in operating and maintenance procedures and in specifying personal protective equipment requirements.

Cryogenic temperatures can cause embrittlement of some materials of construction (e.g., carbon steel) and must be considered in the design of inert gas delivery systems. Controls should be provided to ensure that operational upsets do not allow the migration of cryogenic liquids to piping or equipment not designed to withstand such low temperatures.

The potential for the condensation and fractional distillation of air on the outside of equipment containing cryogenic liquids with boiling points less than that of O₂ must be considered. For example, because N₂ boils at a lower temperature than O₂ (-196 versus -183°C), air can condense on the outside of liquid N₂-bearing piping. The liquid that drops off of the piping will be enriched in O₂ and can pose an enhanced fire or explosion risk in the vicinity of the equipment.

High pressure Cryogenic liquids produce large volumes of gas upon evaporation (for example, 1 volume of liquid N₂ produces 694 volumes of gas at 20°C) (Air Products, *Safe Handling of Cryogenic Liquids*, Safetygram 16, 1999). Containers such as transport and storage vessels must be provided with overpressure relief to address this hazard. An additional concern is the hydrostatic pressure that can be produced if cryogenic liquids are trapped in a liquid-full system. Absent a vapor space to allow liquid expansion, extremely high pressures can be produced; accordingly, pressure relief devices must be installed in sections of equipment where cryogenic liquids might become trapped between closed valves.

Given the large liquid-to-gas expansion ratio, consideration should be given to limiting the quantity of cryogenic liquid stored inside tight enclosures or buildings that could become pressurized. The asphyxiation hazard associated with inert gases was addressed previously.

Portable containers of high-pressure inert gases can operate at pressures of thousands of pounds per square inch. Suitable precautions are required to protect containers and associated regulators and piping from damage. Refer to CGA (CGA, *Safe Handling of Compressed Gases in Containers*, Publication P-1, 2000; CGA, *Precautions for Connecting Compressed Gas Containers to Systems*, Publication SB-10, 2003) and Air Products (Air Products, *Handling, Storage, and Use of Compressed Gas Cylinders*, Safetygram 10, 2000) for guidance.

Air Products (Air Products, *Product Migration of Liquefied Compressed Gases in Manifolder Systems*, Safetygram 38, 2003) provides precautionary guidance with respect to manifolding of liquid-containing cylinders. A temperature difference of only a few degrees between cylinders can cause gas from the warmer cylinder to migrate through the manifold to the cooler cylinder, where it could condense and potentially fill the cylinder. A liquid-filled cylinder could rupture if it was subsequently valved closed.

Static electricity The use of high-pressure CO₂ for inerting poses a concern for potential static electricity hazards. CO₂ converts directly to a solid if the liquid is depressurized below 61 psig (Air Products, *Carbon Dioxide*, Safetygram 18, 1998). Consequently, discharge of liquid CO₂ produces CO₂ "snow" that, when moving at a high velocity, can generate static electric charge. Incendive sparks (5 to 15 mJ at 10 to 20 kV) have been reported (Urban, *Bretherick's Handbook of Reactive Chemical Hazards*, 6th ed., Butterworth-Heinemann Ltd., 1999).

Chemical Incompatibility Hazards While N₂ and CO₂ may act as inerts with respect to many combustion reactions, they are far from being chemically inert. Only the noble gases (e.g., Ar and He) can, for practical purposes, be regarded as true inerts. Frank (Frank, "Inerting for Explosion Prevention," *Proceedings of the 38th Annual Loss Prevention Symposium*, AIChE, 2004) lists a number of incompatibilities for N₂, CO₂, and CO (which can be present in gas streams from combustion-based inert gas generators). Notable incompatibilities for N₂ are lithium metal and titanium metal (which is reported to burn in N₂). CO₂ is incompatible with many metals (e.g., aluminum and the alkali metals), bases, and amines, and it forms carbonic acid in water,

which can corrode some materials. CO is a strong reducing agent and is incompatible with oxidizers, potassium, sodium, some aluminum compounds, and certain metal oxides. Trace metals and residual organic compounds may contaminate gas streams from combustion-based inert gas generators, posing a variety of potential incompatibility and product quality concerns.

Certain polymerization inhibitors added to stabilize monomers require a small concentration of dissolved O₂ to be effective (NFPA

69, *Standard on Explosion Prevention Systems*, National Fire Protection Association, 2002). For example, methyl acrylate and ethyl acrylate are commonly stabilized with hydroquinone monomethyl ether. Industry guidance recommends a minimum concentration of 5 vol % O₂ in the atmosphere above the acrylate to prevent polymerization (Intercompany Committee for the Safety and Handling of Acrylic Monomers, *Acrylate Esters, A Summary of Safety and Handling*, 3d ed., 2002).

INHERENTLY SAFER DESIGN AND OTHER PRINCIPLES

INHERENTLY SAFER AND MORE USER-FRIENDLY DESIGN

Introduction For many years the usual procedure in plant design was to identify the hazards, by one of the systematic techniques described later or by waiting until an accident occurred, and then add protective equipment to control them or to protect people from their consequences. This protective equipment is often complex and expensive and requires regular testing and maintenance. It often interferes with the smooth operation of the plant and is sometimes bypassed. Gradually the industry came to realize that, whenever possible, we should design user-friendly plants that can withstand human error and equipment failure without serious effects on safety (and output and efficiency). When we handle flammable, explosive, toxic, or corrosive materials, we can tolerate only very low failure rates, of people and equipment, rates which it may be impossible or impracticable to achieve consistently for long periods.

The most effective way of designing user-friendly plants is to avoid, when possible, large inventories of hazardous materials in process or storage. "What you don't have, can't leak." This sounds obvious, but until the explosion at Flixborough in 1974, little systematic thought was given to ways of reducing inventories. The industry simply designed a plant and accepted whatever inventory the design required, confident it could be kept under control. Flixborough weakened that confidence, and 10 years later Bhopal almost destroyed it. Plants in which we avoid a hazard, by reducing inventories or avoiding hazardous reactions, are usually called inherently safer.

The principle ways of designing inherently safer plants, and other ways of making plants user-friendly, are summarized below, with examples (Kletz, *Process Plants: A Handbook for Inherently Safer Design*, Taylor & Francis, 1998).

Intensification or Minimization One approach is to use so little hazardous material that it does not matter if it all leaks out. For example, at Bhopal methyl isocyanate (MIC), the material that leaked and killed over 2000 people, was an intermediate that was convenient but not essential to store. Within a few years many companies had reduced their stocks of MIC and other hazardous intermediates.

Intensification is the preferred route to inherently safer design as the plants, being smaller, are also cheaper (Bell, *Loss Prevention in the Process Industries*, Institution of Chemical Engineers Symposium Series no. 34, 1971 p. 50).

Substitution If intensification is not possible, then an alternative is to use a safer material in place of a hazardous one. Thus it is possible to replace flammable solvents, refrigerants, and heat-transfer media by nonflammable or less flammable (high-boiling) ones, hazardous products by safer ones, processes that use hazardous raw materials or intermediates by processes that do not. As an example of the latter, the product manufactured at Bhopal (carbaryl) was made from three raw materials. Methyl isocyanate is formed as an intermediate. It is possible to react the same raw materials in a different order so that a different and less hazardous intermediate is formed.

Attenuation or Moderation Another alternative to intensification is attenuation, or using a hazardous material under the least hazardous conditions. Thus large quantities of liquefied chlorine, ammonia, and petroleum gas can be stored as refrigerated liquids at atmospheric pressure instead of under pressure at ambient tempera-

ture. (Leaks from the refrigeration equipment should also be considered, so there is probably no net gain in refrigerating quantities less than a few hundred tons.) Dyestuffs that form explosive dusts can be handled as slurries.

Limitation of Effects of Failures Effects of failures can be limited by equipment design or change in reaction conditions, rather than by adding protective equipment. For example:

- Heating media such as steam or hot oil should not be hotter than the temperature at which the materials being heated are liable to ignite spontaneously or react uncontrollably.
- Spiral-wound gaskets are safer than fiber gaskets because if the bolts work loose or are not tightened correctly, the leak rate is much lower.
- Tubular reactors are safer than pot reactors as the inventory is usually lower and a leak can be stopped by closing a valve.
- Vapor-phase reactors are safer than liquid-phase ones as the mass flow rate through a hole of a given size is much less. (This is also an example of attenuation.)
- A small, deep diked area around a storage tank is safer than a large shallow one as the evaporation rate is lower and the area of any fire is smaller.
- Changing the order of operations, reducing the temperature, or changing another parameter can prevent many runaway reactions.
- Reduce the frequency of hazardous operations such as sampling or maintenance. We should consider the optimum balance between reliability and maintenance.

Simplification Simpler plants are friendlier than complex ones as they provide fewer opportunities for error and less equipment which can go wrong. Some of the reasons for complication in plant design are

- The need to control hazards. If we can intensify or carry out one of the other actions already discussed, we need less added protective equipment and plants will therefore be simpler.
- A desire for flexibility. Multistream plants with numerous crossovers and valves, so that any item can be used on any stream, have numerous leakage points, and errors in valve settings are easily made.
- Lavish provision of installed spares with the accompanying isolation and changeover valves.
- Continued adherence to traditional rules or practices that are no longer necessary.
- Design procedures that result in a failure to identify hazards until late in design. By this time it is impossible to avoid the hazard, and all we can do is to add complex equipment to control it.

Knock-on Effects Plants should be designed so that those incidents that do occur do not produce knock-on or domino effects. This can be done, e.g., by

- Providing firebreaks, about 15 m wide, between sections, similar to firebreaks in a forest, to restrict the spread of fire.
- Locating equipment that is liable to leak out-of-doors so that leaks of flammable gases and vapors are dispersed by natural ventilation. Indoors a few tens of kilograms are sufficient for an explosion that can destroy the building. Outdoors a few tons are necessary for serious damage. A roof over a piece of equipment, such as a compressor, is acceptable, but walls should be avoided. If leaks of toxic gases are liable to occur, it may be safer to locate the plant indoors, unless leaks will disperse before they reach members of the public or employees on other units.

- Constructing storage tanks so that the roof-wall weld will fail before the base-wall weld, thus preventing spillage of the contents. In general, in designing equipment we should consider the way in which it is most likely to fail and, when possible, locate or design the equipment so as to minimize the consequences.

Making Incorrect Assembly Impossible Plants should be designed so that incorrect assembly is difficult or impossible. For example, compressor valves should be designed so that inlet and exit valves cannot be interchanged; hose connections of different types or sizes should be used for compressed air and nitrogen.

Making Status Clear It should be possible to see at a glance if equipment has been assembled or installed incorrectly or whether it is in the open or shut position. For example:

- Check valves should be marked so that installation the wrong way round is obvious. It should not be necessary to look for a faint arrow hardly visible beneath the dirt.
- Gate valves with rising spindles are friendlier than valves with non-rising spindles, as it is easy to see whether they are open or shut. Ball valves and cocks are friendly if the handles cannot be replaced in the wrong position.
- Figure 8 plates are friendlier than slip plates (blinds) as their position is apparent at a glance. If slip plates are used, their projecting tags should be readily visible, even when the line is insulated. In addition, spectacle plates are easier to fit than slip plates, if the piping is rigid, and are always available on the job. It is not necessary to search for one, as with slip plates.

Tolerance Whenever possible, equipment should tolerate poor installation or operation without failure. Expansion loops in pipework are more tolerant of poor installation than are expansion joints (bellows). Fixed pipes, or articulated arms, if flexibility is necessary, are friendlier than hoses. For most applications, metal is friendlier than glass or plastic.

Bolted joints are friendlier than quick-release couplings. The former are usually dismantled by a fitter after issue of a permit-to-work. One worker prepares the equipment and another opens it up; the issue of the permit provides an opportunity to check that the correct precautions have been taken. In addition, if the joints are unbolted correctly, any trapped pressure is immediately apparent and the joint can be remade or the pressure allowed to blow off. In contrast, many accidents have occurred because operators opened up equipment that was under pressure, without independent consideration of the hazards, using quick-release couplings. There are, however, designs of quick-release coupling which give the operator a second chance.

Low Leak Rate If friendly equipment does leak, it does so at a low rate which is easy to stop or control. Examples already mentioned are spiral-wound gaskets, tubular reactors, and vapor-phase reactors.

Ease of Control Processes with a flat response to change are obviously friendlier than those with a steep response. Processes in which a rise of temperature decreases the rate of reaction are friendlier than those with a positive temperature coefficient, but this is a difficult ideal to achieve in the chemical industry. However, there are a few examples of processes in which a rise in temperature reduces the rate of reaction. For example, in the manufacture of peroxides, water is removed by a dehydrating agent. If magnesium sulfate is used as the agent, a rise in temperature causes release of water by the agent, diluting the reactants and stopping the reaction (Gerrison and van't Land, *I&EC Process Design*, 24, 1985, p. 893).

Software In some programmable electronic systems (PESs), errors are much easier to detect and correct than in others. Accidentally pressing the wrong key should never produce serious consequences. If we press the delete key on our computers, sometimes we are asked if we really want to do so; but stocks and currency have been accidentally sold or bought because someone pressed the wrong key. If we use the term *software* in the wider sense to cover all procedures, as distinct from hardware or equipment, some software is much friendlier than others. For example, if many types of gaskets or nuts and bolts are stocked, sooner or later the wrong type will be installed. It is better, and cheaper in the long run, to keep the number of types stocked to a minimum even though more expensive types than are strictly necessary are used for some applications.

Actions Needed for the Design of Inherently Safer and User-Friendly Plants

1. Designers need to be made aware that there is scope for improving the friendliness of the plants they design.

2. To achieve many of the changes suggested above, it is necessary to carry out much more critical examination and systematic consideration of alternatives during the early stages of design than has been customary in most companies. Two studies are suggested, one at the conceptual or business analysis stage when the process is being chosen and another at the flow sheet stage. For the latter the usual hazard and operability study (HAZOP) questions may be suitable but with one difference. In a normal HAZOP study on a line diagram, if we are discussing "more of temperature," say, we assume that it is undesirable and look for ways of preventing it. In a HAZOP of a flow sheet, we should ask if "more of temperature" would be better. For the conceptual study, different questions are needed.

Many companies will say that they do consider alternatives during the early stages of plant design. However, what is lacking in many companies is a formal, systematic structured procedure of the HAZOP type.

When a new plant is needed, it is usually wanted as soon as possible, and so there is no time to consider and develop inherently safer designs (or other innovations). When we are designing a new plant, we are conscious of all the improvements we could have made if we had had more time. These possible improvements should be noted and work on their feasibility started, ready for the plant after next. Unless we do so, we will never innovate and will ultimately lose to those who do.

3. To achieve the more detailed improvements suggested above, it may be necessary to add a few questions to those asked during a normal HAZOP. For example, what types of valve, gasket, blind, etc. will be used?

INCIDENT INVESTIGATION AND HUMAN ERROR

Although most companies investigate accidents (and many investigate dangerous incidents in which no one was injured), these investigations are often superficial and we fail to learn all the lessons for which we have paid the high price of an accident. The collection of evidence is usually adequate, but often only superficial conclusions are drawn from it. Identifying the causes of an accident is like peeling an onion. The outer layers deal with the immediate technical causes and triggering events while the inner layers deal with ways of avoiding the hazard and with the underlying weaknesses in the management system (Kletz, *Learning from Accidents*, 3d ed., Gulf Professional, 2001).

Dealing with the immediate technical causes of a leak, e.g., will prevent another leak for the same reason. If we can use so little of the hazardous material that leaks do not matter, or a safer material instead, as discussed above, we prevent all significant leaks of this hazardous material. If we can improve the management system or improve our designs, we may be able to prevent many more accidents.

Other points to watch when you are drawing conclusions from the facts are as follows:

1. Avoid the temptation to list causes we can do little or nothing about. For example, a source of ignition should not be listed as the primary cause of a fire or explosion as leaks of flammable gases are liable to ignite even though we remove known sources of ignition. The cause is whatever led to the formation of a flammable mixture of gas or vapor and air. (Removal of known sources of ignition should, however, be included in the recommendations.)

Similarly, human error should not be listed as a cause. See item 7 below.

2. Do not produce a long list of recommendations without any indication of the relative contributions they will make to the reduction of risk or without any comparison of costs and benefits. Resources are not unlimited, and the more we spend on reducing one hazard, the less there is left to spend on reducing others.

3. A named person should be made responsible for carrying out each agreed recommendation, and a completion date agreed with him or her. The report should be brought forward at this time; otherwise, nothing will happen except a repeat of the accident.

4. Avoid the temptation to overreact after an accident and install an excessive amount of protective equipment or complex procedures that

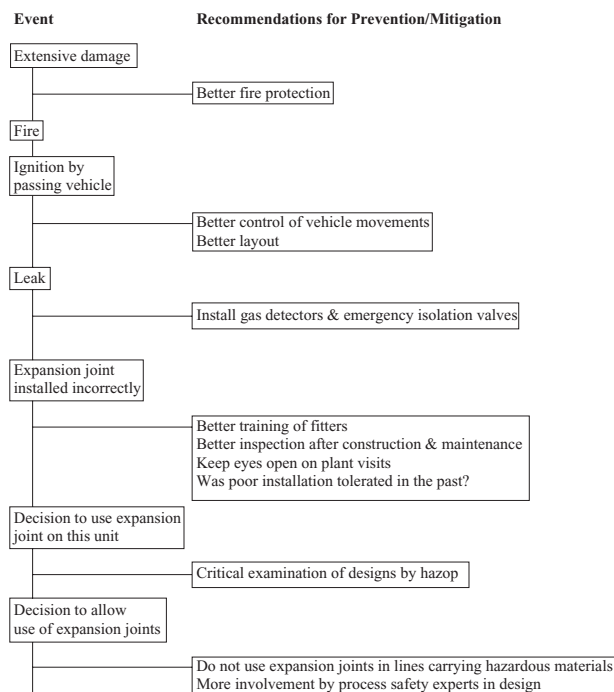


FIG. 23-17 An example of the many ways by which an accident could have been prevented.

are unlikely to be followed after a few years have elapsed. Sometimes an accident occurs because the protective equipment available was not used, but nevertheless the report recommends installation of more protective equipment; or an accident occurs because complex procedures were not followed, and the report recommends extra procedures. It would be better to find out why the original equipment was not used or the original procedures were not followed.

5. Remember that few, if any, accidents have a single cause. In most cases many people had an opportunity to prevent it, from the chemist who developed the process to the operator who closed the wrong valve. Figure 23-17 shows by example the opportunities that were available to prevent a fire or minimize the consequences of an apparently simple incident: an expansion joint (bellows) was incorrectly installed in a pipeline so that it was distorted. After some months it leaked, and a passing vehicle ignited the escaping vapor. Damage was extensive as the surrounding equipment had not been fire-protected to save on costs.

The fitter who installed the expansion joint incorrectly could have prevented the fire. So could the person who was responsible for her or his training and supervision; so could the designers if they had not specified an expansion joint, had carried out a HAZOP, or had consulted experts; so could the author of the company's design standards, and those responsible for the training of designers; those responsible for inspection of workmanship, and anyone who kept his or her eyes open when walking round the plant.

6. When you are reading an accident report, look for the things that are not said. For example, a gland leak on a liquefied flammable gas pump caught fire and caused considerable damage. The report drew attention to the congested layout, the amount of redundant equipment in the area, the fact that a gearbox casing had been made of aluminum, which melted, and several other unsatisfactory features. It did not stress that there had been a number of gland leaks on this pump over the years, that reliable glands are available for liquefied gases at

ambient temperatures, and therefore there was no need to have tolerated a leaky pump on this duty.

7. At one time most accidents were said to be due to human error, and in a sense they all are. If someone, designer, manager, operator, or maintenance worker had done something differently, the accident would not have occurred. However, the term *human error* is not very helpful as different types of error require quite different actions to prevent their happening again. The following classification of errors is recommended as it helps us see the type of action needed to prevent a repeat (Kletz, *An Engineer's View of Human Error*, 3d ed., Institution of Chemical Engineers, United Kingdom, and Taylor and Francis, 2001).

a. Some errors, called mistakes, are due to poor training or instructions—someone did not know what to do. It is a management responsibility to provide good training and instructions and avoid instructions that are designed to protect the writer rather than help the reader. However many instructions we write, problems will arise that are not covered, and so people, particularly operators, should be trained in flexibility, i.e., the ability to diagnose and handle unforeseen situations. If the instructions are hard to follow, can the job be simplified?

b. Some accidents, called violations or noncompliance, occur because someone knew what to do but made a decision not to do it. We should, if possible, simplify the job (if the correct method is difficult, an incorrect one will be used); explain the reasons for the instructions; carry out checks from time to time to see that instructions are being followed; and not turn a blind eye if they are not. While some violations make the job easier, many are made by people who think they have found a better way of doing the job. If instructions are incorrect, a violation can prevent an accident.

The methods of behavioral science can be used to reduce violations. Specially trained members of the workforce keep their eyes open and tactfully draw the attention of fellow workers to violations such as failures to wear protective clothing. These techniques can reduce the incidence of everyday accidents, but they have little or no influence on process safety.

c. Some accidents—mismatches—occur because the job is beyond the physical or mental ability of the person asked to do it, sometimes beyond anyone's ability. We should improve the plant design or method of working.

d. The fourth category is the commonest, a momentary slip or lapse of attention. This happens to everyone from time to time and cannot be prevented by telling people to be more careful, telling them to keep their minds on the job, or better training. In fact, slips and lapses of attention occur only when we are well trained. We put ourselves on autopilot and carry out the task without continually monitoring our progress, though we may check it from time to time. These errors are more likely to occur when we are distracted or stressed. To avoid slips and lapses of attention, all we can do is to change the plant design or method of working so as to remove opportunities for error (or minimize the consequences or provide opportunities for recovery). We should, whenever possible, design user-friendly plants (see above) that can withstand errors (and equipment failures) without suffering serious effects on safety (and output and efficiency). It is more effective to change the behavior of equipment than to try to change the behavior of people.

When an accident report says that an accident was due to human error, the writer usually means an error by an operator or other front line worker. But designers and managers also make errors, not slips or lapses of attention as they usually have time to check their work, as well as mistakes or, less often, violations.

INSTITUTIONAL MEMORY

Most accidents occur not because we do not know how to prevent them but because we do not use the information that is available. The recommendations made after an accident are forgotten when the people involved have left the plant; the procedures they introduced are allowed to lapse, the equipment they installed is no longer used, and the accident happens again. The following actions can prevent or reduce this loss of information.

- Include a note on the reason why in every instruction, code, and standard as well as accounts of accidents that would not have occurred if the instruction, code, or standard had been followed.
- Describe old accidents as well as recent ones in safety bulletins and newsletters, and discuss them at safety meetings.
- Follow up at regular intervals (e.g., during audits) to see that the recommendations made after accidents are being followed, in design as well as operations.
- Make sure that recommendations for changes in design are acceptable to the design organization.
- Remember that the first step down the road to an accident is taken when someone turns a blind eye to a missing blind (or other feature).
- On each unit keep a memory book, a folder of reports on past accidents, which is compulsory reading for new recruits and which others dip into from time to time. It should include relevant reports from other companies but should not include cuts and bruises.
- Never remove equipment before you know why it was installed. Never abandon a procedure before you know why it was adopted.
- When people are moving to other jobs in the company or leaving it, make sure that the remaining employees at all levels have adequate knowledge and experience.
- Include important accidents of the past in the training of undergraduates and company employees. The training should start with accounts of accidents that demonstrate the need for codes, standards, or instructions. Ask audience members to say what *they think* should be done to prevent similar accidents from happening again. More will be remembered after a discussion than after a lecture, and audience members are more likely to be convinced when they have worked out for themselves the actions that should be taken (Kletz, *Lessons from Disaster—How Organizations Have No Memory and Accidents Recur*, Institution of Chemical Engineers, Rugby, United Kingdom, 1993, Chap. 10). Suitable accidents for discussion can be found in books of accident case histories, but local accidents have the greatest impact.
- There are many databases of accidents as well as books of case histories, but they have been little used. We need better retrieval systems so that we can find, more easily than at present, details of past accidents, in our own and other companies, and the recommendations made afterward.

PROCESS SAFETY ANALYSIS

HAZARD ANALYSIS

GENERAL REFERENCES: Alaimo, *Handbook of Chemical Health and Safety*, Oxford University Press, New York, 2001; *Chemical Exposure Index*, 2d ed., AIChE, New York, 1994; Council of the European Union, Directive 96/82/EC, *Prevention of Major Accidents*, December 9, 1996; CPQRA, *Guidelines for Chemical Process Quantitative Risk Analysis*, 2d ed., CCPS-AIChE, New York, 2000; Crawley, Preston, and Tyler, *HAZOP: Guide to Best Practice*, IChemE, Rugby, Warwickshire, U.K., 2002; Crowl and Louvar, *Chemical Process Safety: Fundamentals with Applications*, 2d ed., Prentice-Hall, Englewood Cliffs, N.J., 2002; Dowell, "Managing the PHA Team," *Process Safety Progress* **13**, no. 1, January 1994; *Dow Fire and Explosion Index*, AIChE, New York, January 1994; Harris, *Patty's Industrial Hygiene*, 5th ed., Wiley, New York, 2000; *Guidelines for Design Solutions for Process Equipment Failures*, AIChE, New York, 1997; *HEP Guidelines for Hazard Evaluation Procedures*, 2d ed., AIChE, New York, 1992; Johnson, Rudy, and Unwin, *Essential Practices for Managing Chemical Reactivity Hazards*, AIChE, New York, 2003; Knowlton, *Hazard and Operability Studies*, Chemetics International Co., Ltd., Vancouver, British Columbia, February 1989; *LOPA Layer of Protection Analysis*, AIChE, New York, 2001; Manman, *Lees' Loss Prevention in the Process Industries*, Butterworth-Heinemann, London, 2005; Mosley, Ness, and Hendershot, "Screen Reactive Chemical Hazards Early in Process Development," *Chemical Engineering Progress*, 96(1), pp. 51–65; NOAA, *Chemical Reactivity Worksheet*, Version 1.6, U.S. National Oceanic and Atmospheric Administration, 2004 (<http://response.restoration.noaa.gov/chemaids/react.html>); Skelton, *Process Safety Analysis: An Introduction*, Gulf Publishing, Houston, Tex., 1997; *SVA Guidelines for Analyzing and Managing the Security Vulnerabilities of Fixed Chemical Sites*, AIChE, New York, 2003; Tweeddale, *Managing Risk and Reliability of Process Plants*, Gulf Professional, Houston, Tex., 2003; USEPA, "Risk Management Programs for Chemical Accidental Release Prevention Requirements," 40 CFR 68; U.S. OSHA, "Process Safety Management of Highly Hazardous Chemicals, Explosives and Blasting Agents," 29 CFR 1910.119; Wells, *Hazard Identification and Risk Assessment*, IChemE, Rugby, Warwickshire, U.K., 1996.

Introduction The meaning of *hazard* is often confused with *risk*. *Hazard* is defined as the inherent potential of a material or activity to harm people, property, or the environment. Hazard does not have a probability component.

There are differences in terminology on the meaning of *risk* in the published literature that can lead to confusion. *Risk* has been defined in various ways (CPQRA, 2000, pp. 5, 6). In this edition of the handbook, *risk* is defined as "a measure of human injury, environmental damage, or economic loss in terms of both the incident likelihood and magnitude of the injury, damage, or loss." *Risk* implies a probability of something occurring.

Definitions of Terms Following are some definitions that are useful in understanding the components of hazards and risk.

Accident A specific combination of events or circumstances that leads to an undesirable consequence.

Acute hazard The potential for injury or damage to occur as a result of a short-duration exposure to the effects of an accident.

Cause-consequence A procedure using diagrams to illustrate the causes and consequences of a particular scenario. They are not widely used because, even for simple systems, displaying all causes and outcomes leads to very complex diagrams.

Chemical Exposure Index (CEI) A method of rating the relative potential of acute health hazard to people from possible chemical release incidents, developed by Dow Chemical Company.

Chronic hazard The potential for injury or damage to occur as a result of prolonged exposure to an undesirable condition.

Consequence The direct, undesirable result of an accident, usually measured in health and safety effects, environmental impacts, loss of property, or business costs.

Consequence analysis Once hazards and specific incident scenarios through which those hazards might impact people, the environment, or property have been identified, methods exist for analyzing their consequences (size of vapor cloud, blast damage radius, overpressure expected, etc.). This is independent of frequency or probability.

Domino effect An incident which starts in one piece of equipment and affects other nearby items, such as vessels containing hazardous materials, by thermal blast or fragment impact. This can lead to escalation of consequences or frequency of occurrence. This is also known as a *knock-on effect*.

Event An occurrence involving equipment performance or human action or an occurrence external to the system that causes system upset. An event is associated with an incident, either as a cause or a contributing cause of the incident, or as a response to an initiating event.

Event tree A graphical logic model that identifies and potentially quantifies possible outcomes following an initiating event.

Failure mode and effect analysis (FMEA) A hazard identification technique in which all known failure modes of components or features of a system are considered in turn and undesired outcomes are noted. It is often used in combination with hazard and operability (HAZOP) studies or fault tree analysis.

Fault tree A graphical logic model for representing the combinations of various system states which lead to a particular outcome, known as the *top event*.

Fire and Explosion Index (F&EI) A hazard index developed by Dow Chemical Company used to rank fire and explosion hazards in a chemical process.

Hazard An inherent physical or chemical characteristic that has the potential for causing harm to people, the environment, or property.

HAZOP (HAZard and OPerability study) A formal hazard identification and evaluation procedure based on the application of guide words to identify possible deviations from the intended operation of the process.

Incident The loss of containment of material or energy, e.g., a leak of a flammable and toxic gas.

Interaction matrix A two-dimensional matrix listing all components of interest on the *x* and *y* axes and recording the consequences of mixing of these components for each combination of the components. It is useful for identifying chemical reaction hazards and incompatibilities.

Layer-of-protection analysis (LOPA) A method, based on event tree analysis, of evaluating the effectiveness of independent protection layers in reducing the likelihood or severity of an undesired event.

Process hazard analysis (PHA) Any of a number of techniques for understanding and managing the risk of a chemical process or plant. Examples of PHA techniques include HAZOP, checklists, what-if methods, fault tree analysis, event tree analysis, and others.

Process Hazard Analysis Regulations In the United States, the OSHA rule for Process Safety Management (PSM) of Highly Toxic Hazardous Chemicals, 29 CFR 1910.119, part (e), requires an initial PHA and an update every 5 years for processes that handle listed chemicals or contain over 10,000 lb (4356 kg) of flammable material. The PHA must be done by a team, must include employees such as operators and mechanics, and must have at least one person skilled in the methodology employed. Suggested methodologies from Process Safety Management are listed in Table 23-20. The PHA must consider hazards listed in the PSM Rule, part (e), including information from previous incidents with potential for catastrophic consequences, engineering and administrative controls and consequences of their failure, facility siting, and human factors. Consequences of failure of controls must be considered. The key to good PHA documentation is to do it right away while the information is fresh. Periodic follow-up of the PHA study is needed by management and safety professionals to confirm that all recommendations have been addressed. The PHA must be reviewed as part of the management of change procedures when the facility is modified in any way. (Dowell, 1994, pp. 30–34.)

As required by the Clean Air Act Amendments of 1990, the USEPA mandates a Risk Management Program (RMP) for listed substances (40 CFR 68). RMP requires (1) a hazard assessment that details the potential effects of an accidental release, an accident history of the last 5 years, and an evaluation of worst-case and alternative accidental releases; (2) a prevention program that includes safety precautions and maintenance, monitoring, and employee training measures; and (3) an emergency response program that spells out emergency health care, employee training measures, and procedures for informing the public and response agencies, should an accident occur.

Most countries also have regulations analogous to the U.S. regulations. For example, the European Union issued the “Seveso II” Directive in 1996 (replacing the original 1982 directive) which requires all member states to implement regulations for the control of major accident hazards. Also, in addition to the U.S. government requirements,

many state and local governments have implemented regulations requiring process hazard analysis and risk management.

Hazard Identification and Analysis Tools The hazard and risk assessment tools used vary with the stage of the project from the early design stage to plant operations. Many techniques are available. In the following discussion, they will be categorized as hazard identification and analysis tools, hazard ranking methods, and logic model methods. Reviews done early in projects often result in easier, more effective changes.

Safety, Health, Environmental, and Loss Prevention Reviews Most chemical companies have specific internal protocols defining these reviews, which may have different names or descriptions in different organizations. In most organizations, these reviews are conducted at various stages in the process life cycle, from initial process conceptualization, through laboratory development, scale-up, plant design, start-up, operation, modification, and shutdown. The scope and focus of the review will be different at different stages in development, with reviews early in process development focusing on major hazards and strategies for managing the risks. As the process and plant become clearly defined, the reviews will focus more on details of the design and operation. The purpose of the reviews is to have an independent (from the development, design, or operation team) evaluation of the process and layout from safety, industrial hygiene, environmental, and loss prevention points of view. It is often desirable to combine these reviews to improve the efficiency of the use of time for the reviewers (*HEP* Chaps. 4.1, 4.4, 6.1, 6.4, 13).

Checklists Checklists are simple means of applying experience to designs or situations to ensure that the features appearing in the list are not overlooked. Checklists tend to be general and may not be appropriate to a specific situation. They may not handle adequately the novel design or unusual process. (*HEP*, Chaps. 4.2, 6.2, 16, 20). The *CCPS Design Solutions for Process Equipment Failures* (1997) provides a useful set of checklists for common chemical processing equipment.

What-if At each process step, what-if questions are formulated and answered to evaluate the effects of component failures or procedural errors. This technique relies on the experience level of the questioner. What-if methods are often used in conjunction with checklists (*HEP* Chaps. 4.5, 4.6, 6.5, 6.6, 12, 20).

Failure Mode and Effect Analysis (FMEA) This is a systematic study of the causes of failures and their effects. All causes or modes of failure are considered for each element of a system, and then all possible outcomes or effects are recorded (*HEP* Chaps. 4.8, 6.8, 19).

Reactive Chemistry Reviews The process chemistry is reviewed for evidence of exotherms, shock sensitivity, and other instability, with emphasis on possible exothermic reactions. The purpose of this review is to prevent unexpected and uncontrolled chemical reactions. Reviewers should be knowledgeable people in the field of reactive chemicals and include people from loss prevention, manufacturing, and research. The *CCPS Essential Practices for Managing Chemical Reactivity Hazards* provides a useful protocol for identifying chemical reactivity hazards (Johnson et al., 2003). A series of questions about the chemical handling operations and the materials are used to determine if there are possible reactivity hazards. Figure 23-18 summarizes the CCPS protocol for identifying reaction hazards.

Interaction Matrix (Compatibility Chart) An interaction matrix is a tool for identifying and understanding potential hazards, including reaction hazards, from combinations of materials (*HEP*, 1992, Chaps. 3.3, 11.3). The most common representation is a two-dimensional matrix, listing all components of interest (including, e.g., chemicals; materials of construction; potential contaminants; environmental contaminants such as air, rust, or water; utilities). The consequences of mixing the materials for each row-column intersection are identified. Figure 23-19 is an example of an interaction matrix. Johnson et al. (2003) describe the use of the interaction matrix, and Mosley et al. (2000) provide a specific example. The U.S. National Oceanic and Atmospheric Administration (NOAA) has developed a computer tool, the *Chemical Reactivity Worksheet*, which can generate an interaction matrix for materials in the program's database (NOAA, 2004).

Industrial Hygiene Reviews These reviews evaluate the potential of a process to cause harm to the health of people. The review

TABLE 23-20 Process Hazard Analysis Methods Listed in the OSHA Process Safety Management Rule

- What-if
- Checklist
- What-if/checklist
- Hazard and operability study (HAZOP)
- Failure mode and effect analysis (FMEA)
- Fault tree analysis (FTA)
- An appropriate equivalent methodology

SOURCE: Dowell, 1994, pp. 30–34.

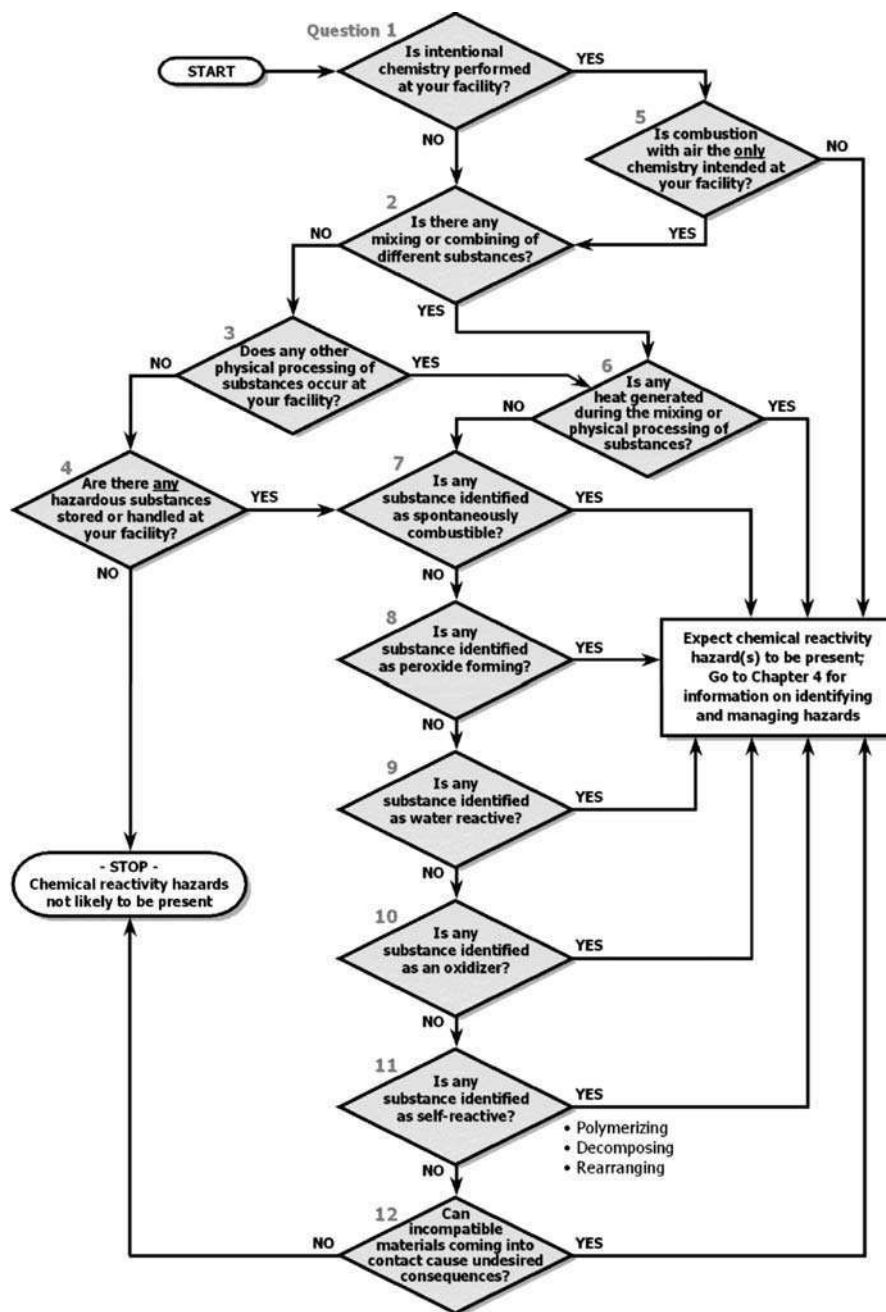


FIG. 23-18 CCPS preliminary screening for chemical reactivity hazards. [From Johnson et al. (2003); copyright AIChE; reproduced with permission.]

normally deals with chronic effects of exposure to chemicals and other harmful agents (e.g., noise, heat, repetitive motion) in the workplace. Chapter 44 of *Patty's Industrial Hygiene* reviews industrial hygiene survey procedures.

Facilities Reviews There are many kinds of facilities reviews that are useful in detecting and preventing process safety problems. They include prestart-up reviews (before the plant operates), new-plant reviews (the plant has started, but is still new), reviews of exist-

ing plants (safety, technology, and operations audits and reviews), management reviews, critical instrument reviews, and hazardous materials transportation reviews.

HAZOP (HEP, 1992, Chaps. 4.7, 6.7, 14, 18; Knowlton, 1989; Lees, 2005; CPQRA, 2000, pp. 583–587). HAZOP is a formal hazard identification and evaluation procedure designed to identify hazards to people, process plants, and the environment. The techniques aim to stimulate, in a systematic way, the imagination of designers and people

SUBSTANCE					
1 Hydrochloric Acid (35%)		1			
2 Sulfuric Acid (90%)	R ¹		2		
3 Acetic Acid	? ⁸	R ²		3	
4 Ethanol	NR ³	R ⁴	NR ¹⁵		4
5 Ethylenediamine	R ⁵	R ⁶	R ⁷	NR ¹²	5
6 Water	R ⁹	R ¹⁰	NR ¹¹	NR ¹³	R ¹⁴ 6

Legend:

R	Reactive under the stated scenario - incompatible
NR	Non-Reactive under the stated scenario - compatible
?	unknown – assume incompatible until further information is obtained

Footnotes/Information Sources:

- 1 Unlikely to be compatible – USCG chart NVC 4-75 indicates a hazard with non-oxidizing acids plus sulfuric acid. Heat of mixing may be significant.
- 2 Unlikely to be compatible – the P-168 chart indicates that gas and heat are formed; USCG chart NVC 4-75 indicates a hazard when combining sulfuric and organic acids.
- 3 Primary alcohols do not react with aq. HCl at ambient temperature.
- 4 Heat of solution followed by reaction to form ethyl hydrogen sulfate.
- 5 Lab experiment 980001 (50/50 mix) resulted in a significant heat of neutralization.
- 6 Lab experiment 980002 (50/50 mix by volume) resulted in a XXX C adiabatic temperature rise.
- 7 Organic acids and amines are generally incompatible due to acid/base neutralization heat.
- 8 The P-168 and USCG charts indicate no hazard; most likely compatible, but lab testing should be performed.
- 9 Heat of mixing may be a concern in some circumstances. The maximum adiabatic temperature rise is XX C (see XYZ Encyclopedia of Chemical Technology).
- 10 Heat of mixing may be a concern in some circumstances. The maximum adiabatic temperature rise is XX C (see XYZ Encyclopedia of Chemical Technology). Violent reaction with splattering if water is added to the acid.
- 11 Lab experiment 98005 showed that mixing acetic acid and water is endothermic at room temperature.
- 12 Lab experiments 98003 and 98008 indicate that the materials do not generate heat or gases when mixed nor when heated to 100 C. Although the USCG chart NVC 4-75 indicates that some alcohols and amines are incompatible, ethylene diamine has been found to be compatible with many alcohols; see Appendix of USCG Guide.
- 13 Plant experience has shown materials to be compatible.
- 14 Mildly exothermic hydrate formation.
- 15 Very slow, nearly thermoneutral, equilibrium-limited esterification at ambient temperature.

FIG. 23-19 Hypothetical compatibility chart. (Copyright ASTM International. Reprinted with permission.)

who operate plants or equipment to identify potential hazards. HAZOP studies assume that a hazard or operating problem can arise when there is a deviation from the design or operating intention. Actions to correct identified hazard or operational scenarios can then be taken before a real incident occurs. The primary goal in performing a HAZOP study is to identify, not analyze or quantify, the hazards in a process. The end product of a HAZOP is a list of concerns and recommendations for prevention of problems, not an analysis of the occurrence, frequency, overall effects, and the definite solution. A HAZOP study is most cost-effective when done during plant design—it is easier and cheaper to change a design than to modify an existing plant. However, HAZOP is a valuable process hazard analysis tool at any stage in the life cycle of a plant.

These studies make use of the combined experience and training of a group of knowledgeable people in a structured setting. Some key HAZOP terms are as follows.

Intention How the part or process is expected to operate.

Guide words Simple words used to qualify the intention in order to guide and stimulate creative thinking and so discover deviations. Table 23-21 describes commonly used guide words.

Deviations Departures from the intention discovered by systematic application of guide words.

Causes Reasons that deviations might occur.

Consequences Results of deviations if they occur.

Safeguards Prevention, mitigation, and control features which already exist in the plant, or which are already incorporated in a new design.

Actions Prevention, mitigation, and control features which do not currently exist and are recommended by the HAZOP team. Actions may also include recommendations for additional study if the HAZOP team does not have sufficient information, or time to understand a concern sufficiently to make a specific recommendation.

TABLE 23-21 Some HAZOP Guide Words Used in Conjunction with Process Parameters

Guide word	Meanings	Comments
No, Not, None	Complete negation of design intentions	No part of intention is achieved and nothing else occurs
More	Quantitative increases	Quantities and relevant physical properties such as flow rates, heat, pressure
Less	Quantitative decreases of any relevant physical parameters	Same as above
As well as	Qualitative increase	All design and operating intentions are achieved as well as some additional activity
Part of	A qualitative decrease	Some parts of the intention are achieved, others are not
Reverse	Logical opposite of intention	Activities such as reverse flow or chemical reaction, or poison instead of antidote
Other than	Complete substitution	No part of intention is achieved; something quite different happens

SOURCE: Knowlton, 1989.

The HAZOP study is not complete until response to actions has been documented. Initial HAZOP planning should establish the management follow-up procedure that will be used.

The guide words are used in conjunction with the process intentions to generate possible deviations from the intended operation (see Table 23-21). Some examples of deviations that might be generated in the course of a HAZOP study include

- No flow
- Reverse flow
- Less flow
- Increased temperature
- Decreased temperature
- Composition change
- Sampling
- Corrosion/erosion

The HAZOP team then determines the specific causes of each deviation; e.g., no flow of a particular material in a specified pipe might include causes such as a manual valve improperly closed, pump stops, pipe plugged with solids, etc. The HAZOP team then determines the consequences of the deviation for each cause and qualitatively decides the magnitude of hazard. The team identifies any existing safeguards in the plant or design and qualitatively judges whether they are adequate. If the team determines that additional safeguards are required, it may recommend specific actions. The team may determine that the issue requires greater study than can be accommodated in the time frame of a HAZOP meeting and recommend more extensive evaluation to determine if further action is needed, and what that action should be.

Many HAZOP studies incorporate a qualitative evaluation of risk to assist the team in evaluating the adequacy of existing safeguards, and the need for additional safeguards. This involves constructing a risk matrix, such as the one shown in Fig. 23-20. The team determines, based on its knowledge of the plant, experience, and engineering judgment, which of the several consequence and likelihood categories in the risk matrix best describe the particular deviation-cause-consequence sequence under consideration. Scenarios with high consequence and high frequency represent a large risk, those with low likelihood and consequence are of low risk. An organization can use the matrix to establish guidelines for which of the boxes in the risk matrix require action.

HAZOP studies may be made on batch as well as continuous processes. For a continuous process, the working document is usually a set of flow sheets or piping and instrument diagrams (P&IDs). Batch processes have another dimension: time. Time is usually not significant with a continuous process that is operating smoothly, although start-up and shutdown must also be considered, when the continuous process will resemble a batch process. For batch processes, the working documents consist not only of the flow sheets or P&IDs but also

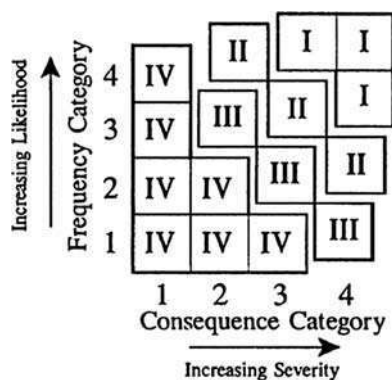


FIG. 23-20 Example of PHA qualitative risk matrix.

TABLE 23-22 HAZOP Guide Words Associated with Time

Guide word	Meaning
No time	Step(s) missed
More time	Step does not occur when it should
Less time	Step occurs before previous step is finished
Wrong time	Flow or other activity occurs when it should not

SOURCE: Knowlton, 1989.

the operating procedures. One method to incorporate this fourth dimension is to use guide words associated with time, such as those described in Table 23-22.

HAZOP studies involve team members, at least some of whom have had experience in the plant design to be studied. These team members apply their expertise to achieve the aims of HAZOP. There are four overall aims to which any HAZOP study should be addressed:

1. Identify as many deviations as possible from the way the design is expected to work, their causes, and problems associated with these deviations.

2. Decide whether action is required, and identify ways in which the problem can be solved.

3. Identify cases in which a decision cannot be made immediately, and decide what information or action is required.

4. Ensure that required actions are followed through.

The team leader is a key to the success of a HAZOP study and should have adequate training for the job. Proper planning is important to success. The leader is actually a facilitator (a discussion leader and one who keeps the meetings on track) whose facilitating skills are just as important as technical knowledge. The leader outlines the boundaries of the study and ensures that the design intention is clearly understood. The leader applies guide words and encourages the team to discuss causes, consequences, and possible remedial actions for each deviation. Prolonged discussions of how a problem may be solved should be avoided. Ideally, the team leader should be accompanied by a scribe or recorder, freeing the leader for full-time facilitating. The scribe should take notes in detail for full recording of as much of the meeting as is necessary to capture the intent of actions and recommendations.

Many companies offer computer tools to help facilitate and document HAZOP studies.

Team size is important. Having fewer than three contributing members, excluding the secretary and leader, will probably reduce team effectiveness. A team size of five to eight, including the leader and scribe, is probably optimum. The time required for HAZOP studies is significant. It has been estimated that each line or node (a node is usually a line or an item of equipment) may require in the range of 30 min for an experienced team, although the time may vary widely depending on the complexity of the system. It should be recognized that the time required for HAZOP studies may not really be additional time for the project as a whole, particularly if started early enough in the design, and may actually save time on the project. It also should make possible smoother start-ups and make the process or plant safer and easier to operate, which will more than pay back the cost of the HAZOP study during the life of the plant.

Hazard Ranking Methods Hazard ranking methods (HEP, 1992, Chaps. 4.3, 6.3) allow the analyst to compare the hazards of several processes, plants, or activities. They can be used to compare alternative chemical process routes, plant designs, plant siting options, or other design choices. Hazard ranking methods can also be useful for prioritizing facilities for additional risk management studies. They generally produce a numerical score for the process being evaluated. The scores generally do not have units and are only meaningful relative to each other in the context of the hazard index being used. Some of the more commonly used hazard ranking methods are briefly discussed.

NFPA Standard System for Identification of Health, Flammability, Reactivity, and Related Hazards (NFPA 704, 2001) This printed material is not the complete and official position of the National Fire Protection Association on the referenced subject, which is represented only by the standard in its entirety.

TABLE 23-23 NFPA 704 System for Identification of Hazards

Degree of hazard	Health hazard color code: blue	Flammability hazard color code: red	Instability hazard color code: yellow
4	Materials that, under emergency conditions, can be lethal	Materials that will rapidly or completely vaporize at atmospheric pressure and normal ambient temperature or that are readily dispersed in air and will burn readily.	Materials that in themselves are readily capable of detonation or explosive decomposition or explosive reaction at normal temperatures and pressures.
3	Materials that, under emergency conditions, can cause serious or permanent injury	Liquids and solids that can be ignited under almost all ambient temperature conditions. Materials in this degree produce hazardous atmospheres with air under almost all ambient temperatures or, though unaffected by ambient temperatures, are readily ignited under almost all conditions.	Materials that in themselves are capable of detonation or explosive decomposition or explosive reaction, but that require a strong initiating source or that must be heated under confinement before initiation.
2	Materials that, under emergency conditions, can cause temporary incapacitation or residual injury	Materials that must be moderately heated or exposed to relatively high ambient temperatures before ignition can occur. Materials in this degree would not, under normal conditions, form hazardous atmospheres with air, but under high ambient temperatures or under moderate heating could release vapor in sufficient quantities to produce hazardous atmospheres with air.	Materials that readily undergo violent chemical change at elevated temperatures and pressures.
1	Materials that, under emergency conditions, can cause significant irritation	Materials that must be preheated before ignition can occur. Materials in this degree require considerable preheating, under all ambient temperature conditions, before ignition and combustion can occur.	Materials that in themselves are normally stable, but that can become unstable at elevated temperatures and pressures.
0	Materials that, under emergency conditions, would offer no hazard beyond that of ordinary combustible materials	Materials that will not burn under typical fire conditions, including intrinsically noncombustible materials such as concrete, stone, and sand.	Materials that in themselves are normally stable, even under fire conditions.

SOURCE: Reprinted with permission from NFPA 704, *Standard System for the Identification of the Fire Hazards of Materials*, National Fire Protection Association, Quincy, Mass., 2001. This printed material is not the complete and official position of the National Fire Protection Association on the referenced subject, which is represented only by the standard in its entirety.

This is a brief summary of NFPA 704 which addresses hazards that may be caused by short-term exposure to a material during handling under conditions of fire, spill, or similar emergencies. This standard provides a simple, easily recognized and understood system of markings. The objective is to provide on-the-spot identification of hazardous materials. The markings provide a general idea of the hazards of a material and the severity of these hazards as they relate to handling, fire protection, exposure, and control. This standard is not applicable to transportation or to use by the general public. It is also not applicable to chronic exposure. For a full description of this standard, refer to NFPA 704.

The system identifies the hazards of a material in four principal categories: health, flammability, reactivity, and unusual hazards such as reactivity with water. The degree of severity of health, flammability, and reactivity is indicated by a numerical rating that ranges from 0 (no hazard) to 4 (severe hazard). Table 23-23 describes the characteristics associated with the various ratings; for a detailed description of the degree of severity ratings, see NFPA 704. The information is presented in a square-on-point (diamond) field of numerical ratings, e.g., as shown in Figs. 23-21 through 23-23. Information is presented as follows:

- Health rating in blue at nine o'clock
- Flammability rating in red at twelve o'clock
- Reactivity hazard rating in yellow at three o'clock
- Unusual hazards at six o'clock

Materials that demonstrate unusual reactivity with water are identified as W, and materials that possess oxidizing properties shall be identified by the letters OX. Other special hazard symbols may be used to identify radioactive hazards, corrosive hazards, substances that are toxic to fish, and so on. The use of this system provides a standard method of identifying the relative degree of hazard that is contained in various tanks, vessels, and pipelines.

Fire and Explosion Index (F&EI) (Dow Fire and Explosion Index Hazard Classification Guide, 1994; Mannan, 2005, pp. 8/13–8/22.)

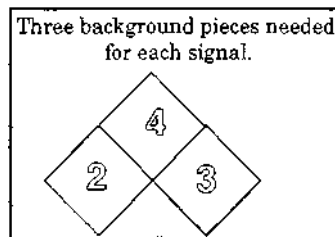


FIG. 23-21 NFPA 704 hazard ratings for use where specified color background is used with numerals of contrasting colors. (NFPA 704, 2001.)

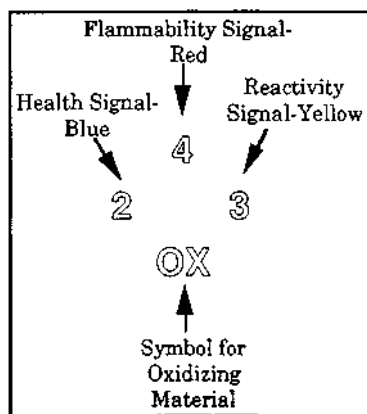


FIG. 23-22 For use where white background is used. (NFPA 704, 2001.)

The F&EI is used to rate the potential of hazard from fires and explosions. Its purpose is to quantify damage from an incident. It identifies equipment that could contribute to an incident and ways to mitigate possible incidents. It is a way to communicate to management the quantitative hazard potential. The F&EI measures realistic maximum loss potential under adverse operating conditions. It is based on quantifiable data. It is designed for flammable, combustible, and reactive materials that are stored, handled, or processed. It does not address frequency (risk) except indirectly, nor does it address specific hazards to people except indirectly. The goals of the F&EI are to raise awareness of loss potential and identify ways to reduce potential severity and potential dollar loss in a cost-effective manner. The index number has significance as a comparison and in calculations to estimate the *maximum probable property damage* (MPPD). It also provides a method for measuring the effect of outage (plant being shut down) on the business. It is easy for users to get credible results with a small amount of training.

Chemical Exposure Index (CEI) (*Chemical Exposure Index*, 1994; Mannan, 2005, pp. 8/22–8/26.) The CEI provides a method of rating the relative potential of acute health hazard to people from possible chemical release incidents. It may be used for prioritizing initial process hazard analysis and establishing the degree of further analysis needed. The CEI also may be used as part of the site review process. The system provides a method of ranking one risk relative to another. It is not intended to define a particular containment system as safe or unsafe, but provides a way of comparing toxic hazards. It deals with acute, not chronic, releases. Flammability and explosion hazards are not included in this index. To develop a CEI, information needs include

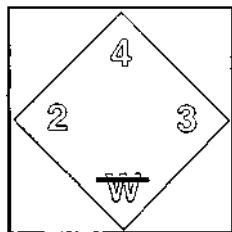


FIG. 23-23 For use where white background is used or for signs or placards. (NFPA 704, 2001.)

- An accurate plot plan of the plant and surrounding area
- A simplified process flow sheet showing containment vessels, major piping, and quantity of chemicals
- Physical, chemical, and toxicological properties of the chemicals
- Process variables such as pressures, temperatures, and quantities of materials

Substance Hazard Index The Substance Hazard Index (SHI) was developed for use by U.S. OSHA for ranking material hazards during the development of the OSHA Process Safety Management regulations. SHI is defined as “the equilibrium vapor concentration (EVC) of a material at 20°C divided by an acute toxicity concentration.” SHI is a measure of the ability of a released material to be transported through the air and impact people. The EVC is defined as the vapor pressure of the material at 20°C $\times 10^6/760$. Different organizations using the SHI may use different toxicity measures, for example, ERPG-3, IDLH, or AEGL.

Consequence-Based Ranking Systems Release consequence modeling can be used to rank potential chemical hazards. For example, the USEPA’s RMP regulations require consequence modeling for a predefined worst-case scenario—release of the entire contents of the largest container of a material in 10 min. EPA provides lookup tables and software (RMPComp) to assist in estimating the hazard distances for materials covered by the RMP regulations.

Logic Model Methods The following tools are most commonly used in quantitative risk analysis, but can also be useful qualitatively to understand the combinations of events which can cause an accident. The logic models can also be useful in understanding how protective systems impact various potential accident scenarios. These methods will be thoroughly discussed in the “Risk Analysis” subsection. Also, hazard identification and evaluation tools discussed in this section are valuable precursors to a quantitative risk analysis (QRA). Generally a QRA quantifies the risk of hazard scenarios which have been identified by using tools such as those discussed above.

- Fault tree analysis
- Event tree analysis
- Cause-consequence diagram
- Layer-of-protection analysis (LOPA)

RISK ANALYSIS

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RISK ESTIMATION REFERENCES: Health and Safety Executive, *Canvey—An Investigation of Potential Hazards from the Operations in the Canvey*

Island/Thurrock Area, HMSO, London, 1978. Rasmussen, *Reactor Safety Study: An Assessment of Accident Risk in U.S. Commercial Nuclear Power Plants*, WASH-1400 NUREG 75/014, Washington, D.C., 1975. Rijnmond Public Authority, *A Risk Analysis of 6 Potentially Hazardous Industrial Objects in the Rijnmond Area—A Pilot Study*, D. Reidel, Boston, 1982. Considine, *The Assessment of Individual and Societal Risks*, SRD Report R-310, Safety and Reliability Directorate, UKAEA, Warrington, 1984. Baybutt, "Uncertainty in Risk Analysis," Conference on Mathematics in Major Accident Risk Assessment, University of Oxford, U.K., 1986.

RISK CRITERIA AND RISK DECISION MAKING: Ale, "The Implementation of an External Safety Policy in the Netherlands," *Intl. Conference on Hazard Identification and Risk Analysis, Human Factors and Reliability in Process Safety*, AIChE, New York, pp. 173–183, 1992; Gibson, "Hazard Analysis and Risk Criteria," *Chemical Engineering Progress*, Nov. 1980, pp. 46–50; Gibson, "The Use of Risk Criteria in the Chemical Industry," *Trans. of IChemE* 71, Pt. B, pp. 117–123, 1993; Health and Safety Executive, *Reducing Risks Protecting People*, HSE's Decision Making Process, HSE Books, London, 2001; Helmers and Schaller, "Calculated Process Risks and Health Management," *Plant Operations Progress*, 1, no. 3, pp. 190–194, 1982; Renshaw, "A Major Accident Prevention Program," *Plant Operations Progress*, 9, no. 3, pp. 194–197, 1990; Pikaar and Seaman, *A Review of Risk Control*, Zoetermeer, Ministerie VROM, Netherlands, 1995; *Tools for Making Acute Risk Decisions with Chemical Process Safety Implications*, CCPS-AIChE, New York, 1995; Hamm and Schwartz, "Issues and Strategies in Risk Decision Making," *Intl. Process Safety Management Conference and Workshop*, San Francisco, AIChE, New York, pp. 351–371, 1993.

Introduction The previous sections dealt with techniques for qualitative hazard analysis only. This section addresses the quantitative methodologies available to analyze and estimate risk, which is a function of both the consequences of an incident and its frequency. The application of these methodologies in most instances is not trivial. A significant allocation of resources is necessary. Therefore, a selection process or risk prioritization process is advised before considering a risk analysis study.

Important definitions are as follows.

Accepted risk The risk is considered tolerable for a given activity by those responsible for managing or regulating the operation of a hazardous facility. The term *acceptable risk* has often been used, but this inevitably raises the question, "Acceptable to whom?" Tweeddale (*Managing Risk and Reliability of Process Plants*, Gulf Professional, Houston, Tex, 2003, p. 70) suggests that *accepted risk* is a better term because it makes it clear that the risk has been accepted by those responsible for the decisions on how to build, operate, and regulate the facility.

Event sequence A specific, unplanned sequence of events composed of initiating events and intermediate events that may lead to an incident.

Frequency The rate at which observed or predicted events occur.

Incident outcome The physical outcome of an incident; e.g., a leak of a flammable and toxic gas could result in a jet fire, a vapor cloud explosion, a vapor cloud fire, or a toxic cloud.

Probability The likelihood of the occurrence of events, the values of which range from 0 to 1.

Probability analysis Way to evaluate the likelihood of an event occurring. By using failure rate data for equipment, piping, instruments, and fault tree techniques, the frequency (number of events per unit time) can be quantitatively estimated.

Probit model A mathematical model of dosage and response in which the dependent variable (response) is a probit number that is related through a statistical function directly to a probability.

Quantitative risk assessment (QRA) The systematic development of numerical estimates of the expected frequency and consequence of potential accidents associated with a facility or an operation. Using consequence and probability analyses and other factors such as population density and expected weather conditions, QRA estimates the fatality rate for a given set of events.

Risk A measure of economic loss or human injury in terms of both incident likelihood (frequency) and the magnitude of the loss or injury (consequence).

Risk analysis The development of an estimate of risk based on engineering evaluation and mathematical techniques for combining

estimates of incident consequences and frequencies. Incidents in the context of the discussion in this chapter are acute events which involve loss of containment of material or energy.

Risk assessment The process by which results of a risk analysis are used to make decisions, either through a relative ranking of risk reduction strategies or through comparison with risk targets. The terms *risk analysis* and *risk assessment* are often used interchangeably in the literature.

A typical hazard identification process, such as a hazard and operability (HAZOP) study, is sometimes used as a starting point for selection of potential major risks for risk analysis. Other selection or screening processes can also be applied. However major risks are chosen, a HAZOP study is a good starting point to develop information for the risk analysis study. A major risk may qualify for risk analysis if the magnitude of the incident is potentially quite large (high potential consequence) or if the frequency of a severe event is judged to be high (high potential frequency) or both. A flowchart which describes a possible process for risk analysis is shown in Fig. 23-24.

The components of a risk analysis involve the estimation of the frequency of an event, an estimation of the consequences, and the selection and generation of the estimate of risk itself.

A risk analysis can have a variety of potential goals:

1. To screen or bracket a number of risks in order to prioritize them for possible future study
2. To estimate risk to employees
3. To estimate risk to the public
4. To estimate financial risk
5. To evaluate a range of risk reduction measures
6. To meet legal or regulatory requirements
7. To assist in emergency planning

The scope of a study required to satisfy these goals will be dependent upon the extent of the risk, the depth of the study required, and the level of resources available (mathematical models and tools and skilled people to perform the study and any internal or external constraints).

A risk analysis can be applied to fixed facilities or transportation movements, although much of the attention today still centers on the former. In a fixed-facility risk analysis, QRA can aid risk management decisions with respect to

1. Chemical processes
2. Process equipment
3. Operating procedures
4. Chemical inventories
5. Storage conditions

In a transportation risk analysis (TRA), the risk parameters are more extensive, but more restrictive in some ways. Examples of risk parameters that could be considered are

1. Alternate modes of transport
2. Routes
3. Travel restrictions
4. Shipment size
5. Shipping conditions (e.g., pressure, temperature)
6. Container design
7. Unit size (e.g., bulk versus drums)

The objective of a risk analysis is to reduce the level of risk wherever practical. Much of the benefit of a risk analysis comes from the discipline which it imposes and the detailed understanding of the major contributors of the risk that follows. There is general agreement that if risks can be identified and analyzed, then measures for risk reduction can be effectively selected.

The expertise required in carrying out a risk analysis is substantial. Although various software programs are available to calculate the frequency of events or their consequences, or even risk estimates, engineering judgment and experience are still very much needed to produce meaningful results. And although professional courses are available in this subject area, there is a significant learning curve required not only for engineers to become practiced risk analysts, but also for management to be able to understand and interpret the results. For these reasons, it may be useful to utilize a consultant organization in this field.

The analysis of a risk—that is, its estimation—leads to the assessment of that risk and the decision-making processes of selecting the

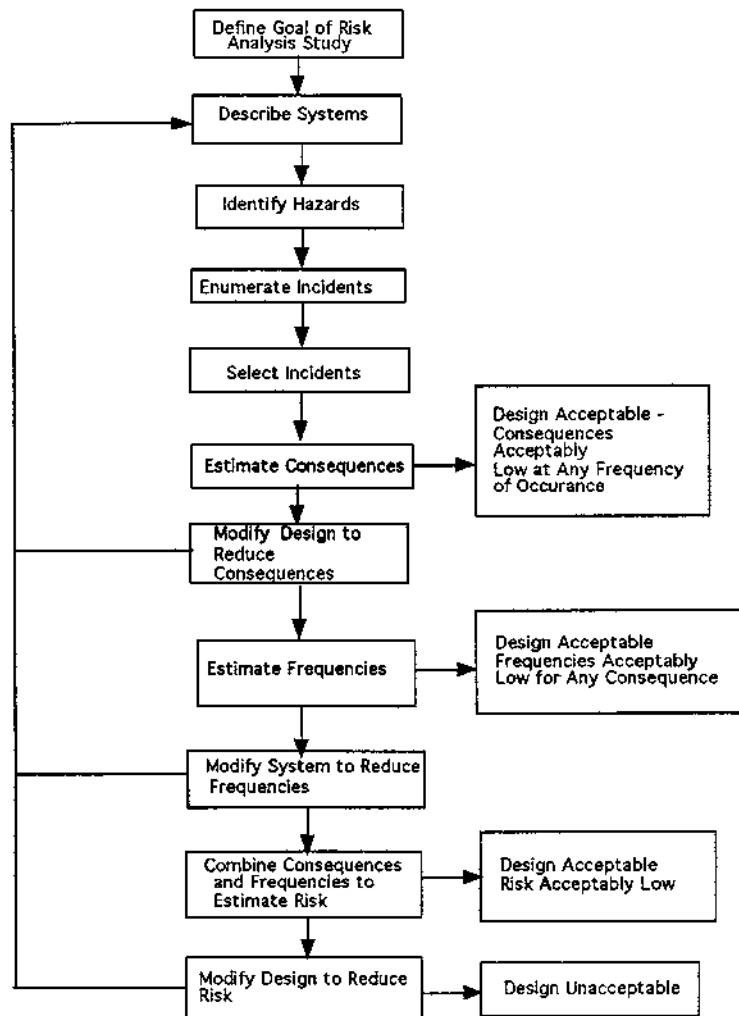


FIG. 23-24 One version of a risk analysis process. (CCPS-AIChE, 1989, p. 13, by permission.)

appropriate level of risk reduction. In most studies this is an iterative process of risk analysis and risk assessment until the risk is reduced to some specified level. The question of how safe is safe enough has to be addressed either implicitly or explicitly in the decision-making process. This subject is discussed in greater detail later on in this section.

Frequency Estimation There are two primary sources for estimates of incident frequencies. These are historical records and the application of fault tree analysis and related techniques, and they are not necessarily applied independently. Specific historical data can sometimes be usefully applied as a check on frequency estimates of various subevents of a fault tree, for example.

The use of historical data provides the most straightforward approach to the generation of incident frequency estimates but is subject to the applicability and the adequacy of the records. Care should be exercised in extracting data from long periods of the historical record over which design or operating standards or measurement criteria may have changed.

An estimate of the total population from which the incident information has been obtained is important and may be difficult to obtain.

Fault tree analysis and other related event frequency estimation techniques, such as event tree analysis, play a crucial role in the risk analysis process. Fault trees are logic diagrams that depict how components and systems can fail. The undesired event becomes the top event and subsequent subevents, and eventually basic causes, are then developed and connected through logic gates. The fault tree is completed when all basic causes, including equipment failures and human errors, form the base of the tree. There are general rules for construction, which have been developed by practitioners, but no specific rules for events or gates to use. The construction of a fault tree is still more of an art than a science. Although a number of attempts have been made to automate the construction of fault trees from process flow diagrams or piping instrumentation diagrams, these attempts have been largely unsuccessful [P. K. Andow, "Difficulties in Fault Tree Synthesis for Process Plant," *IEEE Transactions on Reliability* R-29(1): 2, 1980].

Once the fault tree is constructed, quantitative failure rate and probability data must be obtained for all basic causes. A number of equipment failure rate databases are available for general use. However, specific equipment failure rate data is generally lacking and,

therefore, data estimation and reduction techniques must be applied to generic databases to help compensate for this shortcoming. Accuracy and applicability of data will always be a concern, but useful results from quantifying fault trees can generally be obtained by experienced practitioners.

Human error probabilities can also be estimated using methodologies and techniques originally developed in the nuclear industry. A number of different models are available (Swain, "Comparative Evaluation of Methods for Human Reliability Analysis," GRS Project RS 688, 1988). This estimation process should be done with great care, as many factors can affect the reliability of the estimates. Methodologies using expert opinion to obtain failure rate and probability estimates have also been used where there is sparse or inappropriate data.

In some instances, plant-specific information relating to frequencies of subevents (e.g., a release from a relief device) can be compared against results derived from the quantitative fault tree analysis, starting with basic component failure rate data.

An example of a fault tree logic diagram using AND and OR gate logic is shown in Fig. 23-25.

The logical structure of a fault tree can be described in terms of boolean algebraic equations. Some specific prerequisites to the application of this methodology are as follows.

- Equipment states are binary (working or failed).
- Transition from one state to another is instantaneous.
- Component failures are statistically independent.
- The failure rate and repair rate are consistent for each equipment item.
- After repair, the component is returned to the working state.

Minimal cut set analysis is a mathematical technique for developing and providing probability estimates for the combinations of basic component failures and/or human error probabilities, which are necessary and sufficient to result in the occurrence of the top event.

A number of software programs are available to perform these calculations, given the basic failure data and fault tree logic diagram (AIChE-CCPS, 2000). Other less well known approaches to quantifying fault tree event frequencies are being practiced, which result in gate-by-gate calculations using discrete-state, continuous-time, Markov models (Doelp et al., "Quantitative Fault Tree Analysis, Gate-by-Gate Method," *Plant Operations Progress* 4(3): 227-238, 1984).

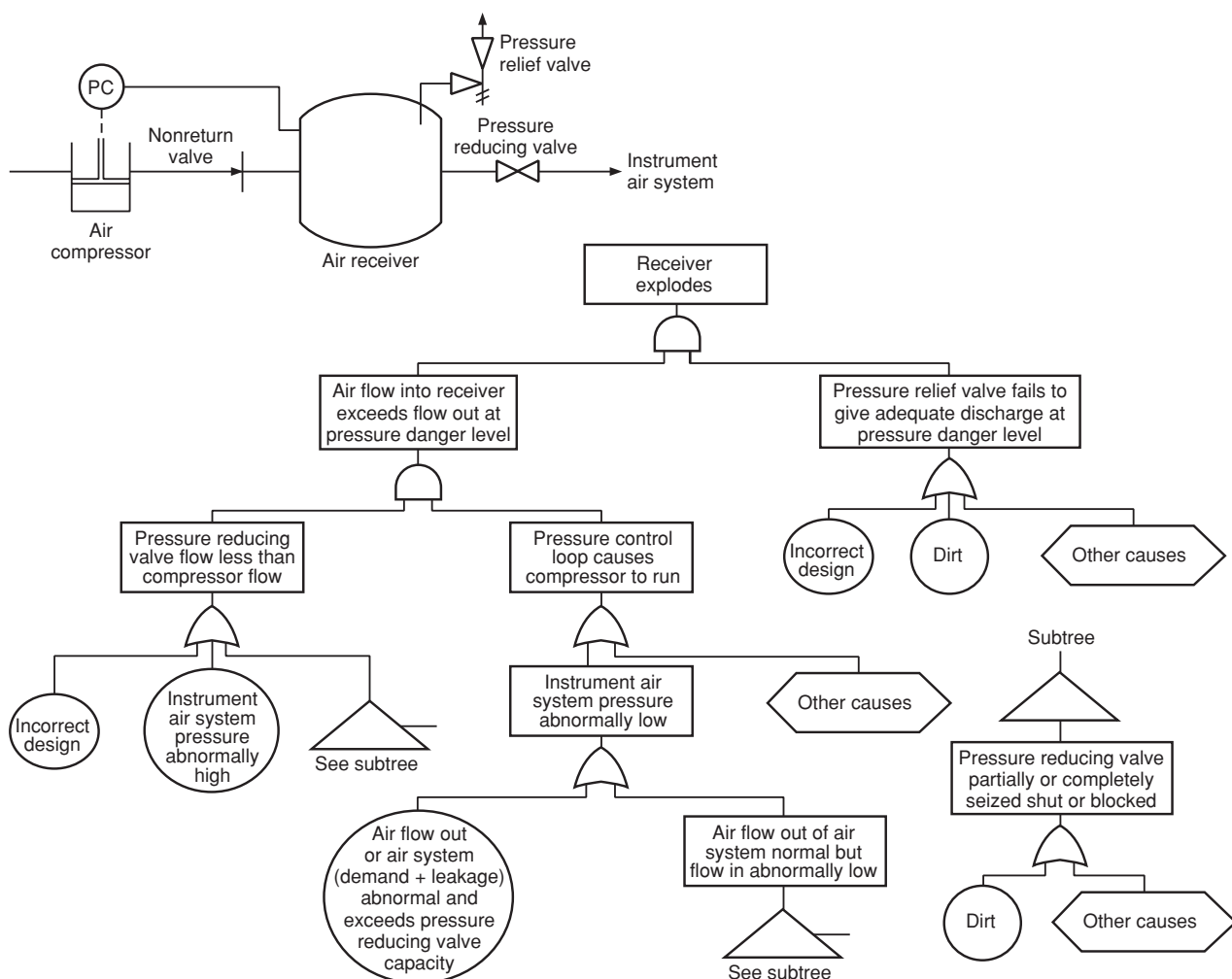


FIG. 23-25 Process drawing and fault tree for explosion of an air receiver. (From Lees, 1980, pp. 200, 201, by permission.)

Identification and quantitative estimation of common-cause failures are general problems in fault tree analysis. Boolean approaches are generally better suited to mathematically handle common-cause failures. The basic assumption is that failures are completely independent events, but in reality dependencies will exist and these are categorized as common cause failures (CCFs). Both qualitative and quantitative techniques can be applied to identify and assess CCFs. An excellent overview of CCF is available (AIChE-CCPS, 2000).

Event tree analysis is another useful frequency estimation technique used in risk analysis. It is a bottom-up logic diagram, which starts with an identifiable event. Branches are then generated, which lead to specific chronologically based outcomes with defined probabilities. Event tree analysis can provide a logic bridge from the top event, such as a flammable release into specific incident outcomes (e.g., no ignition, flash fire, or vapor cloud explosion). Probabilities for each limb in the event tree diagram are assigned and, when multiplied by the starting frequency, produce frequencies at each node point for all the various incident outcome states. The probabilities for all of the limbs at any given level of the event tree must sum to 1.0. Event trees are generally very helpful toward the generation of a final risk estimate. Figure 23-26 shows an event tree for a release of LPG (AIChE-CCPS, 2000).

Layers-of-protection analysis (LOPA) is a semiquantitative methodology for analyzing and assessing risk. It is typically applied after a qualitative hazards analysis has been completed, which provides the LOPA team with a listing of hazard scenarios with associated safeguards for consideration. LOPA uses simplified methods to characterize the process risk based on the frequency of occurrence and consequence severity of potential hazard scenarios. The process risk is compared to the owner/operator risk criteria. When the process risk exceeds the risk criteria, protection layers are identified that reduce the process risk to the risk criteria.

The protection layers are safeguards that are designed and managed to achieve seven core attributes: independence, functionality, integrity, reliability, auditability, access security, and management of change. Protection layers may include inherently safer design, control, supervisory, prevention, mitigation, limitation, barriers, and emergency response systems, depending on the owner/operator's risk management strategy.

In general, risk reduction is accomplished by implementing one or more protective layers, which reduce the frequency and/or consequence of the hazard scenario. LOPA provides specific criteria and restrictions for the evaluation of protection layers, eliminating the subjectivity of qualitative methods at substantially less cost than fully quantitative techniques. LOPA is a rational, defensible methodology that allows a rapid, cost-effective means for identifying the protection layers that lower the frequency and/or the consequence of specific hazard scenarios.

Consequence Estimation Given that an incident (release of material or energy) has been defined, the consequences can be estimated. The general logic diagram in Fig. 23-27 illustrates these calculations for the release of a volatile hazardous substance.

For any specific incident there will be an infinite number of incident outcome cases that can be considered. There is also a wide degree of consequence models which can be applied. It is important, therefore, to understand the objective of the study to limit the number of incident outcome cases to those which satisfy that objective. An example of variables which can be considered is as follows.

- Quality, magnitude, and duration of the release
- Dispersion parameters (wind speed, wind direction, weather stability)
- Ignition probability, ignition sources/location, ignition strength (flammable releases)
- Energy levels contributing to explosive effects (flammable releases)
- Impact of release on people, property, or environment (thermal radiation, projectiles, shock-wave overpressure, toxic dosage)

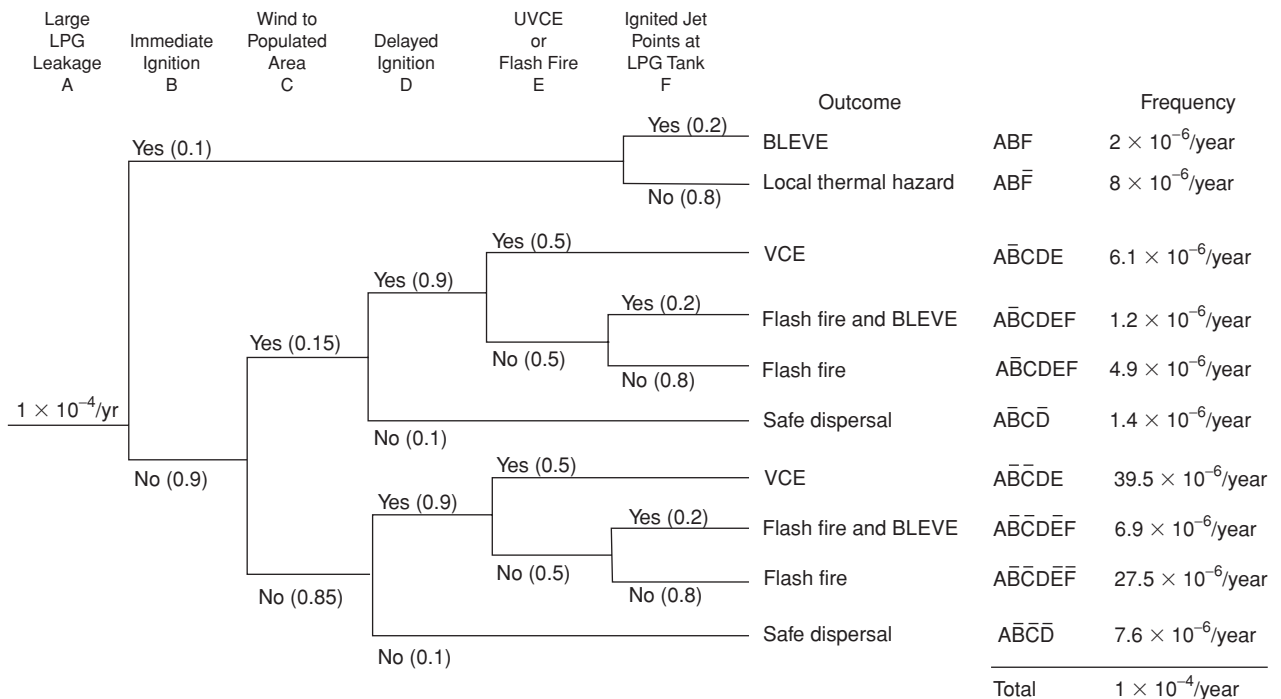


FIG. 23-26 Sample event tree for a release of LPG. (AIChE-CCPS, 2000, p. 329.)

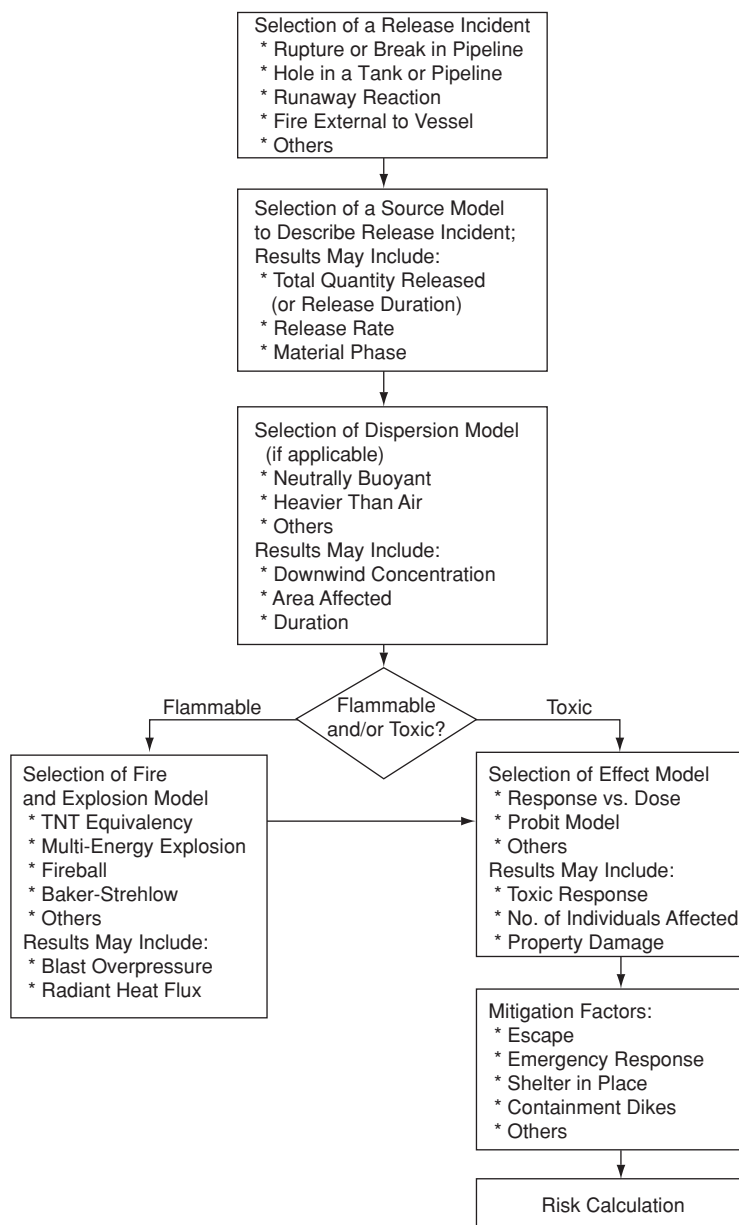


FIG. 23-27 Overall logic diagram for consequence analysis of volatile hazardous substances. (CCPS-AIChE, 1989, p. 60.)

- Mitigation effects (safe havens, evacuation, daytime/nighttime populations)

Probit models have been found generally useful to describe the effects of incident outcome cases on people or property for more complex risk analyses. At the other end of the scale, the estimation of a distance within which the population would be exposed to a concentration of ERPG-2 or higher may be sufficient to describe the impact of a simple risk analysis.

Portions or all of the more complex calculation processes, using specific consequence models, have been incorporated into a few com-

mercially available software packages (AIChE-CCPS, 2000). These programs should be used by risk analysts with extensive engineering experience, as significant judgment will still be required.

The output of these calculation processes is one or more pairs of an incident or incident outcome case frequency and its effect (consequence or impact).

Risk Estimation There are a number of risk measures which can be estimated. The specific risk measures chosen are generally related to the study objective and depth of study, and any preferences or requirements established by the decision makers. Generally, risk

measures can be broken down into three categories: risk indices, individual risk measures, and societal risk measures.

Risk indices are usually single-number estimates, which may be used to compare one risk with another or used in an absolute sense compared to a specific target. For risks to employees the *fatal accident rate* (FAR) is a commonly applied measure. The FAR is a single-number index, which is the expected number of fatalities from a specific event based on 10^5 exposure hours. For workers in a chemical plant the FAR would be calculated for a specific event as

$$\text{FAR} = \frac{10^5}{8760} EP\tau \quad (23-29)$$

where FAR = expected number of fatalities from a specific event based on 10^5 exposure hours

E = frequency of event, yr^{-1}

P = probability of being killed by event at a specific location

τ = fraction of time spent at specific location

However, the worker in a chemical plant could be exposed to other potential events that might result in a fatality. In this case the overall FAR for the worker would be

$$\text{FAR} = \frac{10^5}{8760} \sum_i E_i \sum_j P_{ij} \tau_j \quad (23-30)$$

Each subscript i is a specific event and each subscript j is a specific location.

References are available which provide FAR estimates for various occupations, modes of transportation, and other activities (Kletz, "The Risk Equations—What Risk Should We Run?," *New Scientist*, May 12, pp. 320–325, 1977).

Figure 23-28 is an example of an individual risk contour plot, which shows the expected frequency of an event causing a specified level of harm at a specified location, regardless of whether anyone is present at that location to suffer that level of harm.

The total individual risk at each point is equal to the sum of the individual risks at that point from all incident outcome cases.

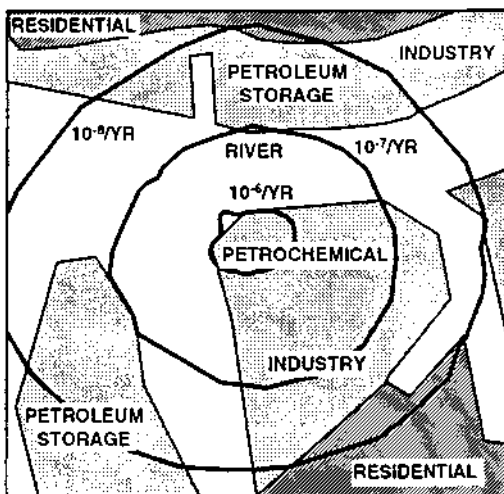


FIG. 23-28 Example of an individual risk contour plot. (CCPS-AIChE, 1989, p. 269.)

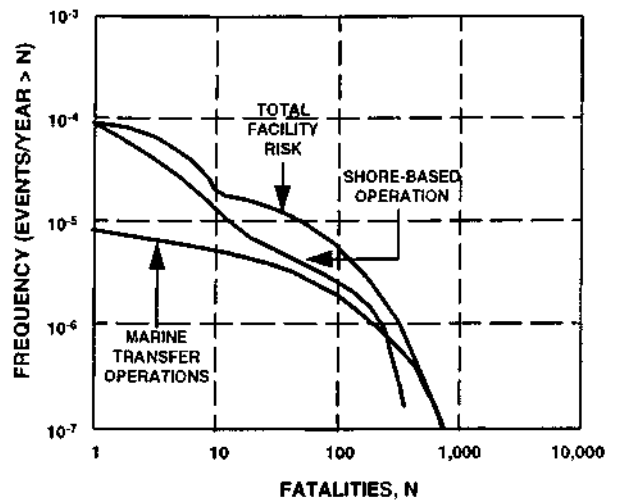


FIG. 23-29 Example of a societal risk F-N curve. (AIChE-CCPS, 1989, p. 4.4.)

$$\text{IR}_{x,y} = \sum_{i=1}^n \text{IR}_{x,y,i} \quad (23-31)$$

where $\text{IR}_{x,y}$ = total individual risk of fatality at geographical location x,y

$\text{IR}_{x,y,i}$ = individual risk of fatality at geographical location x,y from incident outcome case i

n = total number of incident outcome cases

A common form of societal risk measure is an F-N curve, which is normally presented as a cumulative distribution plot of frequency F versus number of fatalities N . An example of this type of measure is shown in Fig. 23-29.

Any individual point on the curve is obtained by summing the frequencies of all events resulting in that number of fatalities or greater. The slope of the curve and the maximum number of fatalities are two key indicators of the degree of risk.

Risk Criteria Once a risk estimate is prepared, alternatives to reduce the risk can be determined; but one always faces the challenge of how low a risk level is low enough. Different countries in the world have established numerical criteria [ALE (1992) and HSE (2001)] primarily as a tool to address the Seveso regulations. In addition, over the years various chemical companies have established numerical targets [Gibson (1980), Helmers (1982), and Renshaw (1990)]. A good survey paper on the use of risk guidelines by both governments and operating companies is provided by Pikaar and Seaman (1995).

Risk Decision Making Risk criteria represent the first step in risk decision making. Efforts must be made to reduce the risk at least to a "tolerable" risk level, which may be dictated by a government or an operating company. Once the tolerable level is reached, then additional risk reductions should still be evaluated. A number of criteria also provide guidance on a level of risk that requires no further risk reduction. In between these two levels is a range of risk where risk reduction options need to be evaluated. How far to continue reducing risk below the tolerable level is up to the individual company and is a function of costs and benefits (degree of risk reduction). At some point additional risk reductions have little benefit. Hamm and Schwartz (1993) summarize some strategies for consideration, and CCPS-AIChE (1995) provides an introduction to a number of decision analysis tools which could be applied.

DISCHARGE RATES FROM PUNCTURED LINES AND VESSELS

Nomenclature

a	Coefficient in nonequilibrium compressibility factor, Eq. (23-74)
A	Cross-sectional area perpendicular to flow, m^2
C_D	Discharge coefficient
C_{DG}	Discharge coefficient for gas flow
C_{DL}	Discharge coefficient for liquid flow
C_p	Heat capacity at constant pressure, $J/(kg \cdot K)$
C_v	Heat capacity at constant volume, $J/(kg \cdot K)$
D	Pipe diameter, m
D_T	Tank diameter, m
f	Fanning friction factor
F_l	Pipe inclination factor defined by Eq. (23-43)
g	Gravitational acceleration, m^2
G	Mass flux, $kg/(m^2 \cdot s)$
H	Specific enthalpy, J/kg
H_{GL}	Heat of vaporization $(H_G - H_L)$ at saturation, J/kg
K	Slip velocity ratio, u_G/u_L
K_e	Number of velocity heads for fittings, expansions, contractions, and bends
L	Length of pipe, m
N	$4fL/D + K_e$, pipe resistance or nonequilibrium compressibility factor
P	Pressure, $Pa = N/m^2$
Q	Heat-transfer rate, W/kg
R	Gas constant, $J/(kgmol \cdot K)$
S	Specific entropy, $J/(kg \cdot K)$
t	Time, s
T	Temperature, K
u	Velocity, m/s
v	Specific volume, m^3/kg
v_{GL}	Specific volume difference, $v_G - v_L$, m^3/kg
w	Mass discharge rate, kg/s
x	Vapor quality, kg vapor/kg mixture or mole fraction of liquid components
y	Mole fraction of vapor components
X_m	Lockhart Martinelli parameter
z	Vertical distance, m

Greek

α	Volume fraction vapor, m^3 vapor/ m^3 mixture
ϵ	Dimensionless specific volume, v/v_0
γ	Heat capacity ratio, C_p/C_v

Greek

η	Pressure ratio, P/P_0
ϕ	Two-phase multiplier, pressure drop for two-phase flow divided by pressure drop for single-phase flow
μ	Single- or two-phase viscosity, $Pa \cdot s$
θ	Overall inclination angle of pipe to horizontal from source to break, deg
ρ	Density, kg/m^3
ω	Slope of dimensionless specific volume to reciprocal dimensionless pressure, defined by Eqs. (23-61a) and (23-61b)

Subscripts

a	Ambient
b	Backpressure
bub	Bubble point, pressure and saturation temperature when first vapor bubbles appear
c	η_{ch}/η_s , ratio of choked pressure to saturation pressure
ch	Choked
C	Condensable components (the contaminant)
d	Discharge
dew	Dew point, pressure and saturation temperature when first (or last) liquid occurs
diff	Differential form
e	Equivalent, for two-phase specific volume with slip
g, G	Gas or vapor
H	Homogeneous flow (slip velocity ratio of unity)
int	Integral form
I	Inert or padding gas component
L	Liquid
max	Maximum
N	Nonequilibrium
ori	Orifice
p	Pipe flow
s	Saturation (or bubble point)
sol	Solids
S	Constant specific entropy
v	Vapor pressure
0	Initial stagnation conditions
1	Point at which backpressure from pipe is felt after entrance from tank
2	Plane at vena contracta or at pipe puncture
$\circ$	Dimensionless

GENERAL REFERENCES: Bragg, "Effect of Compressibility on the Discharge Coefficient of Orifices and Convergent Nozzles," *J. Mech. Eng. Sci.*, **2**: 35-44, 1960. Cheremisinoff and Gupta, eds., *Handbook of Fluids in Motion*, Ann Arbor Science, Ann Arbor, Mich., 1983. Chisholm, *Two-Phase Flow in Pipelines and Heat Exchangers*, George Godwin, New York, in association with the Institution of Chemical Engineers, 1983. Daubert and Danner, "Physical and Thermodynamic Properties of Pure Chemicals, Data Compilation," DIPPR (Design Inst. of Phys. Props. Res.) AIChE, New York, 1989. Diener and Schmidt, "Sizing of Throttling Device for Gas/Liquid Two-Phase Flow, Part 1: Safety Valves, Part 2: Control Valves, Orifices, and Nozzles," *J. Hazardous Mat.*, **23**(4): 335-344, December 2004; **24**(1): 29-37, March 2005. Ely and Huber, "NIST Thermophysical Properties of Hydrocarbon Mixtures Database (STRAPP Program)," Version 1.0 Users' Guide, U.S. Dept. of Commerce, National Institute of Standards and Technology (NIST), Gaithersburg, Md., Feb. 1990. Fisher et al., *Emergency Relief System Design Using DIERS Technology*, AIChE, New York, 1992. Graham, "The Flow of Air-Water Mixtures Through Nozzles," National Engineering Laboratories (NEL) Report no. 308, East Kilbride, Glasgow, 1967. Jobson, "On the Flow of Compressible Fluids Through Orifices," *Proc. Instn. Mech. Engrs.*, **169**(37): 767-776, 1955. "The Two-Phase Critical Flow of One-Component Mixtures in Nozzles, Orifices, and Short Tubes," *Trans. ASME J. Heat Transfer*, **93**(5): 179-187, 1976. Lee and Sommerfeld, "Maximum Leakage Times Through Puncture Holes for Process Vessels of Various Shapes," *J. Hazardous Mat.*, **38**(1) 27-40, July 1994. Lee and Sommerfeld, "Safe Drainage or Leakage Considerations and Geometry in the Design of Process Vessels," *Trans. IChemE*, **72**, part B: 88-89, May 1994. Leung, "A Generalized Correlation for One-Component Homogeneous Equilibrium Flashing Choked Flow," *AIChE J.*, **32**(10): 1743-1746, 1986. Leung, "Similarity Between Flashing and Non-flashing Two-Phase Flow," *AIChE J.*, **36**(5): 797, 1990. Leung, "Size Safety Relief Valves for Flashing Liquids," *Chem. Eng. Prog.*, **88**(2): 70-75, February 1992. Leung, "Two-Phase Flow Discharge in Nozzles and Pipes—A Unified

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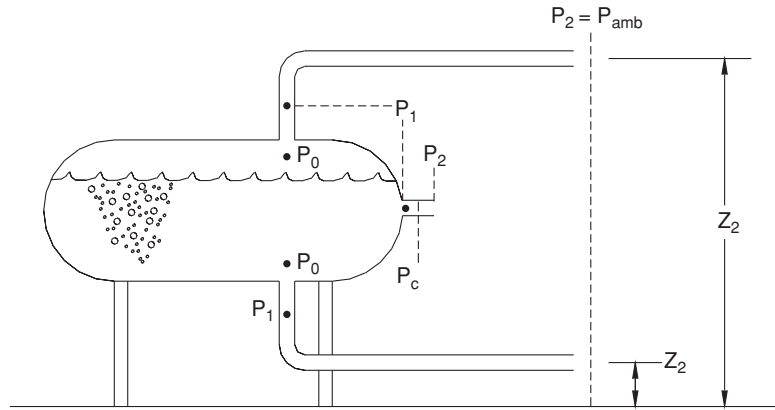


FIG. 23-30 Types of accidental discharge and pressure evaluation points.

Overview Modeling the consequences of accidental releases of hazardous materials begins with the calculation of discharge rates. In the most general case, the discharged material is made up of a volatile flashing liquid and vapor along with noncondensable gases and solid particles. Most of the model development is given in terms of two-phase, vapor + liquid flow, but it can be readily extended to three phases with vapor, liquid, plus suspended solids flow. The solids contribute primarily to the density and heat capacity of the mixture, contributing energy to the flashing and to the subsequent dispersion.

Types of Discharge Hazardous accidental releases can occur from vessels, pipelines, reactors, and distillation columns. The more common accidental events are breaks in a vessel or its associated piping. Figure 23-30 illustrates this. For a vessel, the piping attachments can be from a dip leg or from the bottom or top of the vessel. A puncture in the vapor space of the vessel or a break in a top-attachment pipe, initially at least, discharges vapor plus the padding gas used to maintain vessel pressure. This padding gas can be air for an atmospheric vessel. The discharged vapor can cool upon expansion and condense liquids when the pressure in the jet drops to the dew point pressure P_{dew} .

A puncture in the liquid space of the vessel or a break in the bottom-attachment or dip-leg pipe, initially, at least, discharges liquid plus any solids present without any noncondensable components. The liquid can begin to flash when the pressure drops to the bubble point pressure P_{bub} . If the liquid is extremely volatile, it could totally evaporate when the pressure drops below the dew point, producing vapor plus solids. The initial mass vapor fraction x_0 is zero as is the initial volume fraction α_0 .

For a puncture, break, or pressure relief valve (PRV) opening from a reactor or distillation column, there may be no clear-cut level distinguishing the liquid and vapor phases. That is, the system is initially mixed. In this case, noncondensable gases, condensable vapors, and liquid plus solids are initially discharged. The value of α_0 is nonzero and less than unity, reflecting the contributions of the gases and vapors.

For a blow-down calculation, the conditions change. With a tank vapor space release, as the pressure decreases, the liquid could reach its bubble point and begin to flash. Vapor bubbles in the liquid generate a liquid swell, so a frothy, two-phase interface rises and could drive out all the vapor (along with noncondensables) and begin two-phase flashing flow of liquid without noncondensables. Thus, the discharge calculation would also change from a vapor discharge to a two-phase flashing liquid discharge.

The energy and momentum balance equations are drawn across planes at points 0, 1, and 2, as illustrated for a general case in Fig. 23-31.

Energy Balance Method for Orifice Discharge Solutions can be found for the discharge rate by solving the energy balance and/or the momentum balance. The energy balance solution is quite simple and general, but is sensitive to inaccuracies in physical properties correlations. The equations below apply to orifice flow. A separate momentum balances solution is applied to pipe flow to find the pressure losses. The balances are written across an orifice from a plane 0 inside the tank at stagnant conditions (i.e., far enough away from the orifice that the velocity inside the tank is negligible) to a plane at the backpressure P_b at point 2.

$$H_0 = H_2 + \frac{1}{2} (G^2 v_c^2)_2 + Q \quad (23-32)$$

Rarely, if ever, is the heat-transfer term Q nonnegligible.

Perform an isentropic expansion. That is, as the pressure decreases from P_0 to the backpressure P_b (usually ambient pressure P_a), select intermediate values of pressure P_1 . At each P_1 find the temperature that keeps S constant T_{s1} . Solve for the vapor fraction x , using the entropy balance between planes 0 and 1:

$$S_0 = xS_{G1} + (1 - x - x_{\text{sol}}) S_{L1} + x_{\text{sol}} S_{\text{sol}} \quad (23-33)$$

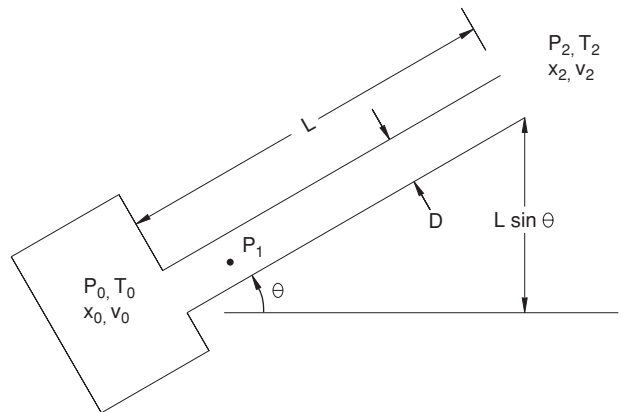


FIG. 23-31 Configuration modeled for pipe flow with elevation change.

23-56 PROCESS SAFETY

Since the mass fraction of solids is constant, the vapor flash fraction is

$$x = \frac{S_0 - [(1 - x_{\text{sol}})S_{L1} - x_{\text{sol}}S_{\text{sol}}]_{T_{s1}}}{(S_{G1} - S_{L1})_{T_{s1}}} \quad (23-34)$$

Find the enthalpy H_2 at T_{s1} from physical property correlations. Velocity is found solving Eq. (23-32) for homogeneous flow (with $Q = 0$):

$$Gv_e = u = \sqrt{2(H_0 - H_2)} \quad (23-35)$$

Find the density ρ from an equation of state at P_1, T_{s1} for the phase densities ρ_G and ρ_L , and the reciprocal of the homogeneous specific volume v_H . Calculate the mass flux from

$$G = u\rho = \frac{u}{v} \quad (23-36)$$

Search with values of P_1 until G is maximized. The choke pressure, P_{ch} , is the value of P_1 that produces a maximum value of mass flux G_{max} . The discharge rate w is given from the mass flux, a discharge coefficient C_D , and the orifice cross-sectional area A as

$$w = C_D A G_{\text{max}} \quad (23-37)$$

This approach is illustrated for two-phase flashing flow with a multicomponent mixture (mole fractions 0.477 allyl alcohol, 0.3404 allyl chloride, 0.1826 dodecane). Figure 23-32 plots the flash curve showing that as pressure decreases, flashing begins at the bubble point of 7.22 bar(a) (104.7 psia) and increases to 30.8 mass %. At the same time, temperature follows the saturation curve down. Figure 23-33 plots the two-phase density, the velocity, and the product of these, the mass flux G . In the plot G is a maximum at 8586 kg/(m²·s) at the choke pressure just below the bubble point. This is so because the two-phase density drops quickly as flashing occurs at reduced pressures, while the velocity increases more slowly. With a discharge coefficient of 0.61 and a 2-in orifice (area of 0.002027 m²) the discharge rate w is 10.6 kg/s.

Momentum Balance in Dimensionless Variables For pipe flow, it is necessary to solve the momentum balance. The momentum balance is simplified by using the following dimensionless variables:

$$\text{Pressure ratio: } \eta = \frac{P}{P_0}$$

$$\text{Mass flux ratio: } G_* = \frac{G}{\sqrt{P_0 \rho_0}} \quad (23-38)$$

$$\text{Specific volume ratio: } \varepsilon = \frac{v}{v_0}$$

The discharge relationships are derived by solving the differential momentum balance over a tank plus pipe:

$$v dP + G^2 v dv + \left(4f \frac{dz}{D} + K_e\right) \frac{1}{2} G^2 v_L^2 \phi_L^2 + g \sin \theta dz = 0 \quad (23-39)$$

The terms represent, respectively, the effect of pressure gradient, acceleration, line friction, and potential energy (static head). The effect of fittings, bends, entrance effects, etc., is included in the term K_e correlated as a number of effective “velocity heads.” The inclination angle θ is the angle to the horizontal from the elevation of the pipe connection to the vessel to the discharge point. The term ϕ_L^2 is the two-phase multiplier that corrects the liquid-phase friction pressure loss to a two-phase pressure loss. Equation (23-39) is written in units of pressure/density.

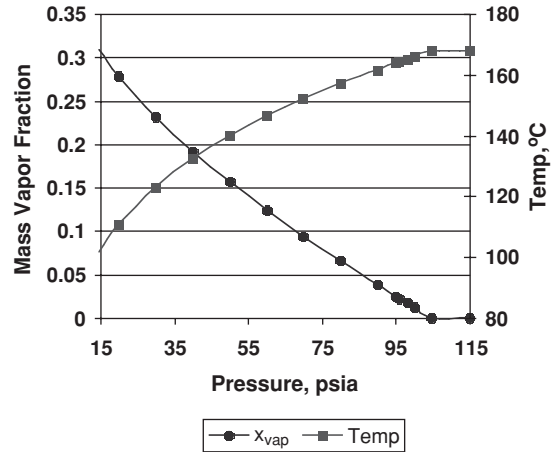


FIG. 23-32 Example flash curve for multicomponent material.

In dimensionless variables, the momentum balance is

$$\varepsilon d\eta + G_*^2 \varepsilon d\varepsilon + N \frac{1}{2} G_*^2 \varepsilon^2 \phi_L^2 + \frac{g \sin \theta dz}{P_0 v_0} = 0 \quad (23-40)$$

where N collects the friction loss terms in terms of the number of velocity heads, or

$$N = 4f_L \frac{dz}{D} + K_e \quad (23-41)$$

The two-phase flow multiplier ϕ_L^2 is discussed in a later section.

The momentum balance for homogeneous flow can be factored to a form that enables integration as

$$-N = \frac{G_{*p}^2 \varepsilon d\varepsilon + \varepsilon d\eta}{\frac{1}{2} G_{*p}^2 \varepsilon^2 + F_I} \quad (23-42)$$

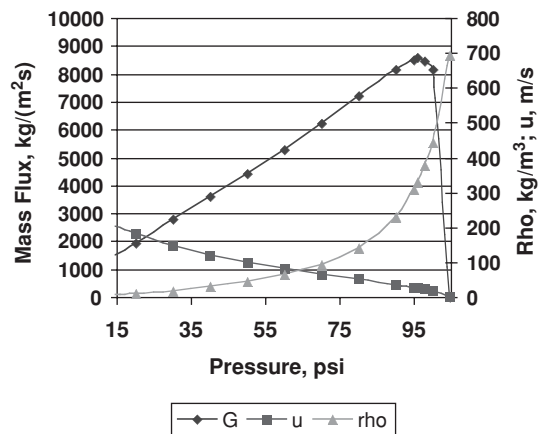


FIG. 23-33 Example of using the energy balance to find the orifice discharge flux.

by defining a pipe inclination factor F_I as

$$F_I = \frac{gD \sin \theta}{4f_{L0} P_0 v_0} \quad (23-43)$$

where F_I is positive for upflow, negative for downflow, and zero for horizontal flow.

Analytical Solutions for Orifice and Pipe Flow Equation (23-42) can be solved analytically for pipe breaks and tank punctures for the following cases:

- Subcooled liquid flow
 - Adiabatic expansion of ideal gases
 - Flashing liquid flow without noncondensable gases ($\alpha_0 = 0$)
 - Subcooled liquid mixed with noncondensable gases ($\alpha_0 > 0$) (frozen flow)
 - Flashing liquid mixed with noncondensable gases (hybrid flow)
- The solutions for all except hybrid flow follow.

Orifice Discharge for Gas Flow The analytic solution for discharge through an orifice of an ideal gas is derived by invoking the equation of state for adiabatic expansion of an ideal gas:

$$\frac{P}{P_0} = \left(\frac{\rho}{\rho_0} \right)^\gamma = \left(\frac{T}{T_0} \right)^{\gamma/(\gamma-1)} \quad (23-44)$$

where

$$\gamma = \frac{C_p}{C_v} \quad (23-45)$$

The solution is

$$G_g^2 = \frac{2\gamma}{\gamma-1} \eta_2^{2/\gamma} (1 - \eta_2^{(\gamma-1)/\gamma}) \quad (23-46)$$

This solution applies for both subsonic and choked flow. If the flow is choked, the exit pressure ratio η_2 is replaced by the choked pressure ratio η_{ch} , given by

$$\eta_{ch} = \left(\frac{2}{\gamma+1} \right)^{\gamma/(\gamma-1)} \quad (23-47)$$

Equation (23-47) must be evaluated to test for choked flow in any event. When η_{ch} from Eq. (23-47) is substituted into Eq. (23-46), the general solution reduces to the choked flow solution:

$$G_g^2 = \gamma \left(\frac{2}{\gamma+1} \right)^{(\gamma+1)/(\gamma-1)} \quad (23-48)$$

The discharge rate is found by using Eqs. (23-37) and (23-38).

Blowdown of Gas Discharge through Orifice An analytic solution is available for blowdown (time-dependent discharge) of an ideal gas from a tank. The time-varying mass of gas in the tank m_T is the product of the tank volume V_T and the density ρ :

$$m_T(t) = V_T \rho(t) \quad (23-49)$$

Differentiating, solving for dt , and integrating gives

$$t = -V_T \int_{P_0}^P \frac{dP}{w} \quad (23-50)$$

where w is the time-varying discharge rate. Typically, all but a small fraction of the mass in a tank is discharged at sonic flow, so a sonic

flow solution is most useful. Transforming P and ρ to temperature T , using Eq. (23-44) allows Eq. (23-50) to be integrated to give a solution in terms of the initial discharge rate w_0 and the initial tank mass m_{T0} :

$$w(t) = w_0 [F(t)]^{(\gamma+1)/(\gamma-1)} \quad (23-51)$$

where

$$F(t) = (1 + At)^{-1} \quad (23-52a)$$

$$A = \frac{w_0 (\gamma + 1)}{2m_{T0}} \quad (23-52b)$$

Pipe and Orifice Flow for Subcooled Liquids Since liquids are essentially incompressible, ϵ is constant at ϵ_0 , and $d\epsilon$ is zero in Eq. (23-42). Recognizing that η_0 and ϵ_0 are unity, we see that integration gives

$$-N \left(\frac{1}{2} G_p^2 \epsilon_0^2 + F_I \right) = \epsilon_0 \int_{\eta_0}^{\eta_2} d\eta = \eta_2 - \eta_0$$

or

$$G_p^2 = 2 \left(\frac{1 - \eta_2}{N} - F_I \right) \quad (23-53)$$

The solution for orifice flow is a special case with N equal to unity (entrance losses only) and F_I equal to zero, giving

$$G_{ori} = \sqrt{2(1 - \eta_2)} \quad (23-54)$$

Numerical Solution for Orifice Flow With orifice flow, the last two terms of the momentum balance (line resistance and potential energy change) are negligible. The momentum balance, Eq. (23-40), reduces to

$$G^2 \epsilon d\epsilon = -\epsilon d\eta \quad (23-55)$$

This equation can be treated in differential form and in integral form. In differential form it becomes

$$G_{diff} = - \left(\frac{d\epsilon}{d\eta} \right)^{-1/2} \quad \text{or} \quad G_{diff} = - \left(\frac{dw}{dP} \right)^{-1/2} \quad (23-56)$$

For the integral form, express Eq. (23-55) as

$$\frac{1}{2} G^2 d\epsilon^2 = -\epsilon d\eta$$

to obtain

$$G \cdot d\epsilon = (-2\epsilon d\eta)^{1/2} \quad (23-57)$$

In general, the limits of integration are from ϵ_0 , η_0 to an arbitrary final point ϵ , η (recognizing that $\epsilon_s = \epsilon_0 = 1$). However, this method works better, using an indefinite integration of $d\epsilon$:

$$G_{int} = \frac{1}{\epsilon} \left(2 \int_{\eta}^{\eta_0} \epsilon d\eta \right)^{1/2} \quad (23-58)$$

or in dimensional form

$$G_{\text{int}} = \frac{1}{v} \left(2 \int_p^{p_0} v dp \right)^{1/2} \quad (23-59)$$

Equations (23-56) and (23-59) are readily evaluated at intermediate pressure points P_1 in the range $P_a < P_1 < P_0$, giving two curves for G . The two curves cross at the choke point, which is also the point of maximum mass flux, as illustrated in Fig. 23-34 for a mixture of isobutane (90.898 mol %) and ethylene (9.102 mol %) initially at 33.0 bara and 357.4 K. The mass flux G found by evaluating Eq. (23-59) is labeled G_{int} in the figure, and the value of G found by evaluating Eq. (23-56) is labeled G_{diff} . The G_{int} curve has a maximum value of 20,637 kg/(m²·s) at the choke pressure of 20.64 bara.

The disadvantage of this solution is that it is sensitive to inaccuracies in the physical properties correlations used to evaluate the flash fraction and specific volumes. Figure 23-34 is found by using DIPPR properties (Daubert and Danner, 1989). The mass flux found does not agree well with that found by using the energy balance method that gives 24,330 kg/(m²·s). In addition, the rightmost branch of the G_{diff} curve is anomalously low and crosses the G_{int} curve spuriously to the right of the maximum flux. By using more accurate physical properties developed by the STRAPP program of NIST (Ely and Huber, 1990), the maximum value of G_{int} agrees with that found by the energy balance method and the right-hand branch of the G_{diff} curve does not fall low and cause this confusion. For this reason the method is not recommended for general use, but provides good confirmations when used with accurate physical properties correlations.

Omega Method Model for Compressible Flows The factored momentum balance, Eq. (23-42), can be analytically integrated after first relating the dimensionless specific volume ε to the dimensionless pressure ratio η . A method to do this, designated the *omega method*, was suggested by Leung (1986):

$$\varepsilon = \begin{cases} \omega \left(\frac{\eta_s}{\eta} - 1 \right) + 1 & \text{if } \frac{\eta_s}{\eta} > 1 \\ 1.0 & \text{if } \frac{\eta_s}{\eta} \leq 1 \end{cases} \quad (23-60)$$

Equation (23-60) represents a linear relationship between the two- or three-phase specific volume and reciprocal pressure (v versus P^{-1} or ε

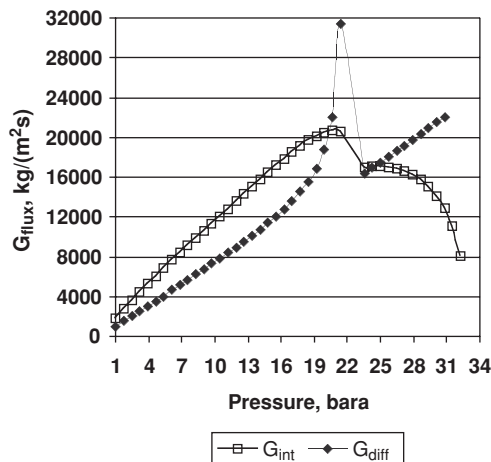


FIG. 23-34 Numerical solution of the momentum balance for orifice flow using Eqs. (23-56) and (23-59).

versus η^{-1}) beginning at the bubble point P_s , where η is η_s . For single components, ω is found by using the Clapeyron equation to give

$$\omega = \frac{C_p T_0 P_s}{v_{L0}} \left[\frac{v_{GL0}(P_s)}{H_{GL0}(P_s)} \right]^2 \quad (23-61a)$$

Alternately, use the slope of the ε versus η^{-1} curve between the bubble point and a second, lower pressure at ε_2 , η_2 to evaluate ω , or

$$\omega = \frac{\varepsilon_2 - 1}{\eta_s/\eta_2} - 1 \quad (23-61b)$$

This is a more convenient formula for multicomponent mixtures.

This value of ω can be called the saturation value or ω_s , since it applies only with flashing liquids (i.e., in the flashing region for pressures less than the bubble point, as seen in Fig. 23-32). A generalization defines omega to apply also with noncondensable gases by using $\alpha_0 = x_{v0} v_{v0}/v_0$

$$\omega = \alpha_0 + (1 - \alpha_0)\omega_s \quad (23-61c)$$

Homogeneous Equilibrium Omega Method for Orifice and Horizontal Pipe Flow The homogeneous equilibrium model (HEM) solution is obtained by substituting Eq. (23-60) for ε and integrating the momentum balance. The solution is given first for horizontal flow, with the flow inclination factor $F_I = 0$. Orifice flow is a special case in which only the orifice solution is needed. For pipe flow, two solutions are needed, the orifice flow solution giving G_{ori} and the pipe flow solution giving G_p . Plane 1 in Fig. 23-31 is at the pressure ratio η_1 where the pipe pressure balance begins and the inlet orifice pressure loss ends. The final pipe solution finds η_1 so that

$$G_{\text{ori}} = G_p \quad (23-62)$$

The integration for both solutions must be conducted over the subcooled region and the flashing region. The complicating factor is that the bubble point pressure ratio η_s could fall either in the orifice flow integration span ($\eta_s > \eta_1$), giving flashing in the orifice (case 1), or in the pipe integration span ($\eta_s < \eta_1$), giving flashing in the pipe (case 2). These two options are illustrated in Fig. 23-35.

Again, over the subcooled region from η_0 to η_s , $d\varepsilon$ is zero and ε is constant at unity, ε_0 . The solutions for the two cases are as follows:

Case 1: *Flashing in Orifice, Two-Phase Flashing Flow in Pipe*

$$G_{\text{ori}}^2 = \frac{2[(1 - \eta_s) + [\omega \eta_s \ln(\eta_s/\eta_{\text{ch}}) - (\omega - 1)(\eta_s - \eta_{\text{ch}})]]}{\varepsilon_{\text{ch}}^2} \quad (23-63)$$

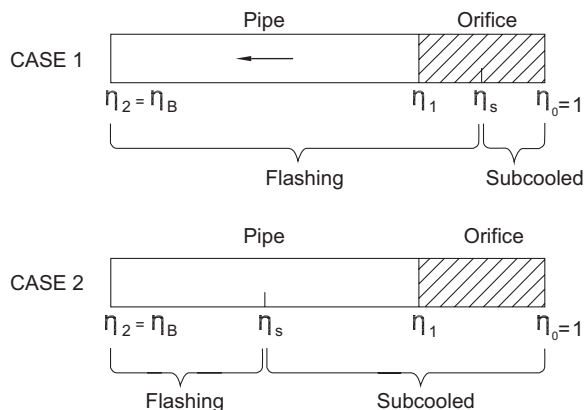


FIG. 23-35 Options for integration range of HEM omega method.

$$G_{cp}^2 = 2 \left\{ \frac{\eta_1 - \eta_2 + \frac{\omega \eta_s}{(1 - \omega)^2} \ln \frac{\eta_2 \epsilon_2}{\eta_1 \epsilon_1}}{N + 2 \ln \left(\frac{\epsilon_2}{\epsilon_1} \right)} \right\} \quad (23-64)$$

Case 2: Subcooled Liquid in Orifice, Flashing in Pipe

$$G_{ori}^2 = 2(\eta_0 - \eta_1) \quad (23-65)$$

$$G_{cp}^2 = 2 \left[\frac{\eta_1 - \eta_s + \frac{\eta_s - \eta_2}{1 - \omega} + \frac{\eta_s - \eta_2}{1 - \omega} + \frac{\omega \eta_s}{(1 - \omega)^2} \ln \left(\frac{\eta_2 \epsilon_2}{\eta_s \epsilon_s} \right)}{N + 2 \ln \left(\frac{\epsilon_2}{\epsilon_s} \right)} \right] \quad (23-66)$$

The choke point is found by the usual maximization relationship:

$$\left(\frac{dG}{d\eta_2} \right)_{\eta_2 = \eta_b} = 0 \quad (23-67)$$

This gives an implicit equation in the choked pressure ratio η_c (short-hand for η_{ch}/η_s):

If $\omega \leq 2$:

$$\eta_c^2 + \omega(\omega - 2)(1 - \eta_c)^2 + 2\omega^2 \ln \eta_c + 2\omega^2(1 - \eta_c) = 0 \quad (23-68a)$$

An explicit equation provides an adequate approximate solution for larger values of ω :

For $\omega \geq 2$:

$$\eta_c = 0.55 + 0.217 \ln \omega - 0.046 (\ln \omega)^2 + 0.004 (\ln \omega)^3 \quad (23-68b)$$

For orifice flow, the definition of choked flow in terms of the back-pressure ratio η_b is

- Choked compressible: $\eta_b \leq \eta_{ch}$ set $\eta = \eta_{ch}$
- Subsonic compressible: $\eta_b > \eta_{ch}$ set $\eta = \eta_b$
- Subcooled liquid: always subsonic

The omega method HEM solution for orifice flow is plotted in Fig. 23-36. The solution for flashing liquids without noncondensables is to the right of $\omega = 1$, and the solution for frozen flow with subcooled liquids plus noncondensables is to the left. The omega method HEM solution for horizontal pipe flow is plotted in Fig. 23-37 as the ratio of pipe mass flux to orifice mass flux.

HEM for Inclined Pipe, Discharge If a pipe leak occurs at an elevation above or below the pump or source tank the elevation change can be idealized between the starting and ending points as shown in Fig. 23-38. That is, elevation changes can be treated as an inclined pipe with a non-zero inclination factor F_i . This is an approximation to the actual piping isometrics, but is often an adequate approximation.

The HEM solution in this case is implicit in G_{cp} :

Case 1: Flashing in Orifice, Two-Phase Flashing Flow in Pipe:

Use Eq. (23-63) for G_{ori} :

$$\begin{aligned} N + \ln \left\{ \frac{X(\eta_2)}{X(\eta_1)} \left[\frac{\eta_1}{\eta_2} \right]^2 \right\} &= \frac{1 - \omega}{c} (\eta_1 - \eta_2) \\ &+ \frac{b(1 - \omega) - c\omega \eta_s}{2c^2} \ln \frac{X(\eta_2)}{X(\eta_1)} \\ &+ \frac{bc\omega \eta_s - (1 - \omega)(b^2 - 2ac)}{2c^2} [I_0(\eta_2) - I_0(\eta_1)] \end{aligned} \quad (26-69)$$

Case 2: Subcooled Liquid in Orifice, Flashing in Pipe: Use Eq. (23-65) for G_{ori} :

$$\begin{aligned} N + \ln \left\{ \frac{X(\eta_2)}{X(\eta_s)} \left[\frac{\eta_s}{\eta_2} \right]^2 \right\} &= \frac{\eta_1 - \eta_s}{\frac{1}{2}G_{cp}^2 + F_i} + \frac{1 - \omega}{c} (\eta_s - \eta_2) \\ &+ \frac{b(1 - \omega) - c\omega \eta_s}{2c^2} \ln \frac{X(\eta_2)}{X(\eta_s)} \\ &+ \frac{bc\omega \eta_s - (1 - \omega)(b^2 - 2ac)}{2c^2} [I_0(\eta_2) - I_0(\eta_s)] \end{aligned} \quad (23-70)$$

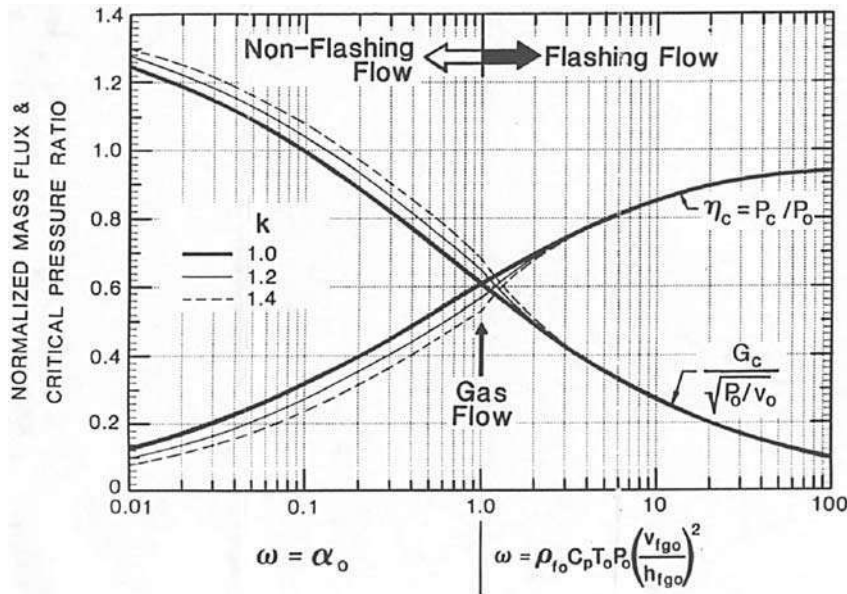


FIG. 23-36 Omega method solution for orifice flow of flashing liquids and for noncondensable gas plus subcooled liquids.

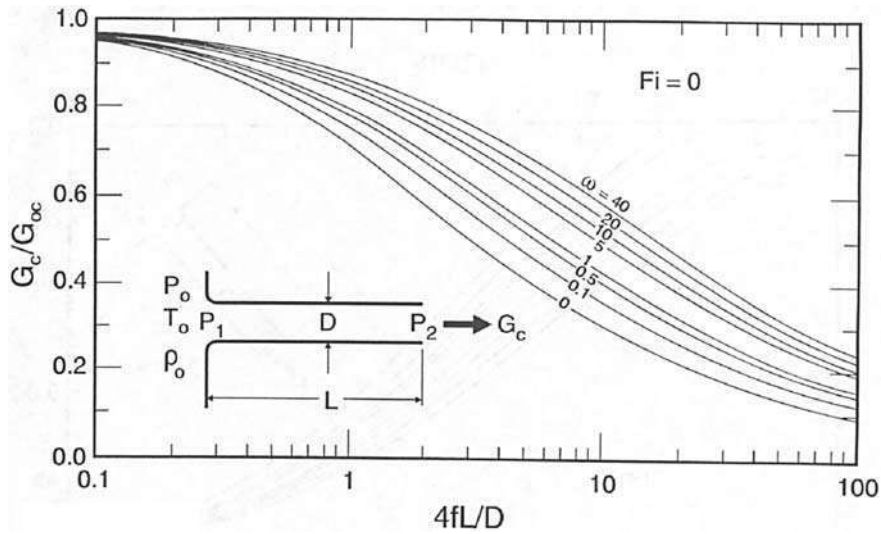


FIG. 23-37 Omega method solution for flashing liquid horizontal pipe flow.

where

$$a = \frac{1}{2} G^2 \omega^2 \eta_s^2 \quad (23-71a)$$

$$b = \frac{1}{2} G^2 2\omega (1 - \omega) \eta_s \quad (23-71b)$$

$$c = \frac{1}{2} G^2 (1 - \omega)^2 + F_I \quad (23-71c)$$

$$q = 4ac - b^2 \quad (23-71d)$$

$$X(\eta) = a + b\eta + c\eta^2 \quad (23-72)$$

$$I_0(\eta) = \int \frac{d\eta}{X(\eta)}$$

If $q > 0$, $F_I > 0$, upflow:

$$I_0(\eta) = \frac{2}{\sqrt{q}} \tan^{-1} \frac{2c\eta + b}{\sqrt{q}} \quad (23-73a)$$

If $q < 0$, $F_I < 0$, downflow:

$$I_0(\eta) = \ln \frac{2c\eta + b - \sqrt{-q}}{2c\eta + b + \sqrt{-q}} \quad (23-73b)$$

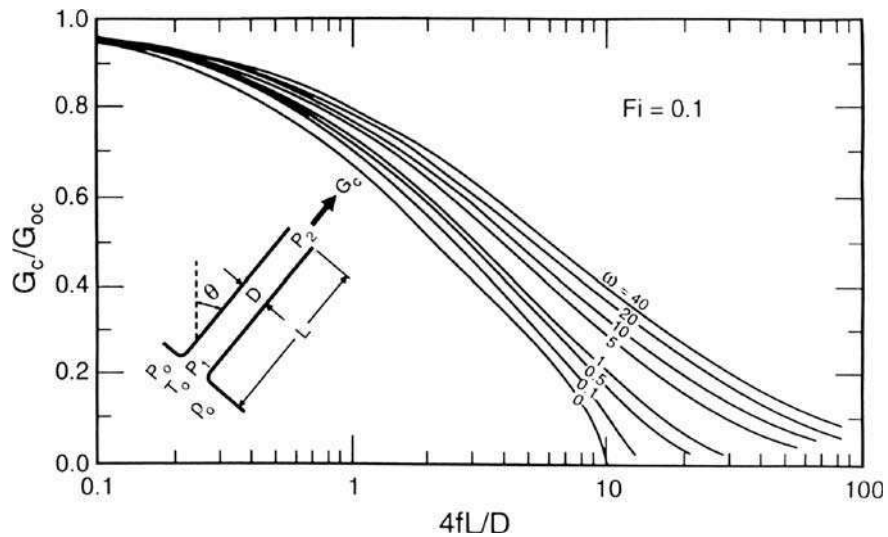


FIG. 23-38 Omega method HEM solution for inclined pipe flow at $F_I = 0.1$.

If $q = 0$, $F_I = 0$, horizontal flow:

$$I_0(\eta) = \frac{-1}{(1 - \omega)\epsilon\eta} \quad (23-73c)$$

The omega method HEM solution for inclined pipe flow is illustrated in Fig. 23-38.

Nonequilibrium Extension of Omega Method The omega method HEM tends to produce discharge rates that are low, particularly for short pipes. To correct this deficiency, Diener and Schmidt proposed a modification they term the *nonequilibrium compressibility factor* N , defined by

$$N = \left[x_0 + C_{PL0} T_0 P_0 \left(\frac{v_{GL0}}{H_{GL0}^2} \right) \ln \left[\frac{1}{\eta_C} \right] \right]^a \quad (23-74)$$

with

$$a = \begin{cases} 0.6 & \text{for orifices, control valves, short nozzles} \\ 0.4 & \text{for pressure relief valves, high - lift control valves} \\ 0 & \text{for long nozzles, office with large area ratio} \end{cases}$$

The choked pressure ratio η_c is found by using Eqs. (23-68a) and (23-68b). The nonequilibrium factor N is used to modify ω :

$$\omega = \alpha_0 + \omega_s N \quad (23-75)$$

Differences between Subcooled and Saturated Discharge for Horizontal Pipes Data by Uchaida and Narai (1966) in Figs. 23-39 and 23-40 illustrate the substantial differences between subcooled and saturated-liquid discharge rates. Discharge rates decrease with increasing pipe length in both cases, but the drop in discharge rate is much more pronounced with saturated liquids. This is so, because the flashed vapor effectively chokes the flow and decreases the two-phase density.

Accuracy of Discharge Rate Predictions Model verification is difficult because agreement with data varies substantially from one set of experimental data to another. The accuracy of the HEM and the NEM correction is illustrated in Fig. 23-41 and compared with data for saturated water by Uchaida and Narai (1966). The NEM correction using the power coefficient a of 0.6 decreases omega and increases the predicted discharge rates so that good predictions are obtained for pipe lengths greater than 0.5 m.

The accuracy of the energy balance method for discharge of flashing liquids through orifices and horizontal pipes is illustrated in Figs. 23-42 and 23-43. The ratio of predicted to experimental mass flux is plotted for saturated water data by Uchaida and Narai and by Sozzi and

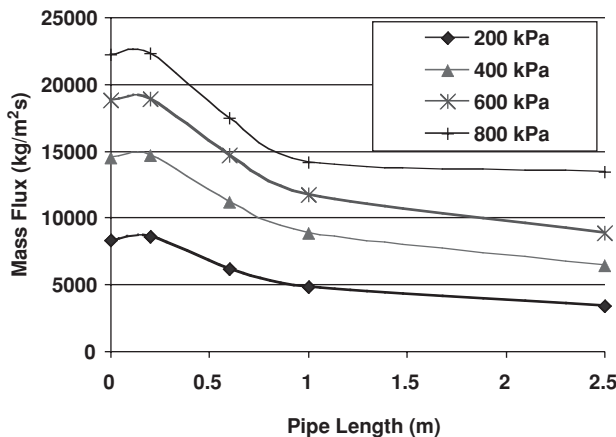


FIG. 23-39 Subcooled water data by Uchaida and Narai.

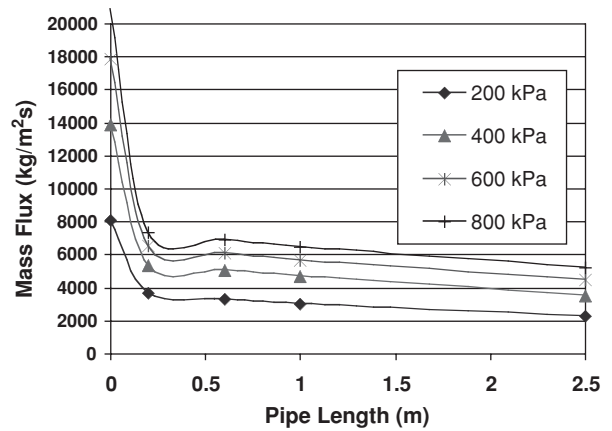


FIG. 23-40 Saturated water data by Uchaida and Narai.

Sutherland (1975). Orifice flow rates are underpredicted by about the same factor with the energy balance method and with the NEM. Discharge predictions for short (0.2-m) pipes are overpredicted by the energy balance method. In this region, the assumption of homogeneous equilibrium is not justified. A model that takes slip velocity into account may improve predictions for short pipes.

ATMOSPHERIC DISPERSION

Nomenclature

$\langle C \rangle_1$	(Ensemble) time-averaged concentration for averaging time t_1 , mass per volume
d_0	Displacement height, length
E	Airborne contaminant mass rate in a plume, mass per time
E_t	Total airborne contaminant mass in a puff, mass
g	Acceleration due to gravity, length squared per time
k	Von Karman's constant (typically 0.4)
L	Monin-Obukhov length, length
p	Index reflecting decrease in time-averaged concentration with averaging time
t_d	Along wind (x direction) dispersion time scale, time
t_h	Time scale for contaminant to pose a particular hazard, time
t_s	Source time scale, length of time for contaminant to become airborne, time
t_t	Time scale for a contaminant cloud to reach hazard endpoint distance x_h , time
u	(Characteristic) wind speed at elevation z , length per time
u_r	Wind speed at reference height z_r , length per time
u^*	Friction velocity, length per time
x_v	Virtual source distance upwind of real source, length
x	(Downwind) distance in wind direction, length
x_h	Hazard endpoint distance, length
z_0	Surface roughness, length
z_r	Reference height, length

Greek symbols

α	Monin-Obukhov length coefficient
ϵ	Height of ground covering, length
ρ	Air-contaminant mixture density at concentration C , mixture mass per mixture volume
ρ_a	Ambient air density, mass per volume
$\sigma_x, \sigma_y, \sigma_z$	Dispersion coefficients in x, y, z directions, length
σ_θ	Wind direction standard deviation, angle

Subscripts

x	Along wind direction
y	Lateral direction
z	Vertical direction

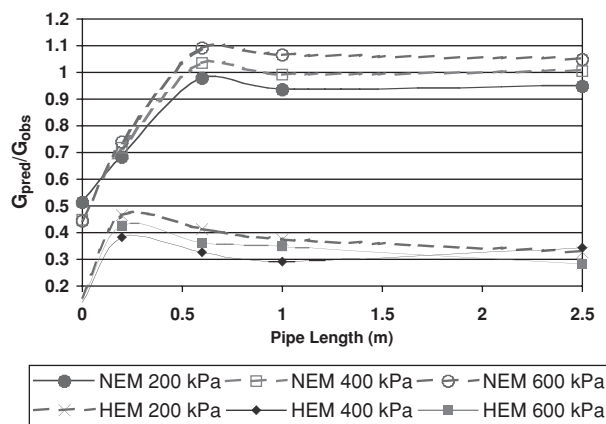


FIG. 23-41 Accuracy of Omega method HEM and NEM correction compared with saturated water data by Uchaida and Narai.

GENERAL REFERENCES: Britter and McQuaid, *Workbook on the Dispersion of Dense Gases*, Health and Safety Executive Report 17/1988, Sheffield, U.K., 1988. Mannan, *Lees' Loss Prevention in the Process Industries*, 3d ed., Chap. 15, Elsevier Butterworth-Heinemann, Oxford, U.K., 2005. Panofsky and Dutton, *Atmospheric Turbulence*, Wiley, New York, 1984. Pasquill and Smith, *Atmospheric Diffusion*, 3d ed., Ellis Horwood Limited, Chichester, U.K., 1983. Seinfeld, *Atmospheric Chemistry and Physics of Air Pollution*, Chaps. 12–15, Wiley, New York, 1986. Turner, *Workbook of Atmospheric Dispersion Estimates*, U.S. Department of Health, Education, and Welfare, 1970.

Introduction Atmospheric dispersion models predict the dilution of an airborne contaminant after its release (and depressurization) to the atmosphere. The discussion in this section focuses on episodic releases representing acute biological, toxic, or flammable hazards. Physical and mathematical dispersion models discussed here are typically used for (1) forensic purposes (in comparison with actual events such as experiments, field trials, or accidents); (2) regulatory purposes, such as estimating the consequences of releases of toxic or flammable materials for siting requirements; or (3) planning purposes. Release consequences may be used to estimate the risk to a facility and workforce or the surrounding population. All such estimates can be used to identify appropriate emergency response or mitigation measures and the priority with which such measures should be considered.

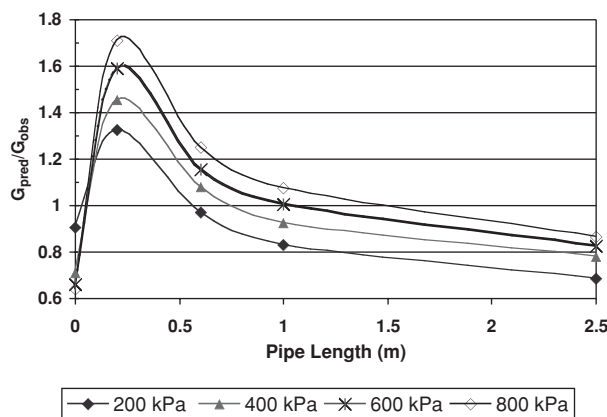


FIG. 23-42 Accuracy of energy balance method compared with saturated water data by Uchaida and Narai.

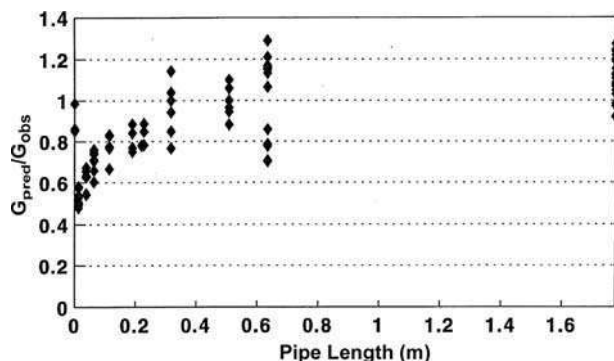


FIG. 23-43 Accuracy of energy balance method for flashing liquid discharge through orifices and horizontal pipes.

Parameters Affecting Atmospheric Dispersion The parameters important to atmospheric dispersion can be divided roughly into three categories: contaminant source; atmospheric and terrain properties; and contaminant interaction with the atmosphere.

Contaminant Source The contaminant source includes such factors as the following:

1. *Rate or total amount of contaminant that becomes airborne.*

For a continuous contaminant release, the contaminant concentration at a fixed downwind distance is roughly proportional to the rate at which the contaminant becomes airborne. Likewise, the contaminant concentration at a fixed downwind distance is roughly proportional to the amount of contaminant released if the release is instantaneous. However, materials released from containment may not immediately become airborne. For example, a liquid below its boiling point stored at atmospheric pressure will form a liquid pool when released. The rate at which the contaminant becomes airborne will depend on the heat-transfer rate to the liquid pool and the mass-transfer rate from the liquid pool to air. However, the same liquid stored at a higher pressure may form an aerosol so that the liquid phase is airborne. If the aerosol droplet size formed in the release is too large to remain suspended, the liquid phase will fall to the ground (rainout), and the resulting liquid pool evaporation will again be dictated by heat- and mass-transfer considerations.

2. *Release momentum.* For jet releases, the amount of air entrained in an unobstructed jet is proportional to the jet velocity. Depending on the orientation of the jet relative to nearby obstructions, the momentum of a jet can be dissipated without significant air entrainment. The degree of initial air entrainment can be an important determinant of the hazard extent, particularly for flammable hazards. It would be (possibly overly) conservative to assume the source momentum is dissipated without air dilution. Explosive releases are high-momentum, instantaneous releases. For explosive releases, a rough first approximation is to assume that the mass of contaminant in the explosion is mixed with 10 times that mass of air.

3. *Release buoyancy.* Contaminant release buoyancy is determined by the initial contaminant temperature and molecular weight, and whether there is a suspended liquid phase as in an aerosol. Lighter-than-air contaminants will rise and be more readily dispersed. Denser-than-air contaminants will tend to stay near ground level, and the atmospheric dispersion of such materials can be importantly determined by buoyancy effects. The degree of importance is typically quantified by a Richardson number (proportional to the density difference with air and the quantity or rate of contaminant release and inversely proportional to the square of the wind speed); the higher the Richardson number, the more important the effect of buoyancy on the atmospheric dispersion (see Denser-than-Air Dispersion Models — Britter-McQuaid below). Denser-than-air contaminants can actually displace the atmospheric flow field. In contrast, contaminants which do not perturb the atmospheric flow field and can be thought of as simply following the airflow are termed *passive* contaminants.

4. *Other source conditions.* The source height and the source area are also important source characteristics. The greatest impact is typically associated with ground-level sources, so elevated sources are not considered in the model discussion below.

Atmospheric and Terrain Parameters In addition to terrain parameters, basic atmospheric parameters are listed below. There can be many other meteorological effects which can be important in some circumstances (e.g., inversion layers) and are beyond the scope of this introduction.

1. *Wind direction.* Wind direction is the most important determinant of the location of hazard zones with notable exceptions involving high-momentum releases or releases where buoyancy is important (particularly for denser-than-air contaminants involving terrain effects such as valleys and slopes especially under low-wind-speed conditions). Near ground level, lateral wind direction variability is much larger than vertical variability (typical of flat-plate boundary layers) and measured with the standard deviation σ_θ which is a function of atmospheric stability.

2. *Atmospheric stability.* Atmospheric stability is often characterized by Pasquill stability class (ranging from A through F). Atmospheric stability classes are broad classifications which are used to characterize the continuum of atmospheric turbulence available to the (mixing) dispersion process. Neutral atmospheric stability (class D) occurs most often and indicates that the vertical momentum flux is not influenced by the vertical heat flux in the lowest atmospheric layers. Unstable atmospheric conditions (instability increasing from classes C through A) indicate the atmospheric vertical heat flux is enhancing the vertical mixing of the atmosphere, such as when ground surface heating due to insolation (incident solar radiation) enhances vertical atmospheric mixing (turbulence). Stable atmospheric conditions (stability increasing from classes E through F) indicate the atmospheric vertical heat flux is suppressing the vertical mixing of the atmosphere, such as when ground surface cooling at night suppresses vertical mixing (turbulence). All atmospheric dispersion models rely in some way on measured atmospheric parameters, and these parameters are quite frequently correlated on the basis of Pasquill stability class. Consequently, atmospheric dispersion model results should be viewed as only representative of the atmospheric conditions present in a real or hypothetical release. Atmospheric stability class is an important determinant of concentration at a fixed downwind distance in dispersion models. For passive contaminants, concentration at a fixed downwind distance can be roughly an order of magnitude higher for F stability in comparison to D stability. Table 23-24 illustrates the relationship between atmospheric stability class and other meteorological conditions. Determination of Pasquill stability class should be made with detailed measurements such as discussed by Golder ("Relations among Stability Parameters in the Surface Layer," *Boundary Layer Meteorol.*, 3, pp. 47-58, 1972).

TABLE 23-24 Typical Atmospheric Stability Classes in Terms of Wind Speed, Insolation, and State of the Sky

Surface wind speed m/s	Insolation			Night	
	Strong	Moderate	Slight	Thinly overcast or ≥4/8 low cloud	≤3/8 cloud
<2	A	A-B	B	—	—
2-3	A-B	B	C	E	F
3-5	B	B-C	C	D	E
5-6	C	C-D	D	D	D
>6	C	D	D	D	D

For A-B, take the average of values for A and B, etc. Pasquill and Smith relate insolation to conditions in England. Seinfeld (*Atmospheric Chemistry and Physics of Air Pollution*, Wiley, New York, 1986) classifies insolation greater than 700 W/m² as strong, less than 350 W/m² as slight, and between these limits as moderate. Night refers to the period from 1 h before sunset to 1 h after dawn. The neutral class D should be used, regardless of wind speed, for overcast conditions during day or night and for all sky conditions during the hour preceding or following the night period.

SOURCE: Pasquill and Smith, *Atmospheric Diffusion*, 3d ed., Ellis Horwood Limited, Chichester, U.K., 1983.

3. *General terrain characteristics.* General terrain characteristics are used to describe features with length scales much smaller than the depth of the contaminant cloud or the height of the characteristic wind speed. General terrain characteristics influencing the shape of the vertical wind speed profile are parameterized with the surface roughness and displacement height. The surface roughness z_0 is roughly proportional to the height of the ground covering ϵ (and $z_0 \sim 0.05\epsilon$ to 0.15ϵ) and can be used to infer the amount of vertical turbulence (mixing) in the atmospheric flow (higher obstacles enhance vertical turbulence above the obstacle height). However, obstacles can also displace the atmospheric flow; the displacement height is the height at which the (average) ambient wind speed is negligible. If the displacement height is negligible (flow blockage is not important), contaminant concentration at a fixed downwind distance decreases with increased surface roughness (an order-of-magnitude increase in surface roughness decreases the concentration by roughly a factor of 3).

4. *Wind speed.* Because wind speed varies with height above the ground, the reference wind speed u_r must be specified at a particular height z_r (typically 10 m provided local flow obstructions are not important at this height). For a continuous contaminant release, the contaminant concentration at a fixed downwind distance is roughly inversely proportional to the wind speed. As indicated above, denser-than-air effects are less important at higher wind speeds. The wind speed profile is affected by atmospheric stability, surface roughness z_0 , and displacement height d_0 (the height below which the wind speed is negligible). The vertical profile of the wind speed is logarithmic

$$u = \frac{u_r}{k} \left[\ln \left(\frac{z + d_0}{z_0} \right) + \alpha \left(\frac{z}{L} \right) \right] \quad (23-76)$$

for neutral and stable conditions when $z \gg z_0$ and d_0 ; k is von Karman's constant (typically 0.4), u_r is the friction velocity (u_r^2 characterizes the atmospheric vertical mixing rate), L is the Monin-Obukhov length (a measure of stability class), and α is the Monin-Obukhov length coefficient ($\alpha = 5.2$ for stable conditions). For D stability, $L = \infty$ so $z/L = 0$, and the second term in brackets is zero. Since Eq. (23-76) must also hold for u_r and z_r , u_r is proportional to u_r (all other things being equal), so reduction of the wind speed by a factor of 2 reduces the vertical atmospheric mixing by a factor of 4.

5. *Flow obstructions.* In contrast to general terrain characteristics, flow obstructions have length scales much larger than the depth or width of the contaminant cloud or the height of the characteristic wind speed. Flow obstructions can increase or decrease contaminant concentration depending on location. Flow obstructions increase concentration by delaying the dispersal of the contaminant cloud; e.g., inside a dike, concentration is higher, but downwind of the dike, concentrations can be smaller; and the downwind side of flow obstructions can temporarily trap higher concentrations. Obstructions also distort the wind speed profile given above.

Contaminant Interaction with the Atmosphere Contaminant interaction with the atmosphere is important for several reasons:

1. There are chemical reactions between the released contaminant and ambient air or surfaces. If the released contaminant reacts, any reacted material can no longer be considered airborne (although the reaction products may also be hazardous), and so chemical reactions effectively reduce the rate or amount of airborne contaminant. Some reactions can be characterized as dry or wet deposition.

2. Phase changes are typically associated with the evaporation of any suspended liquid phase in an aerosol release. As air is mixed with an aerosol, equilibrium constraints cause additional evaporation of the liquid phase which reduces the temperature of the liquid phase (and the vapor phase if thermal equilibrium is maintained).

3. Ground-to-contaminant cloud heat transfer acts to warm the contaminant cloud if the cloud temperature is lower than ambient. Ground-to-cloud heat transfer can be important for cold clouds at ground level because the buoyancy of the contaminant cloud can be significantly reduced for cold clouds with contaminant molecular weight less than that of air. At higher wind speeds, heat transfer is by forced convection, and even though such conditions produce higher heat-transfer coefficients than do low-wind-speed conditions, heat

transfer is typically more important at low-wind-speed conditions because of two effects: (1) At low wind speeds, the amount of air entrainment is reduced so that the heat-transfer driving force is larger, and (2) at low wind speeds, the contact time between the contaminant cloud and the ground is longer.

Atmospheric Dispersion Models Atmospheric dispersion models generally fall into the categories discussed below. Regardless of the modeling approach, models should be verified that the appropriate physical phenomena are being modeled and validated by comparison with relevant data (at field and laboratory scale). The choice of modeling techniques may be influenced by the expected distance to the level of concern.

1. *Physical or wind tunnel models* Wind tunnel models have long been used to study the atmospheric flow around structures such as buildings and bridges to predict pressure loading and local velocities. Wind tunnel measurement of contaminant concentrations for release scenarios can be used to estimate hazard zones. However, wind tunnel models are generally considered to be incapable of simultaneously scaling mechanical turbulence and thermally induced turbulence (verification issue). Wind tunnel experiments can be very useful when considering validation of mathematical models. Wind tunnel models typically do not account for the lateral variation in wind direction.

2. *Empirical models* Empirical models rely on the correlation of atmospheric dispersion data for characteristic release types. Two examples of empirically based models are the Pasquill-Gifford model (for passive contaminants) and the Britter-McQuaid model (for denser-than-air contaminants) both of which are described below. Empirical models can be useful for the validation of other mathematical models but are limited to the characteristic release scenarios considered in the correlation. Selected empirical models are discussed in greater detail below because they can provide a reasonable first approximation of the hazard extent for many release scenarios and can be used as screening tools to indicate which release scenarios are most important to consider.

3. *First principle mathematical models* These models solve the basic conservation equations for mass and momentum in their form as partial differential equations (PDEs) along with some method of turbulence closure and appropriate initial and boundary conditions. Such models have become more common with the steady increase in computing power and sophistication of numerical algorithms. However, there are many potential problems that must be addressed. In the verification process, the PDEs being solved must adequately represent the physics of the dispersion process especially for processes such as ground-to-cloud heat transfer, phase changes for condensed phases, and chemical reactions. Also, turbulence closure methods (and associated boundary and initial conditions) must be appropriate for the dis-

persion processes present, especially for denser-than-air contaminants. Regardless of the algorithm for solving the PDEs, any solution must demonstrate resolution independence (i.e., the numerical solution must be independent of grid spacing or time step). Finally, models should be validated against relevant information for the scenario considered. Despite decreased computational costs, such models still require a significant investment for investigating a release scenario.

4. *Simplified mathematical models* These models typically begin with the basic conservation equations of the first principle models but make simplifying assumptions (typically related to similarity theory) to reduce the problem to the solution of (simultaneous) ordinary differential equations. In the verification process, such models must also address the relevant physical phenomenon as well as be validated for the application being considered. Such models are typically easily solved on a computer with typically less user interaction than required for the solution of PDEs. Simplified mathematical models may also be used as screening tools to identify the most important release scenarios; however, other modeling approaches should be considered only if they address and have been validated for the important aspects of the scenario under consideration.

All mathematical models predict (ensemble) time-averaged cloud behavior for a particular set of release conditions. To illustrate in very broad terms, consider a set of trials (field experiments) with continuous contaminant releases (as plumes) that are conducted under identical atmospheric conditions. Suppose that you could measure the concentration on the plume centerline at a given downwind distance with a reasonably fast concentration sensor. Owing to the turbulent nature of the atmosphere and dispersion process, the measured concentration on the plume centerline at a given downwind location observed in each trial would not be identical. If these measurements are averaged during the period for which the contaminant is present, the average of measurements will not change after a sufficient number of trials; this is an ensemble average. This ensemble average reflects the instantaneous (ensemble average) concentration provided the averaging time of the sensor is sufficiently fast. If one considers the difference between any one measured data set and this ensemble average, the measurements will show peak concentrations higher than the average (mean). Peak-to-mean concentration values depend on many factors, but for many purposes, a peak-to-mean concentration ratio of 2 can be assumed. For this hypothetical example, the concentrations were assumed to be measured on the plume centerline. However, the plume centerline of a passive contaminant does not remain at the same ground location due to variation in the wind direction related to large-scale atmospheric turbulence, as illustrated in Fig. 23-44 for a ground-level release on flat, unobstructed terrain. This effect is termed *plume meander*. (Note that denser-than-air contaminant plumes would exhibit less of this effect because such plumes may actually displace the

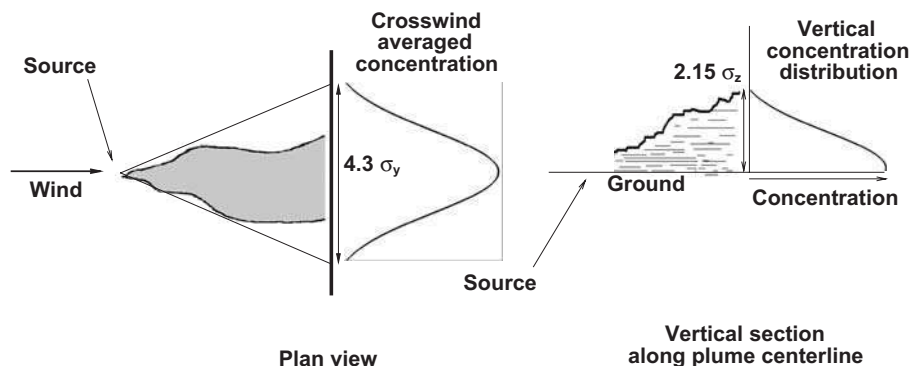


FIG. 23-44 Schematic representation of time-averaged distribution and spread for a continuous plume. σ_y and σ_z are the statistical measures of crosswind and vertical dimensions; $4.3\sigma_y$ is the width corresponding to a concentration 0.1 of the central value when the distribution is of gaussian form (a corresponding cloud height is $2.15\sigma_z$). (Redrawn from Pasquill and Smith, *Atmospheric Diffusion*, 3d ed., Ellis Horwood Limited, Chichester, U.K., 1983).

atmospheric flow field.) For a fixed ground location, the concentration sensor will be at various locations in the plume (effectively moving in and out of the plume). For a constant average wind direction, concentrations at a fixed ground location will again approach an (ensemble) average over a sufficiently long averaging time (10-min averaging time has proved to be standard). In this case, concentrations are (ensemble-averaged) 10-min concentrations. (Other properties of the concentration distribution are considered in the literature.) For the effect of plume meander, the relationship between concentrations for various averaging times for ground-level plumes can be approximated as

$$\frac{\langle C \rangle_2}{\langle C \rangle_1} = \left(\frac{t_1}{t_2} \right)^p \quad (23-77)$$

where $\langle C \rangle_1$ and $\langle C \rangle_2$ are (ensemble) time-averaged concentrations with averaging times t_1 and t_2 and p is an index typically taken to be around 0.2 for passive plumes. In addition to other more sophisticated approaches, some dispersion models adjust the dispersion coefficients to account for the effect of plume meander. Equation (23-77) shows that concentration decreases as averaging time increases. Based on comparison between passive puff and (10-min) plume coefficients, the averaging time associated with puff coefficients is about 20 s, which is the smallest value that should properly be used in Eq. (23-77). There is no correction to puff coefficients or concentrations associated with plume meander. Note that this is only a rough illustration of the processes present which are discussed in much greater detail in the literature.

Basic Scenario Time Scales There are several competing time scales that are important:

1. The source time scale t_s describes the length of time for the contaminant to become airborne; the source time scale is also limited by the inventory of contaminant.

2. The hazard endpoint time scale t_h describes the length of time required for the contaminant to pose a hazard. There are many different time scales associated with various toxicity levels (e.g., TLV-C ceiling limit values are never to be exceeded, TLV-STEL values are not to be exceeded in a 15-min period, etc.). Time scales associated with flammability hazards reflect the maximum local concentration (and also typically including peak-to-mean concentration ratios) and for reasons discussed above are considered representative of dispersion model averaging times of around 20 s.

3. The travel time t_t is the time required for a contaminant cloud to reach the endpoint distance x_e . As a first approximation, $t_t = x_e/(u_r/2)$ for ground-level clouds; for elevated releases, $t_t = x_e/u$ where u is the characteristic wind speed.

The source time scale t_s must be greater than the travel time t_t for a steady-state plume to be possible. Other time scale restrictions are considered for the models discussed here.

Scenario Development and Simulation The typical procedure for assessing the consequences of an airborne contaminant release is as follows:

1. Identify release scenarios by which containment can be lost along with the hazards of that scenario. Hazards may differ depending on the physical state of the released contaminant and the circumstances of the scenario. Hazards arise from the properties of the released material such as biological agents, toxic materials, or flammable materials; flammable or reactive materials may pose an explosion hazard depending on their reactivity and the degree to which the air-contaminant mixture is confined. Scenarios include a description of the applicable atmospheric conditions (which may be dictated by regulatory requirements).

2. Develop an appropriate source model to define the source description (see previous subsection) for each scenario.

3. Use an appropriate atmospheric dispersion model to assess the consequences or risk of each scenario. For screening purposes, atmospheric dispersion models which are less costly may be used to identify the most important scenarios; examples are discussed in the following subsection. More expensive modeling procedures can be applied to the most important scenarios provided such procedures are more appropriate and accurate. Screening methods may also be useful in considering the validity of more complicated models.

4. Determine if the resulting consequences or risk to people and property is acceptable. For unacceptable scenarios, mitigation measures should be applied such as discussed by Prugh and Johnson (*Guidelines for Vapor Release Mitigation*, AIChE, New York, 1988), and amended scenarios should be reassessed. If mitigation measures cannot sufficiently reduce the consequences or risk, the appropriate business and ethical decision would be to discontinue such operations.

Passive Contaminant (Pasquill-Gifford) Dispersion Models

The gaussian dispersion model is based on the assumption of a passive contaminant release (i.e., the released contaminant does not change the atmospheric flow field). Based on the theoretical model of a passive contaminant, the spatial distribution would have a gaussian distribution with characteristic length scales. Using extensive observations of (steady-state) plumes, the Pasquill-Gifford dispersion model correlates characteristic vertical and lateral length scales (or dispersion coefficients σ_z and σ_y , respectively) with atmospheric stability class; other correlations have also been proposed for the plume dispersion coefficients taking other effects into account such as the effect of surface roughness z_0 . Less extensive observations of instantaneously released puffs have been used to characterize the length scales of puffs (with the additional length scale σ_x to characterize the along wind direction); σ_x is often assumed to be approximated by σ_y in puff models. Although the Pasquill-Gifford approach provides for the prediction of the concentration distribution, the discussion below is limited to the maximum predicted concentration since this is most important for hazard assessment purposes.

Pasquill-Gifford plume model At a given downwind distance x , the maximum (average) concentration for a (continuous) passive plume from a point source is

$$\langle C \rangle_1 = \frac{E}{\pi \sigma_y \sigma_z u} \quad (23-78)$$

where E is the mass rate at which the contaminant becomes airborne and u is the characteristic wind speed (typically taken to be u_r). Pasquill-Gifford plume dispersion coefficients as a function of downwind distance and atmospheric stability are available from many sources (Seinfeld, *Atmospheric Chemistry and Physics of Air Pollution*, Wiley, New York, 1986; Mannan, *Less' Loss Prevention in the Process Industries*, 3d ed., Chap. 15, Elsevier Butterworth-Heinemann, Oxford, U.K., 2005; Griffiths, "Errors in the Use of the Briggs Parameterization for Atmospheric Dispersion Coefficients," *Atmospheric Environment*, vol. 28, no. 17, pp. 2861–2865, 1994). Passive dispersion coefficients are typically not provided for distances less than 100 m or greater than a few kilometers because predicted concentrations outside this range must be viewed with some caution (e.g., meteorological conditions may not persist over such large time scales, and at such long distances, large-scale meteorological and terrain features may dictate plume behavior in ways not accounted for by this simple approach).

Note that the predicted values of σ_y and σ_z are sensitive to the *specification* of atmospheric stability. Between D and F stability classes, σ_y for D stability is roughly three times greater than for F stability, and σ_z is roughly two times greater. Since $\langle C \rangle_1$ is inversely proportional to $\sigma_y \sigma_z$, the predicted $\langle C \rangle_1$ for F stability is roughly six times greater than for D stability.

Pasquill-Gifford puff model At a given downwind distance x , the maximum (average) concentration for a (instantaneous) passive puff from a point source is

$$\langle C \rangle = \frac{2E_t}{(2\pi)^{3/2} \sigma_x \sigma_y \sigma_z} \quad (23-79)$$

where E_t is the total contaminant mass that becomes airborne. Pasquill-Gifford puff dispersion coefficients as function of downwind distance and atmospheric stability are available from many sources (e.g., Mannan, *Less' Loss Prevention in the Process Industries*, 3d ed., Chap. 15, Elsevier Butterworth-Heinemann, Oxford, U.K., 2005). As indicated above, σ_x is often assumed to be approximated by σ_y in puff models. As for passive plumes, note that passive puff dispersion coefficients are not provided for distances of less than 100 m (where

near source effects will be important), and predicting concentrations for distances longer than a few kilometers must be viewed with some caution for the same reasons cited above.

Note that the predicted values of σ_y and σ_z are sensitive to the *specification* of atmospheric stability for puffs as well. Between D and F stability classes, σ_y for D stability is roughly three times greater than for F stability, and σ_z is roughly 10 times greater. Since $\langle C \rangle$ is inversely proportional to $\sigma_y^2 \sigma_z$, the predicted $\langle C \rangle$ for F stability is roughly 40 times greater than for D stability.

Virtual sources As indicated above, the gaussian model was formulated for an idealized point source, and such an approach may be unnecessarily conservative (predict an unrealistically large concentration) for a real release. There are formulations for area sources, but such models are more cumbersome than the point source models above. For point source models, methods using a virtual source have been proposed in the past which essentially use the maximum concentration of the real source to determine the location of an equivalent upwind point source that would give the same maximum concentration at the real source. Such an approach will tend to overcompensate and unrealistically reduce the predicted concentration because a real source has lateral and along-wind extent (not a maximum concentration at a point). Consequently, the modeled concentration can be assumed to be bounded above, using the point source formulas in Eq. (23-78) or (23-79), and bounded below by concentrations predicted by using a virtual source approach.

Assuming the source concentration C_s is known, the virtual distance is found by using the known source concentration to find the virtual source distance. For a plume, solve Eq. (23-78) for the product $\sigma_y \sigma_z$, then determine the virtual source distance x_v by iterative solution (or trial and error) using E and C_s . For a puff, solve Eq. (23-79) for the product $\sigma_x \sigma_y \sigma_z$ (or $\sigma_y^2 \sigma_z$); then determine the virtual source distance x_v by iterative solution (or trial and error) using E_t and C_s . The dispersion coefficients at distance x_v will now represent a distance from the real source of $x_v - x_r$.

Passive puff or plume In addition to the restriction on plumes discussed above, there is an along-wind dispersion time scale given by $t_d = 2\sigma_x/u_r$, where σ_x is evaluated at the endpoint distance x_r . The release can usually be considered a plume if $t_s > 2.5 t_d$, where t_s is the source time scale defined above, and the release can be considered a puff if $t_d > t_s$. For $t_d \leq t_s \leq 2.5 t_d$, neither puff nor plume models are entirely appropriate; the predicted concentration is considered the largest of the puff and plume predictions.

Denser-than-Air Contaminant (Britter-McQuaid) Dispersion Models Britter and McQuaid (Workbook on the Dispersion of Dense Gases, Health and Safety Executive Report 17/1988, Sheffield, U.K., 1988) proposed a correlation for estimating the dispersion of denser-than-air contaminants from area sources for plume and puff releases. Their objective was to produce correlations which predicted the distance to a given concentration level within a factor of 2. Their analysis identified the dominant independent variables as (1) density of released contaminant after depressurization to atmospheric pressure ρ_s ; (2) volumetric rate E/ρ_s (or total volume E/ρ_s) of contaminant released; (3) characteristic wind speed u_r (typically taken to be at 10-m elevation z_r); and (4) characteristic source dimension D_s . Based at least in part on the fact that presently available field test data for denser-than-air contaminants do not clearly indicate the importance of these parameters, Britter and McQuaid considered some independent variables to be of lesser importance, and these parameters were not considered in the correlation, including surface roughness, atmospheric stability, and exact source dimensions. (Many models for denser-than-air behavior indicate these parameters to be important.) Other effects were not included in the analysis including dilution due to source momentum and condensation of ambient humidity; such effects may be of crucial importance for contaminants that have a molecular weight less than that of air including, e.g., liquefied natural gas (LNG), ammonia, and hydrogen fluoride.

Britter and McQuaid provide correlations for denser-than-air (continuous) plumes and (instantaneous) puffs released at ambient temperature. Since many materials of practical interest are released below ambient temperature, Britter and McQuaid provide guidance as to how to predict the limiting cases for such releases.

Britter and McQuaid report that the averaging time t_1 for the plume correlation is 10 min, and Eq. (23-77) should be used with $p = 0.12$ for other averaging times (and limited to averaging times no shorter than about 20 s as for releases).

Denser-than-air puff or plume Britter and McQuaid use the ratio of the source duration to the travel time to distinguish between plumes and puffs with a slightly different definition of travel time: $t_t = x_r/(0.4u_r)$. The release can be considered a plume if $t_s > t_t$, where t_s is the source time scale defined above, and the release can be considered a puff if $t_s < t_t/4$. For $t_t/4 \leq t_s \leq t_t$, neither puff nor plume models are entirely appropriate; the predicted concentration is considered the largest of the puff and plume predictions.

Recommended Procedure for Screening Estimates The recommended procedure for making concentration estimates at a specified downwind distance with the simplified models discussed here is as follows:

1. For a given release scenario, estimate the state of the released contaminant after it has depressurized and become airborne (including any initial dilution). The initial mole fraction of hazardous components will be applied to the final reported concentrations and hazardous endpoint concentrations throughout the process. If source momentum is important (as in a jet release or for plume rise), other models are available that can address these considerations. Disregarding the dilution due to source momentum will likely result in higher concentrations downwind, but not always.
2. Consider the important time scales involved, and decide whether a puff or plume model is indicated. If this choice is unclear, assume a plume release.
3. Determine whether denser-than-air behavior is important.
4. When denser-than-air effects are important, use the Britter-McQuaid (plume or puff) models. Otherwise, assume the release is passive and use the Pasquill-Gifford (plume or puff) models. Adjust values for the virtual source correction(s) as appropriate.
5. Adjust values for the averaging time correction for plume predictions with Eq. (23-77). Note that the index p for use in the averaging time correction depends on the model used. If the hazard time scale t_h is different from the model averaging time scale (10 min for plumes), then the predicted concentration should be adjusted to t_h but only if $t_s \geq t_h$; if $t_s < t_h$, then adjust the predicted concentration to t_s .
6. For plume predictions, confirm that plume behavior applies by consideration of appropriate time scales. If plume behavior is not justified, revise the calculations with the puff model and recheck the dispersion time scale. Report the appropriate concentration or distance.

ESTIMATION OF DAMAGE EFFECTS

Nomenclature

A	Projected area of fragment, breach area, or fragment cross-sectional area
a_a	Sound velocity in atmosphere
a_e	Sound velocity in high-pressure gas prior to vessel failure
B	Batch energy availability
d	Fragment diameter
E	Explosion energy available to generate blast and fragment kinetic energy, etc.
E_p	Critical perforation energy ($1/2 MV_f^2$)
E_y	Young's modulus of elasticity
F	Dimensionless initial fragment acceleration
F	$P_e AR/Ma_e = P_e R/ma_e$ for vessel completely shattered into many small fragments
g	Acceleration due to gravity
h	Vessel wall thickness
k	Ratio of vessel outside diameter to internal diameter
L	Length of cylindrical vessel
M	Fragment mass
m	Mass per unit area of vessel shell
N	Length of cylindrical vessel forming rocketing tub fragment
N/m ²	Unit of pressure in SI system, N/m ² ; also called pascal (Pa). One psi = 6.89476 $\times 10^3$ Pa or 6.89476 kPa.
P	Liquid pressure
P_a	Atmospheric pressure

Nomenclature (Concluded)

P_b	Dynamic vessel burst pressure
P_c	Pressure at expanding gas contact surface
P_e	Pressure at vessel failure
P_{inc}	Incident (side-on) blast pressure
P_r	Normally reflected (face-on) blast pressure
R	Vessel radius
r	Fragment radius = $(A/\pi)^{0.5}$
R_g	Range of fragment
t	Steel target thickness
U	$U_f + U_m$
u	Ultimate tensile strength of target steel
U_f	Fluid compression energy
U_m	Elastic strain energy in vessel walls
V	Volume of gas
V_f	Fragment velocity
V_L	Liquid volume
W	Equivalent mass of TNT
w	Unsupported span of steel target
X	Distance from wall of vessel to target

Greek letters

β_r	Fluid compressibility
γ	Ratio of specific heats of gas C_p/C_v
Φ_0	Standard steady-state availability
ν	Poisson's ratio of vessel steel

Subscripts

0	Reference state
1	Initial state
a	Environmental state
$1 \rightarrow a$	Denotes the path from state 1 to the environmental ambient state a

GENERAL REFERENCES: Baker, Cox et al., "Explosion Hazards and Evaluation," *Fundamental Studies in Engineering* 5, Elsevier Science Publishing, New York, 1983. Kinney and Graham, *Explosive Shocks in Air*, 2d ed., Springer-Verlag, New York, 1985. Petes, *Annals, New York Academy of Sciences* 1968, vol. 152, pp. 283-316. Holden, *Assessment of Missile Hazards: Review of Incident Experience Relevant to Major Hazard Plant*, UKAEA SRD/HSE/R477, November 1988. Lees, *Loss Prevention in the Process Industries*, Butterworths, London, 1996. Leslie and Birk, "State of the Art Review of Pressurized Liquefied Gas Container Failure Modes and Associated Projectile Hazards," *Journal of Hazard Materials* 28, 1991, pp. 329-365. ASCE *Structural Analysis and Design of Nuclear Plant Facilities Manual and Reports on Engineering Practice* no. 58, 1980. Pritchard and Roberts, "Blast Effects from Vapour Cloud Explosions: A Decade of Progress," *Safety Science*, vol. 16 (3/4) 1993, pp. 527-548. "Explosions in the Process Industries," Major Haz. Monograph Series, I. Chem. E. (U.K.), 1994.

The availability of energy from an explosion can be approximately calculated in most cases but the method used depends upon the nature of the explosion.

Inert, Ideal Gas-Filled Vessels The energy available for external work following the rapid disintegration of the vessel is calculated by assuming that the gas within the vessel expands adiabatically to atmospheric pressure.

$$E = \frac{P_e V}{(\gamma - 1)} \left\{ \left[1 - \left(\frac{P_a}{P_e} \right)^{(\gamma - 1)/\gamma} \right] + (\gamma - 1) \left(\frac{P_a}{P_e} \right) \left[1 - \left(\frac{P_a}{P_e} \right)^{-1/\gamma} \right] \right\} \quad (23-80)$$

In the case of thick-walled HP vessels, the strain energy in the vessel shell can contribute to the available energy, but for vessels below about 20 MN/m² (200 barg) it is negligible and can be ignored. If a Mollier chart for the gas is available, the adiabatic energy can be measured directly. This is the preferred method, but in many cases the relevant chart is not available.

The available energy is dissipated in several ways, e.g., the strain energy to failure, plastic strain energy in the fragments, kinetic energy of the fragments, blast wave generation, kinetic energy of vessel contents, heat energy in vessel contents, etc. For damage estimation purposes, the energy distribution can be simplified to:

$$E \left(\int p \, dv \right)$$

30% blast 40% fragment kinetic energy 30% other dissipative mechanisms

Blast Characteristics Accurate calculation of the magnitude of the blast wave from an exploding pressure vessel is not possible, but it may be estimated from several approximate methods that are available.

One method of estimating the blast wave parameters is to use the TNT equivalent method, which assumes that the damage potential of the blast wave from a fragmenting pressure vessel can be approximated by the blast from an equivalent mass of trinitrotoluene (TNT). The method is not valid for the region within a few vessel diameters from the vessel. However, a rough approximation can be made outside this region by calculating an equivalent mass of TNT and utilizing its well-known blast properties. The term *equivalent mass* means the mass of TNT which would produce a similar damage pattern to that of the blast from the ruptured vessel. The energy of detonation of TNT is 4.5 MJ/kg (1.5×10^6 ft-lb/lb), so the TNT equivalent mass W is given by $W = 0.3E/4.5$ kg. Standard TNT data (Dept. of the Army, Navy, and Air Force, "Structures to resist the effects of accidental explosions," TM5-1300, NAVFAC P-397, AFM 88-22, U.S. Gov. Printing Office, vol. 2, November 1990, Figs. 2-7 and 2-15; or Kingery and Pannill, Memorandum Report No. 1518, Ballistic Research Laboratories, Aberdeen Proving Ground, U.S., April 1964) can then be used to determine the blast parameters of interest (Fig. 23-45). This method has limitations in the far field where the peak incident overpressure is less than 4 kN/m² (0.5 psi). In this region, local terrain and weather effects become significant.

The blast parameters also depend upon the physical location of the vessel. If the vessel is located close to or on the ground, then *surface-burst* data should be used. In other circumstances where the vessel is high in the air, either *free-air* or *air-burst* blast data may be used. These data are best presented in the form of *height-of-burst* curves (Petes, "Blast and Fragmentation Characteristics," *Ann. of New York Acad. of Sciences*, vol. 152, art. 1, fig. 3, 1968, p. 287). For incident blast pressures of 3×10^5 N/m² (3 bar) or less, using surface-burst data may overestimate the blast pressure by about 33 percent. Generally, pressure vessel ruptures rarely cause ground craters, so no allowance for cratering should be made.

Fragment Formation The way in which a vessel breaks up into several fragments as a consequence of an explosion or metal failure is

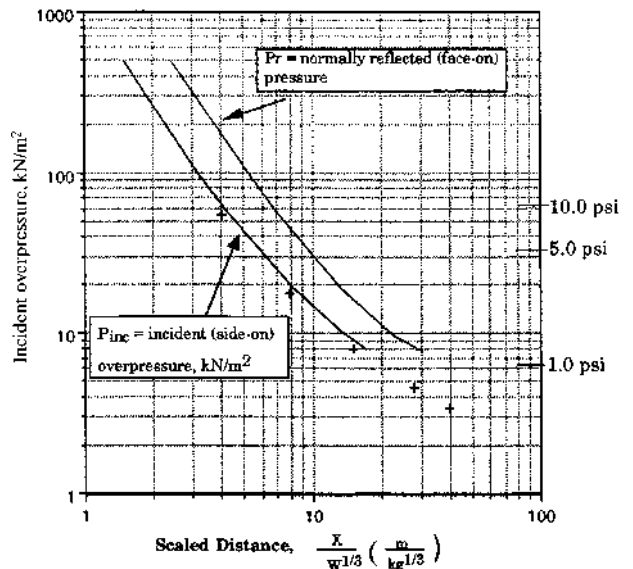


FIG. 23-45 Incident overpressure vs. scaled distance, surface burst. (The "+" points are from Kingery and Pannill, Memo Report 1518 BRL. Adapted from Department of Army, Navy, and Air Force TM5-1300, NAVFAC P-397, AFM 88-22.)

impossible to predict. Consequently, in most cases it is necessary to assume several failure geometries and to assess the effect of each. The number of fragments formed is strongly dependent upon the nature of the explosion and the vessel design. For high-speed explosions—e.g., detonations or condensed phase explosions—the vessel frequently shatters into many fragments, but for slower-speed explosions—e.g., deflagrations and BLEVEs—generally fewer than ten fragments are formed, and frequently less than five.

Initial Fragment Velocity (V_f) The process of energy transfer from the expanding gas to the vessel fragments is not efficient and seldom exceeds 40 percent of the available energy. According to Baum ("The Velocity of Missiles Generated by the Disintegration of Gas Pressurized Vessels and Pipes," *Journal of Press. Vessel Technology*, Trans. ASME, vol. 106, November 1984, pp. 362–368), there is an upper limit to the fragment velocity, which is taken to be the velocity of the contact surface between the expanding high-pressure gas and the surrounding atmospheric air. This is referred to as the zero-mass fragment velocity and, for most industrial low- to medium-pressure vessels, is less than about 1.3 Mach. It is calculated using ideal gas, one-dimensional shock tube theory and is given by the equation for the shock tube contact surface velocity (Wright, *Shock Tubes*, Methuen & Co., London, 1961).

$$\frac{V_f}{a_0} = -\frac{2}{(\gamma-1)} \left[\left(\frac{P_c}{P_e} \right)^{(\gamma-1)/2\gamma} - 1 \right] \quad (23-81)$$

where P_c is determined from the relationship:

$$\frac{a_a(1-\mu_a) \left(\frac{P_c}{P_a} - 1 \right)}{a_e \left[(1+\mu_a) \left(\frac{P_c}{P_a} + \mu_a \right) \right]^{1/2}} = \frac{2}{(\gamma-1)} \left[1 - \left(\frac{P_a}{P_e} \cdot \frac{P_c}{P_a} \right)^{(\gamma-1)/2\gamma} \right] \quad (23-82)$$

$$\text{and} \quad \mu_a = \frac{\gamma_a - 1}{\gamma_a + 1} \quad (23-83)$$

where P_c = pressure at expanding gas contact surface
 a_a = sound velocity at ambient conditions
 a_e = sound velocity in gas prior to vessel failure

The value of a_e may be approximated using physical property data for the specific gas at the temperature and pressure at the start of the expansion. Equation (23-82) must be solved using a trial-and-error method. Most fragments never achieve the zero-mass velocity and their velocity can be assessed using the correlations of Baum ("Disruptive Failure of Pressure Vessels: Preliminary Design Guidelines for Fragments Velocity and the Extent of the Hazard Zone," *J. Pressure Vessel Technology*, Trans. ASME, vol. 110, May 1988, pp. 168–176; Baum, "Rupture of a Gas-Pressurized Cylindrical Pressure Vessel. The Velocity of Rocketing Fragments," *J. Loss Prev. Process Ind.*, vol. 4, January 1991, pp. 73–86; Baum, "Velocity of a Single Small Missile Ejected from a Vessel Containing High Pressure Gas," *J. Loss Prev. Process Ind.*, vol. 6, no. 4, 1993, pp. 251–264).

Baum provides correlations for several vessel failure modes.

Vessel Filled with Reactive Gas Mixtures Most cases of damage arise not from the vessel failing at its normal operating pressure but because of an unexpected exothermic reaction occurring within the vessel. This usually is a decomposition, polymerization, deflagration, runaway reaction, or oxidation reaction. In assessing the damage potential of such incidents, the peak explosion or reaction pressure can often be calculated, and if this peak pressure P_e is then inserted into Eq. (23-80), the available energy can be assessed and the blast and fragment hazard determined. Where the expected peak explosion pressure P_e is greatly in excess of the vessel dynamic burst pressure, it is sufficient to increase the burst pressure to allow for the increase in vessel pressure during the period necessary for both the vessel to rupture and the fragments to be removed from the path of the expanding vessel contents. Where the gas pressure in the vessel is rising

rapidly, the gas may reach a much higher pressure than the estimated dynamic burst pressure of the vessel. This effect is similar to the accumulation on a relief valve. It is, therefore, conservative to assume that the gas reaches the pressure calculated on the assumption of complete reaction. The reaction is assumed to go to completion before the containing vessel fails. However, there are reactions where it is simpler to calculate the energy availability using thermodynamic methods. The maximum energy released in an explosion can be assessed from the change in the Helmholtz free energy ($-\Delta H = -\Delta E + T\Delta S$), but if the required data is not available, it may be necessary to use the Gibbs free energy ($\Delta F = \Delta H - T\Delta S$), which—especially in the case of reactions with little or no molal change, e.g., hydrocarbon/air oxidation—is similar to the Helmholtz energy. It may sometimes be more convenient to calculate the batch energy availability [$B = \phi_0 + \Delta\phi_{1 \rightarrow a} + \Delta(PV)_{1 \rightarrow a} - P_a\Delta V_{1 \rightarrow a}$] (Crowl, "Calculating the Energy of Explosion Using Thermodynamic Availability," *J. Loss Prev. Process Ind.*, **5**, no. 2, 1992, p. 109), which for an ideal gas becomes

$$B_1 = f_0 + f_{1 \rightarrow a} + nRT_1 \left[\left(\frac{P_a}{P_1} \right) - 1 \right] \quad (23-84)$$

The energy partition between blast wave energy and fragment kinetic energy is as described in paragraph 1.

Vessels Completely Filled with an Inert High-Pressure Liquid* A typical example is the pressure testing of vessels with water. The energy available to cause damage is the sum of the liquid compression energy and the strain energy in the vessel shell. The sudden release of this energy on vessel failure generally creates flying fragments but rarely any significant blast effects.

The fluid compression energy up to about 150 MN/m² (22,000 psig) can be estimated from $U_f = \frac{1}{2}\beta_T P^2 V_L$, where β_T is the liquid bulk compressibility, P is the liquid pressure, and V_L is the liquid volume. At higher pressure, this simple equation becomes too conservative and more complex methods of calculating the fluid compression energy are required. The elastic strain energy for cylindrical vessels, ignoring end closures, can be estimated from:

$$U_m = \frac{P^2 V_L}{2E(k^2 - 1)} [3(1 - 2\nu) + 2k^2(1 + \nu)] \quad (23-85)$$

where P = pressure of liquid
 V_L = volume of liquid
 E_y = Young's modulus of elasticity
 ν = Poisson's ratio

Energy available $U = U_f + U_m$.

Only a small fraction of U is available to provide kinetic energy to the fragments. There are few data available, but in five incidents analyzed by High (unpublished data), no fraction was greater than 0.15. The fragment initial velocity can be assessed from $0.15 U = \frac{1}{2}MV_f^2$.

Distance Traveled by Fragments There is no method available to estimate the distance traveled by an irregularly shaped, possibly tumbling, subsonic fragment projected at an unknown angle. A conservative approach is to assume that all the fragments are projected at an angle of 45° to the horizontal and to ignore the aerodynamic effects of drag and/or lift. The range R_g is then given by $R_g = V_f^2/g$, where g = gravitational acceleration.

This is too conservative to provide anything more than an upper bound. Some limited guidance is given by Scilly and Crowther ("Methodology for Predicting Domino Effects from Pressure Vessel Fragmentation," *Proc. Hazards Ident. and Risk Anal., Human Factors and Human Reliability in Process Safety*, Orlando, Fla., 15 Jan 1992, p. 5, sponsored by AIChE and HSE), where the range, for vessels with walls less than 20 mm (0.79 in), is $2.8 P_b$ with the range in meters and P_b as the vessel burst pressure in bars. Other sources are Baker (*Explosion Hazards and Evaluation*, Elsevier, 1983, p. 492) and Chemical

*An excellent review of the necessary precautions to be taken is given by Saville, "Pressure Test Safety," HSE Contract Research Report 168/1998, HMSO, U.K., 1998.

Propulsion Information Agency (*Hazards of Chemical Rockets and Propellants Handbook*, vol. 1, NTIS, Virginia, May 1972, pp. 2-56, 2-60).

Fragment Striking Velocity It is generally impossible to assess the fragment velocity, trajectory, angle of incidence, and fragment attitude at the moment of striking a target; consequently, the conservative view is taken that the fragment strikes the target at right angles, in the attitude to give the greatest penetration, with a velocity equal to the initial velocity.

Damage Potential of Fragments In designing protection for fragment impact, there are two failure modes to be considered: local response and overall response. Local response includes penetration/perforation in the region of the impact. Overall response includes the bending and shear stresses in the total target element; i.e., will the whole target element fail regardless of whether the element is penetrated or perforated?

Local Failure The penetration or perforation of most industrial targets cannot be assessed using theoretical analysis methods, and recourse is made to using one of the many empirical equations. In using the equations, it is essential that the parameters of the empirical equation embrace the conditions of the actual fragment.

The penetrability of a fragment depends on its *kinetic energy density* (KED), given by

$$\text{KED} = \frac{1}{2} \frac{MV_f^2}{A} \quad (23-86)$$

where A is the fragment cross-sectional area. The KED is a useful comparative measure of a fragment's penetrability when comparing like with like. Several equations are given in the following sections.

Ballistics Research Laboratory (BRL) Equation for Steel Targets

$$E_p = 1.4 \times 10^9 (dt)^{1.5} \quad (23-87)$$

where d is the fragment diameter, t is the steel target plate thickness, and E_p is the critical perforation energy in SI units (kg, m, m/s, J), when applied to fragments between 1 kg and 19.8 kg, impacting targets 1 mm to 25 mm (1 in) thick plate with velocities from 10 m/s to 100 m/s. Neilson (*Procedures for the Design of Impact Protection of Off-shore Risers and ESV's*, U.K. AEA [ed.], 1990) found a large scatter in the results, but most were within ± 30 percent.

Stanford Research Institute (SRI) Equation for Steel Targets

$$E = \frac{dut^2}{10.3} \left(42.7 + \frac{w}{t} \right) \quad (23-88)$$

where w is the unsupported span of the target plate (m) and u the ultimate tensile strength of the target steel (N/m²). The parameters for this equation are given by Brown ("Energy Release Protection for Pressurized Systems," part II, "Review of Studies into Impact/Terminal Ballistics," *Applied Mechanics Review*, vol. 39, no. 2, 1986, pp. 177-201) as $0.05 \leq d \leq 0.25$ m, $414 \leq u \leq 482$ MN/m² for a fragment mass between 4.5 and 50 kg.

Overall Response The transition from local to overall response is difficult to define. High-velocity impact implies that the boundary conditions of the target have little influence on the local response (excluding reflected shock waves). If the fragment is small relative to the target, local response will dominate, but fragments that are of the same order of size as the target will produce an overall response. It is often necessary to consider both overall and local response. Low values of KED are associated with overall response. Design methods for dynamically applied loads are given by Newark ("An Engineering Approach to Blast Resistant Design," ASCE New York, 1953), Baker (see General References), or ASCE (*Manual and Reports on Engineering Practice*, no. 58, 1980).

Response to Blast Waves The effect of blast waves upon equipment and people is difficult to assess because there is no single blast wave parameter which can fully describe the damage potential of the blast. Some targets respond more strongly to the peak incident overpressure and others to the impulse ($\int p dt$) of the blast. The blast parameters are usually based on the conservative assumption that the

TABLE 23-25 Damage Effects

Incident, psi	Pressure, kPa	Damage effects
10	70	Damage to most refineries would be severe, although some pumps, compressors, and heat exchangers could be salvaged. All conventional brick buildings would be totally destroyed. Rail wagons (rail cars) overturned. Storage tanks ruptured. Fatalities certain.
5.0	34	Brick buildings severely damaged, 75% external wall collapse. Fired heaters badly damaged. Storage tanks leak from base. Threshold for eardrum damage to people. Domino or knock-on radius. Pipe bridges may move.
2.0	14	Doors and windows removed. Some frame distortion to steel frame buildings and cladding removed. Some electrical/instrument cables broken.*
1.0	7	Lethal glass fragments. Limit for public housing, schools, etc.
0.3	2	About 50% domestic glass broken.

*1% probability electrical cables broken at 2.0 psi inc. 99% probability electrical cables broken at 3.6 psi inc.

blast strikes the target normal to its surface, so that normal reflection parameters are used.

The pressure exerted by the blast wave on the target depends upon the orientation of the target. If the target surface faces the blast, then the target will experience the reflected or face-on blast pressure P_r , but if the target surface is side-on to the blast, then the target will experience the incident or side-on blast pressure P_{inc} . The reflected blast pressure is never less than double the incident pressure and can, for ideal gases, be as high as eight times the incident pressure. For most industrial targets where the incident pressure is less than about 17 kN/m² (25 psi), the reflected pressure is not more than 2.5 times the incident pressure.

Response of Equipment The response of equipment to blast is usually a combination of two effects: one is the displacement of the equipment as a single entity and the other is the failure of the equipment itself. The displacement of the equipment is an important consideration for small, unsecured items—e.g., empty drums, gas cylinders, empty containers. Most damage results from the failure in part or totally of the equipment or containing structure itself.

The blast parameters are usually based on the conservative assumption that the blast strikes the target normal to its surface, so that normal reflection parameters may be used.

The response of a target is a function of the ratio of the blast wave duration and the natural period of vibration of the target (T/T_n). Neither of these parameters can be closely defined.

Calculating the specific response of a specific target can generally be done only approximately. Accuracy is not justified when the blast properties are not well defined. A guide to the damage potential of condensed phase explosive blast is given in Table 23-25 (Scilly and High, "The Blast Effect of Explosions," *Loss Prevention and Safety Promotion in the Process Industries*, European Fed. of Chem. Eng., 337 Event, France, September 1986, table 2). Nuclear data is available (Table 23-26) (Walker, "Estimating Production and Repair Effort in Blast-damaged Petroleum Refineries," *Stanford Research Inst.*, July 1969, fig. 5, p. 45), which is based upon long positive-duration blast (± 6 s). This suggests that the Walker data will be conservative for the much shorter duration blast from accidental industrial explosions.

A blast incident overpressure of 35 kN/m² (5 psi) is often used to define the region beyond which the damage caused will be minor and not lead to significant involvement of plant and equipment beyond the 35 kN/m² boundary.

Response of People The greatest hazard to people from blast is generally from the deceleration mechanism after people have been blown off their feet and they become missiles. This occurs at an incident overpressure of about 27 kN/m² (4.0 psi) for long positive-duration

TABLE 23-26 Blast Overpressure Effects on Vulnerable Refinery Parts

Equipment	Overpressure (psi)																										
	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0	6.5	7.0	7.5	8.0	8.5	9.0	9.5	10.0	12.0	14.0	16.0	18.0	20.0	>20.0	
Control house: steel roof	<i>a</i>	<i>c</i>	<i>d</i>				<i>h</i>																				
Control house: concrete roof	<i>a</i>	<i>e p</i>	<i>d</i>				<i>n</i>													<i>e</i>							
Cooling tower	<i>b</i>		<i>f</i>				<i>o</i>																				
Tank: cone roof		<i>d</i>					<i>k</i>							<i>u</i>													
Instrument cubicle			<i>a</i>				<i>i m</i>						<i>t</i>														
Fired heater				<i>g</i>	<i>i</i>					<i>t</i>																	
Reactor: chemical				<i>a</i>				<i>i</i>					<i>p</i>						<i>t</i>								
Filter				<i>h</i>					<i>i</i>										<i>v</i>		<i>t</i>						
Regenerator						<i>i</i>				<i>i p</i>					<i>t</i>												
Tank: floating roof						<i>k</i>							<i>u</i>												<i>d</i>		
Reactor: cracking							<i>i</i>							<i>i</i>							<i>t</i>						
Pipe supports							<i>p</i>					<i>s o</i>															
Utilities: gas meter								<i>q</i>																			
Utilities: electric transformer								<i>h</i>							<i>l</i>					<i>t</i>							
Electric motor									<i>h</i>									<i>l</i>									<i>v</i>
Blower									<i>q</i>											<i>t</i>							
Fractionation column										<i>r</i>			<i>t</i>														
Pressure vessel: horizontal											<i>p i</i>							<i>t</i>									
Utilities: gas regulator											<i>i</i>									<i>m q</i>							
Extraction column												<i>i</i>								<i>v</i>	<i>t</i>						
Steam turbine															<i>l</i>						<i>m</i>	<i>s</i>					<i>v</i>
Heat exchanger															<i>i</i>			<i>t</i>									
Tank: sphere																<i>i</i>						<i>i</i>	<i>t</i>				
Pressure vessel: vertical																					<i>i</i>	<i>t</i>					
Pump																					<i>i</i>			<i>v</i>			

CODE:

<i>a</i>	Windows and gauges break.	<i>h</i>	Debris-missile damage occurs.	<i>p</i>	Frame deforms.
<i>b</i>	Louvers fall at 0.3–0.5 psi.	<i>i</i>	Unit moves and pipes break.	<i>q</i>	Case is damaged.
<i>c</i>	Switchgear is damaged from roof collapse.	<i>j</i>	Bracing fails.	<i>r</i>	Frame cracks.
<i>d</i>	Roof collapses.	<i>k</i>	Unit uplifts (half-filled).	<i>s</i>	Piping breaks.
<i>e</i>	Instruments are damaged.	<i>l</i>	Power lines are severed.	<i>t</i>	Unit overturns or is destroyed.
<i>f</i>	Inner parts are damaged.	<i>m</i>	Controls are damaged.	<i>u</i>	Unit uplifts (0.9 filled).
<i>g</i>	Brick cracks.	<i>n</i>	Block walls fail.	<i>v</i>	Unit moves on foundation.
		<i>o</i>	Frame collapses.		

SOURCE: F. E. Walker, "Estimating Production and Repair Effort in Blast Damaged Petroleum Refineries," SRI, July 1969.

nuclear weapon blasts. People have more blast resistance than most equipment and can survive incident overpressures of 180 kN/m² (27 psi) (Bowen, Fletcher, and Richmond, DASA-2113, Washington, D.C., October 1968), even for long-duration blasts.

PROJECT REVIEW AND AUDIT PROCESSES

GENERAL REFERENCES: Center for Chemical Process Safety (CCPS), *Guidelines for Hazard Evaluation Procedures, Second Edition with Worked Examples*, AIChE, September 1992. CCPS, *Guidelines for Technical Management of Chemical Process Safety*, AIChE, 1989. CCPS, *Guidelines for Auditing Process Safety Management Systems*, AIChE, 1993.

Introduction Review and audit processes are used in the chemical process industry to evaluate, examine, and verify the design of process equipment, operating procedures, and management systems. These processes assure compliance with company standards and guidelines as well as government regulations. Reviews and audits can encompass the areas of process and personnel safety, environmental and industrial hygiene protection, quality assurance, maintenance procedures, and so on.

A review is a critical examination or evaluation of any operation, procedure, condition, event, or equipment item. Reviews can take many forms and be identified as project reviews, design reviews, safety reviews, pre-start-up reviews, and so on. The following discussion of the review process will deal with project reviews associated with capital projects and focus on the area of process safety.

An audit is a formal, methodical examination and verification of an operation, procedure, condition, event, or series of transactions. The verification element of an audit makes it distinctive from a review. A project review will *recommend* design, procedural, maintenance, and management practices to minimize hazards and reduce risk while meeting company standards and government regulations. An audit will *verify* that the design, the procedures, and the management systems are actually in place, and are being maintained and used as intended. In fact, it is not uncommon for an audit to be done on a review process, to verify that the elements of the review process are being followed.

Project Review Process The scope of capital projects can be large, involving the construction of new plants with new technologies and products, or small, involving minor changes to existing facilities. In either case, project safety reviews can be used to evaluate and examine the process design, operating procedures, and process control scheme for process hazards, conformance to company standards and guidelines, and compliance with government regulations. Some objectives of the review process (CCPS, 1992, p. 53) are: (1) identify equipment or process changes that could introduce hazards, (2) eval-

uate the design basis of control and safety systems, (3) evaluate operating procedures for necessary revisions, (4) evaluate the application of new technology and any subsequent hazards, (5) review the adequacy of maintenance and safety inspections, and (6) evaluate the consequences of process deviations and determine if they are acceptable (CCPS, 1989, p. 46).

The project review process should be integrated with the development of the project from the conceptual stage to the start-up stage (CCPS, 1989, p. 46). Figure 23-46 depicts the various stages of a capital project. The size and complexity of a project will determine if the project progresses through all these stages and, in the same manner, determine the number and type of reviews that are needed. The earlier in a project that a review can be used to identify required changes, the less costly the change will be to implement.

As the project progresses, more information is available; therefore, the review technique used can be different at each stage of the project. The use of various hazard evaluation techniques, such as checklist analyses, relative rankings, what-if analyses, and hazard and operability studies, is documented in *Guidelines for Hazard Evaluation Procedures: Second Edition with Worked Examples* (CCPS, 1992). The need to use more quantitative techniques for hazard evaluation may be identified during these reviews, and become an action item for the project team.

The project review process involves multiple steps that should be defined in management guidelines (CCPS, 1993, pp. 57-61). The steps include: (1) review policy, (2) review scheduling, (3) review technique, (4) review team representation, (5) review documentation, (6) review follow-up, (7) review follow-up verification, and (8) review procedures change management. These steps define how a review, whether it be a safety review, environmental review, pre-start-up review, or whatever, is conducted and how closure of review action items is achieved.

Review Policy The review policy should establish when project safety reviews should be done. All capital projects, large or small, should have one or more safety reviews during the course of the project. The number and types of review should be stated in a management policy. Any reasons for exceptions to the policy should be documented as well. The policy should address not only projects internal to a company, but also any joint ventures or turnkey projects by outside firms.

Review Scheduling A review scheduling procedure should be established that documents who is responsible for initiating the review and when the review(s) should occur during the project. The scheduling needs to balance availability of process information, review technique used, and the impact of potential review action items on project costs (i.e., early enough to minimize the cost of any potential changes

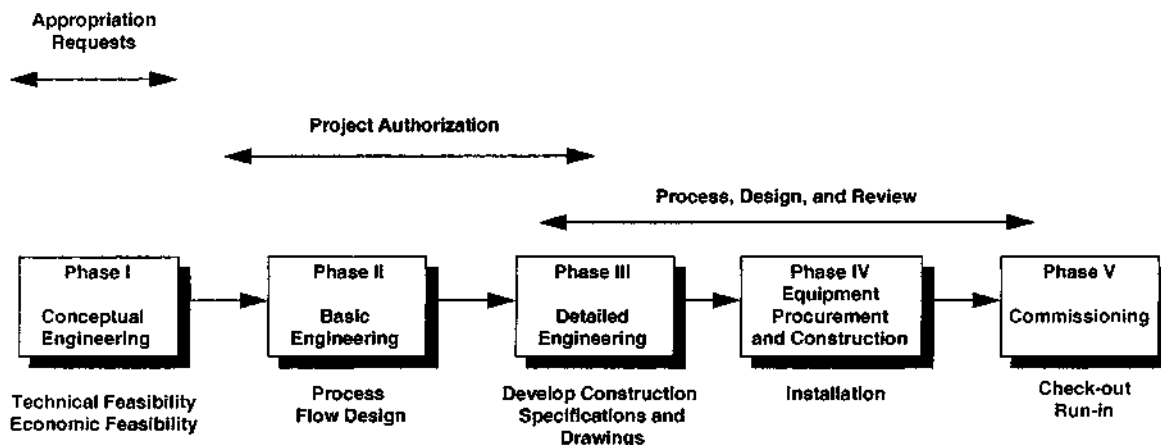


FIG. 23-46 The phases of a capital project. (CCPS, 1989, by permission of AIChE.)

to the process). The actual amount of time needed for the review should also be stated in the procedure. On the basis of the number of project reviews required and the estimated time needed for each review, the project cost estimate should include the cost for project reviews as part of the total cost for the project.

Review Team The project review should be conducted by a functionally diverse team. The team should consist of a team leader to organize and lead the team review, a scribe or secretary to record and issue a review summary with action items, and functional experts in fields relevant to the project such as safety, environmental, and industrial hygiene (CCPS, 1993, p. 58). The team leader should be experienced in the use of the selected review technique with leadership skills and no direct involvement with the project under review. The review procedure should address the minimum requirements for team leaders and team members. Some typical requirements could be years of experience, educational background, and training in the review technique. Responsibilities should be clearly defined for initiating the review, assigning the review team, recording the team findings, and monitoring follow-up of team recommendations.

Review Techniques The review techniques used at the various stages of a project should be selected based on the amount of process information and detail available. Figure 23-47 depicts some typical review techniques at the various stages of a capital project. A detailed description, including the type and amount of process information required, for each review technique can be found in *Guidelines for Hazard Evaluation Procedures: Second Edition with Worked Examples* (CCPS, 1992). The process information required for the review should be defined and documented in the review guidelines. Up-to-date and accurate process information is essential to conducting a successful review.

Review Documentation The project review team leader has the ultimate responsibility for documenting the results of the project

review. This responsibility may be delegated to a team scribe or secretary to record the review minutes and issue a summary report with listed action items. The action items could address exceptions to company or industry standards and government regulations, review team recommendations based on experience and knowledge, and further issues for study that could not be resolved during the review session.

The summary report should have a standard format and could contain a short project scope summary, a listing of review team members by function, a listing of project team members present, a meeting agenda or checklist of topics reviewed by the team, and a list of concerns and action items for project team follow-up. The distribution list for the summary report should be established and include the review team, project team, and any personnel outside the project team who have follow-up responsibilities for any of the action items. Also, include on the distribution list any appropriate management personnel, whether they be project team supervisors, manufacturing managers, or engineering managers. The documentation for the review should be archived in a process plant file with the appropriate records retention time (e.g., the life of the plant).

Review Follow-up An important element (maybe the most important) of the review process is the follow-up to action items. The project review will result in a list of potential concerns and action items, but, without follow-up, the issues will never be resolved and implemented. A person(s) should be assigned to each action item, preferably at the time of the project review. The person(s) assigned should have a combination of knowledge, resources, and authority to do a proper job in following up on the action items (CCPS, 1993, p. 59). The total action item list should not be assigned to one person, since it may overwhelm one individual. Depending on the number of action items generated, prioritizing the action item list may be helpful and a responsibility the review team can assume.

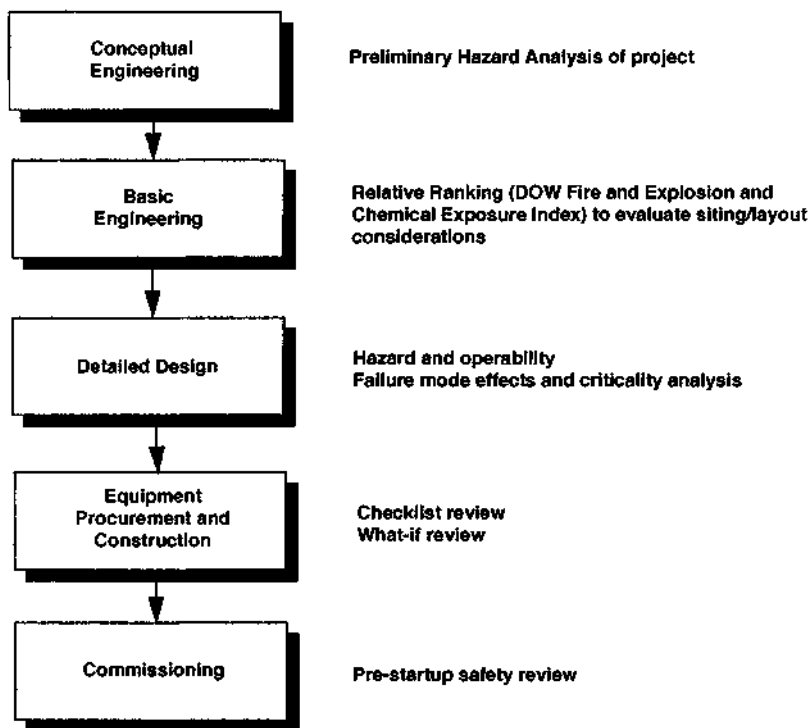


FIG. 23-47 Hazard evaluation at various project stages. (CCPS, 1993, by permission of AIChE.)

Progress on the action items should be documented in periodic progress reports to the review team leader or others assigned that functional responsibility. If no one is assigned the responsibility of tracking this progress, completion of the action list will probably be relegated to a lower priority and not be done.

Changes made to the process as a result of the project review may require a similar review before implementation, especially if the change is significant.

Review Follow-up Verification Responsibility should be assigned to verify that any process changes were actually made in the field. This verification can be done by a review team as part of a process pre-start-up review. It could also be part of the project team management responsibility or assigned to a particular functional (i.e., safety and loss prevention) representative. The closure of the review process is complete once implementation is verified.

On rare occasions, the resolution of project review concerns or action items is a point of contention between review team and project team members. In such a case, a management structure must be in place to arbitrate such disputes.

Review Procedure Change Management The project review process can require changes in policy and procedures at certain times. Therefore, the procedures should provide a management-of-change mechanism for suggesting changes and assign a person responsible for initiating and implementing any necessary changes.

Audit Process Audits in the chemical process industry can be focused on process safety, process safety management, environmental, and health areas. The discussion in this section will focus on the process safety and process safety management area, but it should be recognized that the process can be applied to the other areas as well. "Process safety audits are intended to provide management with increased assurance that operating facilities and process units have

been designed, constructed, operated, and maintained such that the safety and health of employees, customers, communities, and the environment are being properly protected" (CCPS, 1989, p. 133). Process safety management system audits "provide increased assurance that operating units have appropriate systems in place to manage process risk" (CCPS, 1989, p. 130).

The key steps in the audit process are outlined according to pre-audit activities, audit activities, and postaudit activities in Fig. 23-48. These activities are described in detail in *Guidelines for Auditing Process Safety Management Systems* (CCSP, 1993) and will be only briefly discussed in this section.

Preaudit Process Prior to the actual on-site audit, some preliminary activities should take place. These activities include selecting the facilities to be audited, scheduling the audit, selecting the audit team, and planning the audit. The selection criteria may be random, based on potential hazards of the facilities or the value of the facilities from a business standpoint. Audit scheduling must account for the availability of key facility personnel and audit team members, operational mode of the facility (i.e., it should be in normal operation), and the lead time required to obtain background information that may require advance visits to the facility and preaudit interviews. The audit team members should possess the technical training and experience to understand the facilities being audited. They should be knowledgeable in the auditing process and in the appropriate regulations and standards that will apply to the facilities. They should also be impartial and objective about audit findings. The audit plan should define the audit scope (what parts of the facility will be covered, what topics, who will do it, etc.), develop an audit protocol that is a step-by-step guide to how the audit is performed, identify any priority topics for coverage, and develop an employee interview schedule.

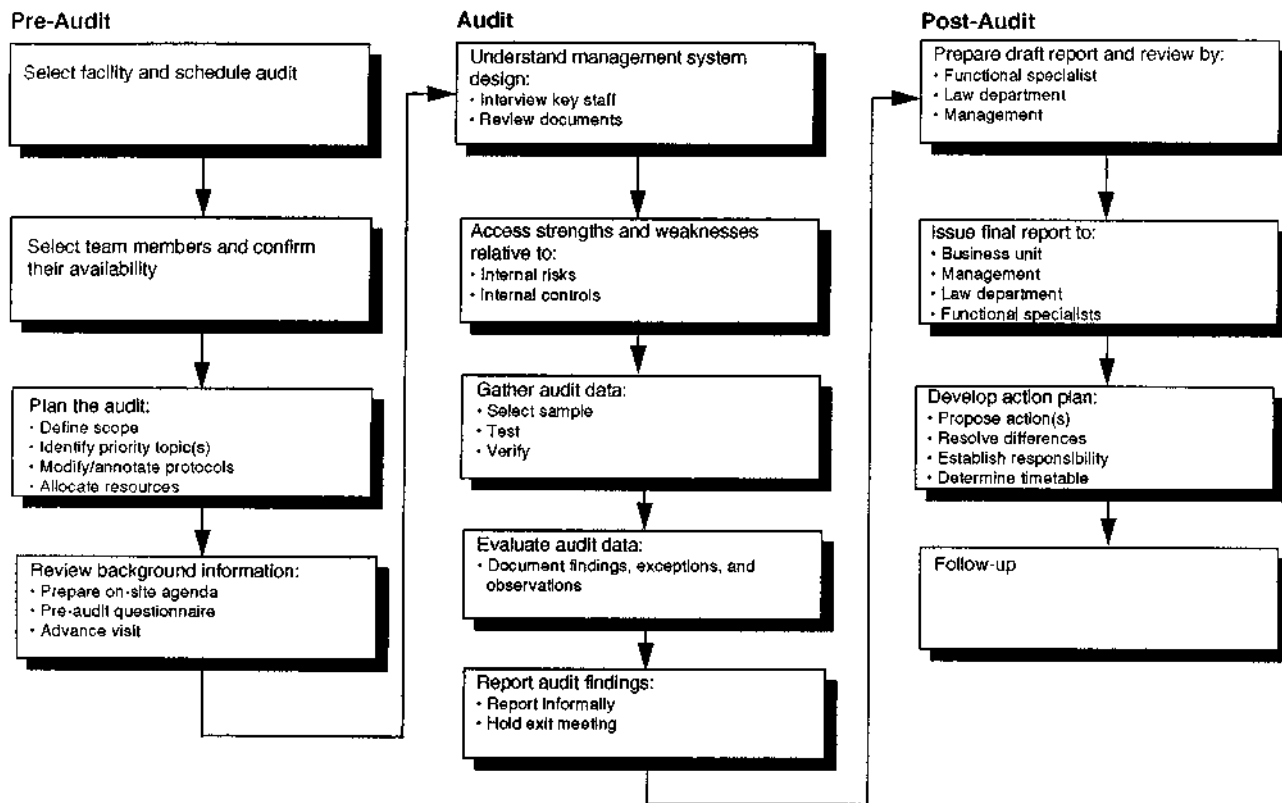


FIG. 23-48 Typical steps in the process safety management audit process. (CCSP, 1993, by permission of AIChE.)

On-Site Audit Process An opening meeting with key facility personnel is held at which the audit team covers the objectives and approach for the audit, and the facility personnel provide an overview of the site operations including site safety rules and a site tour. The on-site audit process should then follow five basic steps that include: (1) understanding management systems, (2) evaluating management systems, (3) gathering audit information, (4) evaluating audit information, and (5) reporting audit findings (CCSP, 1993, p. 17).

An understanding of the management systems in place to control and direct the process safety of the facility can be obtained from reading engineering and administrative standards, guidelines, and procedures that should be available in the background information supplied prior to the on-site audit. Informal procedures and guidelines used by the facility may only be discovered in interviews with staff management and operations management. This understanding of the formal and informal management systems is a critical step in the audit process.

The next process step evaluates the process safety management systems to determine if they are adequate to achieve the desired results, and if they are used as intended. This evaluation is highly subjective on the auditor's part. This step sets the stage for the rest of the audit, guiding the auditor's information gathering and focusing attention on critical areas.

Gathering audit data can be accomplished through observations, documents, and interviews. The data obtained is used to verify and validate that the process safety management systems are implemented and functioning as designed. Data gathering can be aided by the use of audit samples, where a representative number of items are audited to draw a conclusion, and by using self-evaluation questionnaires.

The audit data can now be evaluated, resulting in audit findings (i.e., conclusions both positive and negative). The audit team should confirm that sufficient data has been collected to support each finding. Additional data may need to be gathered if the team decides a preliminary finding needs to be strengthened. The conclusions drawn from the data evaluation should be a team consensus.

The reporting step of the on-site audit should be planned to avoid any surprises to facility personnel. Reporting sessions should be held at the end of each audit day to inform facility personnel of the findings, clear up any misunderstandings of the data, and help redirect

the audit team, if necessary. The on-site audit should end with a well-planned exit or closeout meeting between the audit team and facility personnel. All the findings of the audit team should be presented at this meeting. This verbal report is the opportunity for clarification of any ambiguities and determination of the final disposition of the findings (written audit report, for local attention only, etc.).

Postaudit Process The postaudit process consists of preparation of a draft report, preparation of a final report, development of action plans, and follow-up. A draft report of the audit findings should be prepared shortly after the completion of the on-site audit. The draft report usually undergoes review and comment by facility personnel involved with the audit, experienced auditors not involved with the subject audit, functional specialists, and attorneys. The review of the draft report is done to assure that a clear, concise, and accurate report is issued, and not to modify or change the findings. Once this review procedure is completed, a final report can be issued and distributed based on a distribution list provided by the facility personnel. The final audit report should be issued in a timely manner and meet the time requirement specified in the audit plan.

An action plan should be developed by the appropriate personnel of the audited facility to address any deficiencies stated in the audit report. Action plans should state what is to be done, who is responsible for getting it done, and when it is to be completed. Rationale for not taking any action for any of the stated deficiencies should also be documented. The action plan is an important step in closing the audit process.

It would not be unusual for some action plans to take a long time to complete. When extended implementation time is necessary, a follow-up mechanism should be used to document progress and show that an effort is being made to resolve the issues. Periodic (i.e., quarterly, semiannually) progress reports should be used as a follow-up method to ensure implementation. Future audits of the facility should include confirmation of the implementation of previous audit action plans.

The final audit report, action plans, progress reports, and any closure report should be retained by the facility based on the facility record retention policy. Typically, these items will be retained until future audit documentation replaces them. In some cases, audit records are retained for the life of the plant.

SAFETY EQUIPMENT, PROCESS DESIGN, AND OPERATION

PRESSURE RELIEF SYSTEMS

GENERAL REFERENCES: Center for Chemical Process Safety (CCPS), *Guidelines for Pressure Relief and Effluent Handling Systems*, American Institute of Chemical Engineers, New York, 1998. DIERS Project Manual, *Emergency Relief System Design Using DIERS Technology*, American Institute of Chemical Engineers, New York, 1992. Leung, "Simplified Vent Sizing Equations for Emergency Relief Requirements in Reactors and Storage Vessels," *AIChE J.* 32(10), pp. 1622-1634 (1986a). Leung, "A Generalized Correlation for One-Component Homogeneous Equilibrium Flashing Choked Flow," *AIChE J.* 32(10), pp. 1743-1746 (1986b). Leung, "Easily Size Relief Devices and Piping for Two-Phase Flow," *Chem. Eng. Prog.* 92(12), pp. 28-50 (1996). See more references under Code, Standards, and Guidelines subsection.

Introduction All process designs should attempt to arrive at an inherently safe facility. Incorporating safety features that are intrinsic (built-in) rather than extrinsic (added-on) to the basic design, together with the use of high-integrity equipment and piping, provides the first lines of defense against the dramatic, often catastrophic effects of an overpressure and subsequent rupture. In recent years, many companies have incorporated the principles of depressurizing or instrumental shutdown of key equipment as a means to control a release and avoid the actuation of pressure relief devices. This minimizes the probability of failure of the device, because, once used, the device may no longer be dependable. Since maintenance of relief devices can be sporadic, this redundancy provides yet another layer of safety. However, regardless of the number of lines of defense and depressur-

izing systems in place, overpressure protection must still be provided. Emergency pressure relief systems are intended to provide the last line of protection and thus must be designed for high reliability, even though they will have to function infrequently.

Self-actuated pressure relief systems must be designed to limit the pressure rise that can occur as a result of overcompressing, overfilling, or overheating either an inert or a chemically reactive medium in a closed system. Pressure generation is usually the result of either expansion of a single-phase medium (by material addition and/or heating) or a shift of the phase equilibrium in a multiphase medium (as a result of composition and/or temperature changes, particularly in the case of a reactive system). These mechanisms of pressure generation differ from what is commonly referred to as explosion venting. Events such as dust explosions and flammable vapor deflagrations propagate nonuniformly from a point of initiation, generating pressure or shock waves. Such venting problems are not included in these discussions.

Relief System Terminology Refer to API-RP520 Part I for complete terminology.

Accumulation The rise of pressure above the MAWP of the protected system, usually expressed as a percentage of the gauge MAWP. Maximum allowable accumulations are established by applicable codes for emergency operating and fire contingencies.

Backpressure The pressure existing at the outlet of a relief device. The value under no-flow conditions is superimposed backpressure. The value under flowing conditions consists of both

superimposed backpressure and built-up pressure due to piping pressure drop.

Blowdown The reduction in flowing pressure below the set point required for a PRV to close.

Design pressure The design pressure used to determine the minimum thickness of a vessel component and possibly used in place of MAWP where the latter has not been established. It is equal to or less than the MAWP. It is the pressure specified on the equipment purchase order.

Maximum allowable working pressure (MAWP) The maximum allowed pressure at the top of the vessel in its normal operating position at the operating temperature specified for that pressure.

Overpressure A pressure increase above the set point during relief flow, usually expressed as a percentage of the differential set pressure.

Pressure relief valve (PRV) A pressure relief device designed to open and relieve excess pressure and to reclose after normal conditions have been restored. *PRV* is a generic term applied to *relief valve* (set up for liquid flow), *safety valve* (set up for gas or vapor flow), and *safety relief valve* (set up for either liquid or compressible flow).

Relieving pressure Set pressure plus the overpressure.

Rupture disk A non-reclosing pressure relief device actuated by static differential pressure and designed to function by the bursting of a pressure-containing disk.

Set pressure The inlet gauge pressure at which a PRV will start to open (or a rupture disk will burst) under service conditions of temperature and backpressure.

Codes, Standards, and Guidelines Industry practice is to conform to the applicable regulations, codes, and recommended practices. In many cases, these will provide different guidelines. A suggested approach would be to review all applicable codes, standards, and recommended practices prior to choosing a design basis. The Design Institute for Emergency Relief Systems (DIERS) was established by AIChE to address sizing aspects of relief system for two-phase, vapor-liquid flashing flow regimes. The DIERS Project Manual (*Emergency Relief System Design Using DIERS Technology*, 1982) and the CCPS Guidelines (*Guidelines for Pressure Relief and Effluent Handling Systems*, 1998) are the generally accepted industry standard for two-phase relief venting.

NFPA 30 and API Standard 2000 provide guidance for design of overpressure protection involving storage tanks that operate at or near atmospheric pressure. In particular, NFPA 30 focuses on flammability issues, while API 2000 addresses both pressure and vacuum requirements. The ASME code (Sections I and VIII) and API RP 520 are the primary references for pressure relief device sizing requirements.

Designers of emergency pressure relief systems should be familiar with the following list of regulations, codes of practice, and industry standards and guidelines in the United States.

API RP 520. *Sizing, Selection, and Installation of Pressure-Relieving Devices in Refineries*. Part I, *Sizing and Selection*, 7th ed., January 2000, and Part II, *Installation*, 4th ed., December 1994. American Petroleum Institute, Washington.

API RP 521, 1997. *Guide for Pressure-Relieving and Depressuring Systems*, 4th ed. American Petroleum Institute, Washington.

API STD 526, 1995. *Flanged Steel Pressure Relief Valves*, 4th ed. American Petroleum Institute, Washington.

API STD 2000, 1998. *Venting Atmospheric and Low-Pressure Storage Tanks, Nonrefrigerated and Refrigerated*, 5th ed. American Petroleum Institute, Washington.

API RP 2001, 1984. *Fire Protection in Refineries*. American Petroleum Institute, Washington.

ASME, 2001. *Boiler and Pressure Vessel Code*, Section I, *Power Boilers*, and Section VIII, *Pressure Vessels*. American Society of Mechanical Engineers, New York.

ASME, 1988. *Performance Test Code PTC-25, Safety and Relief Valves*. American Society of Mechanical Engineers, New York.

CCPS, 1993. *Engineering Design for Process Safety*. American Institute of Chemical Engineers, New York.

CCPS, 1998. *Guidelines for Pressure Relief and Effluent Handling Systems*, American Institute of Chemical Engineers, New York.

DIERS, 1992. *Emergency Relief System Design Using DIERS Technology*. DIERS Project Manual. American Institute of Chemical Engineers, New York.

National Board of Boiler and Pressure Vessel Inspectors, 2004. *Pressure Relieving Device Certifications (Red Book NB-18)*.

National Board of Boiler and Pressure Vessel Inspectors, Columbus, Ohio.

NFPA 30, 2000. *Flammable and Combustible Liquids Code*. National Fire Protection Association, Quincy, Mass.

OSHA 1910.106, 2005. *Flammable and Combustible Liquids, Regulations (Standards—29 CFR)*. U.S. Dept. of Labor, Occupational Safety and Health Administration, Washington.

Relief Design Scenarios The most difficult part of designing an adequate emergency pressure relief system lies in determining the emergency events (credible design scenarios) for which to design. The difficulty arises primarily because the identification of credible design scenarios usually involves highly subjective judgments, which are often influenced by economic situations. Unfortunately, there exists no universally accepted list of credible design scenarios. Relief systems must be designed for the credible chain of events that results in the most severe venting requirements (worst credible scenario). Credibility is judged primarily by the number and the time frame of causative failures required to generate the postulated emergency. Only totally independent equipment or human failures should be considered when judging credibility. A failure resulting from another failure is an effect, rather than an independent causative factor. A suggested guideline for assessing credibility as a function of the number and time frames of independent causative events is as follows:

- Any single failure is *credible*.
- Two or more simultaneous failures are *not credible*.
- Two events in sequence are *credible*.
- Three or more events in sequence are *not credible*.

The first step in scenario selection is to identify all the credible emergencies by using the preceding guidelines (or a similar set). This is perhaps best accomplished by identifying all the possible sources of pressure and vacuum. Table 23-27 lists a number of commonly existing pressure and vacuum sources.

Fire The main consequence of fire exposure is heat input causing thermal expansion, vaporization, or thermally induced runaway reaction and decomposition resulting in a pressure rise. An additional result of fire exposure is the possibility of overheating the wall of the equipment in the vapor space where the wall is not cooled by the liquid. In this case, the vessel wall may fail due to the high temperature, even though the relief system is operating. Hence API RP-521 recommends vapor depressurizing facilities for high-pressure services (greater than 17 bar or 250 psig). Guidelines for estimating the heat input from a fire are found in API Recommended Practices, NFPA 30 (for bulk storage tanks), OSHA 1910.106, and corporate engineering standards. In determining the heat input from fire exposure, NFPA allows credit for application of water spray to a vessel; API allows no such credit.

Pressure vessels (including heat exchangers and air coolers) in a plant handling flammable fluids are subject to potential exposure to external fire. A vessel or group of vessels which could be exposed to a

TABLE 23-27 Common Sources of Pressure and Vacuum

Heat Related
• Fire
• Out-of-control heaters and coolers
• Ambient temperature changes
• Runaway chemical reactions
Equipment and Systems
• Pumps and compressors
• Heaters and coolers
• Vaporizers and condensers
• Vent manifold interconnections
• Utility headers (steam, air, water, etc.)
Physical Changes
• Gas absorption (e.g., HCl in water)
• Thermal expansion
• Vapor condensation

pool fire must be protected by pressure relief device(s). Additional protection to reduce the device relief load can be provided by insulation, water spray, drainage, or remote-controlled depressurizing devices. Plant layout should consider spacing requirements, such as those set forth by NFPA, API, Industrial Risk Insurers, or Factory Mutual, and must include accessibility for firefighting personnel and equipment. Several pieces of equipment located adjacent to each other that cannot be isolated by shutoff valves can be protected by a common relief device, providing the interconnecting piping is large enough to handle the required relief load and the relief set pressure is no higher than the minimum MAWP of these pieces of equipment.

Operational Failures A number of scenarios of various operational failures may result in the generation of overpressure conditions:

- **Blocked outlet** Operation or maintenance errors (especially following a plant turnaround) can block the outlet of a liquid or vapor stream from a piece of process equipment, resulting in an overpressure condition.
- **Opening a manual valve** Manual valves which are normally closed to isolate two or more pieces of equipment or process streams can be inadvertently opened, causing the release of a high-pressure stream or resulting in vacuum conditions.
- **Cooling water failure** The loss of cooling water is one of the more commonly encountered causes of overpressurization. Different scenarios should be considered for this event, depending on whether the failure affects a single piece of equipment (or process unit) or is plantwide.
- **Power failure** The loss of power will shut down all motor-driven rotating equipment, including pumps, compressors, air coolers, and vessel agitators.
- **Instrument air failure** The consequences of the loss of instrument air should be evaluated in conjunction with the failure mode of the control valve actuators. It should not be assumed that the correct air failure response will occur on these control valves, as some valves may stick in their last operating position.
- **Thermal expansion** Equipment and pipelines that are liquid-full under normal operating conditions are subject to hydraulic expansion if the temperature increases. Common sources of heat that can result in high pressures due to thermal expansion include solar radiation, steam or other heated tracing, heating coils, and heat transfer from other pieces of equipment.
- **Vacuum** Vacuum conditions in process equipment can develop due to a wide variety of situations, including instrument malfunction, draining or removing liquid with venting, shutting off purge steam without pressurizing with noncondensable vapors, extreme cold ambient temperatures resulting in subatmospheric vapor pressures, and water addition to vessels that have been steam-purged. If vacuum conditions can develop, then either the equipment must be designed for vacuum conditions or a vacuum relief system must be installed.

Equipment Failure Most equipment failures that can lead to overpressure situations involve the rupture or break of internal tubes inside heat exchangers and other vessels and the failure of valves and regulators. Heat exchangers and other vessels should be protected with a relief system of sufficient capacity to avoid overpressure in case of internal failure. API RP 521 presents guidance in determining these requirements, including criteria for deciding when a full tube rupture is likely. In cases involving the failure of control valves and regulators, it is important to evaluate both the fail-open and fail-closed positions.

Runaway Reactions Runaway temperature and pressure in process vessels can occur as a result of many factors, including loss of cooling, feed or quench failure, excessive feed rates or temperatures, contaminants, catalyst problems, and agitation failure. Of major concern is the high rate of energy release in runaway reactions and/or formation of gaseous products, which generally cause a rapid pressure rise in the equipment. To properly assess these effects, the reaction kinetics must be either known or obtained experimentally (see "Reactivity Testing" in the earlier "Chemical Reactivity" subsection for a description of adiabatic calorimeters for runaway reactions). In general, a lower relief set pressure (much below the equipment MAWP) is desirable for these runaway reaction systems in order to relieve the

TABLE 23-28 Summary of Device Characteristics

	Reclosing devices		Nonreclosing devices
	Relief valves	Disk-valve combinations	Rupture disks
Fluid above normal boiling point	+	+	—
Toxic fluids	+	+	—
Corrosive fluids	—	+	+
Cost	—	—	+
Minimum pipe size	—	—	+
Testing and maintenance	—	—	+
Won't fatigue and fail low	+	+	—
Opens quickly and fully	—	—	+

NOTE: + indicates advantageous
— indicates disadvantageous

system at a lower reaction rate, since most reactions are Arrhenius in behavior.

Pressure Relief Devices The most common method of overpressure protection is through the use of safety relief valves and/or rupture disks which discharge into a containment vessel, a disposal system, or directly to the atmosphere. Table 23-28 summarizes some of the device characteristics and the advantages.

Safety Relief Valves (SRVs) Conventional safety relief valves are used in systems where built-up backpressures typically do not exceed 10 percent of the set pressure. The spring setting of the valve is reduced by the amount of superimposed backpressure expected. Higher built-up backpressures can result in a complete loss of continuous valve relief capacity. The designer must examine the effects of other relieving devices connected to a common header on the performance of each valve. Some mechanical considerations of conventional relief valves are presented in the ASME code; however, the manufacturer should be consulted for specific details.

Balanced safety relief valves may be used in systems where built-up and/or superimposed backpressure is high or variable. In general, the capacity of a balanced valve is not significantly affected by backpressures below 30 percent of the set pressure. Most manufacturers recommend keeping the backpressure on balanced valves below 45 to 50 percent of the set pressure. Consult API-526 and valve manufacturers for the maximum outlet pressure limit for bellows-type SRVs.

For both conventional and balanced SRVs, the inlet pressure loss, including the mounting nozzle entrance loss, rupture disk flow resistance, and inlet pipe friction, is recommended to stay below 3 percent of the differential set pressure, or else valve instability may occur, resulting in degraded relief capacity.

Pilot-Operated Relief Valves In a pilot-operated relief valve, the main valve is combined with and controlled by a smaller, self-actuating pressure relief valve. The pilot is a spring-loaded valve that senses the process pressure and opens the main valve by lowering the pressure on the top of an unbalanced piston, diaphragm, or bellows of the main valve. Once the process pressure is lowered to the blowdown pressure, the pilot closes the main valve by permitting the pressure in the top of the main valve to increase. Pilot-operated relief valves are commonly used in clean, low-pressure services and in services where a large relieving area at high set pressures is required. The set pressure of this type of valve can be close to the operating pressure. Pilot-operated valves are frequently chosen when operating pressures are within 5 percent of set pressures and a close tolerance valve is required.

Rupture Disks A rupture disk is a non-reclosing device designed to function by the bursting of a pressure-retaining disk. This assembly consists of a thin, circular membrane usually made of metal, plastic, or graphite that is firmly clamped in a disk holder. When the process reaches the bursting pressure of the disk, the disk ruptures and releases the pressure. Rupture disks can be installed alone or in combination with other types of devices. Once blown, rupture disks do not reseal; thus, the entire contents of the upstream process equipment will be vented. Rupture disks of nonfragmented type are commonly used in series (upstream) with a safety relief valve to prevent corrosive

fluids from contacting the metal parts of the valve. In addition, this combination is a reclosing system.

The burst tolerances of rupture disks are typically about ± 5 percent for set pressures above 2.76 barg (40 psig). Consult API-520-I (Section 2.3.6.1) on the proper selection and burst setting of the rupture disks.

Pressure Vacuum Relief Valves For applications involving atmospheric and low-pressure storage tanks, pressure-vacuum relief valves (PVRVs) are used to provide pressure relief. Such devices are not included in the scope of ASME Section VIII and thus are not Code-certified; hence they are used mostly in non-Coded vessels (<1.03-barg or 15-psig design). These units combine both a pressure and a vacuum relief valve into a single assembly that mounts on a nozzle on top of the tank and are usually sized to handle the normal in-breathing and out-breathing requirements.

Sizing of Pressure Relief Systems A critical point in design is to determine whether the relief system must be sized for single-phase or two-phase relief flow. Two-phase flow frequently occurs during a runaway, but it can also occur in nonreactive systems such as vessels with gas spargers, vessels experiencing high heat input rates, or systems containing known foaming agents such as latex. The *drift flux* methodology (Zuber and Findlay, "Average Volumetric Concentration in Two-Phase Flow Systems," *Trans. ASME J. Heat Transfer*, pp. 453–468, November 1965; Wallis, *One-Dimensional Two-Phase Flow*, McGraw-Hill, New York, 1969) has been extended and applied to both the volumetric heating case (uniform vapor generation throughout the liquid) and wall heating case (vaporization occurring only at vessel wall) in the DIERS (Design Institute for Emergency Relief Systems) study (DIERS, 1992). The DIERS methodology is important as a means of addressing situations, such as two-phase flow, not covered adequately by current ASME and API methods. The recent CCPS Guidelines (CCPS, 1998) is the best source of updated information on these methods. For a top-vented vessel, the important mechanism for the liquid carryover resulting in two-phase relief is *boilover*. The vessel hydrodynamic model based on drift flux formulation is used to estimate the quality (i.e., vapor mass fraction) entering the vent system. The churn-turbulent regime vessel model is generally reserved for nonfoaming and nonviscous liquids. This regime would yield the highest degree of vapor-liquid disengagement. The bubbly regime vessel model is generally applied to foamy liquids and viscous systems. The bubbly regime would yield only limited disengagement in the vessel. Finally, a conservative but often realistic case for runaway reactive systems is the homogeneous (or uniform froth) vessel model. This model assumes no vapor-liquid disengagement inside the vessel, an idealized assumption, but it leads to much simpler sizing equations.

Required Relief Rate The required relief rate is the vent rate W (kg) required to remove the volume being generated within the protected equipment when the equipment is at its highest allowed pressure:

$$W_{\text{req}} = \frac{\text{net volume generation rate}}{\text{specific volume of vent stream}} \quad (23-89)$$

For steady-state design scenarios, the required vent rate, once determined, provides the capacity information needed to properly size the relief device and associated piping. For situations that are transient (e.g., two-phase venting of a runaway reactor), the required vent rate would require the simultaneous solution of the applicable material and energy balances on the equipment together with the in-vessel hydrodynamic model. Special cases yielding simplified solutions are given below. For clarity, nonreactive systems and reactive systems are presented separately.

Nonreactive Systems

Consider constant flow into protected equipment (blocked outlet). For the steady-state design scenario with a constant flow of fluid W_{in} (kg/s) from a pressure source that is above the maximum allowed pressure in the protected equipment, volume is being generated within the equipment at a rate of $W_{\text{in}}/\rho_{\text{in}}$, where ρ_{in} (kg/m³) is the incoming fluid density evaluated at the maximum allowed pressure. Denoting ρ_{out} (kg/m³) as the vent stream fluid density, Eq. (23-89)

then yields the required vent rate:

$$W_{\text{req}} = \frac{W_{\text{in}}}{\rho_{\text{in}}} \rho_{\text{out}} \quad (23-90)$$

Consider constant heat input into protected equipment. If the addition of heat to the equipment does not cause the fluid to boil, then the volume generation rate is the thermal expansion rate of the fluid:

$$W_{\text{req}} = \frac{q\beta}{C_p} \quad (23-91)$$

where q is the heat input rate (J/s), β is the coefficient of volumetric expansion at constant pressure ($^{\circ}\text{C}^{-1}$), and C_p is the constant-pressure specific heat [J/(kg $\cdot^{\circ}\text{C}$)].

The properties are evaluated at maximum allowed pressure conditions. For liquids, β can typically be evaluated from the specific volume change over a 5°C temperature increment. For ideal gases, Eq. (23-91) becomes

$$W_{\text{req}} = \frac{q}{C_p T} \quad (23-92)$$

where T is the absolute temperature (K).

If the fluid is at its boiling point, then volume is generated through the phase change that occurs upon vaporization. For *nonfoamy fluids*, vents sized for all-vapor relief are adequate, even if some initial two-phase venting is predicted. The required vent rate based on the volumetric rate balance criterion of Eq. (23-89) is given for a single-component fluid as (Leung, 1986a)

$$W_{\text{req}} = \frac{q}{h_{fg}} \left(\frac{v_{fg}}{v_g} \right) \quad (23-93)$$

where h_{fg} is the latent heat of vaporization (J/kg), v_g is the vapor specific volume (m³/kg), v_{fg} is the specific volume increase upon vaporization = $v_g - v_f$ (m³/kg), and v_f is the liquid specific volume (m³/kg).

Near the critical region, the property ratio h_{fg}/v_{fg} can be replaced by $T(dP/dT)_{\text{sat}}$ via the Clapeyron relation since both h_{fg} and v_{fg} are approaching zero. Here $(dP/dT)_{\text{sat}}$ is the slope of the vapor-pressure curve and has units of Pa/K. For multicomponent fluids, Eq. (23-93) is evaluated for each major component (> 10 % wt), and the largest single-component venting requirement is used. Refer to CCPS Guidelines (1998) for more complex schemes.

For *foamy fluids*, the homogeneous vessel model with volumetric heating assumption is used (Leung, 1986a):

$$W(T_p - T_s) = \frac{q}{C_p} \left(\ln \frac{m_0 q v_{fg}}{W h_{fg}} - 1 \right) + \frac{W V h_{fg}}{m_0 C_p v_{fg}} \quad (23-94)$$

where W is the vent rate (kg/s), T_p is the temperature at peak pressure ($^{\circ}\text{C}$), T_s is the temperature at set pressure ($^{\circ}\text{C}$), m_0 is the initial mass in system (kg), V is the equipment (vessel) volume (m³), q is the heat input rate (J/s), and C_p is the constant-pressure liquid specific heat [J/(kg $\cdot^{\circ}\text{C}$)].

Note that Eq. (23-94) is implicit in W ; a trial-and-error solution method is required. This equation is applicable for single-component or pseudo-one-component systems. For the latter, h_{fg} is defined to be the enthalpy (heat) required to vaporize a unit mass of liquid at equilibrium vapor composition.

Analytical equations have been presented for both the bubbly regime and churn-turbulent regime; see Leung (Leung, "Overpressure During Emergency Relief Venting in Bubbly and Churn-Turbulent Flow," *AIChE J.* 33(6), pp. 952–958, 1987) and Dalessandro (Dalessandro, "Emergency Venting Requirements for Tempered Systems Considering Partial Vapor-Liquid Separation with Disengagement Parameters Greater than Unity, Part I: Model Development, Part II: Application," *Process Safety Prog.* 23, pp. 1–15 and 86–98, 2004).

Reactive Systems Exothermic runaway (uncontrolled chemical) reactions add considerable complexity. Note that external heating (such as a fire) on a vessel containing unstable compounds would fall into this category. The added complications are due to the following:

1. Reaction rate varies exponentially with temperature.
2. Exothermic heat release rates are time-varying.
3. Volatile or noncondensable gas can be generated (in decomposition).
4. Composition change causes shift in boiling point curve.
5. Viscosity increases due to polymerization.

In dealing with such complexity, DIERS has developed a special bench-scale apparatus commercialized by Fauske & Associates, LLC, as the Vent Sizing Package 2 (VSP2™). The design and operation are described in several references [Fauske and Leung, "New Experimental Technique for Characterizing Runaway Chemical Reactions," *Chem. Eng. Prog.*, **80**(8), pp. 39–46, 1985; Leung, Fauske, and Fisher, "Thermal Runaway Reactions in a Low Thermal Inertia Apparatus," *Thermochimica Acta*, **104**, pp. 13–29, 1986; DIERS, 1992; CCPS, 1998]. This adiabatic calorimeter has been extensively used to obtain the above key information for pressure relief design. Upset scenarios such as external heating, overcharge of reactants, or loss of cooling can be directly simulated in this apparatus. Two special-case solutions are given here.

For boiling liquid reactive systems, an analytical method yielding closed-form integral equations for homogeneous vessel two-phase relief has been used for some time (Leung, 1986a). Simplifying assumptions are that (1) the vent rate is taken as constant with time, (2) average values of heat release rate per unit mass and physical properties can be used, (3) there is constant or decreasing system volatility, and (4) a pseudo-one-component approach is taken. The required vent rate is

$$W_{req} = \frac{m_0 C_p (dT/dt)_{av}}{[(V/m_0)/(h_{fg}/v_{fg})]^{1/2} + (C_p \Delta T)^{1/2}} \quad (23-95)$$

where $\Delta T = T_p - T_s$ is the "overtemperature" (°C) corresponding to the overpressure ΔP selected, and $(dT/dt)_{av}$ is the average temperature rise rate (°C/s).

The average dT/dt is typically an arithmetic average between the value at set pressure and the value at peak allowed pressure. The properties C_p , h_{fg} , v_{fg} either can be evaluated at the set conditions or can be taken as the average values between the set condition and the peak allowed pressure condition. Alternatively, the term h_{fg}/v_{fg} in Eq. (23-95) can be replaced by $T(dP/dT)_{sat}$ via the Clapeyron relation. This holds reasonably well for a multicomponent system of near constant volatility. Such an application permits direct use of the experimental pressure-temperature data obtained from a closed-system runaway VSP2 test. This form of Eq. (23-95) has been used to demonstrate the advantageous reduction in both vent rate and vent area with allowable overpressure (Leung, 1986a).

For partial disengagement venting models (in bubbly and churn-turbulent regimes), refer to publications by Leung (1987) and Dalessandro (2004).

The special case of **gas generating reactive systems** is for a decomposition reaction with negligible gas solubility in the liquid and no significant latent heat effects. Such a reaction will proceed to the maximum rate whether vented or not. The main purpose of venting is to limit the pressure rise in these "gassy" systems. The maximum gas generation rate is usually obtained from the pressure rise rate data in the DIERS-type apparatus. At the peak allowed pressure P_m , the volumetric gas generation rate for the protected equipment is given by

$$Q_{g,max} = \frac{m_0}{m_t} \frac{V_t}{P_m} \left(\frac{dP}{dt} \right)_{max} \quad (23-96)$$

where m_0 is the mass in protected equipment (kg), m_t is the sample mass in test apparatus (kg), V_t is the volume occupied by gas in test apparatus (m^3), P_m is the peak allowed pressure in equipment (Pa, psia), and $(dP/dt)_{max}$ is the maximum pressure rise rate measured in test apparatus (Pa/s, psi/s). Note that consistent pressure units can be used for P_m and dP/dt here. The required vent rate based on the volumetric rate balance criterion of Eq. (23-89) is given by

$$W_{req}^0 = \frac{Q_{g,max}}{V/m_0} \quad (23-97)$$

However, this vent rate assumes homogeneous vessel venting at the peak allowed pressure with no prior loss of mass. A less conservative design would consider the mass loss during the overpressure venting interval [Leung, "Venting of Runaway Reactions with Gas Generation," *AIChE J.*, **38**(5), pp. 723–732, 1992].

Relief System Flow Capacity This subsection establishes the relation between relief system configuration and flow capacity in order to determine the size required for a given vent rate. The mass flow rate W through a given relief system geometry, in general, requires a trial-and-error approach when the system configuration contains more than a single diameter. The generalized approach is to assume a flow rate W —this is usually governed by the limiting area such as the orifice or nozzle in a pressure relief valve (PRV) or the rupture disk (RD)—and calculate the resulting pressure profiles down the system until the final discharge pressure matches the specified value. If choked flow is encountered at any area enlargement location in the vent system, then the pressure just after the choking location is determined by calculating back up the larger pipe from a downstream point of specified pressure.

The treatment of vent flow calculations in most typical relief system configurations involves two classes of computational models: flow through nozzles and frictional flow in pipes.

The flow path in well-formed nozzles [ideal (**frictionless**) flow] follows smoothly along the nozzle contour without flow separation. The effects of small imperfections and small frictional losses are accounted for by correcting the ideal nozzle flow by an empirically determined coefficient of discharge K_d . This applies to PRV geometry. The acceleration of a fluid initially at rest to flowing conditions in an ideal nozzle is given by

$$-\frac{G^2 v}{2} = \int_{P_0}^{P_1} v dP \quad (23-98)$$

where P_0 is the stagnation pressure of the fluid, P_1 is the static pressure at the minimum flow area called the *throat*, G is the mass flux [$kg/(m^2 \cdot s)$], and v is the fluid specific volume (m^3/kg). If the fluid is compressible, the flow will increase to a maximum value as the downstream pressure is reduced, and any further decrease in the downstream pressure will not affect the flow. This maximum flow condition is referred to as the *critical* (or *choked*) condition. The flow and pressure at this condition are related by

$$G_c = \sqrt{\frac{-1}{(dv/dP)_s}} \quad (23-99)$$

The subscript s denotes an isentropic path for ideal nozzle flow. For ideal gas with $Pv^k = \text{constant}$, substitution of this isentropic expansion law into Eq. (23-98) yields the following critical pressure ratio P_c/P_0 and critical flow rate G_c :

$$\frac{P_c}{P_0} = \left(\frac{2}{k+1} \right)^{k/(k-1)} \quad (23-100)$$

$$G_c = \left[k \left(\frac{2}{k+1} \right)^{(k+1)/(k-1)} \right]^{1/2} \sqrt{\frac{P_0}{v_0}} \quad (23-101)$$

where k is the isentropic expansion exponent for ideal gas ($k = C_p/C_v$), P_c is the choking pressure at throat (Pa), P_0 is the stagnation inlet pressure (Pa), and v_0 is the ideal gas specific volume at P_0 (m^3/kg).

Thus for a PRV system, the following relief rate formula (for choked flow) is given in SI units (kg/s , m^2 , N/m^2 , K) in API-520-I (2000):

$$W = K_d K_b A P_0 \left(\frac{M_w}{RT_0 Z} \right)^{1/2} \left[k \left(\frac{2}{k+1} \right)^{(k+1)/(k-1)} \right]^{1/2} \quad (23-102)$$

where K_d is the discharge coefficient of PRV nozzle (orifice), K_b is the capacity correction factor due to backpressure (consult API-520-1), A is the PRV orifice area (m^2), M_w is the molecular weight, R is the gas constant, T_0 is the absolute temperature at inlet (K), and Z is the compressibility. For *nonideal gas* at near thermodynamic critical or supercritical region, special treatments other than simply applying the Z factor as in Eq. (23-102) are necessary [see Leung and Epstein, "A Generalized Critical Flow Model for Non-ideal Gases," *AIChE J.* 34(9), pp. 1568–1572, 1988; and CCPS, 1998].

The treatment for two-phase relief is more involved due to the presence of slip and thermodynamic nonequilibrium between the two phases. However, for flow capacity evaluation, the traditional homogeneous (no slip) equilibrium model (HEM) is recommended (DIERS, 1992; CCPS, 1998). For a two-phase flashing system (both single-component and multicomponent), the P - v relation can be obtained from adiabatic flash calculations—strictly speaking under constant-entropy, but constant-enthalpy flash computations are adequate for use in numerical integration of Eq. (23-98). A particularly useful simplification is the *omega* method by Leung (1986b; see the earlier subsection "Discharge Rates from Punctured Lines and Vessels" for more discussion), where a P - v relation for a two-phase fluid expansion is written as

$$\frac{v}{v_0} = \omega \left(\frac{P_0}{P} - 1 \right) + 1 \quad (23-103)$$

For a single-component or a pseudo-one-component fluid, the dimensionless parameter ω is evaluated at the inlet conditions (subscript 0) from (Leung, 1996)

$$\omega = \alpha_0 \left(1 - 2 \frac{P_0 v_{f0}}{h_{f0}} \right) + \frac{C_{p0} T_0 P_0}{v_0} \left(\frac{v_{f0}}{h_{f0}} \right)^2 \quad (23-104)$$

For multicomponent systems with boiling range greater than 80°C , a single adiabatic flash calculation to 80 to 90 percent of the inlet pressure P_0 yields the two-phase specific volume v_1 at pressure P_1 , and ω is calculated from (Nazario and Leung, "Sizing Pressure Relief Valves in Flashing and Two-Phase Service: An Alternative Procedure," *J. Loss Prev. Process Ind.* 5(5), pp. 263–269, 1992)

$$\omega = \frac{v_1/v_0 - 1}{P_0/P - 1} \quad (23-105)$$

For the case of a noncondensable gas and a nonflashing liquid (a nonflashing two-phase fluid), ω is simply defined by (Leung, 1996)

$$\omega = \frac{\alpha_0}{\Gamma} \quad (23-106)$$

where the two-phase isentropic expansion exponent for a thermal equilibrium process is given by

$$\Gamma = \frac{x C_{pg} + (1-x) C_{pf}}{x C_{vg} + (1-x) C_{vf}} \quad (23-107)$$

Note that for most cases of interest, Γ is close to unity since the flowing gas mass fraction $x \ll 1$. For the case of frozen flow (i.e., no heat transfer between the two phases), Γ would be replaced by k (or C_{pg}/C_{vg}) in Eq. (23-106). However, the difference between the two limiting cases is small (< 10 percent) in terms of the flow capacity [Leung and Epstein, "A Generalized Correlation for Two-Phase Nonflashing Homogeneous Choked Flow," *Trans. ASME J. Heat Transfer* 112 (May), pp. 528–530, 1990].

With the P - v relation given by Eq. (23-103), Eq. (23-98) can be integrated to give a generalized equation for flow through an ideal nozzle

$$\frac{G}{\sqrt{P_0 v_0}} = \frac{[-2[\omega \ln(P/P_0)] + (\omega - 1)(1 - P/P_0)]^{0.5}}{\omega(P_0/P - 1) + 1} \quad (23-108)$$

The maximum (critical) or choked flow condition occurs at the critical pressure ratio P_c/P_0 from solving (Leung, 1986a)

$$\left(\frac{P_c}{P_0} \right)^2 + (\omega^2 - 2\omega) \left(1 - \frac{P_c}{P_0} \right)^2 + 2\omega^2 \ln \frac{P_c}{P_0} + 2\omega^2 \left(1 - \frac{P_c}{P_0} \right) = 0 \quad (23-109)$$

The critical mass flux G_c is given by the dimensionless form

$$\frac{G_c}{\sqrt{P_0 v_0}} = \frac{P_c/P_0}{\sqrt{\omega}} \quad (23-110)$$

If $P_c > P_b$, the backpressure, the flow is choked, and Eq. (23-110) yields the critical mass flux. Otherwise, $P_c < P_b$, flow is not choked, and P is equated to P_b in Eq. (23-108) to obtain the unchoked mass flux.

A relief rate formula for two-phase flow similar to Eq. (23-102) for gas can be written as

$$W = K_d K_b A G \quad (23-111)$$

For the boiling-liquid system, ω is calculated from Eq. (23-104) or Eq. (23-105), while for a gas-generating system, ω is evaluated from Eq. (23-106).

For **pipe flow**, HEM requires solution of the equations of conservation of mass, energy, and momentum. The momentum equation is in differential form, which requires partitioning the pipe into segments and carrying out numerical integration. For constant-diameter pipe, these conservation equations are as follows:

Mass:

$$G = \text{constant} \quad (23-112)$$

Energy:

$$h_0 = h + \frac{G^2 v^2}{2} \quad (23-113)$$

Momentum:

$$v \, dP + G^2 \left(v \, dv + \frac{4f v^2 \, dL}{2D} \right) + g \cos \theta \, dL = 0 \quad (23-114)$$

where h_0 is enthalpy at stagnation (J/kg), h is enthalpy (J/kg), v is fluid specific volume, f is Fanning friction factor, L is flow length (m), g is acceleration due to gravity, and θ is angle of inclination from vertical.

Equation (23-114), the momentum equation, can be solved more conveniently by rewriting it and solving it numerically

$$\Delta L = - \frac{\bar{v} \, \Delta P + G^2 v \, \Delta v}{(2f/D) G^2 v^2 + g \cos \theta} \quad (23-115)$$

where $\bar{v}$ is the average specific volume in the pressure increment ΔP and Δv is the incremental specific volume over ΔP .

In the numerical integration of Eq. (23-115) for a given pipe length L , the following steps are suggested (note that detailed thermodynamic properties are required also):

1. Either G is known or guessed.
2. Increments of pressure are taken from the initial to the final pressure.
3. Both $\bar{v}$ and Δv are obtained for each increment for a constant stagnation enthalpy process satisfying Eq. (23-113).
4. For each ΔP taken, ΔL is computed from Eq. (23-115).
5. Total length of pipe L is $\Sigma \Delta L$.
6. If ΔL is negative, then ΔP is too large.
7. A critical flow condition corresponds to $\Delta L = 0$, and the final pressure corresponds to choked pressure.
8. If $\Sigma \Delta L > L$, then G was guessed too small, and steps 1 through 7 are repeated with a larger G . If $\Sigma \Delta L < L$, then G was guessed too large; steps 1 through 7 are repeated with a smaller G .
9. A converged solution is obtained when $\Sigma \Delta L = L$ to within a given tolerance.

The above calculations can be simplified somewhat by ignoring the kinetic energy term in the energy equation, Eq. (23-113), and simply basing calculations on an isenthalpic flash. Thus a P - v relation can be obtained from such flash calculations prior to performing the above iteration steps. In the special case of turbulent pipe flow horizontal discharge ($\cos \theta = 0$), Eq. (23-114) can be integrated based on the omega P - v relation to yield [Leung and Grolmes, "The Discharge of Two-Phase Flashing Flow in a Horizontal Duct," *AIChE J.* 33(3), pp. 524–527, 1987; also errata, 34(6), p. 1030, 1988]

$$4f \frac{L}{D} = \frac{2}{G^2} \left[\frac{\eta_1 - \eta_2}{1 - \omega} + \frac{\omega}{(1 - \omega)^2} \ln \frac{(1 - \omega)\eta_2 + \omega}{(1 - \omega)\eta_1 + \omega} \right] - 2 \ln \left[\frac{(1 - \omega)\eta_2 + \omega}{(1 - \omega)\eta_1 + \omega} \left(\frac{\eta_1}{\eta_2} \right) \right] \quad (23-116)$$

where $\eta_1 \equiv P_1/P_0$, $\eta_2 \equiv P_2/P_0$, $G^* \equiv G/\sqrt{P_0/v_0}$, P_1 is the absolute static pressure at the inlet of the pipe (N/m^2), P_2 is the absolute static pressure at the outlet of the pipe (N/m^2), and P_0 is the reference pressure typically associated with the stagnation pressure in the upstream equipment. The choking criterion at pipe exit takes the form

$$G_c^* = \frac{\eta_{2c}}{\sqrt{\omega}} \quad (23-117)$$

where subscript c denotes choking. The integrated momentum equation, Eq. (23-116), yields the pressure drop $P_1 - P_2$ in the pipe implicitly as a function of ω , $4fL/D$, and G . For the case of a pipe discharge from a vessel, the inlet acceleration pressure drop $P_0 - P_1$ is given by Eq. (23-108) above with P/P_0 being P_1/P_0 or η_1 . Essentially the inlet pipe section is treated as an ideal nozzle, and any irreversible loss due to sharp entrance effect is included in the overall $4fL/D$ term. Likewise, the velocity head loss associated with bends (elbows) is to be included in the $4fL/D$ term as well. For high-velocity, two-phase discharge, an average f value of 0.005 has been used in most applications (Wallis, 1969).

Choked pipe exit: $\eta_{2c} > \eta_b$ or $P_{2c} > P_b$. Equations (23-109), (23-116), and (23-117) are used to solve for G_c^* , η_1 , and η_2 (likewise, for G_c , P_1 , and P_2) for a given $4fL/D$ and ω value.

Unchoked pipe exit: $\eta_{2c} < \eta_b$ or $P_{2c} < P_b$. Equation (23-116) does not apply. With $\eta_2 = \eta_b$ (or $P_2 = P_b$) for an unchoked exit, only Eqs. (23-108) and (23-116) are needed to solve for G^* and η_1 (likewise, for G and P_1). Under most situations, it suffices to set $\eta_1 = 1$ and to solve for G^* readily by rearranging Eq. (23-116).

This method has been extended to inclined pipe discharge [Leung and Epstein, "The Discharge of Two-Phase Flashing Flow from an Inclined Duct," *Trans. ASME J. Heat Transfer* 112 (May), pp. 524–528, 1990], which together with some useful design charts is presented in the earlier subsection "Discharge Rates from Punctured Lines and Vessels."

EMERGENCY RELIEF DEVICE EFFLUENT COLLECTION AND HANDLING

GENERAL REFERENCES: API RP 521, *Guide for Pressure Relieving and Depressurizing Systems*, 5th ed., American Petroleum Institute, Washington, 2007. API STD 537, *Flare Details for General Refinery and Petrochemical Service*, American Petroleum Institute, Washington, September 2003. AIChE-CCPS, *Guidelines for Pressure Relief and Effluent Handling Systems*, AIChE, New York, 1998. DIERS, *Emergency Relief System Design Using DIERS Technology*, AIChE, New York, 1992. Fthenakis, *Prevention and Control of Accidental Releases of Hazardous Gases*, Van Nostrand-Reinhold, New York, 1993. Grossel and Crowl, *Handbook of Highly Toxic Materials Handling and Management*, Marcel Dekker, New York, 1995. Grossel, *Journal of Loss Prevention in the Process Industries* 3(1): 112–124, 1990. Grossel, *Plant/Operations Progress* 5(3): 129–135, 1986. Keiter, *Plant/Operations Progress* 11(3): 157–163, 1992. McIntosh and Nolan, *Journal of Loss Prevention in the Process Industries* 14(1): 17–26, 2001. McIntosh and Nolan, *Journal of Loss Prevention in the Process Industries* 14(1): 27–42, 2001. Schmidt and Giesbrecht, *Process Safety Progress* 20(1): 6–16, 2001.

Introduction In determining the disposal of an effluent vent stream from an emergency relief device (safety valve or rupture disk), a number of factors must be considered, such as

1. Is the stream single-phase (gas or vapor) or multiphase (vapor-liquid or vapor-liquid-solid)?
2. Is the stream flammable or prone to deflagration?
3. Is the stream toxic?
4. Is the stream corrosive to equipment or personnel?

Some vent streams, such as light hydrocarbons, can be discharged directly to the atmosphere even though they are flammable and explosive. This can be done because the high-velocity discharge entrains sufficient air to lower the hydrocarbon concentration below the lower explosive limit (API RP 521, 2007). Toxic vapors must be sent to a flare or scrubber to render them harmless. Multiphase streams, such as those discharged as a result of a runaway reaction, e.g., must first be routed to separation or containment equipment before final discharge to a flare, a scrubber, or a quench pool tank.

This discussion is organized into three major areas: types of equipment, criteria employed in the selection of equipment, and sizing and design of the equipment.

Types of Equipment Equipment for handling emergency relief discharge streams can be divided into two classes:

1. Vapor/gas/solid-liquid separators
2. Final vapor/gas control or destruction equipment

The two most commonly used types of vapor/gas-liquid separators are

- Gravity separators—horizontal and vertical (also called blowdown drums, knockout drums, or catch tanks)
 - Emergency cyclone separators
- The most commonly used types of final vapor/gas control or destruction equipment are
- Quench pools/catch tanks and quench towers
 - Emergency flare systems
 - Emergency scrubbers (absorbers)

Gravity Separators Three types of gravity separators are commonly used in the chemical process industries (CPI): horizontal blowdown drum/catch tank, vertical blowdown drum/catch tank, and multi-reactor knockout drum/catch tank.

Horizontal blowdown drum/catch tank This type of drum, shown in Fig. 23-49, combines both the vapor-liquid separation and holdup functions in one vessel. Horizontal drums are commonly used where space is plentiful. The two-phase mixture usually enters at one end, and the vapor exits at the other end. To overcome reentrainment of liquid droplets, due to high inlet velocities, various devices and piping arrangements are used to provide a more uniform distribution of vapor-liquid mixtures into the separator, as shown in Fig. 23-50. For two-phase streams with very high vapor flow rates, inlets may be provided at each end, with the vapor outlet at the center of the drum, thus minimizing vapor velocities at the inlet and aiding vapor-liquid separation.

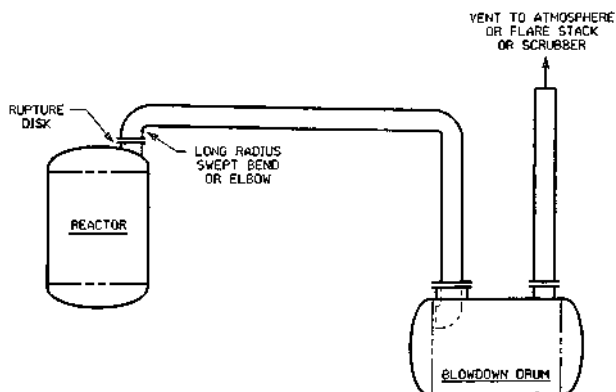


FIG. 23-49 Horizontal blowdown drum/catch tank.

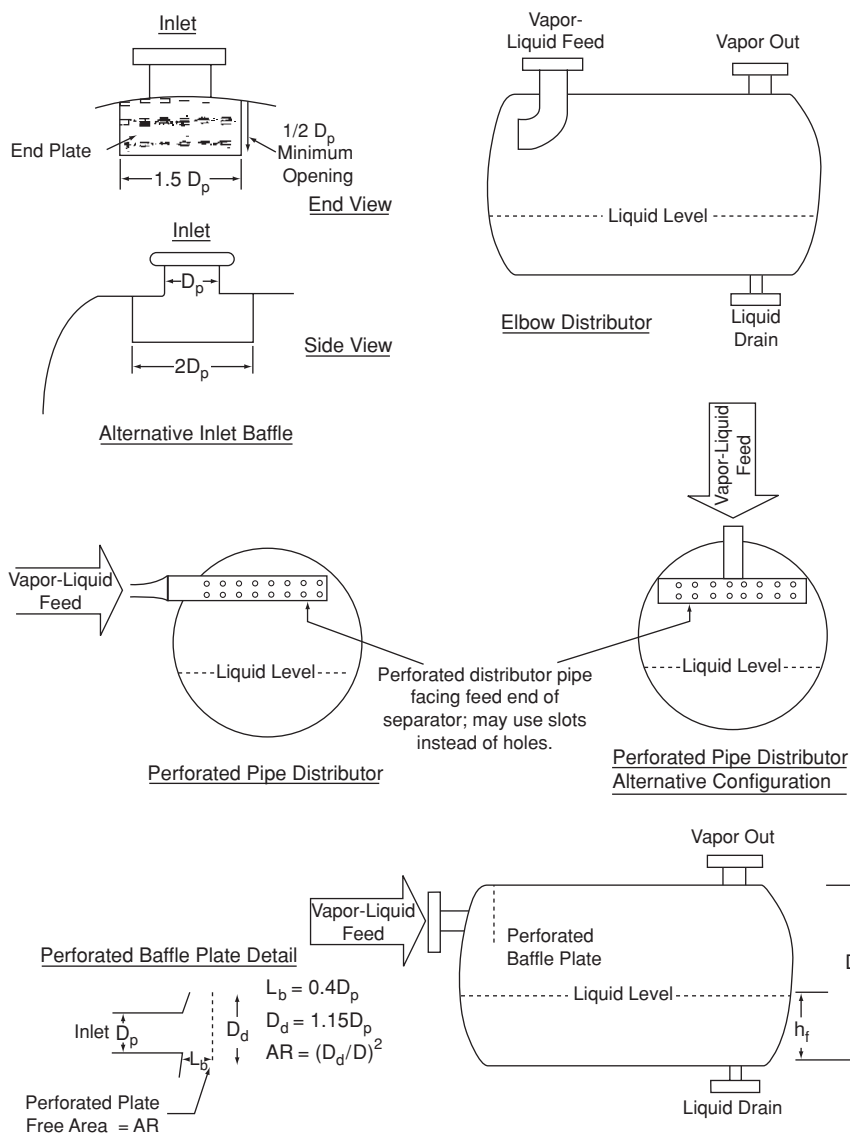


FIG. 23-50 Feed distributor devices for horizontal blowdown drums/catch tanks. (Guidelines for Pressure Relief and Effluent Handling Systems, *Center for Chemical Process Safety (CCPS) of the American Institute of Chemical Engineers (AIChE)*; copyright 1988 AIChE and reproduced with permission.).

Vertical blowdown drum/catch tank This type of drum, shown in Fig. 23-51, performs the same function and operates on similar principles as horizontal separators. These separators are usually used where horizontal space is limited. The two-phase mixture enters the vessel via a nozzle on the vertical shell and is distributed by an inlet baffle chamber.

Multireactor knockout drum/catch tank This interesting system is sometimes used as the containment vessel for a series of closely spaced reactors (Speechly et al., "Principles of Total Containment System Design," presented at I.Chem.E. North West Branch Meeting, 1979).

Emergency Cyclone Separators Two types of emergency cyclone separators are commonly used, those with a separate catch tank and those with a catch tank integral with the cyclone section.

Cyclone separator with separate catch tank This type of blowdown system, shown in Figs. 23-52 and 23-53, is frequently used in chemical plants where plot plan space is limited. The cyclone performs the vapor-liquid separation, while the catch tank accumulates the liquid from the cyclone. This arrangement allows location of the cyclone knockout drum close to the reactor so that the length of the relief device discharge line can be minimized. The cyclone has internals, vital to its proper operation, which are discussed in the following subsections.

Cyclone separator with integral catch tank This type of containment system, depicted in Fig. 23-54, is similar to the aforementioned type, except that the knockout drum and catch tank are combined in one vessel shell. This design is used when the vapor rate is quite high so that the knockout drum diameter is large.

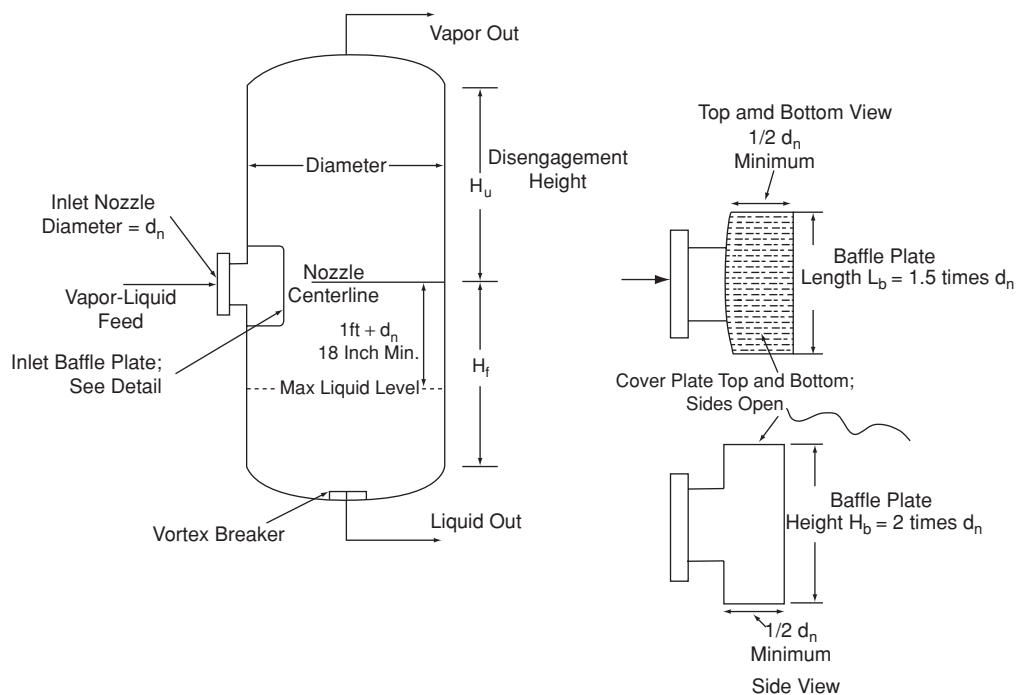


FIG. 23-51 Vertical blowdown drum/catch tank. (Guidelines for Pressure Relief and Effluent Handling Systems, *Center for Chemical Process Safety (CCPS) of the American Institute of Chemical Engineers (AIChE)*; copyright 1988 AIChE and reproduced with permission.).

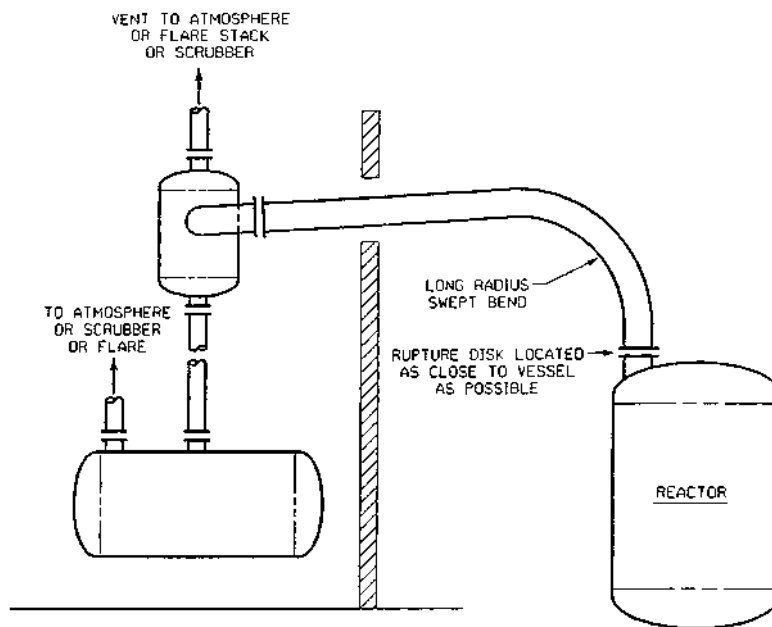


FIG. 23-52 Cyclone separator with separate liquid catch tank.

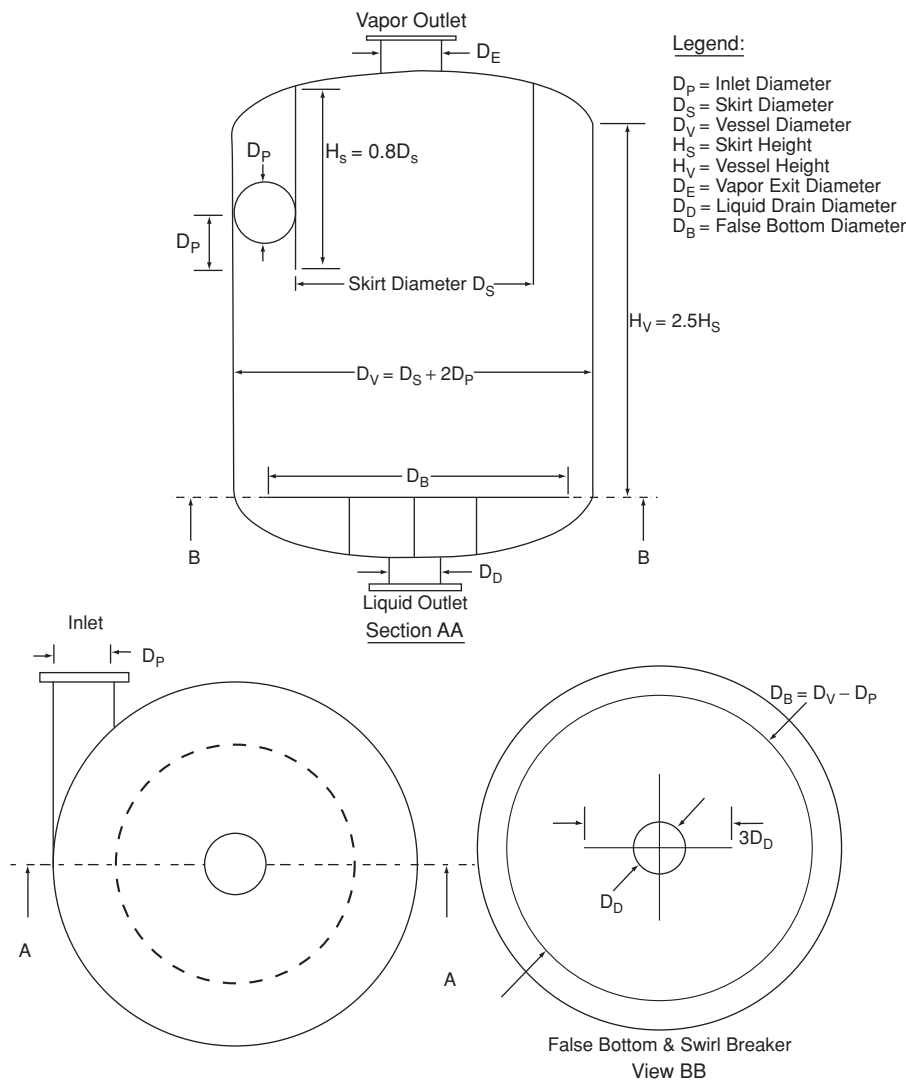


FIG. 23-53 Cyclone separator design details. (Guidelines for Pressure Relief and Effluent Handling Systems, *Center for Chemical Process Safety (CCPS) of the American Institute of Chemical Engineers (AIChE)*; copyright 1988 AIChE and reproduced with permission.).

Quench Pools/Catch Tanks and Quench Towers Two types of quench systems are commonly used: the quench pool/catch tank (also called passive scrubber) and quench tower.

Quench pool/catch tank This type of system, as shown in Fig. 23-55, is used to condense, cool, react with, and/or collect a mixture of liquid and vapors discharging from a relief device by passing them through a pool of liquid in a vessel. Feed vapor and liquid (if present) are sparged into the pool of cool liquid, where the vapors are condensed and the liquid is cooled. If the feed materials are miscible with the pool liquid, they mix with and are diluted by the pool liquid; if not, the condensate, feed liquid, and pool liquid separate into layers after the emergency relief event is over. The condensed vapors, feed liquid, and quench liquid are contained in the vessel until they are sent to final disposal.

Quench pools can be used for the neutralization or killing of reactive materials. A chemical reaction can be quenched by the cooling

effect in the pool, by the dilution effect if the reaction mass is miscible with the pool liquid, or by a killing or neutralizing agent added to the quench liquid. For example, discharge of chlorine or acid mixtures can be neutralized by adding sodium hydroxide to the quench liquid. However, systems involving slow to moderate reaction rates usually require more than one contacting stage, and a quench pool (equal to one stage only) may not be adequate to completely neutralize or kill the reactant in the entering relief device effluent stream. In this case, a multiple-stage emergency scrubber will be needed.

Depending on the relief stream components (with or without non-condensable gases) and quenching efficiency, this arrangement often obviates the need for a subsequent scrubber and/or flare stack. The design of the quencher arm is critical to efficient condensation and avoidance of water hammer. Figure 23-55 is the more conventional passive type quench pool used in the chemical and nuclear industry.

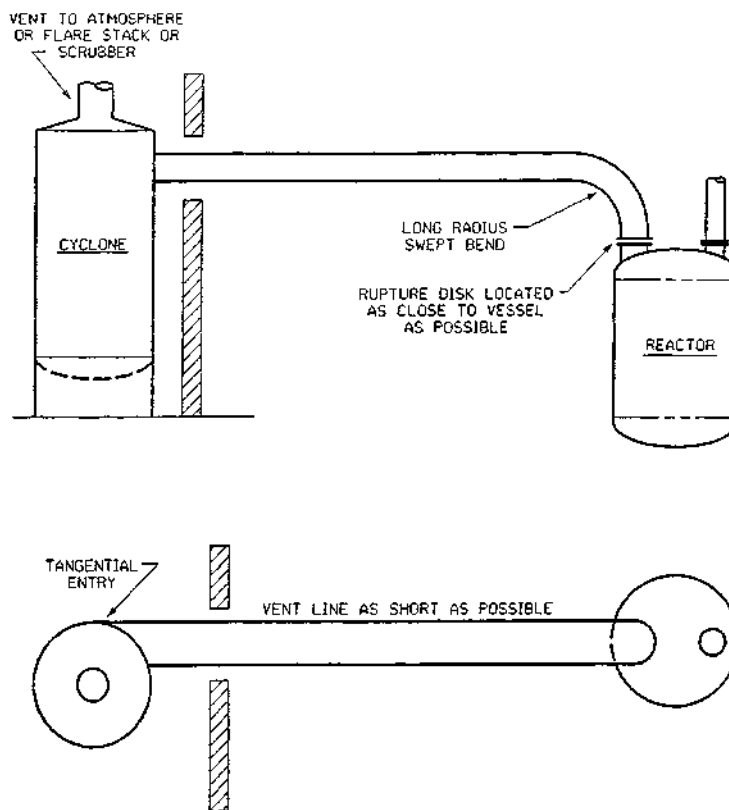


FIG. 23-54 Cyclone separator with integral catch tank.

Quench tower The type shown in Fig. 23-56, with a superimposed baffle plate section, is often used when complete condensation of the incoming vapors is not required. The exiting vapors are usually cooled to 150 to 200°F in the baffle plate section. This type is often used in petroleum refineries.

Emergency Flare Systems Single-phase effluent streams (gases or vapors) from pressure relief devices are often sent directly to emergency flares. The vapor/gas streams from gas-liquid separators are also often sent to emergency flares. Emergency flares are similar in design to normal process flares (continuous flaring), but they are only used to handle emergency releases (intermittent operation).

Flares can be grouped into two major categories, i.e., vertical and horizontal.

Vertical flares These are single-burner combustors generally oriented to fire upward. The discharge point is in an elevated position relative to the surrounding grade and/or nearby equipment. Three types of support methods are used for vertical flares:

- **Self-supported.** A mechanically and structurally designed riser supports the flame burner.
- **Guyed.** An elevated flare with the riser is supported by cables. Cables are attached to the flare riser at one or more elevations to limit the deflection of the structure. The cables (guy-wires) are typically positioned in a triangular plan to provide strong support.
- **Derrick supported.** Support is provided by a steel trussed structure that supports one or more flare risers.

The main components of an elevated flare system are the flare burner with or without smoke suppression capability, pilot(s), pilot igniter(s), support structure, and piping. A number of optional features are available, such as pilot flame detectors; air seals (buoyancy or

velocity type); knockout drum; flame/detonation arrester; liquid seal; smoke suppression control system; blower(s); flow, composition, heat content, or video monitoring; ladders (caged or with safety-climbing system) and platforms; davit for tip removal; aircraft warning lights and painting; radiation heat shields; and rain shields. Detailed information about these components can be found in API STD 537.

Smokeless flaring is required by law in the United States (40 CFR 60.18, Chap. 1) for normal process flares (continuous flaring). However, smokeless flaring is not required by the EPA for emergency flaring, but local conditions and regulations may require smokeless flaring.

Air seals (also called purge reduction or gas seals) are often recommended to prevent air from entering the stack, which could possibly cause flashbacks and explosions. More information about air seals can be found in API RP 521 (2007) and the AIChE-CCPS book *Guidelines for Pressure Relief and Effluent Handling Systems*.

Horizontal flares A horizontal flare can be either an enclosed ground flare or an open field ground flare (also called a matrix flare).

Enclosed ground flares are most commonly used as a supplement to an elevated flare on the same relief system. The primary reason for an enclosed ground flare is to reduce the visual impact of flared gas combustion on a nearby community. They are often used when it is desirable that all or part of a flare load be disposed of in a way that causes the minimum of disturbance to the immediate locality. They offer many advantages in comparison to elevated flares; there is no smoke, no visible flame, no odor, no objectionable noise, and no thermal radiation (heat shield) problems. Enclosed ground flares are typically used for normal process flow (continuous) flaring, but with recent technical advances they are now also used for emergency flaring (AIChE-CCPS, 1998).

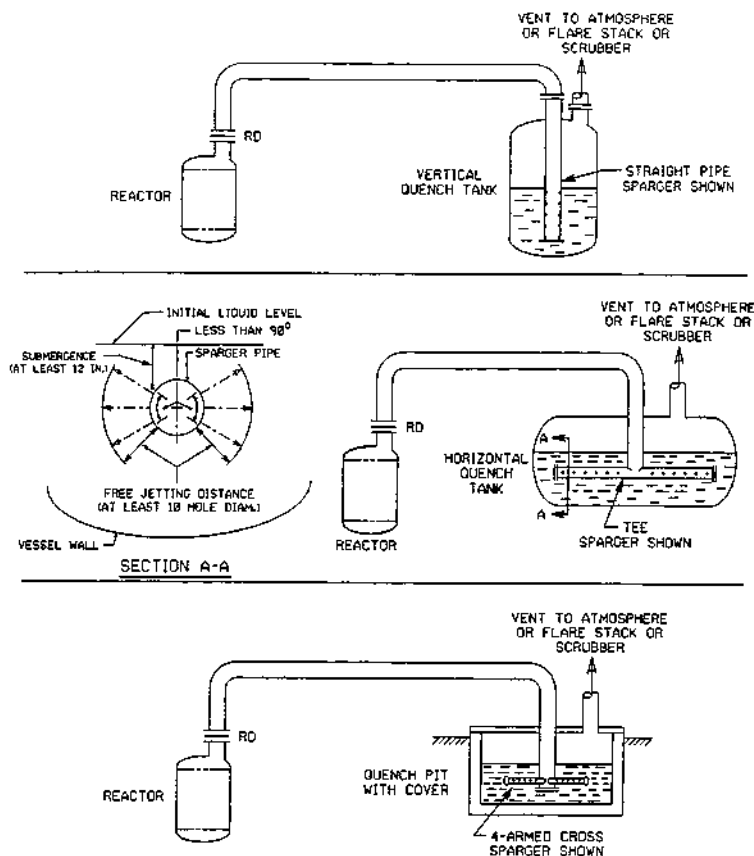


FIG. 23-55 Quench tank/catch tank.

Enclosed ground flares burn the effluent gas in multiple specially designed multitip burners placed as near to the ground as practicable to ensure good operation. The resulting flames are hidden from sight by a surrounding wall or chamber. The top of the chamber is open to the atmosphere, and allowance is made in the bottom of the chamber to permit ingress of air for combustion. Overall heights can vary according to the needs of the specific location. In some locations, a low profile of perhaps 10 m may be mandated, while in others an elevated discharge may be needed to ensure a satisfactory downwind trajectory for the flue gases. It is common for the chamber to be surrounded by a wind fence to modify the effect of crosswinds on the combustion process and to prevent unauthorized access.

An enclosed ground flare system has a number of key components: combustion chamber, burners, piping system, wind fence, and operational and safety controls. More details about enclosed ground flare system components are given in API STD 537.

Enclosed ground flares can be used to burn toxic gases, but since any noncombusted release would be near the ground, special safety dispersion systems and more stringent monitoring and control instrumentation are usually provided in this application.

There is a practical economical limit to the size of enclosed ground flares. If the maximum effluent relief rate exceeds 100 tons/h, the feasibility of an enclosed ground flare becomes questionable, and at 500 tons/h, it is impractical (AIChE-CCPS, 1998).

Open field ground flares (also called **matrix flares**) are sometimes used in lieu of elevated flares, most commonly, although not exclusively, in remote locations with relatively low population density.

A primary reason for use of an open field ground (matrix) flare is to reduce the visual impact of flared gas combustion in the manner of an enclosed ground flare. The open field ground (matrix) flare, however, has a significantly larger capacity than could be practically handled in a furnacelike structure, and the visual shielding is provided by tall fencing located some distance away from the burners themselves so that the fence encloses a small "field."

An open field (matrix) ground flare consists of a number of staged multiburners mounted on a series of horizontal pipes, arranged in manifolds, in a large field or pit. Smokeless or nonsmokeless flaring can be provided. The main structural and physical design issues are concerned with protecting the piping and manifolds inside the flare field from the effects of high radiant heat loads, while still permitting normal thermal expansions. More information about ground flare components is given in API STD 537.

Emergency Scrubbers (Absorbers) Emergency relief effluent streams are often sent to scrubbers (also called absorbers, columns, or towers) for final disposal by absorption of the gas into a solvent. Some gases or vapors can be removed by physical absorption. Other gases or vapors can be removed by chemical absorption (reaction of the vent gas/vapor into a liquid that reacts with it).

A typical emergency scrubber system consists of a scrubbing column (often filled with packing), recirculating liquid pumps, solvent cooler, and in some cases (where the entering effluent gas/vapor is at a low pressure) exhaust blowers (see Fig. 23-57). Redundant equipment and instrumentation is usually provided to ensure reliable scrubber operation at all times.

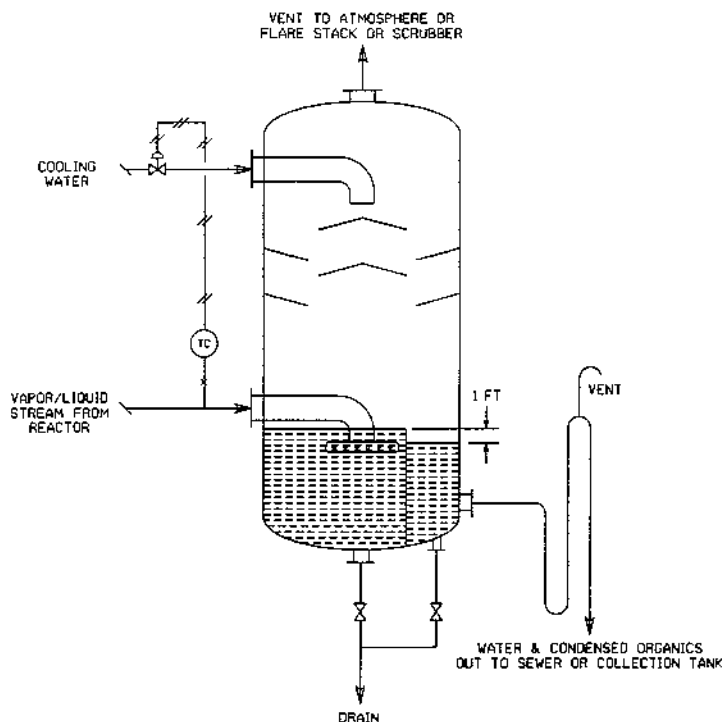


FIG. 23-56 Quench tank with direct-contact baffle tray section.

Equipment Selection Criteria and Guidelines A number of factors should be considered to determine when to select which type of vapor/gas/solid-liquid separator (blowdown drum or cyclone separator) to handle a multiphase stream from a relief device, and which final control or destruction equipment (emergency flare, emergency scrubber, or quench pool/catch tank) should be selected. These factors include the plot plan space available, the operating limitations of each type, and the physicochemical properties of the stream.

The selection of vapor/gas/solid-liquid separators and final control and destruction equipment can be helped by considering the advantages and disadvantages of each, as given below.

Gravity Separators (Horizontal and Vertical Drums/Catch Tanks)

Advantages:

- History of good performance.
 - Passive.
 - Excellent turndown up to the maximum design flow; lower vapor flow improves separation effectiveness.
 - Low pressure drop.
 - Can handle high liquid loading and provide large holdup of liquid.
 - Can be combined with secondary vapor-liquid separator such as vane impingement type.
 - Can handle high-viscosity liquids.
 - Moderate cost.
- Disadvantages:
- May be large.
 - Cannot economically remove low-density (e.g., light hydrocarbon) droplets smaller than about 150 μm .
 - Cannot handle stable foams.
 - Reacting liquids may continue to react in separator.

Cyclones

Advantages:

- Can remove over 99 percent above 20- μm droplet diameter in high liquid-loading situations.

- Some designs able to handle foamy liquids under some conditions.

- Passive.

- May be able to handle streams containing solids or with fouling tendencies.

- Relatively low capital cost.

Disadvantages:

- Performance not as well defined for emergency services as normal process cyclones.

- Little data on handling foamy or high-viscosity mixtures.

- May require higher pressure drop than gravity separators.

Quench Pool Tanks/Catch Tanks

Advantages:

- Passive, but must have system to verify proper liquid level, and composition if pool liquid is used as a neutralizing or kill liquid.
- Can handle two-phase, foamy, and high-viscosity liquids.
- Can handle high-flow-rate discharges.
- Can handle high liquid loading with special sparger design.
- Can contain reaction-stopping or -neutralizing chemicals and antifoam agents.
- Complete containment possible at low levels of noncondensables.
- Can use variety of quench liquids.

Disadvantages:

- Pilot-plant demonstration or scale-up test possibly needed.
- Requires sufficient discharge pressure to provide adequate pressure drop (at least 10 psig) across sparger and to overcome static head of liquid in pool.
- Sparger piping to be supported to control vibration and the suddenly applied load from the relief discharge.
- Some proven sparger designs are proprietary.
- May not be suitable for systems with slow to moderate reaction rates when used as an absorber.
- If vented, entrainment and vapor may be carried out of the vessel.

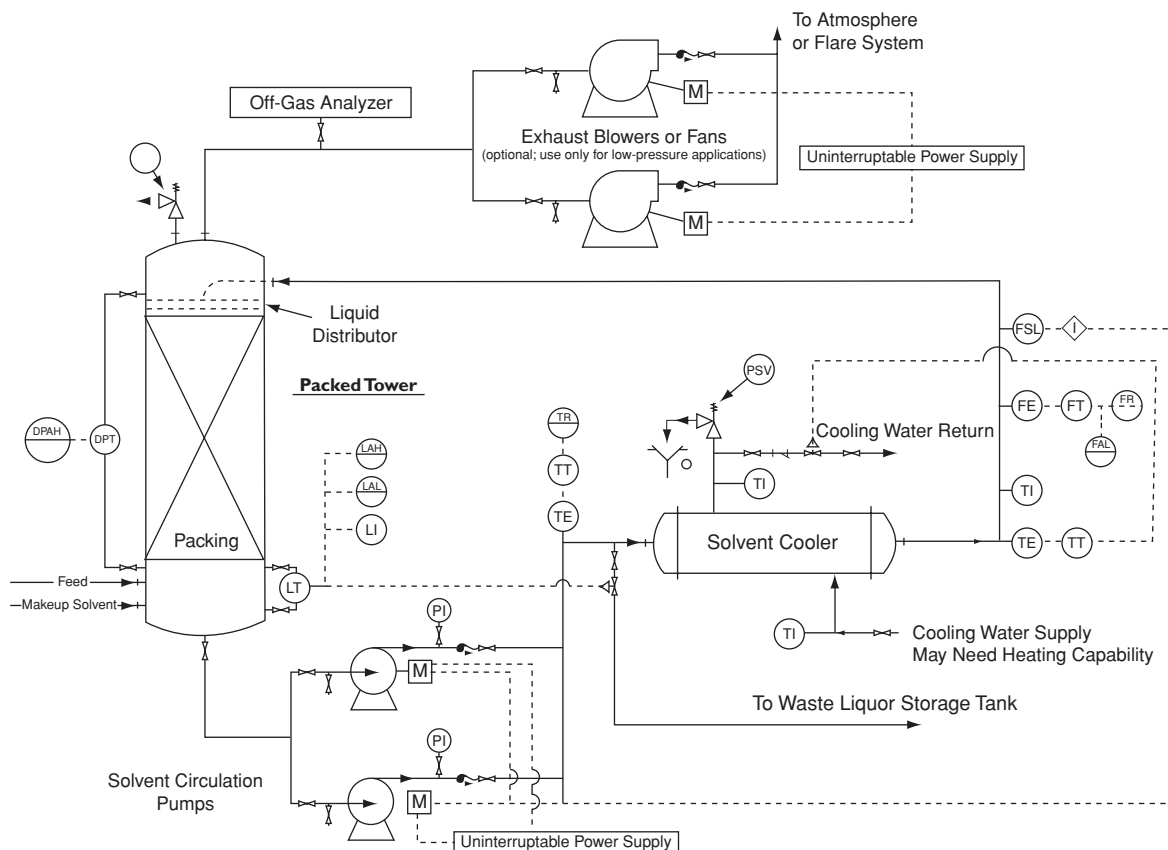


FIG. 23-57 Schematic of an emergency scrubber (absorber) system. (Guidelines for Pressure Relief and Effluent Handling Systems, *Center for Chemical Process Safety (CCPS) of the American Institute of Chemical Engineers (AIChE)*; copyright 1988 AIChE and reproduced with permission.).

- Noncondensables in the feed will lower recovery efficiency, and need to be vented.
- May require large pool to handle lower-boiling materials.

Emergency Flares

Advantages:

- Can destroy over 98 percent of combustibles.
- Experimental tests not required for most chemicals.
- Commercially proven technology.
- Very reliable—onstream over 99 percent.
- Good turndown ratio.
- Smokeless operation not needed for emergency flares (only required for continuous flares).
- No generation of appreciable liquid waste as would be the case for a quench pool or an emergency scrubber, but knockout drum required upstream of the flare to remove liquids and any condensate.
- May be used as final step after other control equipment, e.g., emergency scrubber or quench pool.

Disadvantages:

- May not be suitable for some corrosive chemicals (e.g., most chlorine-containing compounds).
- May be subject to flameout.
- Required minimum heating value of 300 Btu/SCF for nonassisted flares, or 200 Btu/SCF for assisted flares.
- Not passive; require continuous pilot fuel, electricity, oxygen analyzers for collection headers, and instrumentation.

- Require backup systems to monitor flame and provide pilot burning.
- Care needed not to feed foams or liquids into the flare stack.
- Because of maintenance of flare tip on elevated flares (and even some open ground and enclosed ground flares), crane or helicopter required possibly for access. Connection often needed for backup flare system, i.e., portable trailer flare or other temporary flare.

- Elevated flares with excessive light, noise, or odors a possible public nuisance in some situations and locales.

Emergency Scrubbers

Advantages:

- Sound basis for technology.
- High turndown ratio at constant scrubbing liquid flow rate.
- Countercurrent operation that can attain high absorption (removal) efficiency and low outlet concentration of solute.
- Can handle two-phase mixtures.
- May be able to handle unstable foams (to be confirmed by testing).
- Some types of scrubbers that can handle fouling streams, and baffle trays that can handle fouling and solids-containing streams.
- Can be used for neutralization of reactive materials.
- Generally used only for gas/vapor feed, but can handle liquid by use of a separator section in the bottom of the tower.

Disadvantages:

- Possible need for spent solvent to be treated or disposed of after the emergency release.

- Not passive. Utilities and process control instrumentation must be available at all times, and must have a sufficient and reliable supply of fresh solvent for the expected duration of the emergency.
- May need large diameter to avoid flooding.
- Possible restriction of tray towers to nonfouling materials that are free of suspended solids.
- May need continuous flow of scrubbing solvent for the scrubber to be effective at the start of the effluent flow.
- *Guidelines for Pressure Relief and Effluent Handling Systems* (AIChE-CCPS, 1998) contains more information about selection criteria and factors that affect them.

Sizing and Design of Equipment This subsection presents sizing methods and design criteria for the design of vapor/gas/solid-liquid separators for normal vapor-liquid systems (low-viscosity and nonfoamy or unstable foams). They are not applicable usually to high-viscosity (newtonian and nonnewtonian) liquids and/or systems that exhibit surface-active foaming behavior, as insufficient information is available at present relating to the separation efficiency for these types of equipment. Design information is also presented for quench pool tanks which usually can handle high-viscosity liquids as well as stable foams, and for final vapor/gas control or destruction equipment.

Vapor-Liquid Gravity Separator Design Fundamentals The critical factors in the performance of a horizontal separator are the vapor residence time and the settling rate of the liquid droplets. However, two other factors enter into the design—the vapor velocity must be limited to avoid liquid entrainment, and there must be sufficient freeboard within the vessel to allow for a feed distributor. For vertical separators, the design is based on a vapor velocity that must be less than the settling velocity of the smallest droplet that is to be collected, with due allowance for turbulence and maldistribution of the feed. The vapor residence time is a function of the vapor flow rate (mass), vapor density, and volume of vapor space in the separator, based on the following:

- The vapor flow rate (mass) is the sum of the vapor feed to the separator and any vapor generated by flashing of the liquid as the feed enters the separator.
- The vapor density depends on the compositions of the vapor fed and the vapor generated, and the pressure and temperature within the separator. The pressure in the separator is essentially the sum of the pressure drop from the outlet of the separator to the ultimate disposal point of the effluent and the pressure at the downstream discharge location (e.g., the atmosphere).
- The vapor volume is the total volume of the separator minus the volume of the liquid collected; and the volume of liquid is the volume collected (separated) from the feed minus the volume of liquid flashed when the static pressure decreases as the flow rate declines. The maximum volume of liquid collected is also a function of pressure—a greater volume is collected at higher pressure, and time.

Horizontal Blowdown Drum/Catch Tank (See Fig. 23-49.) The two main criteria used in sizing horizontal blowdown drums or catch tanks are as follows:

1. The diameter must be sufficient to provide good vapor-liquid separation.
2. The total volume must be sufficient to hold the estimated amount of liquid carryover from the reactor or process vessel. For a foamy discharge, the holding volume should be greater than the reactor or process vessel volume (to be discussed below).

The recommended method is from *Guidelines for Pressure Relief and Effluent Handling Systems* (AIChE-CCPS, 1998). It is an improvement over the method presented in the 7th edition of this Handbook. The procedure involves calculating a terminal velocity for a selected droplet size, then providing enough residence time in the vapor space to allow the droplets to fall from the top of the vessel to the level of liquid collected. Also, the vapor velocity in the separator must not exceed the value above which liquid may be entrained from the liquid surface in the separator. The tank is treated as a simple horizontal cylinder, neglecting the volume of liquid in the heads.

Other approaches for sizing a horizontal separator (blowdown drum) are given in *Guidelines for Pressure Relief and Effluent Handling Systems* (AIChE-CCPS, 1998) and API RP 521 (2007).

Vertical Blowdown Drum/Catch Tank (See Fig. 23-51.) Some of the basic equations used for sizing horizontal separators can be applied to vertical separators; however, the superficial vapor velocity for vertical separators is based on the total vessel cross-sectional area, and the vapor velocity must be less than the terminal settling velocity for the droplet size selected as the basis of design.

The recommended design procedure is also from *Guidelines for Pressure Relief and Effluent Handling Systems* (AIChE-CCPS, 1998). As with a horizontal separator, the assumed value of K is expected to be suitable for most vertical separators in emergency relief services. However, if it is necessary to remove droplets smaller than 300 to 600 μm , then lower values of K should be used.

Multireactor Knockout Drum/Catch Tank The area needed for vapor disengaging is calculated by the equations given earlier in the section on horizontal blowdown drums. The diameter and length (or height) are determined by considering a number of factors:

1. The length should be sufficient to extend beyond the locations of the reactors discharging into the vessel so as to simplify discharge pipe runs (for a horizontal vessel).
2. The height should not greatly exceed the height of the building (for vertical vessels).
3. The diameter should be sufficient to allow attenuation of the shock wave leaving the deflector plate.
4. The diameter should be sufficient to allow installation of the pipes and deflector plates in such a way as not to interfere directly with one another (particularly important for vertical vessels).
5. The cost of pressure vessels increases as the diameter increases.
6. An upper limit to the diameter is set by the need to transport complete cylindrical sections from the manufacturer to the site.
7. The volume of liquid in the reactor or reactors (assuming more than one vents at the same time) must be determined.

Mechanical Design Considerations The paper by Speechly et al. ("Principles of Total Containment System Design," presented at I. Chem. E. Northwestern Branch Meeting, 1979) discusses a number of pertinent design features:

1. Each vent device discharge pipe is extended into the vessel, and its end is fitted with a deflector device. This disperses the jet stream of solids (catalyst) and liquids discharged and dissipates this force, which should otherwise be exerted on the vessel wall immediately opposite.
2. The deflector device (baffle plate) must be carefully designed as described by Woods (*Proc. Inst. Mech. Engrs.* 180, Part 3j: 245–259, 1965–1966).
3. Isolate the catch tank from both reaction loads and forces generated by thermal expansion of the pipes; the pipes can be designed to enter the vessel through a sliding gland. Depending on layout, vessels that tend to have shorter, stiffer pipes between the building and the vessel may also require flexible bellows to be incorporated in the pipes.
4. There are usually several reactors linked to a single catch tank. To ensure that rupture of a disk on one reactor does not affect the others, each reactor is fitted with a double-rupture disk assembly. The use of double-rupture disks in this application requires installation of a leak detection device in the space between the two disks.

Gravity Separator Safety Considerations and Features A pressure relief device may be required on vapor-liquid gravity separators, based on the following assumptions:

- External fire exposure, if the vapor outlet is inadequately sized for fire relief.
- The vessel outlets could be closed.
- A chemical reaction could continue in the knockout drum/catch tank.

The pressure relief device should be evaluated and sized in accordance with the latest industry standards (i.e., DIERS, API, NFPA). An open passage (nozzle or line) from the separator to the atmosphere may be considered as the pressure relief device provided that it meets the relieving capacity requirements of the ASME Boiler and Pressure Vessel Code (BPVC).

Freeze protection is not normally provided in separators intended solely for emergency release service; however, occasionally, coils or a

jacket are used to keep high-viscosity liquids flowable, or to boil off volatile components. If internal coils or a heating jacket is used, it should be recognized that the source of heat might initiate a chemical reaction or increase its rate. Internal heating coils should be

- Designed for complete drainage.
- Provided with block valves to shut off the source of heat.
- Designed with a generous corrosion allowance.
- Braced to withstand impingement by the relief effluent stream.

Also the calculated liquid level in the separator should be corrected for the volume of the internal coils, and a heel of solvent might be retained in the separator to prevent freezing or to dissolve any components that can solidify or freeze.

Gravity Separator Vessel Design and Instrumentation Some of or all the following features might be appropriate in selected situations [also see API RP 521(2007)]:

- Liquid level indicator.
- High and low level alarms.
- Temperature indicator (with high-temperature alarm if deemed necessary).
- Pressure gauge.
- Pressure relief device.
- Manhole(s) for tank entry and cleaning.
- Pump for transferring the accumulated liquid to subsequent rework, recovery, or disposal.
- Antivortex baffle above the liquid outlet to the pump.
- Bracing of wall stiffeners to allow for possible "jet force" impingement on the vessel wall, and for intermittent buffeting and vibration if two-phase slug flow can occur.

For separators that normally operate at about atmospheric pressure, consider designing the vessel for 50 or 75 psig minimum MAWP to avoid vessel damage by an internal deflagration. For higher normal operating pressures, the design basis should be adjusted accordingly, consistent with the ASME BPVC, NFPA 68, and NFPA 69.

It should be realized that the separator may not be used for long periods, and the process control instrumentation and auxiliaries must receive regular inspection, testing, and maintenance.

Emergency Cyclone Separators Sizing of cyclone separators to handle unsteady-state discharge from emergency pressure relief devices is somewhat different from normal steady-state service (process cyclones). Emergency cyclones are designed usually for higher inlet velocity and pressure drop. Inlet velocity and cyclone pressure drop are limited only by the allowable backpressure on the pressure relief device. Emergency cyclone separator designs may allow some entrained mist or small droplets to leave the cyclone. This can be acceptable if the consequences of the release of the droplets are not significant.

Emergency cyclones often handle flashing vapor-liquid mixtures, and more vapor may be generated from flashing liquid because of the pressure drop in the cyclone; thus the mass flow rate of vapor leaving the cyclone is often greater than that entering. The cyclone size is determined by the outlet vapor volumetric flow rate. Since the discharge flow rate and ratio of vapor to liquid often vary, the cyclone design calculations should be made for conditions occurring at various times during the overpressure event.

Cyclone separator with separate catch tank (See Figs. 23-52 and 23-53.) The sizing of a cyclone knockout drum for emergency relief systems is somewhat different from that of a cyclone separator for normal process service for the following reasons:

1. In normal process service, the superficial vapor velocity at the inlet of tangential-entry vapor-liquid separators is limited to about 120 to 150 ft/s. Higher velocities may lead to
 - a. Excessive pressure drop in the separator and in the inlet piping.
 - b. Generation of fine mist in the inlet piping, which escapes collection in the separator.
2. Inlet velocity restrictions do not apply in the design of separators for emergency relief systems because
 - a. Pressure drop is usually not a penalty.
 - b. Escape of fine mist can usually be tolerated.

Sizing procedure The cyclone is sized by choosing a superficial F factor for the skirt in the range of 8.0 to 5.0 in USCS units (10 to 6 in

SI units). The higher value may be used for waterlike liquids; the lower value for liquids such as molasses. If design F factors exceed the range of 5 to 8, the liquid draining down the skirt is entrained and escapes with the vapor. These F factors were determined in small-scale lab experiments using water and a high-polymer solution as the test liquids. The high-polymer solution had a viscosity that was molasseslike, probably in the range of 1500 cP. There were no liquids of intermediate viscosity used in the tests.

The F factors of 5.0 and 8.0 are conservative in the opinion of the researcher who performed the experiments (private communication from E. I. duPont de Nemours Co., Inc., to the DIERS Project).

The sizing procedure is presented in *Guidelines for Pressure Relief and Effluent Handling Systems* (AIChE-CCPS, 1998). Figure 23-53 shows the dimensions of a cyclone separator designed in accordance with this procedure. If liquid is continuously drained from the cyclone to a separate accumulator, a vortex breaker and false bottom should be used (Fig. 23-53, view BB).

If the liquid contents of the vented vessel are to be retained in the separator for subsequent disposition, the holdup capacity may be increased by increasing the height of the vessel to increase the total volume by an amount equal to the vented liquid volume.

If the pressure drop is too high, it may be necessary to increase the inlet line size for a distance of about 3 to 5 diameters upstream of the cyclone.

A sizing method similar to the one recommended above is proposed by Schmidt and Giesbrecht (2001), based on experimental work done at BASF in Germany.

Cyclone separator with integral catch tank (See Fig. 23-54.) The diameter of the knockout drum is calculated by the criteria given in the preceding subsection and Fig. 23-53. Since the liquid is also to be retained in the vessel, extend the shell height below the normal bottom tangent line to increase the total volume by an amount equal to the volume of the liquid carried over.

Quench Tank (See Figs. 23-55 and 23-56.) There is very little information in the open technical literature on the design of quench tanks in the chemical industry. Most of the information available deals with the design of quench tanks (also called suppression pools) for condensation of steam or steam-water mixtures from nuclear reactor safety valves. Information and criteria from quench tanks in the nuclear industry often can be used for the design of quench tanks in the chemical industry. *Guidelines for Pressure Relief and Effluent Handling Systems* (AIChE-CCPS, 1998) provides more information for chemical industry quench tank design. The following subsections summarize some of this information. Pertinent criteria for quench tank sizing and design are presented below:

Operating pressure There are three modes of operation of a quench tank: atmospheric pressure operation, nonvented operation, and controlled venting operation. Atmospheric operation is usually feasible when the effluent being emitted has a bubble point well above the maximum ambient temperature. A very small quantity of vapor escapes with the air that is displaced as the tank fills with the emergency discharge (typically about 0.2 percent of the reactor contents). Depending on the toxic or flammable properties of the vapor, the vent from the quench tank can be routed to the atmosphere or must be sent to a scrubber or flare.

In nonvented operation, no material is vented to the atmosphere, and this design is used when complete containment of the discharge is required. It is also used when the discharge mixture bubble point is close to or below the maximum ambient temperature and the concentration of noncondensable gas in the feed stream is very low. The tank design pressure is relatively high since the initial air in the tank is compressed by the rising liquid level, adding to the vapor pressure. The designer must take into consideration that the quench tank backpressure must be limited so as not to adversely affect the reactor relief system.

In controlled venting operation, the quench tank pressure is maintained at a desired level by a pressure controller/control valve system or pressure relief valve. This mode of operation is used when the discharge mixture bubble point is close to or below the maximum ambient temperature and it is desired to limit the maximum quench tank pressure.

Quench liquid selection The choice of the appropriate quench liquid depends on a number of factors. Water is usually the first quench liquid to consider. If water is selected as the quench liquid, the tank should be located indoors, if possible, to avoid freezing problems if the facility is in a cold climate. If the tank has to be located outdoors in a cold climate, the addition of antifreeze is preferable to heat-tracing the tank, since overheating of the tank can result from tracing, thus reducing its effectiveness.

If other quench liquids are required, the liquid should have as many of the following properties as possible: compatibility with the discharge effluent, low vapor pressure, high specific heat, low viscosity, low flammability, low freezing point, high thermal conductivity, immiscibility with the discharge effluent, low cost, and ready availability.

Quench liquid quantity A good discussion of the factors determining the quantity of quench liquid required is presented by CCPS (AIChE-CCPS, 1998).

When water is used as the quench medium and the effluent stream is a hydrocarbon or organic, separate liquid phases are often formed. In this case, heat transfer is the predominating mechanism during the quench. To achieve effective heat transfer, there must be a sufficient difference between the quench liquid temperature and the bubble point of the incoming effluent stream. The minimum temperature difference occurs at the end of the discharge, when the quench pool temperature is highest. A rule of thumb, from industry practice, is to allow a 10 to 20°C (18 to 36°F) ΔT . For atmospheric tank operation, the final quench liquid temperature is then set 10 to 20°C (18 to 36°F) below the normal boiling point of the final quench pool mixture. For nonvented or controlled venting operation, the final boiling point is elevated, permitting a greater design temperature rise and the use of less quench liquid. Therefore, the quench pool final temperature must be set 10 to 20°C (18 to 36°F) lower than the saturated temperature of the discharge effluent at the design maximum quench tank pressure.

The minimum capacity of quench liquid can be estimated by a heat balance, given the final quench pool temperature, using an equation given by AIChE-CCPS (1998). An equation is also presented there that includes the heat of reaction. In some cases, an experiment is necessary to confirm that the reaction indeed stops in the quench pool. An equation is also presented for the heat balance needed to calculate the minimum amount of quench liquid when the feed contains noncondensable gases.

It is good practice to provide 10 to 20 percent more quench liquid than the minimum amount calculated.

Quench tank volume The total volume of the quench tank should be equal to the sum of the following volumes:

- Quench liquid required
- Liquid entering in the multiphase effluent stream
- Liquid condensed from vapors in entering effluent stream
- Freeboard for noncondensables (a minimum of 10 percent is recommended)

Quench tank geometry Quench tanks can have any of the following three types of geometry:

- Horizontal cylindrical vessel
- Vertical cylindrical vessel
- Concrete pit (usually rectangular)

Usually, the geometry is determined by space limitations. Both horizontal and vertical cylindrical vessels are designed as pressure vessels and for pressures up to 50 to 75 psig. An L/D ratio of 2 to 3 results in an economic design.

Sparger design The effluent stream should be discharged into the quench liquid by means of a sparger, which breaks it up into small jets to provide good heat and mass transfer. The sparger design must also incorporate the following capabilities:

- Maximize momentum-induced recirculation in the quench pool.
- Provide adequate flow area (cross section for pressure relief without imposing high backpressure).
- Minimize shock due to vapor bubble collapse.
- Minimize unbalanced momentum forces.

Figure 23-55 shows conventional quench tank sparger arrangements. As can be seen in this figure, the sparger can be of the following types:

- Vertical straight-pipe sparger
- Tee sparger
- Four-armed cross-sparger

Alternative sparger designs are shown in AIChE-CCPS (1998).

The following design criteria are recommended:

1. The flow area of the manifold and/or distributor piping should be at least 2 times the total area of the sparger holes. This generally ensures that flow through the sparger holes will be uniform.
2. The number of holes should provide at least 0.2 hole per square foot (2.2 holes per square meter) of pool cross-sectional area, distributed evenly over the cross section of the pool.
3. Holes should be uniformly spaced in the distributor pipe—adjacent holes should be spaced at least 2 to 3 hole diameters apart, center to center.
4. Holes should discharge from the bottom or bottom half of the distributor pipe so that they can serve also as drains for liquid. Separate drain holes may be needed for the manifold pipe and in each distributor pipe.
5. Holes should be as small as possible consistent with consideration of possible plugging or fouling when handling liquids that contain polymers or suspended solids. For effluent streams consisting of only liquid and vapor, hole diameters ranging from $\frac{1}{8}$ to $\frac{1}{2}$ in (3 to 13 mm) are typical. Larger hole diameters (up to 2 in) may be required if the blowdown stream contains solids (polymers and/or catalyst). Smaller holes tend to minimize vibration and water hammer.
6. Sonic hole velocity is desirable in smaller holes and is essential in $\frac{1}{2}$ - to 2-in holes. A minimum sparger pressure drop of 10 psi should be used.
7. The quencher arm should be anchored to prevent pipe whip. It should also extend to the length (for horizontal vessels) or the height (for vertical vessels) of the vessel to evenly distribute the vapors in the pool.

When quenching effluents discharged by safety valves, it is preferable to use a straight, vertical sparger with holes in the end cap as well as in the pipe sidewalls. This is recommended to minimize the possibility of liquid hammer, which can occur more readily in horizontal spargers. The liquid hammer usually occurs for the following reasons: As the relief valve opens for the first time, the pressure spike is cushioned by the air trapped in the vent line. This air is blown out. If the valve recloses, the line may cool, causing slugs of condensate to accumulate. When the valve reopens, the slugs will accelerate to very high velocities and impact any elbows and end caps of the sparger. In severe cases, the sparger-arm end caps can be knocked off. The preceding recommendation avoids turns, and the holes in the end cap provide some relief from the pressure spike.

Many aspects of quench pool design (vessel, spargers, mechanical design loads, etc.) and operation are covered in *Guidelines for Pressure Relief and Effluent Handling Systems* (AIChE-CCPS, 1998), and it should be consulted when a quench pool has to be designed.

Mass-Transfer Contact Section Where there is a strong possibility that not all the incoming vapors will be condensed in the pool (the feed stream contains a large amount of noncondensable gases), a direct-contact mass-transfer section is superimposed on the quench tank. This can be a baffle tray section (as shown in Fig. 23-56) or a packed column section.

The design of direct-contact mass-transfer columns is discussed in detail by Scheiman [*Petro Chemical Engr.* 37(3): 29–33, 1965; *ibid.* 37(4): 75, 78–79] and Fair (*Chem. Eng.*, June 12, 1972).

Emergency Flare Systems A discussion of design criteria, principles, and practices for a number of flare system components is presented below.

Flare tip diameter The flare tip diameter is sized on a velocity basis, but the pressure drop must also be checked. Flare tip velocities can be chosen on the basis of EPA requirements or API recommendations. EPA recommendations are given in the AIChE-CCPS book (1998). API RP 521 states that tips may be based on velocities of 0.5 Mach number or higher, if pressure drop, noise, and other factors permit.

Flare height and thermal radiation The height of an elevated flare is based on the minimum distance from the flare flame to an object whose exposure to thermal radiation must be limited. Industrial flares are normally designed so that personnel in the vicinity are not exposed to a heat intensity greater than 1500 to 2000 Btu/(h·ft²) when flaring at the maximum design rates.

An equation by Hajek and Ludwig (*Petro/Chem. Eng.* pp. C31–38, June 1960; pp. C44–51, July 1960) may be used to determine the distance required between a flare flame and a point of exposure where thermal radiation must be limited. [See API RP 521 (2007).]

A flame under the influence of wind will tilt in the direction that the wind is blowing and will change the location of the flame center. API RP 521 (2007) presents two methods for estimating the required flare height when the flame is tilted.

Other methods for estimating the required flare height are presented in *Guidelines for Pressure Relief and Effluent Handling Systems* (AIChE-CCPS, 1998).

Purge gas requirements Purging is used to prevent the formation of explosive mixtures in flare systems by preventing the admission of air into the system through leaks, backflow of air at the flare tip at very low flows, and back diffusion of air into the flare tip. Combustible gases such as methane or natural gas, or inert gases such as nitrogen or carbon dioxide, are frequently used for purging flare systems. API RP 521 (section 7.3.3.3) presents equations for calculating the purge rate as a function of the stack diameter, the desired oxygen level, and the type of purge gas.

Air (purge reduction) seals Air seals (also called purge reduction or gas seals) are often installed near the top of flare stacks. Others are located near the base of the flare. They are used to prevent air from entering the stack and are often recommended to prevent flashbacks and explosions. They also greatly reduce the amount of purge gas required. There are two basic types of air seals primarily used: diffusion (sometimes called molecular or labyrinth seals) and velocity types.

The diffusion seal consists of a baffled concentric cylinder arrangement, which uses the difference in molecular weight of the flare gas and ambient air to prevent air from entering the stack. Diffusion seals, combined with the purge gas flow rate, are intended to form a secondary line of defense against the entry of air into the flare stack. In a vertical flare stack, gravity exerts a driving force to increase the diffusion of air into the stack. Diffusion seals create an inverted flow field to reverse the gravitational effect and reduce air entry. Use of lighter-than-air purge gas (e.g., methane or natural gas)

creates a pocket of light gas at the top of the air entry path. Use of diffusion seal and lighter-than-air purge gas allows a much lower purge gas rate. Installation of a diffusion seal at the top of the flare, and immediately below the burner tip, reduces the purge gas required to 1/50 of the volume required if the diffusion seal is not used (Niemeyer and Livingston, *Chem. Eng. Prog.*, pp. 39–44, December 1993).

Velocity seals are more recent developments in air seal design. They use conical baffles to redirect and focus the purge gas flow field just below the flare tip to sweep air from the flare stack. Some velocity seal designs can reduce the purge gas flow rate requirement to about 1/10 of the rate needed without the seal. Also, some velocity seal designs reportedly require only about 25 to 33 percent of the purge gas used in diffusion seals (AIChE-CCPS, 1998). More details about air (purge reduction) seals may be found in API RP 521 (2007).

Liquid seal drums Emergency vent streams are usually passed through a liquid seal, commonly water, before going to the flare stack. The liquid seal drum is usually located downstream of the knockout drum, and some vendors' designs include them in the base of the flare stack. A liquid seal drum is used to maintain a positive pressure in the vent header system and upstream system. It also reduces the possibility of flame flashbacks, caused when air is inadvertently introduced into the flare system and the flame front pulls down into the stack; it also acts as a mechanical damper on any explosive shock wave in the flare stack. Figure 23-58 is a schematic of a typical flare stack liquid seal drum, designed per API RP 521 criteria.

A properly designed and operated liquid seal drum should allow gas to pass through the seal with minimum surging in gas flow and/or upstream gas pressure. The design of the liquid seal internals and the design of the vessel can have a significant impact on the ability of the seal to meet the performance objective. For example, a common design for the end of a dipleg pipe uses V notches cut into the end of the pipe wall. This design is less effective than the proprietary designs developed by vendors of liquid seal drums. These proprietary designs, which use alternative design guidelines, may offer

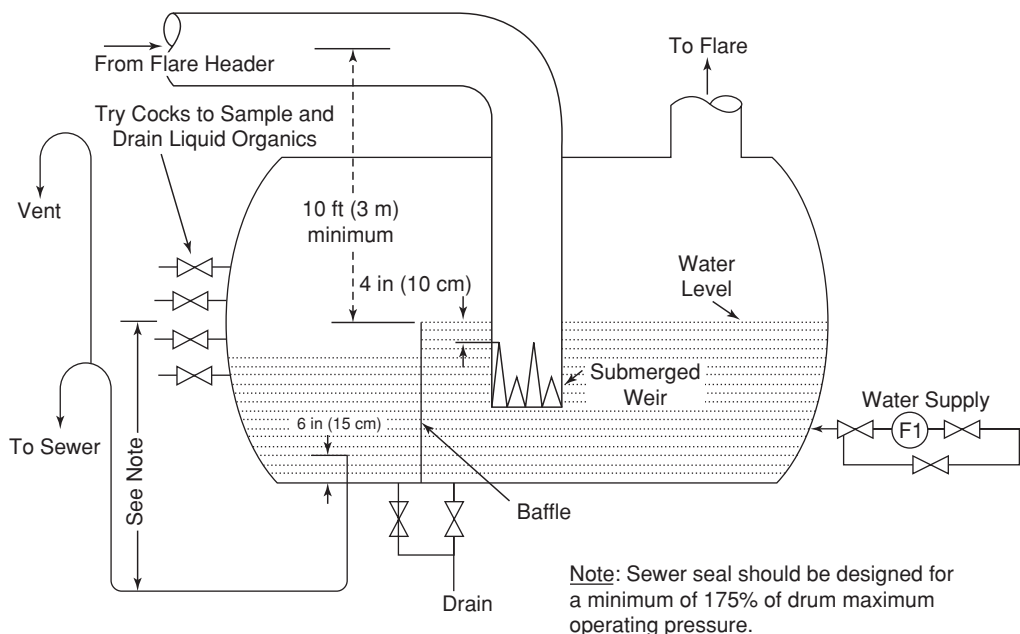


FIG. 23-58 Schematic for typical flare stack seal drum. (Adapted from API RP 521.) (Guidelines for Pressure Relief and Effluent Handling Systems, Center for Chemical Process Safety (CCPS) of the American Institute of Chemical Engineers (AIChE); copyright 1988 AIChE and reproduced with permission.).

economic or operational advantages. Details about several types of proprietary designs for liquid seal drums are presented by Grossel (*Deflagration and Detonation Flame Arresters*, AIChE-CCPS, New York, 2002).

More detailed discussions of flare systems and their components are presented in API RP 521 (2007), API STD 537 (2003), and AIChE-CCPS (1998).

Emergency Scrubbers (Absorbers) Emergency relief discharges are often passed through scrubbers (also called absorbers, or absorption columns or towers) for removal of flammable, corrosive, or toxic chemicals. The removal mechanism in some scrubbers involves physical absorption in a solvent, whereas in others chemical absorption (reactive scrubbing) is required.

Unlike quench pools for treatment of emergency relief effluent, scrubbers are not passive. For reliable performance, all critical components, including instruments, must be working correctly. There must be a reliable and sufficient supply of solvent and utilities (cooling water, electricity, etc.), with all process variables maintained under design conditions at all times. This includes proper liquid levels, liquid concentrations, flow rates, and reliable operation of ancillary equipment (e.g., pumps, blowers, etc.). General practice is to provide redundancy in pumps and blowers and utilities to ensure reliable scrubber operation (see Fig. 23-57). In the preferred mode of operation for hazardous materials, especially toxic gases, the scrubber is in a standby mode, with solvent flow maintained at all times. An alternate mode is to have the scrubber shut down (solvent not circulating) and then started up after detecting a release from the pressure relieving system. The consequences of the delay in starting the scrubber up after it has been shut down must be considered before accepting this mode of operation.

A discussion of emergency scrubber design and operating criteria and practices is given in *Guidelines for Pressure Relief and Effluent Handling Systems* (AIChE-CCPS, 1998).

FLAME ARRESTERS

GENERAL REFERENCES: [1] Howard, W. B., Precautions in Selection, Installation and Use of Flame Arresters, *Chem. Eng. Prog.*, April 1992. [2] Piotrowski, T., Specification of Flame Arresting Devices for Manifolder Low Pressure Storage Tanks, *Plant/Operations Progress*, vol. 10, no. 2, April 1991. [3] Roussakis and Lapp, A Comprehensive Test Method for In-Line Flame Arresters, *Plant/Operations Progress*, vol. 10, no. 2, April 1991. [4] Deflagration and Detonation Flame Arresters, Chapter 13, *Guidelines for Engineering Design for Process Safety*, Center for Chemical Process Safety, American Institute of Chemical Engineers, 1993. [5] Ibid. Chapter 15, Effluent Disposal Systems. [6] Howard, W. B., Flame Arresters and Flashback Preventers, *Plant/Operations Progress*, vol. 1, no. 4, 1982. [7] Thibault, P., Britton, L., and Zhang, F., Deflagration and Detonation of Ethylene Oxide Vapor in Pipelines, *Process Safety Progress*, vol. 19, no. 3, 2000. [8] Grossel, S. S., *Deflagration and Detonation Flame Arresters*, AIChE-CCPS Concept Book, 2002. [9] Britton, L. G., Using Maximum Experimental Safe Gap to Select Flame Arresters, *Process Safety Progress*, vol. 19, no. 3, 2000. [10] Britton, L. G., Using Heats of Oxidation to Evaluate Flammability Hazards, *Process Safety Progress*, vol. 21, no. 1, 2002. [11] Davies, M., and Heidermann, T., Investigation of Common Application Failures Proven by Life Field Testing of Endurance Burning Tested End-of-Line Flame Arresters, American Institute of Chemical Engineers, 37th Loss Prevention Symposium, 2003. [12] Brandes, E., and Redeker, T., Maximum Experimental Safe Gap of Binary and Ternary Mixtures, Fourth International Symposium on Hazards, Prevention and Mitigation of Industrial Explosions, IV ISHPMIE October 2002, pp. 207–213. [13] Förster, H., and Kersten, C., Investigation of Deflagrations and Detonations in Pipes and Flame Arresters by High Speed Framing, 4th International Symposium on Hazards, Prevention and Mitigation of Industrial Explosions (ISHPMIE), Bourges, France, October 21–25, 2002. [14] Lunn, G. A., The Maximum Experimental Safe Gap: The Effect of Oxygen Enrichment and the Influence of Reaction Kinetics, *Journal of Hazardous Materials*, 8 (1984), pp. 261–270, Elsevier Science Publisher B.V. Amsterdam. [15] European Standard EN 12574, *Flame Arrester Performance Requirements, Test Methods and Limits of Use*, European Committee for Standardization, Brussels, 2001.

General Considerations In 2002, the Center for Chemical Process Safety (CCPS) of the American Institute of Chemical Engineers (AIChE) published a Concept Book on this topic [8]. The book was intended to expand and update the coverage given in [4], providing extensive information on the history, technology, practice, and regulatory aspects of deflagration and detonation flame arresters.

Flame arresters are passive devices designed to prevent propagation of gas flames through pipelines. Typical applications are to prevent flames entering a system from outside (such as via a tank vent) or propagating within a system (such as from one tank to another). Flame arrestment is achieved by a permeable barrier, usually a metallic matrix containing narrow channels, which removes heat and free radicals from the flame fast enough to both quench it within the matrix and prevent reignition of the hot gas on the protected side of the arrester. These metallic matrices are known as *elements*. Some preliminary considerations for arrester selection and placement are as follows:

1. Identify the at-risk equipment and the potential ignition sources in the piping system to determine where arresters should be placed and what general type (deflagration or detonation, unidirectional or bidirectional) is needed.

2. Determine the worst-case gas mixture combustion characteristics, system pressure, and permissible pressure drop across the arrester, to help select the most appropriate element design. Not only does element design impact pressure drop, but also the rate of blockage due to particle impact, liquid condensation, and chemical reaction (such as monomer polymerization) can make some designs impractical even if in-service and out-of-service arresters are provided in parallel.

3. The possibility of a stationary flame residing on the arrester element surface should be evaluated, and the need for additional safeguards, should such an event occur, should be evaluated (see the subsection Endurance Burn).

4. Consider any material of construction limitations due to reactive or corrosive stream components, plus the impact of elevated temperature and pressure during flame events.

5. Consider upset conditions that could exceed the test conditions at which the arrester was certified. These include the gas composition with regard to concentration of sensitive constituents such as ethylene or hydrogen, maximum system pressure during an emergency shutdown, plus the maximum temperature and oxygen concentration. Under certain upset conditions such as a high-pressure excursion and/or elevated oxygen concentration [14], there may be no flame arrester available for the task.

6. Consider the type and location of the arrester with respect to ease of maintenance, particularly for large in-line arresters.

These questions address the type of arrester needed, the appropriate location, and the best design with respect to flow resistance, maintainability, and cost. It should be recognized that while flame arrester effectiveness is high, it is not 100 percent. To maximize effectiveness, attention should be given to proper selection, application, and maintenance of the device. Since arresters may fail on demand, it is good engineering practice to conduct layers of protection analysis to determine what additional mitigation may be required (Förster, H., *Flame Arresters—The New Standard and Its Consequences, Proceedings of the International ESMG Symposium, Part 2: Industrial Explosion Protection*, Nürnberg, Germany, March 27–29, 2001). For marine vapor control systems in the United States, flame arrester applications are regulated by the U.S. Coast Guard. For other applications, alternative test protocols and procedures have been developed by different agencies. Some arresters, such as hydraulic arresters and in-line types used to stop decomposition flames, have specialized applications for which general design and testing information is scarce. Where flame arresters are impractical, alternative strategies such as fast-acting valves, vapor suppression, and flammable mixture control should be considered. Dynamic systems such as fast-acting valves and suppression should be tested and certified as suitable for the required task.

Combustion: Deflagrations and Detonations A deflagration is a combustion wave propagating at less than the speed of sound as measured in the unburned gas immediately ahead of the flame front. Flame speed relative to the unburned gas is typically 10 to 100 m/s although owing to expansion of hot gas behind the flame, several hundred meters per second may be achieved relative to the pipe wall. The combustion wave propagates via a process of heat transfer and species diffusion across the flame front, and there is no coupling in time or space with the weak shock front generated ahead of it. Deflagrations typically generate maximum pressures in the range of 8 to 12 times the initial pressure. The pressure peak coincides with the flame front

although a marked pressure rise precedes it; thus the unburned gas is compressed as the deflagration proceeds, depending on the flame speed and vent paths available. The precompression of gas ahead of the flame front (also known as *cascading* or *pressure piling*) establishes the gas conditions in the arrester when the flame enters it, and hence affects both the arrestment process and the maximum pressure generated in the arrester body. A severe deflagration arrestment test involves placing a restricting orifice behind the arrester, which increases the degree of precompression. This is known as *restricted-end* deflagration testing. As discussed later, the *maximum experimental safe gap* (MESG) of the gas mixture may underestimate its tendency to defeat the arrester under such conditions. This is so because the current MESG test method minimizes precompression effects and hence the likelihood of autoignition on the "protected side" of the arrester. It is most important to consider the influences of pressure, temperature, and oxygen concentration on the safe gap. In addition the safe gap of mixtures is influenced by the chemical interaction of the different gases [12, 14].

As the deflagration flame travels through piping, its speed increases due to flow-induced turbulence and compressive heating of the unburned gas ahead of the flame front. Turbulence is especially enhanced by flow obstructions such as valves, elbows, and tees. Once the flame speed has attained the order of 100 m/s, a *deflagration-to-detonation transition* (DDT) can occur, provided that the gas composition is within the detonable limits, which lie inside the flammable limits. The travel distance for this to occur is referred to as the *run-up* distance for detonation. This distance varies with the gas mixture sensitivity and increases with pipe diameter. It is often difficult to estimate the run-up distance at which a DDT may occur in a piping system, and when to specify either a deflagration or a detonation flame arrester. However, some indication of this can be obtained from [13]. If the actual plant piping installation is different from that of the experimental configurations, then it is recommended that tests be conducted simulating the actual plant piping configurations to establish the DDT run-up length. The European Standard EN12 874 [15] limits the use of in-line deflagration arresters to $L/D = 50$ for group D and C applications and $L/D = 30$ for group B. Tabulated run-up distances are generally for straight pipe runs, and DDT can occur for much smaller distances in pipe systems containing flow obstructions. At the instant of transition, a transient state of *overdriven detonation* is achieved and persists for a distance of a few pipe diameters. Overdriven detonations propagate at speeds greater than the speed of sound (as measured in the burned gas immediately behind the flame front), and side-on pressure ratios (at the pipe wall) in the range of 50 to 100 have been measured. The peak pressure is variable depending on the amount of precompression during deflagration. A severe test for detonation-type flame arresters is to arrange for the arrester to encounter a series of overdriven detonations.

After the abnormally high velocities and pressures associated with DDT have decayed, a state of stable detonation is attained. A detonation is a combustion-driven shock wave propagating at the speed of sound as measured in the burned gas immediately behind the flame front. Since the speed of sound in this hot gas is much larger than that in the unburned gas or the ambient air, and the flame front speed is augmented by the burned gas velocity, stable detonations propagate at supersonic velocities relative to an external fixed point. A typical velocity for a stoichiometric gas mixture is 1800 m/s. The wave is sustained by chemical energy released by shock compression and ignition of the unreacted gas. The flame front is coupled in space and time with the shock front with no significant pressure rise ahead of the shock front. The high velocities and pressures associated with detonations require special element design to quench the high-velocity flames plus superior arrester construction to withstand the associated impulse loading. Since this entails narrower and/or longer element channels plus bracing of the element facing, both inherent pressure drop and the possibility of fouling of detonation arresters should be considered.

The problem of flame arrestment, of either deflagrations or detonations, depends on the properties of the gas mixture involved plus the initial temperature, pressure, and oxygen concentration. Gas mixture combustion properties cannot be quantified for direct use in

flame arrester selection, and only general characteristics can be assigned. For this reason flame arrester performance must be demonstrated by realistic testing. Such testing has demonstrated that arresters capable of stopping even overdriven detonations may fail under restricted end deflagration test conditions. It is important to understand the significance of the test conditions addressed and their possible limitations. It is recommended that users request a detailed test report from the flame arrester manufacturer and only an approved arrester tested to a recognized test standard be installed.

Combustion: Gas Characteristics and Sensitivity Combustion thermodynamic calculations allow determination of peak deflagration and detonation pressures, plus stable detonation velocity. The peak pressure calculation may be used to determine combustion product venting requirements although a conservative volume increase of 9:1 may be used for essentially closed systems. Other relevant gas characteristics are entirely experimental. The sensitivity to detonation depends on the detonable range and fundamental burning velocity, although no specific correlations or measures of sensitivity exist based on fundamental properties. National Electrical Code (NEC) groups are commonly used to rank gases for the purposes of flame arrester selection. By this method, group A gases (acetylene) are considered most sensitive to detonation (and most difficult to arrest) while group D gases (such as saturated hydrocarbons) are considered least sensitive and easiest to arrest. As formerly applied, successful testing of an arrester using one gas in a NEC Electrical Group was assumed to mean that the arrester would be suitable for all other gases in that group. As presently applied in various codes, a representative test gas from each group (such as hydrogen in group B, ethylene in group C, or propane in group D) is typically used for arrester certification, and its ability to arrest a different gas or gas mixture is deduced by comparing the respective maximum experimental safe gaps. The arrester is assumed to be suitable for any single gas or mixture having an MESG greater than or equal to that of the representative test gas. Although the arrester may be directly tested using the gas or gas mixture of interest, this is rarely carried out owing to cost. The MESG comparison procedure is currently applied to both deflagration and detonation arresters. Specific testing protocols are specified by the codes used. Since MESGs provide an independent ranking for arrester selection, reference to NEC groups is not essential. Nevertheless, arresters continue to be described in terms of the NEC group of the representative test gas used for certification. For example, an arrester might be described as a "group C (ethylene)" type. A detailed discussion of the use of MESGs is given in [9].

It is cautioned that there have been no systematic studies proving that arrester performance can be directly correlated with MESG, especially if the MESG for a mixture is estimated by using the Le Chatelier rule. An alternative way for calculating the MESG and safe gap of mixtures is given in [12]. Furthermore, major revisions to the MESG test method have caused many "historic" MESG values to increase significantly; the current test method minimizes compression of the unburned gas mixture and may therefore underestimate the likelihood of arrester failure via autoignition.

Corrosion Consideration should be given to possible corrosion of both the element material and the arrester housing, since corrosion may weaken the structure, increase the pressure drop, and decrease the effectiveness of the element. While the housing might be designed to have a corrosion allowance, corrosion of the element must be avoided by proper material specification. Common materials of construction include aluminum, carbon steel, ductile iron and 316 stainless steel housings, and aluminum or 316 stainless steel elements. While special materials such as Hastelloy might be used for situations such as high HCl concentrations, it may be more cost-effective to use a hydraulic arrester made of carbon steel and lined with a suitable polymeric lining in such applications. Also, at least one flame arrester manufacturer can provide flame arresters with PTFE (Teflon) arresting elements and lined housings up to a nominal size of 4 in for very corrosive service [8].

Directionality To select an arrester for any service, the potential sources of ignition must be established in relation to the pipe system and the equipment to be protected. The pipe connecting an arrester with an identified ignition source is the *unprotected side* of the arrester.

The pipe connecting the arrester with at-risk equipment is the *protected side*. If the arrester will encounter a flame arriving only from one direction, a *unidirectional* arrester can be used. If a flame may arrive from either direction a *bidirectional* arrester is needed. The latter either are symmetrically constructed or are certified by testing. Back-to-back use of unidirectional arresters will not usually be cost-effective unless testing reveals a specific advantage such as increased allowable operating pressure during restricted end deflagration testing.

Endurance Burn Under certain conditions a successfully arrested flame may stabilize on the unprotected side of an arrester element. Should this condition not be corrected, the flame will eventually penetrate the arrester as the channels become hot. An endurance burn time can be determined by testing, which specifies that the arrester has withstood a stabilized flame without penetration for a given period. The test should address either the actual or the worst-case geometry since heat transfer to the element will depend on whether the flame stabilizes on the top, bottom, or horizontal face. In general the endurance burn time identified by test should not be regarded as an accurate measure of the time available to take remedial action, since test conditions will not necessarily approximate the worst possible practical case. Temperature sensors may be incorporated at the arrester to indicate a stabilized flame condition and either alarm or initiate appropriate action, such as valve closure. It is very important to install an endurance burning flame arrester in the same way as tested to avoid malfunction [11].

Installation End-of-line arresters should be protected by using appropriate weather hoods or cowls. In-line arresters (notably detonation arresters and in-line deflagration arresters) must be designed to withstand the highest line pressure that might be seen, including upset conditions. The design should be verified by hydrostatic and pneumatic pressure tests. The piping system should be designed with adequate supports and should allow routine access to the arrester for inspection and maintenance.

Maintenance It is important to provide for arrester maintenance by both selection of the most suitable arrester type and judicious location. Inspection and maintenance should be performed on a regular basis depending on experience with the particular arrester in the service involved. It should also be carried out after successful function of the arrester. Some in-line designs allow removal, inspection, and cleaning of the element without the need to expand the line. Unit designs featuring multiple elements in parallel can reduce downtime by extending the period between cleaning. For systems that cannot be shut down during maintenance, parallel arresters incorporating a three-way valve may be used. Detonation arrester elements are especially prone to damage during dismantling, cleaning, and reassembly. Maintenance must be carefully done, avoiding sharp objects that could disable the delicate channels in the element. Spare elements should be available to reduce downtime and provisions made for storing, transporting, and cleaning the elements without damage.

Monitoring The differential pressure across the arrester element can be monitored to determine the possible need for cleaning. The pressure taps must not create a flame path around the arrester. It can be important to provide temperature sensors such as thermocouples at the arrester to detect flame arrival and stabilization. Since arrester function may involve damage to the arrester, the event of successful function (flame arrival) may be used to initiate inspection of the element for damage. If the piping is such that flame stabilization on the element is a realistic concern, action must be taken immediately upon indication of such stabilization (see also Endurance Burn). Such action may involve valve closure to shut off gas flow or inert gas purging.

Operating Temperature and Pressure Arresters are certified subject to maximum operating temperatures and absolute pressures normally seen at the arrester location. Arrester placement in relation to heat sources such as incinerators must be selected so that the allowable temperature is not exceeded, with due consideration for the detonation potential as run-up distance is increased. Flame arrester manufacturers can provide recommended distances from heat sources, such as open flames, to avoid thermal damage to a flame arrester element. If heat tracing is used to prevent condensation of liquids, the same temperature constraint applies. In the case of in-line arresters, there may

be certain upset conditions that produce unusually large system pressures outside the stipulated operating range of the arrester. Since the maximum operating pressure for a detonation arrester may be in the range of 16 to 26 psia, depending on the gas sensitivity and arrester design, it may be impossible to find a suitable arrester to operate during such an upset. The situation may be exacerbated by pressure drop across the device caused by high flow and/or fouling.

Pressure Drop Flow resistance depends on flame arrester channel arrangement and on a time-dependent fouling factor due to corrosion, or accumulation of liquids, particles or polymers, depending on the system involved. Monomer condensation is a difficult problem since inhibitors will usually be removed during monomer evaporation and catalysis might occur over particulates trapped in the element. Pressure drop can be a critical factor in operability, and cleaning may represent a large hidden cost. Sizing for pressure drop must be based on worst-credible-case operating conditions rather than normal operating conditions.

Fouling may be mitigated in a number of ways. First, the least sensitive element design can be selected, and in the case of end-of-line arresters, weather hoods or cowls can be used to protect against water or ice accumulation. Second, a fouling factor (20 percent or greater) might be estimated and an element with a greater tested flow capacity selected to reduce the pressure drop. This should be further increased if liquid condensation might occur. It is important that certified flow curves for the arrester be used rather than calculated curves since the latter can be highly optimistic. Condensation and polymerization may be mitigated by geometry (minimizing liquid accumulation in contact with the element) and provision for drainage. Alternatively, the arrester may be insulated and possibly heat-traced. Drains should not provide flame paths around the arrester or leak in either direction when closed. If heat tracing is used, the temperature must be limited to the certified operating range of the arrester.

In addition to using an arrester element with greater flow capacity, it is common to use two arresters in parallel where frequent cleaning is required, with one arrester in standby. A three-way valve can be used to allow uninterrupted operation during changeover. Where elements have an intrinsically high pressure drop, such as sintered metal elements used in acetylene service, multiple parallel elements can be used.

Deflagration Arresters The two types of deflagration arrester normally encountered are the *end-of-line arrester* (Figs. 23-59 and 23-60) and the *tank vent deflagration arrester*, which is mounted close to the end of the line going direct to the atmosphere. Neither type of arrester is designed to stop in-line deflagrations or detonations. If mounted sufficiently far from the atmospheric outlet of a piping system, which constitutes the unprotected side of the arrester, the flame can accelerate sufficiently to cause these arresters to fail. Failure can occur at high flame speeds even without a run-up to detonation.

If atmospheric tanks are equipped with flame arresters on the vents, fouling or blockage by extraneous material can inhibit gas flow to the degree that the tank can be damaged by underpressure. API standards allow the use of pressure vacuum (P/V) valves without flame arresters for free-venting tanks on the basis that the high vapor velocity in the narrow gap between pressure pallet (platter) and valve body will prevent flashback. However, it is important to ensure the pallet is not missing or stuck open since this will remove the protection. Absence of the pallet was a listed factor in the 1991 Coode Island fire (State Coroner Victoria, Case No. 2755/91, Inquest into Fire at Coode Island on August 21 and 22, 1991, Finding). Whether or not flame arresters are used, proper inspection and maintenance of these vent systems is required.

End-of-line arresters are mounted at the outlet of a pipe system and go directly to the atmosphere, so there is no potential for significant flame acceleration in the pipe. **Tank vent deflagration arresters** are strictly limited by the approval agency, but for group D gases they are typically mounted no more than 20 ft from the end of a straight pipe that vents directly to the atmosphere. The allowed distance must be established by proper testing with the appropriate gas mixture and the pipe diameter involved. Turbulence-promoting irregularities in the flow (bends, tees, elbows, valves, etc.) cannot be used unless testing has addressed the exact geometry. It is essential that run-up to detonation not occur in the

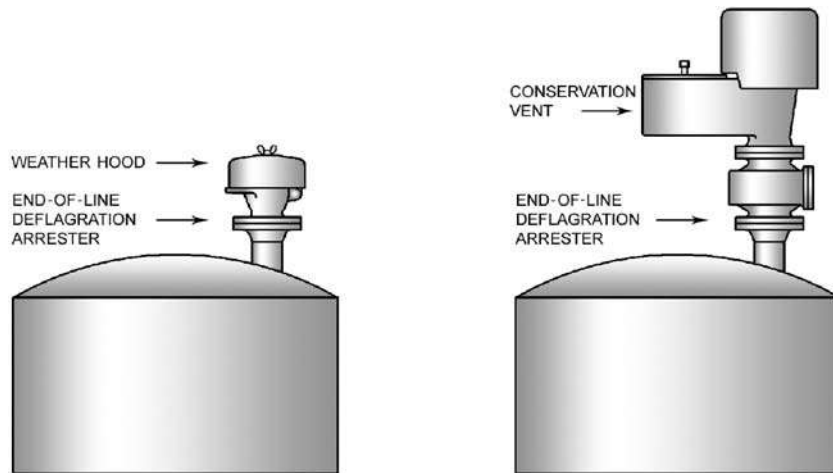


FIG. 23-59 Typical end-of-line deflagration arrester installations. (Courtesy of PROTEGO®.)

available piping system, and run-up distance can be less than 2 ft for some fast-burning gases such as hydrogen in air (group B). Thus the NEC grouping of the gas mixture must be considered. More importantly, it must be emphasized that even if run-up to detonation does not occur, a deflagration arrester can fail if the flame speed is great enough. Thus the run-up distance is not an adequate criterion for acceptable location, and this limitation can be determined only by realistic testing. A number of explosions have occurred due to misapplication of end-of-line or tank vent deflagration arresters where detonation arresters or in-line deflagration arresters should have been used. The latter are described in the next subsection.

Detonation and Other In-Line Arresters If the point of ignition is remote from the arrester location, the arrester is an in-line type such as might be situated in a vapor collection system connecting several tanks. Due to the possibility of DDT, most in-line arresters are designed to stop both deflagrations and detonations (including overdriven detonations) of the specified gas mixture. These are known as **detonation arresters**. Figure 23-61 shows a typical design. Detonation arresters may be further delineated into those types that will arrest only stable detonations and those that will arrest both stable and unstable detonations (i.e., overdriven detonations or DDT events).

In some cases, in-line arresters need to stop deflagrations only. However, in such cases it must be demonstrated that detonations

cannot occur in the actual pipework system; unless the gas mixture is intrinsically not capable of detonation, this requires full-scale testing using the exact pipe geometry to be used in practice, which must not be changed after installation. In certain exceptional cases, an in-line deflagration arrester may be mounted without regard to run-up distance. This can be done only where the system is known to be incapable of detonation. Examples are the decomposition flames of ethylene and ethylene oxide, which are briefly discussed under Special Arrester Types and Alternatives.

Detonation arresters are typically used in conjunction with other measures to decrease the risk of flame propagation. For example, in vapor control systems the vapor is often enriched, diluted, or inerted, with appropriate instrumentation and control [5]. In cases where ignition sources are present or predictable (such as most vapor destruct systems), the detonation arrester is used as a last-resort method anticipating possible failure of vapor composition control. Where vent collection systems have several vapor/oxidant sources, stream compositions can be highly variable and this can be additionally complicated when upset conditions are considered. It is often cost-effective to perform hazard analyses such as HAZOP or fault

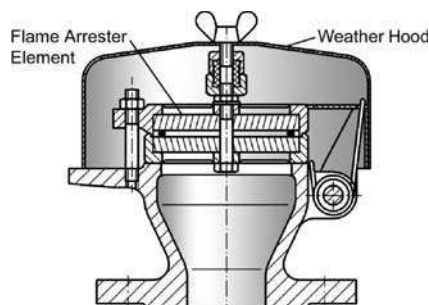


FIG. 23-60 Typical end-of-line deflagration arrester design. (Courtesy of PROTEGO®.)

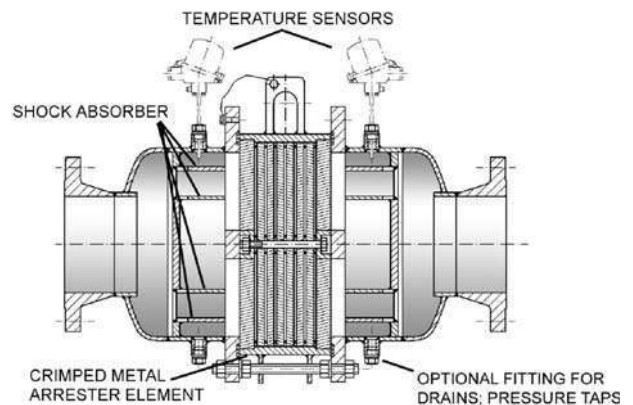


FIG. 23-61 Typical detonation arrester design (crimped ribbon type). (Courtesy of PROTEGO®.)

tree analysis to determine whether such vent streams can enter the flammable region and, if so, what composition corresponds to the worst credible case. Such an analysis is also suitable to assess alternatives to arresters.

Effect of Pipe Diameter Changes Arrester performance can be impaired by local changes in pipe diameter. It was shown that a minimum distance of 120 pipe diameters should be allowed between the arrester and any increase in pipe diameter; otherwise, a marked reduction in maximum allowable operating pressure would occur. This impairment was observed during detonation testing but was most pronounced during restricted-end deflagration testing (Lapp and Vickers, *Int. Data Exchange Symp. on Flame Arresters and Arrestment Technology*, Banff, Alberta, October 1992). As a rule, arresters should be mounted in piping either equal to or smaller than the nominal size of the arrester.

Venting of Combustion Products As gas deflagrates or detonates in the piping system, there is a volume expansion of the products and an associated pressure increase. In some instances where the pipe system volume involved is relatively large, a significant overpressure might be developed in the vapor spaces of connected tanks, especially when vapor space is minimal due to high liquid level. It can be assumed that all the gas on the unprotected side of the arrester is converted to equilibrium products; the pressure is relieved via gas expansion into the entire system volume and to the atmosphere via any vent paths present. If heat losses are neglected by the assumption of high flame speeds or detonation, and atmospheric venting paths are neglected, a conservative approach is that storage vessels be designed with a capacity to handle 9 times the pipe volume on the unprotected side of the arrester. With regard to the high pressures associated with detonations, it has been shown (Lapp, Independent Liquid Terminal Association Conference, Houston, June 23, 1992) that detonation arresters attenuate the peak detonation pressure by up to 96 percent depending on the arrester design, and therefore protect from much of the pressure pulse. To further reduce the pressure pulse, relief devices may be provided at the arrester.

Arrester Testing and Standards Regulatory and approval agencies and insurers impose *acceptance testing* requirements, sometimes as part of certification standards. The user may also request testing to demonstrate specific performance needs just as the manufacturer can help develop standards. These interrelationships have resulted in several new and updated performance test procedures. Listing of an arrester by a testing laboratory refers only to performance under a defined set of test conditions. The flame arrester user should develop specific application requirements based on the service involved and the safety and risk criteria adopted.

As discussed in [8], a variety of test procedures and use guidelines have been developed. General considerations are given in Chapter 9.3 of National Fire Protection Association Standard 69. The Federal Register, 33 CFR, Part 154, contains the USCG requirements for detonation arresters in marine vapor control systems. Other U.S. procedures are given in ASTM F 1273-91, UL 525, FM Procedure Classes 6061 and 7371, plus API Publications 2028 and 2210. For U.S. mining applications, the Mine Safety and Health Administration (MSHA) provides regulation and guidance; e.g., in CFR Title 30, Part 36. The International Maritime Organization (IMO) Standard MSC/Circ. 677 (1994) provides testing procedures for end-of-line deflagration and in-line detonation arresters for use on tanker ships. In Canada, CSA-Z343 is followed while in Europe the CEN Standard EN 12874 [15] has replaced previous European National Standards such as BS 7244.

Deflagration Arrester Testing For end-of-line, tank vent, and in-line deflagration flame arresters, approval agencies may require manufacturers to provide users with data for flow capacity at operating pressures, proof of success during an endurance burn or continuous flame test, evidence of flashback test results (for end-of-line arresters) or explosion test results (for in-line or tank vent arrester applications), hydraulic pressure test results, and results of a corrosion test.

Endurance burn testing generally implies that the ignited gas mixture and flow rate are adjusted to give the worst-case heating (based on temperature observations on the protected side of the element surface), that the burn continues for a specified duration, and flame penetration does not occur. Continuous flame testing implies a gas

mixture and flow rate are established at specified conditions and burn on the flame arrester for a specified duration. The endurance burn test is usually a more severe test than the continuous burn. In both cases the flame arrester attachment configuration and any connecting piping or valves should be installed in the same configuration used for testing. General reference [11] gives additional information.

Flashback tests incorporate a flame arrester on top of a tank with a large plastic bag surrounding the flame arrester. A specific gas mixture (e.g., propane, ethylene, or hydrogen at the most sensitive composition in air) flows through and fills the tank and the bag. Deflagration flames initiated in the bag (three at different bag locations) must not pass through the flame arrester into the protected vessel. On the unprotected side, piping and attachments such as valves are included as intended for installation; a series of tests, perhaps 10, is conducted.

Whatever the application, a user should be aware that not all test procedures are the same, or of the same severity, or use the same rating designations. Therefore, it is important to review the test procedure and determine whether the procedure used is applicable to the intended installation and potential hazard the flame arrester is meant to prevent.

Detonation Arrester Testing Requirements are described by various agencies in the documents listed above (UL 525, etc.). For installations governed by the USCG in Appendix A of 33 CFR, Part 154 (Marine Vapor Control Systems), the USCG test procedures must be followed. These are similar but not identical to those of other agencies listed. The European Union mandates arrester testing by an approved testing laboratory according to the EN 12874 Standard. Reference [8] discusses differences between the requirements of disparate agencies.

Detonation arresters are extensively tested for proof of performance against deflagrations, detonations, and endurance burns. In the United States, arrester manufacturers frequently test detonation arresters according to the USCG protocol; other test standards might alternatively or additionally be met. Under this protocol, the test gas must be selected to have either the same or a lower MESH than the gas in question. Typical MESH benchmark gases are stoichiometric mixtures of propane, hexane, or gasoline in air to represent group D gases having an MESH equal to or greater than 0.9 mm, and ethylene in air to represent group C gases with an MESH no less than 0.65 mm. Commercially available arresters are typically certified for use with one or another of these benchmark gas types. An ethylene-type arrester is selected, should the gas in question have an MESH of less than 0.9 mm but not less than 0.65 mm. Five low and five high overpressure deflagration tests are required with and without a flow restriction on the protected side. Of these 20 tests, the restricted-end condition is usually the most severe and often limits the maximum initial pressure at which the arrester will be suitable. Five detonation tests and five overdriven detonation tests are also required, which may involve additional run-up piping and turbulence promoters to achieve DDT at the arrester. If these tests are successful, an endurance burn test is required. This test does not use propane for group D gases but hexane or gasoline, owing to their lower autoignition temperatures. For group C tests, ethylene can be used for all test stages.

Care must be taken when applying the MESH method [4, 9, 12]. The user has the option to request additional tests to address such concerns and may wish to test actual stream compositions rather than simulate them on the basis of MESH values.

Special Arrester Types and Alternatives

Hydraulic (Liquid Seal) Flame Arresters Hydraulic (liquid seal) flame arresters are most commonly used in large-pipe-diameter systems where fixed-element flame arresters are either cost-prohibitive or otherwise impractical (e.g., very corrosive gas or where the gas contains solid particles that would quickly plug a conventional arrester element). These arresters contain a liquid, usually water-based, to provide a flame barrier. Figure 23-62 shows one design. Realistic tests are needed to ensure performance, as described in EN 12874 [15]. Note that hydraulic flame arresters may fail at high flow rates, producing a sufficiently high concentration of gas bubbles to allow transmission of flame. This is distinct from the more obvious failure mode caused by failure to maintain adequate liquid level.

Alternatives to Arresters Alternatives to the use of flame arresters include fast-acting isolation valves, vapor suppression systems,

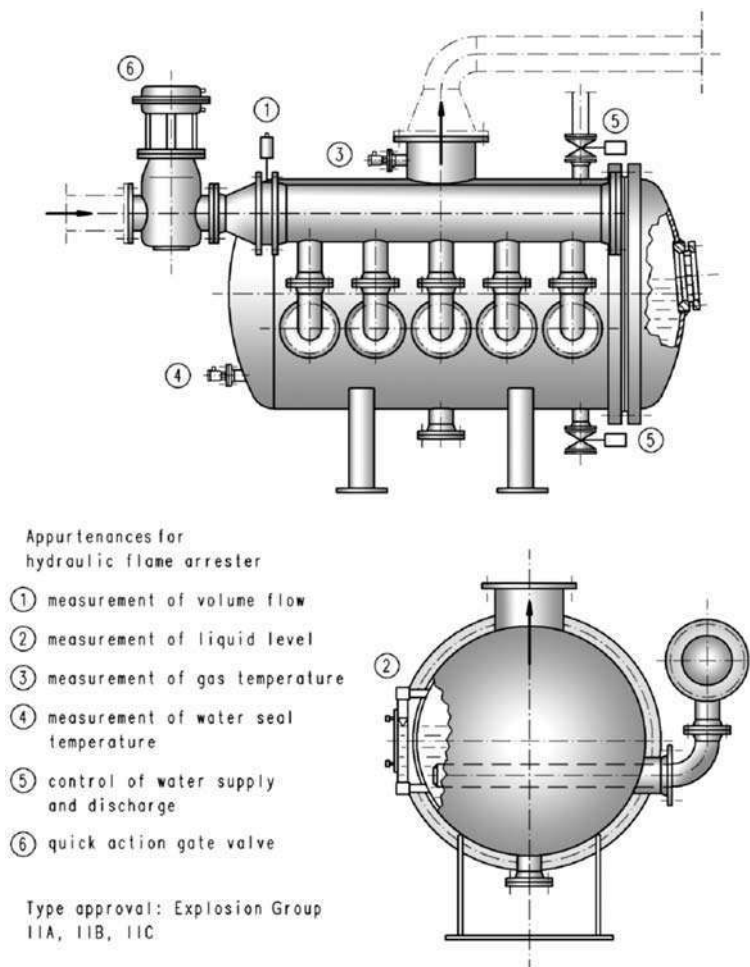


FIG. 23-62 Tested and approved hydraulic (liquid seal) flame arrester. (Courtesy of PROTEGO®.)

velocity-type devices in which gas velocity is designed to exceed flashback velocity, and control of the flammable mixture (NFPA 69 Standard, "Explosion Prevention Systems"). The latter alternative frequently involves reduction of oxygen concentration to less than the limiting oxygen concentration (LOC) of the gas stream.

STORAGE AND HANDLING OF HAZARDOUS MATERIALS

GENERAL REFERENCES: API-620, *Design and Construction of Large, Welded, Low-Pressure Storage Tanks*, American Petroleum Institute, Washington. AP-40, *Air Pollution Engineering Manual*, 2d ed., U.S. Environmental Protection Agency, Office of Air Quality Planning and Standards, 1973. AP-42, *Compilation of Emission Factors for Stationary Sources*, 5th ed., U.S. Environmental Protection Agency, Office of Transportation and Air Quality, 1995. *API Standards*, American Petroleum Institute, Washington. ASME, *Process Piping: The Complete Guide to ASME B31.3*, 2d ed., American Society of Mechanical Engineers, New York, 2004. ASME, *ASME Boiler and Pressure Vessel Code: ASME Code for Pressure Piping*; *ASME General and Safety Standards*; *ASME Performance Test Codes*, American Society of Mechanical Engineers, New York. *Chemical Exposure Index*, 2d ed., AIChE, New York, 1994. *Code of Federal Regulations*, Protection of Environment, Title 40, Parts 53 to 80, Office of the Federal Register, Washington. CGA, *Handbook of Compressed Gases*, 4th ed., Compressed Gas Association, Chantilly, Va., 1999. CCPS, *Guidelines for Chemical Process Quantitative Risk Analysis*, 2d ed., CCPS-AIChE, New York, 2000. CCPS, *Guidelines for Engineering Design for Process Safety*. CCPS, *Guidelines*

for Facility Siting and Layout, CCPS-AIChE, New York, 2003. CCPS, *Guidelines for Process Safety in Batch Reaction Systems*, CCPS-AIChE, New York, 1999. CCPS, *Guidelines for Safe Storage and Handling of High Toxic Hazard Materials*, CCPS-AIChE, New York, 1988. CCPS, *Guidelines for Safe Storage and Handling of Reactive Materials*, CCPS-AIChE, New York, 1995. CCPS, *Guidelines for Mechanical Integrity Systems*, Wiley, New York, 2006. Englund, "Opportunities in the Design of Inherently Safer Chemical Plants," in J. Wei et al., eds., *Advances in Chemical Engineering*, vol. 15, Academic Press, 1990. Englund, "Design and Operate Plants for Inherent Safety," *Chem. Eng. Prog.*, pts. 1 and 2, March and May 1991. Englund, Mallory, and Grinwis, "Preventing Backflow," *Chem. Eng. Prog.*, February 1992. Englund and Grinwis, "Redundancy in Control Systems," *Chem. Eng. Prog.*, October 1992. Fisher et al., "Emergency Relief System Design Using DIERS Technology: The Design Institute for Emergency Relief Systems (DIERS) Project Manual," AIChE, New York, 1992. Grossel and Crowl, *Handbook of Highly Toxic Materials Handling and Management*, Marcel Dekker, New York, 1995. Hendershot, "Alternatives for Reducing the Risks of Hazardous Material Storage Facilities," *Environ. Prog.*, 7, August 1988, pp. 180ff. Kletz, *An Engineer's View of Human Error*, Institution of Chemical Engineers, VCH Publishers, New York, 1991. Kletz, "Friendly Plants," *Chem. Eng. Prog.*, July 1989, pp. 18-26. Kletz, *Plant Design for Safety: A User Friendly Approach*, Hemisphere Publishing, London, 1991. Kohan, *Pressure Vessel Systems: A User's Guide to Safe Operations and Maintenance*, McGraw-Hill, New York, 1987. Mannan, *Lees' Loss Prevention in the Process Industries*, 3d ed., Elsevier, Amsterdam, 2005. Prokop, "The Ashland Tank Collapse," *Hydrocarbon Processing*, May 1988. Russell and Hart, "Underground Storage Tanks, Potential for Economic Disaster," *Chemical Engineering*, March 16, 1987, pp. 61-69. Ventsorb for Industrial Air Purification,

Bulletin 23-56c, Calgon Carbon Corporation, Pittsburgh, Pa., 1986. White and Barkley, "The Design of Pressure Swing Adsorption Systems," *Chem. Eng. Prog.*, January 1989.

Introduction The inherent nature of most chemicals handled in the chemical process industries is that they each have physical, chemical, and toxicological hazards to a greater or lesser degree. This requires that these hazards be contained and controlled throughout the entire life cycle of the facility, to avoid loss, injury, and environmental damage. The provisions that will be necessary to contain and control the hazards will vary significantly depending on the chemicals and process conditions required.

Established Practices Codes, standards, regulatory requirements, industry guidelines, recommended practices, and supplier specifications have all developed over the years to embody the collective experience of industry and its stakeholders in the safe handling of specific materials. These should be the engineer's first resource in seeking to design a new facility.

The ASME Boiler and Pressure Vessel Code, Section VIII, is the standard resource for the design, fabrication, installation, and testing of storage tanks and process vessels rated as pressure vessels (i.e., above 15-psig design). ASME B31.3 is a basic resource for process piping systems.

Examples of established practices and other resources—some of which pertain to the safe storage and handling of specific hazardous chemicals, classes of chemicals, or facilities—include those listed in Table 23-29 from the publications of two U.S. organizations, the NFPA and the Compressed Gas Association (CGA). Other organizations that may have pertinent standards include the International Standards Organization (ISO), the American National Standards Institute (ANSI), ASTM International (Conshohocken, Pa; www.astm.org), and other well-established national standards such as British Standards and Deutsches Institut für Normung e.V. (DIN) standards. Local codes and regulations should be checked for applicability, and the latest version should always be used when employing established practices.

Basic Design Strategies The storage and handling of hazardous materials involve risks that can be reduced to very low levels by good planning, design, and management practices. Facilities that handle hazardous materials typically represent a variety of risks, ranging from small leaks, which require prompt attention, to large releases, which are extremely rare in well-managed facilities but which have the potential for widespread impact (CCPS, 1988). It is essential that good techniques be developed for identifying significant hazards and mitigating them where necessary. Hazards can be identified and evaluated by using approaches discussed in the section on hazard and risk analysis.

Loss of containment due to mechanical failure or improper operation is a major cause of chemical process incidents. The design of storage and piping systems should be based on minimizing the likelihood of loss of containment, with the accompanying release of hazardous materials, and on limiting the amount of the release. An effective emergency response program that can reduce the impacts of a release should be available.

Thus, the basic design strategy for storing and handling hazardous materials can be summarized as follows, with reference to other parts of this section in parentheses:

1. Understand the hazardous properties of the materials to be stored and handled (Flammability, Reactivity, Toxicity, Other Hazards), as well as the physical hazards associated with the expected process design.

2. Reduce or eliminate the underlying hazards as much as is feasible (Inherently Safer and More User-Friendly Design).

3. Evaluate the potential consequences associated with major and minor loss-of-containment events and other possible emergency situations involving the hazardous materials and energies; and take this information into account in the process of site selection and facility layout and the evaluation of the adequacy of personnel, public, and environmental protection (Source Models, Atmospheric Dispersion, Estimation of Damage Effects).

4. Design and build a robust and well-protected primary containment system following codes, standards, regulations, and other established practices (Security).

TABLE 23-29 Examples of Established Practices Related to Storage and Handling of Hazardous Materials

Designation	Title
National Fire Protection Association (Quincy, Mass.; www.nfpa.org)	
NFPA 30	Flammable and Combustible Liquids Code
NFPA 30B	Code for the Manufacture and Storage of Aerosol Products
NFPA 36	Standard for Solvent Extraction Plants
NFPA 45	Standard on Fire Protection for Laboratories Using Chemicals
NFPA 53	Recommended Practice on Materials, Equipment and Systems Used in Oxygen-Enriched Atmospheres
NFPA 55	Standard for the Storage, Use, and Handling of Compressed Gases and Cryogenic Fluids in Portable and Stationary Containers, Cylinders, and Tanks
NFPA 58	Liquefied Petroleum Gas Code
NFPA 59A	Standard for the Production, Storage, and Handling of Liquefied Natural Gas (LNG)
NFPA 68	Guide for Venting of Deflagrations
NFPA 69	Standard on Explosion Prevention System
NFPA 318	Standard for the Protection of Semiconductor Fabrication Facilities
NFPA 326	Standard for the Safeguarding of Tanks and Containers for Entry, Cleaning, or Repair
NFPA 329	Recommended Practice for Handling Releases of Flammable and Combustible Liquids and Gases
NFPA 400	Hazardous Chemical Code
NFPA 430	Code for the Storage of Liquid and Solid Oxidizers
NFPA 432	Code for the Storage of Organic Peroxide Formulations
NFPA 434	Code for the Storage of Pesticides
NFPA 484	Standard for Combustible Metals, Metal Powders, and Metal Dusts
NFPA 490	Code for the Storage of Ammonium Nitrate
NFPA 495	Explosive Materials Code
NFPA 497	Recommended Practice for the Classification of Flammable Liquids, Gases, or Vapors and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas
NFPA 499	Recommended Practice for the Classification of Combustible Dusts and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas
NFPA 654	Standard for the Prevention of Fire and Dust Explosions from the Manufacturing, Processing, and Handling of Combustible Particulate Solids
NFPA 655	Standard for Prevention of Sulfur Fires and Explosions
NFPA 704	Standard System for the Identification of the Hazards of Materials for Emergency Response
Compressed Gas Association (Chantilly, Va.; www.cganet.com)	
CGA G-1	Acetylene
CGA G-2	Anhydrous Ammonia
CGA G-3	Sulfur Dioxide
CGA G-4	Oxygen
CGA G-5	Hydrogen
CGA G-6	Carbon Dioxide
CGA G-8.1	Standard for Nitrous Oxide Systems at Consumer Sites
CGA G-12	Hydrogen Sulfide
CGA G-14	Code of Practice for Nitrogen Trifluoride (EIGA Doc. 92/03)
CGA P-1	Safe Handling of Compressed Gases in Containers
CGA P-8	Safe Practices Guide for Cryogenic Air Separation Plants
CGA P-9	The Inert Gases: Argon, Nitrogen, and Helium
CGA P-12	Safe Handling of Cryogenic Liquids
CGA P-16	Recommended Procedures for Nitrogen Purging of Tank Cars
CGA P-32	Safe Storage and Handling of Silane and Silane Mixtures
CGA P-34	Safe Handling of Ozone-Containing Mixtures Including the Installation and Operation of Ozone-Generating Equipment
CGA S-1.1	Pressure Relief Device Standards—Part 1—Cylinders for Compressed Gases
CGA S-1.2	Pressure Relief Device Standards—Part 2—Cargo and Portable Tanks for Compressed Gases
CGA S-1.3	Pressure Relief Device Standards—Part 3—Stationary Storage Containers for Compressed Gases

NOTE: Always check the latest edition when using established practices.

5. Design and implement a reliable and fault-tolerant basic process control system to ensure the design limitations of the primary containment system are not exceeded.

6. Include provisions for detecting abnormal process conditions and bringing the process to a safe state before an emergency situation occurs (Safety Instrumented Systems).

7. Design, install, and maintain reliable and effective emergency relief systems, as well as mitigation systems such as secondary containment, deluge, and suppression systems, to reduce the severity of consequences in the event an emergency situation does occur (Pressure Relief Systems; Emergency Relief Device Effluent Collection and Handling).

8. Evaluate the risks associated with the process and its safety systems taken as a whole, including consideration of people, property, business, and the environment, that could be affected by loss events; and determine whether the risks have been adequately reduced (Hazard Analysis, Risk Analysis, Source Models, Atmospheric Dispersion, Estimation of Damage Effects).

9. Take human factors into account in the design and implementation of the control system and the facility procedures (Human Error, Key Procedures).

10. Ensure staffing, training, inspections, tests, maintenance, and management of change are all adequate to maintain the integrity of the system throughout the facility lifetime (Key Procedures, Audit Process).

Designers and operating companies will address these items in different ways, according to their established procedures. The steps that are addressed elsewhere in this section are not repeated here.

Site Selection, Layout, and Spacing Facility siting decisions that will have critical, far-reaching implications are made very early in a new facility's life cycle, or in the early planning stages of a site expansion project. The degree of public and regulatory involvement in this decision-making process, as well as the extent of prescriptive requirements and established practices in this area, varies considerably among countries, regions, and companies. Insurance carriers are also generally involved in the process, particularly with regard to fire protection considerations.

From the perspective of process safety, key considerations with respect to site selection, layout, and spacing can be summarized as

- Where on-site personnel (including contractors and visitors), critical equipment, the surrounding public, and sensitive environmental receptors are located with respect to hazardous materials and processes
- Whether the design and construction of control rooms and other occupied structures, as well as detection, warning, and emergency response provisions, will provide adequate protection in the event of a major fire, explosion, or toxic release event

Recommended distances for spacing of buildings and equipment for fire protection were issued as IRI IM.2.5.2, *Plant Layout and Spacing for Oil and Chemical Plants* (Industrial Risk Insurers, Hartford, Conn). These are referenced in "Typical Spacing Tables" included as Appendix A of the CCPS *Guidelines for Facility Siting and Layout* (2003). Other resources pertaining to siting and layout include

- *Dow's Fire & Explosion Index Hazard Classification Guide*, 7th ed. (AIChE, New York, 1994), which gives an empirical radius of exposure and damage factor based on the quantity and characteristics of the material being stored and handled
- API RP 752, "Management of Hazards Associated With Location of Process Plant Buildings," 2d ed. (American Petroleum Institute, Washington, 2003), which gives a risk-based approach to evaluating protection afforded by occupied structures

Storage

Storage Facilities Dating back to at least 1974, when a vapor cloud explosion in Flixborough, U.K., claimed 28 lives and destroyed an entire chemical plant (Mannan, 2005), a major emphasis in the safe storage and handling of hazardous materials has been to reduce hazardous material inventories. Inventory reduction can be accomplished not only by using fewer and smaller storage tanks and vessels but also by eliminating any nonessential intermediate storage vessels and

batch process weigh tanks and generating hazardous materials on demand when feasible. Note, however, that reduction of inventory may require more frequent and smaller shipments and improved management.

There may be more chances for errors in connecting and reconnecting with small shipments. Quantitative risk analysis of storage facilities has revealed solutions that may run counter to intuition. [Schaller, *Plant/Oper. Prog.* 9(1), 1990]. For example, reducing inventories in tanks of hazardous materials does little to reduce risk in situations where most of the exposure arises from the number and extent of valves, nozzles, and lines connecting the tank. Removing tanks from service altogether, on the other hand, generally helps. A large pressure vessel may offer greater safety than several small pressure vessels of the same aggregate capacity because there are fewer associated nozzles and lines. Also, a large pressure vessel is inherently more robust, or it can economically be made more robust by deliberate overdesign than can a number of small vessels of the same design pressure. On the other hand, if the larger vessel has larger connecting lines, the relative risk may be greater if release rates through the larger lines increase the risk more than the inherently greater strength of the vessel reduces it. In transporting hazardous materials, maintaining tank car integrity in a derailment is often the most important line of defense in transportation of hazardous materials.

Safer Storage Conditions The hazards associated with storage facilities can often be reduced significantly by changing storage conditions. The primary objective is to reduce the driving force available to transport the hazardous material into the atmosphere in case of a leak (Hendershot, 1988). Some methods to accomplish this follow.

Dilution Dilution of a low-boiling hazardous material reduces the hazard in two ways:

1. The vapor pressure is reduced. This has a significant effect on the rate of release of material boiling at less than ambient temperature. It may be possible to store an aqueous solution at atmospheric pressure, such as aqueous ammonium hydroxide instead of anhydrous ammonia.
2. In the event of a spill, the atmospheric concentration of the hazardous material will be reduced, resulting in a smaller hazard downwind of the spill.

Refrigeration Loss of containment of a liquefied gas under pressure and at atmospheric temperature causes immediate flashing of a large proportion of the gas. This is followed by slower evaporation of the residue. The hazard from a gas under pressure is normally much less in terms of the amount of material stored, but the physical energy released if a confined explosion occurs at high pressure is large.

Refrigerated storage of hazardous materials that are stored at or below their atmospheric boiling points mitigates the consequences of containment loss in three ways:

1. The rate of release, in the event of loss of containment, will be reduced because of the lower vapor pressure in the event of a leak.
2. Material stored at a reduced temperature has little or no superheat, and there will be little flash in case of a leak. Vaporization will be mainly determined by liquid evaporation from the surface of the spilled liquid, which depends on weather conditions.
3. The amount of material released to the atmosphere will be further reduced because liquid entrainment from the two-phase flashing jet resulting from a leak will be reduced or eliminated.

Refrigerated storage is most effective in mitigating storage facility risk if the material is refrigerated when received.

The economics of storage of liquefied gases are such that it is usually attractive to use pressure storage for small quantities, pressure or semirefrigerated storage for medium to large quantities, and fully refrigerated storage for very large quantities. Quantitative guidelines can be found in Mannan (2005).

It is generally considered that there is a greater hazard in storing large quantities of liquefied gas under pressure than at low temperatures and low pressures. The trend is toward replacing pressure storage by refrigerated low-pressure storage for large inventories. However, it is necessary to consider the risk of the entire system, including the refrigeration system, and not just the storage vessel. The consequences of failure of the refrigeration system must be considered. Each case should be carefully evaluated on its own merits.

Preventing Leaks and Spills from Accumulating under Tanks or Equipment Around storage and process equipment, it is a good idea to design dikes that will not allow toxic and flammable materials to accumulate around the bottom of tanks or equipment in case of a spill. If liquid is spilled and ignites inside a dike where there are storage tanks or process equipment, the fire may be continuously supplied with fuel and the consequences can be severe. It is usually much better to direct possible spills and leaks to an area away from the tank or equipment and provide a firewall to shield the equipment from most of the flames if a fire occurs. Figure 23-63 shows a diking design for directing leaks and spills to an area away from tanks and equipment.

The surface area of a spill should be minimized for hazardous materials that have a significant vapor pressure at ambient conditions, such as acrylonitrile or chlorine. This will make it easier and more practical to collect vapor from a spill or to suppress vapor release with foam or by other means. This may require a deeper nondrained dike area than normal or some other design that will minimize surface area, in order to contain the required volume. It is usually not desirable to cover a diked area to restrict loss of vapor if the spill consists of a flammable or combustible material.

Minimal Use of Underground Tanks The U.S. Environmental Protection Agency's (EPA) Office of Underground Storage Tanks defines underground tanks as those with 10 percent or more of their volume, including piping, underground. An aboveground tank that does not have more than 10 percent of its volume (including piping) underground is excluded from the underground tank regulations. Note, however, that a 5000-gal tank sitting wholly atop the ground but having 1400 ft of 3-in buried pipe or 350 ft of 6-in buried pipe is considered an underground storage tank.

At one time, burying tanks was recommended because it minimized the need for a fire protection system, dikes, and distance separation. At many companies, this is no longer considered good practice. Mounding, or burying tanks above grade, has most of the same problems as burying tanks below ground and is usually not recommended.

Problems with buried tanks include

- Difficulty in monitoring interior and exterior corrosion (shell thickness)
- Difficulty in detecting leaks
- Difficulty of repairing a tank if the surrounding earth is saturated with chemicals from a leak
- Potential contamination of groundwater due to leakage

Government regulations concerning buried tanks have become stricter. This is so because of the large number of leaking tanks that

have been identified as causing adverse environmental and human health problems.

Design of Tanks, Piping, and Pumps Six basic tank designs are used for the storage of organic liquids: (1) fixed-roof, (2) external floating-roof, (3) internal floating-roof, (4) variable vapor space, (5) low-pressure, and (6) high-pressure tanks. The first four tank designs listed are not generally considered suitable for highly toxic hazardous materials.

Low-Pressure Tanks (up to 15 psig) Low-pressure storage tanks for highly hazardous toxic materials should meet, as a minimum, the American Petroleum Institute (API) 620 Standard, "Recommended Rules for the Design and Construction of Large Welded, Low-Pressure Storage Tanks" (API Standards). This standard covers above-ground tanks designed for all pressures less than or equal to 15 psig and metal temperatures less than or equal to 250°F (121°C). There are no specific requirements in API 620 for highly hazardous toxic materials.

API 650, "Welded Steel Tanks for Oil Storage" (API Standards), has limited applicability to storage of highly hazardous toxic materials because it prohibits refrigerated service and limits pressures to 2.5 psig and only if designed for certain conditions. Most API 650 tanks have a working pressure approaching atmospheric pressure, and hence their pressure-relieving devices must generally vent directly to the atmosphere. Its safety factors and welding controls are less stringent than required by API 620.

Horizontal and vertical cylindrical tanks are used to store highly toxic liquids and other hazardous materials at atmospheric pressure. Horizontal, vertical, and spherical tanks are used for refrigerated liquefied gases that are stored at atmospheric pressure. The design pressure of tanks for atmospheric pressure and low-pressure storage at ambient temperature should not be less than 100 percent of the vapor pressure of the material at the maximum design temperature. The maximum design metal temperature to be used takes into consideration the maximum temperature of material entering the tank; the maximum ambient temperature, including solar radiation effects; and the maximum temperature attainable by expected or reasonably foreseeable abnormal chemical reactions.

Since discharges of vapors from highly hazardous materials cannot simply be released to the atmosphere, the use of a weak seam roof is not normally acceptable. It is best that tanks in low-pressure hazardous service be designed and stamped for 15 psig to provide maximum safety, and pressure relief systems must be provided to vent relieved overpressure to equipment that can collect, contain, and treat the effluent.

The minimum design temperature should be the lowest temperature to which the tank will be subjected, taking into consideration the minimum temperature of material entering the tank, the minimum temperature to which the material may be autorefrigerated by rapid evaporation of low-boiling liquids or mechanically refrigerated, the minimum ambient temperature of the area where the tank is located, and any possible temperature reduction by endothermic physical processes or chemical reactions involving the stored material. API 620 provides for installations in areas where the lowest recorded 1-day mean temperature is 50°F (10°C).

While either rupture disks or relief valves are allowed on storage tanks by Code, rupture disks by themselves should not be used on tanks for the storage of toxic or other highly hazardous materials since they do not close after opening and may lead to continuing release of hazardous material to the atmosphere.

The API 620 Code requires a combined pneumatic hydrotest at 125 percent of design tank loading. In tanks designed for low-density liquid, the upper portion is not fully tested. For highly hazardous materials, consideration should be given for hydrotesting at the maximum specified design liquid level. It may be required that the lower shell thickness be increased to withstand a full head of water and that the foundation be designed such that it can support a tank full of water or the density of the liquid, if it is greater than water. Testing in this manner not only tests the containment capability of the tank, but also provides an overload test for the tank and the foundation similar to the overload test for pressure vessels. API 620 also requires radiography.

Proper preparation of the subgrade and grade is extremely important for tanks that are to rest directly on grade. Precautions should be taken to prevent ground freezing under refrigerated tanks, as this can

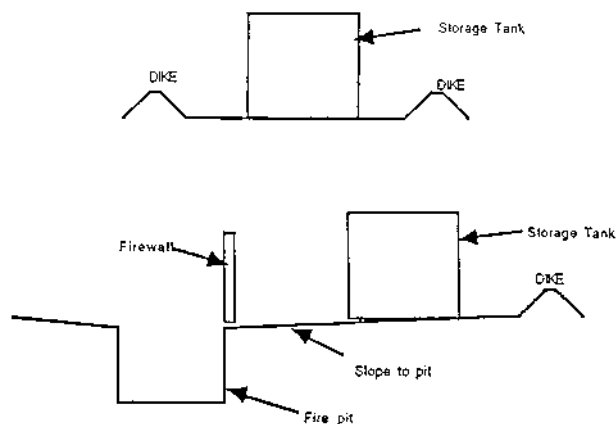


FIG. 23-63 Methods of diking for flammable liquids: (a) traditional diking method allows leaks to accumulate around the tank. In case of fire, the tank will be exposed to flames that can be supplied by fuel from the tank and will be hard to control. (b) In the more desirable method, leaks are directed away from the tank. In case of fire, the tank will be shielded from most flames and fire will be easier to fight. (From Englund, in *Advances in Chemical Engineering*, vol. 15, Academic Press, San Diego, 1990, pp. 73–135, by permission.)

cause the ground to heave and damage the foundation or the tank. Designing for free air circulation under the tank is a method for passive protection from ground freezing.

Steels lose their ductility at low temperatures and can become subject to brittle failure. There are specific requirements for metals to be used for refrigerated storage tanks in API 620, Appendices Q and R.

Corrosive chemicals and external exposure can cause tank failure. Materials of construction should be chosen so that they are compatible with the chemicals and exposure involved. Welding reduces the corrosion resistance of many alloys, leading to localized attack at the heat-affected zones. This may be prevented by the use of the proper alloys and weld materials, in some cases combined with annealing heat treatment.

External corrosion can occur under insulation, especially if the weather barrier is not maintained or if the tank is operating at conditions at which condensation is likely. This form of attack is hidden and may be unnoticed for a long time. Inspection holes and plugs should be installed in the insulation to monitor possible corrosion under the insulation.

Pressure Vessels (above 15 psig) The design of vessels above 15 psig falls within the scope of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code, Section VIII, "Pressure Vessels, Division I," and should be designated as lethal service if required. *Lethal service* means containing substances that are "poisonous gases or liquids of such a nature that a very small amount of the gas or vapor of the liquid mixed or unmixed with air is dangerous to life when inhaled. This class includes substances which are stored under pressure or may generate a pressure if stored in a closed vessel." This is similar to, but not exactly the same as, the same definition as that for "Category M" fluid service of the ASME Pressure Piping Code (see below). Pressure vessels for the storage of highly hazardous materials should be designed in accordance with requirements of the ASME code even if the vessels could be exempted because of high pressure or size. The code requires that the corrosion allowance be adequate to compensate for the more or less uniform corrosion expected to take place during the life of the vessel and not weaken the vessel below design strength.

Venting and Drainage Low-pressure storage tanks are particularly susceptible to damage if good venting practices are not followed. A vent that does not function properly at all times may cause damage to the tank from pressure that is too high or too low. Vapors should go to a collection system, if necessary, to contain toxic and hazardous vents.

Piping Piping systems for toxic fluids fall within Chapter VIII of the ASME Pressure Piping Code, "Piping for Category M Fluid Service." Category M fluid service is defined as "fluid service in which the potential for personnel exposure is judged to be significant and in which a single exposure to a small quantity of a toxic fluid, caused by leakage, can produce serious irreversible harm to persons on breathing or bodily contact, even when prompt restorative measures are taken."

Piping systems should meet the requirements for both Category M fluid service and for "severe cyclic conditions." Piping systems should be subjected to a flexibility analysis, and if they are found to be too rigid, flexibility should be added. Severe vibration pulsations should be eliminated. Expansion bellows, flexible connections, and glass equipment should be avoided. Pipelines should be designed with the minimum number of joints, fittings, and valves. Joints should be flanged or butt-welded. Threaded joints should not be used.

Instrumentation (CCPS, 1986.) Instrument systems are an essential part of the safe design and operation of systems for storing and handling hazardous materials. They are key elements of systems to eliminate the threat of conditions that could result in loss of containment. They are also used for early detection of releases so that mitigating action can be taken before these releases result in serious effects on people in the plant or in the public sector, or on the environment.

Pumps and Gaskets The most common maintenance problem with centrifugal pumps is with the seals. Mechanical seal problems account for most of the pump repairs in a chemical plant, with bearing failures a distant second. The absence of an external motor (on canned pumps) and a seal is appealing to those experienced with mechanical seal pumps.

Sealless pumps are very popular and are widely used in the chemical industry. Sealless pumps are manufactured in two basic types: canned-motor and magnetic-drive. Magnetic-drive pumps have thicker "cans," which hold in the process fluid, and the clearances between the internal rotor and can are greater compared to canned-motor pumps. This permits greater bearing wear before the rotor starts wearing through the can. Many magnetic-drive pump designs now have incorporated a safety clearance, which uses a rub ring or a wear ring to support the rotating member in the event of excessive bearing wear or failure. This design feature prevents the rotating member (outer magnet holder or internal rotating shaft assembly) from accidentally rupturing the can, as well as providing a temporary bearing surface until the problem bearings can be replaced. Because most magnetic-drive pumps use permanent magnets for both the internal and external rotors, there is less heat to the pumped fluid than with canned-motor pumps. Some canned-motor pumps have fully pressure-rated outer shells, which enclose the canned motor; others do not. With magnetic-drive pumps, containment of leakage through the can to the outer shell can be a problem. Even though the shell may be thick and capable of holding high pressures, there is often an elastomeric lip seal on the outer magnetic rotor shaft with little pressure containment capability.

Canned-motor pumps typically have a clearance between the rotor and the containment shell or can, which separates the fluid from the stator, of only 0.008 to 0.010 in (0.20 to 0.25 mm). The can has to be thin to allow magnetic flux to flow to the rotor. It is typically 0.010 to 0.015 in (0.25 to 0.38 mm) thick and made of Hastelloy. The rotor can wear through the can very rapidly if the rotor bearing wears enough to cause the rotor to move slightly and begin to rub against the can. The can may rupture, causing uncontrollable loss of the fluid being pumped.

It should not be assumed that just because there is no seal, sealless pumps are always safer than pumps with seals, even with the advanced technology now available in sealless pumps. Use sealless pumps with considerable caution when handling hazardous or flammable liquids.

Sealless pumps rely on the process fluid to lubricate the bearings. If the wear rate of the bearings in the fluid being handled is not known, the bearings can wear unexpectedly, causing rupture of the can.

Running a sealless pump dry can cause complete failure. If there is cavitation in the pump, hydraulic balancing in the pump no longer functions and excessive wear can occur, leading to failure of the can. *The most common problem with sealless pumps is bearing failure, which occurs either by flashing the fluid in the magnet area because of a drop in flow below minimum flow or by flashing in the impeller eye as it leaves the magnet area.* It is estimated that 9 out of 10 conventional canned-motor pump failures are the result of dry running. Canned pumps are available which, their manufacturer claims, can be operated dry for as long as 48 h.

It is especially important to avoid deadheading a sealless pump. Deadheaded sealless pumps can cause overheating. The bearings may be damaged, and the pump may be overpressurized. The pump and piping systems should be designed to avoid dead spots when pumping monomers. Monomers in dead spots may polymerize and plug the pump. There are minimum flow requirements for sealless pumps. It is recommended that a recirculation system be used to provide internal pump flow whenever the pump operates. Inlet line filters are recommended, but care must be taken not to cause excessive pressure drop on the suction side. Typical inlet filters use sieve openings of 0.0059 in (0.149 mm).

A mistreated sealless pump can rupture with potentially serious results. The can can fail if valves on both sides of the pump are closed and the fluid in the pump expands, either due to heating up from a cold condition or if the pump is started up. If the pump is run dry, the bearings can be ruined. The pump can heat up and be damaged if there is insufficient flow to take away heat from the windings. Sealless pumps, especially canned-motor pumps, produce a significant amount of heat, since nearly all the electric energy lost in the system is absorbed by the fluid being pumped. *If this heat cannot be properly dissipated, the fluid will heat up with possibly severe consequences.* Considerable care must be used when installing a sealless pump to be sure that improper operations cannot occur.

The instrumentation recommended for sealless pumps may seem somewhat excessive. However, sealless pumps are expensive, and they

can be made to last for a long time, compared to conventional centrifugal pumps where seals may need to be changed frequently. Most failures of sealless pumps are caused by running them dry and damaging the bearings. Close monitoring of temperature is necessary in sealless pumps. Three temperature sensors (resistance temperature devices, or RTDs) are recommended: (1) in the internal fluid circulation loop, (2) in the magnet, or shroud, area, and (3) in the pump case area.

It is very important that sealless pumps be flooded with liquid before starting, to avoid damage to bearings from imbalance or overheating. Entrained gases in the suction can cause immediate imbalance problems and lead to internal bearing damage. Some type of liquid sensor is recommended. Sealless pumps must not be operated deadheaded (pump liquid full with inlet and/or outlet valves closed). Properly installed and maintained, sealless pumps, both canned-motor and magnetic-drive, offer an economical and safe way to minimize hazards and leaks of hazardous liquids.

Loss-of-Containment Causes The list in Table 23-30 indicates four basic ways in which containment can be lost. These cause categories can be used both as a checklist of considerations during the design process and as a starting point for evaluating the adequacy of safeguards as part of a process hazard and risk analysis.

Maintaining the Mechanical Integrity of the Primary Containment System The second main category in the above list pertains to containment failure under design operating conditions due to imperfections in the equipment. This group of causes is the main focus of a facility's mechanical integrity (MI) program. The MI program should also detect other imperfections such as previous periods

TABLE 23-30 Summary of Loss-of-Containment Causes in the Chemical Industry

I. Containment lost via an "open-end" route to atmosphere	
A.	Due to genuine process relief or dumping requirements
B.	Due to maloperation of equipment in service; e.g., spurious relief valve operation
C.	Due to operator error; e.g., drain or vent valve left open, misrouting of materials, tank overfilled, unit opened up under pressure
II. Containment failure under design operating conditions due to imperfections in the equipment	
A.	Imperfections arising prior to commissioning and not detected before start-up
B.	Imperfections due to equipment deterioration in service and not detected before the effect becomes significant
C.	Imperfections arising from routine maintenance or minor modifications not carried out correctly, e.g., poor workmanship, wrong materials
III. Containment failure under design operating conditions due to external causes	
A.	Impact damage, such as by cranes, road vehicles, excavators, machinery associated with the process
B.	Damage by confined explosions due to accumulation and ignition of flammable mixtures arising from small process leaks, e.g., flammable gas buildup in analyzer houses, in enclosed drains, around submerged tanks
C.	Settlement of structural supports due to geologic or climatic factors or failure of structural supports due to corrosion, etc.
D.	Damage to tank trucks, railcars, containers, etc., during transport of materials on- or off-site
E.	Fire exposure
F.	Blast effects from a nearby explosion (unconfined vapor cloud explosion, bursting vessel, etc.), such as blast overpressure, projectiles, structural damage
G.	Natural events (acts of God) such as windstorms, earthquakes, floods, lightning
IV. Containment failure due to deviations in plant conditions beyond design limits	
A.	Overpressurizing of equipment
B.	Underpressurizing of non-vacuum-rated equipment
C.	High metal temperature (causing loss of strength)
D.	Low metal temperature (causing cold embrittlement and overstressing)
E.	Wrong process materials or abnormal impurities (causing accelerated corrosion, chemical attack of seals or gaskets, stress corrosion cracking, embrittlement, etc.)

SOURCE: Summarized from Appendix A of Prugh and Johnson, *Guidelines for Vapor Release Mitigation*, CCPS-AIChE, New York, 1988.

of operating outside design limits, or improper process materials or impurities that cause accelerated corrosion, chemical attack of seals or gaskets, stress corrosion cracking, embrittlement, etc. MI programs include quality assurance of the initial construction and of maintenance materials; routine preventive maintenance activities; regular inspections and nondestructive testing (NDT) of vessels, tanks, and piping to detect corrosion, pitting, erosion, cracking, creep, etc.; functional testing of standby equipment including alarms, safety instrumented systems, and emergency relief systems; and correcting problems that are identified while inspecting, testing, or maintaining the equipment and instrumentation. CCPS (2006) provides guidance on developing and implementing a mechanical integrity program.

Release Detection and Mitigation Mitigation means reducing the severity of consequences of an emergency situation such as a major release, fire, and/or explosion. The choice of mitigation strategies will depend on the nature of the hazardous materials and energies that can be released and the degree to which risk reduction is needed to ensure people, property, and the environment are adequately protected. The latter will be affected by the proximity of populations and sensitive environments surrounding the facility. An unstaffed remote natural gas facility will obviously not warrant the same mitigation measures as a facility using large quantities of high-toxic-hazard materials with other industry or residences nearby.

To be effective, a mitigation strategy will need to be capable of

- Detecting either an incipient or an actual emergency situation
- Deciding on and initiating the proper course of action to mitigate the situation
- Reducing the severity of consequences at the source, in transit, and/or at the receptor locations
- Preventing domino effects that could have even more severe consequences

Each of these steps might be performed either by direct action of operations or emergency response personnel or by automatic systems. An example of the latter might be an array of toxic or flammable gas detectors that might trip an emergency shutdown system that closes remotely actuated block valves and vents off the process pressure to a flare if two adjacent sensors read above a predetermined vapor concentration.

Mitigation measures can also be *passive* safeguards, meaning that they require no human intervention and no engineered sensing and actuation system to work. Examples of passive mitigation measures are secondary containment systems, blast-resistant and fire-resistant structures, insulated or low-heat-capacity spill surfaces to reduce the rate of evaporation, and an increased distance between the hazardous materials and energies and the sensitive receptors.

SAFETY INSTRUMENTED SYSTEMS

REFERENCES: *Guidelines for Safe and Reliable Instrumented Protective Systems*, American Institute of Chemical Engineers, New York, 2007; ISA TR84.00.04, *Guidelines for the Implementation of ANSI/ISA 84.00.01-2004 (IEC 61511)*, Instrumentation, Systems, and Automation Society, N.C., 2005; ANSI/ISA 84.00.01-2004, *Functional Safety: Safety Instrumented Systems for the Process Industry Sector*, Instrumentation, Systems, and Automation Society, N.C., 2004; IEC 61511, *Functional Safety: Safety Instrumented Systems for the Process Industry Sector*, International Electrotechnical Commission, Geneva, Switzerland, 2003.

GENERAL REFERENCES: *Guidelines for Hazard Evaluation Procedures, Second Edition with Worked Examples*, American Institute of Chemical Engineers, New York, 1992; *Layer of Protection Analysis: A Simplified Risk Assessment Approach*, American Institute of Chemical Engineers, New York, 2001; ISA TR84.00.02, *Safety Instrumented Functions (SIF)—Safety Integrity Level (SIL) Evaluation Techniques*, Instrumentation, Systems, and Automation Society, N.C., 2002.

Glossary

Basic process control system (BPCS) System that responds to input signals from the process, its associated equipment, other programmable systems, and/or an operator and generates output signals,

causing the process and its associated equipment to operate in the desired manner. The BPCS is commonly referred to as the *control system*.

Compensating measures Planned means for managing process risk during periods of process operation with known faults or problems that increase risk.

Control layer Protection layer that is used to maintain the process within the normal operating limits, such as standard operating procedures, basic process control system, and process alarms.

Core attribute Fundamental underlying property of a protection layer. The core attributes are independence, functionality, integrity, reliability, auditability, management of change, and access security.

Independent protection layer An IPL is a device, system, or action that is capable of preventing a hazard scenario from proceeding to the undesired consequence regardless of the initiating cause occurrence (or its consequences) or the failure of any other protection layer.

Safety instrumented function (SIF) A safety function allocated to the safety instrumented system with a safety integrity level necessary to achieve the desired risk reduction for an identified process hazard.

Safety instrumented system (SIS) Any combination of separate and independent devices (sensors, logic solvers, final elements, and support systems) designed and managed to achieve a specified safety integrity level. An SIS may implement one or more safety instrumented functions.

Safety integrity level (SIL) Discrete level (one out of a possible four SIL categories) used to specify the probability that a safety instrumented function will perform its required function under all operational states within a specified time.

Introduction The chemical processing industry relies on many types of instrumented systems, e.g., the basic process control systems (BPCSs) and safety instrumented system (SIS). The BPCS controls the process on a continuous basis to maintain it within prescribed control limits. Operators supervise the process and, when necessary, take action on the process through the BPCS or other independent operator interface. The SIS detects the existence of unacceptable process conditions and takes action on the process to bring it to a safe state. In the past, these systems have also been called emergency shutdown systems, safety interlock systems, and safety critical systems.

In 1993, the Center for Chemical Process Safety (CCPS) published *Guidelines for Safe Automation of Chemical Processes* (referred to henceforth as *Safe Automation*). *Safe Automation* provides guidelines for the application of automation systems used to control and shut down chemical and petrochemical processes. The popularity of one of the hazard and risk analysis methods presented in *Safe Automation* led to the publication of the 2001 Concept Series book from CCPS, *Layer of Protection Analysis: A Simplified Risk Assessment Approach*. This method builds upon traditional process hazards analysis techniques. It uses a semiquantitative approach to define the required performance for each identified protective system.

The Instrumentation, Systems, and Automation Society (ISA) published the Standard ANSI/ISA 84.01-1996, documenting the good engineering practice for the design, operation, maintenance, and testing of SIS. The standard established a numerical benchmark for the SIS performance known as the safety integrity level (SIL) and provided requirements on how to design and manage the SIS to achieve the target SIL.

Safe Automation and ANSI/ISA 84.01-1996 served as significant technical references for the first international standard, IEC 61511, issued by the International Electrotechnical Commission (IEC). In the United States, IEC 61511 was accepted by ISA as ISA 84.00.01-2004, replacing the 1996 standard. In 2004, the European Committee for Electrotechnical Standardization (CENELEC) and the American National Standards Institute (ANSI) recognized IEC 61511 as a consensus standard for the process industry. IEC 61511 covers the complete process safety management life cycle. With its adoption, this standard serves as the primary driving force behind the work processes followed to achieve and maintain safe operation using safety instrumented systems.

It is important that personnel understand how to achieve safe operation, but not at the exclusion of other important considerations, such as reliability, operability, and maintainability. The chemical industry has also found significant benefit to plant productivity and operability when SIS work processes are used to design and manage other instrumented protective systems (IPS), such as those mitigating potential economic and business losses. The CCPS book (2007) *Guidelines for Safe and Reliable Instrumented Protective Systems* discusses the activities and quality control measures necessary to achieve safe and reliable operation throughout the IPS lifecycle.

Hazard and Risk Analysis Consideration should be given to identifying process hazards as early as possible in the process equipment design, so that measures can be taken to reduce or eliminate the hazards. Inherently safer design strategies, such as minimize, substitute, moderate, and simplify, should be implemented.

When it is no longer practical to reduce the risk further by process design modification, protection layers are used to mitigate the remaining process risk. IPLs must meet the necessary rigor associated with seven core attributes: independence, functionality, integrity, reliability, auditability, access security, and management of change. There are two critical activities to be completed during the risk assessment phase. First, the safety functions (i.e., those functions that detect and respond to process hazards) are identified by using an accepted hazard and risk analysis (H&RA) methodology. Second, each safety function is allocated to a protection layer that is designed and managed to achieve the required risk reduction.

An H&RA involves a review of the process design and its control, operation, and maintenance practices. The review is conducted by a multidisciplinary team with expertise in the design and operation of the process unit. The team uses a systematic screening process to determine how deviations from normal operation lead to process hazards. The H&RA identifies areas where the process risk is too high, requiring the implementation of safety functions. The team's objective is to reduce the risk to below the owner/operator's risk criteria.

Process risk is defined by the frequency of the occurrence and the potential consequence severity of the process hazard. To define the frequency, the initiating causes (e.g., single causes or multiple causes and conditions) are identified for each process hazard, and their frequency of occurrence is estimated. The consequence severity is the logical conclusion to the propagation of the process hazard if no protection layers are implemented as barriers to the event.

The gap between the process risk and the owner/operator's risk criteria establishes the requirements for risk reduction. The risk gap can be managed by a single safety function or by multiple functions allocated to protection layers. The team defines the risk reduction that must be provided by each safety function and allocates the safety function to a protection layer that is designed and managed to achieve the allocated risk reduction.

When the safety function is allocated to the SIS, it is a safety instrumented function (SIF). The risk reduction allocated to the SIF defines its target safety integrity level (SIL). This target is related to the SIF probability of failure on demand (PFD), e.g., SIL 1 (PFD range: 0.01 to 0.1), SIL 2 (PFD range: 0.001 to 0.01), SIL 3 (PFD range: 0.0001 to 0.001), and SIL 4 (PFD range: 0.00001 to 0.0001).

The identification of safety functions continues until the process risk associated with the hazard is reduced to meet the risk criteria. When there is insufficient risk reduction provided by the current or planned design, the team makes recommendations for process design changes (e.g., inherently safer design), improvement to existing functions, or the design and implementation of new functions. These recommendations are generally prioritized based on the magnitude of the gap between the mitigated process risk (i.e., risk considering the presence of existing functions) and the risk criteria.

Design Basis In the design phase, the project team works together to create an SIS design basis that achieves the risk reduction strategy established in the risk reduction phase. This strategy relies, in part, on the implementation of SIFs to address identified process risk. The SIF uses dedicated devices, including process sensors that detect the process hazard, a logic solver that decides what to do, and final elements that take action on the process. Often, a single logic solver

implements multiple SIFs, so the potential for common-cause failures between SIFs should be considered during design.

The SIS is normally designed to fail-safe on loss of power and takes action only when the process demands that it do so. These demands often occur when safe operating limits are exceeded due to BPCS failures. Therefore, the SIS is designed and managed to be independent of the BPCS in terms of its hardware and software and its user interfaces, such as operator, maintenance, and engineering interfaces.

Systematic errors can occur anywhere in the design and implementation process or during the operational life of an SIS device. These errors put the SIS on the path to failure in spite of the design elements incorporated to achieve robust hardware and software systems. Systematic errors are minimized using work processes that address potential human errors in the SIS design and management (e.g., programming errors or hardware specification errors).

Random hardware failure can occur throughout the device life as components age in the environmental conditions of the process unit. These failures can cause a device to fail dangerously; i.e., it cannot perform as required. These failures are estimated by examining the dangerous failure modes of each device and their frequency of occurrence. The resulting failure rate is used to estimate the PFD of the SIS considering its specific devices, redundancy, diagnostics, common-cause failure potential, and proof test interval. The PFD is then compared to the target SIL assigned during the risk assessment phase to determine whether the design is adequate.

The design basis includes the process requirements specification and the safety requirements specification. The process requirements specification is typically developed by process engineering, with input from operations personnel. The process requirements are provided to the instrumentation, electrical, or controls systems personnel to develop the safety requirements specification with input from operations and maintenance personnel.

Process Requirements Specification Process engineering uses the H&RA findings, process design information, and operations input to

- Define safe state, including safety and nonsafety actions.
- Define reliability requirements necessary to achieve desired process unit uptime performance.
- Define operability requirements for modes of operation, such as start-up, reduced rates, maintenance modes, and shutdown.
- Identify windows of opportunity for SIS testing.
- Define process-related parameters.
- Define human-related parameters.

Guidance can be found in the CCPS book (2007) *Guidelines for Safe and Reliable Instrumented Protective Systems* related to the development of the process requirements specification.

Safety Requirements Specification The instrumentation and electrical (I&E) requirements are developed to meet the intent of any H&RA findings and the process requirements. The design documentation should establish a clear connection between each process hazard and the design of its SIFs. I&E personnel should meet with the process engineering representative responsible for the process requirements to ensure that the intent is understood.

I&E design focuses on achieving the target SIL through careful selection of the devices (e.g., user approved for safety), use of redundancy, on-line diagnostics, and frequent proof testing. The ISA technical report TR84.00.04 gives extensive guidance on design requirements for the hardware and software systems used to implement the SIS. Application-specific standards by organizations such as American Society of Mechanical Engineers (ASME), American Petroleum Institute (API), and the National Fire Protection Association (NFPA) may provide additional requirements and guidance.

There is often quite a bit of give and take between the process requirements and I&E requirements in the early stages of the project. For example, the ideal process measurement may not be practical in the existing installation. At all times, it should be recognized that the goal of the design is to prevent the process hazard from propagating to an incident.

Engineering, Installation, Commissioning, and Validation (EICV) This phase involves the physical realization of the design basis, which is developed in response to process risk identified in an H&RA study. The bulk of the work in this phase is not a process

engineering effort. Detailed engineering, installation, and commissioning is generally an I&E function. However, this is where the assumptions and requirements developed by the process engineer are put into practice and validated.

Validation of the SIS functionality is performed as part of a site acceptance test (SAT). Validation involves a full functional test that demonstrates the SIS actually works in the real-world installation. It proves the SIS devices execute the logic according to the specification and ensures that the SIS and its devices interact as intended with other systems, such as the BPCS and operator interface. From a systematic error standpoint, the SAT also provides an opportunity for a first-pass validation of the procedures developed for the operating basis (see next subsection).

Pre-start-up Safety Review (PSSR) approval of the SIS establishes the point where the SIS design and construction is considered complete. All documentation should be formally updated to as-built status, incorporating any modifications made since the last formal drawing or document revision. Once the PSSR has approved the SIS for process unit start-up, formal management of change procedures should be followed to address proposed modification to the SIS or its associated documentation. Any deviation from the approved design basis should be reviewed and approved by appropriate parties prior to change implementation.

Operating Basis As the SIS engineering design nears completion, the resources and skills of plant operations should be considered. At some point, the SIS is turned over to operations and maintenance personnel, who must be trained on the new SIS and on their responsibilities. Consequently, thought should be given to the content and depth of the information that must be communicated to various personnel. This is especially important as the responsibility for the SIS transitions from the project team to operations and maintenance control.

The process engineer is responsible for defining the content of SIS operating procedures, which should cover SIS specific information (e.g., set points, SIS actions, and the hazard that is being prevented with SIS), the correct use of bypasses and resets, the operator response to SIS alarms and trips, when to execute a manual shutdown, and provisions for operation with detected faults (e.g., compensating measures). These procedures, along with analogous ones developed by maintenance and reliability engineering for maintenance activities, make up the backbone of the operating basis.

Since a device can fail at any time during its life, periodic proof tests are performed to demonstrate the functionality of the SIS. Proof tests are covered by operation and maintenance procedures that ensure that the test is done correctly, consistently, and safely and that the device is returned to a fully operational state after test. Each test serves as an opportunity for personnel to see the SIS in action and to validate the procedures associated with its operation.

Proof testing is required for all SISs. It is used to demonstrate that the devices are operating as specified and are maintained in "as good as new" condition. Failures found during testing indicate gaps in the mechanical integrity program, necessitating root-cause investigation and corrective action.

SECURITY

Definition of Terms (American Petroleum Institute/National Petroleum Refiner's Association, *Security Vulnerability Assessment Methodology for the Petroleum Industry*, 2004.)

Adversary Any individual, group, organization, or government that conducts activities, or has the intention and capability to conduct activities detrimental to critical assets. An adversary could include intelligence services of host nations, or third-party nations, political and terrorist groups, criminals, rogue employees, and private interests. Adversaries can include site insiders, site outsiders, or the two acting in collusion.

Alert levels A progressive, qualitative measure of the likelihood of terrorist actions, from negligible to imminent, based on government or company intelligence information. Different security measures may be implemented at each alert level based on the level of threat to the facility.

Asset Any person, environment, facility, material, information, business reputation, or activity that has a positive value to an owner. The asset may have value to an adversary, as well as an owner, although

the nature and magnitude of those values may differ. Assets in the SVA include the community and the environment surrounding the site.

Asset category Assets may be categorized in many ways. Among these are by people, hazardous materials (used or produced), information, environment, equipment, facilities, activities and operations, and company reputation.

Countermeasures An action taken or a physical capability provided whose principal purpose is to reduce or eliminate one or more vulnerabilities. The countermeasure may also affect the threat(s) (intent and/or capability) as well as the asset's value. The cost of a countermeasure may be monetary, but may also include nonmonetary costs such as reduced operational effectiveness, adverse publicity, unfavorable working conditions, and political consequences.

Cyber security Protection of critical information systems including hardware, software, infrastructure, and data from loss, corruption, theft, or damage.

Delay A countermeasures strategy that is intended to provide various barriers to slow the progress of an adversary in penetrating a site, to prevent an attack or theft, or in leaving a restricted area to assist in apprehension and prevention of theft.

Detection A countermeasures strategy that is intended to identify an adversary attempting to commit a security event or other criminal activity in order to provide real-time observation as well as postincident analysis of the activities and identity of the adversary.

Deterrence A countermeasures strategy that is intended to prevent or discourage the occurrence of a breach of security by means of fear or doubt. Physical security systems such as warning signs, lights, uniformed guards, cameras, and bars are examples of countermeasures that provide deterrence.

Hazard A situation with the potential for harm.

Intelligence Information to characterize specific or general threats including the motivation, capabilities, and activities of adversaries.

Intent A course of action that an adversary intends to follow.

Likelihood of adversary success (LAS) The potential for causing a catastrophic event by defeating the countermeasures. LAS is an estimate that the security countermeasures will thwart or withstand the attempted attack, or if the attack will circumvent or exceed the existing security measures. This measure represents a surrogate for the conditional probability of success of the event.

Physical security Security systems and architectural features that are intended to improve protection. Examples include fencing, doors, gates, walls, turnstiles, locks, motion detectors, vehicle barriers, and hardened glass.

Response The act of reacting to detected or actual criminal activity either immediately following detection or after the incident.

Risk The potential for damage to or loss of an asset. Risk, in the context of process security, is the potential for a catastrophic outcome to be realized. Examples of the catastrophic outcomes that are typically of interest include an intentional release of hazardous materials to the atmosphere, or the theft of hazardous materials that could later be used as weapons, or the contamination of hazardous materials that may later harm the public, or the economic costs of the damage or disruption of a process.

Risk assessment Risk (R) assessment is the process of determining the likelihood of an adversary (T) successfully exploiting vulnerability (V) and the resulting degree of consequences (C) on an asset. A risk assessment provides the basis for rank-ordering risks and thus establishing priorities for the application of countermeasures.

Security layers of protection Also known as concentric "rings of protection," a concept of providing multiple independent and overlapping layers of protection in depth. For security purposes, this may include various layers of protection such as countersurveillance, counterintelligence, physical security, and cyber security.

Security vulnerability assessment (SVA) The process of determining the likelihood of an adversary successfully exploiting vulnerability, and the resulting degree of damage or impact. SVAs are not quantitative risk analyses, but are performed qualitatively using the best judgment of security and safety professionals. The determination of risk (qualitatively) is the desired outcome of the SVA, so that it provides the basis for rank ordering of the security-related risks and thus establishing priorities for the application of countermeasures.

Target attractiveness An estimate of the value of a target to an adversary based on the factors shown below. Experience has shown that, particularly for terrorist attacks, certain targets better accomplish the objectives of the adversaries than do others. Since the SVA is a risk-based analytical approach, consideration must be given to these factors in defining the threat and in determining the need for any enhanced countermeasures.

- Potential for mass casualties and fatalities
- Extensive property damage
- Proximity to national assets or landmarks
- Possible disruption or damage to critical infrastructure
- Disruption of the national, regional, or local economy
- Ease of access to target
- Media attention or possible interest of the media
- Company reputation and brand exposure

Threat Any indication, circumstance, or event with the potential to cause the loss of, or damage to, an asset. Threat can also be defined as the intention and capability of an adversary to undertake actions that would be detrimental to critical assets.

Threat categories Adversaries may be categorized as occurring from three general areas:

- Insiders
- Outsiders
- Insiders working in collusion with outsiders

Vulnerability Any weakness that can be exploited by an adversary to gain access to an asset. Vulnerabilities can include, but are not limited to, building characteristics; equipment properties; personnel behavior; locations of people, equipment, and buildings; or operational and personnel practices.

Introduction Prior to September 11, 2001, known as 9/11, chemical process safety activities primarily focused on accidental release risks and excluded most considerations of intentional releases. Security was provided mostly for lesser threats than such extreme acts of violence, and terrorism was generally not provided for except in high-security areas of the world. Exceptions to this included general concerns for sabotage. This was due to a perception that these risks were managed adequately, and that the threat of a terrorist attack, particularly on U.S. chemical manufacturing facilities or transportation system, was remote.

Following 9/11 it became apparent that the threat of intentional harm to infrastructure, especially where hazardous materials were manufactured, stored, processed, or transported, had to be considered a credible concern. Security for the chemical industry took on increased emphasis as a result, and such organizations as the American Institute of Chemical Engineers recognized the paradigm shift and published guidelines on analyzing these threats.^{*} The concerns of international terrorism have spread to many countries around the world, and addressing this concern is now a permanent part of the requirements of the chemical engineering profession. Chemical engineers now must include *chemical process security* as a critical element of the management of a process facility.

Chemical process security management has as its objectives

1. To minimize the risk of harm to the public or employees from intentional acts against a process facility
2. To protect the assets (including employees) of the process facility to maintain the ongoing integrity of the operation and to preserve the value of the investment

Process security and process safety have many parallels and use many common programs and systems for achieving their ends. Process security management requires a systems approach to develop a comprehensive security program, which shares many common elements with process safety management.

Chemical process security includes, but goes beyond, traditional *physical security*. Physical security includes such considerations as guards, barriers, surveillance equipment, and other physical system considerations. Physical security is an element of chemical process security, but physical security alone is not always adequate to address the new challenges of security against extreme acts of violence, such as terrorism.

^{*}*Guidelines for Managing and Analyzing the Security Vulnerabilities of Fixed Chemical Sites*, American Institute of Chemical Engineers, August 2002.

Effective chemical process security must also consider integration of broader process elements including technology, chemical usage and quantities, procedures, administrative controls, training, and cyber interface with those traditional physical security elements.

The chemical engineer has an opportunity to influence these considerations in all stages of a process life cycle, including concept, engineering, construction, commissioning, operations, modification, and decommissioning. Security issues that are recognized in the concept and design phases of a project may allow for cost-effective considerations that can eliminate or greatly minimize security risks. For example, if a buffer zone can be provided between the public areas and a plant fence, and then again between a plant fence and critical process equipment, those two zones can effectively provide such benefits as

- **Detection zone(s)**, given they are free of obstacles and have sufficient depth to allow for adversaries to be detected while attempting unlawful entry
- **Standoff zone(s)**, given they have sufficient depth to keep adversaries from using explosives or standoff weapons effectively from the perimeter
- **Delay zone(s)** allowing intervening force the time to respond or time for operators to take evasive action before an adversary reaches a target following detection

A chemical engineer may have a choice of inherent safety variables, such as quantity stored or process temperatures and pressures, or process safety measures such as emergency isolation valves or containment systems, all of which may greatly reduce the vulnerabilities or the consequences of intentional loss. These are in addition to traditional security measures, which may include physical security, background checks, administrative controls, access controls, or other protective measures. For a more complete discussion of the options, refer to the AIChE Center for Chemical Process Safety *Guidelines for Analyzing and Managing the Security Vulnerabilities of Fixed Chemical Sites*⁸ and other references.⁹

Threats of Concern Terrorist acts can be the most problematic to defend against since they may be more extreme or malevolent than other crimes focused on monetary gains or outcomes with less malicious intent. Plus terrorists may use military tactics not often provided for in base chemical facility design. Chemical facility security must be considered in context with local and national homeland security and law enforcement activities, as well as with emergency response capabilities. There is a practical limit to the ability of a chemical site to prevent or mitigate a terrorist act. Above a certain level of threat, the facility needs to rely on law enforcement and military services to provide physical security against extreme acts of intentional harm. The security posture must be risk-based, and so extremely robust security measures are not always applicable or necessary.

The acts of concern for terrorism can be generally defined as involving the four motives shown in Table 23-31.

Other adversaries that must be considered as applicable include those capable and interested in perpetrating a full spectrum of security acts. These may include outside parties or insiders or a combination of the two working in collusion.

The threats that are applicable and the adversaries that may be culpable are characterized to understand their capabilities, intent, and therefore potential targets and tactics. The targets and acts of interest to various adversaries will vary with the group. For example, a terrorist may be interested in destroying a process through violent means, such as by the use of an explosive device. An activist may be interested only in a nonviolent protest or in causing some limited physical damage, but not in harming the environment or the public in the process. The various

TABLE 23-31 Security Issues of Concern with Example Applications to Terrorism

Security motives of concern*	Example terrorist means and objectives
Intentional loss of containment	By causing a release of chemicals to the atmosphere and potential toxic release, fire, or explosion to harm the public, workers, or the environment, or to destroy the facility
Theft of chemicals	For their eventual reuse as primary or secondary improvised weapons against a third party
Contamination or spoilage of a process	To cause immediate or delayed harm to people or the environment, or to cause severe economic injury
Degradation of the asset	By causing mechanical damage or physical or cyber disruption, for purposes of causing severe direct or indirect economic damages

*Adapted from *Guidelines for Managing and Analyzing the Security Vulnerabilities of Fixed Chemical Sites*, American Institute of Chemical Engineers, August 2002.

adversaries and strategies of interest form the basis of the vulnerability assessment, which is the foundation of a chemical process security management system specific to address the anticipated threats.

Overall Objectives of Terrorism Terrorists attempt to cause change to accomplish their goals by creating fear and uncertainty in the population they are targeting through the use of violent acts. The underlying goals include fundamentalist objectives, such as purity of religion or idealistic goals, but they may include power struggles, such as trying to overthrow a government, or reparations, such as revenge for past actions. The reason for a chemical plant being targeted may be that it serves an adversary of the terrorist (economic or military significance) or that it can be weaponized to cause third-party harm (health and safety consequences from intentional release of hazardous materials).

Security Vulnerability Assessment A security vulnerability assessment is intended to identify security vulnerabilities from a wide range of threats ranging from vandalism to terrorism. With the recognition of threats, consequences, and vulnerabilities, the risk of security events can be evaluated, and a security management system can be organized that will effectively mitigate those risks.

SVA Methodologies There are several SVA techniques and methods available to the industry, all of which share common elements. The following is a list of some available SVA methodologies published by various governments, private, and trade and professional organizations. Some are merely chapters or sections of documents that address security or risk assessment/risk management in broader terms. Some are SVA or VA publications by themselves. Some of these "methods" are complete, systematic analytical techniques, and others are mere checklists.

- American Institute of Chemical Engineers Center for Chemical Process Safety: *Guidelines for Analyzing and Managing the Security Vulnerabilities of Fixed Chemical Sites*, 2002.
- American Petroleum Institute/National Petroleum Refiner's Association, *Security Vulnerability Assessment Methodology for the Petroleum Industry*, 2003.
- National Institute of Justice, *Chemical Facility Vulnerability Assessment Methodology*, July 2002 (Sandia VAM).
- Synthetic Organic Chemical Manufacturers Association, Inc. (SOCMA), *Manual on Chemical Site Security Vulnerability Analysis Methodology and Model*, 2002.

One approach to conducting an SVA is shown in Fig. 23-64. This methodology was published by the American Institute of Chemical Engineers, Center for Chemical Process Safety, in 2002. The CCPS SVA is founded on a risk-based approach to managing chemical facility security. To begin the process, companies may perform an enterprise-level screening methodology to sort out significant risks among multiple sites and to determine priorities for analysis and implementation of any recommended changes. The screening, if performed, would result in a prioritized list of sites and forms the foundation of the choice of specific SVAs required. The book covers how to integrate chemical security

⁸*Guidelines for Managing and Analyzing the Security Vulnerabilities of Fixed Chemical Sites*, American Institute of Chemical Engineers, August 2002.

⁹*Counterterrorism and Contingency Planning Guide*, special publication from Security Management Magazine and American Society for Industrial Security, 2001; Dalton, D., *Security Management: Business Strategies for Success*, Butterworth-Heinemann Publishing, Newton, Mass., 1995; Walsh, Timothy J., and Richard J. Healy, eds., *Protection of Assets Manual*, Merritt Co., Santa Monica, Calif. (four-volume loose-leaf reference manual, updated monthly).

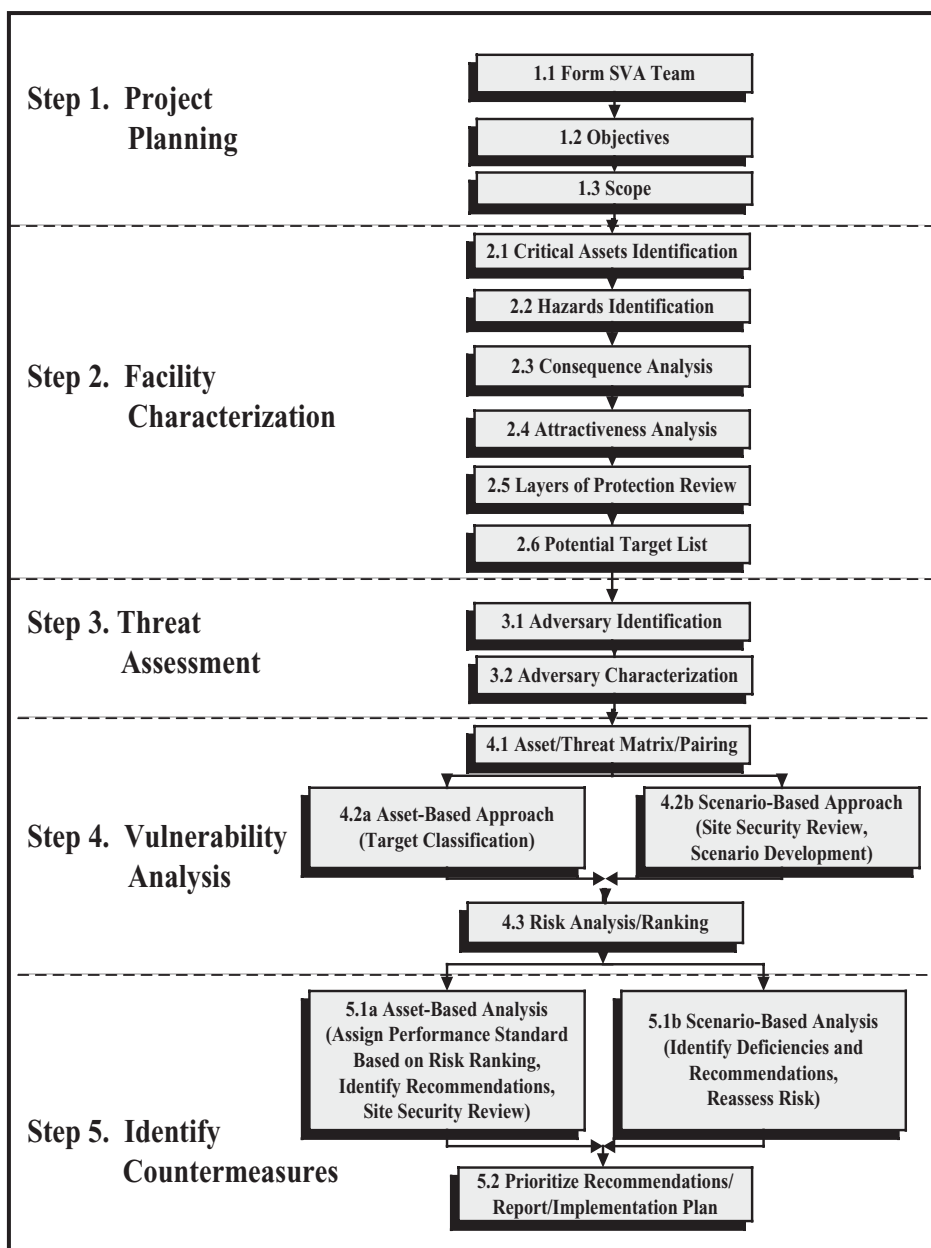


FIG. 23-64 CCPS SVA process.

management and process safety management strategies into a comprehensive process safety and security strategy. Security risk reduction opportunities during the process life cycle are explained, as well as various process risk management strategies (including inherent safety) that are applicable.

In the appendices, the book contains a set of tools including an enterprise-level screening tool, reference information available to conduct the CCPS SVA, and a workbook with worksheets for conducting the CCPS SVA, examples of enhanced security measures, and checklists for assessing security measures at a site.

Defining the Risk to Be Managed For the purposes of an SVA, the definition of risk is shown in Fig. 23-65. The risk that is being

analyzed for the SVA is defined as an expression of the likelihood that a defined threat will target and successfully attack a specific security vulnerability of a particular target or combination of targets to cause a given set of consequences. This is contrasted with the usual accidental risk definitions. The risk variables are defined as shown in Table 23-32.

A challenge for security vulnerability analysis is that the accurate prediction of the frequency and location of terrorist acts is not considered credible. As such, the analyst has a choice of assuming a frequency of a certain attack or assuming the attack frequency is 1, thereby focusing solely on the conditional likelihood of success of the adversary who attempts an attack. While the latter approach provides a baseline for making decisions about vulnerability, it does not fully

Intentional release risk is a function of • Consequences of a successful attack against an asset • Likelihood of a successful attack against an asset	Accidental release risk is a function of • Consequences of an accidental event • Likelihood of the occurrence of the event
Likelihood is a function of • The attractiveness to the adversary of the asset • The degree of threat posed by the adversary • The degree of vulnerability of the asset	Likelihood is a function of • The probability of an event cascading from initiating event to the consequences of interest and the frequency of the events over a given period

FIG. 23-65 Intentional release vs. accidental release.

answer the question of cost/benefit of any countermeasures. Certain crimes other than terrorism may be more predictable or frequent, allowing for statistical analysis to help frame the risks and justify countermeasure expenditures. Due to this limitation, the factor of attractiveness is considered along with consequences, threat, and vulnerability, to determine the priorities for and design of security measures for the industry.

Security Strategies A basic premise is that not all security risks can be completely prevented. Appropriate strategies for managing security can vary widely depending on the circumstances including the type of facility and the threats facing the facility. As a result, it is difficult to prescribe security measures that apply to all facilities in all industries. Instead, it is suggested to use the SVA as a means of identifying, analyzing, and reducing vulnerabilities. The specific situations must be evaluated individually by local management using best judgment of applicable practices. Appropriate security risk management decisions must be made commensurate with the risks. This flexible approach recognizes that there isn't a uniform approach to security in the chemical process industry, and that resources are best applied to mitigate high-risk situations primarily.

Security strategies for the process industries are generally based on the application of four key concepts against each threat: deterrence, detection, delay, and response.*

**Managing and Analyzing the Security Vulnerabilities of Fixed Chemical Sites*, AIChE, August 2002.

A complete security design includes these four concepts in layers of protection or a defense in depth arrangement. The most critical assets should be placed in the center of conceptual concentric levels of increasingly more stringent security measures. In the concept of rings of protection, the spatial relationship between the location of the target asset and the location of the physical countermeasures is important.

In the case of malicious acts, the layers or rings of protection must be particularly robust because the adversaries are intentionally attempting to breach the protective features and can be counted on to use whatever means are available to be successful. This could include explosions or other initiating events that result in widespread common-cause failures. Some particularly motivated adversaries might commit suicide while attempting to breach the security layers of protection.

Countermeasures and Security Risk Management Concepts Countermeasures are actions taken to reduce or eliminate one or more vulnerabilities. Countermeasures include hardware, technical systems, software, interdictive response, procedures, and administrative controls.

Security risk reduction at a site can include the following strategies:

- Physical security
- Cyber security
- Crisis management and emergency response plans
- Policies and procedures
- Information security
- Intelligence
- Inherent safety

TABLE 23-32 SVA Risk Variables

Consequences	Potential Impact of the Event
Likelihood	Likelihood is a function of the chance of being targeted for attack, and the conditional chance of mounting a successful attack (both planning and executing), given the threat and existing security measures. This is a function of three variables below.
Threat	Threat, is a function of the adversary's existence, intent, motivation, capabilities, and known patterns of potential adversaries. Different adversaries may pose different threats to various assets within a given facility.
Vulnerability	This is weakness that can be exploited by an adversary to gain access and damage or steal an asset or disrupt a critical function. This is a variable that indicates the likelihood of a successful attack, given the intent to attack an asset.
Target attractiveness	Target attractiveness is a surrogate measure for likelihood of attack. This factor is a composite estimate of the perceived value of a target to the adversary and the adversary's degree of interest in attacking the target.

SOURCE: *Managing and Analyzing the Security Vulnerabilities of Fixed Chemical Sites*, AIChE, August 2002.

TABLE 23-33 American Chemistry Council's Responsible Care® Security Code Process Security Management System*

1.	Leadership commitment
2.	Analysis of threats, vulnerabilities, and consequences
3.	Implementation of security measures
4.	Information and cyber security
5.	Documentation
6.	Training, drills, and guidance
7.	Communications, dialogue, and information exchange
8.	Response to security threats
9.	Response to security incidents
10.	Audits
11.	Third-party verification
12.	Management of change
13.	Continuous improvement

*Site Security Guidelines for the U.S. Chemical Industry, American Chemistry Council, October 2001.

Security Management System A comprehensive process security management system must include management program elements that integrate and work in concert with other management systems to control security risks. The 13 management practices shown in Table 23-33 are an example of a management system developed by the American Chemistry Council.

The purpose of a security management system is to ensure the ongoing, integrated, and systematic application of security principles and programs to protect personnel and assets in a dynamic security environment to ensure the continuity of the operation and supporting or dependent infrastructure. Traditional industrial facility security management tended to focus on protection of persons and property from crime (e.g., theft of property, workplace violence) and crime prevention, response, and investigation. While that is still an element of facility security, a management system allows incorporation of broader security concerns relating to intentional attack on fixed assets, such as by terrorists. To develop and implement a security management system not only provides a more thorough, dynamic, risk-based, and proactive approach, but also allows security management to be integrated into a facility's overall EH&S management systems.

The American Chemistry Council's Responsible Care® Security Code is designed to encourage continuous improvement in security performance by using a risk-based approach to identify, assess, and address vulnerabilities; prevent or mitigate incidents; enhance training and response capabilities; and maintain and improve relationships with key stakeholders. As a condition of membership in the council, each member company must implement the Security Code for facilities, transportation and value chain, and cyber security.

KEY PROCEDURES

Safety by design should always be our aim but is often impossible or too expensive, and then we have to rely on procedures. Key features of all procedures are as follows:

- They should be as simple as possible and described in simple language, so as to help the reader rather than protect the writer.
- They should be explained to and discussed with those who will have to carry them out, not just sent to them through the post.
- Regular checks and audits should be made to confirm that they are being carried out correctly.

Many accidents have occurred because the three procedures discussed below were unsatisfactory or were not followed.

Preparation of Equipment for Maintenance The essential feature of this procedure is a permit-to-work system: The operating team members prepare the equipment and write down on the permit the work to be done, the preparation carried out, the remaining hazards, and the precautions necessary. The permit is then accepted by the person or group who will carry out the work and is returned when the work is complete. The permit system will not make maintenance 100 percent safe, but it reduces the chance that hazards will be overlooked, lists ways of controlling them, and informs those doing the job what precautions they should take. The

system should cover such matters as who is authorized to issue and accept permits-to-work, the training they should receive (not forgetting deputies), and the period of time for which permits are valid. It should also cover the following:

Isolation of the Equipment under Maintenance Poor or missing isolation has been the cause of many serious accidents. Do not rely on valves except for quick jobs; use blinds or disconnection and blanking unless the job is so quick that blinding (or disconnection) would take as long and be as hazardous as the main job. Valves used for isolation (including isolation while fitting blinds or disconnecting) should be locked shut (e.g., by a padlock and chain). Blinds should be made to the same standard (pressure rating and material of construction) as the plant. Plants should be designed so that blinds can be inserted without difficulty; i.e., there should be sufficient flexibility in the pipework, or a slip-ring or figure 8 plate should be used. Electricity should be isolated by locking off or removal of fuses. Do not leave the fuses lying around for anyone to replace. Always try out electrical equipment after defusing to check that the correct fuses have been withdrawn.

Identification of the Equipment Many accidents have occurred because maintenance workers opened up the wrong equipment. Equipment that is under repair should be numbered or labeled unambiguously. Temporary labels should be used if there are no permanent ones. Pointing out the correct equipment is not sufficient. "The pump you repaired last week is leaking again" is a recipe for an accident.

Freeing from Hazardous Materials Equipment that is to be repaired should be freed as far as possible from hazardous materials. Gases can be removed by sweeping out with nitrogen (if the gases are flammable) or air; water-soluble liquids, by washing with water; and oils, by steaming. Some materials such as heavy oils and materials that polymerize are very difficult or impossible to remove completely. Tests should be carried out to make sure that the concentration of any hazardous material remaining is below an agreed level. Machinery should be in the lowest energy state. Thus the forks of fork lift trucks should be lowered, and springs should not be compressed or extended. For some machinery the lowest energy state is less obvious. Do not work under heavy suspended loads.

Special Jobs Certain jobs, such as entry to vessels and other confined spaces, hot work, and responsibilities of contractors, raise special problems.

Handover Permits should be handed over (and returned when the job is complete) person to person. They should not be left on the table for people to sign when they come in.

Change of Intent If there is a change in the work to be done, the permit should be returned and a new one issued (Crowl and Grossel, eds., *Handbook of Toxic Materials Handling and Management*, Marcel Dekker, 1994, Chap. 12).

Control of Modifications to Plants, Processes, and Organization Many accidents have occurred when such modifications had unforeseen and unsafe side effects (Sanders, *Chemical Process Safety Learning from Case Histories*, 3d ed., Gulf Professional, 2005). No such modifications should therefore be made until they have been authorized by a professionally qualified person who has made a systematic attempt to identify and assess the consequences of the proposal, by hazard and operability study or a similar technique. When the modification is complete, the person who authorized it should inspect it to make sure that the design intention has been followed and that it "looks right." What does not look right is usually wrong and should at least be checked.

- Unauthorized modifications are particularly liable to occur
- During start-ups, as changes may be necessary to get the plant on-line.
 - During maintenance, as the maintenance workers may be tempted to improve the plant as well as repair it. They may suggest modifications, but should put the plant back as it was unless a change has been authorized.
 - When the modification is cheap and no financial authorization is necessary. Many seemingly trivial modifications have had tragic results.
 - When the modification is temporary. Twenty-eight people were killed by the temporary modification at Flixborough, one of the

most famous of all time (Mannan, *Lees' Loss Prevention in the Process Industries*, 3d ed., Elsevier, Amsterdam, 2005, Appendix A1; Kletz, *Learning from Accidents*, 3d ed., Gulf Professional, Boston, 2001, Chap.8).

- When one modification leads to another, and then another (Kletz, *Plant/Operations Progress*, vol. 5, 1986, p. 136).
- When organizations are changed often, especially when staffing is reduced. Such changes should be studied as thoroughly as changes to equipment or processes.

Inspection and Testing of Protective Equipment All protective equipment should be scheduled for regular inspection and for testing if failure is latent (hidden); e.g., we do not know if an interlock, trip, alarm, or relief valve is in working order unless we test it. The frequency of testing or inspection depends on the failure rate and the length of time considered tolerable if it fails. Relief valves fail about once per 100 years on average, and testing every 1 or 2 years is usually adequate. Protective systems based on instruments, such as trips and alarms, fail more often, about once every couple of years on average; so more frequent testing is necessary, about once per month. Pressure systems (vessels and pipework) on noncorrosive duties can go for many years between inspections, but on some duties they may have to be inspected annually or even more often.

All protective equipment should be designed so that it can be tested or inspected, and access should be provided. Audits should include a check that the tests are carried out and the results acted on.

The supervisor, manager, or engineer responsible should be reminded when a test or inspection is due, and senior managers should be informed if it has not been carried out by the due date. Test and inspection schedules should include guidance on the methods to be used and the features that should be inspected. For example, if the time of response is critical, it should be checked.

Test results should be displayed for all to see, e.g., on a board in the control room.

Tests should be like real life. For example, a high-temperature trip failed to work despite regular testing. It was removed from its case before testing so the test did not disclose that the pointer rubbed against the case. This prevented it from indicating a high temperature.

Operators sometimes regard tests and inspections as a nuisance, interfering with the smooth operation of the plant. Operator training should emphasize that protective equipment is there for their protection and they should "own" it.

Key Performance Indicators Preparation for maintenance, the control of modifications, and the testing of protective equipment are examples of key performance indicators; i.e., taken together, they indicate the quality of the plant's and company's process safety. If they are below standard, the plant is at risk. The usual measure of safety, the lost-time accident (LTA) rate, does not measure process safety. Many companies that had a low LTA rate and assumed that their process safety was therefore under control have experienced serious fires and explosions.

SECTION 24

Energy Resources,
Conversion,
and Utilization

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



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Energy Resources, Conversion, and Utilization*

Walter F. Podolski, Ph.D. *Chemical Engineer, Electrochemical Technology Program, Argonne National Laboratory; Member, American Institute of Chemical Engineers (Section Editor)*

David K. Schmalzer, Ph.D., P.E. *Fossil Energy Program Manager, Argonne National Laboratory; Member, American Chemical Society, American Institute of Chemical Engineers (Fuels, Liquid Petroleum Fuels, Gaseous Fuels)*

Vincent Conrad, Ph.D. *Group Leader, Technical Services Development, CONSOL Energy Inc.; Member, Spectroscopy Society of Pittsburgh, Society for Analytical Chemistry of Pittsburgh, Society for Applied Spectroscopy. (Solid Fuels)*

Douglas E. Lowenhaupt, M.S. *Group Leader, Coke Laboratory, CONSOL Energy Inc.; Member, American Society for Testing and Materials, Iron and Steel Making Society, International Committee for Coal Petrology (Solid Fuels)*

Richard A. Winschel, B.S. *Director, Research Services, CONSOL Energy Inc.; Member, American Chemical Society, Technical Committee of the Coal Utilization Research Council (Solid Fuels)*

Edgar B. Klunder, Ph.D. *Project Manager, National Energy Technology Laboratory, U.S. Department of Energy (Coal Conversion)*

Howard G. McIlvried III, Ph.D. *Consulting Engineer, Science Applications International Corporation, National Energy Technology Laboratory (Coal Conversion)*

Massood Ramezan, Ph.D., P.E. *Program Manager, Science Applications International Corporation, National Energy Technology Laboratory (Coal Conversion)*

Gary J. Stiegel, P.E., M.S. *Technology Manager, National Energy Technology Laboratory, U.S. Department of Energy (Coal Conversion)*

Rameshwar D. Srivastava, Ph.D. *Principal Engineer, Science Applications International Corporation, National Energy Technology Laboratory (Coal Conversion)*

John Winslow, M.S. *Technology Manager, National Energy Technology Laboratory, U.S. Department of Energy (Coal Conversion)*

Peter J. Loftus, D.Phil. *Principal, ENVIRON International Corp.; Member, American Society of Mechanical Engineers (Heat Generation, Thermal Energy Conversion and Utilization, Energy Recovery)*

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24-2 ENERGY RESOURCES, CONVERSION, AND UTILIZATION

Charles E. Benson, M.Eng. *Principal, ENVIRON International Corp.; Treasurer, American Flame Research Committee; Member, Combustion Institute (Heat Generation, Thermal Energy Conversion and Utilization, Energy Recovery)*

John M. Wheeldon, Ph.D. *Electric Power Research Institute (Fluidized-Bed Combustion)*

Michael Krumpelt, Ph.D. *Manager, Fuel Cell Technology, Argonne National Laboratory; Member, American Institute of Chemical Engineers, American Chemical Society, Electrochemical Society (Electrochemical Energy Conversion)*

(Francis) Lee Smith, Ph.D., M.Eng. *Principal, Wilcrest Consulting Associates, Houston, Texas; Member, American Institute of Chemical Engineers, Society of American Value Engineers, Water Environment Federation, Air and Waste Management Association (Energy Recovery, Economizers, Turbine Inlet Cooling)*

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Nomenclature and Units

Symbol	Definition	SI units	U.S. Customary System units	Acronym	Definition
A	Area specific resistance	Ω/m^2	Ω/ft^2	AFBC	atmospheric fluidized bed combustion
c	Heat capacity	$\text{J}/(\text{kg}\cdot\text{K})$	$\text{Btu}/(\text{lb}\cdot^\circ\text{F})$	AFC	alkaline fuel cell
E	Activation energy	J/mol	$\text{Btu}/\text{lb mol}$	AGC-21	Advanced Gas Conversion Process
E	Electrical potential	V	V	BGL	British Gas and Lurgi process
f	Fugacity	kPa	psia	COE	cost of electricity
F	Faraday constant	C/mol	$\text{C}/\text{lb mol}$	COED	Char Oil Energy Development Process
ΔG	Free energy of reaction	J/mol	$\text{Btu}/\text{lb mol}$	DOE	U.S. Department of Energy
ΔH	Heat of reaction	J/mol	$\text{Btu}/\text{lb mol}$	EDS	Exxon Donor Solvent Process
i	Current density	A/m^2	A/ft^2	FBC	fluidized bed combustion
k	Rate constant	$\text{g}/(\text{h}\cdot\text{cm}^3)$	$\text{lb}/(\text{h}\cdot\text{ft}^3)$	HAO	hydrogenated anthracene oil
K	Latent heat of vaporization	kJ/kg	Btu/lb	HPO	hydrogenated phenanthrene oil
P	Pressure	kPa	psia	HRI	Hydrocarbon Research, Inc.
Q	Heating value	kJ/kg	Btu/lb	HTI	Hydrocarbon Technologies, Inc.
R	Gas constant	$\text{J}/\text{mol}\cdot\text{K}$	$\text{Btu}/\text{lb mol}\cdot^\circ\text{R}$	IGCC	integrated gasification combined-cycle
s	Relative density	Dimensionless	Dimensionless	KRW	Kellogg-Rust-Westinghouse process
T	Temperature	K	$^\circ\text{F}$	MCFC	molten carbonate fuel cell
U	Fuel utilization	percent	percent	MTG	methanol-to-gasoline process
V	Molar gas volume	m^3/mol	$\text{ft}^3/\text{lb mol}$	OTFT	once-through Fischer-Tropsch process
Z	Compressibility factor	Dimensionless	Dimensionless	PAFC	phosphoric acid fuel cell
Greek Symbol				PC	pulverized coal
ϵ	Energy conversion efficiency	Percent	Percent	PEFC	polymer electrolyte fuel cell
Acronyms and Unit Prefixes				PFBC	pressurized fluidized bed combustion
Symbol	Name	Value		Quad	10^{15} Btu
E	Exa	10^{18}		SASOL	South African operation of synthetic fuels plants
G	Giga	10^9		SMDS	Shell Middle Distillate Synthesis Process
K	Kilo	10^3		SNG	synthetic natural gas
M	Mega	10^6		SOFC	solid oxide fuel cell
P	Peta	10^{15}		SRC	solvent-refined coal
T	Tera	10^{12}			
Z	Zetta	10^{21}			

INTRODUCTION

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Energy is usually defined as the capacity to do work. Nature provides us with numerous sources of energy, some difficult to utilize efficiently (e.g., solar radiation and wind energy), others more concentrated or energy dense and therefore easier to utilize (e.g., fossil fuels). Energy sources can be classified also as *renewable* (solar and nonsolar) and *nonrenewable*. Renewable energy resources are derived in a number of ways: gravitational forces of the sun and moon, which create the tides; the rotation of the earth combined with solar energy, which generates the currents in the ocean and the winds; the decay of radioactive minerals and the interior heat of the earth, which provide geothermal energy; photosynthetic production of organic matter; and the direct heat of the sun. These energy sources are called renewable because they are either continuously replenished or, for all practical purposes, are inexhaustible.

Nonrenewable energy sources include the fossil fuels (natural gas, petroleum, shale oil, coal, and peat) as well as uranium. Fossil fuels are both energy dense and widespread, and much of the world's industrial, utility, and transportation sectors rely on the energy contained in them. Concerns over global warming notwithstanding, fossil fuels will remain the dominant fuel form for the foreseeable future. This is so for two reasons: (1) the development and deployment of new technologies able to utilize renewable energy sources such as solar, wind, and biomass are uneconomic at present, in most part owing to the diffuse or intermittent nature of the sources; and (2) concerns persist over storage and/or disposal of spent nuclear fuel and nuclear proliferation.

Fossil fuels, therefore, remain the focus of this section; their principal use is in the generation of heat and electricity in the industrial, utility, and commercial sectors, and in the generation of shaft power in transportation. The material in this section deals primarily with the conversion of the chemical energy contained in fossil fuels to heat and electricity. Material from *Perry's Chemical Engineers' Handbook*, 7th ed., Sec. 27, has been updated and condensed. Recent improvements in materials and manufacturing methods have brought fuel cells closer to being economic for stationary and transportation power generation, but additional advances are required for broad adoption.

FUELS

RESOURCES AND RESERVES

Proven worldwide energy resources are large. The largest remaining known reserves of crude oil, used mainly for producing transportation fuels, are located in the Middle East, along the equator, and in the former Soviet Union. U.S. proven oil reserves currently account for only about 3 percent of the world's total. Large reserves of natural gas exist in the former Soviet Union and the Middle East. Coal is the most abundant fuel on earth and the primary fuel for electricity in the United States, which has the largest proven reserves. Annual world consumption of energy is still currently less than 1 percent of combined world reserves of fossil fuels. The resources and reserves of the principal fossil fuels in the United States—coal, petroleum, and natural gas—follow.

ZJ*			
Fuel	Proven reserves	Discovered conventionally reservoired	Total estimated resource
Coal	6.48	12.5	96.2
Petroleum	0.16	0.5	1.2
Natural gas	0.21	0.4	1.6

*ZJ = 10²¹ J. (To convert to 10¹⁸ Btu, multiply by 0.948.)

The energy content of fossil fuels in commonly measured quantities is as follows.

Energy content		
Bituminous and anthracite coal	30.2 MJ/kg	26 × 10 ⁶ Btu/US ton
Lignite and subbituminous coal	23.2 MJ/kg	20 × 10 ⁶ Btu/US ton
Crude oil	38.5 MJ/L	5.8 × 10 ⁶ Btu/bbl
Natural-gas liquids	25.2 MJ/L	3.8 × 10 ⁶ Btu/bbl
Natural gas	38.4 MJ/m ³	1032 Btu/ft ³

1 bbl = 42 US gal = 159 L = 0.159 m³

SOLID FUELS

Coal

GENERAL REFERENCES: Lowry (ed.), *Chemistry of Coal Utilization*, Wiley, New York, 1945; suppl. vol., 1963; 2d suppl. vol., Elliott (ed.), 1981. Van Krevelen, *Coal*, Elsevier, Amsterdam, 1961. *Annual Book of ASTM Standards*, sec. 5, ASTM International, West Conshohocken, Pa., 2004. *Methods of Analyzing and Testing Coal and Coke*, U.S. Bureau of Mines Bulletin, 638, 1967.

Origin Coal originated from the arrested decay of the remains of trees, bushes, ferns, mosses, vines, and other forms of plant life, which flourished in huge swamps and bogs many millions of years ago during prolonged periods of humid, tropical climate and abundant rainfall. The precursor of coal was peat, which was formed by bacterial and chemical action on the plant debris. Subsequent actions of heat, pressure, and other physical phenomena metamorphosed the peat to the various ranks of coal as we know them today. Because of the various degrees of the metamorphic changes during this process, coal is not a uniform substance; no two coals are ever the same in every respect.

Classification Coals are classified by rank, i.e., according to the degree of metamorphism in the series from lignite to anthracite. Table 24-1 shows the classification system described in ASTM D 388-99 (2004) (ASTM International, op. cit.). The heating value on the moist *mineral-matter-free* (mmf) basis, and the fixed carbon, on the dry mmf basis, are the bases of this system. The lower-rank coals are classified according to the heating value, kJ/kg (Btu/lb), on a moist mmf basis. The agglomerating character is used to differentiate between adjacent groups. Coals are considered agglomerating if the coke button remaining from the test for volatile matter will support a specified weight or if the button swells or has a porous cell structure.

The Parr formulas, Eqs. (24-1) to (24-3), are used for classifying coals according to rank. The Parr formulas are employed in litigation cases.

$$F' = \frac{100(F - 0.15S)}{100 - (M + 1.08A + 0.55S)} \quad (24-1)$$

TABLE 24-1 Classification of Coals by Rank*

Class/group	Fixed carbon limits (dry, mineral-matter-free basis), %		Volatile matter limits (dry, mineral-matter-free basis), %		Gross calorific value limits (moist, mineral-matter-free basis)†				Agglomerating character
					MJ/kg		Btu/lb		
	Equal or greater than	Less than	Greater than	Equal or less than	Equal or greater than	Less than	Equal or greater than	Less than	
Anthracitic:									
Meta-anthracite	98	—	—	2	—	—	—	—	Nonagglomerating
Anthracite	92	98	2	8	—	—	—	—	
Semianthracite‡	86	92	8	14	—	—	—	—	
Bituminous:									
Low-volatile bituminous coal	78	86	14	22	—	—	—	—	Commonly agglomerating¶
Medium-volatile bituminous coal	69	78	22	31	—	—	—	—	
High-volatile A bituminous coal	—	69	31	—	32.6§	—	14,000§	—	
High-volatile B bituminous coal	—	—	—	—	30.2§	32.6	13,000§	14,000	
High-volatile C bituminous coal	—	—	—	—	26.7	30.2	11,500	13,000	
					24.4	26.7	10,500	11,500	Agglomerating
Subbituminous:									
Subbituminous A coal	—	—	—	—	24.4	26.7	10,500	11,500	Nonagglomerating
Subbituminous B coal	—	—	—	—	22.1	24.4	9,500	10,500	
Subbituminous C coal	—	—	—	—	19.3	22.1	8,300	9,500	
Lignitic:									
Lignite A	—	—	—	—	14.7	19.3	6,300	8,300	
Lignite B	—	—	—	—	—	14.7	—	6,300	

Adapted, with permission, from D388-99, Standard Classification of Coals by Rank; copyright ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428.

*This classification does not apply to certain coals, as discussed in source.

†Moist refers to coal containing its natural inherent moisture but not including visible water on the surface of the coal.

‡If agglomerating, classify in low-volatile group of the bituminous class.

§Coals having 69 percent or more fixed carbon on the dry, mineral-matter-free basis shall be classified according to fixed carbon, regardless of gross calorific value.

¶It is recognized that there may be nonagglomerating varieties in these groups of the bituminous class and that there are notable exceptions in the high-volatile C bituminous group.

$$V' = 100 - F' \quad (24-2)$$

$$Q' = \frac{100(Q - 50S)}{100 - (M + 1.08A + 0.55S)} \quad (24-3)$$

where M , F , A , and S are weight percentages, on a moist basis, of moisture, fixed carbon, ash, and sulfur, respectively; Q and Q' are calorific values (Btu/lb), on a moist non-mmF basis and a moist mmF basis, respectively. (Btu/lb = 2326 J/kg)

Composition and Heating Value Coal analyses are reported on several bases, and it is customary to select the basis best suited to the application. The **as-received** basis represents the weight percentage of each constituent in the sample as received in the laboratory. The sample itself may be coal as fired, as mined, or as prepared for a particular use. The **moisture-free** (dry) basis is a useful basis because performance calculations can be easily corrected for the actual moisture content at the point of use. The **dry, ash-free** basis is frequently used to approximate the rank and source of a coal. For example, the heating value of coal from a given deposit is remarkably constant when calculated on this basis.

The composition of coal is reported in two different ways: the proximate analysis and the ultimate analysis, both expressed in weight percent. The *proximate analysis* [ASTM D 3172-89 (2002), ASTM International, op. cit.] is the determination by prescribed methods of moisture, volatile matter, fixed carbon, and ash. The moisture in coal consists of inherent moisture, also called equilibrium moisture, and surface moisture. Free moisture is that moisture lost when coal is air-dried under standard low-temperature conditions. The *volatile matter* is the portion of coal which, when the coal is heated in the absence of air under prescribed conditions, is liberated as gases and vapors. Volatile matter does not exist by itself in coal, except for a little absorbed methane, but results from thermal decomposition of the coal substance. *Fixed carbon*, the residue left after the volatile matter is driven off, is calculated by subtracting from 100 the percentages of moisture, volatile matter, and ash of the proximate analysis. In addition to carbon, it may contain several tenths of a percent of hydrogen and oxygen, 0.4 to 1.0 percent nitrogen, and about half of the sulfur that was in the coal. *Ash* is the inorganic residue that remains after the coal has been burned under specified conditions, and it is composed largely of compounds of silicon, aluminum, iron, and calcium, and minor amounts of compounds of magnesium, sodium, potassium, phosphorous, sulfur, and titanium. Ash may vary considerably from the original mineral matter, which is largely kaolinite, illite, montmorillonite, quartz, pyrites, and calcite. The *ultimate analysis* [ASTM D 3176-89 (2002), ASTM International, op. cit.] is the determination by prescribed methods of the ash, carbon, hydrogen, nitrogen, sulfur, and (by difference) oxygen. Other, minor constituent elements are also sometimes determined, most notably chlorine.

The **heating value**, or **calorific value**, expressed as kJ/kg (Btu/lb), is the heat produced at constant volume by the complete combustion of a unit quantity of coal in an oxygen-bomb calorimeter under specified conditions (ASTM D 5865-04, ASTM International, op. cit.). The result includes the latent heat of vaporization of the water in the combustion products and is called the gross heating or **high heating value** (HHV) Q_h . And Q_h in Btu/lb ($\times 2.326 = \text{kJ/kg}$) on a dry basis can be approximated by a formula developed by the Institute of Gas Technology:

$$Q_h = 146.58C + 568.78H + 29.4S - 6.58A - 51.53(O + N) \quad (24-4)$$

where C , H , S , A , O , and N are the weight percentages on a dry basis of carbon, hydrogen, sulfur, ash, oxygen, and nitrogen, respectively. The heating value when the water is not condensed is called the **low heating value** (LHV) Q_l and is obtained from

$$Q_l = Q_h - K \cdot W \quad (24-5)$$

where W = weight of water in the combustion products/weight of fuel burned. The factor K is the latent heat of vaporization at the partial pressure of the vapor in the gas, which, at 20°C, is 2395 kJ/kg (1030 Btu/lb) of water. Thus,

$$Q_l = Q_h - \%H \cdot 214 \quad \text{kJ/kg} \quad (24-6)$$

$$Q_l = Q_h - \%H \cdot 92.04 \quad \text{Btu/lb} \quad (24-7)$$

where $\%H$ = weight percent hydrogen in the coal and all values are on an as-determined (including moisture) basis.

Sulfur Efforts to abate atmospheric pollution have drawn considerable attention to the sulfur content of coal, since the combustion of coal results in the discharge to the atmosphere of sulfur oxides. Sulfur occurs in coal in three major forms: as organic sulfur (20 to 80 percent of the sulfur), which is a part of the coal substance; as pyrite (FeS_2); and as sulfate (<5 percent of the sulfur in unoxidized coals). Organic sulfur is chemically bound to the coal substance, and severe treatment is necessary to break the chemical bonds to remove the sulfur. There is no existing economical method that will remove organic sulfur. Pyritic sulfur can be partially removed by using standard coal-washing equipment. The degree of pyrite removal depends on the size of the coal and the size and distribution of the pyrite particles.

The sulfur content of U.S. coals varies widely, ranging from a low of 0.2 percent to as much as 7 percent by weight, on a dry basis. The estimated remaining (as of 1997) recoverable U.S. coal reserves of all ranks, by sulfur content and major producing region, are shown in Table 24-2. The values in the table are in units of 10^9 metric tonnes (Pg) as reported in *U.S. Coal Reserves: 1997 Update*, DOE/EIA-0529(97). Extensive data on sulfur and sulfur reduction potential, including washability, in U.S. coals are given in *Sulfur and Ash Reduction Potential and Selected Chemical and Physical Properties of United States Coal* (U.S. Dept. of Energy, DOE/PETC, TR-90/7, 1990; TR-91/1 and TR-91/2, 1991).

Mercury Impending regulations limiting the emission of mercury from coal-fired furnaces have created great interest in knowing the concentration of mercury in various coals, determining the fate of mercury during combustion, and developing methods to control mercury emissions to the atmosphere. Most commercially produced U.S. coals have mercury contents of about 0.05 to 0.2 mg/kg, dry basis, although numerous individual coals fall outside that range. Depending on the specific coal properties, furnace conditions, and type and configuration of pollution control devices with which the furnace is equipped, less than 10 percent or more than 90 percent of the coal mercury will be emitted to the atmosphere with the flue gas. The chemical form of the vaporous mercury in the flue gas, whether elemental or ionic, is important, because ionic mercury is easier to capture with available technology than is elemental mercury. The chemical form is apparently influenced by the chlorine content of the coal: coals with higher chlorine contents (0.05 percent and higher) tend to produce mainly ionic mercury in their flue gases, and thus the mercury emissions from these coals are more easily controlled.

Coal-Ash Characteristics and Composition When coal is to be burned, used in steel making, or gasified, it is often important to know the ash fusibility, or the temperatures at which formed pyramids of ash attain certain defined stages of fusing in either a mildly reducing or an oxidizing atmosphere; these temperatures are known as the initial deformation, softening, hemispherical, and fluid temperatures. The procedure for determining the fusibility of coal ash is prescribed by ASTM D 1857-04 (ASTM International, op. cit.). The ash fusibility temperatures are most often used as indicators of the tendency of the ash to form sintered or fused masses, which can impede gas flow through a grate and impair heat flow through furnace heat-transfer surfaces, or as indicators of the flowability of ash in slag-tap and cyclone furnaces. However, because ash fusibility is not an infallible index of ash behavior in practice, care is needed in using fusibility data for designing and operating purposes. There is an excellent discussion on this subject in *Steam: Its Generation and Use* (40th ed., Babcock & Wilcox Co., New York, 1992).

TABLE 24-2 Estimate of Recoverable U.S. Coal Reserves by Sulfur Ranges and Regions (Pg)

	Low sulfur*	Medium sulfur*	High sulfur*	Total
Appalachia	10.6	18.4	21.1	50.2
Interior	0.7	9.1	52.6	62.4
West	79.6	49.5	8.0	137.1
U.S. total	90.9	77.0	81.7	249.6

*Sulfur content ranges defined in units of kg S/kg (lb S/MBtu) as follows: low sulfur, < 0.29 (< 0.60); medium sulfur, 0.29 to 0.80 (0.61 to 1.67); high sulfur, > 0.80 (> 1.67).

The composition of coal ash varies widely. Calculated as oxides, the composition (percent by weight) varies as follows:

SiO ₂	20–60
Al ₂ O ₃	10–35
Fe ₂ O ₃	5–35
CaO	1–20
MgO	0.3–4
TiO ₂	0.5–2.5
P ₂ O ₅	0.01–1
Na ₂ O and K ₂ O	1–4
SO ₃	0.1–12

Knowledge of the composition of coal ash is useful for estimating and predicting the fouling and corrosion of heat-exchange surfaces in pulverized-coal-fired furnaces and in coke making. Multiple correlations for ash composition and ash fusibility are discussed in *Coal Conversion Systems Technical Data Book* (part IA, U.S. Dept. of Energy, 1984). The slag viscosity-temperature relationship provided in that reference for completely melted slag is

$$\text{Log viscosity} = \frac{10^7 M}{(T - 150)^2 + C} \quad (24-8)$$

where viscosity is in poise ($\times 0.1 = \text{Pa}\cdot\text{s}$), $M = 0.00835(\text{SiO}_2) + 0.00601(\text{Al}_2\text{O}_3) - 0.109$, $C = 0.0415(\text{SiO}_2) + 0.0192(\text{Al}_2\text{O}_3) + 0.0276$ (equivalent Fe_2O_3) + $0.0160(\text{CaO}) - 3.92$, and $T = \text{temperature, K}$.

The oxides in parentheses are the weight percentages of these oxides when $\text{SiO}_2 + \text{Al}_2\text{O}_3 + \text{Fe}_2\text{O}_3 + \text{CaO} + \text{MgO}$ are normalized to 100 percent.

Physical Properties The *free-swelling index* (FSI) measures the tendency of a coal to swell when burned or gasified in fixed or fluidized beds. Coals with a high FSI (greater than 4) can usually be expected to cause difficulties in such beds. Details of the test are given by the ASTM D 720–91 (2004) (American Society for Testing and Materials, op. cit.) and U.S. Bureau of Mines Report of Investigations 3989.

The *Hardgrove grindability index* (HGI) indicates the ease (or difficulty) of grinding coal and is complexly related to physical properties such as hardness, fracture, and tensile strength. The Hardgrove machine is usually employed (ASTM D 409–02, American Society for Testing and Materials, op. cit.). It determines the relative grindability or ease of pulverizing coal in comparison with a standard coal, chosen as 100 grindability. The FSI and HGI of some U.S. coals are given in Bureau of Mines Information Circular 8025 for FSI and HGI data for 2812 and 2339 samples, respectively.

The *bulk density* of broken coal varies according to the specific gravity, size distribution, and moisture content of the coal and the amount of settling when the coal is piled. Following are some useful approximations of the bulk density of various ranks of coal.

	kg/m ³	lb/ft ³
Anthracite	800–930	50–58
Bituminous	670–910	42–57
Lignite	640–860	40–54

Size stability refers to the ability of coal to withstand breakage during handling and shipping. It is determined by dropping a sample of coal onto a steel plate in a specified manner and comparing the size distribution before and after the test, as in ASTM D 440–86 (2002) (ASTM International, op. cit.). A complementary property, *friability*, is the tendency of coal to break during repeated handling, and it is determined by the standard tumbler test, as in ASTM D 441–86 (2002) (ASTM International, op. cit.).

Spier's *Technical Data on Fuels* gives the *specific heat* of dry, ash-free coal as follows.

	kJ/(kg·K)	Btu/(lb·°F)
Anthracite	0.92–0.96	0.22–0.23
Bituminous	1.0–1.1	0.24–0.25

The relationships between specific heat and water content and between specific heat and ash content are linear. Given the specific heat on a dry, ash-free basis, it can be corrected to an as-received

basis. The specific heat and enthalpy of coal to 1366 K (2000°F) are given in *Coal Conversion Systems Technical Data Book* (part 1A, U.S. Dept. of Energy, 1984).

The *mean specific heat* of coal ash and slag, which is used for calculating heat balances on furnaces, gasifiers, and other coal-consuming systems, follows.

Temperature range		Mean specific heat	
K	°F	kJ/(kg·K)	Btu/(lb·°F)
273–311	32–100	0.89	0.21
273–1090	32–1500	0.94	0.22
273–1310	32–1900	0.97	0.23
273–1370	32–2000	0.98	0.24
273–1640	32–2500	1.1	0.27

Coke Coke is the solid, cellular, infusible material remaining after the carbonization of coal, pitch, petroleum residues, and certain other carbonaceous materials. The varieties of coke generally are identified by prefixing a word to indicate the source, if other than coal, (e.g., *petroleum coke*), the process by which a coke is manufactured (e.g., vertical slot oven coke), or the end use (e.g., blast furnace coke). The carbonization of coal into coke involves a complex set of physical and chemical changes. Some of the physical changes are softening, devolatilization, swelling, and resolidification. Some of the accompanying chemical changes are cracking, depolymerization, polymerization, and condensation. More detailed theoretical information is given in the general references listed in the beginning of the section on coal.

High-Temperature Coke (1173 to 1423 K or 1652 to 2102°F.) Essentially all coal-derived coke produced in the United States is high-temperature coke for metallurgical applications; its production comprises nearly 5 percent of the total bituminous coal consumed in the United States. About 90 percent of this type of coke is made in slot-type by-product recovery ovens, and the rest is made in heat recovery ovens. Blast furnaces use about 90 percent of the production, the rest going mainly to foundries and gas plants. The ranges of chemical and physical properties of metallurgical coke used in the United States are given in Table 24-3. Blast furnaces use about 90 percent of the production, the rest going mainly to foundries and gas plants.

The typical by-product yields per U.S. ton (909 kg) of dry coal from high-temperature carbonization in ovens with inner-wall temperatures from 1273 to 1423 K (1832 to 2102°F) are: coke, 653 kg (1437 lb); gas, 154 kg (11,200 ft³); tar, 44 kg (10 gal); water, 38 kg (10 gal); light oil, 11 kg (3.3 gal); and ammonia, 2.2 kg (4.8 lb).

Foundry Coke This coke must meet specifications not required of blast furnace coke. The volatile matter should not exceed 1.0 percent, the sulfur should not exceed 0.7 percent, the ash should not exceed 8.0 percent, and the size should exceed 100 mm (4 in).

Low- and Medium-Temperature Coke (773 to 1023 K or 932 to 1382°F.) Cokes of this type are no longer produced in the United States to a significant extent. However, there is some interest in low-temperature carbonization as a source of both hydrocarbon liquids and gases to supplement petroleum and natural-gas resources.

Pitch Coke and Petroleum Coke Pitch coke is made from coal-tar pitch, and petroleum coke is made from petroleum residues from petroleum refining. Pitch coke has about 1.0 percent volatile matter, 1.0 percent ash, and less than 0.5 percent sulfur on the as-received basis. There are two kinds of petroleum coke: delayed coke and fluid coke. Delayed coke is produced by heating a gas oil or heavier feedstock to

TABLE 24-3 Chemical and Physical Properties of Metallurgical Cokes Used in the United States

Property	Range of values
Volatile matter	0.5–1.0%, dry basis
Ash content	8–12%, dry basis
Sulfur content	0.6–1.0%, dry basis
Stability factor	55–65
Hardness factor	60–68
Strength after reaction	55–65
Apparent specific gravity (water = 1.0)	0.8–0.99

TABLE 24-4 Waste Fuel Analysis

Type of waste	Heating value, Btu/lb	Percentage composition by weight					Density, lb/ft ³
		Volatiles	Moisture	Ash	Sulfur	Dry combustible	
Paper	7,572	84.6	10.2	6.0	0.20		
Wood	8,613	84.9	20.0	1.0	0.05		
Rags	7,652	93.6	10.0	2.5	0.13		
Garbage	8,484	53.3	72.0	16.0	0.52		
Coated fabric: rubber	10,996	81.2	1.04	21.2	0.79	78.80	23.9
Coated felt: vinyl	11,054	80.87	1.50	11.39	0.80	88.61	10.7
Coated fabric: vinyl	8,899	81.06	1.48	6.33	0.02	93.67	10.1
Polyethylene film	19,161	99.02	0.15	1.49	0	98.51	5.7
Foam: scrap	12,283	75.73	9.72	25.30	1.41	74.70	9.1
Tape: resin-covered glass	7,907	15.08	0.51	56.73	0.02	43.27	9.5
Fabric: nylon	13,202	100.00	1.72	0.13	0	99.87	6.4
Vinyl scrap	11,428	75.06	0.56	4.56	0.02	95.44	23.4

SOURCE: From Hescheles, *MECAR Conference on Waste Disposal*, New York, 1968; and *Refuse Collection Practice*, 3d ed., American Public Works Association, Chicago, 1966.

To convert British thermal units per pound to joules per kilogram, multiply by 2326; to convert pounds per cubic foot to kilograms per cubic meter, multiply by 16.02.

755 to 811 K (900 to 1000°F) and spraying it into a large vertical cylinder where cracking and polymerization reactions occur. Fluid coke is made in a fluidized-bed reactor where preheated feed is sprayed onto a fluidized bed of coke particles. Coke product is continuously withdrawn by size classifiers in the solids loop of the reactor system. Petroleum coke contains many of the impurities from its feedstock; thus, the sulfur content is usually high, and appreciable quantities of vanadium may be present. Ranges of composition and properties are as follows:

Composition and properties	Delayed coke	Fluid coke
Volatile matter, wt %	8–18	3.7–7.0
Ash, wt %	0.05–1.6	0.1–2.8
Sulfur, wt %	—	1.5–10.0
Grindability index	40–60	20–30
True density, g/cm ³	1.28–1.42	1.5–1.6

Most petroleum coke is used for fuel, but some premium delayed coke known as “needle coke” is used to make anodes for the aluminum industry. That coke is first calcined to less than 0.5 percent volatiles at 1300 to 1400°C before it is used to make anodes.

Other Solid Fuels

Coal Char Coal char is, generically, the nonagglomerated, nonfusible residue from the thermal treatment of coal; however, it is more specifically the solid residue from low- or medium-temperature carbonization processes. Char is used as a fuel or a carbon source. Chars have compositions intermediate between those of coal and coke: the volatile matter, sulfur content, and heating values of the chars are lower, and the ash content is higher, than those of the original coal.

Peat Peat is partially decomposed plant matter that has accumulated in a water-saturated environment. It is the precursor of coal but is not classified as coal. Peat is used extensively as a fuel primarily in Ireland and the former Soviet Union, but in the United States, its main use is in horticulture and agriculture. Although analyses of peat vary widely, a typical high-grade peat has 90 percent water, 3 percent fixed carbon, 5 percent volatile matter, 1.5 percent ash, and 0.10 percent sulfur. The moisture-free heating value is approximately 20.9 MJ/kg (9000 Btu/lb).

Wood Typical higher heating values are 20 MJ/kg (8600 Btu/lb) for oven-dried hardwood and 20.9 MJ/kg for oven-dried softwood. These values are accurate enough for most engineering purposes. U.S. Department of Agriculture Handbook 72 (revised 1974) gives the specific gravity of the important softwoods and hardwoods, useful if heating value on a volume basis is needed.

Charcoal Charcoal is the residue from the destructive distillation of wood. It absorbs moisture readily, often containing as much as 10 to 15 percent water. In addition, it usually contains about 2 to 3 percent ash and 0.5 to 1.0 percent hydrogen. The heating value of charcoal is about 27.9 to 30.2 MJ/kg (12,000 to 13,000 Btu/lb).

Solid Wastes and Biomass The generation of large quantities of solid wastes is a significant feature of affluent societies. In the United

States in 2001 the rate was about 2 kg (4.4 lb) per capita per day, or nearly 208 Tg (229 M short tons) per year. Table 24-4 shows that the composition of miscellaneous refuse is fairly uniform, but size and moisture variations cause major difficulties in efficient, economical disposal. The fuel value of municipal solid wastes is usually sufficient to enable self-supporting combustion, leaving only the incombustible residue and reducing by 90 percent the volume of waste consigned to landfill. The heat released by the combustion of waste can be recovered and utilized, although this is not always economically feasible.

Wood, wood scraps, bark, and wood product plant waste streams are major elements of biomass, industrial, and municipal solid waste fuels. In 1991, about 1.7 EJ (1.6×10^{15} Btu [quads]) of energy were obtained from wood and wood wastes, representing about 60 percent of the total biomass-derived energy in the United States. *Bagasse* is the solid residue remaining after sugarcane has been crushed by pressure rolls. It usually contains from 40 to 50 percent water. The dry bagasse has a heating value of 18.6 to 20.9 MJ/kg (8000 to 9000 Btu/lb). Tire-derived fuel (TDF), which is produced by shredding and processing waste tires and which has a heating value of 30.2 to 37.2 MJ/kg (13,000 to 16,000 Btu/lb), is an important fuel for use in cement kilns and as a supplement to coal in steam raising.

LIQUID FUELS

Liquid Petroleum Fuels The discussion here focuses on burner fuels rather than transportation fuels. There is overlap, particularly for fuels in the distillate or “gas oil” range. Other factors such as the Tier II gasoline specifications, the ultralow-sulfur diesel specifications, and the gradual reduction in crude quality impact refining and blending practices for burner fuels. The principal liquid fuels are made by fractional distillation of crude petroleum (a mixture of hydrocarbons and hydrocarbon derivatives ranging from methane to heavy bitumen). As many as one-quarter to one-half of the molecules in crude may contain sulfur atoms, and some contain nitrogen, oxygen, vanadium, nickel, or arsenic. Desulfurization, hydrogenation, cracking (to lower molecular weight), and other refining processes may be performed on selected fractions before they are blended and marketed as fuels. Viscosity, gravity, and boiling ranges are given in Table 24-5.

Specifications The American Society for Testing and Materials has developed specifications (*Annual Book of ASTM Standards*, Conshohocken, Pa., updated annually) that are widely used to classify fuels. Table 24-5 shows fuels covered by ASTM D 396, Standard Specification for Fuel Oils. D 396 omits kerosenes (low-sulfur, clean-burning No. 1 fuels for lamps and freestanding flueless domestic heaters), which are covered separately by ASTM D 3699.

In drawing contracts and making acceptance tests, refer to the pertinent ASTM standards. *ASTM Standards* contain specifications (classifications) and test methods for burner fuels (D 396), motor and aviation gasolines (D 4814-03 and D 910-03), diesel fuels (D 975-03), and aviation and gas-turbine fuels (D 1655-03 and D 2880-03).

TABLE 24-5 Detailed Requirements for Fuel Oils^a

Property	ASTM Test Method ^b	No. 1 Low Sulfur ^c	No. 1 ^c	No. 2 Low Sulfur ^c	No. 2 ^c	Grade No. 4 (Light) ^c	No. 4	No. 5 (Light)	No. 5 (Heavy)	No. 6
Flash point, °C, min	D 93—Proc. A D 93—Proc. B	38 —	38 —	38 —	38 —	38 —	— 55	— 55	— 55	— 60
Water and sediment, % vol, max	D 2709 D 95 + D 473	0.05 —	0.05 —	0.05 —	0.05 —	— (0.50) ^d	— (0.50) ^d	— (1.00) ^d	— (1.00) ^d	— (2.00) ^d
Distillation temperature, °C	D 86	—	—	—	—	—	—	—	—	—
10% volume recovered, max		215	215	—	—	—	—	—	—	—
90% volume recovered, min		—	—	282	282	—	—	—	—	—
90% volume recovered, max		288	288	338	338	—	—	—	—	—
Kinematic viscosity at 40°C, mm ² /s	D 445	—	—	—	—	—	—	—	—	—
min		1.3	1.3	1.9	1.9	1.9	>5.5	—	—	—
max		2.1	2.1	3.4	3.4	5.5	24.0 ^e	—	—	—
Kinematic viscosity at 100°C, mm ² /s	D 445	—	—	—	—	—	—	5.0	9.0	15.0
min		—	—	—	—	—	—	8.9 ^e	14.9 ^e	50.0 ^e
max		—	—	—	—	—	—	—	—	—
Ramsbottom carbon residue on 10% distillation residue, % mass, max	D 524	0.15	0.15	0.35	0.35	—	—	—	—	—
Ash, % mass, max	D 482	—	—	—	—	0.05	0.10	0.15	0.15	—
Sulfur, % mass max ^f	D 129 D 2622	— 0.05	— 0.50	— 0.05	— 0.50	—	—	—	—	—
Copper strip corrosion rating, max, 3 h at 50°C	D 130	No. 3	No. 3	No. 3	No. 3	—	—	—	—	—
Density at 15°C, kg/m ³	D 1298	—	—	—	—	>876 ^g	—	—	—	—
min		850	850	876	876	—	—	—	—	—
max		—	—	—	—	—	—	—	—	—
Pour point, °C, max ^h	D 97	-18	-18	-6	-6	-6	-6	—	—	ⁱ

Adapted, with permission, from D396-06, Standard Specification for Fuel Oils; copyright ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428.

^a It is the intent of these classifications that failure to meet any requirement of a given grade does not automatically place an oil in the next lower grade unless in fact it meets all requirements of the lower grade. However, to meet special operating conditions, modifications of individual limiting requirements may be agreed upon among the purchaser, seller, and manufacturer.

^b The test methods indicated are the approved referee methods. Other acceptable methods are indicated in Sections 2 and 5.1 of ASTM D 396.

^c Under U.S. regulations, Grades No. 1, No. 1 Low Sulfur, No. 2, No. 2 Low Sulfur, and No. 4 (Light) are required by 40 CFR Part 80 to contain a sufficient amount of the dye Solvent Red 164 so its presence is visually apparent. At or beyond terminal storage tanks, they are required by 26 CFR Part 48 to contain the dye Solvent Red 164 at a concentration spectrally equivalent to 3.9 lb per thousand barrels of the solid dye standard Solvent Red 26.

^d The amount of water by distillation by Test Method D 95 plus the sediment by extraction by Test method D 473 shall not exceed the value shown in the table. For Grade No. 6 fuel oil, the amount of sediment by extraction shall not exceed 0.50 mass percent, and a deduction in quantity shall be made for all water and sediment in excess of 1.0 mass percent.

^e Where low sulfur fuel oil is required, fuel oil falling in the viscosity range of a lower numbered grade down to and including No. 4 can be supplied by agreement between the purchaser and supplier. The viscosity range of the initial shipment shall be identified, and advance notice shall be required when changing from one viscosity range to another. This notice shall be in sufficient time to permit the user to make the necessary adjustments.

^f Other sulfur limits may apply in selected areas in the United States and in other countries.

^g This limit ensures a minimum heating value and also prevents misrepresentation and misapplication of this product as Grade No. 2.

^h Lower or higher pour points can be specified whenever required by conditions of storage or use. When a pour point less than -18°C is specified, the minimum viscosity at 40°C for Grade No. 2 shall be 1.7 mm²/s and the minimum 90% recovered temperature shall be waived.

ⁱ Where low sulfur fuel oil is required, Grade No. 6 fuel oil will be classified as Low Pour (+15°C max) or High Pour (no max). Low Pour fuel oil should be used unless tanks and lines are heated.

ASTM D 4057-95 contains procedures for sampling bulk oil in tanks, barges, etc.

Fuel specifications from different sources may differ in test limits on sulfur, density, etc., but the same general categories are recognized worldwide: kerosene-type vaporizing fuel, distillate (or gas oil) for atomizing burners, and more viscous blends and residuals for commerce and heavy industry.

Foreign specifications are generally available from the American National Standards Institute, New York; United States federal specifications, at Naval Publications and Forms, Philadelphia. The International Association for Stability, Handling and Use of Liquid Fuels maintains a web site, www.iash.net, with extensive references to fuel standards.

Equipment manufacturers and large-volume users often write fuel specifications to suit particular equipment, operating conditions, and economics. Nonstandard test procedures and restrictive test limits should be avoided; they reduce the availability of fuel and increase its cost.

Bunker-fuel specifications for merchant vessels were described by ASTM D 2069, Standard Specification for Marine Fuels, which was withdrawn in 2003. Specifications under ASTM D-396 or foreign specifications may be substituted as appropriate.

Chemical and Physical Properties Petroleum fuels contain paraffins, isoparaffins, naphthenes, and aromatics, plus organic sulfur, oxygen, and nitrogen compounds that were not removed by refining. Olefins are absent or negligible except when created by severe refining. Vacuum-tower distillate with a final boiling point equivalent to 730 to 840 K (850 to 1050°F) at atmospheric pressure may contain from 0.1 to

0.5 ppm vanadium and nickel, but these metal-bearing compounds do not distill into No. 1 and 2 fuel oils.

Black, viscous residuum directly from the still at 410 K (390°F) or higher serves as fuel in nearby furnaces or may be cooled and blended to make commercial fuels. Diluted with 5 to 20 percent distillate, the blend is No. 6 fuel oil. With 20 to 50 percent distillate, it becomes No. 4 and No. 5 fuel oils for commercial use, as in schools and apartment houses. Distillate-residual blends also serve as diesel fuel in large stationary and marine engines. However, distillates with inadequate solvent power will precipitate asphaltenes and other high-molecular-weight colloids from *visbroken* (severely heated) residuals. A blotter test, ASTM D 4740-02, will detect sludge in pilot blends. Tests employing centrifuges, filtration (D 4870-99), and microscopic examination have also been used.

No. 6 fuel oil contains from 10 to 500 ppm vanadium and nickel in complex organic molecules, principally porphyrins. These cannot be removed economically, except incidentally during severe hydrodesulfurization (Amero, Silver, and Yanik, *Hydrodesulfurized Residual Oils as Gas Turbine Fuels*, ASME Pap. 75-WA/GT-8). Salt, sand, rust, and dirt may also be present, giving No. 6 a typical ash content of 0.01 to 0.5 percent by weight.

Ultimate analyses of some typical fuels are shown in Table 24-6.

The hydrogen content of petroleum fuels can be calculated from density with the following formula, with an accuracy of about 1 percent for petroleum liquids that contain no sulfur, water, or ash:

$$H = 26 - 15s \quad (24-9)$$

TABLE 24-6 Typical Ultimate Analyses of Petroleum Fuels

Composition, %	No. 1 fuel oil (41.5° API)	No. 2 fuel oil (33° API)	No. 4 fuel oil (23.2° API)	Low sulfur, No. 6 F.O. (12.6° API)	High sulfur, No. 6 (15.5° API)
Carbon	86.4	87.3	86.47	87.26	84.67
Hydrogen	13.6	12.6	11.65	10.49	11.02
Oxygen	0.01	0.04	0.27	0.64	0.38
Nitrogen	0.003	0.006	0.24	0.28	0.18
Sulfur	0.09	0.22	1.35	0.84	3.97
Ash	<0.01	<0.01	0.02	0.04	0.02
C/H Ratio	6.35	6.93	7.42	8.31	7.62

NOTE: The C/H ratio is a weight ratio.

where H = percent hydrogen and s = relative density at 15°C (with respect to water), also referred to as specific gravity.

Relative density is usually determined at ambient temperature with specialized hydrometers. In the United States these hydrometers commonly are graduated in an arbitrary scale termed *degrees API*. This scale relates inversely to relative density s (at 60°F) as follows (see also the abscissa scale of Fig. 24-1):

$$\text{Degrees API} = \frac{141.5}{s} - 131.5 \quad (24-10)$$

For practical engineering purposes, relative density at 15°C (288 K), widely used in countries outside the United States, is considered equivalent to specific gravity at 60°F (288.6 K). With the adoption of

SI units, the American Petroleum Institute favors absolute density at 288 K instead of degrees API.

The hydrogen content, heat of combustion, specific heat, and thermal conductivity data herein were abstracted from Bureau of Standards Miscellaneous Publication 97, *Thermal Properties of Petroleum Products*. These data are widely used, although other correlations have appeared, notably that by Linden and Othmer (*Chem. Eng.* 54[4, 5], April and May, 1947).

Heat of combustion can be estimated within 1 percent from the relative density of the fuel by using Fig. 24-1. Corrections for water and sediment must be applied for residual fuels, but they are insignificant for clean distillates.

Pour point ranges from 213 K (−80°F) for some kerosene-type jet fuels to 319 K (115°F) for waxy No. 6 fuel oils. Cloud point (which is not measured on opaque fuels) is typically 3 to 8 K higher than pour point unless the pour has been depressed by additives. Typical petroleum fuels are practically newtonian liquids between the cloud point and the boiling point and at pressures below 6.9 MPa (1000 psia).

Fuel systems for No. 1 (kerosene) and No. 2 fuel oil (diesel, home heating oil) are not heated. Systems for No. 6 fuel oil are usually designed to preheat the fuel to 300 to 320 K (90 to 120°F) to reduce viscosity for handling and to 350 to 370 K (165 to 200°F) to reduce viscosity further for proper atomization. No. 5 fuel oil may also be heated, but preheating is usually not required for No. 4. (See Table 24-5.) Steam or electric heating is employed as dictated by economics, climatic conditions, length of storage time, and frequency of use. Pressure relief arrangements are recommended on sections of heated pipelines when fuel could be inadvertently trapped between valves.

The kinematic viscosity of a typical No. 6 fuel oil declines from 5000 mm²/s (0.054 ft²/s) at 298 K (77°F) to about 700 mm²/s (0.0075 ft²/s) and 50 mm²/s (0.000538 ft²/s) on heating to 323 K (122°F) and 373 K (212°F), respectively. Viscosity of 1000 mm²/s or less is required for manageable pumping. Proper boiler atomization requires a viscosity between 15 and 65 mm²/s.

Thermal expansion of petroleum fuels can be estimated as volume change per unit volume per degree. ASTM-IP Petroleum Measurement Tables (ASTM D 1250 IP 200) are used for volume corrections in commercial transactions.

Heat capacity (specific heat) of petroleum liquids between 0 and 205°C (32 and 400°F), having a relative density of 0.75 to 0.96 at 15°C (60°F), can be calculated within 2 to 4 percent of the experimental values from the following equations:

$$c = \frac{1.685 + (0.039 \times ^\circ\text{C})}{\sqrt{s}} \quad (24-11)$$

$$c' = \frac{0.388 + (0.00045 \times ^\circ\text{F})}{\sqrt{s}} \quad (24-12)$$

where c is heat capacity, kJ/(kg·°C) or kJ/(kg·K), and c' is heat capacity, Btu/(lb·°F). Heat capacity varies with temperature, and the arithmetic average of the values at the initial and final temperatures can be used for calculations relating to the heating or cooling of oil.

The thermal conductivity of liquid petroleum products is given in Fig. 24-2. Thermal conductivities for asphalt and paraffin wax in their solid states are 0.17 and 0.23 W/(m·K), respectively, for temperatures above 273 K (32°F) (1.2 and 1.6 Btu/[h·ft²][°F/in]).

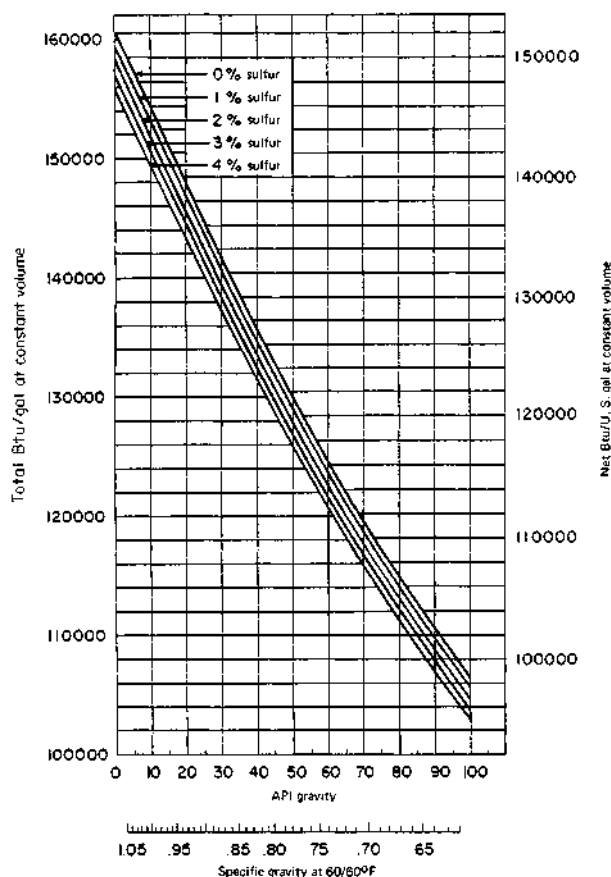


FIG. 24-1 Heat of combustion of petroleum fuels. To convert Btu/U.S. gal to kJ/m³, multiply by 278.7.

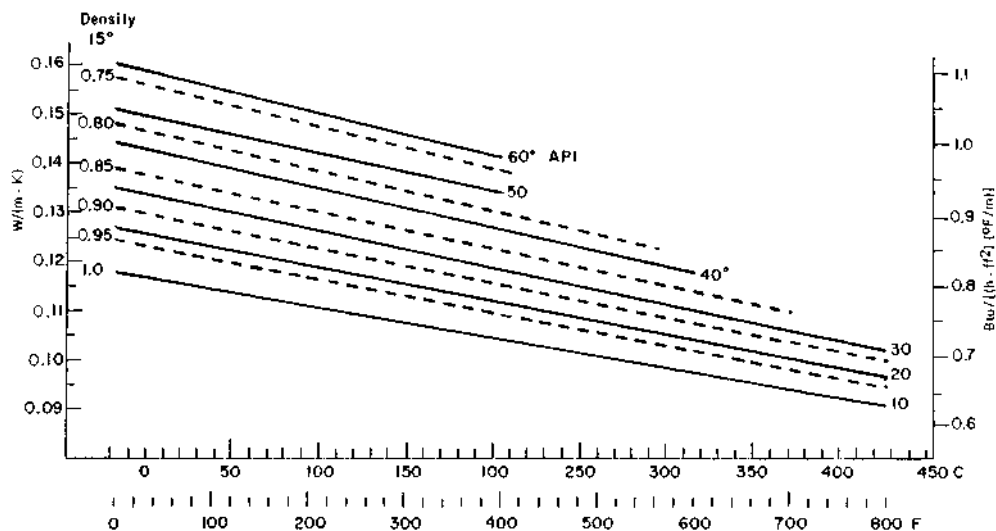


FIG. 24-2 Thermal conductivity of petroleum liquids. The solid lines refer to density expressed as degrees API; the broken lines refer to relative density at 288 K (15°C). ($K = [^{\circ}\text{F} + 459.7]/1.8$)

Commercial Considerations Fuels are sold in gallons and in multiples of the 42-gal barrel (0.159 m^3) in the United States, while a weight basis is used in other parts of the world. Transactions exceeding about 20 to 40 m^3 (5000 to 10,000 U.S. gal) usually involve volume corrections to 288 K (60°F) for accounting purposes. Fuel passes through an air eliminator and mechanical meter when loaded into or dispensed from trucks. Larger transfers such as pipeline, barge, or tanker movements are measured by fuel depth and strapping tables (calibration tables) in tanks and vessels, but positive-displacement meters that are proved (calibrated) frequently are gaining acceptance. After an appropriate settling period, water in the tank bottom is measured with a plumb bob or stick smeared with water-detecting paste.

Receipts of tank-car quantities or larger are usually checked for gravity, appearance, and flash point to confirm product identification and absence of contamination.

Safety Considerations Design and location of storage tanks, vents, piping, and connections are specified by state fire marshals, underwriters' codes, and local ordinances. In NFPA 30, *Flammable and Combustible Liquids Code*, 2003 (published by the National Fire Protection Association, Quincy, Ma.), liquid petroleum fuels are placed in Class I through Class III B based on their flash point, boiling point, and vapor pressure.

NFPA 30 details the design features and safe placement of handling equipment for flammable and combustible liquids.

Crude oils with flash points below 311 K (100°F) have been used in place of No. 6 fuel oil. Different pumps may be required because of low fuel viscosity.

Nonpetroleum Liquid Fuels

Tar Sands Canadian tar sands either are strip-mined and extracted with hot water or employ steam-assisted gravity drainage (SAGD) for in situ recovery of heavy oil (bitumen). The bitumen is processed into naphtha, kerosene, and gasoline fractions (which are hydrotreated), in addition to gas (which is recovered). Current production of syncrude from Canadian tar sands is about 113,000 T/d (790,000 B/d) with expected increases to about 190,000 T/d (1.7 MB/d) by 2010.

Oil Shale Oil shale is nonporous rock containing organic kerogen. Raw shale oil is extracted from mined rock by pyrolysis in a surface retort, or in situ by partial combustion after breaking up the rock with explosives. Pyrolysis cracks the kerogen, yielding raw shale oil high in nitrogen, oxygen, and sulfur. **Shale oil** has been hydrotreated

and refined in demonstration tests into relatively conventional fuels. Refining in petroleum facilities is possible with significant pretreatment or by incorporating upgrading units into the refinery.

Coal-Derived Fuels Liquid fuels derived from coal range from highly aromatic coal tars to liquids resembling petroleum. Raw liquids from different hydrogenation processes show variations that reflect the degree of hydrogenation achieved. Also, the raw liquids can be further hydrogenated to refined products. Properties and cost depend on the degree of hydrogenation and the boiling range of the fraction selected. A proper balance between fuel upgrading and equipment modification is essential for the most economical use of coal liquids in boilers, industrial furnaces, diesels, and stationary gas turbines.

Coal-tar fuels are high-boiling fractions of crude tar from pyrolysis in coke ovens and coal retorts. Grades range from free-flowing liquids to pulverizable pitch. Low in sulfur and ash, they contain hydrocarbons, phenols, and heterocyclic nitrogen and oxygen compounds. Being more aromatic than petroleum fuels, they burn with a more luminous flame. From 288 to 477 K (60 to 400°F) properties include:

Heat capacity	1.47–1.67 kJ/(kg·K) (0.35–0.40 Btu/[lb·°F])
Thermal conductivity	0.14–0.15 W/(m·K) (0.080–0.085 Btu/[h·ft·°F])
Heat of vaporization	349 kJ/kg (150 Btu/lb)
Heat of fusion	Nil

Table 24-7 shows representative data for liquid fuels from tar sands, oil shale, and coal.

GASEOUS FUELS

RESERVE AND PRODUCTION INFORMATION: DOE/EIA-0216(2003), November 2004; U.S. Crude Oil, Natural Gas, and Natural Gas Liquids Reserves: 2003 Annual Report, Table 2, Table 8, Table 12; *Annual Energy Review 2003*, DOE/EIA-0364, September 2004; *Natural Gas Annual 2003*, DOE/EIA-0131(03), December 2004, Table 9.

Natural Gas Natural gas is a combustible gas that occurs in porous rock of the earth's crust and is often found with accumulations of crude oil or coal. Natural gas termed *dry* has less than $0.013 \text{ dm}^3/\text{m}^3$ ($0.1 \text{ gal}/1000 \text{ ft}^3$) of gasoline. Above this amount, it is termed *wet*.

Proven reserves of conventionally reservoired natural gas in the United States total about 5.35 Tm^3 (189 Tft³). An additional 0.53 Tm^3 (18.7 Tft³) of proven reserves are in coal bed methane deposits. Production in 2003 was about 0.54 Tm^3 (19.1 Tft³), over 75 percent

TABLE 24-7 Characteristics of Typical Nonpetroleum Fuels

	Conventional coal-tar fuels from retorting ^a		Typical coal-derived fuels with different levels of hydrogenation ^b				Synthetic crude oils, by hydrogenation	
	CTF 50	CTF 400	Minimal	Mild	Mild ^c	Severe	Oil shale	Tar sands ^d
Distillation range, °C			175–280	280–500	160–415	175–400	125–495	
Density, kg/m ³ , 15°C	1.018	1.234	0.974	1.072	0.964	0.9607	0.914	0.864
lb U.S. gal, 60°F	8.5	10.3	8.1	8.9	8.0	8.0	7.6	7.2
Viscosity, mm ² /s	2–9	9–18	3.1–3.4	50–90	3.6	—	2.18	
	At 38°C	At 121°C	At 38°C	At 38°C	At 38°C	—	At 38°C	
Ultimate analysis, %								
Carbon	87.4	90.1	86.0	89.1	87.8	89.6	89.0	87.1
Hydrogen	7.9	5.4	9.1	7.5	9.7	10.1	11.1	12.69
Oxygen	3.6	2.4	3.6–4.3	1.4–1.8	2.4	0.3	0.5	0.04
Nitrogen	0.9	1.4	0.9–1.1	1.2–1.4	0.6	0.04	0.09	0.07
Sulfur	0.2	0.7	<0.2	0.4–0.5	0.07	0.004	0.04	0.10
Ash ^e	Trace	0.15	<0.001	^f				
C/H ratio, weight	11.0	16.5	9.4	11.9	9.1	8.9	8.0	6.9
Gross calorific value, MJ/kg	38.4–40.7	36.8–37.9						
Btu/lb	16,500–17,500	15,800–16,300						

^aCTF 50 and 400 indicate approximate preheat temperature, °F, for atomization of fuel in burners (terminology used in British Standard B.S. 1469).

^bProperties depend on distillation range, as shown, and to a lesser extent on coal source.

^cUsing recycle-solvent process.

^dTar sands, although a form of petroleum, are included in this table for comparison.

^eInorganic mineral constituents of coal tar fuel:

5 to 50 ppm: Ca, Fe, Pb, Zn (Na, in tar treated with soda ash)

0.05 to 5 ppm: Al, Bi, Cu, Mg, Mn, K, Si, Na, Sn

Less than 0.05 ppm: As, B, Cr, Ge, Ti, V, Mo

Not detected: Sb, Ba, Be, Cd, Co, Ni, Sr, W, Zr

^fInherent ash is “trace” or “<0.1%,” although entrainment in distillation has given values as high as 0.03 to 0.1%.

from nonassociated gas wells. Conventional proven reserves have declined about 0.03 Tm³ (0.9 Tft³) per year from 1977 through 2003. Net gas imports in 2003 were 0.09 Tm³ (3.3 Tft³), about 15 percent of consumption. Imports as LNG were 0.4 Tft³, about 2 percent of gas consumption.

Natural gas consists of hydrocarbons with a very low boiling point.

Methane is the main constituent, with a boiling point of 119 K (–245°F). Ethane, with a boiling point of 184 K (–128°F) may be present in amounts up to 10 percent; propane, with a boiling point of 231 K (–44°F), up to 3 percent. Butane, pentane, hexane, heptane, and octane may also be present. Physical properties of these hydrocarbons are given in Sec. 2.

Although there is no single composition that may be called “typical” natural gas, Table 24-8 shows the range of compositions in large cities in the United States.

TABLE 24-8 Analysis of Natural Gas*

	Range†	
	Low	High
Composition, vol %		
Methane	86.3	95.2
Ethane	2.5	8.1
Propane	0.6	2.8
Butanes	0.13	0.66
Pentanes	0	0.44
Hexanes plus	0	0.09
CO ₂	0	1.1
N ₂	0.31	2.47
He	0.01	0.06
Heating value	38.15(1024)	40.72(1093)
MJ/m ³ (Btu/ft ³)		
Specific gravity	0.586	0.641
Ref.: Air at 288 K (60°F)		

*Adapted from *Gas Engineers Handbook*, American Gas Association, Industrial Press, New York, 1965.

†Ranges are the high and low values of annual averages reported by 13 utilities (1954 data).

Commodity natural gas is substantially free of sulfur compounds; the terms *sweet* and *sour* are used to denote the absence or presence of H₂S. Some wells, however, deliver gas containing levels of hydrogen sulfide and other sulfur compounds (e.g., thiophenes, mercaptans, and organic sulfides) that must be removed before transfer to commercial pipelines. Pipeline-company contracts typically specify maximum allowable limits of impurities; H₂S and total sulfur compounds seldom exceed 0.023 and 0.46 g/m³ (1.0 and 20.0 gr/100 std ft³), respectively. The majority of pipeline companies responding to a 1994 survey limited H₂S to less than 0.007 g/m³ (0.3 gr/100 std ft³), but a slightly smaller number continued specifying 0.023 g/m³, in accord with an American Gas Association 1971 recommendation.

Supercompressibility of Natural Gas All gases deviate from the simple gas laws to a varying extent. This deviation is called *supercompressibility* and must be taken into account in gas measurement, particularly at high line pressure. For example, since natural gas is more compressible under high pressure at ordinary temperatures than is called for by Boyle's law, gas purchased at an elevated pressure gives a greater volume when the pressure is reduced than it would if the gas were ideal.

The supercompressibility factor may be expressed as

$$Z = (RT/PV)^{1/2} \quad (24-13)$$

where Z = supercompressibility factor; R = universal gas constant, 8.314 kPa·m³/(kmol·K); T = gas temperature, K; P = gas pressure, kPa; V = molar gas volume, m³/kmol.

For determining supercompressibility factors of natural gas mixtures, see *Manual for the Determination of Supercompressibility Factors for Natural Gas*, American Gas Association, New York, 1963; and A.G.A Gas Measurement Committee Report No. 3, 1969.

Liquefied Natural Gas The advantages of storing and shipping natural gas in liquefied form (LNG) derive from the fact that 0.035 m³ (1 ft³) of liquid methane at 111 K (–260°F) equals about 18 m³ (630 ft³) of gaseous methane. One cubic meter (264 U.S. gal) weighs 412 kg (910 lb) at 109 K (–263°F). The heating value is about 24 GJ/m³ (86,000 Btu/U.S. gal). The heat of vaporization of LNG at 0.1 MPa is 232 MJ/m³ of liquid. On a product gas basis, the heat required is about 0.3 kJ/m³ (10 Btu/std ft³) of gas produced.

LNG is actively traded in international commerce. In 2003 LNG was about 13 percent of gas imports to the United States but only 2 percent of gas consumption. The Energy Information Administration (EIA) projects (Annual Energy Outlook 2004) that gas imports will grow to 5.5 Tft³ in 2010 and to 7.2 Tft³ in 2025, with nearly all the increased volume being LNG.

Specialized ships are used to transport LNG. Receiving terminals have storage tankage and reevaporization facilities. Several new terminals have been proposed, but none has advanced to construction at this writing.

Liquefied Petroleum Gas The term *liquefied petroleum gas* (LPG) is applied to certain specific hydrocarbons which can be liquefied under moderate pressure at normal temperatures but are gaseous under normal atmospheric conditions. The chief constituents of LPG are propane, propylene, butane, butylene, and isobutane. LPG produced in the separation of heavier hydrocarbons from natural gas is mainly of the paraffinic (saturated) series. LPG derived from oil-refinery gas may contain varying low amounts of olefinic (unsaturated) hydrocarbons.

LPG is widely used for domestic service, supplied either in tanks or by pipelines. It is also used to augment natural gas deliveries on peak days and by some industries as a standby fuel.

Other Gaseous Fuels

Hydrogen Hydrogen is used extensively in the production of ammonia and chemicals, in the refining of petroleum, in the hydrogenation of fats and oils, and as an oven reducing atmosphere. It is also used as a fuel in industrial cutting and welding operations. There are no resources of uncombined hydrogen as there are of the other fuels. It is made industrially by the steam reforming of natural gas; as the by-product of industrial operations such as the thermal cracking of hydrocarbons; and, to a small extent, by the electrolysis of water.

Hydrogen is seen as the ultimate nonpolluting form of energy; when electrochemically combined with oxygen in fuel cells, only water, heat, and electricity are produced. Means for transforming the world's fossil energy economy into a hydrogen economy are being considered as a long-term option. Hydrogen can be stored in gaseous, liquid, or solid forms; however, currently available technologies are not suited to meet mass energy market needs. Technologies for economically producing, storing, and utilizing hydrogen are being researched in the United States, Europe, and Japan.

Acetylene Acetylene is used primarily in operations requiring high flame temperature, such as welding and metal cutting. To transport acetylene, it is dissolved in acetone under pressure and drawn into small containers filled with porous material.

Miscellaneous Fuels A variety of gases have very minor market shares. These include reformed gas, oil gases, producer gas, blue water gas, carbureted water gas, coal gas, and blast-furnace gas. The heating values of these gases range from 3.4 to 41 MJ/m³ (90 to 1100 Btu/ft³). They are produced by pyrolysis, the water gas reaction, or as by-products of pig-iron production.

Hydrogen sulfide in manufactured gases may range from approximately 2.30 g/m³ (100 gr/100 ft³) in blue and carbureted water gas to several hundred grains in coal- and coke-oven gases. Another important sulfur impurity is carbon disulfide, which may be present in amounts varying from 0.007 to 0.07 percent by volume. Smaller amounts of carbon oxysulfide, mercaptans, and thiophene may be found. However, most of the impurities are removed during the purification process and either do not exist in the finished product or are present in only trace amounts.

FUEL AND ENERGY COSTS

Fuel costs vary widely both geographically and temporally. Oil and gas markets have been highly volatile in recent years while steam coal markets have not. Much combustion equipment is designed for a specific fuel, limiting the potential for fuel switching to take advantage of price trends. The costs given in Table 24-9 are U.S. averages not necessarily applicable to a specific location; they do provide fuel cost trends.

TABLE 24-9 Time-Price Relationships for Fossil Fuels

Year	Bituminous coal, \$/Mg (\$/U.S. ton)	Wellhead natural gas, \$/1000 m ³ (\$/1000 scf)	Crude oil, domestic first purchase price, \$/m ³ (\$/bbl)
1975	21.81 (19.79)	15.54 (0.44)	48.24 (7.67)
1985	33.93 (30.78)	88.64 (2.51)	151.52 (24.09)
1995	28.17 (25.56)	54.74 (1.55)	91.96 (14.62)
2000	26.51 (24.05)	129.96 (3.68)	168.06 (26.72)
2003	29.29 (26.57)	175.87 (4.98)	173.35 (27.56)

SOURCE: *Annual Energy Review 2003*, DOE/EIA-0384, September 2004, Tables 7.8, 6.7, and 5.18, respectively.

COAL CONVERSION

Coal is the most abundant fossil fuel, and it will be available long after petroleum and natural gas are scarce. However, because liquids and gases are more desirable than solid fuels, technologies have been, and continue to be, developed to economically convert coal into liquid and gaseous fuels.

Bodley, Vyas, and Talwalker (*Clean Fuels from Coal Symposium II*, Institute of Gas Technology, Chicago, 1975) presented the chart in Fig. 24-3, which shows very simply the different routes from coal to clean gases and liquids.

Coal Gasification

GENERAL REFERENCES: *Fuel Gasification Symp.*, 152d American Chemical Society Mtg., Sept. 1966. *Chemistry of Coal Utilization*, suppl. vol., Lowry (ed.), Wiley, New York, 1963; and 2d suppl. vol., Elliott (ed.), 1981. *Coal Gasification Guidebook: Status, Applications, and Technologies*, Electric Power Research Institute, EPRI TR-102034, Palo Alto, Calif., 1993. *Riegel's Handbook of Industrial Chemistry*, 10th ed., Kent (ed.), Chap. 17, 2003. *Gasification* by Higman and van der Burgt, Elsevier, 2003. "The Case for Gasification" by Stiegel and Ramezan, EM, Dec. 2004, pp. 27-33.

Background The advantages of gaseous fuels have resulted in an increased demand for gas and led to the invention of advanced processes for coal gasification. Converting coal to combustible gas has been practiced commercially since the early 19th century. Chapter 17 of *Riegel's Handbook of Industrial Chemistry*, 10th ed., provides a good summary of the early history of coal gasification. Coal-derived gas was distributed in urban areas of the United States for residential and commercial uses until its displacement by lower-cost natural gas, starting in the 1940s. At about that time, development of oxygen-based gasification processes was initiated. An early elevated-pressure gasification process, developed by Lurgi Kohle u Mineralöltechnik GmbH, is still in use. The compositions of gases produced by this and a number of more recent gasification processes are listed in Table 24-10.

Theoretical Considerations The chemistry of coal gasification can be approximated by assuming coal is only carbon and considering the most important reactions involved (see Table 24-11). Reaction (24-14), the combustion of carbon with oxygen, which can be assumed to go to completion, is highly exothermic and supplies most of the thermal energy for the other gasification reactions. The oxygen used in the gasifier may come from direct feeding of air or may be high-purity oxygen from an air separation unit. Endothermic reactions (24-16) and (24-17), which represent the conversion of carbon to combustible gases, are driven by the heat energy supplied by reaction (24-14).

Hydrogen and carbon monoxide produced by the gasification reaction react with each other and with carbon. The hydrogenation of carbon to produce methane, reaction (24-15), is exothermic and contributes heat energy. Similarly, methanation of CO, reaction (24-19), can also contribute heat energy. These reactions are affected by the water-gas-shift reaction (24-18), the equilibrium of which controls the extent of reactions (24-16) and (24-17).

Several authors have shown [cf. Gumz, *Gas Producers and Blast Furnaces*, Wiley, New York, 1950; Elliott and von Fredersdorff, *Chemistry of Coal Utilization*, 2d suppl. vol., Lowry (ed.), Wiley, New

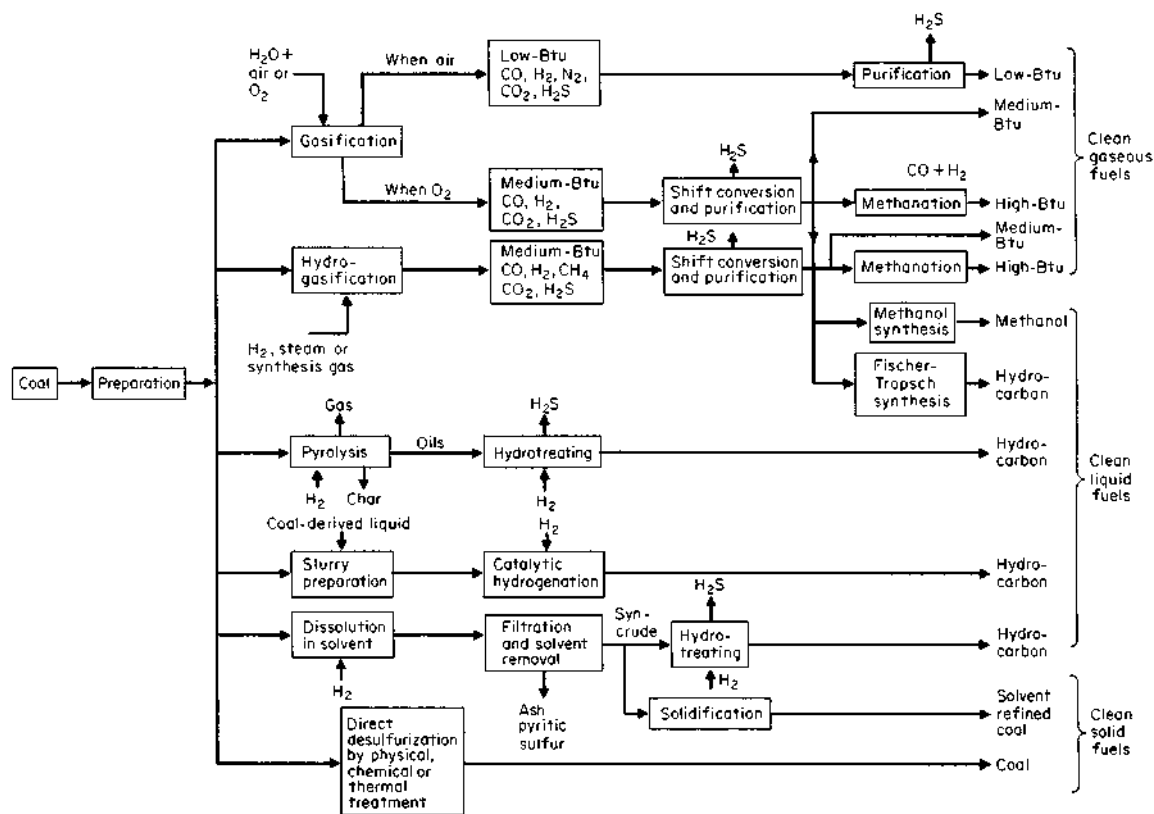


FIG. 24-3 The production of clean fuels from coal. (Based on W. Bodle, K. Vyas, and A. Talwalkar; Clean Fuels from Coal Symposium II, Institute of Gas Technology, Chicago, 1975.)

York, 1963] that there are three fundamental gasification reactions: the Boudouard reaction (24-17), the heterogeneous water-gas-shift reaction (24-18), and the hydrogasification reaction (24-15). The equilibrium constants for these reactions are sufficient to calculate equilibrium

for all the reactions listed. Unfortunately, it is not possible to calculate accurate gas composition by using reactions (24-14) to (24-19). One reason is that not all reactions may be in equilibrium. Another reason is that other reactions are taking place. Since gasifica-

TABLE 24-10 Coal-Derived Gas Compositions

Gasifier technology	Sasol/Lurgi ^a	Texaco/GE Energy [†]	BGL [‡]	E-Gas/ConocoPhillips	Shell/Uhde [§]
Type of bed	Moving	Entrained	Moving	Entrained	Entrained
Coal feed form	Dry coal	Coal slurry	Dry coal	Coal slurry	Dry coal
Coal type	Illinois No. 6	Illinois No. 6	Illinois No. 6	Illinois No. 6	Illinois No. 5
Oxidant	Oxygen	Oxygen	Oxygen	Oxygen	Oxygen
Pressure, MPa (psia)	0.101 (14.7)	4.22 (612)	2.82 (409)	2.86 (415)	2.46 (357)
Ash form	Slag	Slag	Slag	Slag	Slag
Composition, vol %					
H ₂	52.2	30.3	26.4	33.5	26.7
CO	29.5	39.6	45.8	44.9	63.1
CO ₂	5.6	10.8	2.9	16.0	1.5
CH ₄	4.4	0.1	3.8	1.8	0.03
Other hydrocarbons	0.3	—	0.2	—	—
H ₂ S	0.9	1.0	1.0	1.0	1.3
COS	0.04	0.02	0.1	0.1	0.1
N ₂ + Ar	1.5	1.6	3.3	2.5	5.2
H ₂ O	5.1	16.5	16.3	—	2.0
NH ₃ + HCN	0.5	0.1	0.2	0.2	0.02
HCl	—	0.02	0.03	0.03	0.03
H ₂ S : COS	20 : 1	42 : 1	11 : 1	10 : 1	9 : 1

^aRath, "Status of Gasification Demonstration Plants," *Proc. 2d Annu. Fuel Cells Contract Review Mtg.*, DOE/METC-9090/6112, p. 91.

[†]Coal Gasification Guidebook: Status, Applications, and Technologies, Electric Power Research Institute, EPRI TR-102034, 1993, p. 5-28.

[‡]Ibid., p. 5-58.

[§]Ibid., p. 5-48.

TABLE 24-11 Chemical Reactions in Coal Gasification

Reaction	Reaction heat, kJ/(kg-mol)	Process	Number
Solid-gas reactions			
$C + O_2 \rightarrow CO_2$	+393,790	Combustion	(24-14)
$C + 2H_2 \rightarrow CH_4$	+74,900	Hydrogasification	(24-15)
$C + H_2O \rightarrow CO + H_2$	-175,440	Steam-carbon	(24-16)
$C + CO_2 \rightarrow 2CO$	-172,580	Boudouard	(24-17)
Gas-phase reaction			
$CO + H_2O \rightarrow H_2 + CO_2$	+2,853	Water-gas shift	(24-18)
$CO + 3H_2 \rightarrow CH_4 + H_2O$	+250,340	Methanation	(24-19)
Pyrolysis and hydropyrolysis			
CH_x	$\left(1 - \frac{x}{4}\right)C + \left(\frac{x}{4}\right)CH_4$	Pyrolysis	(24-20)
$CH_x + m H_2$	$\left[1 - \left(\frac{x+2m}{4}\right)\right]C + \left(\frac{x+2m}{4}\right)CH_4$	Hydropyrolysis	(24-21)

tion of coal always involves elevated temperatures, thermal decomposition (pyrolysis) takes place as coal enters the gasification zone of the reactor. Reaction (24-20) treats coal as a hydrocarbon and postulates its thermal disintegration to produce carbon (coke) and methane. Reaction (24-21) illustrates the stoichiometry of hydrogasifying part of the carbon to produce methane.

These reactions can be used to estimate the effect of changes in operating parameters on gas composition. As temperature increases, endothermic reactions are favored over exothermic reactions. Methane production will decrease, and CO production will be favored as reactions are shifted in the direction in which heat absorption takes place. An increase in pressure favors reactions in which the number of moles of products is less than the number of moles of reactants. At higher pressure, production of CO_2 and CH_4 will be favored.

It is generally believed that oxygen reacts completely in a very short distance from the point at which it is mixed or comes in contact with coal or char. The heat evolved acts to pyrolyze the coal, and the char formed then reacts with carbon dioxide, steam, and other gases formed by combustion and pyrolysis. The assumption made in Table 24-11 that the solid reactant is carbon is probably close to correct, but the type of char formed affects the kinetics of gas-solid reactions. The overall reaction is probably rate-controlled below 1273 K (1832°F). Above this temperature, pore diffusion has an overriding effect, and at very high temperatures surface-film diffusion probably controls. For many gasification processes, the reactivity of the char is quite important and can depend on feed coal characteristics, the method of heating, the rate of heating, and particle-gas dynamics.

The importance of these concepts can be illustrated by the extent to which the pyrolysis reactions contribute to gas production. In a moving-bed gasifier, the particle is heated through several distinct thermal zones. In the initial heat-up zone, coal carbonization or devolatilization dominates. In the successively hotter zones, char devolatilization, char gasification, and fixed carbon combustion are the dominant processes. About 17 percent of total gas production occurs during coal devolatilization, and about 23 percent is produced during char devolatilization. The balance is produced during char gasification and combustion.

Gasifier Types and Characteristics The three main types of gasifier reactors, moving bed, fluidized bed, and entrained bed, as shown in Fig. 24-4, are all in commercial use. The moving bed is sometimes referred to as a "fixed" bed, because the coal bed is kept at a constant height. These gasifiers can differ in many ways: size, type of coal fed, feed and product flow rates, residence time, and reaction temperature. Gas compositions from the gasifiers discussed below are listed in Table 24-10.

Moving bed Depending on the temperature at the base of the coal bed, the ash can either be dry or in the form of molten slag. If excess steam is added, the temperature can be kept below the ash fusion point, in which case the coal bed rests on a rotating grate which allows the dry ash to fall through for removal. To reduce steam usage, a slagging bottom gasifier was developed by British Gas and Lurgi (BGL) in which the ash is allowed to melt and drain off through a slag tap. This gasifier has over twice the capacity per unit of cross-section

area of the dry-bottom gasifier. The BGL technology is offered commercially by Allied Syngas and Advantica.

Fluidized bed The problem of coal agglomeration is eliminated by a fluidized-bed gasifier developed by GTI. The U-Gas gasification process uses an agglomerating-ash fluidized-bed gasifier in which crushed limestone can be injected with the coal for sulfur capture. Char and ash that exit the gasifier with the product gas are recycled to the hot agglomerating and jetting zone, where temperatures are high enough to pyrolyze fresh coal introduced at that point, gasify the char, and soften the ash particles. The ash particles stick together and fall to the base of the gasifier, where they are cooled and removed. The agglomerating fluid-bed gasifier can be blown by either air or oxygen. Pressurized operation has several advantages: slightly higher methane formation, resulting in higher heating value of the gas; increased heat from the methanation reactions, which reduces the amount of oxygen needed; reduced heat losses through the wall and, consequently, improved efficiency; and higher capacity.

Entrained bed The primary example of an oxygen-blown, dry-feed, entrained-flow gasifier is the Shell gasifier. An advantage of Shell coal gasification technology is its ability to process a range of coals, with a wide variety of coals (from brown coal to anthracite) having been successfully tested. As with other entrained-flow gasifiers, disadvantages of the Shell process include a high oxygen requirement and a high waste heat recovery duty. However, the ability to feed dry coal reduces the oxygen requirement below that of single-stage entrained-flow gasifiers that use slurry feed and makes the Shell gasifier somewhat more efficient. The penalty for this small efficiency improvement is a more complex coal-feeding system. Uhde and Shell are marketing this technology.

Two slurry-fed, entrained-flow gasifiers are the Texaco gasifier (now owned by GE) and the E-Gas gasifier (now owned by ConocoPhillips). The Texaco gasifier is similar to the Shell gasifier, except that the coal is fed as a slurry. Reactor exit gas is cooled either by direct water injection or by a radiative cooler directly below the reactor. The E-Gas gasifier differs from other systems in that it uses a two-stage reactor. The bulk of the feed slurry and all the oxygen are sent to the first (horizontal) stage, where the coal is gasified. Hot gas flows into the second (vertical) stage, where the remainder of the coal slurry is injected. Hot fuel gas is cooled in a fire-tube boiler fuel gas cooler.

Gasification-Based Liquid Fuels and Chemicals Liquid fuels and chemicals from gasification-based synthesis gas are described in the coal liquefaction section following this section. While the downstream areas of power system and indirect liquefaction plants will differ markedly, the gasification sections will be quite similar and are described in this section.

Gasification-Based Power Systems An important driving force for coal gasification process development is the environmental superiority of gasification-based power generation systems, generally referred to as integrated gasification combined-cycle (IGCC) power production (Fig. 24-5). Coal is crushed prior to being fed to a reactor, where it is gasified through contact with steam and air or oxygen. Partial oxidation produces the high-temperature [1033 to 2255 K

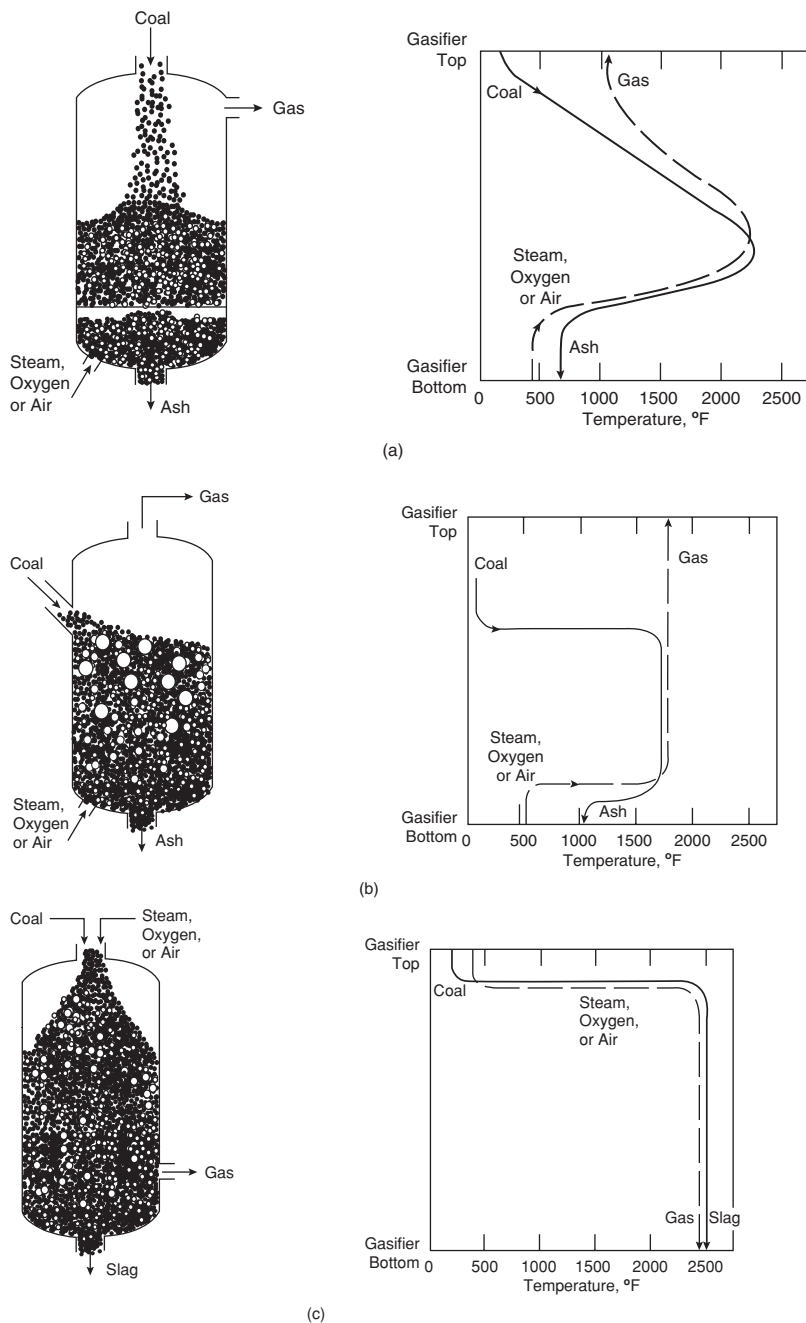


FIG. 24-4 Gasifier types and temperature profiles: (a) fixed bed (dry ash); (b) fluidized bed; (c) entrained flow. (This figure was published in N. Holt and S. Alpert, "Integrated Gasification Combined-Cycle Power," vol. 7, pp. 897–905, in *Encyclopedia of Physical Science and Technology*, 3d ed. Copyright Elsevier, 2002.)

(1400 to 3600°F), depending on the type of gasifier] reducing environment necessary for gasification. The product fuel gas passes through heat recovery and cleanup, where particulates (dust) and sulfur are removed. After cleanup, the fuel gas, composed primarily of hydrogen and carbon oxides, is burned with compressed air and expanded through a gas turbine to generate electricity. Heat is recovered from the turbine's hot exhaust gas to produce steam (at subcritical conditions), which is expanded in a steam turbine for additional electric power generation.

All three basic gasifier types could be incorporated into IGCC plant designs, although to date only entrained-flow gasifiers have actually been deployed. With each gasifier type, the oxidant can be air or oxygen, and the coal can be fed dry or in a slurry. The composition of the fuel gas, as well as its pressure and temperature, is determined by the design of the gasifier and the gas cleanup system.

There are several features of IGCC power systems that contribute to their improved thermal efficiency and environmental superiority compared to a conventional pulverized-coal fired power plant. First, the

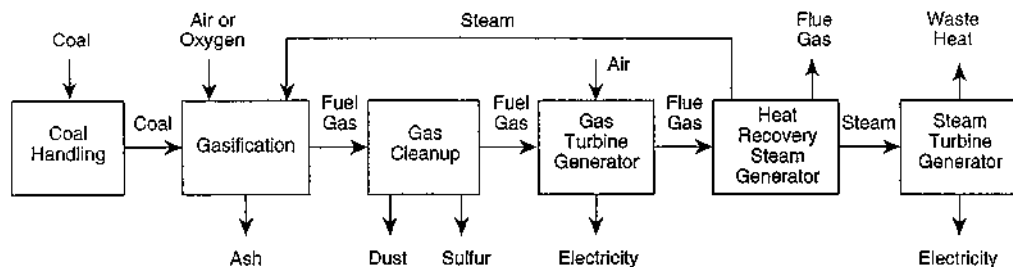


FIG. 24-5 Integrated gasification combined-cycle block diagram.

mass flow rate of gas from a gasifier is about one-fourth that from a combustor, because the gasifier is oxygen-blown and operates substoichiometrically, while the combustor is air-blown and operates with excess air. Because of the elevated operating pressure and the lack of nitrogen dilution, the volumetric gas flow to the sulfur removal system is actually only 0.5 percent to 1 percent of the volumetric flow to the flue gas desulfurization unit; this lowers the capital cost of the gas cleanup system.

Furthermore, the sulfur in coal-derived fuel gas is mainly present as hydrogen sulfide, which is much more easily recovered than the sulfur dioxide in flue gas. Not only can hydrogen sulfide be easily converted to elemental sulfur, a more valuable by-product than the calcium sulfate produced when lime is used to remove sulfur dioxide from flue gas, but also neither lime nor limestone is required. Nitrogen in the coal is largely converted to nitrogen gas or ammonia, which is easily removed by water washing, thus reducing nitrogen oxide emissions when the fuel gas is burned. Carbon dioxide can also be scrubbed from the fuel gas, and if even further reductions in carbon emissions are required, the carbon monoxide in the fuel gas can be converted to hydrogen and carbon dioxide before CO_2 removal. Finally, it has been estimated that the cost of mercury removal from an IGCC system would be only about one-tenth the cost for a conventional power plant.

Another advantage is that the IGCC system generates electricity by both combustion (Brayton cycle) and steam (Rankine cycle) turbines. The inclusion of the Brayton topping cycle improves efficiency compared to a conventional power plant's Rankine cycle-only generating system. Typically, about two-thirds of the power generated comes from the Brayton cycle and one-third from the Rankine cycle.

Current Status There have been substantial advancements in the development of gasification-based power systems during the last two decades. Programs are in place in the United States to support demonstration projects and for conducting research to improve efficiency, cost effectiveness, and environmental performance of IGCC power generation. Two areas of research that are likely to produce significant improvements in coal gasification technology are air separation (oxygen production) and fuel gas cleanup (removal of sulfur, mercury, particulates, and other pollutants).

Gasification technology is being widely used throughout the world. A study conducted in 2004 indicated that there were 156 gasification projects worldwide. Total capacity of the projects in operation was 45,000 MW (thermal) with another 25,000 MW (thermal) in various stages of development. As discussed later, in addition to producing fuel gas for power production, synthesis gas production by gasification is the first step in the indirect liquefaction of coal. Furthermore, gasification of carbonaceous, hydrogen-containing fuels is an effective method of thermal hydrogen production and is considered to be a key technology in the transition to a hydrogen economy. Therefore, the possibility exists for the coproduction of electric power and liquid fuels while sequestering carbon dioxide. Such an option could allow a gasifier in an IGCC system to operate at full capacity at all times, producing fuel gas at times of peak power demand and a mix of fuel gas and synthesis gas at other times.

In 2005, there were four coal-based IGCC power plants in operation in the world: Tampa Electric in Polk County, Florida, based on a Texaco (now GE) gasifier; Wabash repowering project in Indiana, based on E-Gas (now ConocoPhillips E-Gas); and Buggenum in The Netherlands and Puertollano in Spain, both based on Shell gasifiers (N.A.H. Holt, "IGCC Technology—Status, Opportunities, and Issues," *EM*, Dec. 2004, pp. 18–26).

Cost of Gasification-Based Power Systems Comparing power options is complicated by the many different parameters that must be considered in making a cost determination: coal cost; coal properties, including sulfur and moisture contents; ambient temperature; degree of process integration; gas turbine model; and gas cleanup method. These, and many other factors, have a significant impact on cost.

While comparison of absolute costs among different power systems is difficult, the costs of the component units are usually within given ranges. For an oxygen-blown IGCC power system, the breakdown of the capital cost for the four component units is: air separation plant, 10 to 15 percent; gasifier including gas cleanup, 30 to 40 percent; combined-cycle power unit, 40 to 45 percent; and balance of plant, 5 to 10 percent. The breakdown of the cost of electricity is: capital charge, 52 to 56 percent; operating and maintenance, 14 to 17 percent; and fuel, 28 to 32 percent.

One of the main challenges to the development and deployment of IGCC power plants has been that the capital cost was significantly higher than that of a natural-gas-fired generating unit, thereby negating the fuel cost savings. However, if natural gas prices remain substantially above \$4.75/GJ (\$5.00/10⁶ Btu), that should no longer be the case. Moreover, capital costs of IGCC power plants are likely to decline considerably as more of these facilities are built, standard designs are developed, and economies of scale are realized.

Coal Liquefaction

GENERAL REFERENCES: *Riegel's Handbook of Industrial Chemistry*, 10th ed., Kent (ed.), Ch. 17 Coal Technology, Kluwer Academic/Plenum Publishers, New York, 2003. *Chemistry of Coal Utilization*, suppl. vol., Lowry (ed.), Wiley, New York, 1963, and 2d suppl. vol., Elliott (ed.), 1981. Wu and Storch, *Hydrogenation of Coal and Tar*, U.S. Bur. Mines Bull. 633, 1968. Srivastava, McIlvried, Gray, Tomlinson, and Klunder, *American Chemical Society Fuel Chemistry Division Preprints*, Chicago, 1995. Dry, *The Fischer-Tropsch Synthesis, Catalysis Science and Technology*, vol. 1, Springer-Verlag, New York, 1981. Anderson, *The Fischer-Tropsch Synthesis*, Academic Press, New York, 1984. Sheldon, *Chemicals from Synthesis Gas*, D. Reidel Publishing Co., Dordrecht, Netherlands, 1983. Rao, Stiegel, Cinquigrane, and Srivastava, "Iron-Based Catalyst for Slurry-Phase Fischer-Tropsch Process: Technology Review," *Fuel Processing Technology*, **30**, 83–151 (1992). Wender, "Reactions of Synthesis Gas," *Fuel Processing Technology*, **48**, 189–297 (1996).

Background Coal liquefaction denotes the process of converting solid coal to a liquid fuel. The primary objective of any coal liquefaction process is to increase the hydrogen-to-carbon molar ratio. For a typical bituminous coal, this ratio is about 0.8, while for light petroleum it is about 1.8. A secondary objective is to remove sulfur, nitrogen, oxygen, and ash so as to produce a nearly pure hydrocarbon. There are several ways to accomplish liquefaction: (1) pyrolysis, (2) direct hydrogenation of the coal at elevated temperature and pressure, (3) hydrogenation of coal slurried in a solvent, and (4) gasification of coal to produce synthesis gas (a mixture of hydrogen and carbon monoxide, also referred to as syngas) followed by the use of Fischer-Tropsch (F-T) chemistry to produce liquid products. The first three of these approaches are generally referred to as direct liquefaction, in that the coal is directly converted to a liquid. The fourth approach is termed indirect liquefaction, because the coal is first converted to an intermediate product.

Pyrolysis In pyrolysis, coal is heated in the absence of oxygen to drive off volatile components, leaving behind a solid residue enriched in carbon and known as char or coke. Most coal pyrolysis operations are for the purpose of producing metallurgical coke, with the liquids

promoted by increasing the temperature, hydrogen partial pressure, and residence time. However, if the temperature is too high, gas yield is increased and coking can occur. Solvent-to-coal ratio is important. If the ratio is too low, there will be insufficient hydrogen transfer activity; on the other hand, if the ratio is high, a larger reactor will be necessary to provide the required residence time. Typical operating conditions are:

Temperature	670–730 K (750–850°F)
Pressure	10.3–20.7 MPa (1500–3000 psia)
Residence time	0.5 h
Solvent-to-coal ratio	1.5–2 kg/kg (1.5–2 lb/lb)

For the most highly developed processes, maf coal conversion can be as high as 90 to 95 % with a C₄+ distillate yield of 60 to 75 wt % and a hydrogen consumption of 5 to 7 wt %. When an external catalyst is used, it is typically some combination of cobalt, nickel, and molybdenum on a solid acid support, such as silica alumina. In slurry hydrogenation processes, catalyst life is typically fairly short because of the large number of potential catalyst poisons present in the system.

Several variations of the slurry hydrogenation process, depicted in Fig. 24-6 and discussed below, were tested at pilot-plant scale. Table 24-12 presents typical operating conditions and yields for these processes.

The *SRC-I process*, developed by the Pittsburg & Midway Coal Mining Co. in the early 1960s, was not really a liquefaction process; rather, it was designed to produce a solid fuel for utility applications. Only enough liquid was produced to keep the process in solvent balance. The bottoms product was subjected to filtration or solvent extraction to remove ash and then solidified to produce a low-ash, low-sulfur substitute for coal. However, the value of the product was not high enough to make this process economically viable.

The *SRC-II process*, developed by Gulf Oil Corp., was an improved version of the SRC-I process, designed to produce more valuable liquid products, rather than a solid. The major difference between SRC-II and the typical process described above was the recycle of a portion of the fractionator bottoms to the slurry feed tank. This increased the ash content of the reactor feed. This ash, particularly the iron pyrites in the ash, acted as a catalyst and improved product yield and quality.

The *Exxon Donor Solvent (EDS) Process*, developed by the Exxon Research and Engineering Co., differed from the typical process in that, before being recycled, the solvent was hydrogenated in a fixed-bed reactor using a hydrotreating catalyst, such as cobalt or nickel molybdate. Exxon found that use of this hydrogen donor solvent with carefully controlled properties improved process performance. Exxon developed a solvent index, based on solvent properties, which correlated with solvent effectiveness.

The *H-Coal Process*, based on H-Oil technology, was developed by Hydrocarbon Research, Inc. (HRI). The heart of the process was a three-phase, ebullated-bed reactor in which catalyst pellets were fluidized by the upward flow of slurry and gas through the reactor. The reactor contained an internal tube for recirculating the reaction mixture to the bottom of the catalyst bed. Catalyst activity in the reactor was maintained by the withdrawal of small quantities of spent catalyst and the addition of fresh catalyst. The addition of a catalyst to the reactor is the main feature which distinguishes the H-Coal Process from the typical process.

Two-stage liquefaction is an advanced process concept that provides higher yields of better quality products by carrying out the coal solution and the hydrogenation/hydrocracking steps in separate reactors whose conditions are optimized for the reaction that is occurring. Either or both reactors may be catalytic. Slurry catalysts have been tested, in addition to the more conventional supported catalysts, as a means of simplifying reactor design and removing process constraints. The U.S. Department of Energy and its private sector collaborators, Hydrocarbon Technologies, Inc., and others, have advanced the development of the two-stage direct liquefaction process to commercialization status during the last two decades. Coal-derived product quality has been improved dramatically (less than 50 ppm nitrogen content, for example) through the addition of in-line fixed-bed hydrotreating of the product stream.

In *coal-oil coprocessing*, coal is slurried in petroleum residuum rather than in recycle solvent, and both the coal and petroleum components are converted to high-quality fuels in the slurry reactor. This variation offers the potential for significant cost reduction by eliminating or

TABLE 24-12 Direct Liquefaction Process Conditions and Product Yields

Developer	Gulf ^a	Gulf	Exxon	HRI	SCS, ^b EPRI, Amoco	HTI
Process	SRC-I	SRC-II	EDS	H-Coal	Two-stage	Two-stage
Coal type	Kentucky 9 & 14	Illinois No. 6	Illinois No. 6	Illinois No. 6	Illinois No. 6	Illinois No. 6
Operating conditions						
Nominal reactor residence time, h	0.5	0.97	0.67			
Coal space velocity per stage, kg/(h·m ³) (lb/(h·ft ³))				530 (33.1)	825 ^c (51.7)	310 (19.4)
Temperature, K (°F)	724 (842)	730 (855)	722 (840)	726 (847)	700 (800)	692 (787)
Total pressure, MPa (psia)	10.3 (1500)	13.4 (1950)	10.3 (1500)			19.2 (2790)
H ₂ partial pressure, MPa (psia)	9.7 (1410)	12.6 (1830)		12.6 (1827)	18.3 (2660)	
Catalyst type	Coal minerals	Coal minerals	Coal minerals	Supported catalyst (Co/Mo)	AKZO-AO-60 (Ni/Mo)	AKZO-AO-60 (Ni/Mo)
Catalyst replacement rate, kg/kg (lb/US ton) mf coal					1.5 × 10 ⁻³ (3.0)	2.3 × 10 ⁻³ (4.5)
Product yields, wt % maf coal						
H ₂	-2.4	-5.0	-4.3	-5.9	-6.0	-7.2
H ₂ O	—	—	12.2 ^d	8.3	9.7	9.8
H ₂ S, CO ₂ , NH ₃	—	9.6	4.2 ^e	5.0	5.2	5.2
C ₁ -C ₃	3.7 ^f	13.9 ^f	7.3	11.3	6.5	5.6
C ₄ + distillate	13.5 ^g	46.8 ^g	38.8	53.1	65.6	73.3
Bottoms ^h	68.4	30.7	41.8	28.2	19.0	13.3
Unreacted coal ⁱ	5.4	4.6	—	6.4	7.0	5.0
Distillate end point, K (°F)	727 (850)	727 (850)	911 (1180)	797 (975)	797 (975)	524 (975)

^a In partnership with Pittsburg & Midway Coal Mining Co.

^b Southern Company Services, Inc., prime contractor for Wilsonville Facility.

^c Coal space velocity is based on settled catalyst volume.

^d CO₂ is included.

^e CO₂ is excluded.

^f C₄ is included.

^g C₄ is excluded.

^h Unreacted coal is included.

ⁱ "Unreacted coal" is actually insoluble organic matter remaining after reaction.

reducing recycle streams. More importantly, fresh hydrogen requirements are reduced, because the petroleum feedstock has a higher initial hydrogen content than coal. As a result, plant capital investment is reduced substantially. Other carbonaceous materials, such as municipal waste, plastics, cellulose, and used motor oils, might also serve as cofeedstocks with coal in this technology.

Commercial Operations The world's only commercial-scale direct coal liquefaction plant, located in the Inner Mongolia Autonomous Region of China, was dedicated in 2004. The plant is scheduled to begin production in 2007. The first train of the first phase of the Shenhua Direct Coal Liquefaction Plant will liquefy 2,100,000 Mg/a (2,315,000 ton/yr) of coal from the Shangwan Mine in the Shenhua coal field of Inner Mongolia. The plant will use a combination of technologies developed in the United States, Japan, and Germany with modifications and enhancements developed in China. The first train will use a two-stage reactor system and include an in-line hydrotreater and produce 591,900 Mg/a (652,460 ton/yr) of diesel; 174,500 Mg/a (192,350 ton/yr) of naphtha; 70,500 Mg/a (77,710 ton/yr) of LPG; and 8300 Mg/a (9150 ton/yr) of ammonia. When completed, the plant will include 10 trains producing approximately 10,000,000 Mg/a (11,000,000 ton/yr) of oil products.

Unlike the processes described above, *indirect liquefaction* is not limited to coal but may be performed using any carbonaceous feed, such as natural gas, petroleum residues, petroleum coke, coal, and biomass. Figure 24-7 presents a simplified process flow diagram for a typical indirect liquefaction process using coal as the feedstock. The syngas is produced in a gasifier (see the description of coal gasifiers earlier in this section), which partially combusts the coal or other feed at high temperature [1500 to 1750 K (2200 to 2700°F)] and moderate pressure [2 to 4 MPa (300 to 600 psia)] with a mixture of oxygen (or air) and steam. In addition to H₂ and CO, the raw synthesis gas contains other constituents, such as CO₂, H₂S, NH₃, N₂, H₂O, and CH₄, as well as particulates and, with some gasifiers, tars.

The syngas leaving the gasifier is cooled and passed through particulate removal equipment. Following this, depending on the requirements of the syngas conversion process, it may be necessary to adjust the H₂/CO ratio. Modern high-efficiency gasifiers typically produce syngas with a H₂/CO molar ratio between 0.45 and 0.7, which is lower than the stoichiometric ratio of about 2 for F-T synthesis or methanol production. Some F-T catalysts, particularly iron catalysts, possess water-gas shift conversion activity and permit operation with a low H₂/CO ratio [see reaction (24-25)]. Others, such as cobalt catalysts, possess little shift activity and require adjustment of the H₂/CO ratio before the syngas enters the synthesis reactor.

After shift conversion (if required), acid gases (CO₂ and H₂S) are scrubbed from the synthesis gas. A guard chamber is sometimes used to remove the last traces of H₂S, since F-T catalysts are generally very sensitive to sulfur poisoning. The cleaned gas is sent to the synthesis reactor, where it is converted at moderate temperature and pressure, typically 498 to 613 K (435 to 645°F) and 1.5 to 6.1 MPa (220 to 880 psia). Products, whose composition depends on operating conditions, the catalyst employed, and the reactor design, include saturated hydrocarbons (mainly straight chain paraffins from methane through *n*-C₅₀ and higher), oxygenates (methanol, higher alcohols, ethers), and olefins.

Fischer-Tropsch Synthesis The best-known technology for producing hydrocarbons from synthesis gas is the Fischer-Tropsch synthesis. This technology was first demonstrated in Germany in 1902 by Sabatier and Senderens when they hydrogenated carbon monoxide (CO) to methane, using a nickel catalyst. In 1926 Fischer and Tropsch were awarded a patent for the discovery of a catalytic technique to convert synthesis gas to liquid hydrocarbons similar to petroleum.

The basic reactions in the Fischer-Tropsch synthesis are:

Paraffins formation:

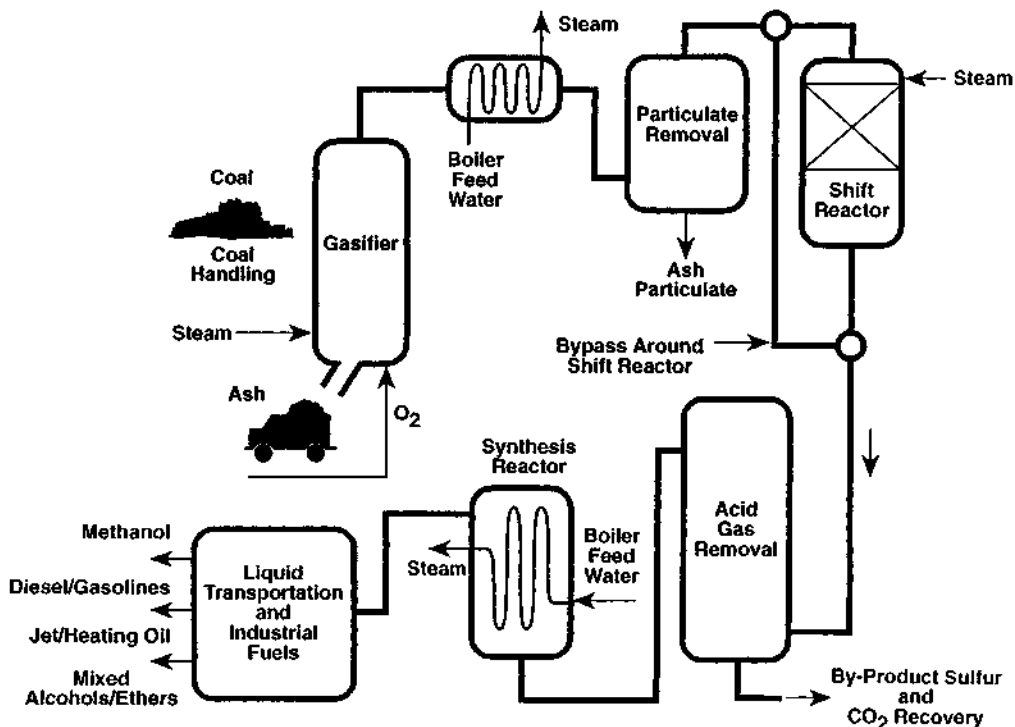
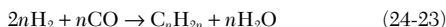


FIG. 24-7 Indirect liquefaction of coal.

Olefins formation:



Alcohols formation:



Other reactions may also occur during the Fischer-Tropsch synthesis, depending on the catalyst employed and the conditions used:

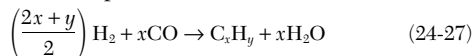
Water-gas shift:



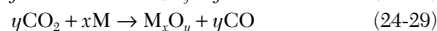
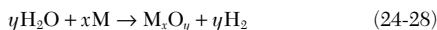
Boudouard disproportionation:



Surface carbonaceous deposition:



Catalyst oxidation-reduction:



Bulk carbide formation:



where M represents a catalytic metal atom.

The production of hydrocarbons using traditional F-T catalysts is governed by chain growth (polymerization) kinetics. The theoretical equation describing the distribution of hydrocarbon products, commonly referred to as the Anderson-Schulz-Flory (ASF) equation, is

$$\log \frac{W_n}{n} = n \log \alpha + \log \frac{(1-\alpha)^2}{\alpha} \quad (24-31)$$

where W_n is the weight fraction of products with carbon number n , and α is the chain growth probability, i.e., the probability that a carbon chain on the catalyst surface will grow by adding another carbon atom rather than desorb from the catalyst surface and terminate. In deriving Eq. (24-31), α is assumed to be independent of chain length. However, α is dependent on temperature, pressure, H_2/CO ratio, and catalyst composition. As α increases, the average carbon number of the product also increases. When α equals 0, methane is the only product formed. As α approaches 1, the product becomes predominantly wax. In practice, α is not really independent of chain length. Methane production, particularly with cobalt catalysts, is typically higher than predicted; and C_2 yield is often lower. Some investigators have found a significant deviation from the ASF distribution for higher-carbon-number products and have proposed a dual alpha mechanism to explain their results.

Figure 24-8 provides a graphical representation of Eq. 24-32 showing the weight fraction of various products as a function of α . This figure shows that there is a particular α that will maximize the yield of any desired product, such as gasoline or diesel fuel. Based on the ASF equation, the weight fraction of material between carbon numbers m and n inclusive is given by

$$W_{mn} = m\alpha^{m-1} - (m-1)\alpha^m - (n+1)\alpha^n + n\alpha^{n+1} \quad (24-32)$$

The α to maximize the yield of the carbon number range from m to n is given by

$$\alpha_{\text{opt}} = \left(\frac{m^2 - m}{n^2 + n} \right)^{1/(n-m+1)} \quad (24-33)$$

Additional gasoline and diesel fuel can be produced through further refining, such as hydrocracking or catalytic cracking of the wax product or polymerization of light olefins.

F-T Catalysts The patent literature is replete with recipes for the production of F-T catalysts, with most formulations being based on iron, cobalt, or ruthenium, typically with the addition of some promoter(s). Nickel is sometimes listed as a F-T catalyst, but nickel has too much hydrogenation activity and produces mainly methane. In practice, because of the cost of ruthenium, commercial plants use either cobalt-based or iron-based catalysts. Cobalt is usually deposited on a refractory oxide support, such as alumina, silica, titania, or zirconia. Iron is typically not supported and may be prepared by precipitation.

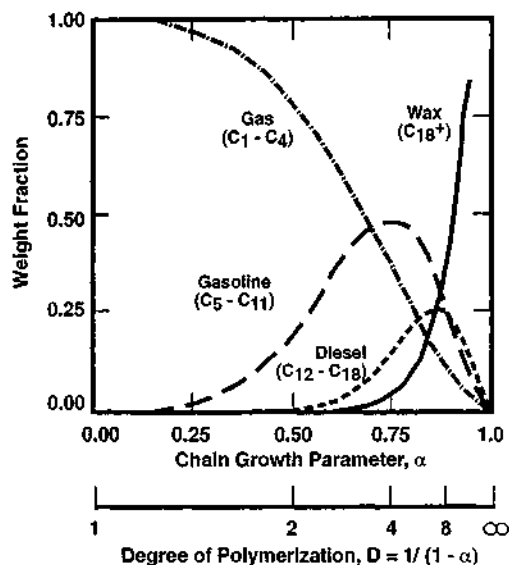
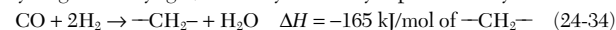


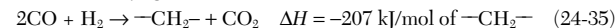
FIG. 24-8 Product yield in Fischer-Tropsch synthesis.

Of the large number of promoters listed in patents, rhenium or one of the noble metals may be used to promote cobalt, and iron is often promoted with potassium.

Reactor Design The F-T reaction is highly exothermic and, for hydrogen-rich syngas, can be symbolically represented by



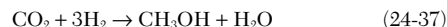
For CO-rich syngas, the overall reaction is



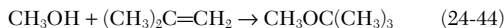
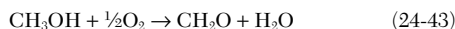
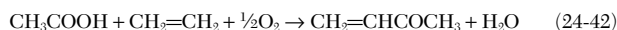
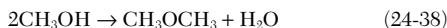
Because of the high heat release, adequate heat removal from the reactor is critical. High temperatures result in high yields of methane as well as coking and sintering of the catalyst. Three types of reactors (tubular fixed bed, fluidized bed, and slurry) provide good temperature control, and all three types are being used for synthesis gas conversion. The first plants used tubular or plate-type fixed-bed reactors. Later, SASOL, in South Africa, used fluidized-bed reactors, and most recently, slurry reactors have come into use. Fluid bed reactors are limited to the production of products that are volatile at reactor operating conditions. Nonvolatile products accumulate on the catalyst and destroy its fluidizing properties.

F-T reactor operations can be classified into two categories: high-temperature, 613 K (645°F), or low-temperature, 494-544 K (430 to 520°F). The Synthol reactor developed by SASOL is typical of high-temperature operation. Using an iron-based catalyst, this reactor produces a very good gasoline having high olefinicity and a low boiling range. The olefin fraction can readily be oligomerized to produce diesel fuel. Low-temperature operation, typical of fixed-bed reactors, produces a much more paraffinic and straight-chain product. The chain growth parameter can be tailored to give the desired product selectivity. The primary diesel fraction, as well as the diesel-range product from hydrocracking of the wax, is an excellent diesel fuel.

Chemicals from Syngas A wide range of products can be produced from syngas. These include such chemicals as methanol, ethanol, isobutanol, dimethyl ether, dimethyl carbonate, and many other chemicals. Typical methanol-producing reactions are

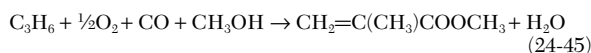


Once methanol is produced, it can be converted to an extensive range of materials. The following reactions illustrate some of the chemicals of major importance that can be made from methanol. Among these are dimethyl ether, acetic acid, methyl acetate, acetic anhydride, vinyl acetate, formaldehyde, and methyl tertiarybutyl ether (MTBE).



Reaction (24-38) can occur in parallel with the methanol-producing reactions, thereby overcoming the equilibrium limitation on methanol formation. Higher alcohols can also be formed, as illustrated by reaction (24-24), which can generate either linear or branched alcohols, depending on the catalyst used and the operating conditions. The production of methyl acetate, reaction (24-40), from synthesis gas is currently being practiced commercially. Following methanol synthesis, one-half of the methanol is reacted with carbon monoxide to form acetic acid, which is reacted with the rest of the methanol to form methyl acetate.

Methyl acrylate and methyl methacrylate, which are critical to the production of polyesters, plastics, latexes, and synthetic lubricants, can also be produced. For example, methyl methacrylate can be produced from propylene and methanol:



Commercial Operations The only commercial indirect coal liquefaction plants for the production of transportation fuels are operated by SASOL in South Africa. Construction of the original plant was begun in 1950, and operations began in 1955. This plant employed both fixed-bed (Arge) and entrained-bed (Synthol) reactors. The fixed-bed reactors have been converted to natural gas, and the Synthol reactors have been replaced by advanced fixed-fluidized bed reactors. Two additional plants that employ dry-ash Lurgi Mark IV coal gasifiers and entrained-bed (Synthol) reactors for synthesis gas conversion were constructed with start-ups in 1980 and 1983. In addition to producing a significant fraction of South Africa's transportation fuel requirements, these plants produce more than 120 other products from coal.

SASOL and others, including Exxon, Statoil, Air Products and Chemicals, Inc., and the U.S. Department of Energy, have engaged in the development of slurry bubble column reactors for F-T and oxygenate synthesis. SASOL commissioned a 5-m-diameter slurry reactor in 1993, which doubled the wax capacity of the SASOL I facility. The development work on slurry reactors shows that they have several advantages over competing reactor designs: (1) excellent heat-transfer capability resulting in nearly isothermal reactor operations, (2) high catalyst and reactor productivity, (3) ease of catalyst addition and withdrawal, (4) simple construction, and (5) ability to process hydrogen-lean synthesis gas successfully. Because of the small particle size of the catalyst used in slurry reactors, effective separation of catalyst from the products can be difficult but is crucial to successful operation.

The United States has two commercial facilities that convert coal to fuels or chemicals via a syngas intermediate. The Great Plains Synfuels

Plant, located in Beulah, N. Dak., and operated by Dakota Gasification Company (DGC), produces synthetic natural gas (SNG) from North Dakota lignite. Fourteen Lurgi dry-ash gasifiers in the plant convert approximately 15,400 Mg/d (17,000 U.S. ton/day) of lignite into syngas which is methanated to about $4.7 \times 10^6 \text{ N}\cdot\text{m}^3$ ($166 \times 10^6 \text{ std ft}^3$) of pipeline-quality gas. Aromatic naphtha and tar oil are also produced in the gasification section. The plant operates at 120 percent of its original design capacity. In addition to SNG, a wide assortment of other products are produced and sold (anhydrous ammonia, ammonium sulfate, phenol, cresylic acid, naphtha, krypton and xenon gases, liquid nitrogen, and carbon dioxide).

Eastman Chemical Company has operated a coal-to-methanol plant in Kingsport, Tenn., since 1983. Two Texaco gasifiers (one is a backup) process 34 Mg/h (37 U.S. ton/h) of coal to synthesis gas. Using ICI methanol technology, the synthesis gas is converted to methanol, which is an intermediate in the production of methyl acetate and acetic acid. The plant produces about 225,000 Mg/a (250,000 U.S. ton/yr) of acetic anhydride. As part of the DOE Clean Coal Technology Program, Air Products and Chemicals, Inc., and Eastman Chemical Company constructed and operated a 9.8 Mg/h (260 U.S. ton/d) slurry-phase reactor for the conversion of synthesis gas to methanol.

Despite the success of SASOL, most of the commercial interest in Fischer-Tropsch synthesis technology is based on natural gas, as this represents a way to bring remote gas deposits to market using conventional tankers. In 1985, Mobil commercialized its Methanol-to-Gasoline (MTG) technology in New Zealand, natural gas being the feedstock. This fixed-bed process converted synthesis gas to 4000 Mg/d (4400 U.S. ton/day) of methanol; the methanol could then be converted to 2290 m³/d (14,400 bbl/d) of gasoline. Owing to economic factors, the plant was used primarily for the production of methanol; it has been shut down due to an insufficient gas supply.

Shell Gas B.V. constructed a 1987 m³/d (12,500 bbl/d) F-T plant in Malaysia that started operations in 1994. The Shell Middle Distillate Synthesis (SMDS) process uses natural gas as the feedstock to fixed-bed reactors containing cobalt-based catalyst. The heavy hydrocarbons from the F-T reactors are converted to distillate fuels by hydrocracking and hydroisomerization. The quality of the products is very high, the diesel fuel having a cetane number in excess of 75 with no sulfur.

The largest F-T facility based on natural gas is the Moss gas plant located in Mossel Bay, South Africa. Natural gas is converted to synthesis gas in a two-stage reformer and subsequently converted to hydrocarbons by SASOL's Synthol technology. The plant, commissioned in 1992, has a capacity of 7155 m³/d (45,000 bbl/d).

In addition to these commercial facilities, several companies have smaller-scale demonstration facilities, generally with capacities of a few hundred barrels per day of liquid products. For example, Exxon Research and Engineering Company developed a process for converting natural gas to high-quality refinery feedstock, the AGC-21 Advanced Gas Conversion Process. The technology involves three highly integrated process steps: fluid-bed synthesis gas generation; slurry-phase Fischer-Tropsch synthesis; and mild fixed-bed hydroisomerization. The process was demonstrated in the early 1990s with a slurry-phase reactor having a diameter of 1.2 m (4 ft) and a capacity of about 32 m³/d (200 bbl/d).

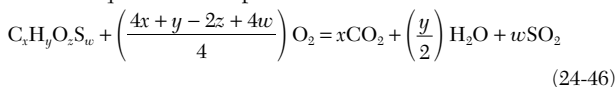
HEAT GENERATION

GENERAL REFERENCES: Stultz and Kitto (eds.), *Steam: Its Generation and Use*, 40th ed., Babcock and Wilcox, Barberton, Ohio, 1992. North American Combustion Handbook, 3d ed., vols. I and II, North American Manufacturing Company, Cleveland, Ohio, 1996. Singer (ed.), *Combustion: Fossil Power Systems*, 4th ed., Combustion Engineering, Inc., Windsor, Conn., 1991. Cuenca and Anthony (eds.), *Pressurized Fluidized Bed Combustion*, Blackie Academic & Professional, London, 1995. Basu and Fraser, *Circulating Fluidized Bed Boilers: Design and Operations*, Butterworth and Heinemann, Boston, 1991. *Proceedings of International FBC Conference(s)*, ASME, New York, 1991, 1993, 1995. *Application of FBC for Power Generation*, Electric Power Research Institute, EPRI PR-101816, Palo Alto, Calif., 1993. Boyen, *Thermal Energy Recovery*, 2d ed., Wiley, New York, 1980.

COMBUSTION BACKGROUND

Basic Principles

Theoretical Oxygen and Air for Combustion The amount of oxidant (oxygen or air) just sufficient to burn the carbon, hydrogen, and sulfur in a fuel to carbon dioxide, water vapor, and sulfur dioxide is the *theoretical* or *stoichiometric oxygen or air* requirement. The chemical equation for complete combustion of a fuel is



x , y , z , and w being the number of atoms of carbon, hydrogen, oxygen, and sulfur, respectively, in the fuel. For example, 1 mol of methane (CH_4) requires 2 mol of oxygen for complete combustion to 1 mol of carbon dioxide and 2 mol of water. If air is the oxidant, each mol of oxygen is accompanied by 3.76 mol of nitrogen.

The volume of theoretical oxygen (at 0.101 MPa and 298 K) needed to burn any fuel can be calculated from the ultimate analysis of the fuel as follows:

$$24.45 \left(\frac{C}{12} + \frac{H}{4} - \frac{O}{32} + \frac{S}{32} \right) = m^3 \text{O}_2 / \text{kg fuel} \quad (24-47)$$

where C , H , O , and S are the decimal weights of these elements in 1 kg of fuel. (To convert to ft^3 per lb of fuel, multiply by 16.02.) The mass of oxygen (in kg) required can be obtained by multiplying the volume by 1.31. The volume of theoretical air can be obtained by using a coefficient of 116.4 in Eq. (24-47) in place of 24.45.

Figure 24-9 gives the theoretical air requirements for a variety of combustible materials on the basis of fuel higher heating value (HHV). If only the fuel lower heating value is known, the HHV can be calculated from Eq. (24-5). If the ultimate analysis is known, Eq. (24-4) can be used to determine HHV.

Excess Air for Combustion More than the theoretical amount of air is necessary in practice to achieve complete combustion. This excess air is expressed as a percentage of the theoretical air amount. The *equivalence ratio* is defined as the ratio of the actual fuel-air ratio to the stoichiometric fuel-air ratio. Equivalence ratio values less than 1.0 correspond to fuel-lean mixtures. Conversely, values greater than 1.0 correspond to fuel-rich mixtures.

Products of Combustion For lean mixtures, the *products of combustion* (POC) of a sulfur-free fuel consist of carbon dioxide, water vapor, nitrogen, oxygen, and possible small amounts of carbon monoxide and unburned hydrocarbon species. Figure 24-10 shows the effect of fuel-air ratio on the flue gas composition resulting from the combustion of natural gas. In the case of solid and liquid fuels, the POC may also include solid residues containing ash and unburned carbon particles.

Equilibrium combustion product compositions and properties may be readily calculated using thermochemical computer codes which minimize the Gibbs free energy and use thermodynamic databases containing polynomial curve-fits of physical properties. Two widely used versions are those developed at NASA Lewis (Gordon and McBride, NASA SP-273, 1971) and at Stanford University (Reynolds, *STANJAN Chemical Equilibrium Solver*, Stanford University, 1987).

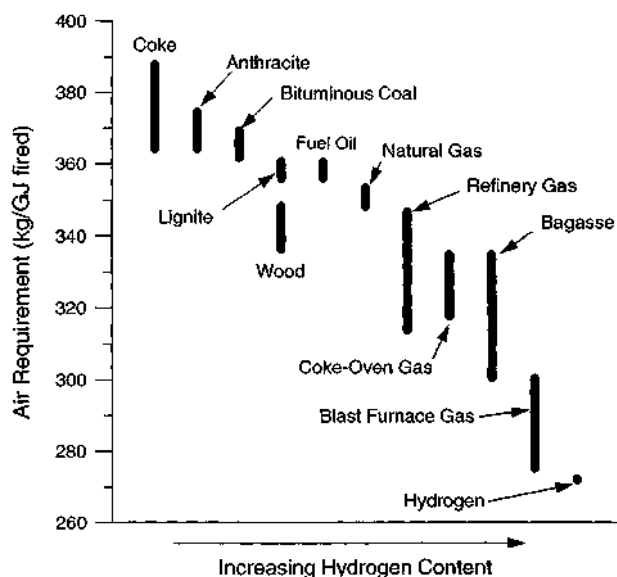


FIG. 24-9 Combustion air requirements for various fuels at zero excess air. To convert from kg air/GJ fired to lb air/10⁶ Btu fired, multiply by 2.090.

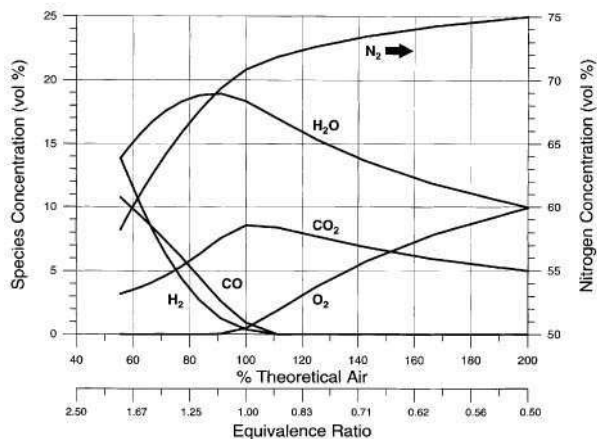


FIG. 24-10 Effect of fuel-air ratio on flue-gas composition for a typical U.S. natural gas containing 93.9% CH_4 , 3.2% C_2H_6 , 0.7% C_3H_8 , 0.4% C_4H_{10} , 1.5% N_2 and 1.1% CO_2 by volume.

Flame Temperature The heat released by the chemical reaction of fuel and oxidant heats the POC. Heat is transferred from the POC, primarily by radiation and convection, to the surroundings, and the resulting temperature in the reaction zone is the flame temperature. If there is no heat transfer to the surroundings, the flame temperature equals the theoretical, or adiabatic, flame temperature.

Figure 24-11 shows the *available heat* in the products of combustion for various common fuels. The available heat is the total heat

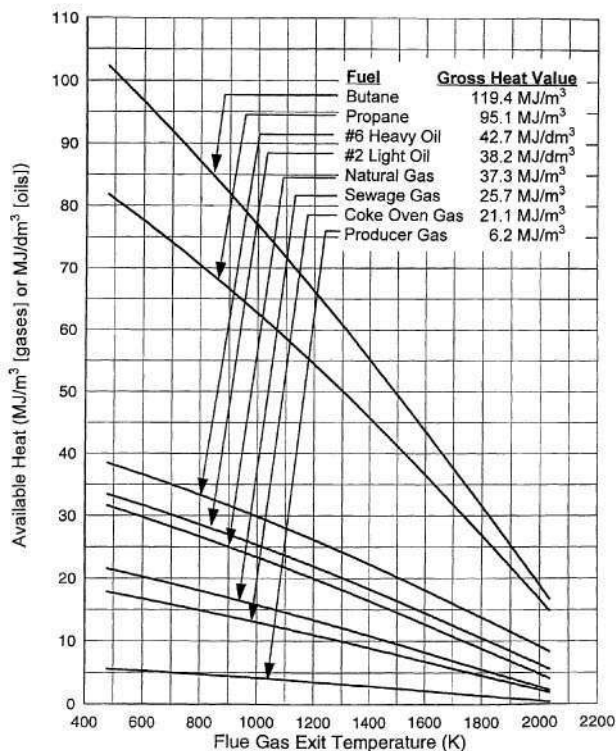


FIG. 24-11 Available heats for some typical fuels. The fuels are identified by their gross (or higher) heating values. All available heat figures are based upon complete combustion and fuel and air initial temperature of 288 K (60°F). To convert from MJ/Nm³ to Btu/ft³, multiply by 26.84. To convert from MJ/dm³ to Btu/gal, multiply by 3585.

TABLE 24-13 Combustion Characteristics of Various Fuels*

Fuel	Minimum ignition temp., K/°F	Calculated flame temperature, † K/°F		Flammability limits, % fuel gas by volume in air		Maximum flame velocity, m/s and ft/s		% theoretical air for max. flame velocity
		in air	in O ₂	lower	upper	in air	in O ₂	
Acetylene, C ₂ H ₂	578/581	2905/4770	3383/5630	2.5	81.0	2.67/8.75	—	83
Blast furnace gas	—	1727/2650	—	35.0	73.5	—	—	—
Butane, commercial	753/896	2246/3583	—	1.86	8.41	0.87/2.85	—	—
Butane, n-C ₄ H ₁₀	678/761	2246/3583	—	1.86	8.41	0.40/1.3	—	97
Carbon monoxide, CO	882/1128	2223/3542	—	12.5	74.2	0.52/1.7	—	55
Carbureted water gas	—	2311/3700	3061/5050	6.4	37.7	0.66/2.15	—	90
Coke oven gas	—	2261/3610	—	4.4	34.0	0.70/2.30	—	90
Ethane, C ₂ H ₄	745/882	2222/3540	—	3.0	12.5	0.48/1.56	—	98
Gasoline	553/536	—	—	1.4	7.6	—	—	—
Hydrogen, H ₂	845/1062	2318/4010	3247/5385	4.0	74.2	2.83/9.3	—	57
Hydrogen sulfide, H ₂ S	565/558	—	—	4.3	45.5	—	—	—
Mapp gas, (allene) C ₃ H ₄	728/850	—	3200/5301	3.4	10.8	—	4.69/15.4	—
Methane, CH ₄	905/1170	2191/3484	—	5.0	15.0	0.45/1.48	4.50/14.76	90
Methanol, CH ₃ OH	658/725	2177/3460	—	6.7	36.0	—	0.49/1.6	—
Natural gas	—	2214/3525	2916/4790	4.3	15.0	0.30/1.00	4.63/15.2	100
Producer gas	—	1927/3010	—	17.0	73.7	0.26/0.85	—	90
Propane, C ₃ H ₈	739/871	2240/3573	3105/5130	2.1	10.1	0.46/1.52	3.72/12.2	94
Propane, commercial	773/932	2240/3573	—	2.37	9.50	0.85/2.78	—	—
Propylene, C ₃ H ₆	—	—	3166/5240	—	—	—	—	—
Town gas (brown coal)	643/700	2318/3710	—	4.8	31.0	—	—	—

*For combustion with air at standard temperature and pressure. These flame temperatures are calculated for 100 percent theoretical air, disassociation considered. Data from *Gas Engineers Handbook*, Industrial Press, New York, 1965.

†Flame temperatures are theoretical—calculated for stoichiometric ratio, dissociation considered.

released during combustion minus the flue-gas heat loss (including the heat of vaporization of any water formed in the POC).

Flammability Limits There are both upper (or rich) and lower (or lean) limits of flammability of fuel-air or fuel-oxygen mixtures. Outside these limits, a self-sustaining flame cannot form. Flammability limits for common fuels are listed in Table 24-13.

Flame Speed Flame speed is defined as the velocity, relative to the unburned gas, at which an adiabatic flame propagates normal to itself through a homogeneous gas mixture. It is related to the combustion reaction rate and is important in determining burner flashback and blow-off limits. In a premixed burner, the flame can *flash back* through the flameholder and ignite the mixture upstream of the burner head if the mixture velocity at the flameholder is lower than the flame speed. Conversely, if the mixture velocity is significantly higher than the flame speed, the flame may not stay attached to the flameholder and is said to *blow off*. Flame speed is strongly dependent on fuel/air ratio, passing from nearly zero at the lean limit of flammability through a maximum and back to near zero at the rich limit of flammability. Maximum flame speeds for common fuels are provided in Table 24-13.

Pollutant Formation and Control in Flames Key combustion-generated air pollutants include nitrogen oxides (NO_x), sulfur oxides (principally SO₂), particulate matter, carbon monoxide, and unburned hydrocarbons.

Nitrogen Oxides Three reaction paths, each having unique characteristics (see Fig. 24-12), are responsible for the formation of NO_x during combustion processes: (1) *thermal* NO_x, which is formed by the combination of atmospheric nitrogen and oxygen at high temperatures; (2) *fuel* NO_x, which is formed from the oxidation of fuel-bound nitrogen; and (3) *prompt* NO_x, which is formed by the reaction of fuel-derived hydrocarbon fragments with atmospheric nitrogen. (NO_x is used to refer to NO + NO₂. NO is the primary form in combustion products [typically 95 percent of total NO_x]. NO is subsequently oxidized to NO₂ in the atmosphere.)

Thermal NO_x The formation of thermal NO_x is described by the Zeldovich mechanism:



The first of these reactions is the rate-limiting step. Assuming that O and O₂ are in partial equilibrium, the NO formation rate can be expressed as follows:

$$\frac{d[\text{NO}]}{dt} = A[\text{N}_2][\text{O}_2]^{1/2} \exp\left(\frac{-E}{RT}\right) \quad (24-51)$$

As indicated, the rate of NO formation increases exponentially with temperature, and, of course, oxygen and nitrogen must be available for thermal NO_x to form. Thus, thermal NO_x formation is rapid in high-temperature lean zones of flames.

Fuel NO_x Fuel-bound nitrogen (FBN) is the major source of NO_x emissions from combustion of nitrogen-bearing fuels such as heavy oils, coal, and coke. Under the reducing conditions surrounding the burning droplet or particle, the FBN is converted to fixed nitrogen species such as HCN and NH₃. These, in turn, are readily oxidized to form NO if they reach the lean zone of the flame. Between 20 and 80 percent of the bound nitrogen is typically converted to NO_x, depending on the design of the combustion equipment. With prolonged exposure (order of 100 ms) to high temperature and reducing conditions, however, these fixed nitrogen species may be converted to molecular nitrogen, thus avoiding the NO formation path.

Prompt NO_x Hydrocarbon fragments (such as C, CH, CH₂) may react with atmospheric nitrogen under fuel-rich conditions to yield fixed nitrogen species such as NH, HCN, H₂CN, and CN. These, in turn, can be oxidized to NO in the lean zone of the flame. In most flames, especially those from nitrogen-containing fuels, the prompt mechanism is responsible for only a small fraction of the total NO_x. Its control is important only when attempting to reach the lowest possible emissions.

NO_x emission control It is preferable to minimize NO_x formation through control of the mixing, combustion, and heat-transfer processes rather than through postcombustion techniques such as selective catalytic reduction. Four techniques for doing so, illustrated in Fig. 24-13, are air staging, fuel staging, flue-gas recirculation, and lean premixing.

Air staging Staging the introduction of combustion air can control NO_x emissions from all fuel types. The combustion air stream is split to create a fuel-rich primary zone and a fuel-lean secondary zone. The rich primary zone converts fuel-bound nitrogen to molecular nitrogen and suppresses thermal NO_x. Heat is removed prior to addition of the secondary combustion air. The resulting lower flame temperatures (below 1810 K [2800°F]) under lean conditions reduce the rate of formation of thermal NO_x. This technique has been widely applied to furnaces and boilers and it is the preferred approach for burning liquid and solid fuels. Staged-air burners are typically capable of reducing NO_x emissions by 30 to 60 percent, relative to uncontrolled levels. Air staging can also be accomplished by use of overfire air systems in boilers.

Fuel staging Staging the introduction of fuel is an effective approach for controlling NO_x emissions when burning gaseous fuels. The first combustion stage is very lean, resulting in low thermal and prompt NO_x. Heat is removed prior to injection of the secondary fuel. The secondary fuel entrains flue gas prior to reacting, further reducing flame temperatures. In addition, NO_x reduction through reburning

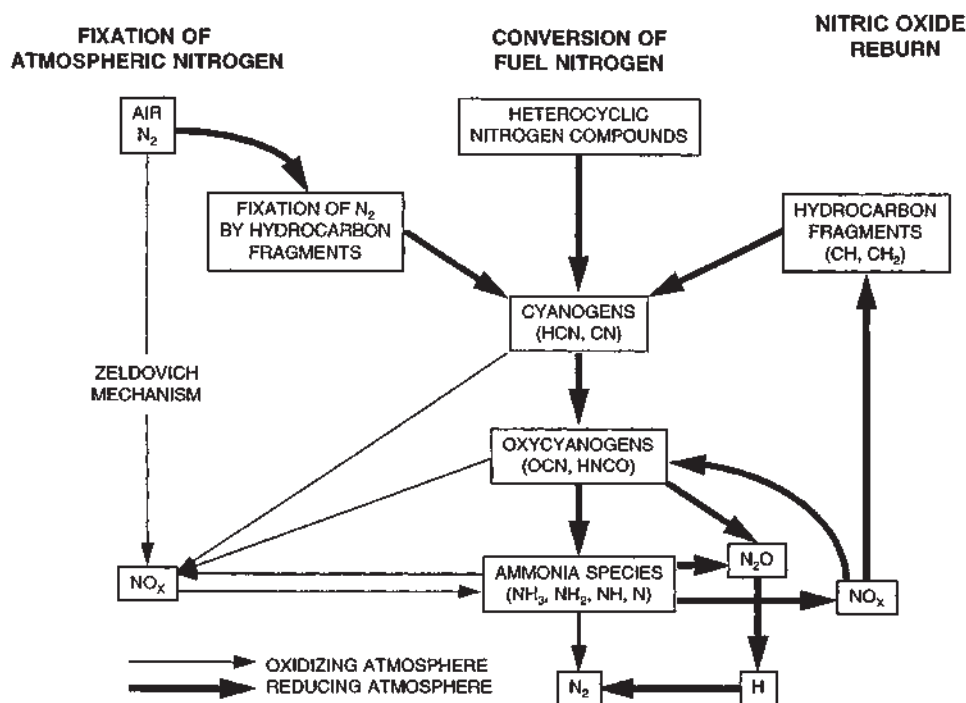


FIG. 24-12 Nitrogen oxide formation pathways in combustion.

reactions may occur in the staged jets. This technique is the favored approach for refinery- and chemical plant-fired heaters utilizing gaseous fuels. Staged-fuel burners are typically capable of reducing NO_x emissions by 40 to 70 percent, relative to uncontrolled levels.

Flue gas recirculation Flue gas recirculation, alone or in combination with other modifications, can significantly reduce thermal NO_x . Recirculated flue gas is a diluent that reduces flame temperatures. External and internal recirculation paths have been applied: internal recirculation can be accomplished by jet entrainment using either combustion air or fuel jet energy; external recirculation requires a fan or a jet pump (driven by the combustion air). When combined with staged-air or staged-fuel methods, NO_x emissions from gas-fired burners can be reduced by 50 to 90 percent. In some applications, external flue-gas recirculation can decrease thermal efficiency. Condensation in the recirculation loop can cause operating problems and increase maintenance requirements.

Lean premixing Very low NO_x emissions can be achieved by pre-mixing gaseous fuels (or vaporized liquid fuels) with air and reacting at high excess air. The uniform and very lean conditions in such systems favor very low thermal and prompt NO_x . However, achieving such low emissions requires operating near the lean stability limit. This is an attractive NO_x control approach for gas turbines, where operation at high excess air does not incur an efficiency penalty. In this application, NO_x emissions have been reduced by 75 to 95 percent.

Sulfur Oxides Sulfur occurs in fuels as inorganic minerals (primarily pyrite, FeS_2), organic structures, sulfate salts, and elemental sulfur. Sulfur contents range from parts per million in pipeline natural gas, to a few tenths of a percent in diesel and light fuel oils, to 0.5 to 5 percent in heavy fuel oils and coals. Sulfur compounds are pyrolyzed during the volatilization phase of oil and coal combustion and react in the gas phase to form predominantly SO_2 and some SO_3 . Conversion of fuel sulfur to these oxides is generally high (85 to 90 percent) and is relatively independent of combustion conditions. From 1 to 4 percent of the SO_2 is further oxidized to SO_3 , which is highly reactive and extremely hygroscopic. It combines with water to form sulfuric acid

aerosol, which can increase the visibility of stack plumes. It also elevates the dew point of water so that, to avoid back-end condensation and resulting corrosion, the flue-gas discharge temperature must be raised to about 420 K (300°F), reducing heat recovery and thermal efficiency. This reaction is enhanced by the presence of fine particles, which serve as condensation nuclei. Some coals may contain ash with substantial alkali content. In combustion of these fuels, the alkali may react to form condensed phase compounds (such as sulfates), thereby reducing the amount of sulfur emitted as oxides. Reductions in SO_2 emissions may be achieved either by removing sulfur from the fuel before and/or during combustion, or by postcombustion flue-gas desulfurization (wet scrubbing using limestone slurry, for example).

Particulates Combustion-related particulate emissions may consist of one or more of the following types, depending on the fuel.

Mineral matter derived from ash constituents of liquid and solid fuels can vaporize and condense as sub-micron-size aerosols. Larger mineral matter fragments are formed from mineral inclusions which melt and resolidify downstream.

Sulfate particles formed in the gas phase can condense. In addition, sulfate can become bound to metals and can be adsorbed on unburned carbon particles.

Unburned carbon includes unburned char, coke, cenospheres, and soot.

Particles of char are produced as a normal intermediate product in the combustion of solid fuels. Following initial particle heating and devolatilization, the remaining solid particle is termed *char*. Char oxidation requires considerably longer periods (ranging from 30 ms to over 1 s, depending on particle size and temperature) than the other phases of solid fuel combustion. The fraction of char remaining after the combustion zone depends on the combustion conditions as well as the char reactivity.

Cenospheres are formed during heavy oil combustion. In the early stages of combustion, the oil particle is rapidly heated and evolves volatile species, which react in the gas phase. Toward the end of the volatile-loss phase, the generation of gas declines rapidly and the

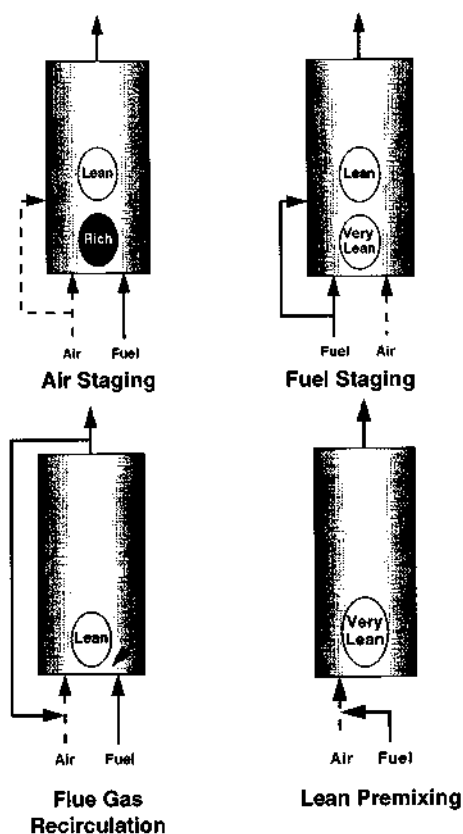


FIG. 24-13 Combustion modifications for NO_x control.

droplet (at this point, a highly viscous mass) solidifies into a porous coke particle known as a *cenosphere*. This is called *initial coke*. For the heaviest oils, the initial coke particle diameter may be 20 percent larger than the initial droplet diameter. For lighter residual oils, it may be only one third of the original droplet diameter. After a short interval, the initial coke undergoes contraction to form *final coke*. Final coke diameter is ~80 percent of the initial droplet diameter for the heaviest oils. At this time the temperature of the particle is approximately 1070 to 1270 K (1470 to 1830°F). Following coke formation, the coke particles burn out in the lean zone, but the heterogeneous oxidation proceeds slowly. Final unburned carbon levels depend on a balance between the amount of coke formed and the fraction burned out. Coke formation tends to correlate with fuel properties such as asphaltene content, C:H ratio, or Conradson Carbon Residue. Coke burnout depends on combustion conditions and coke reactivity. Coke reactivity is influenced by the presence of combustion catalysts (e.g., vanadium) in the cenospheres.

Formation of *soot* is a gas-phase phenomenon that occurs in hot, fuel-rich zones. Soot occurs as fine particles (0.02 to 0.2 μm), often agglomerated into filaments or chains which can be several millimeters long. Factors that increase soot formation rates include high C:H ratio, high temperature, very rich conditions, and long residence times at these conditions. Pyrolysis of fuel molecules leads to soot precursors such as acetylene and higher analogs and various polyaromatic hydrocarbons. These condense to form very small (< 2 nm) particles. The bulk of solid-phase material is generated by *surface growth*—attachment of gas-phase species to the surface of the particles and their incorporation into the particulate phase. Another growth mechanism is *coagulation*, in which particles collide and coalesce. Soot particle formation and growth

is typically followed by soot oxidation to form CO and CO_2 . Eventual soot emission from a flame depends on the relative balance between the soot-formation and oxidation reactions.

Carbon Monoxide Carbon monoxide is a key intermediate in the oxidation of all hydrocarbons. In a well-adjusted combustion system, essentially all the CO is oxidized to CO_2 and final emission of CO is very low indeed (a few parts per million). However, in systems which have low temperature zones (for example, where a flame impinges on a wall or a furnace load) or which are in poor adjustment (for example, an individual burner fuel-air ratio out of balance in a multiburner installation or a misdirected fuel jet which allows fuel to bypass the main flame), CO emissions can be significant. The primary method of CO control is good combustion system design and practice.

Unburned Hydrocarbons Various unburned hydrocarbon species may be emitted from hydrocarbon flames. In general, there are two classes of unburned hydrocarbons: (1) small molecules that are the intermediate products of combustion (for example, formaldehyde) and (2) larger molecules that are formed by pyro-synthesis in hot, fuel-rich zones within flames, e.g., benzene, toluene, xylene, and various polycyclic aromatic hydrocarbons (PAHs). Many of these species are listed as Hazardous Air Pollutants (HAPs) in Title III of the Clean Air Act Amendment of 1990 and are therefore of particular concern. In a well-adjusted combustion system, emission of HAPs is extremely low (typically, parts per trillion to parts per billion). However, emission of certain HAPs may be of concern in poorly designed or maladjusted systems.

COMBUSTION OF SOLID FUELS

There are three basic modes of burning solid fuels, each identified with a furnace design specific for that mode: in suspension, in a bed at rest* on a grate (fuel-bed firing), or in a fluidized bed. Although many variations of these generic modes and furnace designs have been devised, the fundamental characteristics of equipment and procedure remain intact. They will be described briefly.

Suspension Firing Suspension firing of pulverized coal (PC) is commoner than fuel-bed or fluidized-bed firing of coarse coal in the United States. This mode of firing affords higher steam-generation capacity; is independent of the caking characteristics of the coal, and responds quickly to load changes. Pulverized coal firing accounts for approximately 55 percent of the power generated by electric utilities in the United States. It is rarely used on boilers of less than 45.4 Mg/h (100,000 lb/h) steam capacity because its economic advantage decreases with size.

A simplified model of PC combustion includes the following sequence of events: (1) on entering the furnace, a PC particle is heated rapidly, driving off the volatile components and leaving a char particle; (2) the volatile components burn independently of the coal particle; and (3) on completion of volatiles combustion, the remaining char particle burns. While this simple sequence may be generally correct, PC combustion is an extremely complex process involving many interrelated physical and chemical processes.

Devolatilization The volatiles produced during rapid heating of coal can include H_2 , CH_4 , CO, CO_2 , and C_2 - C_4 hydrocarbons, as well as tars, other organic compounds, and reduced sulfur and nitrogen species. The yield of these various fractions is a function of both heating rate and final particle temperature. The resulting char particle may be larger in diameter than the parent coal particle, owing to swelling produced by volatiles ejection. The particle density also decreases.

Char oxidation dominates the time required for complete burnout of a coal particle. The heterogeneous reactions responsible for char oxidation are much slower than the devolatilization process and gas-phase reaction of the volatiles. Char burnout may require from 30 ms to over 1 s, depending on combustion conditions (oxygen level, temperature), and char particle size and reactivity. Char reactivity depends on parent coal type. The rate-limiting step in char burnout can be chemical reaction or gaseous diffusion. At low temperatures or for very large particles, chemical reaction is the rate-limiting step. At

*The burning fuel bed may be moved slowly through the furnace by the vibrating action of the grate or by being carried on a traveling grate.

higher temperatures boundary-layer diffusion of reactants and products is the rate-limiting step.

Pulverized-Coal Furnaces In designing and sizing PC furnaces, particular attention must be given to the following fuel-ash properties:

- Ash fusion temperatures, including the spread between initial deformation temperature and fluid temperature
- Ratio of basic (calcium, sodium, potassium) to acidic (iron, silicon, aluminum) ash constituents, and specifically iron-to-calcium ratio
- Ash content
- Ash friability

These characteristics influence furnace plan area, furnace volume, and burning zone size required to maintain steam production capacity for a given fuel grade or quality.

Coal properties influence pulverizer capacity and the sizing of the air heater and other heat-recovery sections of a steam generator. Furnace size and heat-release rates are designed to control slagging characteristics. Consequently, heat-release rates in terms of the ratio of net heat input to plan area range from 4.4 MW/m^2 ($1.4 \times 10^6 \text{ Btu/[h}\cdot\text{ft}^2]$) for severely slagging coals to 6.6 MW/m^2 ($2.1 \times 10^6 \text{ Btu/[h}\cdot\text{ft}^2]$) for low-slagging fuels.

The various burner and furnace configurations for PC firing are shown schematically in Fig. 24-14. The U-shaped flame, designated as *fantail vertical firing* (Fig. 24-14a), was developed initially for pulverized coal before the advent of water-cooled furnace walls. Because a large percentage of the total combustion air is withheld from the fuel stream until it projects well down into the furnace, this type of firing is well suited for solid fuels that are difficult to ignite, such as those with less than 15 percent volatile matter. Although this configuration is no longer used in central-station power plants, it may find favor again if low-volatile chars from coal-conversion processes are used for steam generation or process heating.

Modern central stations use the other burner-furnace configurations shown in Fig. 24-14, in which the coal and air are mixed rapidly in and close to the burner. The primary air, used to transport the pulverized coal to the burner, comprises 10 to 20 percent of the total combustion air. The secondary air comprises the remainder of the total air and mixes in or near the burner with the primary air and coal. The velocity of the mixture leaving the burner must be high enough to prevent flashback in the primary air-coal piping. In practice, this velocity is maintained at about 31 m/s (100 ft/s).

In *tangential firing* (Fig. 24-14b), the burners are arranged in vertical banks at each corner of a square (or nearly square) furnace and directed toward an imaginary circle in the center of the furnace. This results in the formation of a large vortex with its axis on the vertical centerline. The burners consist of an arrangement of slots one above the other, admitting, through alternate slots, primary air-fuel mixture and secondary air. It is possible to tilt the burners upward or downward, the maximum inclination to the horizontal being 30° , enabling the operator to selectively utilize in-furnace heat-absorbing surfaces, especially the superheater.

The circular burner shown in Fig. 24-15 is widely used in horizontally fired furnaces and is capable of firing coal, oil, or gas in capacities as high as 174 GJ/h ($1.65 \times 10^8 \text{ Btu/h}$). In such burners the air is often swirled to create a zone of reverse flow immediately downstream of the burner centerline, which provides for combustion stability.

Low- NO_x burners are designed to delay and control the mixing of coal and air in the main combustion zone. A typical low- NO_x air-staged burner is illustrated in Fig. 24-16. This combustion approach can reduce NO_x emissions from coal burning by 40 to 50 percent. Because of the reduced flame temperature and delayed mixing in a low- NO_x burner, unburned carbon emissions may increase in some applications and for some coals. *Overfire air* is another technique for

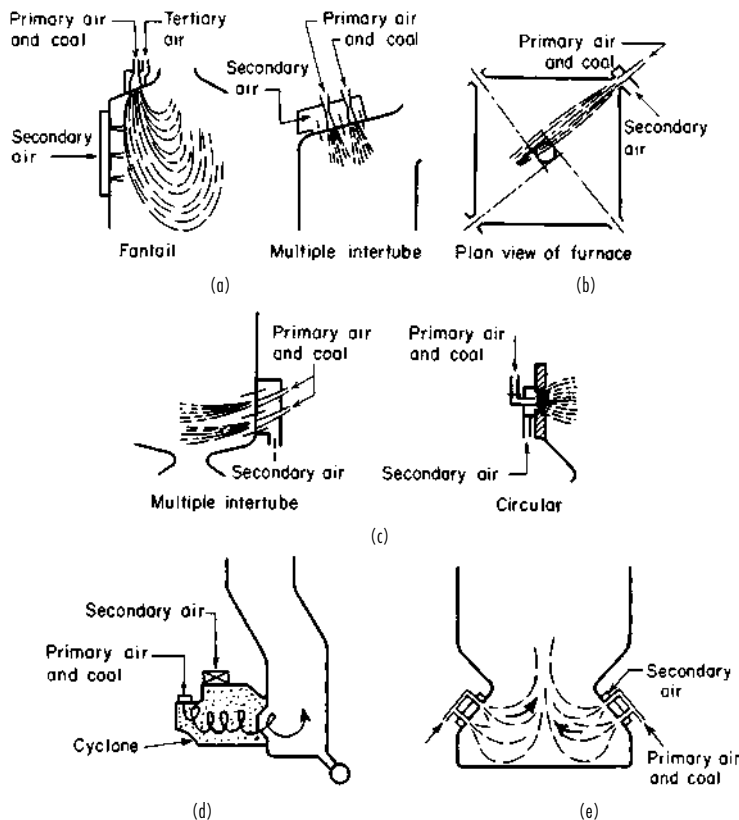


FIG. 24-14 Burner and furnace configurations for pulverized-coal firing: (a) vertical firing; (b) tangential firing; (c) horizontal firing; (d) cyclone firing; (e) opposed-inclined firing.

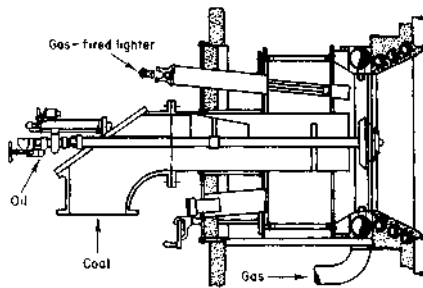


FIG. 24-15 Circular burner for pulverized coal, oil, or gas. (From Marks' Standard Handbook for Mechanical Engineers, 8th ed., McGraw-Hill, New York, 1978.)

staging the combustion air to control NO_x emissions when burning coal in suspension-firing systems. Overfire air ports are installed above the top level of burners on wall- and tangential-fired boilers. Use of overfire air can reduce NO_x emissions by 20 to 30 percent. *Reburn* is a NO_x control strategy that involves diverting a portion of the fuel from the burners to a second combustion zone (reburn zone) above the main burners. Completion air is added above the reburn zone to complete fuel burnout. The reburn fuel can be natural gas, oil, or pulverized coal, though natural gas is used in most applications. In this approach, the stoichiometry in the reburn zone is controlled to be slightly rich (equivalence ratio of ~ 1.15), under which conditions a portion (50 to 60 percent) of the NO_x is converted to molecular nitrogen.

Pulverizers The pulverizer is the heart of any solid-fuel suspension-firing system. Air is used to dry the coal, transport it through the pulverizer, classify it, and transport it to the burner, where the transport air provides part of the air for combustion. The pulverizers themselves are classified according to whether they are under positive or negative pressure and whether they operate at slow, medium, or high speed.

Pulverization occurs by impact, attrition, or crushing. The capacity of a pulverizer depends on the grindability of the coal and the fineness desired, as shown by Fig. 24-17. Capacity can also be seriously reduced by excessive moisture in the coal, but it can be restored by

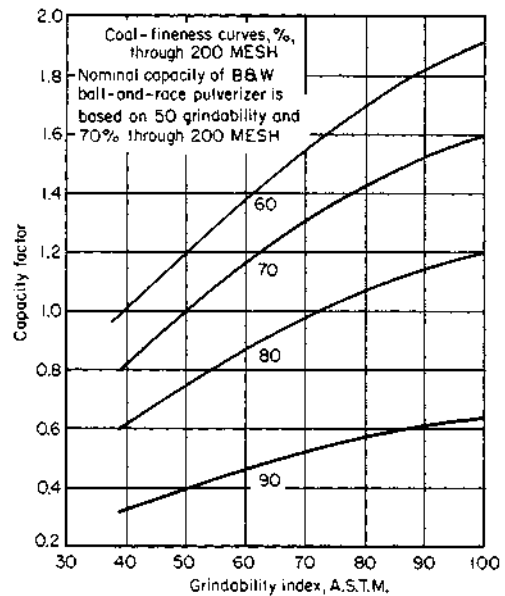


FIG. 24-17 Variation of pulverizer capacity with the grindability of the coal and the fineness to which the coal is ground. (Babcock & Wilcox Co.)

increasing the temperature of the primary air. Figure 24-18 indicates the temperatures needed. For PC boilers, the coal size usually is 65 to 80 percent through a 200-mesh screen, which is equivalent to $74 \mu\text{m}$.

Cyclone Furnaces In *cyclone firing* (Fig. 24-14d) the coal is not pulverized but is crushed to 4-mesh (4.76-mm) size and admitted tangentially with primary air to a horizontal cylindrical chamber, called a *cyclone furnace*, which is connected peripherally to a boiler furnace. Secondary air also is admitted, so that almost all of the coal burns within the chamber. The combustion gas then flows into the boiler furnace. In the cyclone furnace, finer coal particles burn in suspension and the

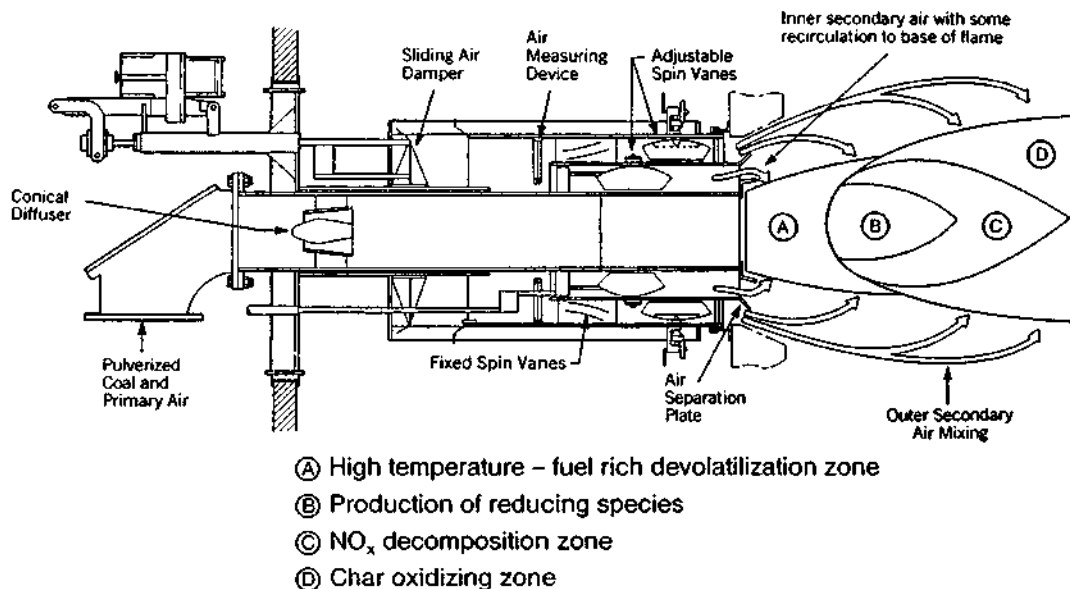


FIG. 24-16 Low- NO_x pulverized coal burner. (Babcock & Wilcox Co.)

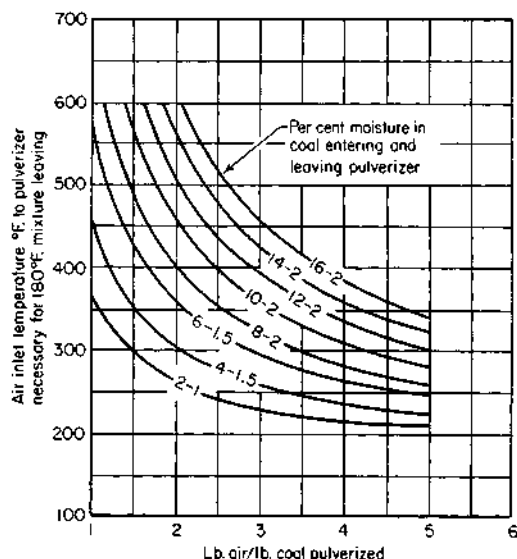


FIG. 24-18 Effect of moisture in coal on pulverizer capacity. Sufficient drying can be accomplished to restore capacity if air temperatures are high enough. [$K = (°F + 459.7)/1.8$] (Combustion Engineer, *Combustion Engineering Inc.*, New York, 1966.)

coarser ones are thrown centrifugally to the chamber wall, where most of them are captured in a sticky wall coating of molten slag. The secondary air, admitted tangentially along the top of the cyclone furnace, sweeps the slag-captured particles and completes their combustion. A typical firing rate is about $18.6 \text{ GJ}/(\text{h}\cdot\text{m}^3)$ ($500,000 \text{ Btu}/(\text{h}\cdot\text{ft}^3)$). The slag drains continuously into the boiler furnace and thence into a quenching tank. Figure 24-19 shows a cyclone furnace schematically.

Fuel-Bed Firing Fuel-bed firing is accomplished with mechanical stokers, which are designed to achieve continuous or intermittent fuel feed, fuel ignition, proper distribution of the combustion air, free release of the gaseous combustion products, and continuous or intermittent disposal of the unburned residue. These aims are met with two classes of stokers, distinguished by the direction of fuel feed to the bed: underfeed and overfeed. Overfeed stokers are represented by two types, distinguished by the relative directions of fuel and air flow (and also by the manner of fuel feed): crossfeed, also termed mass-burning, and spreader. The principles of these three methods of fuel-bed firing are illustrated schematically in Fig. 24-20.

Underfeed Firing Both fuel and air have the same relative direction in the underfeed stoker, which is built in single-retort and multiple-retort designs. In the *single-retort*, side-dump stoker, a ram pushes coal into the retort toward the end of the stoker and upward toward the tuyere blocks, where air is admitted to the bed. This type of stoker will handle most bituminous coals and anthracite, preferably in the size range 19 to 50 mm ($\frac{3}{4}$ to 2 in) and no more than 50 percent through a 6-mm ($\frac{1}{4}$ -in) screen. Overfire air or steam jets are frequently used in the bridgwall at the end of the stoker to promote turbulence.

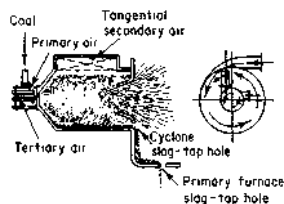


FIG. 24-19 Cyclone furnace. (From Marks' Standard Handbook for Mechanical Engineers, 9th ed., McGraw-Hill, New York, 1987.)

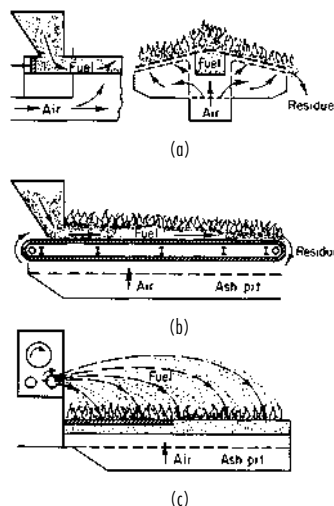


FIG. 24-20 Basic types of mechanical stokers: (a) underfeed; (b) crossfeed; (c) overfeed (spreader stoker).

In the *multiple-retort stoker*, rams feed coal to the top of sloping grates between banks of tuyeres. Auxiliary small sloping rams perform the same function as the pusher rods in the single retort. Air is admitted along the top of the banks of tuyeres, and on the largest units the tuyeres themselves are given a slight reciprocating action to agitate the bed further. This type of stoker operates best with caking coals having a relatively high ash-softening temperature. Coal sizing is up to 50 mm (2 in) with 30 to 50 percent through a 6-mm ($\frac{1}{4}$ -in) screen.

Overfeed Firing: Crossfeed (Mass-Burning) Stokers Crossfeed stokers are also termed mass-burning stokers because the fuel is dumped by gravity from a hopper onto one end of a moving grate, which carries it into the furnace and down its length. Because of this feature, crossfeed stokers are commonly called *traveling-grate* stokers. The grate may be either of two designs: *bar grate* or *chain grate*. Alternatively, the burning fuel bed may be conveyed by a vibratory motion of the stoker (*vibrating-grate*).

The fuel flows at right angles to the air flow. Only a small amount of air is fed at the front of the stoker, to keep the fuel mixture rich, but as the coal moves toward the middle of the furnace, the amount of air is increased, and most of the coal is burned by the time it gets halfway down the length of the grate. Fuel-bed depth varies from 100 to 200 mm (4 to 8 in), depending on the fuel, which can be coke breeze, anthracite, or any noncaking bituminous coal.

Overfeed Firing: Spreader Stokers Spreader stokers burn coal (or other fuel) by propelling it into the furnace. A portion of the coal burns in suspension (the percentage depending on the coal fineness), while the rest burns on a grate. In most units, coal is pushed off a plate under the storage hopper onto revolving paddles (either overthrow or underthrow) which distribute the coal on the grate (Fig. 24-20c). The angle and speed of the paddles control coal distribution. The largest coal particles travel the farthest, while the smallest ones become partially consumed during their trajectory and fall on the forward half of the grate. The grate may be stationary or traveling. The fuel and air flow in opposite directions.

Some spreaders use air to transport the coal to the furnace and distribute it, while others use mechanical means to transport the coal to a series of pneumatic jets.

The performance of spreader stokers is affected by changes in coal sizing. The equipment can distribute a wide range of fuel sizes, but it distributes each particle on the basis of size and weight. Normal size specifications call for 19-mm ($\frac{3}{4}$ -in) nut and slack with not more than 30 percent less than 6.4 mm ($\frac{1}{4}$ in).

Typically, approximately 30 to 50 percent of the coal is burned in suspension. If excessive fines are present, more coal particles will be

carried out of the furnace and burned in suspension, and very little ash will be available to provide a protective cover for the grate surface. On the other hand, if sufficient fines are not present, not all the fuel will be burned on the grate, resulting in derating of the unit and excessive dumping of live coals to the ash hopper.

Excess air is usually 30 to 40 percent for stationary and dumping grates, while traveling grates are operated with from 22 to 30 percent excess air. Preheated air can be supplied for all types of grates but the temperature is usually limited to 395 to 422 K (250 to 300°F) to prevent excessive slagging of the fuel bed.

Overfire air nozzles are located in the front wall underneath the spreaders and in the rear wall from 0.3 to 0.9 m (1 to 3 ft) above the grate level. These nozzles use air directly from a fan or inspirate air with steam to provide turbulence above the grate for most effective mixing of fuel and air. They supply about 15 percent of the total combustion air.

Comparison of Suspension and Fuel-Bed Firing A major factor to consider when comparing a stoker-fired boiler with a PC boiler is the reduction in efficiency due to carbon loss. The carbon content of the ash passing out of a spreader stoker furnace varies from 30 to 50 percent. Overall efficiency of the stoker can be increased by reburning the ash: it is returned to the stoker grate by gravity or a pneumatic feed system. A continuous-ash-discharge spreader-stoker-fired unit will typically have a carbon loss of 4 to 8 percent, depending on the amount of ash reinjection. A properly designed PC boiler, on the other hand, can maintain an efficiency loss due to unburned carbon of less than 0.4 percent.

A difference between these firing methods may also be manifested in the initial fuel cost. For efficient operation of a spreader-stoker-fired boiler, the coal must consist of a proper mixture of coarse and fine particles. Normally, double-screened coal is purchased because less expensive run-of-mine coal does not provide the optimum balance of coarse and fine material.

An advantage of a stoker-fired furnace is its easy adaptability to firing almost any unsized solid fuels. Bark, bagasse, or refuse can normally be fired on a stoker to supplement the coal with a minimum amount of additional equipment. Thus, such supplementary waste fuels may be able to contribute a higher percentage of the total heat input in a stoker-fired furnace than in a PC furnace without expensive equipment modifications.

Fluidized-Bed Combustion The principles of gas-solid fluidization and their application to the chemical process industry are treated in Sec. 17. Their general application to combustion is reviewed briefly here, and their more specific application to fluidized-bed boilers is discussed later in this section.

In fluidized-bed combustion (FBC), fuel is burned in a bed of particles supported in an agitated state by an upward flow of air introduced via an air distributor. The bed particles may be sand or ash derived from the fuel, or spent sorbent (limestone or dolomite) if in-bed sulfur capture is included in the design. The fluidizing action promotes good solids mixing and gas-solids contacting that allow high combustion efficiency to be achieved at temperatures significantly lower than those of a pulverized coal (PC) furnace (typically 1116 K [1550°F] compared to over 1589 K [2400°F]). These lower temperatures also result in reduced slagging and fouling problems and significantly lower NO_x formation. This latter benefit, in conjunction with in-bed sulfur capture capability, constitutes one of the great advantages of fluidized-bed combustion: in situ pollution control. Having this control built into the furnace reduces the extent of back-end clean up required for a given emissions standard.

There are two types of FBC unit being sold commercially, *bubbling* and *circulating*, both operating at atmospheric pressure. Typical superficial fluidizing velocities are as follows.

Bubbling FBC 1.5 to 2.7 m/s (5 to 9 ft/s)

Circulating FBC 3.7 to 7.3 m/s (12 to 24 ft/s)

Pressurized FBC (PFBC) designs, bubbling and circulating, have been under development since the 1970s, but all work has now been curtailed. Work on an advanced PFBC design, incorporating partial coal gasification and char combustion, has also been curtailed. Several commercial bubbling PFBC units were built in the 1990s, covering the range 80 to 340 MWe. However, no new PFBC plants have been

announced since 1996, and consequently the technology is dropped from this edition of the Handbook.

Bubbling Beds A large proportion of the noncombustible feedstock—sand, ash, or sorbent-derived material—remains in the combustor forming the bed. Bed depth is maintained by draining off excess material. Most of the gas in excess of that required for minimum fluidization appears as bubbles (voids), and these carry particles upward in their wake, resulting in the rapid vertical mixing of the bed material that promotes even temperatures and good gas-solids contacting.

Early bubbling FBC units were designed to burn coal, and the heat released was removed by heat transfer to in-bed tubes and/or to the water-wall tubes used to enclose the furnace. These surfaces experienced high rates of metal loss through the combined effects of erosion and abrasion. Protective measures such as plasma-sprayed coatings and metal fins to disrupt the solids flow pattern were used. These were effective for only short periods before requiring replacement, and so maintenance requirements were high.

The two largest bubbling-bed FBC units built, 160 and 350 MWe, entered into service in the late 1980s burning bituminous coal. However, for coal applications, the overall performance of circulating fluidized-bed units has proved superior, and bubbling-bed designs of this size are no longer marketed. The bubbling technology is now mainly used for burning high-moisture biomass fuels for which in-bed heat-transfer surfaces are not required. These units are usually designed for generating capacities below 120 MWe, and typical fuels include wood wastes, paper mill and sewage sludges, and peat.

The boilers are of water-wall construction with the walls in the bed region lined with refractory to protect against erosion and abrasion damage. The heat released in the bed is removed from the flue gas by the exposed water walls and by convection pass tubing set in the gas path after the boiler. The coal boilers included multiclones to collect elutriated carbon and recycle it back to the boiler to increase combustion efficiency. However, because biomass is more reactive, recycle is not required.

To accommodate the wide variations in fuel moisture content while maintaining close bed temperature control, two control strategies are used.

- Operate the bed substoichiometrically so that bed temperature varies with the fluidizing air (primary combustion air). There is an equal and opposite change in the air fed above the bed (secondary combustion air) to complete the combustion process, the total air remaining constant. As the fuel moisture increases, the primary air-flow rate increases, increasing the fluidizing velocity and releasing more heat in the bed to evaporate the additional water and keep the bed temperature constant. The bed operates less substoichiometrically, and in-bed combustion efficiency increases. At the same time the secondary air is reduced, and as there is less above-bed combustion, the upper furnace temperature falls. The control action is reversed when the fuel moisture content decreases. The split of primary to secondary airflow ranges from above 80:20 for high-moisture fuels to below 50:50 for low-moisture fuels.
- The above strategy results in variations in fluidizing velocity that might produce poor fluidization at some operating conditions with a resulting reduction in combustion efficiency. To maintain fluidizing velocity constant, recirculated flue gas is mixed with the primary combustion air to lower its oxygen content while keeping the combined fluidization gas flow rate constant. Air may be fed above the bed to complete combustion, but this is controlled independently of the air passing to the bed. As the fuel moisture increases, less flue gas is recirculated, the oxygen content of the primary combustion air and in-bed combustion both increase, and secondary air decreases. The control action is reversed when the fuel moisture content decreases.

Circulating Beds These fluidized beds operate at higher velocities, and virtually all the solids are elutriated from the furnace. The majority of the elutriated solids, still at combustion temperature, are captured by reverse-flow cyclone(s) and recirculated to the foot of the combustor. The foot of the combustor is a potentially very erosive region, as it contains large particles not elutriated from the bed, and they are being fluidized at high velocity. Consequently, the lower reaches of the combustor do not contain heat-transfer tubes and the water walls are protected with refractory. Some combustors have

experienced damage at the interface between the water walls and the refractory, and measures similar to those employed in bubbling beds have been used to protect the tubes in this region.

The circulating-bed design is the leading FBC technology and is the only design currently offered at sizes above 120 MWe. Units as large as 320 MWe have been built incorporating subcritical steam conditions with reheat and supercritical designs of up to 600 MWe are available. The fuels used in these designs are coals or opportunity fuels such as petroleum coke, although one 240-MWe unit is designed to operate solely on biomass. Circulating FBC boilers as small as 25 MWe have been built to fire either coal or biomass. However, as the size gets smaller, the circulating FBC becomes less competitive for biomass firing, and as indicated earlier, the bubbling FBC design is favored for this application.

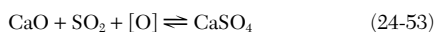
The heat released in the furnace is removed from the flue gas by the exposed upper water-wall tubes. As the units increase in size, more heat-transfer surface is required than is provided by the walls. Surface can be added by wrapping horizontal tubing over the walls of the upper furnace, or by added wing walls, sections of water wall extending short distances into the furnace enclosure. In some designs, tubes are extended across the upper furnace where, although the fluidizing velocity is still high, the erosion potential is low because the solids are finer and their concentration is lower. In some designs, heat is removed from the recirculated solids by passing them through a bubbling-bed heat exchanger before returning them to the furnace. The potential for erosion or abrasion of the tubes in this bed is low as the recirculating solids are mostly less than 500 μm and are well fluidized at velocities of only around 0.6 m/s [2 ft/s].

The combustion air is introduced at two levels, 60 to 70 percent being introduced through the distributor and 30 to 40 percent above the bed. This staged entry results in the lower reaches operating substoichiometrically, which helps to reduce NO_x emissions but tends to reduce the fluidizing velocity at the base of the combustor. To compensate for this and to increase solids mixing by increasing the local gas velocity, the portion of the combustor below the secondary air entry points is tapered.

Fuel Flexibility An advantage of FBC designs is fuel flexibility: a single unit can burn a wider range of fuels than a PC furnace, thus offering owners an improved bargaining position to negotiate lower fuel prices. Among the fuels fired are bituminous and subbituminous coals, anthracite culm, lignite, petroleum coke, refuse-derived fuel, biomass, industrial and sewage sludges, and shredded tires. But fuel flexibility can be achieved only if the unit is designed for the range of fuels intended to be burned. For example, to maintain the same firing rate, a feed system designed for a certain fuel must be capable of feeding a lower calorific fuel at a higher rate. Similarly, to maintain the same degree of sulfur capture, feeders must be capable of delivering sorbent over a range of rates matching the sulfur contents of the fuels likely to be fed. This increase in operating flexibility increases the capital cost, offsetting at least partially the economic benefits of reduced fuel pricing.

Sulfur Emissions Sulfur present in a fuel is released as SO_2 , a known contributor to acid rain deposition. By adding limestone or dolomite to a fluidized bed, much of this can be captured as calcium sulfate, a dry nonhazardous solid. As limestone usually contains over 40 percent calcium, compared to only 20 percent in dolomite, it is the preferred sorbent, resulting in lower transportation costs for the raw mineral and the resulting ash product. Moreover, the high magnesium content of the dolomite makes the ash unsuitable for some building applications and so reduces its potential for utilization. Whatever sorbent is selected, for economic reasons it is usually from a source local to the FBC plant. If more than one sorbent is available, plant trials are needed to determine the one most suitable, as results from laboratory-scale reactivity assessments are unreliable.

At atmospheric pressure, calcium carbonate almost completely calcines to free lime, and it is this that captures the sulfur dioxide. As the free lime is not completely sulfated, the resulting sorbent ash is very alkaline, consisting primarily of CaSO_4 and CaO , with small amounts of CaCO_3 .



The sulfation reaction has an optimum at a mean bed temperature of around 1116 K (1550°F). Units are usually designed to operate at this

temperature, but changes in operating conditions to achieve power output and accommodate changes in fuel composition often result in the plant operating at above or below the optimum temperature. To compensate for the ensuing reduction in sulfur capture, more sorbent is fed to the bed.

Emissions standards are becoming increasingly more stringent, and environmental permits for recent CFB projects have required sulfur capture efficiencies greater than 98 percent. Achieving this with only in-bed sulfur capture requires Ca/S molar ratios as high as 3 (3 mol of calcium in the sorbent for each mole of sulfur in the coal). For a 3 percent sulfur coal and a sorbent with 38 percent calcium, the sorbent-to-coal weight ratio is 0.3. To reduce the sorbent demand for the same sulfur capture efficiency, a back-end dry scrubber is used to remove additional SO_2 from the flue gas. This has been demonstrated to lower the Ca/S molar ratio to 1.9, a sorbent-to-coal weight ratio of 0.19. For a coal with 12 percent ash, this 37 percent reduction in sorbent feed rate reduces the amount of ash sent to disposal by 17 percent. This reduction in solids handling requirements improves process economics primarily by lowering the operating costs of the plant. The capital cost remains roughly the same, the cost of the dry scrubber being mainly offset by the reduction in solids handling equipment.

The sorbent used in the dry scrubber can be either lime or reactivated ash captured by the baghouse. The baghouse ash contains unreacted sorbent particles with a surface coating of calcium sulfate that prevents SO_2 reaching the CaO at their core. Hydrating these solids causes the material to swell and crack, thus exposing the unreacted CaO , which when recycled back to the boiler is available to react with additional SO_2 . As lime is more expensive than limestone, using the baghouse ash is the preferred approach. It also eliminates the need to provide separate handling facilities for lime.

Nitrogen Oxide Emissions FBC units achieve excellent combustion and sulfur emission performance at relatively modest combustion temperatures in the range of 1060 to 1172 K [1450 to 1650°F]. At these temperatures no atmospheric nitrogen is converted to NO_x . The prime variables influencing NO_x formation are excess air, mean bed temperature, nitrogen content of the fuel, and Ca/S molar ratio. With respect to the latter, high sorbent feed rates increase the free lime content, and this catalyzes NO_2 formation. Typical NO_x emissions, consisting of around 90 percent NO and 10 percent NO_2 , are in the range 86 to 129 mg/MJ [0.2 to 0.3 lb/MBtu]. These values have been reduced to as low as 21 mg/MJ [0.05 lb/MBtu] by injecting ammonia into the boiler freeboard to promote selective noncatalytic reduction (SNCR) reactions. This is a less costly approach than the selective catalytic reduction (SCR) units required for PC plants.

Because the operating temperature is lower, FBC units release more N_2O than do PC units. Nitrous oxide is a greenhouse gas that absorbs 270 times more heat per molecule than carbon dioxide and as such is likely to come under increased scrutiny in the future. The emissions at full load from coal-fired units are around 65 mg/MJ [0.15 lb/MBtu], but these increase as load is reduced and furnace temperature falls. Measurements from biomass-fired FBCs have not been made. Combustion processes do not contribute greatly to current U.S. N_2O emissions; agriculture and motor vehicles account for 86 percent of the total.

Particulate Emissions To meet environmental regulations, FBC boilers use a back-end particulate collector, such as a baghouse or an electrostatic precipitator (ESP). Compared to PC units, because of its sulfur sorbent content, the ash from a circulating CFB has higher resistivity and is finer because the flue gas path contains cyclones. Both factors result in a reduction in ESP collection efficiency, although lowering the gas velocity and providing additional collection fields can compensate for this albeit at increased capital cost. Bubbling FBC designs fired on biomass do not normally feed sulfur sorbents and do not include dust collectors in the flue gas path. For these applications suitably designed ESPs are often included. In general, however, baghouses are the preferred particulate collection devices.

FBC ash is irregular, whereas PC ash, because it melts at the elevated operating temperatures, is spherical. This difference in shape influences baghouse design in three ways: (1) FBC ash does not flow from the collection hoppers as readily and special attention has to be given to their design; (2) FBC ash forms a stronger cake, requiring more frequent and more robust cleaning mechanisms, e.g., shake-deflate and

pulse-jet technologies; and (3) this more robust action in conjunction with the more abrasive, irregular particles results in filter bags being more prone to failure in FBC systems. Careful selection of bag materials (synthetic felts generally perform best) and good installation and maintenance practices minimize the latter problem.

Mercury Emissions The U.S. Environmental Protection Agency (EPA) has proposed new rules that require power plants to reduce their mercury emissions by 70 percent by 2018. Tests on operating fluidized beds show that inherent mercury capture from the flue gas for bituminous coals is over 90 percent, but for lignites capture is closer to 50 percent. The reasons for this difference are not fully understood but could be explained by the higher reactivity of lignite resulting in lower unburned carbon carryover. Carbon is known to adsorb elemental and ionic mercury, and a low carbon content will result in reduced mercury capture. The combustion efficiency for less reactive bituminous coal is lower, resulting in more carbon carryover and possibly accounting for the higher mercury capture. Recent tests have shown that halide-impregnated activated carbons injected into flue gas streams can capture over 90 percent of the mercury present. So the low inherent mercury capture associated with lignite can be compensated for by activated carbon injection ahead of the dust collection device used.

COMBUSTION OF LIQUID FUELS

Oil is typically burned as a suspension of droplets generated by atomizing the fuel. As the droplets pass from the atomizer into the flame zone, they are heated both by radiation from the flame and by convection from the hot gases that surround them, and the lighter fuel components vaporize. The vapors mix with surrounding air and ignite. Depending on the fuel type, the fuel droplet may be completely vaporized or it may be partially vaporized, leaving a residual char or coke particle.

Fuel oils can contain a significant amount of sulfur: in the case of high-sulfur No. 6, it may be as much as 4 percent (Table 24-6). SO_2 is the principal product of sulfur combustion with stoichiometric or leaner fuel-air mixtures, but with the excess air customarily used for satisfactory combustion, SO_3 can form and then condense as sulfuric acid at temperatures higher than the normally expected dew point. Thus air preheaters and other heat recovery equipment in the flue-gas stream can be endangered. Figure 24-21 shows the maximum safe upper limits for dew points in the stacks of furnaces burning sulfur-containing oil and emitting unscrubbed flue gas.

Atomizers Atomization is the process of breaking up a continuous liquid phase into discrete droplets. Figure 24-22 shows the idealized process by which the surface area of a liquid sheet is increased until it forms droplets. Atomizers may be classified into two broad groups (see Fig. 24-23), pressure atomizers, in which fuel oil is injected at high pressure, and twin-fluid atomizers, in which fuel oil is injected at moderate pressure and a compressible fluid

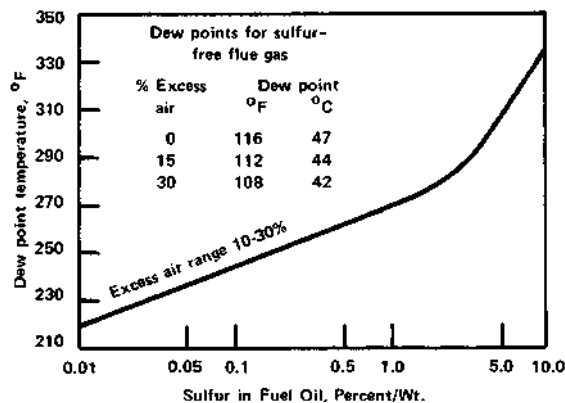


FIG. 24-21 Maximum flue-gas dew point versus percent of sulfur in typical oil fuels. ($K = [^{\circ}\text{F} + 459.7]/1.8$)

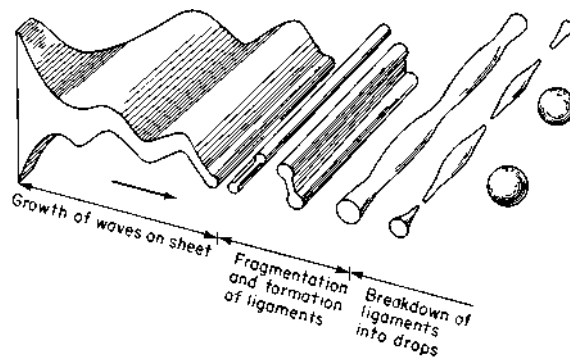


FIG. 24-22 Idealized process of drop formation by breakup of a liquid sheet. (After Dombrowski and Johns, *Chem. Eng. Sci.* 18:203, 1963.)

(steam or air) assists in the atomization process. Low oil viscosity (less than $15 \text{ mm}^2/\text{s}$) is required for effective atomization (i.e., small droplet size). Light oils, such as No. 2 fuel oil, may be atomized at ambient temperature. However, heavy oils must be heated to produce the desired viscosity. Required preheats vary from approximately 373 K (212°F) for No. 6 oil to 623 K (480°F) for vacuum bottoms.

Pressure Atomizers The commonest type of pressure atomizer is the swirl-type (Fig. 24-24). Entering a small cup through tangential orifices, the oil swirls at high velocity. The outlet forms a dam around the open end of the cup, and the oil spills over the dam in the form of a thin conical sheet, which subsequently breaks up into thin filaments and then droplets. Depending on the fuel viscosity, operating pressures range from 0.69 to 6.9 MPa (100 to 1000 psia) and the attainable

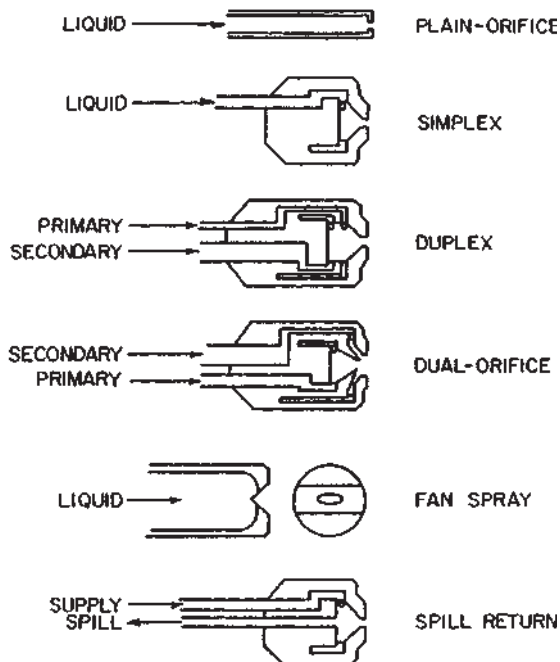


FIG. 24-23a Common types of atomizers: pressure atomizers. (From Lefebvre, *Atomization and Sprays*, Hemisphere, New York, 1989. Reproduced with permission. All rights reserved.)

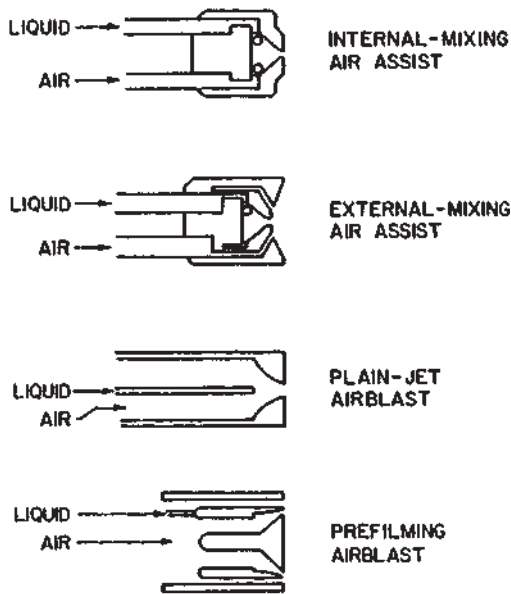


FIG. 24-23b Common types of atomizers: twin-fluid atomizers. (From Lefebvre, *Atomization and Sprays*, Hemisphere, New York, 1989. Reproduced with permission. All rights reserved.)

turndown ratio is approximately 4:1. Pressure atomization is most effective for lighter fuel oils.

Twin-Fluid Atomizers In a twin-fluid atomizer, the fuel stream is exposed to a stream of air or steam flowing at high velocity. In the internal-mixing configuration (Fig. 24-25), the liquid and gas mix inside the nozzle before discharging through the outlet orifice. In the external-mixing nozzle, the oil stream is impacted by the high-velocity gas stream outside the nozzle. The internal type requires lower flows of secondary fluid. In industrial combustion systems, steam is the preferred atomizing medium for these nozzles. In gas turbines, compressed air is more readily available. Maximum oil pressure is about 0.69 MPa (100 psia), with the steam or air pressure being maintained about 0.14 to 0.28 MPa (20 to 40 psia) in excess of the oil pressure. The mass flow of atomizing fluid varies from 5 to 30 percent of the fuel flow rate, and represents only a modest energy consumption. Turndown performance is better than for pressure atomizers and may be as high as 20:1.

A well-designed atomizer will generate a cloud of droplets with a mean size of about 30 to 40 μm and a top size of about 100 μm for light oils such as No. 2 fuel oil. Mean and top sizes are somewhat larger than this for heavier fuel oils.

Oil Burners The structure of an oil flame is shown in Fig. 24-26, and Fig. 24-27 illustrates a conventional circular oil burner for use in boilers. A combination of stabilization techniques is used, typically including swirl. It is important to match the droplet trajectories to the combustion aerodynamics of a given burner to ensure stable ignition and good turndown performance.

Many oil burners are designed as combination gas/oil burners. An example of a modern low- NO_x oil/gas forced-draft burner is shown in Fig. 24-28. This is an air-staged design, with the air divided into primary, secondary, and tertiary streams. An air-staged natural draft process heater oil/gas burner is illustrated in Fig. 24-29.

Emissions of unburned carbon (primarily coke cenospheres) may be reduced by (1) achieving smaller average fuel droplet size (e.g., by heating the fuel to lower its viscosity or by optimizing the atomizer geometry), (2) increasing the combustion air preheat temperature, or (3) firing oils with high vanadium content (vanadium appears to catalyze the burnout of coke).

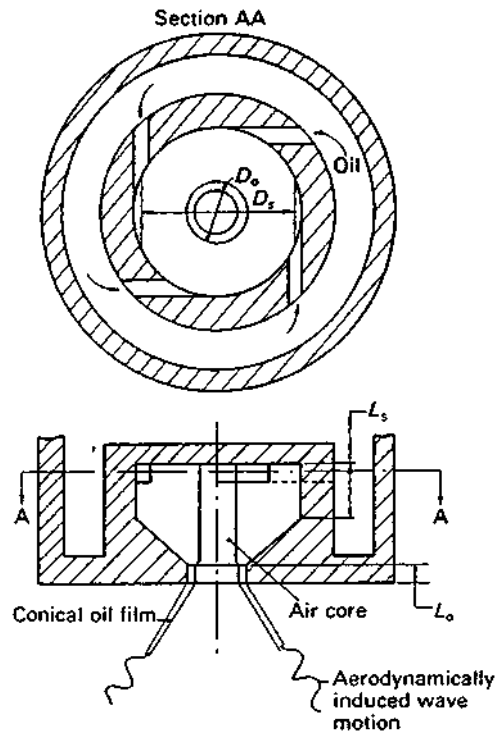


FIG. 24-24 Swirl pressure-jet atomizer. (From Lefebvre, *Atomization and Sprays*, Hemisphere, New York, 1989. Reproduced with permission. All rights reserved.)

COMBUSTION OF GASEOUS FUELS

Combustion of gas takes place in two ways, depending upon when gas and air are mixed. When gas and air are mixed before ignition, as in a Bunsen burner, burning proceeds by hydroxylation. The hydrocarbons and oxygen form hydroxylated compounds that become aldehydes; the addition of heat and additional oxygen breaks down the aldehydes to H_2 , CO , CO_2 , and H_2O . Inasmuch as carbon is converted to aldehydes in the initial stages of mixing, no soot can be developed even if the flame is quenched.

Cracking occurs when oxygen is added to hydrocarbons after they have been heated, decomposing the hydrocarbons into carbon and hydrogen, which, when combined with sufficient oxygen, form CO_2 and H_2O . Soot and carbon black are formed if insufficient oxygen is present or if the combustion process is arrested before completion.

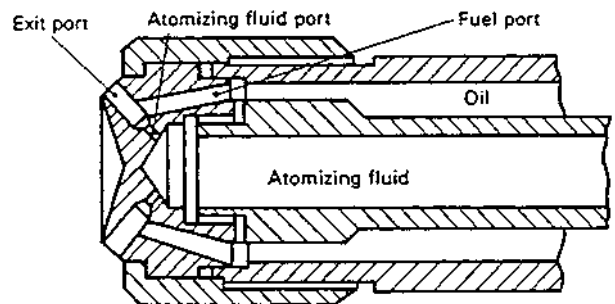


FIG. 24-25 Y-jet twin-fluid atomizer. (From Lefebvre, *Atomization and Sprays*, Hemisphere, New York, 1989. Reproduced with permission. All rights reserved.)

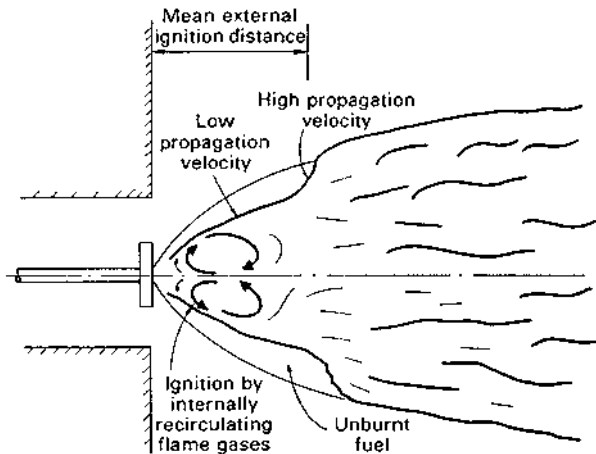


FIG. 24-26 Structure of typical oil flame. (From *Lawn, Principles of Combustion Engineering for Boilers*, Academic Press, London, 1987. Reprinted with permission.)

Gas Burners Gas burners may be classified as premixed or non-premixed. Many types of flame stabilizer are employed in gas burners. Bluff body, swirl, and combinations thereof are the predominant stabilization mechanisms.

Fully Premixed Burners A fully premixed burner includes a section for completely mixing the fuel and air upstream of the burner.

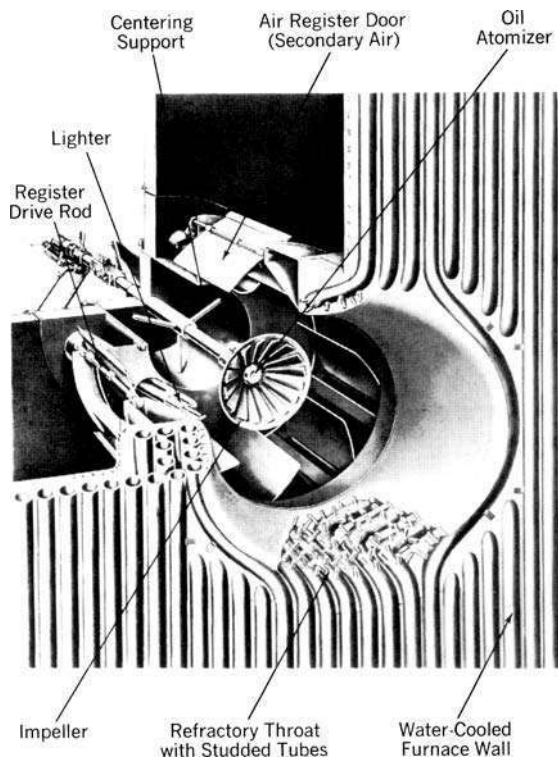


FIG. 24-27 Circular register burner with water-cooled throat for oil firing. (Babcock & Wilcox Co.)

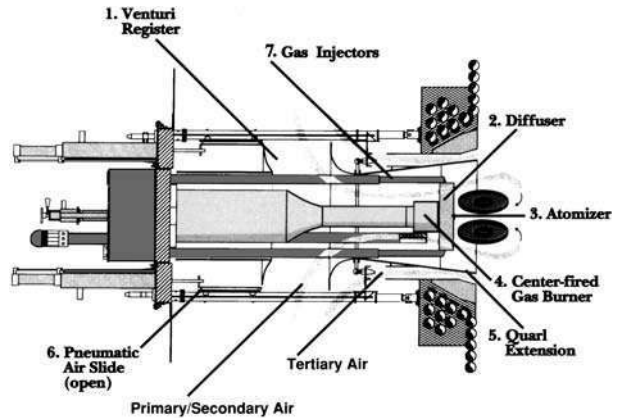


FIG. 24-28 Low-NO_x combination oil/gas forced-draft boiler burner. (Todd Combustion, Inc.)

The burner proper consists essentially of a flame holder. The porting that admits the mixture to the combustion chamber is designed to produce a fairly high velocity through a large number of orifices to avoid the possibility of the flame flashing back through the flame holder and igniting the mixture upstream of the burner.

Surface combustion devices are designed for fully premixing the gaseous fuel and air and burning it on a porous radiant surface. The close coupling of the combustion process with the burner surface results in low flame temperatures and, consequently, low NO_x formation. Surface materials can include ceramic fibers, reticulated ceramics, and metal alloy mats. This approach allows the burner shape to be customized to match the heat transfer profile with the application.

Partially Premixed Burners These burners have a premixing section in which a mixture that is flammable but overall fuel-rich is generated. Secondary combustion air is then supplied around the flame holder. The fuel gas may be used to aspirate the combustion air or vice versa, the former being the commoner. Examples of both are provided in Figs. 24-30 and 24-31.

Nozzle-Mix Burners The most widely used industrial gas burners are of the nozzle-mix type. The air and fuel gas are separated until they are rapidly mixed and reacted after leaving the ports. These burners allow a wide range of fuel-air ratios, a wide variety of flame shapes, and multifuel firing capabilities. They can be used to generate special atmospheres by firing at very rich conditions (50 percent excess fuel) or very lean conditions (1000 percent excess air). By changing nozzle shape and degree of swirl, the flame profile and mixing rates can be varied widely, from a rapid-mixing short flame ($L/D = 1$), to a conventional flame ($L/D = 5$ to 10), to a slow-mixing long flame ($L/D = 20$ to 50).

Staged Burners As was pointed out earlier under "Pollutant Formation and Control in Flames," the proper staging of fuel or air in the combustion process is one technique for minimizing NO_x emissions. Gas burners that achieve such staging are available.

Air-staged burners Low-NO_x air-staged burners for firing gas (or oil) are shown in Fig. 24-28. A high-performance, low-NO_x burner for high-temperature furnaces is shown in Fig. 24-32. In this design, both air-staging and external flue-gas recirculation are used to achieve extremely low levels of NO_x emissions (approximately 90 percent lower than conventional burners). The flue gas is recirculated by a jet-pump driven by the primary combustion air.

Fuel-staged burners Use of fuel-staged burners is the preferred combustion approach for NO_x control because gaseous fuels typically contain little or no fixed nitrogen. Figure 24-33 illustrates a fuel-staged natural draft refinery process heater burner. The fuel is split into primary (30 to 40 percent) and secondary (60 to 70 percent) streams. Furnace gas may be internally recirculated by the primary gas jets for additional NO_x control. NO_x reductions of 80 to 90 percent have been achieved by staging fuel combustion.

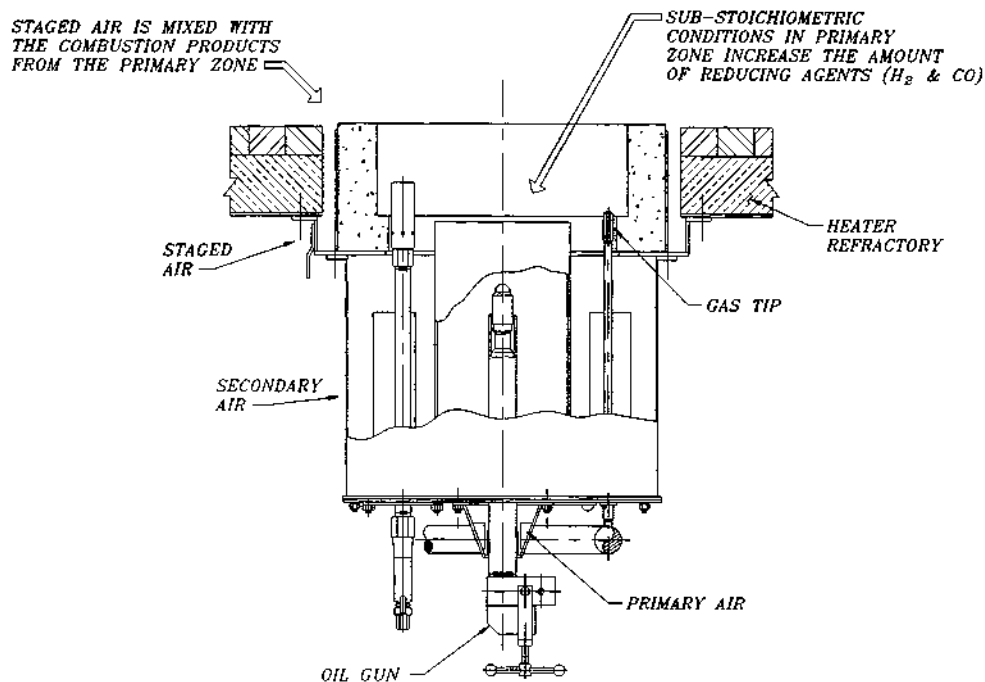


FIG. 24-29 Air-staged natural-draft combination oil/gas burner. (Callidus Technologies, Inc.)

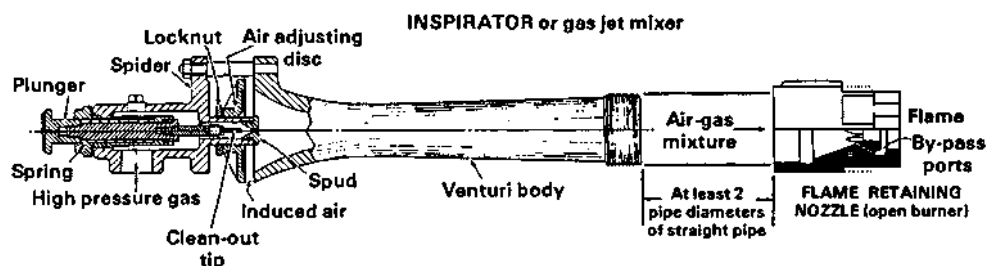


FIG. 24-30 Inspirator (gas-jet) mixer feeding a large port premix nozzle of the flame retention type. High-velocity gas emerging from the spud entrains and mixes with air induced in proportion to the gas flow. The mixture velocity is reduced and pressure is recovered in the venturi section. (From North American Combustion Handbook, 3d ed., North American Manufacturing Company, Cleveland, 1996.)

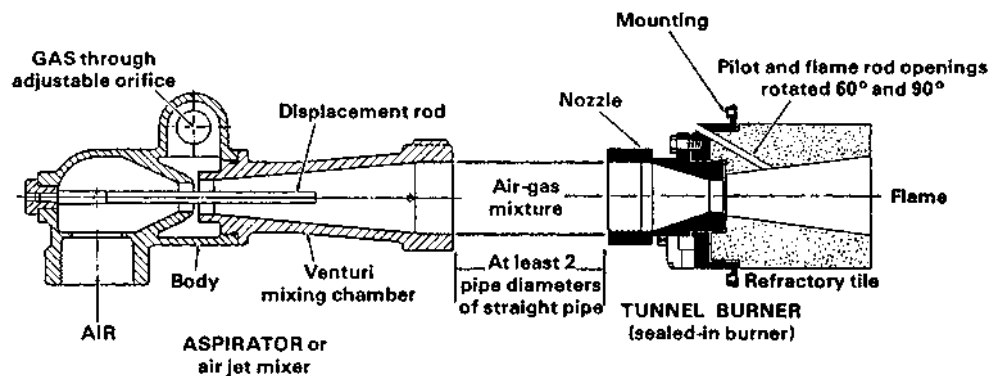


FIG. 24-31 Aspirator (air-jet) mixer feeding a sealed-in large port premix tunnel burner. Blower air enters at lower left. Gas from an atmospheric regulator is pulled into the air stream from the annular space around the venturi throat in proportion to the air flow. (From North American Combustion Handbook, 3d ed., North American Manufacturing Company, Cleveland, 1996.)

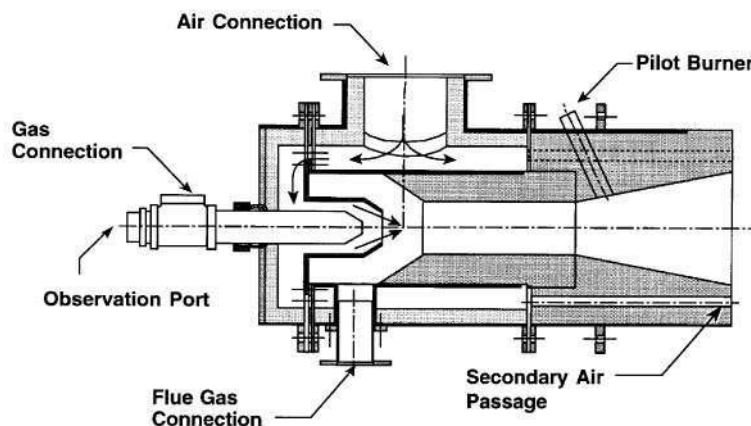


FIG. 24-32 Low- NO_x burner with air-staging and flue-gas recirculation for use in high-temperature furnaces. (Hauck Manufacturing Company. Developed and patented by the Gas Research Institute.)

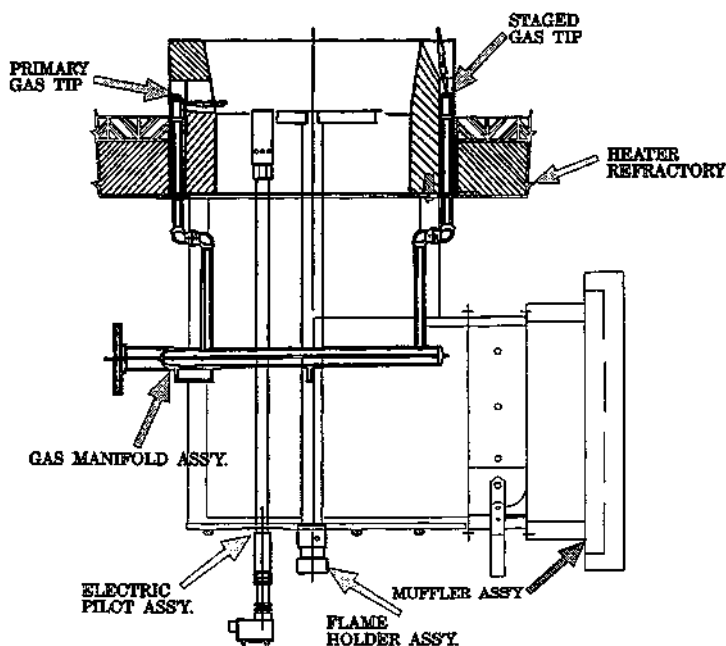


FIG. 24-33 Low- NO_x fuel-staged burner for a natural draft refinery process heater. (Callidus Technologies, Inc.)

THERMAL ENERGY CONVERSION AND UTILIZATION

BOILERS

Steam generators are designed to produce steam for process requirements, for process needs along with electric power generation, or solely for electric power generation. In each case, the goal is the most efficient and reliable boiler design for the least cost. Many factors influence the selection of the type of steam generator and its design, and some of these will be treated later in discussions of industrial and utility boilers.

Figure 24-34 shows the chief operating characteristics of a range of boilers, from small-scale heating systems to large-scale utility boilers.

In the industrial market, boilers have been designed to burn a wide range of fuels and operate at pressures up to 12.4 MPa (1800 psia) and steaming rates extending to 455,000 kg/h (1,000,000 lb/h). High-capacity shop-assembled boilers (package boilers) range in capacity from 4545 kg/h (10,000 lb/h) to about 270,000 kg/h (600,000 lb/h).

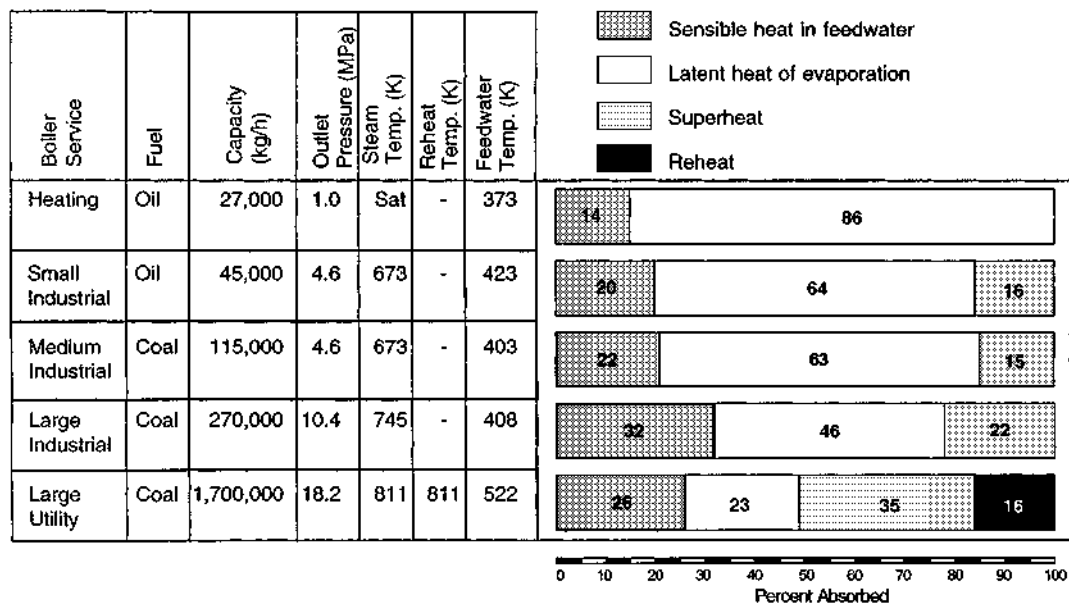


FIG. 24-34 Heat absorption distribution for various types of boilers. (Adapted from Singer, Combustion—Fossil Power, 4th ed., Combustion Engineering, Inc., Windsor, Conn., 1991.)

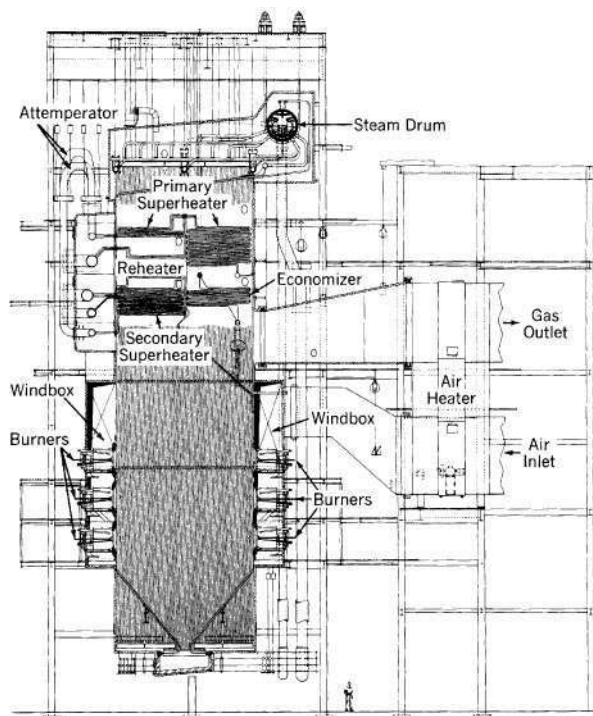


FIG. 24-35 Shop-assembled radiant boiler for natural gas or oil. (Babcock & Wilcox Co.)

These units are designed for operation at pressures up to 11.4 MPa (1650 psia) and 783 K (950°F). Figure 24-35 shows a gas- or liquid-fuel-fired unit. While most shop-assembled boilers are gas- or oil-fired, designs are available to burn pulverized coal. A field-erected coal-fired industrial boiler is shown in Fig. 24-36.

Boilers designed for service in electric power utility systems operate at both subcritical-pressure (pressures below 22.1 MPa [3205 psia]) and supercritical-pressure steam conditions. Subcritical-pressure boilers range in design pressures up to about 18.6 MPa (2700 psia) and in steaming capacities up to about 2955 Mg/h (6,500,000 lb/h). Supercritical-pressure boilers have been designed to operate at pressures up to 34.5 MPa (5000 psia). The 24.1 MPa (3500 psia) cycle has been firmly established in the utility industry, and boilers with steaming capacities up to 4227 Mg/h (9,300,000 lb/h) and superheat and reheat temperatures of 814 K (1005°F) are in service. The furnace of a large coal-fired steam generator absorbs half of the heat released, so that the gas temperature leaving the furnace is about 1376 K (2000°F).

Boiler Design Issues Boiler design involves the interaction of many variables: water-steam circulation, fuel characteristics, firing systems and heat input, and heat transfer. The furnace enclosure is one of the most critical components of a steam generator and must be conservatively designed to assure high boiler availability. The furnace configuration and its size are determined by combustion requirements, fuel characteristics, emission standards for gaseous effluents and particulate matter, and the need to provide a uniform gas flow and temperature entering the convection zone to minimize ash deposits and excessive superheater metal temperatures. Discussion of some of these factors follows.

Circulation and Heat Transfer Circulation, as applied to a steam generator, is the movement of water or steam or a mixture of both through the heated tubes. The circulation objective is to absorb heat from the tube metal at a rate that assures sufficient cooling of the furnace-wall tubes during all operating conditions, with an adequate margin of reserve for transient upsets. Adequate circulation prevents excessive metal temperatures or temperature differentials that would cause failures due to overstressing, overheating, or corrosion.

The rate of heat transfer from the tubes to the fluid depends primarily on turbulence and the magnitude of the heat flux itself. Turbulence is a

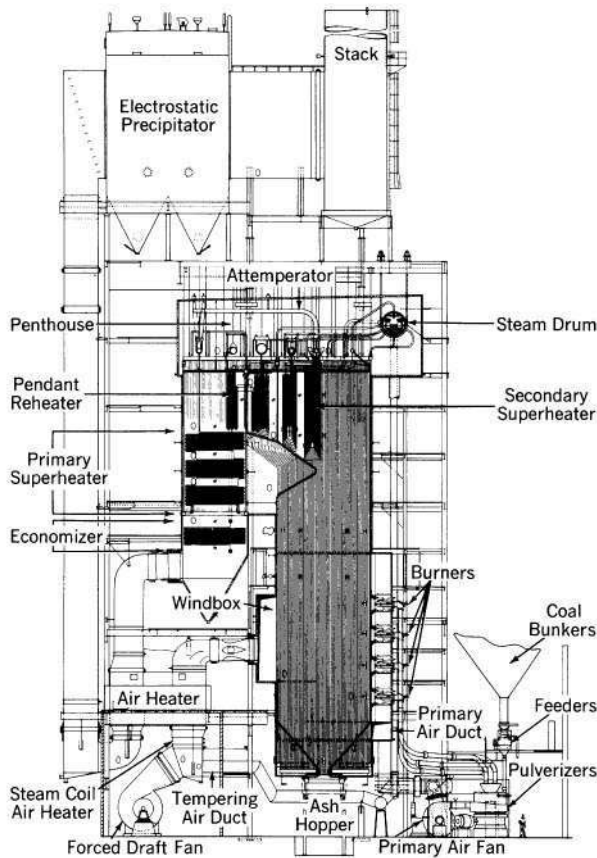


FIG. 24-36 Field-erected radiant boiler for pulverized coal. (Babcock & Wilcox Co.)

function of mass velocity of the fluid and tube roughness. Turbulence has been achieved by designing for high mass velocities, which ensure that nucleate boiling takes place at the inside surface of the tube. If sufficient turbulence is not provided, *departure from nucleate boiling* (DNB) occurs. DNB is the production of a film of steam on the tube surface that impedes heat transfer and results in tube overheating and possible failure.

Satisfactory performance is obtained with tubes having helical ribs on the inside surface, which generate a swirling flow. The resulting centrifugal action forces the water droplets toward the inner tube surface and prevents the formation of a steam film. The internally rifled tube maintains nucleate boiling at much higher steam temperature and pressure and with much lower mass velocities than those needed in smooth tubes. In modern practice, the most important criterion in drum boilers is the prevention of conditions that lead to DNB.

Utility Steam Generators

Steam-Generator Circulation System Circulation systems for utility application are generally classified as natural circulation and forced or pump-assisted circulation in drum-type boilers, and as once-through flow in subcritical- and supercritical-pressure boilers.

Natural circulation in a boiler circulation loop depends only on the difference between the mean density of the fluid (water) in the downcomers and the mean density of the fluid (steam-water mixture) in the heated furnace walls. The actual circulating head is the difference between the total gravity head in the downcomer and the integrated gravity heads in the upcoming legs of the loop containing the heated tubes. The circulating head must balance the

sum of the losses due to friction, shock, and acceleration throughout the loop.

In a once-through system, the feedwater entering the unit absorbs heat until it is completely converted to steam. The total mass flow through the waterwall tubes equals the feedwater flow and, during normal operation, the total steam flow. As only steam leaves the boiler, there is no need for a steam drum.

Fuel Characteristics Fuel choice has a major impact on boiler design and sizing. Because of the heat transfer resistance offered by ash deposits in the furnace chamber in a coal-fired boiler, the mean absorbed heat flux is lower than in gas- or oil-fired boilers, so a greater surface area must be provided.

In addition, coal characteristics have a major impact on the design and operation of a coal-fired boiler. Coals having a low volatile-matter content usually require higher ignition temperatures and those with less than 12 to 14 percent volatile matter may require supplementary fuel to stabilize ignition. Generally, western U.S. coals are more reactive than others and, consequently, easier to ignite, but because of high moisture content they require higher air temperatures for the mills for drying the coal to achieve proper pulverization. Extremely high-ash coal also may present problems in ignition and stabilization. The ash constituents and the quantity of ash will have a decided influence on sizing the furnace. Accordingly, a thorough review of coal characteristics is needed to establish the effect on the design and operation of a boiler.

Superheaters and Reheaters A superheater raises the temperature of the steam generated above the saturation level. An important function is to minimize moisture in the last stages of a turbine to avoid blade erosion. With continued increase of evaporation temperatures and pressures, however, a point is reached at which the available superheat temperature is insufficient to prevent excessive moisture from forming in the low-pressure turbine stages. This condition is resolved by removing the vapor for reheat at constant pressure in the boiler and returning it to the turbine for continued expansion to condenser pressure. The thermodynamic cycle using this modification of the Rankine cycle is called the *reheat cycle*.

Economizers Economizers improve boiler efficiency by extracting heat from the discharged flue gases and transferring it to feedwater, which enters the steam generator at a temperature appreciably lower than the saturation-steam temperature.

Industrial Boilers Industrial boilers are steam generators that provide power, steam, or both to an industrial plant, in contrast to a utility boiler in a steam power plant. A common configuration is a stationary water-tube boiler in which some of the steam is generated in a convection-section tube bank (also termed a *boiler bank*). In the original industrial boilers, in fact, almost all of the boiling occurred in that section, but now many industrial steam generators of 180,000 kg/h (397,000 lb/h) and greater capacity are radiant boilers. The boiler steam pressure and temperature and feedwater temperature determine the fraction of total heat absorbed in the boiler bank. For a typical coal-fired boiler producing about 90,000 kg/h (198,000 lb/h):

% total steam in boiler bank	Boiler pressure		Steam temperature		Feedwater temperature	
	MPa	psia	K	°F	K	°F
45	1.4	200	460	369	389	241
30	4.1	600	672	750	389	241
16	10.3	1500	783	950	450	351
10	12.4	1800	811	1000	450	351

The thicker plate for operation at higher pressures increases the cost of the boiler. As a result, it is normally not economical to use a boiler bank for heat absorption at pressures above 10.7 MPa (1550 psia).

Industrial boilers are employed over a wide range of applications, from large power-generating units with sophisticated control systems, which maximize efficiency, to small low-pressure units for space or process heating, which emphasize simplicity and low capital cost. Although their usual primary function is to provide energy in the form of steam, in some applications steam generation is incidental to a process objective, e.g., a chemical recovery unit in the paper industry,

TABLE 24-14 Solid-Waste Fuels Burned in Industrial Boilers

Waste	HHV, kJ/kg*
Bagasse	8374–11,630
Furfural residue	11,630–13,956
Bark	9304–11,630
General wood wastes	10,467–18,608
Coffee grounds	11,397–15,119
Nut hulls	16,282–18,608
Rich hulls	12,095–15,119
Corncocks	18,608–19,306
Rubber scrap	26,749–45,822
Leather	27,912–45,822
Cork scrap	27,912–30,238
Paraffin	39,077
Cellophane plastics	27,912
Polyvinyl chloride	40,705
Vinyl scrap	40,705
Sludges	4652–27,912
Paper wastes	13,695–18,608

*To convert kilojoules per kilogram to British thermal units per pound, multiply by 4.299×10^{-1} .

a carbon monoxide boiler in an oil refinery, or a gas-cooling waste-heat boiler in an open-hearth furnace. It is not unusual for an industrial boiler to serve a multiplicity of functions. For example, in a paper-pulp mill, the chemical-recovery boiler is used to convert black liquor into useful chemicals and to generate process steam. At the same plant, a bark-burning unit recovers heat from otherwise wasted material and also generates power.

Industrial boilers burn oil, gas, coal, and a wide range of product and/or waste fuels, some of which are shown in Tables 24-4 and 24-14. Natural gas has become the principal fuel of choice, accounting for approximately 80 percent of all the energy fired in industrial boilers across a range of manufacturing industries (Table 24-15). Coal is the second most prevalent fuel, accounting for about 15 percent of the energy fired. Waste fuels, however, are increasing in importance.

An excellent brief exposition of industrial boilers is presented as Chapter 8 of *Combustion—Fossil Power*, Singer (ed.), 4th ed., Combustion Engineering, Windsor, Conn., 1991.

Design Criteria Industrial-boiler designs are tailored to the fuels and firing systems involved. Some of the more important design criteria include:

- Furnace heat-release rates, both W/m^3 and W/m^2 of effective projected radiant surface ($\text{Btu}/[\text{h}\cdot\text{ft}^3]$ and $\text{Btu}/[\text{h}\cdot\text{ft}^2]$)
- Heat release on grates
- Flue-gas velocities through tube banks
- Tube spacings

TABLE 24-16 Typical Design Parameters for Industrial Boilers

Furnace		Heat-release rate, W/m^2 of EPRS†	
Natural gas-fired		630,800	
Oil-fired		551,900–630,800	
Coal: pulverized coal		220,780–378,480	
Spreader stoker		252,320–410,020	
Stoker, coal-fired		Grate heat-release rate, W/m^2	
Continuous-discharge spreader		2,050,000–2,207,800	
Dump-grade spreader		1,419,300–1,734,700	
Overfeed traveling grate		1,261,000–1,734,700	
Flue-gas velocity: type		Baffled	
Fuel-fired	Single-pass Boiler, m/s	Boiler, m/s	Economizer, m/s
Gas or distillate oil	30.5	30.5	30.5
Residual oil	30.5	22.9	30.5
Coal (not lignite)			
Low-ash	19.8–21.3	15.2	15.2–18.3
High-ash	15.2	NA†	12.2–15.2

*To convert watts per square meter to British thermal units per hour-foot, multiply by 0.317.

†Effective projected radiant surface.

‡Not available.

Table 24-16 gives typical values or ranges of these criteria for gas, oil, and coal. The furnace release rates are important, for they establish maximum local absorption rates within safe limits. They also have a bearing on completeness of combustion and therefore on efficiency and particulate emissions. Limiting heat release on grates (in stoker firing) will minimize carbon loss, control smoke, and avoid excessive fly ash.

Limits on flue-gas velocities for gas- or oil-fired industrial boilers are usually determined by the need to limit draft loss. For coal firing, design gas velocities are established to minimize fouling and plugging of tube banks in high-temperature zones and erosion in low-temperature zones.

Convection tube spacing is important when the fuel is residual oil or coal, especially coal with low ash-fusion or high ash-fouling tendencies. The amount of the ash and, even more important, the characteristics of the ash must be specified for design.

Natural-circulation and convection boiler banks are the basic design features on which a line of standard industrial boilers has been developed to accommodate the diverse steam, water, and fuel requirements of the industrial market.

Figure 24-37 shows the amount of energy available for power by using a fire-tube boiler, an industrial boiler, and subcritical- and

TABLE 24-15 Fuel Consumption in Boilers in Various Industries

Industry	Annual energy consumption (PJ/a)				
	Total	Residual fuel oil	Distillate fuel oil	Natural gas	Coal
Chemicals	539	—	—	451	88
Food	323	8	6	237	72
Paper	212	29	2	143	38
Petroleum and coal products	169	6	2	159	2
Transportation equipment	56	—	1	48	7
Plastics and rubber products	54	5	1	48	—
Primary metals	40	—	—	40	—
Beverage and tobacco products	34	—	—	29	5
Computer and electronic products	29	1	1	27	—
Wood products	18	1	1	15	1
Textile mills	15	—	1	—	14
Total	1,489	50	15	1,197	227

SOURCE: 2002 *Manufacturing Energy Consumption Survey*, Energy Information Administration, U.S. Dept. of Energy.

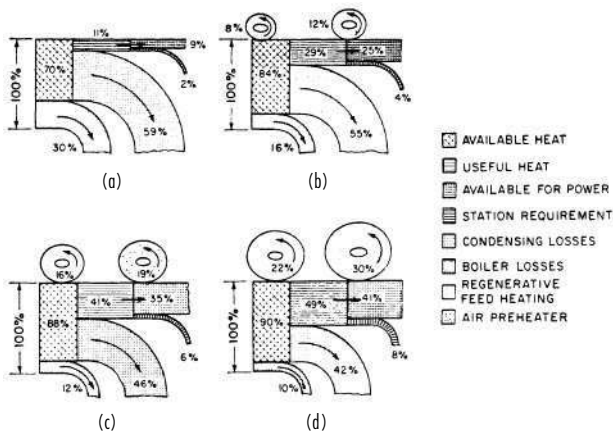


FIG. 24-37 Sankey diagrams for various types of boilers: (a) fire-tube boiler; (b) industrial boiler; (c) subcritical-pressure boiler; (d) supercritical-pressure boiler.

supercritical-pressure boilers. Condensing losses decrease substantially, and regeneration of air and feedwater becomes increasingly important in the most advanced central-station boilers.

The boiler designer must proportion heat-absorbing and heat-recovery surfaces in a way to make the best use of heat released by the fuel. Water walls, superheaters, and reheaters are exposed to convection and radiant heat, whereas convection heat transfer predominates in air preheaters and economizers. The relative amounts of these surfaces vary with the size and operating conditions of the boiler.

Package Boilers In a fire-tube boiler, the hot combustion products flow through tubes immersed in the boiler water, transferring heat to it. In a water-tube boiler, combustion heat is transferred to water flowing through tubes which line the furnace walls and boiler passages. The greater safety of water-tube boilers has long been recognized, and they have generally superseded fire-tube configurations except for small package boiler designs. Fire-tube package boilers range from a few hundred to 18,200 kg/h (40,000 lb/h) steaming capacity. A fire-tube boiler is illustrated in Figs. 24-38 and 24-39. Water-tube package boilers range from a few hundred to 270,000 kg/h (600,000 lb/h) steaming capacity. A water-tube package boiler is illustrated in Fig. 24-40. The majority of water-tube package boilers use natural circulation and are designed for pressurized firing. The most

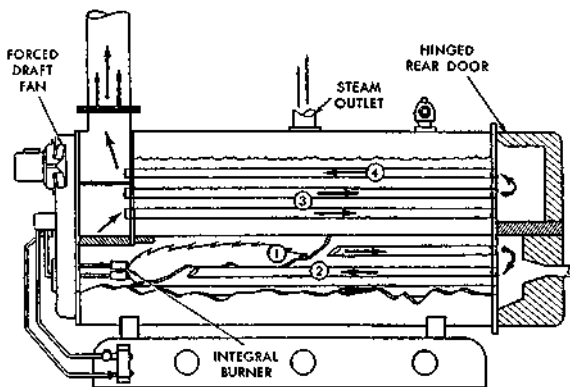


FIG. 24-38 A four-pass packaged fire-tube boiler. Circled numbers indicate passes. (From Cleaver Brooks, Inc. Reproduced from Gas Engineer's Handbook, Industrial Press, New York, 1965, with permission.)

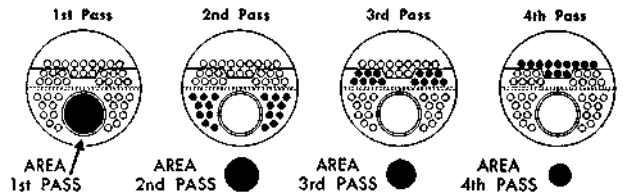


FIG. 24-39 Location and relative size of each of four passes of the flue gas through a fire-tube boiler. (From Cleaver Brooks, Inc. Reproduced from Gas Engineer's Handbook, Industrial Press, New York, 1965, with permission.)

significant advantage of shop-assembled or package boilers is the cost benefit associated with use of standard designs and parts.

Package boilers can be shipped complete with fuel-burning equipment, controls, and boiler trim. It may be necessary to ship the larger units in sections, however, and a shop-assembled boiler with a capacity greater than about 109,000 kg/h (240,000 lb/h) is deliverable only by barge. (For a more detailed discussion of shop-assembled boilers, see Singer, 1991, pp. 8.36–8.42.)

Fluidized-Bed Boilers Although the furnace of a fluid-bed boiler has a unique design, the system as a whole consists mainly of standard equipment items adapted to suit process requirements. The systems for feedstock preparation and feeding (biomass, coal, sorbent, and sand), ash removal, and ash disposal are very similar to those found in PC boiler plants, the main difference being that the top size of the material being handled is greater. The water-wall boiler enclosure and convection pass tubing are also similar to the designs found in PC boilers. The fluidized-bed plant includes particulate removal equipment such as cyclones, baghouses, and ESPs, with designs similar to those found in other solids-handling process plants.

Bubbling FBCs A simplified schematic of a bubbling FBC plant is presented in Fig. 24-41. This design is used primarily for burning biomass, although coal may be cofired to maintain heat release rates when feeding high-moisture feedstocks such as sludges.

The biomass is fed overbed through multiple feed chutes using air jets to help distribute the fuel over the surface of the bed. Variable-speed screw conveyors are usually used to meter the fuel feed rate and control steam output. Feedstocks such as bark and waste wood are chipped to a top size of 25 mm (1 in) to ensure complete combustion. The bed usually consists of sand around 1 m (3 ft) deep. This serves to retain the fuel in the furnace, extending its in-furnace residence time and increasing combustion efficiency. It also provides a heat sink to help maintain bed temperature during periods of fluctuating fuel moisture content.

Biomass fuels, especially forestry products and waste wood, can contain oversized material such as rocks and other debris. If this material is allowed to accumulate, the quality of fluidization deteriorates, resulting in poor fuel mixing and reduced combustion efficiency. To avoid this mal-operation, units typically employ an open-bottom floor design that allows easy solids removal while still ensuring even distribution of the fluidizing air. The fluidizing air (primary combustion air) is fed into large bore pipes extending across the width of the furnace and enters the boiler through a series of nozzles set in the pipes. The pipes are spaced to allow material to drain from all areas of the bed.

The bed temperature control strategies discussed earlier in this section result in the bed operating substoichiometrically. This helps lower the formation of NO_x but increases the amount of CO formed and the amount of unburned fuel leaving the bed. To complete the combustion process, secondary air is introduced through multiple ports above the bed with the upper furnace height designed to achieve 2- to 3-s flue gas residence time. The velocity of the secondary air should be sufficient to penetrate across the furnace and facilitate thorough mixing with the flue gas leaving the surface of the bed. Although the unburned carbon leaving the furnace is low, the CO is still relatively high at 43 to 128 mg/MJ (0.10 to 0.25 lb/MBtu), which may be a problem in nonattainment areas.

Ammonia injected into the upper furnace is used to promote SNCR reactions to reduce NO_x emissions. The optimum temperature for

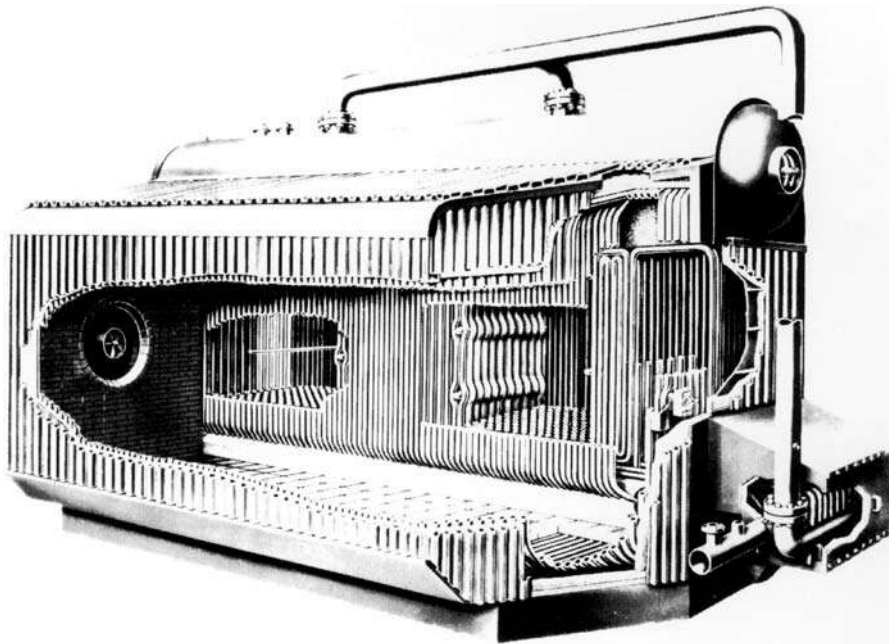


FIG. 24-40 A D-type shop-assembled water-tube boiler. (Combustion Engineering, Inc.)

these reactions is around 830°C (1520°F), but the upper furnace temperature varies depending upon the extent of secondary combustion air usage. To ensure that the ammonia enters into a region at the optimum temperature, multiple levels of injection ports are provided.

Circulating FBCs The circulating FBC is the more widely deployed than the bubbling-bed design and at far larger sizes. A simplified schematic for a design with an external heat exchanger is presented in Fig. 24-42. As of 2005 there were over 20 units in the range of 200 to 320 MWe worldwide, and a similar number were in the planning stages.

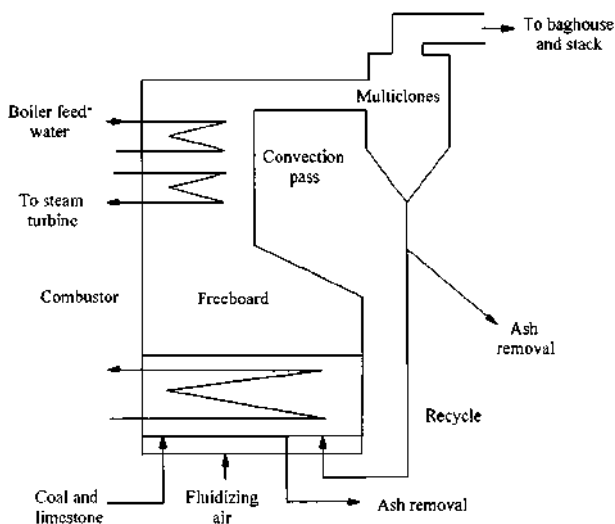


FIG. 24-41 Simplified flow diagram for bubbling AFBC (with underbed feed system).

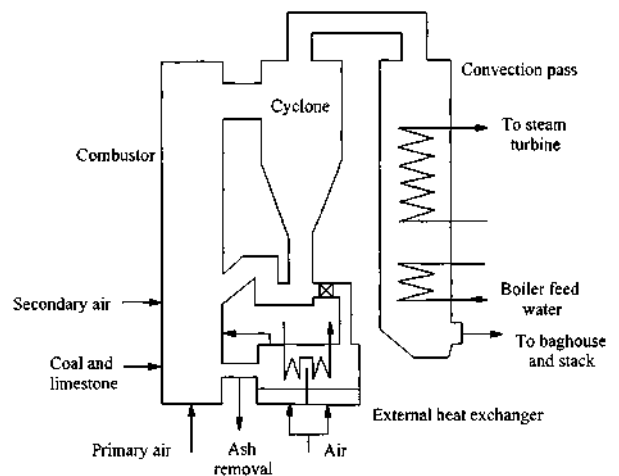


FIG. 24-42 Simplified flow diagram for circulating AFBC (with external heat exchanger).

The most widely used fuels are low-grade coals with high ash and moisture contents, making them unsuitable for use in PC plants. The coal is crushed to a top size of 12 mm (1/2 in), without drying, and fed by gravity into the lower refractory-lined portion of the boiler. The feed points are close to the pressure balance point, and the low backpressure greatly reduces the sealing requirements. In some designs the coal is also introduced into the cyclone ash return lines. The larger units average a feed point for every 23 m^2 (250 ft^2) of freeboard cross section. The highly turbulent bed facilitates good mixing of the coal across the combustor. The sorbent is prepared to a top size of 1 mm (0.04 in) and dried so that it can be pneumatically fed to the combustor also using one feed point for every 23 m^2 (250 ft^2) of freeboard cross section. Load control

is achieved primarily by reducing the coal feed rate, with a corresponding reduction in airflow, to reduce the heat release rate. As the heat-transfer area remains unchanged, the furnace temperature falls.

Almost all the particulate matter leaving the boiler is collected by cyclones and recycled to the base of the unit. The number of cyclones varies in different design concepts, but for the larger units one cyclone is provided for 100 to 130 MWe of generating capacity, providing one recycle point for 210 to 290 m² (720 to 950 ft²) of freeboard cross section. The collected particulate matter is returned against a backpressure of 0.02 MPa (3 psi), through a J-valve. The recycle ratio can be as high as 40:1, corresponding to a relatively long mean particle residence time and accounting for the high performance of circulating units. For bituminous coal combustion, efficiency is close to 99 percent, but like the biomass-fired bubbling beds, CO levels are relatively high at around 43 mg/MJ (0.1 lb/MBtu).

PROCESS HEATING EQUIPMENT

Many major energy-intensive industries depend on direct-fired or indirect-fired equipment for drying, heating, calcining, melting, or chemical processing. This subsection discusses both direct- and indirect-fired equipment, with the greater emphasis on indirect firing for the process industries.

Direct-Fired Equipment Direct-fired combustion equipment transfers heat by bringing the flame and/or the products of combustion into direct contact with the process stream. Common examples are rotary kilns, open-hearth furnaces, and submerged-combustion evaporators. Table 24-17 gives the average energy consumption rates for various industries and processes that use direct heat. Section 12 of this handbook describes and illustrates rotary dryers, rotary kilns, and hearth furnaces. Forging, heat treating, and metal milling furnaces are discussed by Mawhinney (*Marks' Standard Handbook for Mechanical Engineers*, 9th ed., McGraw-Hill, New York, 1987, pp. 7.47–7.52). Other direct-fired furnaces are described later in this section.

Indirect-Fired Equipment (Fired Heaters) Indirect-fired combustion equipment (fired heaters) transfers heat across either a metallic or refractory wall separating the flame and products of combustion from the process stream. Examples are heat exchangers (discussed in Sec. 11), steam boilers, fired heaters, muffle furnaces, and melting pots. Steam boilers have been treated earlier in this section, and a subsequent subsection on industrial furnaces will include muffle furnaces.

TABLE 24-17 Average Energy Consumption for Various Industries Using Direct Heat

Industry	Product/process	Energy consumption per unit of product	
		GJ/Mg	10 ⁶ Btu/US ton
Paper	Kraft process	20.9	18.0
	Integrated plant/paper*	34.2	29.5
	Integrated plant/paperboard*	18.8	16.2
Glass	Flat glass	17.3	14.9
	Container glass	18.1	15.6
	Pressed/blown	31.6	27.2
Clay/ceramics	Portland cement	4.6	4.0
	Lime	5.5	4.7
	Mineral wool	42.7	36.8
Steel	Blast furnace and steel mills	20.7	17.8
Nonferrous metals	Primary copper	34.2	29.5
	Secondary copper	4.6	4.0
	Primary lead	25.1	21.6
	Secondary lead	0.8	0.7
	Primary zinc	69.6	60.0
	Secondary zinc	5.0	4.3
	Primary aluminum	78.9	68.0
	Secondary aluminum	5.2	4.5

SOURCE: *Manufacturing Consumption of Energy*, Energy Information Administration, U.S. Dept. of Energy, 1991.

*Mixture of direct and indirect firing.

Fired heaters differ from other indirect-fired processing equipment in that the process stream is heated by passage through a coil or tubebank enclosed in a furnace. Fired heaters are classified by function and by coil design.

Function Berman [*Chem. Eng.* **85**(14):98–104, June 19, 1978] classifies fired heaters into the following six functional categories, providing descriptions that are abstracted here.

Column reboilers heat and partially vaporize a recirculating stream from a fractionating column. The outlet temperature of a reboiler stream is typically 477 to 546 K (400 to 550°F).

Fractionator-feed preheaters partially vaporize charge stock from an upstream unfired preheater en route to a fractionating column. A typical refinery application: a crude feed to an atmospheric column enters the fired heater as a liquid at 505 K (450°F) and leaves at 644 K (700°F), having become 60 percent vaporized.

Reactor-feed-stream preheaters heat the reactant stream(s) for a high-temperature chemical reaction. The stream may be single-phase/single-component [example: steam being superheated from 644 to 1089 K (700 to 1500°F) for styrene-manufacture reactors]; single-phase/multicomponent [example: preheating the feed to a catalytic reformer, a mixture of hydrocarbon vapors and recycle hydrogen, from 700 to 811 K (800 to 1000°F) under pressure as high as 4.1 MPa (600 psia)]; or multiphase/multicomponent [example: a mixture of hydrogen gas and liquid hydrocarbon heated from 644 to 727 K (700 to 850°F) at about 20 MPa (3000 psia) before it enters a hydrocracker].

Heat-transfer-fluid heaters maintain the temperature of a circulating liquid heating medium (e.g., a paraffinic hydrocarbon mixture, a Dowtherm, or a molten salt) at a level that may exceed 673 K (750°F).

Viscous-liquid heaters lower the viscosity of very heavy oils to pumpable levels.

Fired reactors contain tubes or coils in which an endothermic reaction within a stream of reactants occurs. Examples include steam/hydrocarbon reformers, catalyst-filled tubes in a combustion chamber; pyrolyzers, coils in which alkanes (from ethane to gas oil) are cracked to olefins; in both types of reactor the temperature is maintained up to 1172 K (1650°F).

Coil Design Indirect-fired equipment is conventionally classified by tube orientation: vertical and horizontal. Although there are many variations of each of these two principal configurations, they all are embraced within seven major types, as follows.

A *simple vertical cylindrical* heater has vertical tubes arrayed along the walls of a combustion chamber fired vertically from the floor. This type of heater does not include a convection section and is inexpensive. It has a small footprint but low efficiency, and it is usually selected for small-duty applications [0.5 to 21 GJ/h (0.5 to 20 10⁶ Btu/h)].

Vertical cylindrical; cross-tube convection heaters are similar to the preceding type except for a horizontal convective tube bank above the combustion chamber. The design is economical with a high efficiency and is usually selected for higher-duty applications: 11 to 210 GJ/h (10 to 200 10⁶ Btu/h).

The *arbor (wicket)* heater is a substantially vertical design in which the radiant tubes are inverted U's connecting the inlet and outlet terminal manifolds in parallel. An overhead crossflow convection bank is usually included. This type of design is good for heating large gas flows with low pressure drop. Typical duties are 53 to 106 GJ/h (50 to 100 10⁶ Btu/h).

In the *vertical-tube single-row double-fired* heater, a single row of vertical tubes is arrayed along the center plane of the radiant section that is fired from both sides. Usually this type of heater has an overhead horizontal convection bank. Although it is the most expensive of the fired heater designs, it provides the most uniform heat transfer to the tubes. Duties are 21 to 132 GJ/h (20 to 125 10⁶ Btu/h) per cell (twin-cell designs are not unusual).

Horizontal-tube cabin heaters position the tubes of the radiant-section-coil horizontally along the walls and the slanting roof for the length of the cabin-shaped enclosure. The convection tube bank is placed horizontally above the combustion chamber. It may be fired from the floor, the side walls, or the end walls. As in the case of its vertical cylindrical counterpart, its economical design and high efficiency make it the most popular horizontal-tube heater. Duties are 11 to 105 GJ/h (10 to 100 10⁶ Btu).

In the *horizontal-tube box heater with side-mounted convection tube bank*, the radiant-section tubes run horizontally along the walls and the flat roof of the box-shaped heater, but the convection section is placed in a box of its own beside the radiant section. Firing is horizontal from the end walls. The design of this heater results in a relatively expensive unit justified mainly by its ability to burn low-grade high-ash fuel oil. Duties are 53 to 210 GJ/h (50 to 200 10^6 Btu/h).

Vertical cylindrical helical coil heaters are hybrid designs that are classified as vertical heaters, but their in-tube characteristics are like those of horizontal heaters. There is no convection section. In addition to the advantages of simple vertical cylindrical heaters, the helical coil heaters are easy to drain. They are limited to small-duty applications: 5 to 21 GJ/h (5 to 20 10^6 Btu/h).

Schematic elevation sections of a vertical cylindrical, cross-tube convection heater; a horizontal-tube cabin heater; and a vertical cylindrical, helical-coil heater are shown in Fig. 24-43. The seven basic designs and some variations of them are pictured and described in the reference cited above and by R. K. Johnson [*Combustion* 50(5): 10-16, November 1978].

The design of both radiant and convection sections of fired heaters, along with some equipment descriptions and operating suggestions, are discussed by Berman in *Encyclopedia of Chemical Processing and Design* [McKetta (ed.), vol. 22, Marcel Dekker, 1985, pp. 31-69]. He also treats construction materials, mechanical features, and operating points in three other *Chemical Engineering* articles [all in vol. 85 (1978): no. 17, July 31, pp. 87-96; no. 18, August 14, pp. 129-140; and no. 20, Sept. 11, pp. 165-169].

INDUSTRIAL FURNACES

Industrial furnaces serve the manufacturing sector and can be divided into two groups. Boiler furnaces, which are the larger group and are used solely to generate steam, were discussed earlier in the subsection

on industrial boilers. Furnaces of the other group are classified as follows: by (1) source of heat (fuel combustion or electricity), (2) function (heating without change of phase or with melting), (3) process cycle (batch or continuous), (4) mode of heat application (direct or indirect), and (5) atmosphere in furnace (air, protective, or reactive, including vacuum). Each will be discussed briefly.

Source of Heat Industrial furnaces are either fuel-fired or electric, and the first decision that a prospective furnace user must make is between these two. Although electric furnaces are uniquely suited to a few applications in the chemical industry (manufacture of silicon carbide, calcium carbide, and graphite, for example), their principal use is in the metallurgical and metal-treatment industries. In most cases the choice between electric and fuel-fired is economic or custom-dictated, because most tasks that can be done in one can be done equally well in the other. Except for an occasional passing reference, electric furnaces will not be considered further here. The interested reader will find useful reviews of them in *Kirk-Othmer Encyclopedia of Chemical Technology* (4th ed., vol. 12, articles by Cotchen, Sommer, and Walton, pp. 228-265, Wiley, New York, 1994) and in *Marks' Standard Handbook for Mechanical Engineers* (9th ed., article by Lewis, pp. 7.59-7.68, McGraw-Hill, New York, 1987).

Function and Process Cycle Industrial furnaces are enclosures in which process material is heated, dried, melted, and/or reacted. Melting is considered a special category because of the peculiar difficulties that may be associated with a solid feed, a hot liquid product, and a two-phase mixture in between; it is customary, therefore, to classify furnaces as heating or melting.

Melting Furnaces: The Glass Furnace Most melting furnaces, electric or fuel-fired, are found in the metals-processing industry, but a notable exception is the glass furnace. Like most melting furnaces, a glass furnace requires highly radiative flames to promote heat transfer to the feed charge and employs regenerators to conserve heat from the high-temperature process [greater than 1813 K (2300°F)].

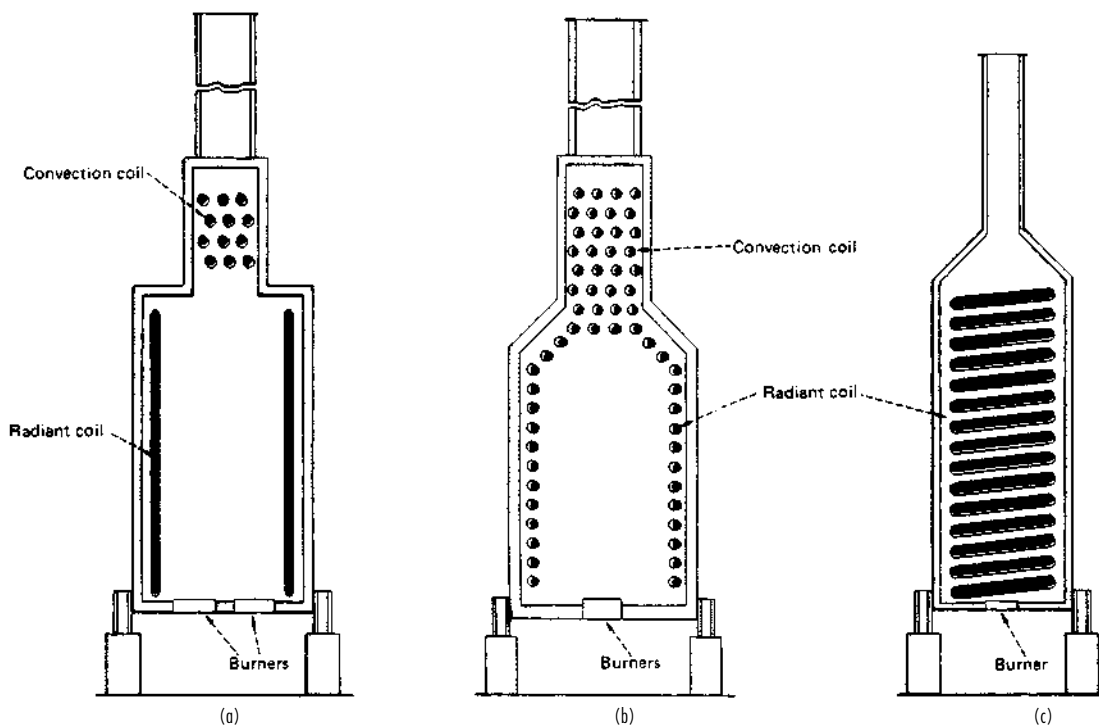


FIG. 24-43 Representative types of fired heaters: (a) vertical-tube cylindrical with cross-flow-convection section; (b) horizontal-tube cabin; (c) vertical cylindrical, helical coil. (From Berman, *Chem. Eng.* 85:98-104, June 19, 1978.)

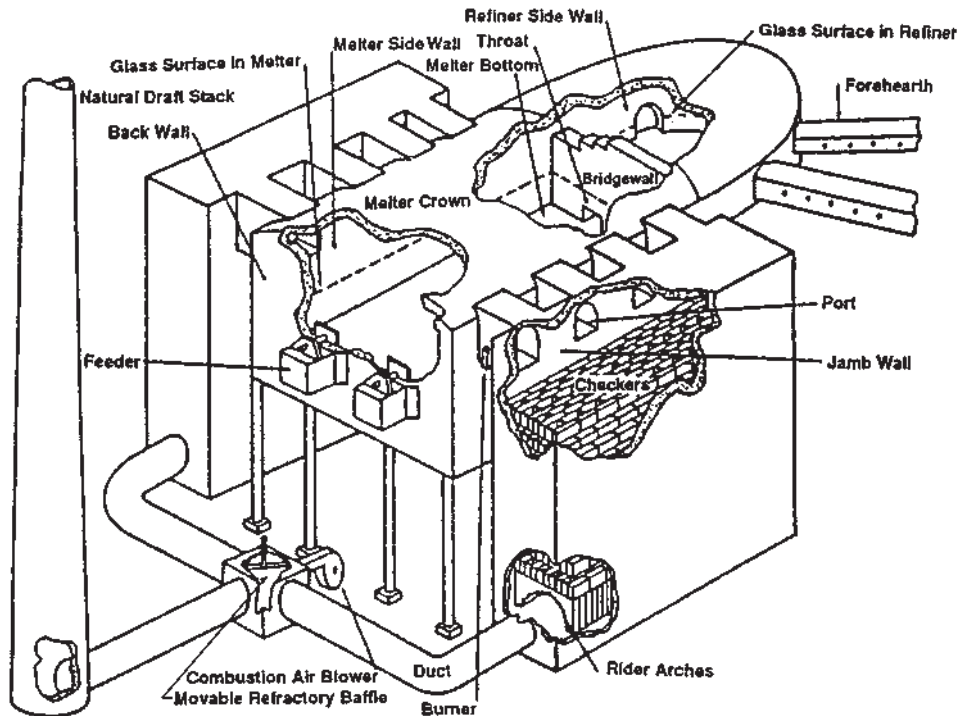


FIG. 24-44 Side-port continuous regenerative glass melting furnace.

A typical side-port continuous regenerative glass furnace is shown in Fig. 24-44. Side-port furnaces are used in the flat and container glass industries. The burners are mounted on both sides of the furnace and the sides fire alternately. Refractory-lined flues are used to recover the energy of the hot flue gas. The high temperature of the flue gas leaving the furnace heats a mass of refractory material called a *checker*. After the checker has reached the desired temperature, the gas flow is reversed and the firing switches to the other side of the furnace. The combustion air is then heated by the hot checker and can reach 1533 K (2300°F). The cycle of airflow from one checker to the other is reversed approximately every 15 to 30 min.

The glass melt is generally 1 to 2 m (3 to 6 ft) deep, the depth being limited by the need for proper heat transfer to the melt. Container glass furnaces are typically 6 to 9 m (20 to 30 ft) wide and 6 to 12 m (20 to 40 ft) long. Flat glass furnaces tend to be longer, typically over 30 m (100 ft), because of the need for complete reaction of the batch ingredients and improved quality (fewer bubbles). They typically have a melting capacity of 450 to 630 Mg/day (500 to 750 U.S. ton/d), compared to a maximum of 540 Mg/day (600 U.S. ton/d) for container and pressed/blown glass furnaces.

Though the stoichiometric chemical energy requirement for glass-making is only some 2.3 GJ/Mg (2×10^6 Btu/U.S. ton) of glass, the inherently low thermal efficiency of regenerative furnaces means that, in practice, at least 7 GJ/Mg (6×10^6 Btu/U.S. ton) is required. Of this total, some 40 percent goes to batch heating and the required heat of reactions, 30 percent is lost through the furnace structure, and 30 percent is lost through the stack. The smaller furnaces used in pressed/blown glass melting are less efficient, and energy consumption may be as high as 17.4 TJ/Mg (15×10^6 Btu/U.S. ton).

Industrial furnaces may be operated in batch or continuous mode.

Batch Furnaces This type of furnace is employed mainly for the heat treatment of metals and for the drying and calcination of ceramic articles. In the chemical process industry, batch furnaces may be used for the same purposes as batch-tray and truck dryers when the drying or process temperature exceeds 600 K (620°F). They are employed

also for small-batch calcinations, thermal decompositions, and other chemical reactions which, on a larger scale, are performed in rotary kilns, hearth furnaces, and shaft furnaces.

Continuous Furnaces These furnaces may be used for the same general purposes as are the batch type, but usually not on small scale. The process material may be carried through the furnace by a moving conveyor (chain, belt, roller), or it may be pushed through on idle rollers, the motion being sustained by an external pusher operating on successively entering cars or trays, each pushing the one ahead along the entire length of the furnace and through the exit flame curtains or doors.

Furnace Atmosphere and Mode of Heating

Direct Heating Industrial furnaces may be directly or indirectly heated, and they may be filled with air or a protective atmosphere, or under a vacuum. Direct heating is accomplished by the hot combustion gases being inside the furnace and therefore in direct contact with the process material. Thus, the material is heated by radiation and convection from the hot gas and by reradiation from the heated refractory walls of the chamber. Three styles of direct firing are illustrated in Fig. 24-45. *Simple direct firing* is used increasingly because of its simplicity and because of improved burners. The *overhead* design allows the roof burners to be so placed as to provide optimum temperature distribution in the chamber. *Underfiring* offers the advantage of the charge's being protected from the flame. The maximum temperature in these direct-heated furnaces is limited to about 1255 K (1800°F) to avoid prohibitively shortened life of the refractories in the furnace.

Indirect Heating If the process material cannot tolerate exposure to the combustion gas or if a vacuum or an atmosphere other than air is needed in the furnace chamber, indirect firing must be employed. This is accomplished in a muffle^a furnace or a radiant-tube furnace (tubes carrying the hot combustion gas run through the furnace).

^aA muffle is an impenetrable ceramic or metal barrier between the firing chamber and the interior of the furnace. It heats the process charge by radiation and furnace atmosphere convection.

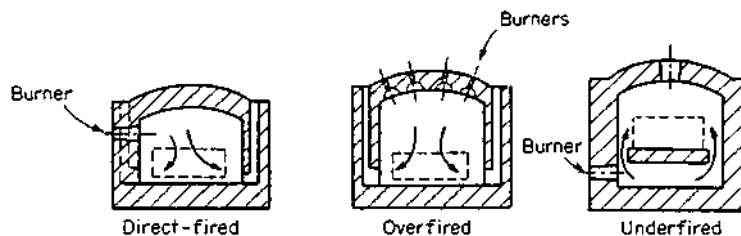


FIG. 24-45 Methods of firing direct-heated furnaces. (From Marks' Standard Handbook for Mechanical Engineers, 9th ed., McGraw-Hill, New York, 1987. Reproduced with permission.)

Atmosphere Protective atmosphere within the furnace chamber may be essential, especially in the heat treatment of metal parts. Mawhinney (in *Marks' Standard Handbook for Mechanical Engineers*, 9th ed., McGraw-Hill, New York, 1987, p. 752) lists pure hydrogen, dissociated ammonia (a hydrogen/nitrogen mixture), and six other protective reducing gases with their compositions (mixtures of hydrogen, nitrogen, carbon monoxide, carbon dioxide, and sometimes methane) that are codified for and by the metals-treatment industry. In general, any other gas or vapor that is compatible with the temperature and the lining material of the furnace can be provided in an indirect-fired furnace, or the furnace can be evacuated.

COGENERATION

Cogeneration is an energy conversion process wherein heat from a fuel is simultaneously converted to useful thermal energy (e.g., process steam) and electric energy. The need for either form can be the primary incentive for cogeneration, but there must be opportunity for economic captive use or sale of the other. In a chemical plant the need for process and other heating steam is likely to be the primary; in a public utility plant, electricity is the usual primary product.

Thus, a cogeneration system is designed from one of two perspectives: it may be sized to meet the process heat and other steam needs of a plant or community of industrial and institutional users, so that the electric power is treated as a by-product which must be either used on site or sold; or it may be sized to meet electric power demand, and the rejected heat used to supply needs at or near the site. The latter approach is the likely one if a utility owns the system; the former if a chemical plant is the owner.

Industrial use of cogeneration leads to small, dispersed electric-power-generation installations—an alternative to complete reliance on large central power plants. Because of the relatively short distances over which thermal energy can be transported, process-heat generation is characteristically an on-site process, with or without cogeneration.

Cogeneration systems will not match the varying power and heat demands at all times for most applications. Thus, an industrial cogeneration system's output frequently must be supplemented by the separate on-site generation of heat or the purchase of utility-supplied

electric power. If the on-site electric power demand is relatively low, an alternative option is to match the cogeneration system to the heat load and contract for the sale of excess electricity to the local utility grid.

Fuel saving is the major incentive for cogeneration. Since all heat-engine-based electric power systems reject heat to the environment, that rejected heat can frequently be used to meet all or part of the local thermal energy needs. Using reject heat usually has no effect on the amount of primary fuel used, yet it leads to a saving of all or part of the fuel that would otherwise be used for the thermal-energy process. Heat engines also require a high-temperature thermal input, usually receiving the working fluid directly from a heating source; but in some situations they can obtain the input thermal energy as the rejected heat from a higher-temperature process. In the former case, the cogeneration process employs a heat-engine topping cycle; in the latter case, a bottoming cycle is used.

The choice of fuel for a cogeneration system is determined by the primary heat-engine cycle. Closed-cycle power systems which are externally fired—the steam turbine, the indirectly fired open-cycle gas turbine, and closed-cycle gas turbine systems—can use virtually any fuel that can be burned in a safe and environmentally acceptable manner: coal, municipal solid waste, biomass, and industrial wastes are burnable with closed power systems. Internal combustion engines, on the other hand, including open-cycle gas turbines, are restricted to fuels that have combustion characteristics compatible with the engine type and that yield combustion products clean enough to pass through the engine without damaging it. In addition to natural gas, butane, and the conventional petroleum-derived liquid fuels, refined liquid and gaseous fuels derived from shale, coal, or biomass are in this category. Direct-coal-fired internal combustion engines have been an experimental reality for decades but are not yet a practical reality technologically or economically.

There are at least three broad classes of application for topping-cycle cogeneration systems:

- Utilities or municipal power systems supplying electric power and low-grade heat (e.g., 422 K [300°F]) for local district heating systems
- Large residential, commercial, or institutional complexes requiring space heat, hot water, and electricity

TABLE 24-18 Cogeneration Characteristics for Heat Engines

Engine type	Size range, MWe/unit	Efficiency at design point	Part-load efficiency	Multifuel capability	Maximum temperature of recoverable heat, °F (°C)*	Recoverable heat, Btu/kWh†	Typical power-to-heat ratio
Steam turbine							
Extraction-condensing type	30–300	0.25–0.30	Fair	Excellent	200 (93)–600 (315)‡	11,000–35,000	0.1–0.3
Backpressure type	20–200	0.20–0.25	Fair	Excellent	200 (93)–600 (315)‡	17,000–70,000	0.05–0.2
Combustion gas turbines	10–100	0.25–0.30	Poor	Poor	1000 (538)–1200 (649)	3000–11,000	0.3–0.45
Indirectly fired gas turbines							
Open-cycle turbines	10–85	0.25–0.30	Poor	Good	700 (371)–900 (482)	3500–8500	0.4–1.0
Closed-cycle turbines	5–350	0.25–0.30	Excellent	Good	700 (371)–900 (482)	3500–8500	0.4–1.0
Diesel engines	0.05–25	0.35–0.40	Good	Fair to poor	500 (260)–700 (371)	4000–6000	0.6–0.85

*°C + 273 = K.

†1 Btu = 1055 J.

‡Saturated steam.

- Large industrial operations with on-site needs for electricity and heat in the form of process steam, direct heat, and/or space heat.

Typical Systems All cogeneration systems involve the operation of a heat engine for the production of mechanical work which, in nearly all cases, is used to drive an electric generator. The commonest heat-engine types appropriate for topping-cycle cogeneration systems are:

- Steam turbines (backpressure and extraction configurations)
- Open-cycle (combustion) gas turbines
- Indirectly fired gas turbines: open cycles and closed cycles
- Diesel engines

Each heat-engine type has unique characteristics, making it better suited for some cogeneration applications than for others. For example, engine types can be characterized by:

- Power-to-heat ratio at design point
- Efficiency at design point
- Capacity range
- Power-to-heat-ratio variability
- Off-design (part-load) efficiency
- Multifuel capability

The major heat-engine types are described in terms of these characteristics in Table 24-18.

ELECTROCHEMICAL ENERGY CONVERSION

Electricity has become as indispensable as heat to the functioning of industrialized society. The source of most of the electricity used is the energy of the fuels discussed earlier in this section: liberated by combustion as heat, it drives heat engines which, in turn, drive electrical generators.

In some instances, however, part of the chemical energy bound in relatively high-enthalpy compounds can be converted directly to electricity as these reactants are converted to products of lower enthalpy (galvanic action). A process in the opposite direction also is possible for some systems: an electric current can be absorbed as the increased chemical energy of the higher-enthalpy compounds (electrolytic action). The devices in which electrochemical energy conversion processes occur are called cells.

Galvanic cells in which stored chemicals can be reacted on demand to produce an electric current are termed *primary cells*. The discharging reaction is irreversible and the contents, once exhausted, must be replaced or the cell discarded. Examples are the dry cells that activate small appliances. In some galvanic cells (called *secondary cells*), however, the reaction is reversible: that is, application of an electrical potential across the electrodes in the opposite direction will restore the reactants to their high-enthalpy state. Examples are rechargeable batteries for household appliances, automobiles, and many industrial applications. Electrolytic cells are the reactors upon which the electrochemical process, electroplating, and electrowinning industries are based.

Detailed treatment of the types of cells discussed above is beyond the scope of this handbook. For information about electrolytic cells, interested readers are referred to Fuller, Newman, Grotheer, and King ("Electrochemical Processing," in *Kirk-Othmer Encyclopedia of Chemical Technology*, 4th ed., vol. 9, Wiley, New York, 1994, pp. 111–197) and for primary and secondary cells, to Crompton (*Battery Reference Book*, 2d ed., Butterworth-Heinemann, Oxford, U.K., 1995). Another type of cell, however, a galvanic cell to which the reactants of an exothermic reaction are fed continuously, in which they react to liberate part of their enthalpy as electrical energy, and from which the products of the reaction are discharged continuously, is called a *fuel cell*. Fuel cell systems for generating electricity in a variety of applications are being commercialized by a number of companies. The rest of this section is devoted to a discussion of fuel cell technology.

FUEL CELLS

GENERAL REFERENCES: W. Vielstich, A. Lamm, H. A. Gasteiger, eds., *Handbook of Fuel Cells*, John Wiley & Sons, 2003. *Fuel Cell Handbook* (Rev. 7), U.S. Department of Energy, DOE/NETL-2004/1206. Appleby and Foulkes, *Fuel Cell Handbook*, Kreger Publishing Co., Molabar, Fla., 1993. Kinoshita and Cairns, "Fuel Cells," in *Kirk-Othmer Encyclopedia of Chemical Technology*, 4th ed., vol. 11, Wiley, New York, 1994, p. 1098. Liebhafsky and Cairns, *Fuel Cells and Fuel Batteries*, Wiley, New York, 1968. Linden (ed.), *Handbook of Batteries and Fuel Cells*, McGraw-Hill, New York, 1984.

Background Energy conversion in fuel cells is direct and simple when compared to the sequence of chemical and mechanical steps in

heat engines. A fuel cell consists of an anode, an electrolyte, and a cathode. On the anode, the fuel is oxidized electrochemically to positively charged ions. On the cathode, oxygen molecules are reduced to oxide or hydroxide ions. The electrolyte serves to transport either the positively charged ions from anode to cathode or the negatively charged ions from cathode to anode. Figure 24-46 is a schematic representation of the reactions in a fuel cell operating on hydrogen and air with a hydrogen-ion-conducting electrolyte. The hydrogen flows over the anode, where the molecules are separated into ions and electrons. The ions migrate through the ionically conducting but electronically insulating electrolyte to the cathode, and the electrons flow through the outer circuit energizing an electric load. The electrons combine eventually with oxygen molecules flowing over the surface of the cathode and hydrogen ions migrating across the electrolyte, forming water, which leaves the fuel cell in the depleted air stream.

A fuel cell has no moving parts. It runs quietly, does not vibrate, and does not generate gaseous pollutants. The idea of the fuel cell is generally credited to Sir William Grove, who lived in the nineteenth century. It took over 100 years for the first practical devices to be built, in the U.S. space program, as the power supply for space capsules and the space shuttle. Commercialization of terrestrial fuel cell systems has only recently begun. Having lower emissions and being more efficient than heat engines, fuel cells may in time become the power source for a broad range of applications, beginning with utility power plants, including civilian and military transportation, and reaching into portable electronic devices.

This slow realization of the concept is due to the very demanding materials requirements for fuel cells. The anodes and cathodes have to be good electronic conductors and must have electrocatalytic properties to facilitate the anodic and cathodic reactions. In addition, the anodes and cathodes must be porous to allow the fuel and oxidant gases to diffuse to the reaction sites, yet they must be mechanically strong enough to support the weight of the fuel cell stacks. The electrolyte must be chemically stable in hydrogen and oxygen, and must

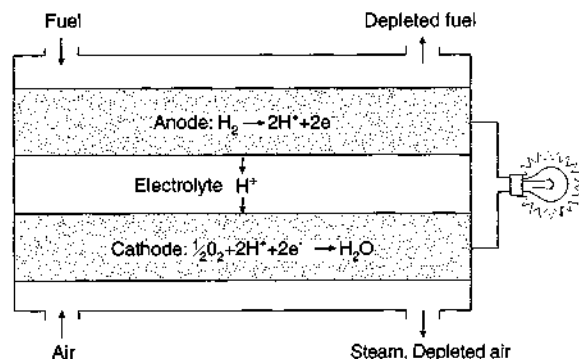


FIG. 24-46 Fuel cell schematic.

have an ionic conductivity of at least 0.1 S/cm. Five classes of electrolytes have been found to meet these requirements: potassium hydroxide, phosphoric acid, perfluorinated sulfonic acid resins, molten carbonates, and oxide-ion-conducting ceramics. Consequently, five types of fuel cell based on these electrolytes have been developed.

Fuel Cell Efficiency The theoretical energy conversion efficiency of a fuel cell ϵ° is given by the ratio of the free energy (Gibbs function) of the cell reaction at the cell's operating temperature ΔG_t to the enthalpy of reaction at the standard state ΔH° , both quantities being based on a mole of fuel:

$$\epsilon^\circ = \frac{\Delta G_t}{\Delta H^\circ} \quad (24-54)$$

The enthalpy of reaction is always taken at a temperature of 298 K (77°F), but the product water can be either liquid or gaseous. If it is liquid, the efficiency is based on the higher heating value (HHV), but if the product is gaseous, the efficiency is based on the lower heating value (LHV). If the fuel cell runs on hydrogen and oxygen at 373 K (212°F), the theoretical conversion efficiency is 91 percent LHV or 83 percent HHV. The theoretical efficiency of fuel cells as given in Eq. (24-54) is equivalent to the Carnot efficiency of heat engines with the working medium absorbing heat at the flame temperature of the fuel and rejecting it at 298 K. Owing to materials and engineering limitations, heat engines cannot operate at the Carnot limit. Fuel cells can run at efficiencies near the theoretical values but only at low power density (power produced per unit of active fuel cell area). At higher power densities, the efficiency of fuel cells is constrained by electrical resistances within the bulk and at the interfaces of the materials, and by gas diffusion losses.

When no net current is flowing, the equilibrium potential of the fuel cell is given by the Nernst equation:

$$E^\circ = \frac{-\Delta G_t}{nF} \quad (24-55)$$

where E° is the electrochemical equilibrium potential, V ; n is the number of electrons transferred in the cell reaction (equivalents), and F is the Faraday constant. If the units of ΔG_t are J/mol, F has the value 96,487 C/mol-equiv. The potential depends on the chemical species of the fuel and the operating temperature. For hydrogen and oxygen, variation of the equilibrium cell potential with temperature is shown in Table 24-19.

When current is flowing, the actual cell operating potential is given by:

$$E = E^\circ - (a_{an} + a_{ca}) - (b_{an} + b_{ca}) \frac{RT}{nF} \ln i - Ai \quad (24-56)$$

where a and b are characteristic constants for the electrochemical reactions at the electrode materials; the subscripts an and ca refer to the anode and the cathode, respectively; R is the gas constant; T is the cell temperature; A is the area-specific resistance of the fuel cell; and i is the current density (current flow per unit of active fuel cell area) in the cell.

Graphs of operating potential versus current density are called *polarization curves*, which reflect the degree of perfection that any particular fuel cell technology has attained. High cell operating potentials are the result of many years of materials optimization. Actual polarization curves will be shown below for several types of fuel cell.

The actual efficiency of an operating fuel cell is given by:

$$\epsilon = \frac{-nFE}{\Delta H^\circ} U_f \quad (24-57)$$

where U_f is the electrochemical fuel utilization (amount of fuel converted divided by amount fed to the cell). For pure hydrogen the fuel utilization can be 1.0, but for gas mixtures it is often 0.85. Equations (24-56) and (24-57) show that the efficiency of fuel cells is not constant, but depends on the current density. The more power that is drawn, the lower the efficiency.

When the fuel gas is not pure hydrogen and air is used instead of pure oxygen, additional adjustment to the calculated cell potential becomes necessary. Since the reactants in the two gas streams practically become depleted between the inlet and exit of the fuel cell, the cell potential is decreased by a term representing the log mean fugacities, and the operating cell efficiency becomes

$$\epsilon_{fc} = \frac{nFU_f}{\Delta H^\circ} \left[E^\circ - \sum a - \sum b \frac{RT}{nF} \ln i - Ai \right] - \frac{RTU_f}{\Delta H^\circ} [v_f \ln(\log \text{ mean } \hat{f}_f) + v_{ox} \ln(\log \text{ mean } \hat{f}_{ox})] \quad (24-58)$$

The quantities v_f and v_{ox} are the stoichiometric coefficients for the fuel cell reaction, and $\hat{f}_f$ and $\hat{f}_{ox}$ are the fugacities of fuel and oxygen in their respective streams.

Further, as the current density of the fuel cell increases, a point is inevitably reached where the transport of reactants to or products from the surface of the electrode becomes limited by diffusion. A *concentration polarization* is established at the electrode, which diminishes the cell operating potential. The magnitude of this effect depends on many design and operating variables, and its value must be obtained empirically.

Design Principles An individual fuel cell will generate an electrical potential of about 1 V or less, as discussed above, and a current that is proportional to the external load demand. For practical applications, the voltage of an individual fuel cell is obviously too small, and cells are therefore stacked up as shown in Fig. 24-47. Anode/electrolyte/cathode assemblies are electrically connected in series by inserting a bipolar plate between the cathode of one cell and the anode of the next. The bipolar plate must be impervious to the fuel and oxidant gases, chemically stable under reducing and oxidizing conditions, and an excellent electronic conductor. In addition, it is often used to distribute the gases to the anode and cathode surfaces through flow channels cut or molded into it.

The number of fuel cells that are stacked is determined by the desired electrical potential. For 110-V systems it can be about 200 cells. Since a typical fuel cell is about 5 mm (0.2 in) thick, a 200-cell stack assembly (including the end hardware that keeps the unit under compression) is about 2 m (6 ft) tall. The reactant and product gas streams are supplied and removed from the stack by external or internal manifolding. Externally manifolded stacks have shallow trays on each of the four sides to supply the fuel and air and to remove the depleted gases and reaction products. The manifolds are mechanically clamped to the stacks and sealed at the edges. These manifold seals must be gastight, electrically insulating, and able to tolerate thermal expansion mismatches between the stack and the manifold materials as well as dimensional changes due to aging.

Alternatively, reactant and product gases can be distributed to and removed from individual cells through internal pipes in a design analogous to that of filter presses. Care must be exercised to assure an even flow distribution between the entry and exit cells. The seals in internally manifolded stacks are generally not subject to electrical, thermal, and mechanical stresses, but are more numerous than in externally manifolded stacks.

Because fuel cells generate an amount of excess heat consistent with their thermodynamic efficiency, they must be cooled. In low-temperature fuel cells, the cooling medium is generally water or oil, which flows through cooling plates interspaced throughout the stack. In high-temperature cells, heat is removed by the reactant air stream and also by the endothermic fuel reforming reactions in the stack.

TABLE 24-19 Thermodynamic Values for $H_2 + \frac{1}{2}O_2 = H_2O$ (g)

Temperature, K	Enthalpy of reaction (ΔH°), kJ/mol	Free energy of reaction (ΔG°), kJ/mol	Equilibrium cell potential (E°), V
300	-241.8	-228.4	1.18
500	-243.9	-219.2	1.14
700	-245.6	-208.8	1.06
900	-247.3	-197.9	1.03
1100	-248.5	-187.0	0.97
1300	-249.4	-175.7	0.91

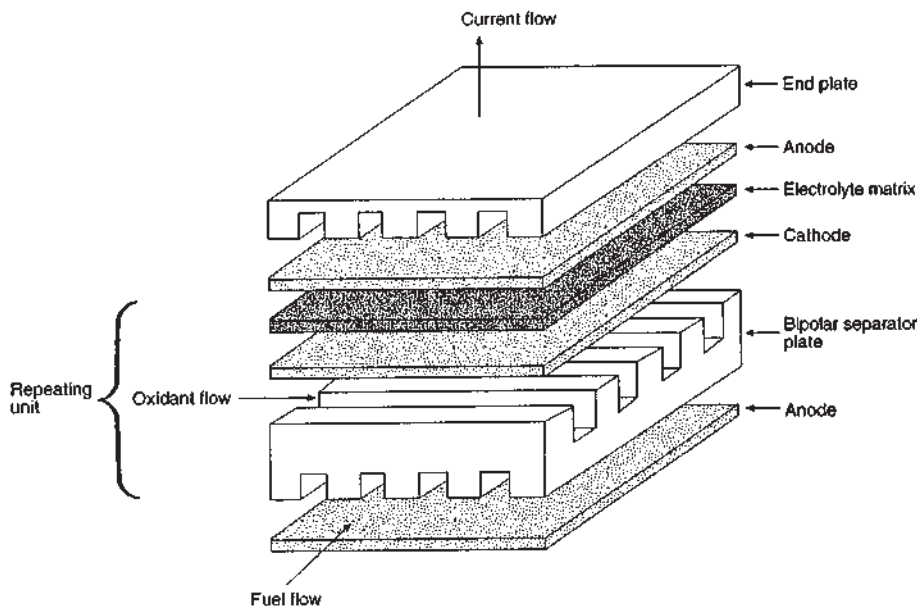


FIG. 24-47 Stacking of individual fuel cells.

Types of Fuel Cells The five major types of fuel cell are listed in Table 24-20. Each has unique chemical features. The *alkaline fuel cell* (AFC) has high power density and has proven itself as a reliable power source in the U.S. space program, but the alkaline electrolyte reacts with carbon dioxide, which is present in reformed hydrocarbon fuels and air. The *polymer electrolyte fuel cell* (PEFC) and the *phosphoric acid fuel cell* (PAFC) are tolerant of carbon dioxide, but both are sensitive to carbon monoxide (PEFC much more so than PAFC), which is adsorbed onto the platinum catalyst and renders it inactive. Therefore, these three types of fuel cell require pure hydrogen as fuel; and if the hydrogen has been obtained by reforming a fuel such as natural gas, the hydrogen-rich fuel stream must be purified before being introduced into the fuel cell. The *molten carbonate fuel cell* (MCFC) and the *solid oxide fuel cell* (SOFC) can tolerate carbon monoxide and carbon dioxide and can operate on hydrocarbon fuels with minimal fuel processing, but they operate at elevated temperatures.

The operating temperature also affects the fuel cell operating potential in more than one way. A high operating temperature accelerates reaction rates but lowers the thermodynamic equilibrium potential. These effects balance one another, and, in practice, the operating point of any fuel cell is usually between 0.7 and 0.8 V. The cell reactions for the five types of fuel cell are summarized in Table 24-21. It is important to note that in cells with acidic electrolytes (PAFC and PEFC) the product water evolves on the air electrode, but in the alkaline ones (AFC, MCFC, and SOFC) it is generated on the fuel electrode. This has consequences for the system design. Whenever water needs to be recovered for generating hydrogen from hydrocarbon fuels, a condenser is required in PAFC and PEFC systems. In others, the depleted fuel can be recycled.

TABLE 24-20 Fuel Cell Characteristics

Type of fuel cell	Electrolyte	Operating temperature		Coolant medium
		K	°C	
Alkaline	KOH	363	90	Water
Polymer	$\text{CF}_3(\text{CF}_2)_n\text{OCF}_2\text{SO}_3^-$	353	80	Water
Phosphoric acid	H_3PO_4	473	200	Steam/water
Molten carbonate	$\text{Li}_2\text{CO}_3\text{-K}_2\text{CO}_3$	923	650	Air
Solid oxide	$\text{Zr}_{0.92}\text{Y}_{0.08}\text{O}_{1.96}$	1273	1000	Air

Following is a summary of the materials, operating characteristics, and mode of construction for each type of fuel cell.

Alkaline Fuel Cell The electrolyte for NASA's space shuttle orbiter fuel cell is 35 percent potassium hydroxide. The cell operates between 353 and 363 K (176 and 194°F) at 0.4 MPa (59 psia) on hydrogen and oxygen. The electrodes contain platinum-palladium and platinum-gold alloy powder catalysts bonded with polytetrafluoroethylene (PTFE) latex and supported on gold-plated nickel screens for current collection and gas distribution. A variety of materials, including asbestos and potassium titanate, are used to form a micro-porous separator that retains the electrolyte between the electrodes. The cell structural materials, bipolar plates, and external housing are usually nickel-plated to resist corrosion. The complete orbiter fuel cell power plant is shown in Fig. 24-48.

Typical polarization curves for alkaline fuel cells are shown in Fig. 24-49. It is apparent that the alkaline fuel cell can operate at about 0.9 V and 500 mA/cm² current density. This corresponds to an energy

TABLE 24-21 Fuel Cell Reaction Electrochemistry

Type of fuel cell	Conducting ion	Anode reaction	Cathode reaction
Alkaline	OH^-	$\text{H}_2 + 2\text{OH}^- \rightarrow 2\text{H}_2\text{O} + 2\text{e}^-$	$\frac{1}{2}\text{O}_2 + \text{H}_2\text{O} + 2\text{e}^- \rightarrow 2\text{OH}^-$
Polymer	H^+	$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$	$\frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}$
Phosphoric acid	H^+	$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$	$\frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}$
Molten carbonate	CO_3^{2-}	$\text{H}_2 + \text{CO}_3^{2-} \rightarrow \text{H}_2\text{O} + \text{CO}_2 + 2\text{e}^-$	$\frac{1}{2}\text{O}_2 + \text{CO}_2 + 2\text{e}^- \rightarrow \text{CO}_3^{2-}$
Solid oxide	O^{2-}	$\text{H}_2 + \text{O}^{2-} \rightarrow \text{H}_2\text{O} + 2\text{e}^-$	$\frac{1}{2}\text{O}_2 + 2\text{e}^- \rightarrow \text{O}^{2-}$

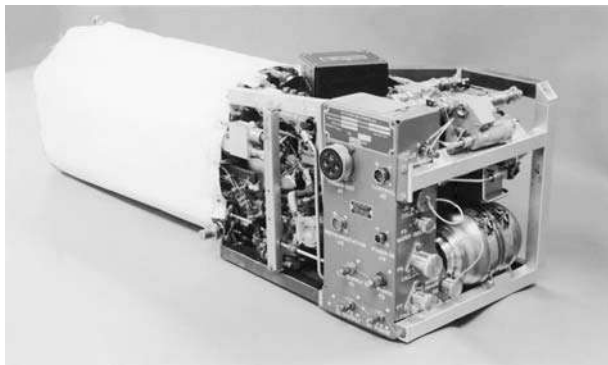


FIG. 24-48 Orbiter power plant. (International Fuel Cells.)

conversion efficiency of about 60 percent HHV. The space shuttle orbiter power module consists of three separate units, each measuring 0.35 by 0.38 by 1 m (14 by 15 by 40 in), weighing 119 kg (262 lb), and generating 15 kW of power. The power density is about 100 W/L and the specific power, 100 W/kg.

Polymer Electrolyte Fuel Cell The PEFC, also known as the *proton-exchange-membrane fuel cell* (PEMFC), is of great interest for transportation applications because it is capable of high power density and it can deliver about 40 percent of its nominal power at room temperature. These features have made the PEFC a candidate to replace internal combustion engines in automobiles. Methanol, ethanol, hydrogen, natural gas, dimethyl ether, and common transportation fuels such as gasoline are being considered as fuel, but hydrogen is currently preferred. The motivation for developing fuel-cell-powered vehicles is a dramatic reduction in environmentally harmful emissions and high fuel economy.

The electrolyte is a perfluorosulfonic acid ionomer, commercially available under the trade name of Nafion™. It is in the form of a membrane about 0.17 mm (0.007 in) thick, and the electrodes are bonded directly onto the surface. The electrodes contain very finely divided platinum or platinum alloys supported on carbon powder or fibers. The bipolar plates are made of graphite-filled polymer or metal.

Typical platinum catalyst loadings needed to support the anodic and cathodic reactions are currently 0.2 to 0.5 mg/cm² of active cell area. Owing to the cost of platinum, substantial efforts have been made to reduce the catalyst loading.

To be ionically conducting, the fluorocarbon ionomer must be “wet”: under equilibrium conditions, it will contain about 20 percent water. The operating temperature of the fuel cell must be

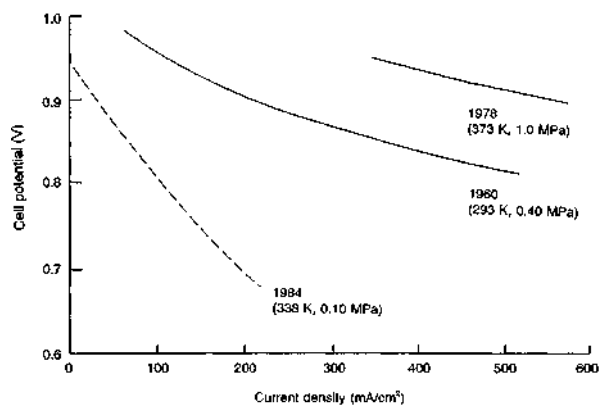


FIG. 24-49 Polarization curves for alkaline fuel cells.

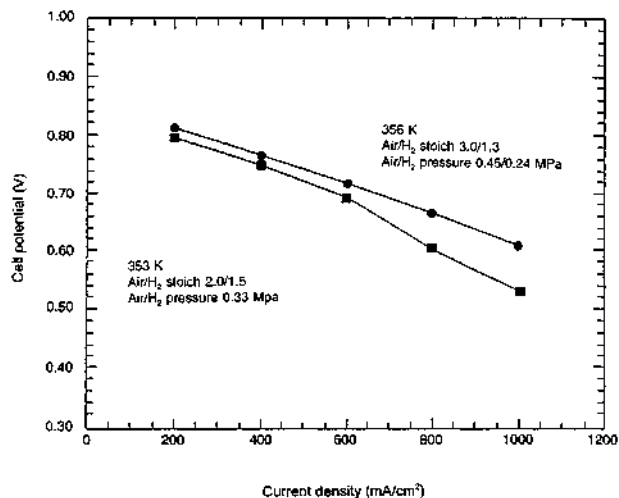


FIG. 24-50 Polarization curves for PEFC stacks.

less than 373 K (212°F), therefore, to prevent the membrane from drying out.

Being acidic, fluorocarbon ionomers can tolerate carbon dioxide in the fuel and air streams; PEFCs, therefore, are compatible with hydrocarbon fuels. However, the platinum catalysts on the fuel and air electrodes are extremely sensitive to carbon monoxide: only a few parts per million are acceptable. Catalysts that are more tolerant to carbon monoxide are being explored. Typical polarization curves for PEFCs are shown in Fig. 24-50.

As mentioned, the primary motivation for the PEFC development was the anticipated applicability in transportation. However, the economics of stationary use are more forgiving, and commercialization of the technology will likely begin as grid-independent power supplies. Figure 24-51 shows a 5-kW PEFC system operating on natural gas.



FIG. 24-51 Natural gas-fueled 5-kW PEFC system. (Courtesy of Plug Power.)

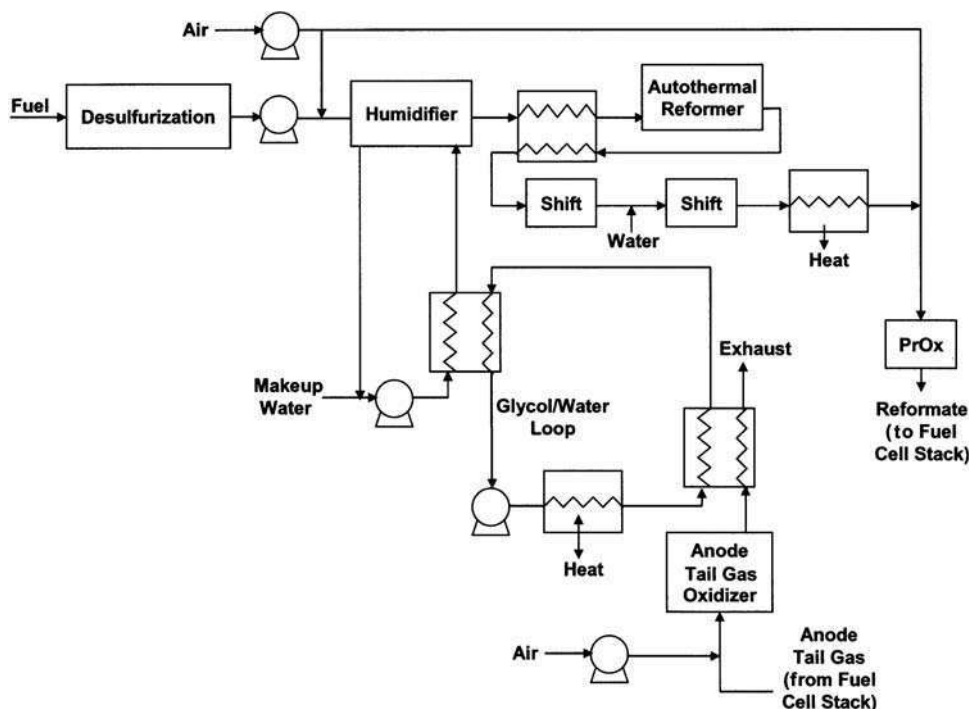


FIG. 24-52 System diagram for reformer-based PEFC system.

The fuel cell stack is visible in the center as a black box, and the fuel processing train is seen on the left. The smaller box on the right contains the power conditioning equipment. Figure 24-52 shows the system diagram for the unit. Natural gas is desulfurized and then mixed with air and steam before entering the autothermal reformer that converts the methane to a mixture of hydrogen, carbon monoxide, and carbon dioxide. Typically this reaction occurs over a catalyst at about 973 K [700°C]. To decrease the carbon monoxide content of this gas, it is cooled to about 623 K [350°C] and reacted with additional water in the so-called shift reactors. Because of the sensitivity of the anode catalyst to carbon monoxide, a preferential oxidation of the remaining carbon monoxide is done before the hydrogen-rich gas is fed to the fuel cell.

About 70 percent of the hydrogen is converted to electricity in the fuel cell stack, and the tail gas is then burned to generate heat which is transferred via a glycol/water loop to the steam generator. The same loop passes through the cooling plates or the fuel cell to remove heat.

A particular version of the PEFC is the **direct methanol fuel cell** (DMFC). As the name implies, an aqueous solution of methanol is used as fuel instead of the hydrogen-rich gas, eliminating the need for reformers and shift reactors. The major challenge for the DMFC is the “crossover” of methanol from the anode compartment into the cathode compartment through the membrane that poisons the electrodes by CO. Consequently, the cell potentials and hence the system efficiencies are still low. Nevertheless, the DMFC offers the prospect of replacing batteries in consumer electronics and has attracted the interest of this industry.

Phosphoric Acid Fuel Cell In this type of fuel cell, the electrolyte is 93 to 98 percent phosphoric acid contained in a matrix of silicon carbide. The electrodes consist of finely divided platinum or platinum alloys supported on carbon black and bonded with PTFE latex. The latter provides enough hydrophobicity to the electrodes to prevent flooding of the structure by the electrolyte. The carbon support of the air electrode is specially formulated for oxidation resistance at 473 K (392°F) in air and positive potentials.

The bipolar plate material of the PAFC is graphite. A portion of it has a carefully controlled porosity that serves as a reservoir for phosphoric

acid and provides flow channels for distribution of the fuel and oxidant. The plates are electronically conductive but impervious to gas crossover.

PAFC systems are commercially available from UTC Power as 200-kW stationary power sources operating on natural gas. The stack cross section is 1 m² (10.8 ft²). It is about 2.5 m (8.2 ft) tall and rated for a 40,000-h life. It is cooled with water/steam in a closed loop with secondary heat exchangers. Fuel processing is similar to that in a PEFC system, but a preferential oxidizer is not needed. These systems are intended for on-site power and heat generation for hospitals, hotels, and small businesses.

Molten Carbonate Fuel Cell The electrolyte in the MCFC is a mixture of lithium/potassium or lithium/sodium carbonates, retained in a ceramic matrix of lithium aluminate. The carbonate salts melt at about 773 K (932°F), allowing the cell to be operated in the 873 to 973 K (1112 to 1292°F) range. Platinum is no longer needed as an electrocatalyst because the reactions are fast at these temperatures. The anode in MCFCs is porous nickel metal with a few percent of chromium or aluminum to improve the mechanical properties. The cathode material is lithium-doped nickel oxide.

The bipolar plates are made from either Type 310 or Type 316 stainless steel, which is coated on the fuel side with nickel and aluminized in the seal area around the edge of the plates. Both internally and externally manifolded stacks have been developed.

In MCFCs, the hydrogen fuel is generated from such common fuels as natural gas or liquid hydrocarbons by steam reforming; the fuel processing function can be integrated into the fuel cell stack because the operating temperature permits reforming using the waste heat. An added complexity in MCFCs is the need to recycle carbon dioxide from the anode side to the cathode side to maintain the desired electrolyte composition. (At the cathode, carbon dioxide reacts with incoming electrons and oxygen in air to regenerate the carbonate ions that are consumed at the anode.) The simplest way is to burn the depleted fuel and mix it with the incoming air. This works well but dilutes the oxygen with the steam generated in the fuel cell. A steam condenser and recuperative heat exchanger can be added to eliminate the steam, but at increased cost.

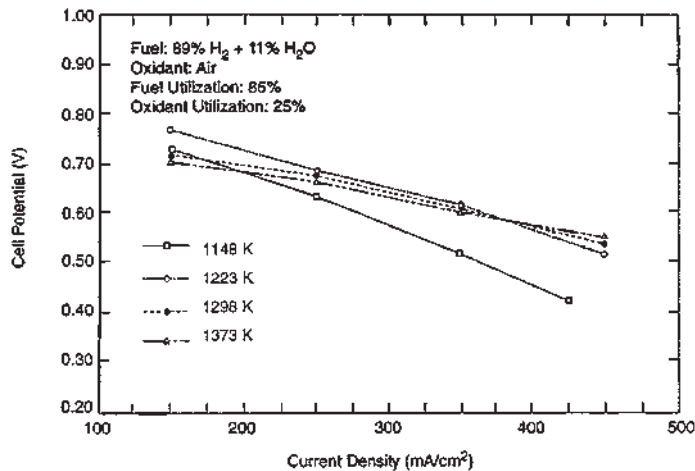


FIG. 24-53 Polarization curves at different temperatures for 50-cm active length thin-wall SOFCs.

The fuel cell must be cooled with either water or air, and the heat can be converted to electricity in a bottoming cycle. The dc electrical output of the stack is usually converted to ac and stepped up or down in voltage, depending on the application. Analogous to PAFCs, MCFC stacks are about 1 m² (10.8 ft²) in plan area and quite tall. A stack generates 200 to 300 kW.

Solid Oxide Fuel Cell In SOFCs the electrolyte is a ceramic oxide ion conductor, such as yttrium-doped zirconium oxide. The conductivity of this material is 0.1 S/cm at 1273 K (1832°F); it decreases to 0.01 S/cm at 1073 K (1472°F), and by another order of magnitude at 773 K (932°F). Because the resistive losses need to be kept below about 50 mV, the operating temperature of the SOFC depends on the thickness of the electrolyte. For a thickness of 100 μm or more, the operating temperature is 1273 K (1832°F) but fuel cells with thin electrolytes can operate between 973 and 1073 K (1292 and 1472°F).

The anode material in SOFCs is a cermet (metal/ceramic composite material) of 30 to 40 percent nickel in zirconia, and the cathode is

lanthanum manganite doped with calcium oxide or strontium oxide. Both of these materials are porous and mixed ionic/electronic conductors. The bipolar separator typically is doped lanthanum chromite, but a metal can be used in cells operating below 1073 K (1472°F). The bipolar plate materials are dense and electronically conductive.

Typical polarization curves for SOFCs are shown in Fig. 24-53. As discussed earlier, the open-circuit potential of SOFCs is less than 1 V because of the high temperature, but the reaction overpotentials are small, yielding almost linear curves with slopes corresponding to the resistance of the components.

SOFCs can have a planar geometry similar to PEFCs, but the leading technology is tubular, as shown in Fig. 24-54. The advantage of the tubular arrangement is the absence of high-temperature seals.

Like MCFCs, SOFCs can integrate fuel reforming within the fuel cell stack. A prereformer converts a substantial amount of the natural gas using waste heat from the fuel cell. Compounds containing sulfur (e.g., thiophene, which is commonly added to natural gas as an odorant)

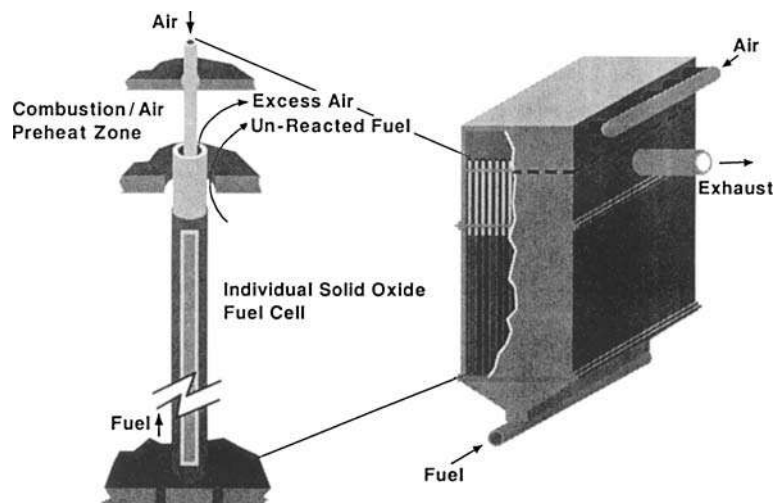


FIG. 24-54 Configuration of the tubular SOFC. (Courtesy of Westinghouse Electric Corporation.)

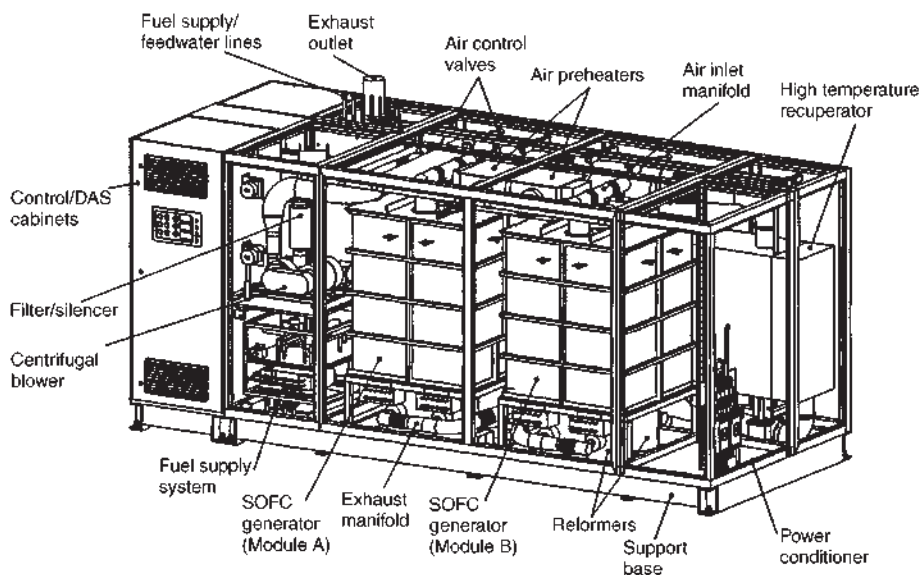


FIG. 24-55 SOFC 25-kW system package. (Courtesy of Westinghouse Electric Corporation.)

must be removed before the reformer. Typically, a hydrodesulfurizer combined with a zinc oxide absorber is used.

The desulfurized natural gas is mixed with the recycled depleted fuel stream containing steam formed in the fuel cell. About 75 percent of the methane is converted to hydrogen and carbon monoxide in the prereformer. The hydrogen-rich fuel is then passed over the

fuel cell anode, where 85 percent is converted to electricity. The balance is burned with depleted air in the combustion zone.

The hot combustion gas preheats the fresh air and the prereformer, and can be used further to generate steam. The system is cooled with 200 to 300 percent excess air. A 25-kW SOFC generator system is shown in Fig. 24-55.

ENERGY RECOVERY

Most processing energy enters and then leaves the process as energy, separate from the product. The energy enters as electricity, steam, fossil fuels, etc. and then leaves, released to the environment as heat, through "coolers," hot combustion flue gases, waste heat, etc. Recovering heat to be used elsewhere in the process is important to increase process efficiency and minimize cost. Minimizing the total annualized costs for this flow of energy through the process is a complex engineering task in itself, separate from classic process design. Since these costs include the costs for getting energy into and out of the process, they should be evaluated together, as elements integrated within a larger system. Such a holistic system evaluation impacts how the overall project will be designed (utilities supply, reaction and separations design, pinch analyses, 3D process layout, plot plan, etc.). Therefore, evaluation and selection of the process energy technology system should be performed at the start of the project design cycle, during **technology selection VIP** (see the subsection "Value-Improving Practices" in Sec. 9), when the **potential to influence project costs exists at its maximum value**.

Following the 1970s energy crisis, enhanced technology systems have been developed which can **significantly reduce the annualized costs for process energy**. Several of these technologies are presented below, because they are broadly applicable, have a rapid payback, and can make a significant reduction in overall annualized energy costs. Wet surface air coolers (WSACs), an evaporative cooling technology, are presented in Sec. 12.

ECONOMIZERS

GENERAL REFERENCES: "Latest Advances in the Understanding of Acid Dew-point Corrosion: Corrosion and Stress Corrosion Cracking in Combustion Gas

Condensates," W.M.M. Huijbregts and R.G.I. Leferink, *Anti-Corrosion Methods and Materials*, vol. 51, no. 3, 2004, pp. 173-188 (<http://www.hbscc.nl/>); "Get Acid Dew Points of Flue Gas," A.G. Okkes, *Hydrocarbon Processing*, July 1987, pp. 53-55; Lahtvee, T., Schaus, O., *Study of Materials to Resist Corrosion in Condensing Gas-Fired Furnaces*, Final Report to Gas Research Institute, GRI-80/0133, February, 1982; Ball, D., et al., *Condensing Heat Exchanger Systems for Oil-Fired Residential/Commercial Furnaces and Boilers: Phase I and II*, US DOE BNL-51617, 1982; Razgaitis, R., et al., *Condensing Heat Exchanger Systems for Residential/Commercial Furnaces: Phase III*, US DOE BNL-51770, 1984; Razgaitis, R., et al., *Condensing Heat Exchanger Systems for Residential/Commercial Furnaces and Boilers: Phase IV*, BNL-51943, 1985; Butcher, T.A., Park, N., and Litzke, W., "Condensing Economizers: Thermal Performance and Particulate Removal Efficiencies," in *ASME Two Phase Flow and Heat Transfer, HTD*, vol. 197, 1992 (for U.S. DOE reports see: <http://www.osti.gov/energycitations>).

Economizers improve boiler thermal efficiency by recovering heat from the combustion flue gases exhausted from the steam boiler section. The recovered heat is used to heat colder streams (heat sinks), before ultimate discharge of the waste gas to atmosphere. This recovered heat displaces the need to burn additional fuel to heat these same streams.

Normally, after being heated, these streams are used in the boiler area (deaerator feedwater, cold return condensate, boiler feedwater, RO feedwater) or in the combustion chamber (air preheat). However, economizers can be used to recover and supply heat elsewhere, such as hot process water or hot utility water, especially as used in the food processing and pulp/paper industries. Additionally, recovered flue gas waste heat can be used indirectly; i.e., remote process streams can be heated locally with hot steam condensate, and then the cooled return steam condensate can be reheated in the flue gas economizer. An

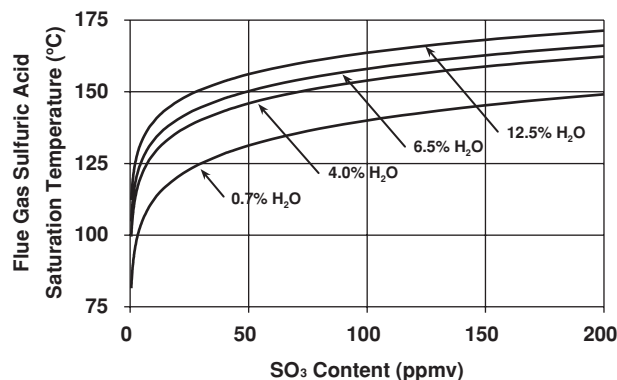


FIG. 24-56 Calculated sulfuric acid dew points, as a function of SO_3 content, for various flue gas water vapor concentrations. (Courtesy W. M. M. Huijbregts, 2004.)

extension of these concepts is provided by the application of using hot water to vaporize LNG: hot-water-based liquid is used to vaporize the process stream [LNG: stored near 122 K (-151°C), returning near 273 K (0°C)] to the hot water heater. Before entering the hot water heater, the cooled stream recovers flue gas waste heat in a condensing economizer.

Acid Dew Point For fossil fuels, the acid dew point temperature is that temperature at which the actual mixed acid vapor pressure equals the mixed acid vapor saturation pressure. The mixed acid dew point can be approximated by the sulfuric acid dew point (Fig. 24-56). It can be described as a function of the SO_3 and water content of the flue gas (Huijbregts). These concentrations result from the sulfur, hydrogen, and free water content of the fuel; the relative humidity of the air; and the amount of excess air used. Using the equation of Verhoff, where T is degrees K and P is mm Hg (see Okkes, A.G.):

$$T_{\text{dew}}(\text{SO}_3) = 1000 / [2.276 - 0.0294 \ln(P_{\text{H}_2\text{O}}) - 0.0858 \ln(P_{\text{SO}_3}) + 0.0062 \ln(P_{\text{H}_2\text{O}} P_{\text{SO}_3})] \quad (24-59)$$

The corrosiveness of flue gas condensate is further complicated by the presence of other components (Cl^- , NO_3^- , etc.). The sources of these components can be either the fuel or the combustion air (salt, ammonia, Freon, chlorine, chlorinated VOCs, etc.), usually producing a more corrosive condensate.

Water Dew Point For flue gas, the water dew point is that temperature at which the actual water vapor pressure equals the water saturation vapor pressure. Cooling the flue gas below this temperature will result in the formation of liquid water [or ice, below 273 K (0°C)]. For example, burning natural gas with 3 percent excess oxygen (15 percent excess air), the flue gas water dew point would be (Fig. 24-57) ~ 330 K (56.7°C).

Boiler Thermal Efficiency Traditionally, boiler thermal efficiency is calculated $Q_{\text{OUT}}/Q_{\text{IN}}$, where Q_{IN} is the LHV (lower heating value) of the fuel. A rule of thumb for economizers is that boiler efficiency increases by ~ 1 percent for every 22°C (40°F) drop in temperature of the dry flue gas. These two statements do not reveal the considerable quantity of **additional heat**, available to be recovered through **condensation of the water vapor** in the flue gas, which is lost to atmosphere with hot flue gas. Based on fuel HHV (higher heating value), the total latent heat loss can be substantial: an additional 9.6 percent (natural gas), 8.0 percent (propane), 6.5 percent (heating oil).

Conventional Economizers Conventional economizers can be constructed from relatively inexpensive materials, such as low-alloy carbon steels, if they will be operated dry on the gas side, with flue gas side metal temperatures above the acid dew point. This practice is done to protect the economizer from corrosion, caused by the acidic flue gas condensate. Conventional economizers can also be con-

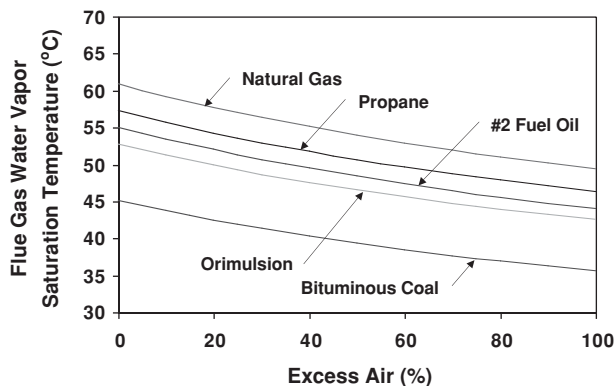


FIG. 24-57 Calculated flue gas water vapor dew points, for different fuel types, as a function of excess air [Orimulsion (28.3% water), Pittsburgh Seam 8 (5% water)]. (Courtesy T. A. Butcher, US DOE; www.bnl.gov.)

structed from more expensive materials, and can be operated below the acid dew point, but above the water dew point. This practice permits greater heat recovery, but with a generally lower payback. A compromise practice for operation below the acid dew point is to use less expensive but less corrosion-resistant materials, accepting an accelerated rate of corrosion, and periodically replacing the damaged heat-transfer surfaces when needed. Nevertheless, when high-sulfur fuel is burned (oil, coal, etc.), typically the water inlet feed to a conventional economizer is preheated to a temperature above the anticipated acid dew point.

Condensing Economizers Flue gas condensing waste heat economizers are designed to operate **below the flue gas water dew point**. This temperature can range from about 316 K (43°C) to 333 K (60°C), depending on the amount of hydrogen and water in the fuel, the amount of excess combustion air used, and the humidity of the air. [Higher flue gas water dew points can be encountered for other industrial applications, such as product driers, fryers (food processing), waste water incinerators, etc.] Such economizers recover **flue gas sensible heat** as well as **water vapor latent heat** from the hot flue gas. Fuel consumption is reduced in proportion to the efficiency increase.

Condensing economizers are constructed from inexpensive, but durable, corrosion-resistant materials. Extensive materials testing has been performed for operation in this service, including for coal combustion (Lahtvee, Ball, Razgaitis, and Butcher). The metallurgy for the tube-side liquid is determined by the liquid chemistry requirements (usually water-based liquid): 304 stainless steel is typical.

For gas-side materials, one available technology employs Teflon-covered metal tubing and Teflon tube sheets. This technology is often operated across both the acid and water dew points, and can accept inlet gas temperatures to 533 K (260°C). Typical applications may achieve a cold-end ΔT below 45°C (80°F), improve the boiler thermal efficiency by ~ 10 percent (LHV basis), and have a simple payback of 2 to 3 years, based on **fuel avoidance** (Figs. 24-58 and 24-59).

A second technology employs metallic finned tubing, extruded over the water tubing. Aluminum 1000 series fins are preferred, for heat-transfer reasons in natural gas applications, but stainless steel (or other material) fins are used for higher temperatures and/or more corrosive flue gas. This second technology both is less expensive and has better heat transfer (per ft^2). Consequently, for the same payback the cold-end approach can be lower, and the water outlet temperature and the boiler efficiency improvement higher. Flue gas condensate from combustion of natural gas typically has a pH of ~ 4.3 , and aluminum fins are suitable. For more acidic (or erosive) flue gas conditions, other metallurgy (Incoloy® 825 and Hastelloy®), or a Hersite or equivalent coating, may be used to prevent corrosion damage (Fig. 24-60).

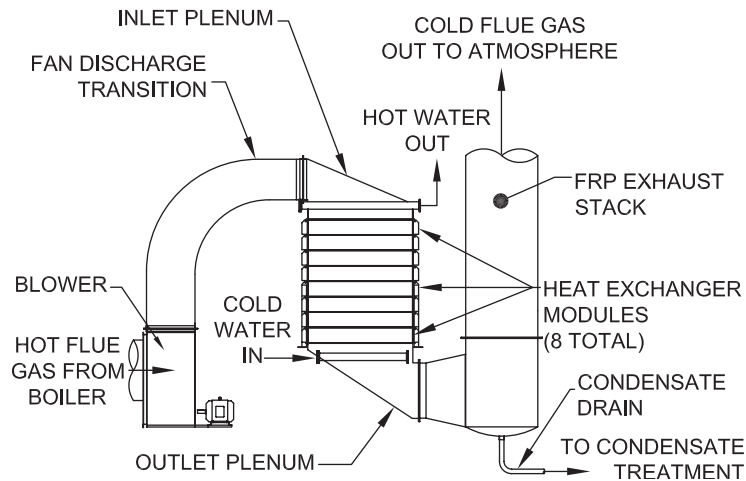


FIG. 24-58 Standard equipment arrangement, flue gas condensing economizer waste heat recovery system (flow: left to right). The ID fan draws hot flue gas from the boiler, propelling it into the top of the condensing economizer. (Courtesy CHX Condensing Heat Exchanger Co.; www.chxheat.com.)

Flue gas condensate at different temperatures, compositions and relative corrosivity condenses and exists at different positions within the condensing economizer. These positions are not fixed in space or time, but move back and forth, in response to changing load conditions

in either stream. Condensing economizers are typically equipped with water spray nozzles for periodic washdown of the flue gas side, to be used (infrequently) for natural gas combustion, but more frequently for services having heavier pollutant loading, such as oil, coal, etc. Over

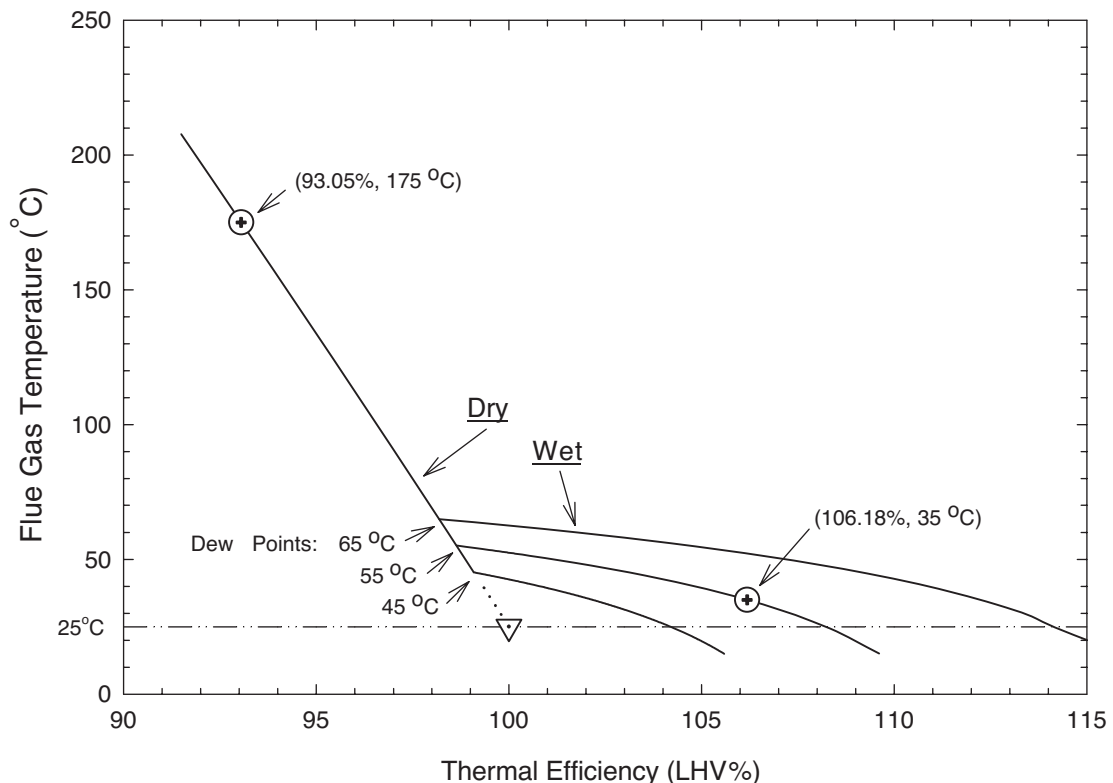


FIG. 24-59 Characteristic curves for boiler thermal efficiency as a function of flue gas effluent temperature and flue gas water dew points. Based on the LHV of a fuel, and stoichiometric reaction, 100 percent efficiency would be achieved if sufficient combustion heat were recovered and removed, so that the temperature of the effluent flue gas was reduced to 25°C. For a flue gas with a 55°C dew point, recovering additional heat via condensation by cooling from 175 to 35°C (as shown) would increase the overall efficiency by more than 13 percent. (Courtesy Combustion & Energy Systems, Ltd.; www.condexenergy.com.)

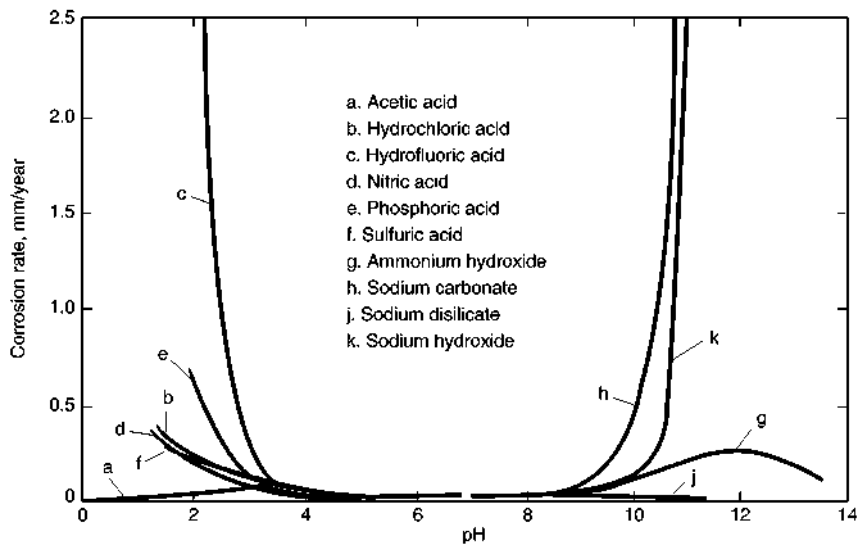


FIG. 24-60 Effect of corrosion on 1100-H14 aluminum alloy by various chemical solutions. Observe the minimal corrosion in the pH range from 4.0 to 9.0. The low corrosion rates in acetic acid, nitric acid, and ammonium hydroxide demonstrate that the nature of the individual ions in solution is more important than the degree of acidity or alkalinity. (With permission from ASM International; www.asminternational.org. Courtesy of Combustion & Energy Systems, Ltd.; www.condexenergy.com.)

200 such heat exchangers have been installed, some in service for more than 20 years (2005). This technology is suitable for heat recovery applications of any magnitude.

Several **environmental benefits** are gained through employment of this condensing technology. Burning less fuel proportionally reduces **collateral combustion emissions** (NO_x , SO_x , CO , CO_2 and particulates, including $\text{PM}_{2.5}$). Additionally, **flue gas pollutant removal** occurs in the condensing economizer, as has been extensively investigated, characterized, and modeled by the U.S. DOE (Butcher et al.) including applications burning coal and Orimulsion. Typically, the condensate will contain most (by mass, >90 percent) of the highly dissociated inorganic matter (H_2SO_4 , HCl , HNO_3 , HNO_2 , NH_3 , salts, etc.) and the **larger-diameter particulates** (>10 μm) and a lower but substantial fraction (>60 percent) of the **smaller-diameter particulates** (<5 μm). Such gross pollutant removal can be a cost effective first stage for a traditional air pollution control system, by reducing the volume of the flue gas to be treated (water content and temperature) as well as reducing the concentration of the pollutants in the flue gas to be treated downstream. Unlike spray quenching, such indirect quench-cooling segregates the flue gas pollutants from the cooling system fluid, generating a much smaller, concentrated waste water stream for subsequent waste treatment. Although pollutant removal percentages are high, they are functions of the specific real-time mass- and heat-transfer conditions within the economizer. Condensate treatment from boiler economizers normally is neutralization, often using the boiler blowdown, and release to the sewer and sewage treatment.

REGENERATORS

Storage of heat is a temporary operation since perfect thermal insulators are unknown; thus, heat is absorbed in solids or liquids as sensible or latent heat to be released later at designated times and conditions. The collection and release of heat can be achieved in two modes: on a batch basis, as in the checkerbrick regenerator for blast furnaces, or on a continuous basis, as in the Ljungstrom air heater.

Checkerbrick Regenerators Preheating combustion air in open-hearth furnaces, ingot-soaking pits, glass-melting tanks, by-product coke ovens, heat-treating furnaces, and the like has been universally carried out in regenerators constructed of fireclay, chrome, or silica bricks of various shapes. Although many geometric arrangements have been used in practice, the so-called basketweave design has been adopted in most applications.

Blast-Furnace Stoves Blast-furnace stoves are used to preheat the air that is blown into a blast furnace. A typical blast furnace, producing 1500 Mg (1650 U.S. ton) of pig iron per day, will be blown with 47.2 m^3/s (100,000 std ft^3/min) of atmospheric air preheated to temperatures ranging in normal practice from 755 to 922 K (900 to 1200°F). A set of four stoves is usually provided, each consisting of a vertical steel cylinder 7.3 m (24 ft) in diameter, 33 m (108 ft) high, topped with a spherical dome. Characteristic plan and elevation sections of a stove are shown in Fig. 24-61. The interior comprises three regions: in the cylindrical portion, (1) a side combustion chamber,

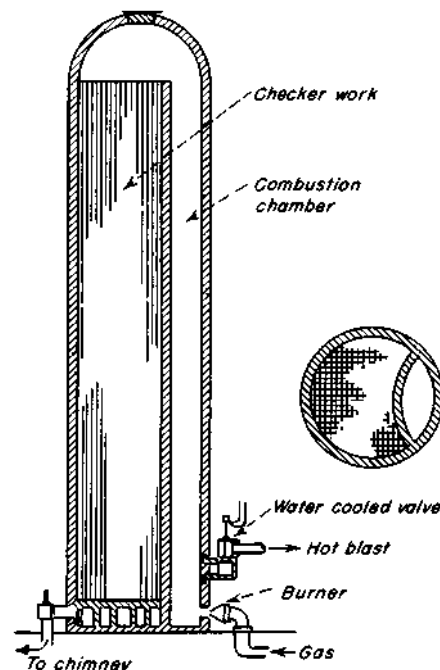


FIG. 24-61 Blast-furnace stove.

lens-shaped in cross section, bounded by a segment of the stove wall and a mirror-image bridgewall separating it from (2) the chamber of the cylinder that is filled with heat-absorbing checkerbrick, and (3) the capping dome, which constitutes the open passage between the two chambers.

The heat-exchanging surface in each stove is just under $11,500 \text{ m}^2$ ($124,000 \text{ ft}^2$). In operation, each stove is carried through a two-step 4-h cycle. In a 3-h *on-gas* step, the checkerbricks in a stove are heated by the combustion of blast-furnace gas. In the alternating *on-wind* 1-h step, they are cooled by the passage of cold air through the stove. At any given time, three stoves are simultaneously on gas, while a single stove is on wind.

At the start of an on-wind step, about one-half of the air, entering at 366 K (200°F), passes through the checkerbricks, the other half being bypassed around the stove through the cold-blast mixer valve. The gas passing through the stove exhausts at 1366 K (2000°F). Mixing this with the unheated air produces a blast temperature of 811 K (1000°F). The temperature of the heated air from the stove falls steadily throughout the on-wind step. The fraction of total air volume bypassed through the mixer valve is continually decreased by progressively closing this valve, its operation being automatically regulated to maintain the exit gas temperature at 811 K . At the end of 1 h of on-wind operation, the cold-blast mixer valve is closed, sending the entire blast through the checkerbricks.

Open-Hearth and Glass-Tank Regenerators These contain checkerbricks that are modified considerably from those used in blast-furnace stoves because of the higher working temperatures, more drastic thermal shock, and dirtier gases encountered. Larger bricks form flue cross sections five times as large as the stove flues, and the percentage of voids in the checkerbricks is 51 percent, in contrast to 32 percent voids in stoves. The vertical height of the flues is limited by the elevation of the furnace above plant level. Short flues from 3 to 4.9 m (10 to 16 ft) are common in contrast to the 26- to 29-m (85- to 95-ft) flue lengths in blast-furnace stoves.

As a result of the larger flues and the restricted surface area per unit of gas passed, regenerators employed with this type of furnace exhibit much lower efficiency than would be realized with smaller flues. In view of the large amount of iron oxide contained in open-hearth exhaust gas and the alkali fume present in glass-tank stack gases, however, smaller checkerbrick dimensions are considered impractical.

Ljungstrom Heaters A familiar continuous regenerative-type air heater is the Ljungstrom heater (Fig. 24-62). The heater assembly consists of a slow-moving rotor embedded between two peripheral housings separated from one another by a central partition. Through one side of the partition a stream of hot gas is being cooled, and, through the other side a stream of cold gas is being heated. Radial and circumferential seals sliding on the rotor limit the leakage between the streams. The rotor is divided into sectors, each of which is tightly packed with metal plates and wires that promote high heat-transfer rates at low pressure drop.

These heaters are available with rotors up to 6 m (20 ft) in diameter. Gas temperatures up to 1255 K (1800°F) can be accommodated. Gas face velocity is usually around 2.5 m/s (500 ft/min). The rotor height depends on service, efficiency, and operating conditions but usually is between 0.2 and 0.91 m (8 and 36 in). Rotors are driven by small motors with rotor speed up to 20 r/min. Heater effectiveness can be as high as 85 to 90 percent heat recovery. Ljungstrom-type heaters are used in power-plant boilers and also in the process industries for heat recovery and for air-conditioning and building heating.

Regenerative Burners In these systems a compact heat storage regenerator (containing ceramic balls, for example) is incorporated into the burner. Operating in pairs, one burner fires while the other exhausts: combustion air is preheated in the regenerator of the firing burner and furnace gas gives up heat to the regenerator in the exhausting burner (see Fig. 24-63). Burner operations are switched periodically. Such systems can yield combustion air preheats between 933 K (1220°F) and 1525 K (2282°F) for furnace temperature between 1073 K (1472°F) and 1723 K (2642°F), respectively.

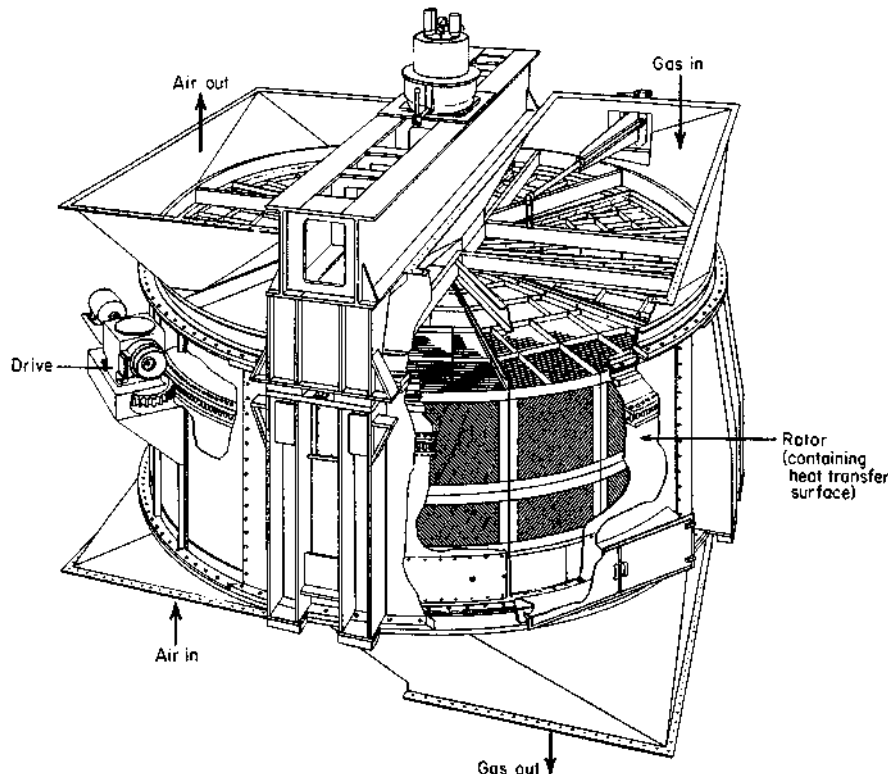


FIG. 24-62 Ljungstrom air heater.

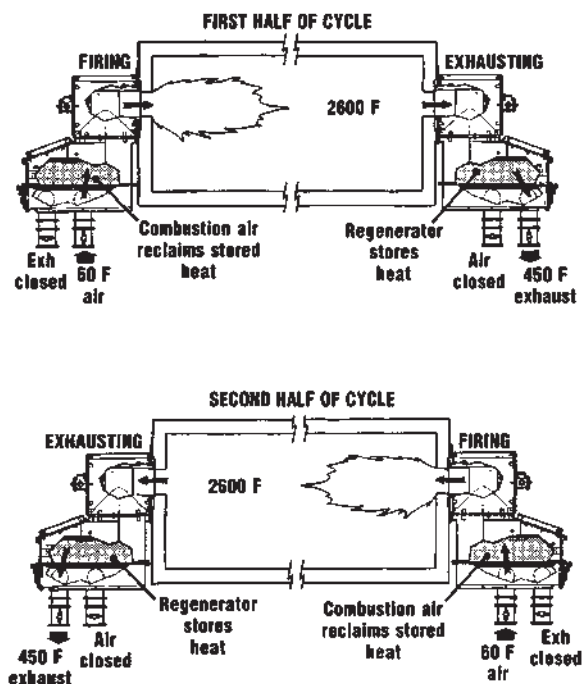


FIG. 24-63 Schematic of a regenerative burner system. (North American Manufacturing Co.)

Corresponding fuel savings compared to cold-air firing will vary approximately from 30 to 70 percent.

Miscellaneous Systems Many other systems have been proposed for transferring heat regeneratively, including the use of high-temperature liquids and fluidized beds for direct contact with gases, but other problems which limit industrial application are encountered. These systems are covered by methods described in Secs. 11 and 12 of this handbook.

RECUPERATORS

Regenerators are by nature intermittent or cycling devices, although, as set forth previously, the Ljungstrom design avoids interruption of the fluid stream by cycling the heat-retrieval reservoir between the hot and cold fluid streams. Truly continuous counterparts of regenerators exist, however, and they are called *recuperators*.

The simplest configuration for a recuperative heat exchanger is the metallic radiation recuperator. An inner tube carries the hot exhaust gases and an outer tube carries the combustion air. The bulk of the heat transfer from the hot gases to the surface of the inner tube is by radiation, whereas that from the inner tube to the cold combustion air is predominantly by convection.

Shell-and-tube heat exchangers (see Sec. 11) may also be used as recuperators; convective heat transfer dominates in these recuperators. For applications involving higher temperatures, ceramic recuperators have been developed. These devices can allow operation at up to 1823 K (2822°F) on the gas side and over 1093 K (1508°F) on the air side. Early ceramic recuperators were built of furnace brick and cement, but the repeated thermal cycling caused cracking and rapid deterioration of the recuperator. Later designs have used various approaches to overcome the problems of leakage and cracking. In one configuration, for example, silicon carbide tubes carry the combustion air through the waste gas, and flexible seals are used in the air headers. In this manner, the seals are maintained at comparatively low temperatures and the leakage rate can be reduced to a few percent of the total flow.

TURBINE INLET (AIR) COOLING

GENERAL REFERENCES: See "Bibliography" at <http://www.turbineinletcooling.org/resources>.

Turbine inlet cooling (TIC) can increase gas turbine (GT) power output on a hot day by 10 to 30 percent, while improving (reducing) the turbine heat rate (kJ/kW_e) by as much as 5 percent. By increasing the air compressor inlet air density, turbine inlet air cooling is the most cost-effective method for increasing turbine gross power output, for fixed-altitude GTs.

GTs are "constant-volume machines," such that a fixed-speed GT air compressor section draws a nearly constant volumetric flow of inlet air, independent of ambient air conditions. Air density drops with increased altitude (reduced barometric pressure), increased ambient temperature, and, to a lesser degree, increased water content (specific humidity).

Increasing the air density increases the GT inlet air mass flow. For a given stoichiometric fuel-to-air-ratio and a given combustion temperature, increased air mass flow allows increased fuel flow, resulting in increased GT power output. Additionally, compressor efficiency increases with decreased air temperatures, resulting in less parasitic compressor shaft work consumed and greater net turbine power output. Therefore, TIC increases net incremental power output faster than incremental fuel consumption, resulting in improved overall fuel efficiency (reduced heat rate); see Fig. 24-64.

Evaporative Technologies These TIC technologies cool the inlet air by vaporizing water in direct contact with the inlet air, using the latent heat of vaporization of water. These technologies either spray water directly into the airstream as a fog or mist or evaporate the water from fixed wetted media placed in the airstream.

Performance of these technologies is typically an 85 percent approach to the ambient wet-bulb temperature (T_{WB}) from the ambient dry-bulb temperature (T_{DB}), limited to a 3°F approach to the ambient T_{WB} . For locations with high T_{DB} and low T_{WB} , the cooling benefit can be significant. For locations with humid conditions, the relative cooling achievable by evaporative technologies can be very limited. Additionally, on a given day, the relative changes in hourly T_{DB} and T_{WB} will result in variable GT peak power output.

Evaporative TIC technologies have the minimum installed and operating costs. The operating electric parasitic load is generally considered insignificant. Water chemistry management is crucial to prevent maintenance problems caused by dissolved or suspended matter in the water, fouling either the evaporation media or the turbines themselves.

Refrigeration Technologies These TIC technologies indirectly cool the inlet air, typically using either chilled water or a vaporizing refrigerant, in a finned coil heat exchanger placed in the incoming airstream. Refrigeration technologies can achieve and maintain much lower air compressor inlet temperatures, largely independent of changing ambient conditions (Fig. 24-65). Best design practices (ice formation / consequent GT damage) have restricted the OEM recommended minimum inlet air cooling temperatures to the 5° to 10°C [41° to 50°F] range. Refrigeration can be supplied by either absorption or mechanical chillers. Absorption chillers employ either LiBr or ammonia absorption. Electrically driven centrifugal packaged chiller systems, using chilled water as a secondary refrigerant, account for the majority of refrigeration machine types. R-123, R-134a, and R-717 are the most common refrigerants offered by vendors. Several installations have used direct refrigeration through R-22 or ammonia (R-717) vaporization in the TIC air coil, with refrigerant condensation in a wet surface air cooler (WSAC) or an air-cooled condenser. Site-specific sources of cooling (the heat sink), such as LNG vaporization, have also been used. The water condensate generated by cooling inlet air below the ambient dew point can be employed as makeup water for a cooling tower or a WSAC.

While refrigeration technologies can provide a near-constant GT inlet air temperature, and thus a nearly constant GT power output, refrigeration TIC systems are significantly more expensive than evaporative systems. However, the incremental net power output from the refrigeration system, when measured on a unit cost basis (\$ per kW_e),

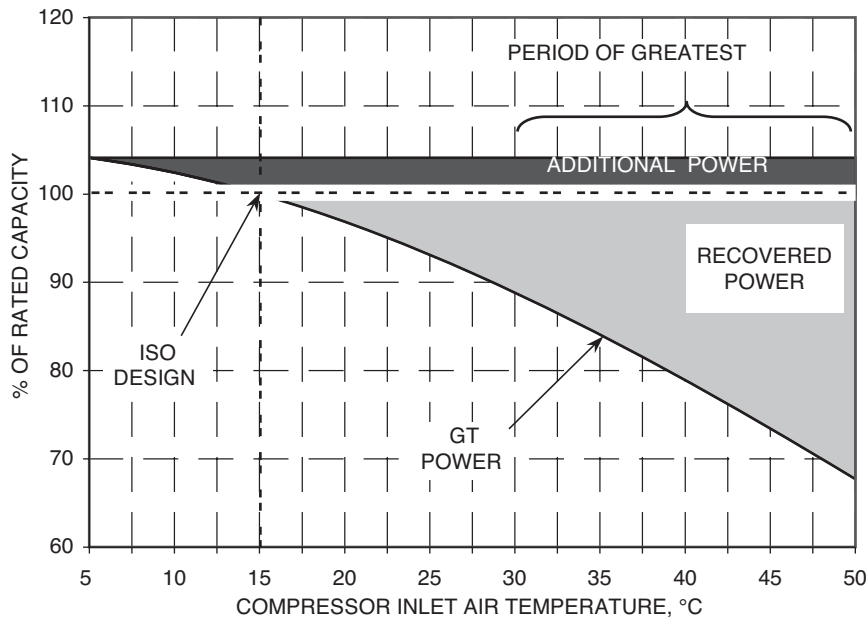


FIG. 24-64 Effect of inlet air ambient temperature on the power output of a typical GT. If ambient air at 95°F (35°C) were cooled to 50°F (10°C), the gross GT power output would be increased by approximately 22 percent, and the gross heat rate improved by 3.7 percent. Operated at its ISO conditions [15°C (59°F) at sea level], GT rated performance is 100 percent. (*Turbine Air Systems; www.tas.com.*)

costs much less than the unit cost of the GT itself, thereby decreasing the overall unit cost of the GT plant. Refrigeration technologies require a significant energy input to operate. In the case of electric-driven chillers, the parasitic electric load can consume 10 percent to 15 percent of the GT gross incremental increase in power. Therefore, the net overall plant incremental output, after allowance for the refrigeration system parasitic load, is used for evaluating the economics of these systems.

Thermal Energy Storage (TES) A TES system consists of an insulated cold storage tank and a chiller, generating ice or chilled water, which is accumulated in the tank. A TES system is typically installed when TIC is required for only a limited number of operating hours per day. The **principal TES benefit** is that the *net* electrical peak output of the power plant is maximized, by shifting the parasitic electric power loss (chiller power demand) from the period of peak demand (midday) to the off-peak hours (night). A **second benefit** is a reduced overall capital cost for a smaller chiller (usually half-sized) because the smaller chiller is operated either continuously (partial storage) or over a nighttime period (full storage). A **third benefit** is

the reduced operating costs, stemming from the ability to use lower-value off-peak power for the chilling load. A **fourth benefit** is that the chiller will typically perform better during the nighttime off-peak period, due to lower ambient temperatures, reducing the energy required for refrigerant compression. A **fifth benefit** is the ability to increase or decrease generation capacity in a short time period, when called upon, to stabilize the transmission system or grid the plant is tied into. However, a TES site footprint may be larger, to include the TES storage tank.

Summary Evaporative technologies have the lowest capital and operating costs, but their benefit is limited by ambient conditions. They are most effective where the peak T_{WB} is lowest, especially at times of peak demand, as are found in hot, arid regions. Refrigeration technologies can achieve and maintain constant lower temperatures, independent of ambient conditions. However, they have higher initial capital and operating costs. Although refrigeration technologies are favored for use in warm, humid regions, where both the peak daytime T_{WB} and air conditioning demand are highest, they have been selected and installed in more northern temperate latitudes (Canada) and arid climates as well. TES and other hybrids, which combine various technologies, are used to minimize the overall annualized costs for TIC systems. The unit costs for TIC technologies are substantially lower than are the unit costs for new GTs. Therefore, the installation of a new TIC technology to an existing GT system may offer the lowest-cost choice for increasing the output capability of that older system. For new installations, the inclusion of an optimal TIC technology with the new GT should result in minimizing the annualized cost for output power. Typical unit costs for TIC options range from \$25 (evaporative) to \$200 (refrigeration) per kW_e of incremental power output, vs. \$300 to \$500 per kW_e for a new single cycle GT. Selection of the most suitable TIC option for any given application (new or retrofit) requires a careful evaluation of (1) the local power requirements and revenues for power generation, (2) the local meteorological conditions, and (3) the capabilities and costs of different TIC technology options, offered by the various TIC vendors (see www.turbineinletcooling.org).

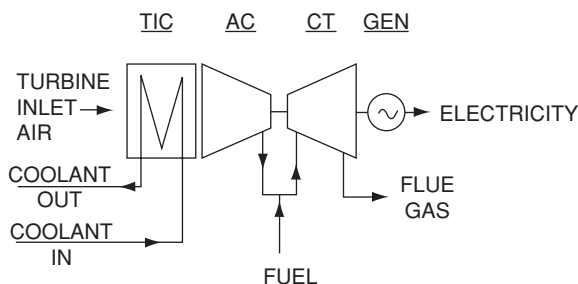


FIG. 24-65 Mechanical refrigeration TIC systems utilizing chilled water for cooling turbine inlet air, and cooling towers to reject the waste heat into the environment, account for the majority of refrigeration TIC systems sold.

SECTION 25

Materials of
Construction

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



OLIVER W. SIEBERT, KEVIN M. BROOKS
LAURENCE J. CRAIGIE, F. GALEN HODGE
L. THEODORE HUTTON, THOMAS M. LARONGE
DANIEL H. POPE, SIMON J. SCOTT
JOHN G. STOECKER II

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Materials of Construction*

Oliver W. Siebert, P.E., B.S.M.E. *Affiliate Professor of Chemical Engineering, Washington University, St. Louis, Mo.; Director, North Central Research Institute; President and Principal, Siebert Materials Engineering, Inc.; Registered Professional Engineer (California, Missouri); Fellow, American Institute of Chemical Engineers; Fellow, American Society of Mechanical Engineers (Founding Member and Chairman RTP Corrosion Resistant Equipment Committee; Lifetime Honorary Member RTP Corrosion Resistant Equipment Committee); Fellow, National Association of Corrosion Engineers International (Board of Directors; presented International NACE Conference Plenary Lecture; received NACE Distinguished Service Award); Fellow, American College of Forensic Examiners; Life Member, American Society for Metals International; Life Member, American Welding Society; Life Member, Steel Structures Painting Council; granted three patents for welding processes; Sigma Xi, Pi Tau Sigma, Tau Beta Pi (Section Editor, Corrosion)*

Kevin M. Brooks, P.E., B.S.Ch.E. *Vice President Engineering and Construction, Koch Knight LLC; Registered Professional Engineer (Ohio) (Inorganic Nonmetallics)*

Laurence J. Craigie, B.S.Chem. *Composite Resources, LLC; industry consultant in regulatory, manufacturing, and business needs for the composite industry; Member, American Society of Mechanical Engineers (Chairman RTP Corrosion Resistant Equipment Committee); Member, American Society of Testing and Materials; Member, National Association of Corrosion Engineers International; Member, Composite Fabricators of America (received President's Award) (Reinforced Thermosetting Plastic)*

F. Galen Hodge, Ph.D. (Materials Engineering), P.E. *Associate Director, Materials Technology Institute; Registered Professional Corrosion Engineer (California); Fellow, American Society for Metals International; Fellow, National Association of Corrosion Engineers International (Metals)*

L. Theodore Hutton, B.S.Mech.&Ind.Eng. *Senior Business Development Engineer, ARKEMA, Inc.; Member, American Welding Society [Chairman Committee G1A; Vice Chairman B-2 (Welding Thermoplastics)]; Member, American Society of Mechanical Engineers (Chairman BPE Polymer Subcommittee); Member, National Fire Protection Association; Member, German Welding Society; Member, American Glovebox Society (Chairman Standards Committee); Member, American Rotomolding Society; author, ABC's of PVDF Rotomolding; Editor, Plastics and Composites Welding Handbook; holds patent for specialized Kynar PVDF material for radiation shielding (Organic Thermoplastics)*

Thomas M. Laronge, M.S.Phys.Chem. *Director, Thomas M. Laronge, Inc.; Member, Cooling Technology Institute (Board of Directors; President; Editor-in-Chief, CTI Journal); Member, National Association of Corrosion Engineers International (received NACE International Distinguished Service Award; presented International NACE Conference Plenary Lecture); Phi Kappa Phi (Failure Analysis)*

J. Ian Munro, P.E., B.A.Sc.E.E. *Senior Consultant, Corrosion Probes, Inc.; Registered Professional Engineer (Ontario, Canada); Member, National Association of Corrosion Engineers International; Member, The Electrochemical Society; Member, Technical Association of Pulp & Paper Industry (Anodic Protection)*

Daniel H. Pope, Ph.D. (Microbiology) *President and Owner, Bioindustrial Technologies, Inc.; Member, National Association of Corrosion Engineers International; Sigma Xi (Microbiologically Influenced Corrosion)*

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Simon J. Scott, B.S.Ch.E. *President and Principal, Scott & Associates; Member, American Society of Mechanical Engineers (Vice Chairman RTP Corrosion Resistant Equipment Committee, Composite Structures); Member, National Association of Corrosion Engineers International; Director, American Composites Manufacturing Association (Organic Plastics)*

John G. Stoecker II, B.S.M.E. *Principal Consultant, Stoecker & Associates; Member, National Association of Corrosion Engineers International; Member, American Society for Metals International; author/editor of two handbooks on microbiologically influenced corrosion published by NACE International (Microbiologically Influenced Corrosion)*

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INTRODUCTION

The selection of materials of construction for the equipment and facilities to produce any and all chemicals is a Keystone subject of chemical engineering. The chemical products desired cannot be manufactured without considering the selection of the optimum materials of construction used as the containers for the safe, economical manufacture, and required product quality, i.e., production, handling, transporting, and storage of the products desired. Therefore, within this Section, the selection of materials of construction [for use within the chemical process industries (CPI), and by their consumers] is guided by the general subjects addressed herein,

properties unique to the materials of construction, corrosion of those materials by those chemicals, effect of the products of corrosion upon the product quality, etc. In the cases where specific (and time-sensitive) materials data are needed, that instructive information is to be found in the current reports, technical papers, handbooks (and other texts), etc., of the various *other* engineering disciplines, e.g., American Society of Metals, ASM; American Society for Testing and Materials, ASTM; American Society of Mechanical Engineers, ASME; National Association Corrosion Engineers, NACE; Society of Plastics Industry, SPI.

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INTRODUCTION*

The metallurgical extraction of the metals from their ore is the noted chemical reaction of removing the metal from its “stable” compound form (as normally found in nature) to become an “unstable,” artificial form (as used by industry to make tools, containers, equipment, etc.). That instability (of those refined metallic compounds) is the desire of

those metals to return to their (original) more stable, natural state. This is, in effect, the (oversimplified) explanation of the corrosion of artificial metallic things. In its simplest form, iron ore exists in nature as one of several iron oxide (or sulfur, etc.) compounds. For example, when refined iron and/or steel is exposed to oxygenated moisture (recall, this is an electrochemical reaction), thus an electrolyte (e.g. water) is required along with oxygen, and what is formed is iron rust (the same compounds as are the stable state/forms of iron in nature). Those (electrochemical) reactions are called *corrosion of metals*; later it is shown that this very necessary distinction is made to fit that electrochemical definition; i.e., only metals *corrode*, whereas nonmetallic materials may deteriorate (or in other ways be destroyed or weakened), but not corroded.

When a metallic material of construction (MOC) is selected to contain, transport, and/or to be exposed to a specific chemical, unless we make a correct, viable, and optimum MOC selection, the life expectancy of those facilities, in a given chemical exposure, can be very short. For the inexperienced in this field, the direct capital costs of the MOC facet of the production of chemicals, the funds spent to maintain these facilities (sometimes several times those initial capital costs), the indirect costs that are associated with outages and loss of production, off-quality product (because of equipment and facility maintenance) as well as from contamination of the product, etc., are many times not even considered, let alone used as one of the major criteria in the selection of that MOC as well as its costs to keep the plant running, i.e., a much overlooked cost figure in the CPI. To emphasize the magnitude and overall economic nature of the direct and indirect (nonproductive) costs/losses that result from the action of corrosion of our metallic facilities, equipment, and the infrastructures, within the United States, Congress has mandated that a survey of the costs of corrosion in the United States be conducted periodically.

The most recent study was conducted by CC Technologies Laboratories, Inc. (circa 1999 to 2001), with support by the Federal Highways Administration and the National Association of Corrosion Engineers, International. The results of the study show that the (estimated) total annual direct costs of corrosion in the United States are \$276 billion, i.e., about 3.1 percent of the U.S. Gross Domestic Product (GDP). That

*Includes information excerpted from papers noted, with the courtesy of ASM, ASTM, and NACE International.

loss to the economy is greater than the GDP of many smaller countries. For example, almost 50 percent of the U.S. steel production is used to compensate for the loss of corroded manufacturing facilities and products; in turn, the petroleum industry spends upward of \$2 million per day due to the corrosion of underground installations, e.g., tanks, piping, and other structures. None of those figures include any indirect costs resulting from corrosion, found to be about as great as the direct costs shown in the study. These indirect costs are difficult to come by because they include losses to the customers and other users and result in a major loss to the overall economy itself due to loss of productivity; at the same time, there are innumerable losses that can only be guessed at. In addition to those economic losses, other factors, e.g., health and safety, are without a method to quantify. The details of this study can be found in the supplement to the July 2002 NACE journal *Materials Performance*, "Corrosion Costs and Preventive Strategies, in the United States," which summarized the FHWA-funded study. It is interesting to note that a similar government-mandated study reported a decade ago in the Seventh Edition of *Perry's Chemical Engineers' Handbook* listed that annual loss at \$300 billion; the earlier evaluation technique was to numerically update (extrapolate) the results of earlier studies, i.e., not nearly so sophisticated as was this 2000 study. A study (similar to the year 2000 U.S. evaluation) was conducted by Dr. Rajan Bhaskaran, of Tamilnadu, India, who has proposed a technique to quantify the global costs of corrosion, both direct and indirect. That global study was published by the American Society for Metals, ASM, in the *ASM Handbook*, vol. 13B, December 2005.

The editors of the "Materials of Construction" section expect that the reader knows little about corrosion; thus, an attempt has been made to present information to engineers of all backgrounds.

A word of caution: Metals, materials in general, chemicals used to study metals in the laboratory, chemicals used for corrosion protection, and essentially any chemicals should be (1) used in compliance with all applicable codes, laws, and regulations; (2) handled by trained and experienced individuals in keeping with workmanlike environmental and safety standards; and (3) disposed only using allowable methods and in allowable quantities.

FLUID CORROSION

In the selection of materials of construction for a particular fluid system, it is important first to take into consideration the **characteristics of the system**, giving special attention to all factors that may influence corrosion. Since these factors would be peculiar to a particular system, it is impractical to attempt to offer a set of hard and fast rules that would cover all situations.

The **materials** from which the system is to be fabricated are the second important consideration; therefore, knowledge of the characteristics and general behavior of materials when exposed to certain environments is essential.

In the absence of factual corrosion information for a particular set of fluid conditions, a reasonably good selection would be possible from data based on the resistance of materials to a very similar environment. These data, however, should be used with considerable reservations. Good practice calls for applying such data for preliminary screening. Materials selected thereby would require further study in the fluid system under consideration.

FLUID CORROSION: GENERAL

Introduction Corrosion is the destructive attack upon a metal by its environment or with sufficient damage to its properties, such that it can no longer meet the design criteria specified. Not all metals and their alloys react in a consistent manner when in contact with corrosive fluids. One of the common intermediate reactions of a metal surface is achieved with oxygen, and those reactions are variable and complex. Oxygen can sometimes function as an electron acceptor and cause cathodic depolarization by removing the "protective" film of hydrogen from the cathodic area. In other cases, oxygen can form protective oxide films. The long-term stability of these films also varies: some are soluble in the environment, others form more stable and inert passive films. Electrochemically, a metal surface is in the *active*

state (the anode), i.e., in which the metal tends to corrode, or is being corroded. When a metal is *passive*, it is in the cathodic state, i.e., the state of a metal when its behavior is much more noble (resists corrosion) than its position in the emf series would predict. Passivity is the phenomenon of an (electrochemically) unstable metal in a given electrolyte remaining observably unchanged for an extended period of time.

Metallic Materials Pure metals and their alloys tend to enter into **chemical** union with the elements of a corrosive medium to form stable compounds similar to those found in nature. When metal loss occurs in this way, the compound formed is referred to as the **corrosion product** and the metal surface is spoken of as being **corroded**.

Corrosion is a complex phenomenon that may take any one or more of several forms. It is usually confined to the metal surface, and this is called **general corrosion**. But it sometimes occurs along defective and/or weak grain boundaries or other lines of weakness because of a difference in resistance to attack or local electrolytic action.

In most aqueous systems, the corrosion reaction is divided into an anodic portion and a cathodic portion, occurring simultaneously at discrete points on metallic surfaces. Flow of electricity from the anodic to the cathodic areas may be generated by local cells set up either on a single metallic surface (because of local point-to-point differences on the surface) or between dissimilar metals.

Nonmetallics As stated, corrosion of metals applies specifically to chemical or electrochemical attack. The deterioration of plastics and other nonmetallic materials, which are susceptible to swelling, crazing, cracking, softening, and so on, is essentially **physiochemical** rather than electrochemical in nature. Nonmetallic materials can either be rapidly deteriorated when exposed to a particular environment or, at the other extreme, be practically unaffected. Under some conditions, a nonmetallic may show evidence of gradual deterioration. However, it is seldom possible to evaluate its chemical resistance by measurements of weight loss alone, as is most generally done for metals.

FLUID CORROSION: LOCALIZED

Pitting Corrosion Pitting is a form of corrosion that develops in highly localized areas on the metal surface. This results in the development of cavities or pits. They may range from deep cavities of small diameter to relatively shallow depressions. Pitting examples: aluminum and stainless alloys in aqueous solutions containing chloride. **Inhibitors** are sometimes helpful in preventing pitting.

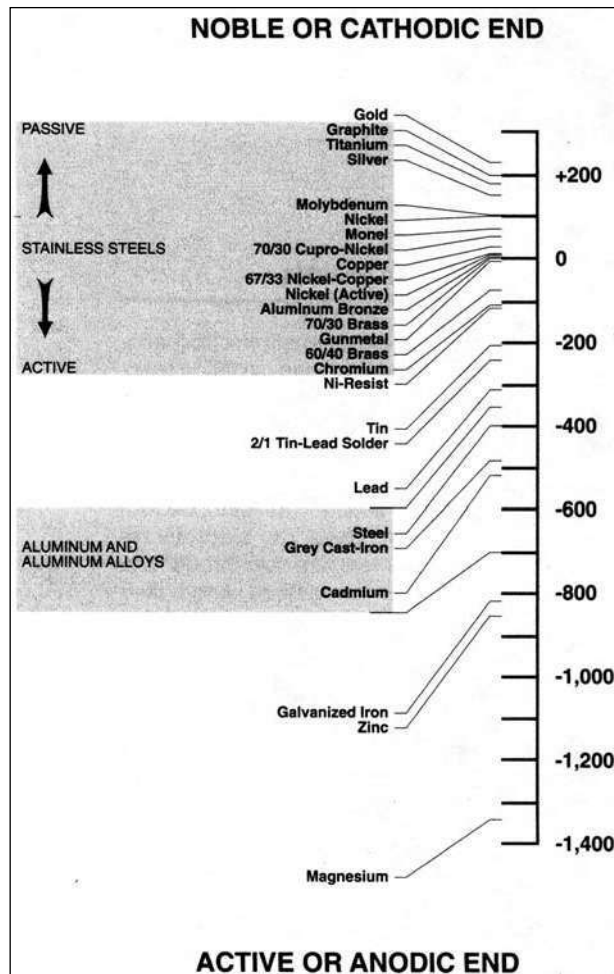
Crevice Corrosion Crevice corrosion occurs within or adjacent to a crevice formed by contact with another piece of the same or another metal or with a nonmetallic material. When this occurs, the intensity of attack is usually more severe than on surrounding areas of the same surface.

This form of corrosion can result because of a deficiency of oxygen in the crevice, acidity changes in the crevice, buildup of ions in the crevice, or depletion of an inhibitor.

Oxygen-Concentration Cell The oxygen-concentration cell is an electrolytic cell in which the driving force to cause corrosion results from a difference in the amount of oxygen in solution at one point as compared with another. Corrosion is accelerated where the oxygen concentration is least, for example, in a stuffing box or under gaskets. This form of corrosion will also occur under solid substances that may be deposited on a metal surface and thus shield it from ready access to oxygen. Redesign or change in mechanical conditions must be used to overcome this situation.

Galvanic Corrosion Galvanic corrosion is the corrosion rate above normal that is associated with the flow of current to a less active metal (cathode) in contact with a more active metal (anode) in the same environment. Table 25-1 shows the **galvanic** series of various metals. It should be used with caution, since exceptions to this series in actual use are possible. However, as a general rule, when dissimilar metals are used in contact with each other and are exposed to an electrically conducting solution, combinations of metals that are as close as possible in the galvanic series should be chosen. Coupling two metals widely separated in this series generally will produce accelerated attack on the more active metal. Often, however, protective oxide films and other effects will tend to reduce galvanic corrosion. Galvanic corrosion can, of course, be prevented by **insulating** the metals from

TABLE 25-1 Practical Galvanic Series of Metals and Alloys



This is a composite galvanic series from a variety of sources and is not necessarily representative of any one particular environment.

each other. For example, when plates are bolted together, specially designed plastic washers can be used.

Potential differences leading to galvanic-type cells can also be set up on a single metal by differences in temperature, velocity, or concentration (see subsection "Crevice Corrosion").

Area effects in galvanic corrosion are very important. An unfavorable area ratio is a large cathode and a small anode. Corrosion of the anode may be 100 to 1,000 times greater than if the two areas were the same. This is the reason why stainless steels are susceptible to rapid pitting in some environments. Steel rivets in a copper plate will corrode much more severely than a steel plate with copper rivets.

Intergranular Corrosion Selective corrosion in the grain boundaries of a metal or alloy without appreciable attack on the grains or crystals themselves is called intergranular corrosion. When severe, this attack causes a loss of strength and ductility out of proportion to the amount of metal actually destroyed by corrosion.

The **austenitic stainless steels** that are not stabilized or that are not of the extra-low-carbon types, when heated in the temperature range of 450 to 843°C (850 to 1,550°F), have chromium-rich compounds (chromium carbides) precipitated in the grain boundaries. This causes grain-boundary impoverishment of chromium and makes the affected metal susceptible to intergranular corrosion in many environments. Hot nitric acid is one environment which causes severe

intergranular corrosion of austenitic stainless steels with grain-boundary precipitation. Austenitic stainless steels stabilized with niobium (columbium) or titanium to decrease carbide formation or containing less than 0.03 percent carbon are normally not susceptible to grain-boundary deterioration when heated in the given temperature range. Unstabilized austenitic stainless steels or types with normal carbon content, to be immune to intergranular corrosion, should be given a solution anneal. This consists of heating to 1,090°C (2,000°F), holding at this temperature for a minimum of 1 h/in of thickness, followed by rapidly quenching in water (or, if impractical because of large size, rapidly cooling with an air-water spray).

Stress-Corrosion Cracking Corrosion can be accelerated by stress, either residual internal stress in the metal or externally applied stress. Residual stresses are produced by deformation during fabrication, by unequal cooling from high temperature, and by internal structural rearrangements involving volume change. Stresses induced by rivets and bolts and by press and shrink fits can also be classified as residual stresses. Tensile stresses at the surface, usually of a magnitude equal to the yield stress, are necessary to produce stress-corrosion cracking. However, failures of this kind have been known to occur at lower stresses.

Virtually every alloy system has its specific environment conditions which will produce stress-corrosion cracking, and the time of exposure required to produce failure will vary from minutes to years. Typical examples include cracking of cold-formed brass in ammonia environments, cracking of austenitic stainless steels in the presence of chlorides, cracking of Monel in hydrofluosilicic acid, and caustic embrittlement cracking of steel in caustic solutions.

This form of corrosion can be prevented in some instances by eliminating high stresses. Stresses developed during fabrication, particularly during welding, are frequently the main source of trouble. Of course, temperature and concentration are also important factors in this type of attack.

Presence of **chlorides** does not generally cause cracking of austenitic stainless steels when temperatures are below about 50°C (120°F). However, when temperatures are high enough to concentrate chlorides on the stainless surface, cracking may occur when the chloride concentration in the surrounding media is a few parts per million. Typical examples are cracking of heat-exchanger tubes at the crevices in rolled joints and under scale formed in the vapor space below the top tube sheet in vertical heat exchangers. The cracking of stainless steel under insulation is caused when chloride-containing water is concentrated on the hot surfaces. The chlorides may be leached from the insulation or may be present in the water when it enters the insulation. Improved design and maintenance of insulation weatherproofing, coating of the metal prior to the installation of insulation, and use of chloride-free insulation are all steps which will help to reduce (but not eliminate) this problem.

Serious stress-corrosion-cracking failures have occurred when chloride-containing hydrotest water was not promptly removed from stainless-steel systems. Use of potable-quality water and complete draining after test comprise the most reliable solution to this problem. Use of chloride-free water is also helpful, especially when prompt drainage is not feasible.

In handling caustic, as-welded steel can be used without developing caustic-embrittlement cracking if the temperature is below 50°C (120°F). If the temperature is higher and particularly if the concentration is above 30 percent, cracking at and adjacent to non-stress-relieved welds frequently occurs.

Liquid-Metal Corrosion Liquid metals can also cause corrosion failures. The most damaging are liquid metals which penetrate the metal along grain boundaries to cause catastrophic failure. Examples include mercury attack on aluminum alloys and attack of stainless steels by molten zinc or aluminum. A fairly common problem occurs when galvanized-structural-steel attachments are welded to stainless piping or equipment. In such cases it is mandatory to remove the galvanizing completely from the area which will be heated above 260°C (500°F).

Erosion Erosion of metal is the **mechanical** destruction of a metal by abrasion or attrition caused by the flow of liquid or gas (with or without suspended solids); **in no manner is this metal loss an electrochemical corrosion mechanism** (see Velocity Accelerated

Corrosion, below). The use of harder materials and changes in velocity or environment are methods employed to prevent erosion attack.

Velocity Accelerated Corrosion This phenomenon is sometimes (**incorrectly**) referred to as erosion-corrosion or velocity corrosion. It occurs when damage is accelerated by the **fluid exceeding its critical flow velocity at that temperature, in that metal. For that system, this is an undesirable** removal of corrosion products (such as oxides) which would otherwise tend to stifle the corrosion reaction.

Corrosion Fatigue Corrosion fatigue is a reduction by corrosion of the ability of a metal to withstand **cyclic or repeated stresses**. The surface of the metal plays an important role in this form of damage, as it will be the most highly stressed and at the same time subject to attack by the corrosive media. Corrosion of the metal surface will lower fatigue resistance, and stressing of the surface will tend to accelerate corrosion.

Under cyclic or repeated stress conditions, rupture of protective oxide films that prevent corrosion takes place at a greater rate than that at which new protective films can be formed. Such a situation frequently results in formation of anodic areas at the points of rupture; these produce pits that serve as stress-concentration points for the origin of cracks that cause ultimate failure.

Cavitation Formation of transient voids or vacuum bubbles in a liquid stream passing over a surface is called cavitation. This is often encountered around propellers, rudders, and struts and in pumps. When these bubbles collapse on a metal surface, there is a severe impact or explosive effect that can cause considerable mechanical damage, and corrosion can be greatly accelerated because of the destruction of protective films. Redesign or a more resistant metal is generally required to avoid this problem.

Fretting Corrosion This attack occurs when metals slide over each other and cause mechanical damage to one or both. In such a case, frictional heat oxidizes the metal and this oxide then wears away; or the mechanical removal of protective oxides results in exposure of fresh surface for corrosive attack. Fretting corrosion is minimized by using harder materials, minimizing friction (via lubrication), or designing equipment so that no relative movement of parts takes place.

Hydrogen Attack At elevated temperatures and significant hydrogen partial pressures, hydrogen will penetrate carbon steel, reacting with the carbon in the steel to form methane. The pressure generated causes a loss of ductility (hydrogen embrittlement) and failure by cracking or blistering of the steel. The removal of the carbon from the steel (decarburization) results in decreased strength. Resistance to this type of attack is improved by alloying with molybdenum or chromium. Accepted limits for the use of carbon and low-alloy steels are shown in the so-called Nelson curves; see American Petroleum Institute (API) Publication 941, *Steels for Hydrogen Service at Elevated Temperatures and Pressures in Petroleum Refineries and Petrochemical Plants*.

Hydrogen damage can also result from hydrogen generated in electrochemical corrosion reactions. This phenomenon is most commonly observed in solutions of specific weak acids. H_2S and HCN are the most common, although other acids can cause the problem. The atomic hydrogen formed on the metal surface by the corrosion reaction diffuses into the metal and forms molecular hydrogen at microvoids in the metal. The result is failure by embrittlement, cracking, and blistering.

FLUID CORROSION: STRUCTURAL

Graphitic Corrosion Graphitic corrosion usually involves **gray cast iron** in which metallic iron is converted into corrosion products, leaving a residue of intact graphite mixed with iron-corrosion products and other insoluble constituents of cast iron.

When the layer of graphite and corrosion products is impervious to the solution, corrosion will cease or slow down. If the layer is porous, corrosion will progress by galvanic behavior between graphite and iron. The rate of this attack will be approximately that for the maximum penetration of steel by pitting. The layer of graphite formed may also be effective in reducing the galvanic action between cast iron and more noble alloys such as bronze used for valve trim and impellers in pumps.

Low-alloy cast irons frequently demonstrate a superior resistance to graphitic corrosion, apparently because of their denser structure and the development of more compact and more protective graphitic coatings. Highly alloyed **austenitic cast irons** show considerable superiority over gray cast irons to graphitic corrosion because of the more noble potential of the austenitic matrix plus more protective graphitic coatings.

Carbon steels heated for prolonged periods at temperatures above 455°C (850°F) may be subject to the segregation of carbon, which is transformed into graphite. When this occurs, the structural strength of the steel will be affected. Killed steels or low-alloy steels of chromium and molybdenum or chromium and nickel should be considered for elevated-temperature services.

Parting, or Dealloying, Corrosion* This type of corrosion occurs when only one component of an alloy is selectively removed by corrosion or leaching. The most common type of parting or dealloying is dezincification of a copper zinc brass, i.e., such as the parting of zinc from the brass, leaving a copper residue (see below). Various kinds of selective dissolution have been named after the alloy family that has been affected, usually on the basis of the dissolved metal (except in the case of graphitic corrosion; see "Graphitization" above). Similar selective corrosion also may lead to terms such as *denickelification* and *demolybdenumization*, etc. The element removed is always anodic to the alloy matrix. While the color of the damaged alloy may change, there is no [apparent (macro)] evidence of a loss of metal, shape, or dimensions and generally, even the original surface and contour remains. That said, the affected metal becomes lighter and porous and loses its original mechanical properties.

Dezincification Dezincification is corrosion of a brass alloy containing zinc in which the principal product of corrosion is metallic copper. This may occur as plugs filling pits (plug type) or as continuous layers surrounding an unattacked core of brass (general type). The mechanism may involve overall corrosion of the alloy followed by redeposition of the copper from the corrosion products or selective corrosion of zinc or a high-zinc phase to leave copper residue. This form of corrosion is commonly encountered in brasses that contain more than 15 percent zinc and can be either eliminated or reduced by the addition of small amounts of **arsenic, antimony, or phosphorus** to the alloy.

Microbiologically Influenced Corrosion (MIC)[†] This brief review is presented from a practical, industrial point of view. Subjects include materials selection, operational, and other considerations that real-world facilities managers and engineers and others charged with preventing and controlling corrosion need to take into account to prevent or minimize potential MIC problems. As a result of active research by investigators worldwide in the last 30 years, MIC is now recognized as a problem in most industries, including the petroleum production and transportation, gas pipeline, water distribution, fire protection, storage tank, nuclear and fossil power, chemical process, and pulp and paper industries.

A seminal summary of the evolutionary study leading to the discovery of a unique type of MIC, the final identification of the mechanism, and its control can be found in Daniel H. Pope, "State-of-the-Art Report on Monitoring, Prevention and Mitigation of Microbiologically Influenced Corrosion in the Natural Gas Industry," Report No. 96-0488, Gas Research Institute.

Microbiologically influenced corrosion is defined by the National Association of Corrosion Engineers as any form of corrosion that is influenced by the presence and/or activities of microorganisms. Although MIC appears to many humans to be a new phenomenon, it is not new to the microbes themselves. Microbial transformation of metals in their elemental and various mineral forms has been an essential part of material cycling on earth for billions of years. Some forms of metals such as reduced iron and manganese serve as energy sources for microbes, while oxidized forms of some metals can substitute for

*Additional reference material came from "Dealloying Corrosion Basics," *Materials Performance*, vol. 33, no. 5, p. 62, May 2006, adapted by NACE from *Corrosion Basics—An Introduction*, by L. S. Van Dellen (ed.), NACE, Houston, Tex., 1984, pp. 105–107.

[†]Excerpted from papers by Daniel H. Pope, John G. Stoecker II, and Oliver W. Siebert, courtesy of NACE International and the Gas Research Institute.

oxygen as electron acceptors in microbial metabolism. Other metals are transformed from one physical and chemical state to another as a result of exposure to environments created by microbes performing their normal metabolic activities. Of special importance are microbial activities which create oxidizing, reducing, acidic, or other conditions under which one form of a metal is chemically transformed to another. It is important to understand that the microbes are simply doing "what comes naturally." Unfortunately when microbial communities perform their natural activities on metals and alloys which would rather be in less organized and more natural states (minerals), corrosion often results.

Most microbes in the real world, especially those associated with surfaces, live in communities consisting of many different types of microbes, each of which can perform a variety of biochemical reactions. This allows microbial communities to perform a large variety of different reactions and processes which would be impossible for any single type of microbe to accomplish alone. Thus, e.g., even in overtly aerobic environments, microbial communities and the metal surfaces underlying them can have zones in which little or no oxygen is present. The result is that aerobic, anaerobic, fermentative, and other metabolic-type reactions can *all* occur in various locations within a microbial community. When these conditions are created on an underlying metal surface, then physical, chemical, and electrochemical conditions are created in which a variety of corrosion mechanisms can be induced, inhibited, or changed in their forms or rates. These include oxygen concentration cell corrosion, ion concentration cell corrosion, under-deposit acid attack corrosion, crevice corrosion, and under-deposit pitting corrosion. Note, however, that all these corrosion processes are electrochemical.

Most practicing engineers are not, and do not need to become, experts in the details of MIC. What is needed is to recognize that MIC-type corrosion can affect almost any metal or alloy exposed to MIC-related microbes in untreated waters, and therefore many types of equipment and structures are at risk. It is critical that MIC be properly diagnosed, or else mitigation methods designed to control MIC may be misapplied, resulting in failure to control the corrosion problem, unnecessary cost, and unnecessary concerns about exposure of the environment and personnel to potentially toxic biological control agents. Fortunately better tools are now available for monitoring and detection of MIC (see the later subsections on laboratory and field corrosion testing, both of which address the subject of MIC). Microbiological, chemical, metallurgical, and operational information is all useful in the diagnosis of MIC and should be used if available. All types of information should conform to the diagnosis of MIC—the data should not be in conflict with one another.

Bacteria, as a group, can grow over very wide ranges of pH, temperature, and pressure. They can be obligate aerobes (require oxygen to survive and grow), microaerophiles (require low oxygen concentrations), facultative anaerobes (prefer aerobic conditions but will live under anaerobic conditions), or obligate anaerobes (will grow only under conditions where oxygen is absent). It should be emphasized that most anaerobes will survive aerobic conditions for quite a while, and the same is true for aerobes in anaerobic conditions. Most MIC-related bacteria are heterotrophic and as a group may use as food almost any available organic carbon molecules, from simple alcohols or sugars to phenols and petroleum products, to wood or various other complex polymers. Unfortunately some MIC-related microbial communities can also use some biocides and corrosion inhibitors as food stuffs. Other microbes are autotrophs (fix CO_2 , as do plants). Some microbes use inorganic elements or ions (e.g., NH_3 , NO_2 , CH_3 , H_2 , S , H_2S , Fe^{2+} , Mn^{2+} , etc.), as sources of energy. Although microbes can exist in extreme conditions, most require a limited number of organic molecules, moderate temperatures, moist environments, and near-neutral bulk environmental pH.

Buried Structures There has been no dramatic improvement in the protection of buried structures against MIC over the last several decades. Experience has been that coating systems, by themselves, do not provide adequate protection for a buried structure over the years; for best results, a properly designed and maintained cathodic protection (CP) system *must* be used in conjunction with a protective coating (regardless of the quality of the coating, as applied) to control

MIC and other forms of corrosion. Adequate levels of CP (the level of CP required is dependent on local environmental conditions, e.g., soil pH, moisture, presence of sealing chemicals) provide caustic environment protection at the holes (holidays) in the coating that are sure to develop with time due to one cause or another. The elevated pH (>10.0) produced by adequate CP discourages microbial growth and metabolism and tends to neutralize acids which are produced as a result of microbial metabolism and corrosion processes. Proper levels of CP, if applied uniformly to the metal surface, also raise the electrochemical potential of the steel to levels at which it does not want to corrode. Areas of metal surface under disbonded coating, under preexisting deposits (including those formed due to microbial actions), and other materials acting to insulate areas of the pipe and "holidays" from achieving adequate CP will often not be protected and may suffer very rapid under-deposit, crevice, and pitting corrosion. In short, adequate CP must be applied *before* MIC communities have become established under disbonded coating or in holidays. Application of CP after MIC processes and sites have been established may *not* stop MIC from occurring.

The user of cathodic protection must also consider the material being protected with regard to caustic cracking; a cathodic potential driven to the negative extreme of -0.95 V for microbiological protection purposes can cause caustic cracking of a steel structure. The benefits and risks of cathodic protection must be weighed for each material and each application.

Backfilling with limestone or other alkaline material is an added step to protect buried structures from microbiological damage. Providing adequate drainage to produce a dry environment both above and below ground in the area of the buried structure will also reduce the risk of this type of damage.

Corrosion of buried structures has been blamed on the sulfate-reducing bacteria (SRB) for well over a century. It was easy to blame the SRB for the corrosion as they smelled very bad (rotten egg smell). It is now known that SRB are one component of the MIC communities required to get corrosion of most buried structures.

Waters Water is required at a MIC "site" to allow microbial growth and corrosion reactions to occur. Most surfaces exposed to natural or industrial environments have large numbers of potential MIC-related microbes associated with them. Most natural and industrial waters (even "ultrapure," distilled, or condensate waters) contain large numbers of microbes. Since the potential to participate in MIC is a property of a large percentage of known microbes, it is not surprising that the potential for developing MIC is present in most natural and industrial environments on earth. Many industries assumed that they were protected against MIC as they used ultrapure waters, in which they assumed microbes were kept in check by the lack of organic food sources for the microbes. However, as several early cases of MIC in the chemical process industry demonstrated, MIC was capable of causing rapid and severe damage to stainless steel welds which had come into contact only with potable drinking water. Since that time, numerous cases of MIC have been reported in breweries; pharmaceutical, nuclear, and computer chip manufacturing; and other industries using highly purified waters. Many other cases of MIC have been documented in metals in contact with "normally treated" municipal waters.

Hydrostatic Testing Waters Microbes capable of causing MIC are present in most waters (even those treated by water purveyors to kill pathogens) used for hydrostatic (safety) testing of process equipment and for process batch waters. Use of these waters has resulted in a large number of documented cases of MIC in a variety of industries. Guidelines for treatment and use of hydrotest waters have been adopted by several industrial and government organizations in an effort to prevent this damage. Generally, good results have been reported for those who have followed this practice. Unfortunately, this can be an expensive undertaking where the need cannot be totally quantified (and thus justified to management). Cost-cutting practices which either ignore these guidelines or follow an adulteration of proven precautions can lead to major MIC damage to equipment and process facilities.

Untreated natural freshwaters from wells, lakes, or rivers commonly contain high levels of MIC-related microbes. These waters should not be used without appropriate treatment. Most potable

waters are treated sufficiently to prevent humans from having contact with waterborne pathogenic bacteria, but *are not* treated with sufficient disinfectant to kill all MIC-related microbes in the water.

They should not be used for hydrotesting or other such activities without appropriate treatment. Biocide treatment of hydrotest waters should be very carefully chosen to make sure that the chemicals are compatible with the materials in the system to be tested and to prevent water disposal problems (most organic biocides cause disposal problems). Use of inexpensive, effective, and relatively accepted biocides such as chlorine, ozone, hydrogen peroxide and iodine should be considered where compatibility with materials and other considerations permit. (For example, use of relatively low levels of iodine in hydrotest waters might be acceptable in steel pipes while stainless steels are much better tested using waters treated with nonhalogen, but oxidizing biocides such as hydrogen peroxide.) Obviously, chlorine must be used only with great care because of the extensive damage it will cause to the 300-series austenitic stainless steels. In all cases, as soon as the test is over, the water *must be completely* drained and the system thoroughly dried so that no vestiges of water are allowed to be trapped in occluded areas. The literature abounds with instructions as to the proper manner in which to accomplish the necessary and MIC-safe testing procedures. Engineering personnel planning these test operations should avail themselves of that knowledge.

Materials of Construction MIC processes are those processes by which manufactured materials deteriorate through the presence and activities of microbes. These processes can be either direct or indirect. Microbial biodeterioration of a great many materials (including concretes, glasses, metals and their alloys, and plastics) occurs by diverse mechanisms and usually involves a complex community consisting of many different species of microbes.

The corrosion engineers' solution to corrosion problems sometimes includes upgrading the materials of construction. This is a natural approach, and since microbiological corrosion most often results in crevice or under-deposit attack, this option is logical. Unfortunately, with MIC, the use of more materials traditionally thought to be more corrosion-resistant can lead to disastrous consequences. The occurrence and severity of any particular case of MIC are dependent upon the types of microbes involved, the local physical environment (temperature, water flow rates, etc.) and chemical environment (pH, hardness, alkalinity, salinity, etc.) and the type of metals or alloys involved. As an example, an upgrade from type 304 to 316 stainless steel does not always help. Kobrin reported biological corrosion of delta ferrite stringers in weld metal. Obviously, this upgrade was futile; type 316 stainless steel can contain as much as or more delta ferrite than does type 304. Kobrin also reported MIC of nickel, nickel-copper alloy 400, and nickel-molybdenum alloy B heat-exchanger tubes. Although the alloy 400 and alloy B were not pitted as severely as the nickel tubes, the use of higher alloys did not solve the corrosion problem.

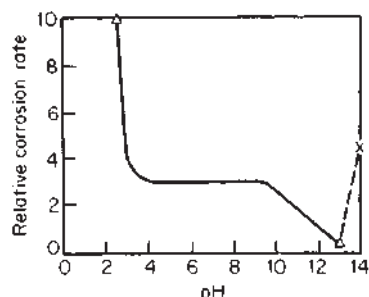
In the past, copper was believed to be toxic to *most* microbiological species. Although this may be true in a test tube under laboratory conditions, it is not generally true in the real world. In this real world, microbial communities excrete slime layers which tend to sequester the copper ions and prevent their contact with the actual microbial cells, thus preventing the copper from killing the microbes. Many cases of MIC in copper and copper alloys have been documented, especially of heat-exchange tubes, potable water, and fire protection system piping.

At this stage of knowledge about MIC, only titanium, zirconium, and tantalum appear to be immune to microbiological damage.

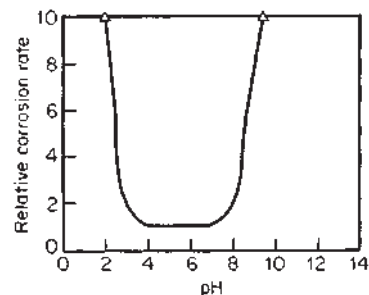
FACTORS INFLUENCING CORROSION

Solution pH The corrosion rate of most metals is affected by pH. The relationship tends to follow one of three general patterns:

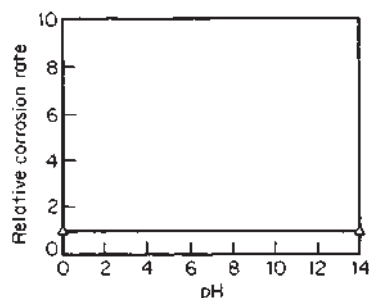
1. Acid-soluble metals such as iron have a relationship as shown in Fig. 25-1a. In the middle pH range (≈ 4 to 10), the corrosion rate is controlled by the rate of transport of oxidizer (usually dissolved O_2) to the metal surface. Iron is weakly amphoteric. At very high temperatures such as those encountered in boilers, the corrosion rate increases with increasing basicity, as shown by the dashed line.



(a)



(b)



(c)

FIG. 25-1 Effect of pH on the corrosion rate. (a) Iron. (b) Amphoteric metals (aluminum, zinc). (c) Noble metals.

2. Amphoteric metals such as aluminum and zinc have a relationship as shown in Fig. 25-1b. These metals dissolve rapidly in either acidic or basic solutions.

3. Noble metals such as gold and platinum are not appreciably affected by pH, as shown in Fig. 25-1c.

Oxidizing Agents In some corrosion processes, such as the solution of zinc in hydrochloric acid, hydrogen may evolve as a gas. In others, such as the relatively slow solution of copper in sodium chloride, the removal of hydrogen, which must occur so that corrosion may proceed, is effected by a reaction between hydrogen and some oxidizing chemical such as oxygen to form water. Because of the high rates of corrosion which usually accompany hydrogen evolution, metals are rarely used in solutions from which they evolve hydrogen at an appreciable rate. As a result, most of the corrosion observed in practice occurs under conditions in which the oxidation of hydrogen to form water is a necessary part of the corrosion process. For this reason, oxidizing agents are often powerful accelerators of corrosion, and in many cases the **oxidizing power of a solution** is its most important single property insofar as corrosion is concerned.

Oxidizing agents that accelerate the corrosion of some materials may also retard corrosion of others through the formation on their surface

of oxides or layers of adsorbed oxygen which make them more resistant to chemical attack. This property of chromium is responsible for the principal corrosion-resisting characteristics of the stainless steels.

It follows, then, that oxidizing substances, such as dissolved air, may accelerate the corrosion of one class of materials and retard the corrosion of another class. In the latter case, the behavior of the material usually represents a balance between the power of oxidizing compounds to preserve a protective film and their tendency to accelerate corrosion when the agencies responsible for protective-film breakdown are able to destroy the films.

Temperature The rate of corrosion tends to increase with rising temperature. Temperature also has a secondary effect through its influence on the solubility of air (oxygen), which is the most common oxidizing substance influencing corrosion. In addition, temperature has specific effects when a temperature change causes phase changes which introduce a corrosive second phase. Examples include condensation systems and systems involving organics saturated with water.

Velocity Most metals and alloys are protected from corrosion, not by *nobility* [a metal's inherent resistance to enter into an electrochemical reaction with that environment, e.g., the (intrinsic) inertness of gold to (almost) everything but aqua regia], but by the formation of a *protective film* on the surface. In the examples of film-forming protective cases, the film has similar, but more limiting, specific assignment of that exemplary-type resistance to the exposed environment (not nearly so broad-based as noted in the case of gold). Velocity-accelerated corrosion is the accelerated or increased rate of deterioration or attack on a metal surface because of relative movement between a corrosive fluid and the metal surface, i.e., the instability (velocity sensitivity) of that protective film.

An increase in the velocity of relative movement between a corrosive solution and a metallic surface frequently tends to accelerate corrosion. This effect is due to the higher rate at which the corrosive chemicals, including oxidizing substances (air), are brought to the corroding surface and to the higher rate at which corrosion products, which might otherwise accumulate and stifle corrosion, are carried away. The higher the velocity, the thinner will be the films which corroding substances must penetrate and through which soluble corrosion products must diffuse.

Whenever corrosion resistance results from the formation of layers of insoluble corrosion products on the metallic surface, the effect of high velocity may be to prevent their normal formation, to remove them after they have been formed, and/or to preclude their reformation. All metals that are protected by a film are sensitive to what is referred to as its *critical velocity*; i.e., the velocity at which those conditions occur is referred to as the *critical velocity* of that chemistry/temperature/velocity environmental corrosion mechanism. When the critical velocity of that specific system is exceeded, that effect allows corrosion to proceed unhindered. This occurs frequently in small-diameter tubes or pipes through which corrosive liquids may be circulated at high velocities (e.g., condenser and evaporator tubes), in the vicinity of bends in pipelines, and on propellers, agitators, and centrifugal pumps. Similar effects are associated with cavitation and mechanical erosion.

Films Once corrosion has started, its further progress very often is controlled by the nature of films, such as passive films, that may form or accumulate on the metallic surface. The classical example is the thin **oxide film** that forms on stainless steels.

Insoluble corrosion products may be completely impervious to the corroding liquid and, therefore, completely protective; or they may be quite permeable and allow local or general corrosion to proceed unhindered. Films that are nonuniform or discontinuous may tend to localize corrosion in particular areas or to induce accelerated corrosion at certain points by initiating electrolytic effects of the concentration-cell type. Films may tend to retain or absorb moisture and thus, by delaying the time of drying, increase the extent of corrosion resulting from exposure to the atmosphere or to corrosive vapors.

It is agreed generally that the characteristics of the **rust films** that form on steels determine their resistance to atmospheric corrosion. The rust films that form on low-alloy steels are more protective than those that form on unalloyed steel.

In addition to films that originate at least in part in the corroding metal, there are others that originate in the corrosive solution. These

include various salts, such as carbonates and sulfates, which may be precipitated from heated solutions, and insoluble compounds, such as "beer stone," which form on metal surfaces in contact with certain specific products. In addition, there are films of oil and grease that may protect a material from direct contact with corrosive substances. Such oil films may be applied intentionally or may occur naturally, as in the case of metals submerged in sewage or equipment used for the processing of oily substances.

Other Effects Stream **concentration** can have important effects on corrosion rates. Unfortunately, corrosion rates are seldom linear with concentration over wide ranges. In equipment such as distillation columns, reactors, and evaporators, concentration can change continuously, making prediction of corrosion rates rather difficult. Concentration is important during plant shutdown; presence of moisture that collects during cooling can turn innocuous chemicals into dangerous corrosives.

As to the effect of time, there is no universal law that governs the reaction for all metals. Some corrosion rates remain constant with time over wide ranges, others slow down with time, and some alloys have increased corrosion rates with respect to time. Situations in which the corrosion rate follows a combination of these paths can develop. Therefore, extrapolation of corrosion data and corrosion rates should be done with utmost caution.

Impurities in a corrodent can be good or bad from a corrosion standpoint. An impurity in a stream may act as an inhibitor and actually retard corrosion. However, if this impurity is removed by some process change or improvement, a marked rise in corrosion rates can result. Other impurities, of course, can have very deleterious effects on materials. The chloride ion is a good example; small amounts of chlorides in a process stream can break down the passive oxide film on stainless steels. The effects of impurities are varied and complex. One must be aware of what they are, how much is present, and where they come from before attempting to recommend a particular material of construction.

HIGH-TEMPERATURE ATTACK

Physical Properties The suitability of an alloy for high-temperature service [425 to 1,100°C (800 to 2,000°F)] is dependent upon properties inherent in the alloy composition and upon the conditions of application. Crystal structure, density, thermal conductivity, electrical resistivity, thermal expansivity, structural stability, melting range, and vapor pressure are all physical properties basic to and inherent in individual alloy compositions.

Of usually high relative importance in this group of properties is **expansivity**. A surprisingly large number of metal failures at elevated temperatures are the result of excessive thermal stresses originating from constraint of the metal during heating or cooling. Such constraint in the case of hindered contraction can cause rupturing.

Another important property is alloy **structural stability**. This means freedom from formation of new phases or drastic rearrangement of those originally present within the metal structure as a result of thermal experience. Such changes may have a detrimental effect upon strength or corrosion resistance or both.

Mechanical Properties Mechanical properties of wide interest include creep, rupture, short-time strengths, and various forms of ductility, as well as resistance to impact and fatigue stresses. Creep strength and stress rupture are usually of greatest interest to designers of stationary equipment such as vessels and furnaces.

Corrosion Resistance Possibly of greater importance than physical and mechanical properties is the ability of an alloy's chemical composition to resist the corrosive action of various hot environments. The forms of high-temperature corrosion which have received the greatest attention are **oxidation** and **scaling**.

Chromium is an essential constituent in alloys to be used above 550°C (1,000°F). It provides a tightly adherent oxide film that materially retards the oxidation process. Silicon is a useful element in imparting oxidation resistance to steel. It will enhance the beneficial effects of chromium. Also, for a given level of chromium, experience has shown oxidation resistance to improve as the nickel content increases.

Aluminum is not commonly used as an alloying element in steel to improve oxidation resistance, as the amount required interferes with

both workability and high-temperature-strength properties. However, the development of high-aluminum surface layers by various methods, including spraying, cementation, and dipping, is a feasible means of improving heat resistance of low-alloy steels.

Contaminants in fuels, especially alkali-metal ions, vanadium, and sulfur compounds, tend to react in the combustion zone to form molten fluxes which dissolve the protective oxide film on stainless steels, allowing oxidation to proceed at a rapid rate. This problem is becoming more common as the high cost and short supply of natural gas and distillate fuel oils force increased usage of residual fuel oils and coal.

COMBATING CORROSION

Material Selection The objective is to select the material which will most economically fulfill the process requirements. The best source of data is well-documented experience in an identical process unit. In the absence of such data, other data sources such as experience in pilot units, corrosion-coupon tests in pilot or bench-scale units, laboratory corrosion-coupon tests in actual process fluids, or corrosion-coupon tests in synthetic solutions must be used. The data from such alternative sources (which are listed in decreasing order of reliability) must be properly evaluated, taking into account the degree to which a given test may fail to reproduce actual conditions in an operating unit. Particular emphasis must be placed on possible composition differences between a static laboratory test and a dynamic plant as well as on trace impurities (chlorides in stainless-steel systems, for example) which may greatly change the corrosiveness of the system. The possibility of severe localized attack (pitting, crevice corrosion, or stress-corrosion cracking) must also be considered.

Permissible **corrosion rates** are an important factor and differ with equipment. Appreciable corrosion can be permitted for tanks and lines if anticipated and allowed for in design thickness, but essentially no corrosion can be permitted in fine-mesh wire screens, orifices, and other items in which small changes in dimensions are critical.

In many instances use of **nonmetallic materials** will prove to be attractive from an economic and performance standpoint. These should be considered when their strength, temperature, and design limitations are satisfactory.

Proper Design Design considerations with respect to minimizing corrosion difficulties should include the desirability for free and complete drainage, minimizing crevices, and ease of cleaning and inspection. The installation of baffles, stiffeners, and drain nozzles and the location of valves and pumps should be made so that free drainage will occur and washing can be accomplished without holdup. Means of access for inspection and maintenance should be provided whenever practical. Butt joints should be used whenever possible. If lap joints employing fillet welds are used, the welds should be continuous.

The use of dissimilar metals in contact with each other should generally be minimized, particularly if they are widely separated in their nominal positions in the galvanic series (see Table 28-1a). If they are to be used together, consideration should be given to insulating them from each other or making the anodic material area as large as possible.

Equipment should be supported in such a way that it will not rest in pools of liquid or on damp insulating material. Porous insulation should be weatherproofed or otherwise protected from moisture and spills to avoid contact of the wet material with the equipment. Specifications should be sufficiently comprehensive to ensure that the desired composition or type of material will be used and the right condition of heat treatment and surface finish will be provided. Inspection during fabrication and prior to acceptance is desirable.

Altering the Environment Simple changes in environment may make an appreciable difference in the corrosion of metals and should be considered as a means of combating corrosion. **Oxygen** is an important factor, and its removal or addition may cause marked changes in corrosion. The treatment of boiler feedwater to remove oxygen, for instance, greatly reduces the corrosiveness of the water on steel. Inert-gas purging and blanketing of many solutions, particularly acidic media, generally minimize corrosion of copper and nickel-base alloys by minimizing air or oxygen content. Corrosiveness of acid media to stainless alloys, on the other hand, may be reduced by aeration because of the formation of passive oxide films. Reduction in

temperature will almost always be beneficial with respect to reducing corrosion if no corrosive phase changes (condensation, for example) result. Velocity effects vary with the material and the corrosive system. When pH values can be modified, it will generally be beneficial to hold the acid level to a minimum. When acid additions are made in batch processes, it may be beneficial to add them last so as to obtain maximum dilution and minimum acid concentration and exposure time. Alkaline pH values are less critical than acid values with respect to controlling corrosion. Elimination of moisture can and frequently does minimize, if not prevent, corrosion of metals, and this possibility of environmental alteration should always be considered.

Inhibitors The use of various substances or inhibitors as additives to corrosive environments to decrease corrosion of metals in the environment is an important means of combating corrosion. This is generally most attractive in closed or recirculating systems in which the annual cost of inhibitor is low. However, it has also proved to be economically attractive for many once-through systems, such as those encountered in petroleum-processing operations. Inhibitors are effective as the result of their controlling influence on the cathode- or anode-area reactions.

Typical examples of inhibitors used for minimizing corrosion of iron and steel in aqueous solutions are the chromates, phosphates, and silicates. Organic sulfide and amine materials are frequently effective in minimizing corrosion of iron and steel in acid solution.

The use of inhibitors is not limited to controlling corrosion of iron and steel. They frequently are effective with stainless steel and other alloy materials. The addition of copper sulfate to dilute sulfuric acid will sometimes control corrosion of stainless steels in hot dilute solutions of this acid, whereas the uninhibited acid causes rapid corrosion.

The effectiveness of a given inhibitor generally increases with an increase in **concentration**, but inhibitors considered practical and economically attractive are used in quantities of less than 0.1 percent by weight.

In some instances the amount of inhibitor present is critical in that a deficiency may result in localized or pitting attack, with the overall results being more destructive than when none of the inhibitor is present. Considerations for the use of inhibitors should therefore include review of experience in similar systems or investigation of requirements and limitations in new systems.

Cathodic Protection This electrochemical method of corrosion control has found wide application in the protection of carbon steel underground structures such as pipe lines and tanks from external soil corrosion. It is also widely used in water systems to protect ship hulls, offshore structures, and water-storage tanks.

Two methods of providing cathodic protection for minimizing corrosion of metals are in use today. These are the sacrificial-anode method and the impressed-emf method. Both depend upon making the metal to be protected the cathode in the electrolyte involved.

Examples of the sacrificial-anode method include the use of zinc, magnesium, or aluminum as anodes in electrical contact with the metal to be protected. These may be anodes buried in the ground for protection of underground pipe lines or attachments to the surfaces of equipment such as condenser water boxes or on ship hulls. The current required is generated in this method by corrosion of the sacrificial-anode material. In the case of the impressed emf, the direct current is provided by external sources and is passed through the system by use of essentially nonsacrificial anodes such as carbon, noncorrodible alloys, or platinum buried in the ground or suspended in the electrolyte in the case of aqueous systems.

The requirements with respect to current distribution and anode placement vary with the resistivity of soils or the electrolyte involved.

Anodic Protection* Corrosion of metals, and their alloys, exposed to a given environment requires at least two separate electrochemical (anodic and cathodic) reactions. The corrosion rate is determined at the intersection of these two reactions (see Fig. 25-2a).

Certain metal-electrolyte combinations exhibit active-passive behavior. Carbon steel in concentrated sulfuric acid is a classic example. The surface condition of a metal that has been forced inactive is termed *passive*.

*Citing of these several publications about anodic protection is noted with the courtesy of NACE International.

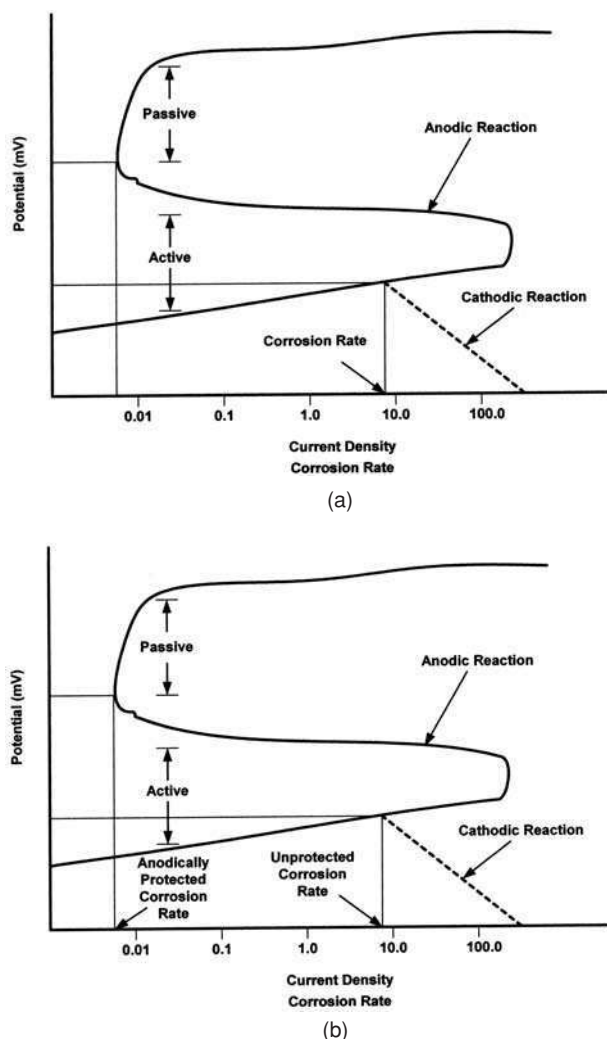


FIG. 25-2 (a) Active-passive behavior. (b) Application of anodic protection.

Anodic protection (AP) is an important method for controlling corrosion when/whereby the corroding (anodic surface) of the metal can be passivated by discharging a current from the surface of that metal, whereby the resulting corrosion product is a protective film. This technique has been known and practiced for nearly 50 years. This electrochemical anodic corrosion protection method relies on an automatic potential controlled current source (potentiostat) to maintain the metal or alloy in a noncorroding (passive) state. See Fig. 25-2b [and Fig. 25-14, the application of the corrosion behavior diagram to select the low-current region (LCR) in the design of an AP system].

The development of improved control instrumentation [e.g., of cathode location (placements), etc.] and many years of proven AP applications in the field have made AP the preferred method of controlling corrosion of uncoated steel equipment handling hot, concentrated sulfuric acid, stainless steel in even hotter exposures, and even steel in nitric acid.

Routinely, AP is used to control corrosion of carbon steel exposed to caustic-sulfide and caustic-aluminate solutions encountered in the pulp and paper and aluminum industries.

With the added benefit of increased product purity (reduced iron contamination), near-zero operational costs speak to AP as the most underapplied corrosion control systems of the ages.

A more recent advancement of AP has come from the application of a controlled cathodic current which can be utilized to shift the corrosion potential back to the passive zone. This (refinement) technique is usually termed the *cathodic potential adjustment protection* (CPAP).

See the NACE Papers: Oliver W. Siebert, "Correlation of Laboratory Electrochemical Investigations with Field Applications of Anodic Protection," *Materials Performance*, vol. 20, no. 2, pp. 38-43, February 1981; "Anodic Protection," *Materials Performance*, vol. 28, no. 11, p. 28, November 1989, adapted by NACE from "Corrosion Basics—An Introduction." (Houston, Tex.: NACE, 1984, pp. 105-107); J. Ian Munro and Winston W. Shim, "Anodic Protection—Its Operation and Applications," vol. 41, no. 5, pp. 22-24, May 2001; and a two-part series, J. Ian Munro, "Anodic Protection of White and Green Kraft Liquor Tankage, Part I, Electrochemistry of Kraft Liquors," and Part II, "Anodic Protection Design and System Operation," *Materials Performance*, vol. 42, no. 2, pp. 22-26, February 2002, and vol. 42, no. 3, pp. 24-28, March 2002.

Coatings and Linings The use of nonmetallic coatings and lining materials in combination with steel or other materials has and will continue to be an important type of construction for combating corrosion.

Organic coatings of many kinds are used as linings in equipment such as tanks, piping, pumping lines, and shipping containers, and they are often an economical means of controlling corrosion, particularly when freedom from metal contamination is the principal objective. One principle that is now generally accepted is that thin nonreinforced paintlike coatings of less than 0.75-mm (0.03-in) thickness should not be used in services for which full protection is required in order to prevent rapid attack of the substrate metal. This is true because most thin coatings contain defects or holidays and can be easily damaged in service, thus leading to early failures due to corrosion of the substrate metal even though the coating material is resistant. Electrical testing for continuity of coating-type linings is always desirable for immersion-service applications in order to detect holiday-type defects in the coating.

The most dependable barrier linings for corrosive services are those which are bonded directly to the substrate and are built up in multiple-layer or laminated effects to thicknesses greater than 2.5 mm (0.10 in). These include flake-glass-reinforced resin systems and elastomeric and plasticized plastic systems. Good surface preparation and thorough inspections of the completed lining, including electrical testing, should be considered as minimum requirements for any lining applications.

Linings of this type are slightly permeable to many liquids. Such permeation, while not damaging to the lining, may cause failure by causing disbonding of the lining owing to pressure buildup between the lining and the steel.

Ceramic or carbon-brick linings are frequently used as facing linings over plastic or membrane linings when surface temperatures exceed those which can be handled by the unprotected materials or when the membrane must be protected from mechanical damage. This type of construction permits processing of materials that are too corrosive to be handled in low-cost metal constructions.

Glass-Lined Steel By proprietary methods, special glasses can be bonded to steel, providing an impervious liner 1.5 to 2.5 mm (0.060 to 0.100 in) thick. Equipment and piping lined in this manner are routinely used in severely corrosive acid services. The glass lining can be mechanically damaged, and careful attention to details of design, inspection, installation, and maintenance is required to achieve good results with this system.

Metallic Linings for Severe/Corrosive Environments The cladding of steel with an alloy is another approach to this problem. There are a number of cladding methods in general use. In one, a sandwich is made of the corrosion-resistant metal and carbon steel by hot rolling to produce a **pressure weld** between the plates.

Another process involves **explosive bonding**. The corrosion-resistant metal is bonded to a steel backing metal by the force generated by properly positioned explosive charges. Relatively thick sections of metal can be bonded by this technique into plates.

In a third process, a **loose liner** is fastened to a carbon steel shell by welds spaced so as to prevent collapse of the liner. A fourth method is **weld overlay**, which involves depositing multiple layers of alloy weld metal to cover the steel surface.

All these methods require careful design and control of fabrication methods to assure success.

Metallic Linings for Mild Environments Zinc coatings applied by various means have good corrosion resistance to many atmospheres. Such coatings have been extensively used on steel. Zinc has the advantage of being anodic to steel and therefore will protect exposed areas of steel by electrochemical action.

Steel coated with tin (**tinplate**) is used to make food containers. Tin is more noble than steel; therefore, well-aerated solutions will galvanically accelerate attack of the steel at exposed areas. The comparative absence of air within food containers aids in preserving the tin as well as the food. Also the reversible potential which the tin-iron couple undergoes in organic acids serves to protect exposed steel in food containers.

Cadmium, being anodic to steel, behaves quite similarly to zinc in providing corrosion protection when applied as a coating on steel. Tests of zinc and cadmium coatings should be conducted when it becomes necessary to determine the most economical selection for a particular environment.

Lead has a good general resistance to various atmospheres. As a coating, it has had its greatest application in the production of terneplate, which is used as a roofing, cornicing, and spouting material.

Aluminum coatings on steel will perform in a manner similar to zinc coatings. Aluminum has good resistance to many atmospheres; in addition, being anodic to steel, it will galvanically protect exposed areas. Aluminum-coated steel products are quite serviceable under high-temperature conditions, for which good oxidation resistance is required.

General Workflow for Minimizing or Controlling Corrosion
In general, the process of reducing and controlling corrosion of metals requires the minimization of the progress of electrochemical deterioration. To accomplish this goal without first changing the selection of material(s), the engineer should generate and follow a punchlist of items the goal of which is to remove essentially any material differences/gradients in the environment of the corroding material(s). Since concentration differences/gradients constitute chemical stress or stress risers that tend to drive corrosion reactions, reducing or eliminating such material differences/gradients constitutes reducing or eliminating stress on the corrosion reaction. Removing the chemical stress places the metal "at rest" with respect to the progress of corrosion.

CORROSION-TESTING METHODS*

The primary purpose of materials selection is to provide the optimum equipment for a process application in terms of materials of construction, design, and corrosion-control measures. *Optimum* here means that which comprises the best combination of cost, life, safety, and reliability.

The selection of materials to be used in design dictates a basic understanding of the behavior of materials and the principles that govern such behavior. If proper design of suitable materials of construction is incorporated, the equipment should deteriorate at a uniform and anticipated gradual rate, which will allow scheduled maintenance or replacement at regular intervals. If localized forms of corrosion are characteristic of the combination of materials and environment, the materials engineer should still be able to predict the probable life of equipment, or devise an appropriate inspection schedule to preclude unexpected failures. The concepts of predictive, or at least preventive, maintenance are minimum requirements to proper materials selection. This approach to maintenance is certainly intended to minimize the possibility of unscheduled production shutdowns because of corrosion failures, with their attendant possible financial losses, hazard to personnel and equipment, and resultant environmental pollution.

Chemical processes may involve a complex variety of both inorganic and organic chemicals. Hard and fast rules for selecting the appropriate materials of construction can be given when the composition is known, constant, and free of unsuspected contaminants; when the rel-

evant parameters of temperature, pressure, velocity, and concentration are defined; and when the mechanical and environmental degradation of the material is uniform, that is, free of localized attack. For example, it is relatively simple to select the materials of construction for a regimen of equipment for the storage and handling of cold, concentrated sulfuric acid. On the other hand, the choice of suitable materials for producing phosphoric acid by the digestion of phosphate rock with sulfuric acid is much more difficult because of the diversity in kind and concentration of contaminants, the temperatures of the reactions, and the strength of sulfuric and phosphoric acid used or formed. Probably the best way to approach the study of materials selection is to categorize the types of major chemicals that might be encountered, describe their inherent characteristics, and generalize about the corrosion characteristics of the prominent materials of construction in such environments.

The background information that materials selection is based on is derived from a number of sources. In many cases, information as to the corrosion resistance of a material in a specific environment is not available and must be derived experimentally. It is to this need that the primary remarks of this subsection are addressed.

Unfortunately, there is no standard or preferred way to evaluate an alloy in an environment. While the chemistry of the operating plant environment can sometimes be duplicated in the laboratory, factors of velocity, hot and cold wall effects, crevice, chemical reaction of the fluid during the test, stress levels of the equipment, contamination with products of corrosion, trace impurities, dissolved gases, and so forth also have a controlling effect on the quality of the answer. Then, too, the progress of the corrosion reaction itself varies with time. Notwithstanding, immersion testing remains the most widely used method for selecting materials of construction.

There is no standard or preferred way to carry out a corrosion test; the method must be chosen to suit the purpose of the test. The principal types of tests are, in decreasing order of reliability:

1. Actual operating experience with full-scale plant equipment exposed to the corroding medium.
2. Small-scale plant-equipment experience, under either commercial or pilot-plant conditions.
3. Sample tests in the field. These include coupons, stressed samples, electrical-resistance probes exposed to the plant corroding medium, or samples exposed to the atmosphere, to soils, or to fresh, brackish, or saline waters. Samples for viable microbes involved in MIC must be processed immediately in the field into appropriate growth media.
4. Laboratory tests on samples exposed to "actual" plant liquids or simulated environments should be done only when testing in the actual operating environment cannot be done. When MIC is a factor in the test, microbial communities from the actual environment of interest must be used. Pure cultures of single types of microbes cannot provide conditions present in the actual operating environment.

Plant or field corrosion tests are useful for

1. Selection of the most suitable material to withstand a particular environment and to estimate its probable durability in that environment
2. Study of the effectiveness of means of preventing corrosion

CORROSION TESTING: LABORATORY TESTS

Metals and alloys do not respond alike to all the influences of the many factors that are involved in corrosion. Consequently, it is impractical to establish any universal standard laboratory procedures for corrosion testing except for inspection tests. However, some details of laboratory testing need careful attention in order to achieve useful results.

In the selection of materials for the construction of a chemical plant, resistance to the corroding medium is often the determining factor; otherwise, the choice will fall automatically on the cheapest material mechanically suitable. Laboratory corrosion tests are frequently the quickest and most satisfactory means of arriving at a preliminary selection of the most suitable materials to use. Unfortunately, however, it is not yet within the state of the art of laboratory tests to predict with accuracy the behavior of the selected material under plant-operating conditions. The outstanding difficulty lies not so much in carrying out the test as in interpreting the results and translating

*Includes information from papers by Oliver W. Siebert, John G. Stoecker II, and Ann Van Orden, courtesy of NACE International; Oliver W. Siebert and John R. Scully, courtesy of ASTM; John R. Scully and Robert G. Kelly, courtesy of ASM; and Metal Samples Company, Division of Alabama Specialty Products Company, Munford, Alabama.

them into terms of plant performance. A laboratory test of the conventional type gives mainly one factor—the chemical resistance of the proposed material to the corrosive agent. There are numerous other factors entering into the behavior of the material in the plant, such as dissolved gases, velocity, turbulence, abrasion, crevice conditions, hot-wall effects, cold-wall effects, stress levels of metals, trace impurities in corrodent that act as corrosion inhibitors or accelerators, and variations in composition of corrodent.

Immersion Test One method of determining the chemical-resistance factor, the so-called **total-immersion test**, represents an unaccelerated method that has been found to give reasonably concordant results in approximate agreement with results obtained on the large scale when the other variables are taken into account. Various other tests have been proposed and are in use, such as salt-spray, accelerated electrolytic, alternate-immersion, and aerated-total-immersion; but in view of the numerous complications entering into the translation of laboratory results into plant results the simplest test is considered the most desirable for routine preliminary work, reserving special test methods for special cases. The total-immersion test serves quite well to eliminate materials that obviously cannot be used; further selection among those materials which apparently can be used can be made on the basis of a knowledge of the properties of the materials concerned and the working conditions or by constructing larger-scale equipment of the proposed materials in which the operating conditions can be simulated.

The National Association of Corrosion Engineers (NACE) TMO169-95 "Standard Laboratory Corrosion Testing of Metals for the Process Industries," and ASTM G31 "Recommended Practice for Laboratory Immersion Corrosion Testing of Metals" are the general guides for immersion testing. Small pieces of the candidate metal are exposed to the medium, and the loss of mass of the metal is measured for a given period of time. Immersion testing remains the best method to eliminate from further consideration those materials that obviously cannot be used. This technique is frequently the quickest and most satisfactory method of making a preliminary selection of the best candidate materials.

Probably the most serious disadvantage of this method of corrosion study is the assumed average-time weight loss. The corrosion rate could be high initially and then decrease with time (it could fall to zero). In other cases the rate of corrosion might increase very gradually with time or it could cycle or be some combination of these things.

The description that follows is based on these standards.

Test Piece* The size and the shape of specimens will vary with the purpose of the test, nature of the material, and apparatus used. A large surface-to-mass ratio and a small ratio of edge area to total area are desirable. These ratios can be achieved through the use of rectangular or circular specimens of minimum thickness. Circular specimens should be cut preferably from sheet and not bar stock to minimize the exposed end grain.

A circular specimen of about 32-mm (1.25-in) diameter is a convenient shape for laboratory corrosion tests. With a thickness of approximately 3 mm ($\frac{1}{8}$ in) and an 8- or 11-mm- ($\frac{5}{16}$ - or $\frac{7}{8}$ -in-) diameter hole for mounting, these specimens will readily pass through a 45/50 ground-glass joint of a distillation kettle. The total surface area of a circular specimen is given by the equation:

$$A = \frac{\pi}{2} (D^2 - d^2) + t\pi D + t\pi d$$

where t = thickness, D = diameter of the specimen, and d = diameter of the mounting hole. If the hole is completely covered by the mounting support, the final term ($t\pi d$) in the equation is omitted.

Rectangular coupons [50 by 25 by 1.6 or 3.2 mm (2 by 1 by $\frac{1}{16}$ or $\frac{1}{8}$ in)] may be preferred as corrosion specimens, particularly if interface or liquid-line effects are to be studied by the laboratory test.

Shapes of typical (commercially available) test coupons are shown in Fig. 25-3: circular in Fig. 25-3a, rectangular in Fig. 25-3b, welded rectangular in Fig. 25-3c, and horseshoe stressed in Fig. 25-3d. Many desired shapes of coupons are also available in various sizes and can be obtained in any size, shape, material of construction, and surface finish

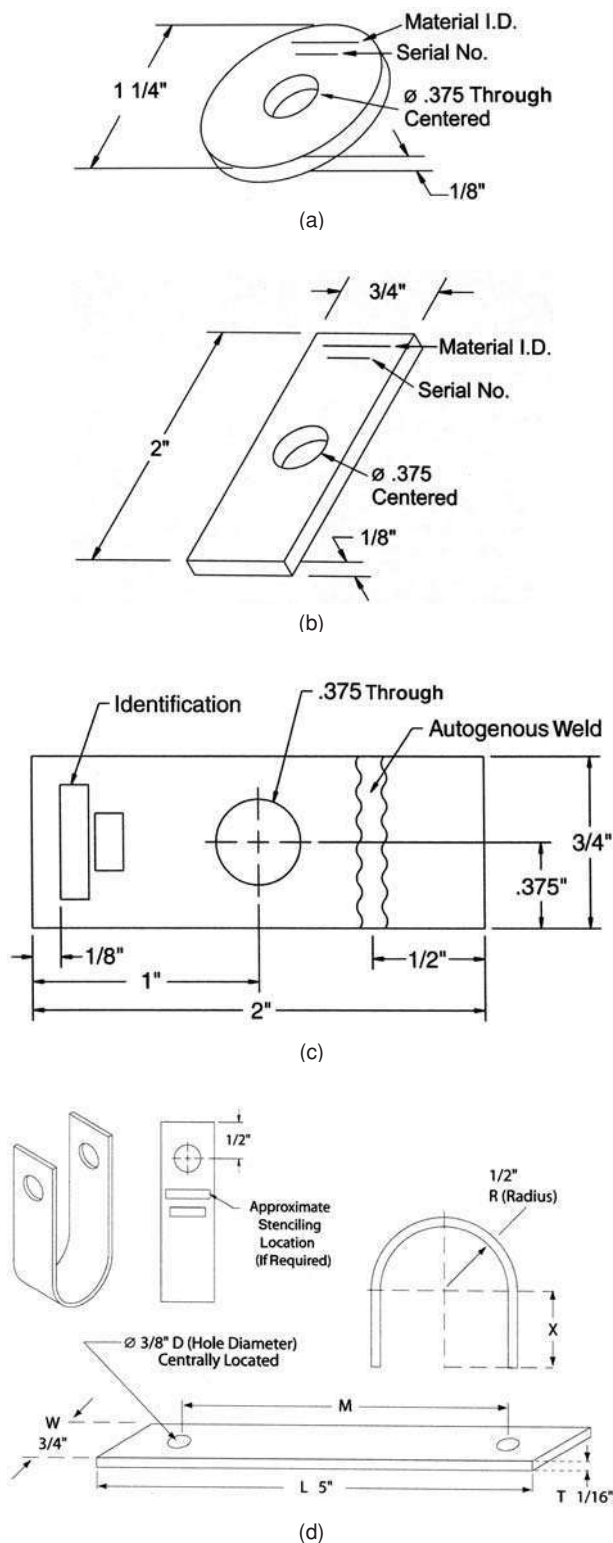


FIG. 25-3 Typical commercially available test coupons: (a) circular; (b) rectangular; (c) welded rectangular; (d) horseshoe stressed.

*Coupons and racks/holders as well as availability information are courtesy of Metal Samples, Munford, Ala.

25-14 MATERIALS OF CONSTRUCTION

required to fit unique laboratory test equipment. There are also a series of somewhat generic racks and holders for mounting corrosion coupons so that they may be installed, exposed, and recovered for examination, after the plant exposures. Several typical pieces of hardware are shown in Fig. 25-4: a pipeline insertion rack in Fig. 25-4a, a spool rack for general equipment exposures in Fig. 25-4b, and a flat bar rack for attachment to accessories within equipment in Fig. 25-4c.

All specimens should be measured carefully to permit accurate calculation of the exposed areas. An area calculation accurate to plus or minus 1 percent is usually adequate.

More uniform results may be expected if a substantial layer of metal is removed from the specimens to eliminate variations in condition of the original metallic surface. This can be done by chemical treatment (pickling), electrolytic removal, or grinding with a coarse abrasive paper or cloth, such as No. 50, using care not to work-harden the surface. At least 2.5×10^{-3} mm (0.0001 in) or 1.5 to 2.3 mg/cm² (10 to

15 mg/in²) should be removed. If clad alloy specimens are to be used, special attention must be given to ensure that excessive metal is not removed. After final preparation of the specimen surface, the specimens should be stored in a desiccator until exposure if they are not used immediately.

Specimens should be finally degreased by scrubbing with bleach-free scouring powder, followed by thorough rinsing in water and in a suitable solvent (such as acetone, methanol, or a mixture of 50 percent methanol and 50 percent ether), and air-dried. For relatively soft metals such as aluminum, magnesium, and copper, scrubbing with abrasive powder is not always needed and can mar the surface of the specimen. The use of towels for drying may introduce an error through contamination of the specimens with grease or lint. The dried specimen should be weighed on an analytic balance.

Apparatus A versatile and convenient apparatus should be used, consisting of a kettle or flask of suitable size (usually 500 to 5,000 mL), a

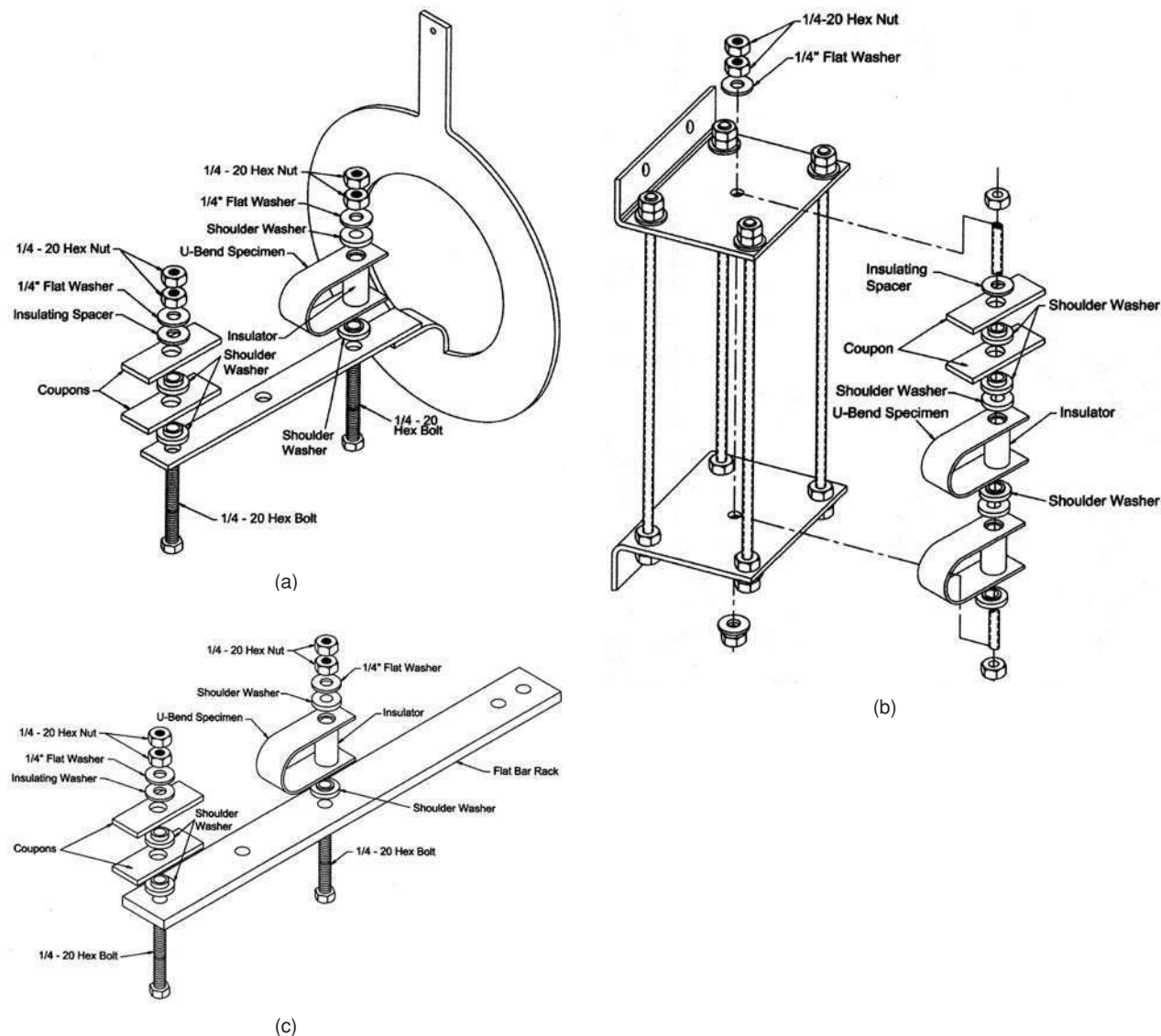


FIG. 25-4 Corrosion racks used to expose corrosion samples in operating production equipment: (a) inside pipes; (b) inside process vessels; (c) to be bolted onto baffles and brackets with process vessels.

reflux condenser with atmospheric seal, a sparger for controlling atmosphere or aeration, a thermowell and temperature-regulating device, a heating device (mantle, hot plate, or bath), and a specimen-support system. If agitation is required the apparatus can be modified to accept a suitable stirring mechanism such as a magnetic stirrer. A typical resin-flask setup for this type of test is shown in Fig. 25-5. Open-beaker tests should *not* be used because of evaporation and contamination.

In more complex tests, provisions might be needed for continuous flow or replenishment of the corrosive liquid while simultaneously maintaining a controlled atmosphere.

Heat flux apparatus for testing materials for heat-transfer applications is shown in Fig. 25-6. Here the sample is at a higher temperature than the bulk solution.

If the test is to be a guide for the selection of a material for a particular purpose, the limits of controlling factors in service must be determined. These factors include oxygen concentration, temperature, rate of flow, pH value, and other important characteristics.

The **composition of the test solution** should be controlled to the fullest extent possible and be described as thoroughly and as accurately as possible when the results are reported. Minor constituents should not be overlooked because they often affect corrosion rates. Chemical content should be reported as percentage by weight of the solution. Molarity and normality are also helpful in defining the concentration of chemicals in the test solution. The composition of the test solution should be checked by analysis at the end of the test to determine the extent of change in composition, such as might result from evaporation.

Temperature of Solution Temperature of the corroding solution should be controlled within $\pm 1^\circ\text{C}$ ($\pm 1.8^\circ\text{F}$) and must be stated in the report of test results.

For tests at ambient temperatures, the tests should be conducted at the highest temperature anticipated for stagnant storage in summer months. This temperature may be as high as 40 to 45°C (104 to 113°F) in some localities. The variation in temperature should be reported also (e.g., $40^\circ\text{C} \pm 2^\circ\text{C}$).

Aeration of Solution Unless specified, the solution should not be aerated. Most tests related to process equipment should be run with the natural atmosphere inherent in the process, such as the vapors of the boiling liquid. If aeration is used, the specimens should not be located in the direct air stream from the sparger. Extraneous effects can be encountered if the air stream impinges on the specimens.

Solution Velocity The effect of velocity is not usually determined in laboratory tests, although specific tests have been designed for this purpose. However, for the sake of reproducibility some velocity control is desirable.

Tests at the boiling point should be conducted with minimum possible heat input, and boiling chips should be used to avoid excessive turbulence and bubble impingement. In tests conducted below the boiling point, thermal convection generally is the only source of liquid velocity. In test solutions of high viscosities, supplemental controlled stirring with a magnetic stirrer is recommended.

Volume of Solution Volume of the test solution should be large enough to avoid any appreciable change in its corrosiveness through either exhaustion of corrosive constituents or accumulation of corrosion products that might affect further corrosion.

A suitable volume-to-area ratio is 20 mL (125 mL) of solution/cm² (in²) of specimen surface. This corresponds to the recommendation of ASTM Standard A262 for the Huey test. The preferred volume-to-area ratio is 40 mL/cm² (250 mL/in²) of specimen surface, as stipulated in ASTM Standard G31, Laboratory Immersion Testing of Materials.

Method of Supporting Specimens The supporting device and container should not be affected by or cause contamination of the test solution. The method of supporting specimens will vary with the apparatus used for conducting the test but should be designed to insulate the specimens from each other physically and electrically and to insulate the specimens from any metallic container or supporting device used with the apparatus.

Shape and form of the specimen support should assure free contact of the specimen with the corroding solution, the liquid line, or the vapor phase, as shown in Fig. 25-5. If clad alloys are exposed, special procedures are required to ensure that only the cladding is exposed (unless the purpose is to test the ability of the cladding to protect cut edges in the test solution). Some common supports are glass or

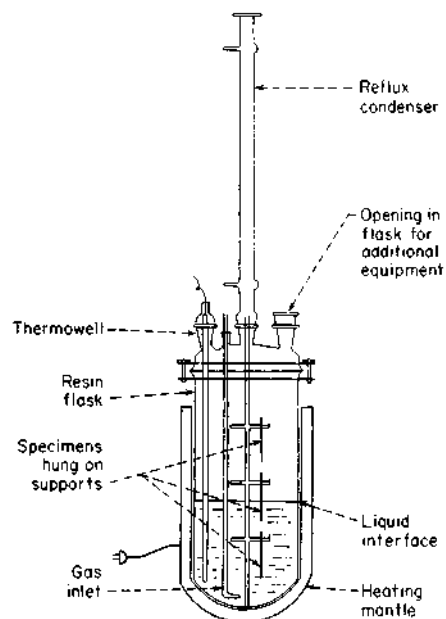


FIG. 25-5 Laboratory-equipment arrangement for corrosion testing. (Based on NACE Standard TMO169-95.)

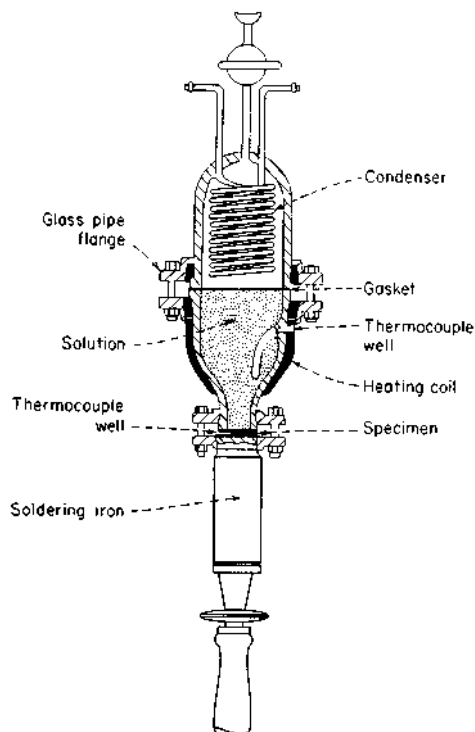


FIG. 25-6 Laboratory setup for the corrosion testing of heat-transfer materials.

ceramic rods, glass saddles, glass hooks, fluorocarbon plastic strings, and various insulated or coated metallic supports.

Duration of Test Although the duration of any test will be determined by the nature and purpose of the test, an excellent procedure for evaluating the effect of time on corrosion of the metal and also on the corrosiveness of the environment in laboratory tests has been presented by Wachter and Treseder [*Chem. Eng. Prog.*, 315-326 (June 1947)]. This technique is called the **planned-interval test**. Other procedures that require the removal of solid corrosion products between exposure periods will not measure accurately the normal changes of corrosion with time.

Materials that experience severe corrosion generally do not need lengthy tests to obtain accurate corrosion rates. Although this assumption is valid in many cases, there are exceptions. For example, lead exposed to sulfuric acid corrodes at an extremely high rate at first while building a protective film; then the rate decreases considerably, so that further corrosion is negligible. The phenomenon of forming a protective film is observed with many corrosion-resistant materials, and therefore short tests on such materials would indicate high corrosion rates and would be completely misleading.

Short-time tests also can give misleading results on alloys that form passive films, such as stainless steels. With borderline conditions, a prolonged test may be needed to permit breakdown of the passive film and subsequently more rapid attack. Consequently, tests run for long periods are considerably more realistic than those conducted for short durations. This statement must be qualified by stating that corrosion should not proceed to the point at which the original specimen size or the exposed area is drastically reduced or the metal is perforated.

If anticipated corrosion rates are moderate or low, the following equation gives a suggested test duration:

$$\text{Duration of test, } h = \frac{78,740}{\text{corrosion rate, mm/y}} \\ = \frac{2000}{\text{corrosion rate, mils/y}}$$

Cleaning Specimens after Test Before specimens are cleaned, their appearance should be observed and recorded. Locations of deposits, variations in types of deposits, and variations in corrosion products are extremely important in evaluating localized corrosion such as pitting and concentration-cell attack.

Cleaning specimens after the test is a vital step in the corrosion-test procedure and, if not done properly, can give rise to misleading test results. Generally, the cleaning procedure should remove all corrosion products from specimens with a minimum removal of sound metal. Set rules cannot be applied to cleaning because procedures will vary with the type of metal being cleaned and the degree of adherence of corrosion products.

Mechanical cleaning includes scrubbing, scraping, brushing, mechanical shocking, and ultrasonic procedures. Scrubbing with a bristle brush and a mild abrasive is the most widely used of these methods; the others are used principally as supplements to remove heavily encrusted corrosion products before scrubbing. Care should be used to avoid the removal of sound metal.

Chemical cleaning implies the removal of material from the surface of the specimen by dissolution in an appropriate chemical agent. Solvents such as acetone, carbon tetrachloride, and alcohol are used to remove oil, grease, or resin and are usually applied prior to other methods of cleaning. Various chemicals are chosen for application to specific materials; some of these treatments in general use are outlined in the NACE standard.

Electrolytic cleaning should be preceded by scrubbing to remove loosely adhering corrosion products. One method of electrolytic cleaning that has been found to be useful for many metals and alloys is as follows:

Solution: 5 percent (by weight) H_2SO_4
Anode: carbon or lead
Cathode: test specimen
Cathode current density: 20 A/dm² (129 A/in²)
Inhibitor: 2 cm³ organic inhibitor per liter
Temperature: 74°C (165°F)
Exposure period: 3 min

Precautions must be taken to ensure good electrical contact with the specimen, to avoid contamination of the solution with easily reducible metal ions, and to ensure that inhibitor decomposition has not occurred. Instead of using 2 mL of any proprietary inhibitor, 0.5 g/L of inhibitors such as diorthotolyl thiourea or quinoline ethiodide can be used.

Whatever treatment is used to clean specimens after a corrosion test, its effect in removing metal should be determined, and the weight loss should be corrected accordingly. A "blank" specimen should be weighed before and after exposure to the cleaning procedure to establish this weight loss.

Evaluation of Results After the specimens have been reweighed, they should be examined carefully. Localized attack such as pits, crevice corrosion, stress-accelerated corrosion, cracking, or intergranular corrosion should be measured for depth and area affected.

Depth of localized corrosion should be reported for the actual test period and not interpolated or extrapolated to an annual rate. The rate of initiation or propagation of pits is seldom uniform. The size, shape, and distribution of pits should be noted. A distinction should be made between those occurring underneath the supporting devices (concentration cells) and those on the surfaces that were freely exposed to the test solution. An excellent discussion of pitting corrosion has been published [*Corrosion*, 25t (January 1950)].

The specimen may be subjected to simple bending tests to determine whether any **embrittlement** has occurred.

If it is assumed that localized or internal corrosion is not present or is recorded separately in the report, the **corrosion rate** or penetration can be calculated alternatively as

$$\frac{\text{Weight loss} \times 534}{(\text{Area})(\text{time})(\text{metal density})} = \text{mils/y (mpy)}$$

$$\frac{\text{Weight loss} \times 13.56}{(\text{Area})(\text{time})(\text{metal density})} = \text{mm/y (mmpy)}$$

where weight loss is in mg, area is in in² of metal surface exposed, time is in hours exposed, and density is in g/cm³. Densities for alloys can be obtained from the producers or from various metal handbooks.

The following **checklist** is a recommended guide for reporting all **important information and data**:

- Corrosive media and concentration (changes during test)
- Volume of test solution
- Temperature (maximum, minimum, and average)
- Aeration (describe conditions or technique)
- Agitation (describe conditions or technique)
- Type of apparatus used for test
- Duration of each test (start, finish)
- Chemical composition or trade name of metals tested
- Form and metallurgical conditions of specimens
- Exact size, shape, and area of specimens
- Treatment used to prepare specimens for test
- Number of specimens of each material tested and whether specimens were tested separately or which specimens were tested in the same container

Method used to clean specimens after exposure and the extent of any error expected by this treatment

Actual weight losses for each specimen

Evaluation of attack if other than general, such as crevice corrosion under support rod, pit depth and distribution, and results of microscopic examination or bend tests

Corrosion rates for each specimen expressed as millimeters (mils) per year

Effect of Variables on Corrosion Tests It is advisable to apply a factor of safety to the results obtained, the factor varying with the degree of confidence in the applicability of the results. Ordinarily, a factor of from 3 to 10 might be considered normal.

Among the more important points that should be considered in attempting to base plant design on laboratory corrosion-rate data are the following.

Galvanic corrosion is a frequent source of trouble on a large scale. Not only is the use of different metals in the same piece of equipment dangerous, but the effect of cold working may be sufficient to establish potential differences of objectionable magnitude between different

parts of the same piece of metal. The mass of metal in chemical apparatus is ordinarily so great and the electrical resistance consequently so low that a very small voltage can cause a very high current. Welding also may leave a weld of a different physical or chemical composition from that of the body of the sheet and cause localized corrosion.

Local variations in temperature and crevices that permit the accumulation of corrosion products are capable of allowing the formation of **concentration cells**, with the result of accelerated local corrosion.

In the laboratory, the **temperature** of the test specimen is that of the liquid in which it is immersed, and the measured temperature is actually that at which the reaction is taking place. In the plant (heat being supplied through the metal to the liquid in many cases), the temperature of the film of (corrosive) liquid on the inside of the vessel may be a number of degrees higher than that registered by the thermometer. As the relation between temperature and corrosion is a logarithmic one, the rate of increase is very rapid. Like other chemical reactions, the speed ordinarily increases twofold to threefold for each 10°C temperature rise, the actual relation being that of the equation $\log K = A + (B/T)$, where K represents the rate of corrosion and T the absolute temperature. This relationship, although expressed mathematically, must be understood to be a qualitative rather than strictly a quantitative one.

Cold walls, as in coolers or condensers, usually have somewhat decreased corrosion rates for the reason just described. However, in some cases, the decrease in temperature may allow the formation of a more corrosive second phase, thereby increasing corrosion.

The effect of **impurities** in either structural material or corrosive material is so marked (while at the same time it may be either accelerating or decelerating) that for reliable results the actual materials which it is proposed to use should be tested and not types of these materials. In other words, it is much more desirable to test the actual plant solution and the actual metal or nonmetal than to rely upon a duplication of either. Since as little as 0.01 percent of certain organic compounds will reduce the rate of solution of steel in sulfuric acid 99.5 percent and 0.05 percent bismuth in lead will increase the rate of corrosion over 1000 percent under certain conditions, it can be seen how difficult it would be to attempt to duplicate here all the significant constituents.

Electrical Resistance The measurement of corrosion by electrical resistance is possible by considering the change in resistance of a thin metallic wire or strip sensing element (probe) as its cross section decreases from a loss of metal. Since small changes in resistance are encountered as corrosion progresses, changes in temperature can cause enough change in the wire resistance to complicate the results. Commercial equipment, such as the Corrosometer®, have a protected reference section of the specimen in the modified electrical Wheatstone bridge (Kelvin) circuit to compensate for these temperature changes. Since changes in the resistance ratio of the probe are not linear with loss of section thickness, compensation for this variable must be included in the circuit. In operation, the specimen probe is exposed to the environment and instrument readings are periodically recorded. The corrosion rate is the loss of metal averaged between any two readings.

The corrosion rate can be studied by this method over very short periods of time, but not instantaneously. The environment does not have to be an electrolyte. Studies can be made in corrosive gas exposures. The main disadvantage of the technique is that local corrosion (pitting, crevice corrosion, galvanic, stress corrosion cracking, fatigue, and so forth) will probably not be progressively identified. If the corrosion product has an electrical conductivity approaching that of the lost metal, little or no corrosion will be indicated. The same problem will result from the formation of conducting deposits on the specimen.

The electrical-resistance measurement has nothing to do with the electrochemistry of the corrosion reaction. It merely measures a bulk property that is dependent upon the specimen's cross-section area. Commercial instruments are available (Fig. 25-7).

Advantages of the electrical-resistance technique are:

1. A corrosion measurement can be made without having to see or remove the test sample.
2. Corrosion measurements can be made quickly—in a few hours or days, or continuously. This enables sudden increases in corrosion rate to be detected. In some cases, it will be possible then to modify the process to decrease the corrosion.

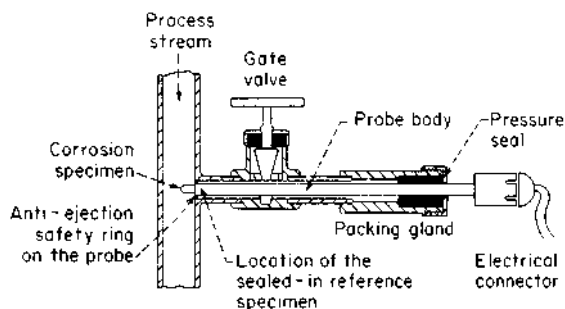


FIG. 25-7 Typical retractable corrosion probe.

3. The method can be used to monitor a process to indicate whether the corrosion rate is dependent on some critical process variable.

4. Corrodent need not be an electrolyte (in fact, need not be a liquid). The method can detect low corrosion rates that would take a long time to detect with weight-loss methods.

Limitations of the technique are:

1. It is usually limited to the measurement of uniform corrosion only and is not generally satisfactory for localized corrosion.

2. The probe design includes provisions to compensate for temperature variations. This feature is not totally successful. The most reliable results are obtained in constant-temperature systems.

EMF versus pH (Pourbaix) Diagrams Potential (EMF) versus pH equilibrium (Pourbaix) diagrams derived from physical property data about the metal and its environment provide a basis for the expression of a great amount of thermodynamic data about the corrosion reaction. These relatively simple diagrams graphically represent the thermodynamics of corrosion in terms of electromotive force, that is, an indication of oxidizing power and pH, or acidity. As an aid in corrosion prediction, their usefulness lies in providing direction for establishing a corrosion study program.

Figure 25-8 is a typical Pourbaix diagram. Generally, the diagrams show regions of immunity (the metal), passivity (the surface film), and corrosion (metallic ions). While of considerable qualitative usefulness, these diagrams have important limitations. Since they are calculated from thermodynamic properties, they represent equilibrium conditions and do not provide kinetic information. Thus, while they show conditions where corrosion will not occur, they do not necessarily indicate under what conditions corrosion will occur. To determine the quantitative value of corrosion, kinetic rate measurement would still be required. Pourbaix diagrams were developed for the study of pure metals. Since few engineering structures are made of pure metals, it is important to extend this technique to include information on the passive behavior of alloys of engineering interest. Values of the open circuit corrosion potential (OCP) or a controlled potential occur, if a steady site potential can be used in conjunction with the solution pH and these diagrams to show what component is stable in the system defined by a given pH and potential. Theoretical diagrams so developed estimate the corrosion product in various regions. The use of computers to construct diagrams for alloy systems provides an opportunity to mathematically overcome many of the limitations inherent in the pure metal system.

A potentiokinetic electrochemical hysteresis method of diagram construction has led to consideration of three-dimensional Pourbaix diagrams for alloy systems useful in alloy development, evaluation of the influence of crevices, prediction of the tendency for dealloying, and the inclusion of kinetic data on the diagram is useful in predicting corrosion rather than just the absence of damage. These diagrams are kinetic, not thermodynamic, expressions. The two should not be confused as being the same reaction, as they are not.

Tafel Extrapolation Corrosion is an electrochemical reaction of a metal and its environment. When corrosion occurs, the current that flows between individual small anodes and cathodes on the metal surface causes the electrode potential for the system to change. While

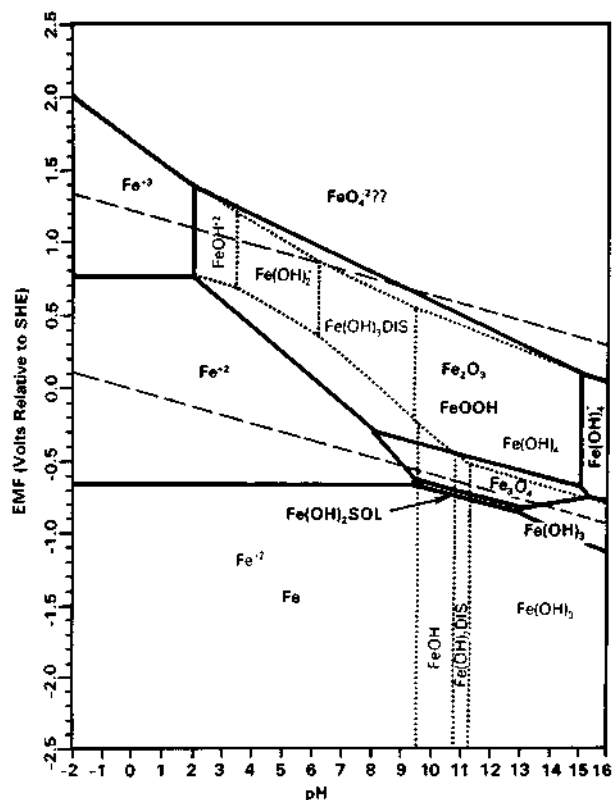


FIG. 25-8 EMF-pH diagram for an iron-water system at 25°C. All ions are at an activity of 10^{-6} .

this current cannot be measured, it can be evaluated indirectly on a metal specimen with an inert electrode and an external electrical circuit. Polarization is described as the extent of the change in potential of an electrode from its equilibrium potential caused by a net current flow to or from the electrode, galvanic or impressed (Fig. 25-9).

Electrochemical techniques have been used for years to study fundamental phenomenological corrosion reactions of metals in corrosive environments. Unfortunately, the learning curve in the reduction of these electrochemical theories to practice has been painfully slow. However, a recent survey has shown that many organizations in the chemical process industries are now adding electrochemical methods to their materials selection techniques. Laboratory electrochemical tests of metal/environment systems are being used to show the degree of compatibility and describe the limitations of those relationships. The general methods being used include electrical resistance, Tafel extrapolation, linear polarization, and both slow and rapid-scan potentiodynamic polarization. Depending upon the study technique used, it has also been possible to indicate the tendency of a given system to suffer local pitting or crevice attack or both. These same tools have been the basis of design protection of less-noble structural metals.

To study the anode reaction of a specimen in an environment, sufficient current is applied to change the freely corroding potential of the metal in a more electropositive direction with respect to the inert electrode (acting as a cathode). The opposite of this so-called anodic polarization is cathodic polarization. Polarization can be studied equally well by varying the potential and measuring the resultant changes in the current. Both of the theoretical (true) polarization curves are straight lines when plotted on a semilog axis. The corrosion current can be measured from the intersection of the corrosion potential and

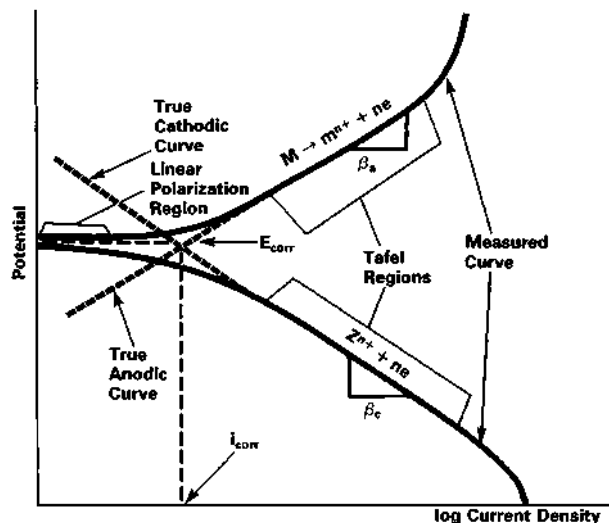


FIG. 25-9 Tafel extrapolation and linear polarization curves.

either the anodic or cathodic curve. The corrosion rate of the system is a function of that corrosion current. Experimentally derived curves are not fully linear because of the interference from the reactions between the anodes and cathodes in the region close to the corrosion potential. IR losses often obscure the Tafel behavior. Away from the corrosion potential, the measured curves match the theoretical (true) curves. The matching region of the measured curves is called the **Tafel region**, and their (Tafel) slopes are constant. Corrosion rates can be calculated from the intersection of the corrosion potential and the extrapolation of the Tafel region. The main advantage of the technique is that it is quick; curves can be generated in about an hour.

The technique is of limited value where more than one cathodic reduction reaction occurs. In most cases it is difficult to identify a sufficient linear segment of the Tafel region to extrapolate accurately. Since currents in the Tafel region are one to two orders of magnitude larger on the log scale than the corrosion current, relatively large currents are required to change the potentials from what they are at the corrosion potential. The environment must be a conductive solution. The Tafel technique does not indicate local attack, only an average, uniform corrosion rate.

The primary use of this laboratory technique today is as a quick check to determine the order of magnitude of a corrosion reaction. Sometimes the calculated rate from an immersion test does not "look" correct when compared to the visual appearance of the metal coupon. While the specific corrosion rate number determined by Tafel extrapolation is seldom accurate, the method remains a good confirmation tool.

Linear Polarization Some of the limitations of the Tafel extrapolation method can be overcome by using the linear-polarization technique to determine the corrosion rate. A relationship exists between the slope of the polarization curves E/I (with units of resistance, linear polarization is sometimes termed *polarization resistance*) and instantaneous corrosion rates of a freely corroding alloy. The polarization resistance is determined by measuring the amount of applied current needed to change the corrosion potential of the freely corroding specimen by about 10-mV deviations. The slope of the curves thus generated is directly related to the corrosion rate by Faraday's law. Several instruments are available that are used in linear polarization work. The main advantage is that each reading on the instrument can be translated directly into a corrosion rate.

As with all electrochemical studies, the environment must be electrically conductive. The corrosion rate is directly dependent on the Tafel slope. The Tafel slope varies quite widely with the particular corroding system and generally with the metal under test. As with the

Tafel extrapolation technique, the Tafel slope generally used is an assumed, more or less average value. Again, as with the Tafel technique, the method is not sensitive to local corrosion.

The amount of externally applied current needed to change the corrosion potential of a freely corroding specimen by a few millivolts (usually 10 mV) is measured. This current is related to the corrosion current, and therefore the corrosion rate, of the sample. If the metal is corroding rapidly, a large external current is needed to change its potential, and vice versa.

The measuring system consists of four basic elements:

1. **Electrodes.** Test and reference electrodes and, in some cases, an auxiliary electrode.
2. **Probe.** It connects the electrodes in the corrodent on the inside of a vessel to the electrical leads.
3. **Electrical leads.** They run from the probe to the current source and instrument panel.
4. **Control system.** Current source (batteries), ammeter, voltmeter, instrument panel, and so on.

Commercial instruments have either two or three electrodes. Also, there are different types of three-electrode systems. The application and limitations of the instruments are largely dependent upon these electrode systems.

Potentiodynamic Polarization Not all metals and alloys react in a consistent manner in contact with corrosive fluids. One of the common intermediate reactions of a metal (surface) is with oxygen, and those reactions are variable and complex. Oxygen can sometimes function as an electron acceptor; that is, oxygen can act as an oxidizing agent, and remove the "protective" film of hydrogen from the cathodic area, **cathodic depolarization**. The activation energy of the oxygen/hydrogen reaction is very large. This reaction does not normally occur at room temperature at any measureable rate. In other cases, oxygen can form protective oxide films. The long-term stability of these oxides also varies; some are soluble in the environment, others form more stable and inert or passive films.

Because corrosion is an electrochemical process, it is possible to evaluate the overall reaction by the use of an external electrical circuit called a **potentiostat**. When corrosion occurs, a potential difference exists between the metal and its ions in solution. It is possible to electrically control this potential; changes in potential cause changes in current (corrosion). Oxidation is a reaction with a loss of electrons (anodic—the reacting electrode is the anode); reduction is a reaction with a gain of electrons (cathodic—the reacting electrode is the cathode). Rather than allowing the electrons being evolved from the corrosion reaction to combine with hydrogen, these electrons can be removed by internal circuitry, and sent through a potentiostat, causing a cathodic (or anodic) reaction to occur at a platinum counter electrode. This is always true for the external polarization method; it is not unique for a potentiostat.

It is now well established that the activity of pitting, crevice corrosion, and stress-corrosion cracking is strongly dependent upon the corrosion potential (i.e., the potential difference between the corroding metal and a suitable reference electrode). By using readily available electronic equipment, the quantity and direction of direct current required to control the corrosion potential in a given solution at a given selected value can be measured. A plot of such values over a range of potentials is called a polarization diagram. By using proper experimental techniques, it is possible to define approximate ranges of corrosion potential in which pitting, crevice corrosion, and stress-corrosion cracking will or will not occur. With properly designed probes, these techniques can be used in the field as well as in the laboratory.

The potentiostat has a three-electrode system: a reference electrode, generally a saturated calomel electrode (SCE); a platinum counter, or auxiliary, electrode through which current flows to complete the circuit; and a working electrode that is a sample of interest (Fig. 25-10). The potentiostat is an instrument that allows control of the potential, either holding constant at a given potential, stepping from potential to potential, or changing the potential anodically or cathodically at some linear rate.

In the study of the anode/cathode polarization behavior of a metal/environment system, the potentiostat provides a plot of the relationship of current changes resulting from changes in potential most often presented as a plot of log current density versus potential, or Evans

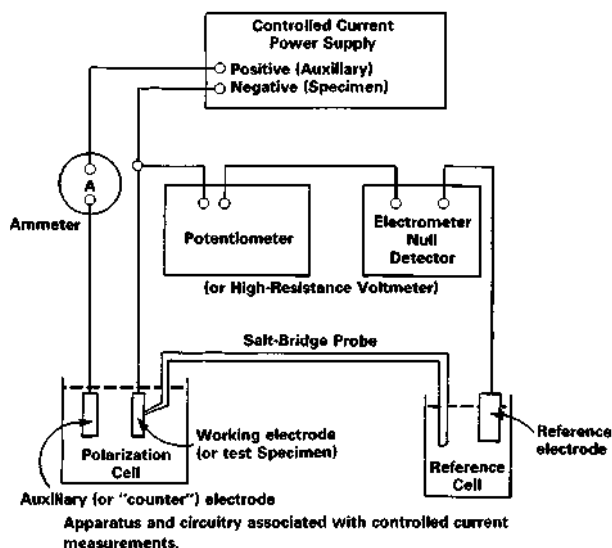


FIG. 25-10 The potentiostat apparatus and circuitry associated with controlled potential measurements of polarization curves.

diagram. A typical active/passive metal anodic polarization curve is seen in Fig. 25-11, generally showing the regions of active corrosion and passivity and a transpassive region.

Scan Rates Sweeping a range of potentials in the anodic (more electropositive) direction of a potentiodynamic polarization curve at a high scan rate of about 60 V/h (high from the perspective of the corrosion engineer, slow from the perspective of a physical chemist) is to indicate regions where intense anodic activity is likely. Second, for otherwise identical conditions, sweeping at a relatively slow rate of

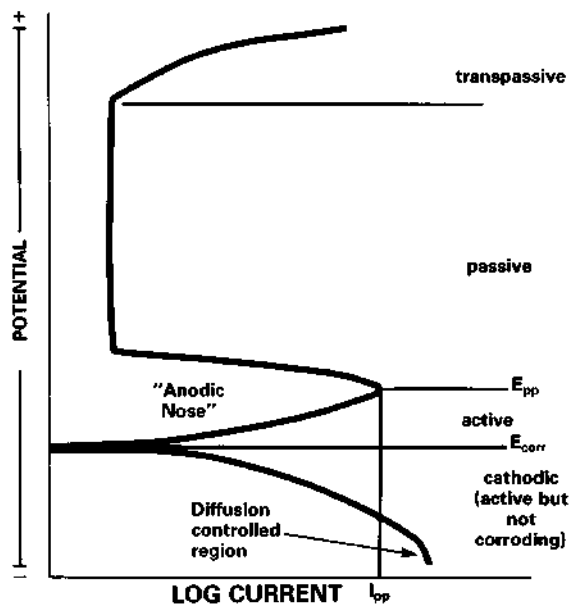


FIG. 25-11 Typical electrochemical polarization curve for an active/passive alloy (with cathodic trace) showing active, passive, and transpassive regions and other important features. (NOTE: E_{pp} = primary passive potential, E_{corr} = freely corroding potential.)

potential change of about 1 V/h will indicate regions wherein relative inactivity is likely. The rapid sweep of the potential range has the object of minimizing film formation, so that the currents observed relate to relatively film-free or thin-film conditions. The object of the slow sweep rate experiment is to allow time for filming to occur. A zero scan rate provides the opportunity for maximum stability of the metal surface, but at high electropositive potentials the environment could be affected or changed. A rapid scan rate compromises the steady-state nature of the metal surface but better maintains the stability of the environment. Whenever possible, corrosion tests should be conducted using as many of the techniques available, potentiodynamic polarization at various scan rates, crevice, stress, velocity, and so forth. An evaluation of these several results, on a holistic basis, can greatly reduce or temper their individual limitations.

Slow-Scan Technique In ASTM G5 "Polarization Practice for Standard Reference Method for Making Potentiostatic and Potentiodynamic Anodic Polarization Measurements," all oxygen in the test solution is purged with hydrogen for a minimum of 0.5 h before introducing the specimen. The test material is then allowed to reach a steady state of equilibrium (open circuit corrosion potential, E_{corr}) with the test medium before the potential scan is conducted. Starting the evaluation of a basically passive alloy that is already in its "stable" condition precludes any detailed study of how the metal reaches that protected state (the normal intersection of the theoretical anodic and cathodic curves is recorded as a zero applied current on the ASTM potentiostatic potential versus applied current diagram). These intersections between the anodic and cathodic polarization curves are the condition where the total oxidation rate equals the total reduction rate (ASTM G3 "Recommended Practice for Conventions Applicable to Electrochemical Measurements in Corrosion Testing").

Three general reaction types compare the activation-control reduction processes. In Fig. 25-12, in Case 1, the single reversible corrosion potential (anode/cathode intersection) is in the active region. A wide range of corrosion rates is possible. In Case 2, the cathodic curve intersects the anodic curve at three potentials, one active and two passive. If the middle active/passive intersection is not stable, the lower and upper

intersections indicate the possibility of very high corrosion rates. In Case 3, corrosion is in the stable, passive region, and the alloys generally passivate spontaneously and exhibit low corrosion rates. Most investigators report that the ASTM method is effective for studying Case 1 systems. An alloy-medium system exhibiting Case 2 and 3 conditions generally cannot be evaluated by this conventional ASTM method.

The potentiodynamic polarization electrochemical technique can be used to study and interpret corrosion phenomena. It may also furnish useful information on film breakdown or repair.

Rapid-Scan Corrosion Behavior Diagram (CBD) Basically, all the same equipment used in the conductance of an ASTM G5 slow-scan polarization study is used for rapid-scan CBDs (that is, a standard test cell, potentiostat, voltmeters, log converters, X-Y recorders, and electronic potential scanning devices). The differences are in technique: the slow scan is run at a potential sweep rate of about 0.6 V/h; the rapid-scan CBDs at about 50 V/h.

Different from the slow-scan technique, which is generally limited to Case 1 alloy/medium systems, the rapid-scan technique allows full anodic polarization study of alloys showing all Case 1, 2, and 3 behavior (Fig. 25-12). In Case 1, the single reversible corrosion potential (the anode/cathode intersection) is in the active region. A wide range of corrosion rates is possible. In Case 2, the cathodic curve intersects the anodic curve at three potentials, one in the active region and two in the passive. Since the middle active/passive intersection is not stable, the intersections indicate the possibility of very high corrosion rates depending on the environment or even slight changes to the exposure/environment system. In Case 3, the curves intersect in the most stable, passive region; the alloys generally passivate spontaneously and exhibit low corrosion rates. Case 2 exhibits the most corrosion, is difficult to study, and presents the most risk for materials of construction selection. Anything that can change the oxygen solubility of the oxidizing agent can alter the corrosion reaction.

The CBD diagram can provide various kinds of information about the performance of an alloy/medium system. The technique can be used for a direct calculation of the corrosion rate as well as for indicating the conditions of passivity and tendency of the metal to suffer local pitting and crevice attack. There are *benefits* from using the rapid-scan technique (the so-called corrosion behavior diagram, Fig. 25-13), and some *limitations* (when compared to the aforementioned slow-scan technique). Since *none* of the electrochemical corrosion test techniques that are currently in vogue, be they the slow-scan, rapid-scan, cyclic potentiodynamic polarization, EIS/AC impedance, or frequency modulation, are, in and of themselves, a true evaluation, measurement, and portrayal of the corrosion reaction being studied; only during the so-called open-circuit potential are we viewing real-life corrosion. Then why are these several electrochemical test methods used? Because (of those various test methods) each provides some valuable information that is more accurate (or not available) than that obtained when using some different test method. The more experienced the electrochemist in understanding these differences, the more closely she or he is able to approach accuracy in her or his corrosion understanding and thus results. For example, in designing an anodic protection system, one of the most critically sensitive parameters is the optimum protection potential at which the corrosion rate of the metal is able to be maintained at its lowest value (the true, lowest anodic protection control is not necessarily the whole length of the passive area of the E versus $\log I$ curve; see Fig. 25-11). Thus (conservatively), the control set point should be selected at the midpoint of the lowest corrosion rate (LCR) of the CBD, i.e., the region of protective oxide formation. If for no other reason than that, the CBD may be a test technique of choice for that anodic protection study. When using electrochemical test methods, many electrochemists try to employ a variety of test methods to arrive at the optimum selection of choice for any MOC study.

Those desiring a more detailed review of the subject of electrochemistry (and/or corrosion testing using electrochemistry) are directed to additional reference material from the following: Perry's *Chemical Engineers' Handbook*, 7th ed., Sec. 28, O. W. Siebert and J. G. Stoecker, "Materials of Construction," pp. 28-11 to 28-20, 1997; J. G. Stoecker, O. W. Siebert, and P. E. Morris, "Practical Applications

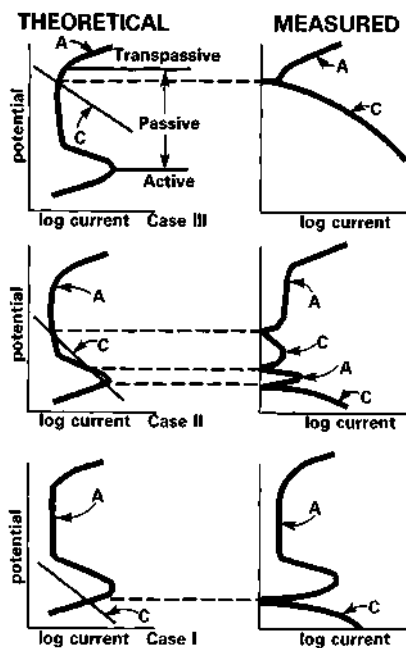


FIG. 25-12 Six possible types of behavior for an active/passive alloy in a corrosive environment.

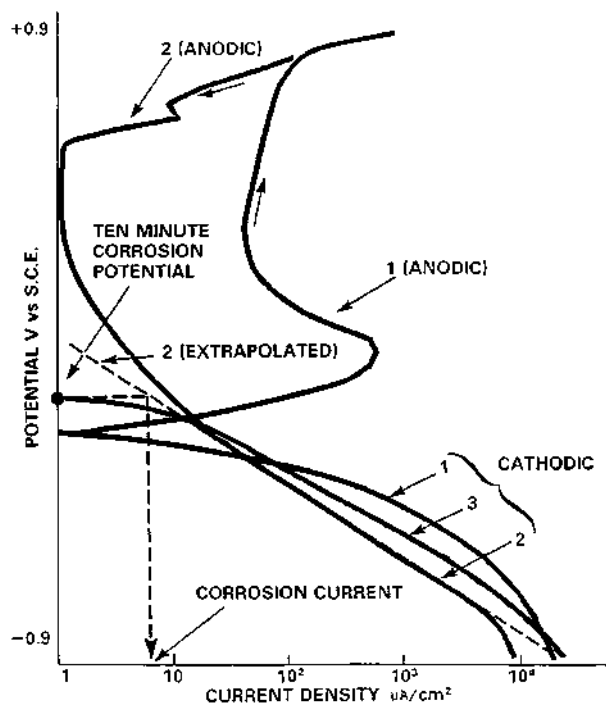


FIG. 25-13 Corrosion behavior diagram (CBD).

of Potentiodynamic Polarization Curves in Materials Selection," *Materials Performance*, vol. 22, no. 11, pp. 13-22, November 1983; O. W. Siebert, "Correlation of Laboratory Electrochemical Investigations with Field Application of Anodic Protection," *Materials Performance*, vol. 20, no. 2, pp. 38-43, February 1981; and the assorted historical literature of Stern, Geary, Evans, Sudbury, Riggs, Pourbaix, and Edeleanu, and other studies referred to in their publications.

Crevice Corrosion Prediction The most common type of localized corrosion is the occluded mode crevice corrosion. Pitting can, in effect, be considered a self-formed crevice. A crevice must be wide enough to permit liquid entry, but sufficiently narrow to maintain a stagnant zone. It is nearly impossible to build equipment without mechanical crevices; on a microlevel, scratches can be sufficient crevices to initiate or propagate corrosion in some metal/environment systems. The conditions in a crevice can, with time, become a different and much more aggressive environment than those on a nearby, clean, open surface. Crevices may also be created by factors foreign to the original system design, such as deposits, corrosion products, and so forth. In many studies, it is important to know or to be able to evaluate the crevice corrosion sensitivity of a metal to a specific environment and to be able to monitor a system for predictive maintenance.

Historically, the immersion test technique involved the use of a crevice created by two metal test specimens clamped together or a metal specimen in contact with an inert plastic or ceramic. The Materials Technology Institute of the Chemical Process Industries, Inc. (MTI) funded a study that resulted in an electrochemical cell to monitor crevice corrosion. It consists of a prepared crevice containing an anode that is connected through a zero-resistance ammeter to a freely exposed cathode. A string bridge provides a solution path that is attached externally to the cell. The electrochemical cell is shown in Fig. 25-14. A continuous, semiquantitative, real-time indication of crevice corrosion is provided by the magnitude of the current flowing between an anode and a cathode, and a qualitative signal is provided by shifts in electrode potential. Both the cell current and electrode potential produced by the test correlate well with the initiation and

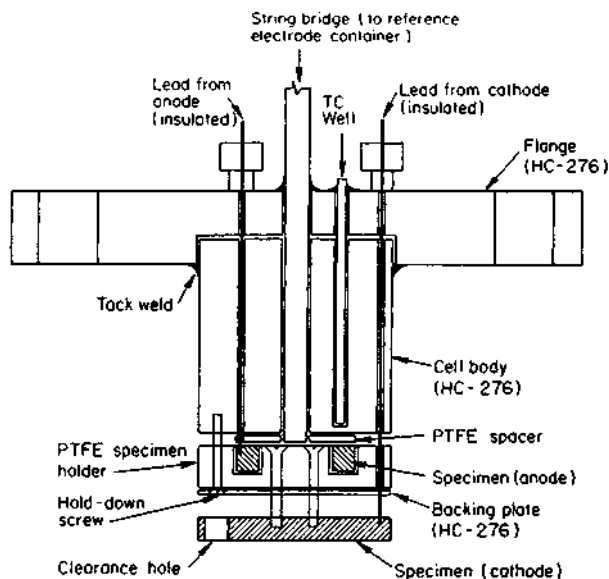


FIG. 25-14 Schematic diagram of the electrochemical cell used for crevice corrosion testing. Not shown are three hold-down screws, gas inlet tube, and external thermocouple tube.

propagation of crevice corrosion. During development of the MTI test technique, results were compared with crevice corrosion produced by a grooved TFE Teflon® plastic disk sandwich-type crevice cell. In nearly every instance, corrosion damage on the anode was similar in severity to that produced by the sandwich-type cell.

Velocity For corrosion to occur, an environment must be brought into contact with the metal surface and the metal atoms or ions must be allowed to be transported away. Therefore, the rate of transport of the environment with respect to a metal surface is a major factor in the corrosion system. Changes in velocity may increase or decrease attack depending on its effect involved. A varying quantity of dissolved gas may be brought in contact with the metal, or velocity changes may alter diffusion or transfer of ions by changing the thickness of the boundary layer at the surface. The boundary layer, which is not stagnant, moves except where it touches the surface. Many metals depend upon the development of a protective surface for their corrosion resistance. This may consist of an oxide film, a corrosion product, an adsorbed film of gas, or other surface phenomena. The removal of these surfaces by effect of the fluid velocity exposes fresh metal, and as a result, the corrosion reaction may proceed at an increasing rate. In these systems, corrosion might be minimal until a so-called **critical velocity** is attained where the protective surface is damaged or removed and the velocity is too high for a stable film to reform. Above this critical velocity, the corrosion may increase rapidly.

The NACE Landrum Wheel velocity test, originally TM0270-72, is typical of several mechanical-action immersion test methods to evaluate the effects of corrosion. Unfortunately, these laboratory simulation techniques did not consider the fluid mechanics of the environment or metal interface, and service experience very seldom supports the test predictions. A rotating cylinder within a cylinder electrode test system has been developed that operates under a defined hydrodynamics relationship (Figs. 25-15 and 25-16). The assumption is that if the rotating electrode operates at a shear stress comparable to that in plant geometry, the mechanism in the plant geometry may be modeled in the laboratory. Once the mechanism is defined, the appropriate relationship between fluid flow rate and corrosion rate in the plant equipment as defined by the mechanism can be used to predict the expected corrosion

*See Review Paper by David C. Silverman, courtesy of NACE International.

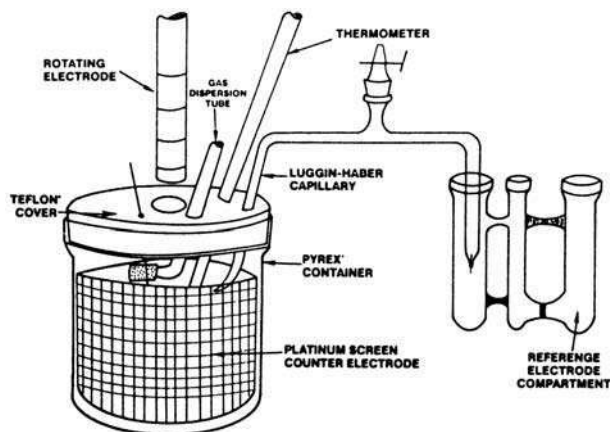


FIG. 25-15 Rotating cylinder electrode apparatus.

rate. If fluid velocity does affect the corrosion rate, the degree of mass transfer control, if that is the controlling mechanism (as opposed to activation control), can be estimated. Conventional potentiodynamic polarization scans are conducted as described previously. In other cases, the corrosion potential can be monitored at a constant velocity until steady state is attained. While the value of the final corrosion potential is virtually independent of velocity, the time to reach steady state may be dependent on velocity. The mass-transfer control of the corrosion potential can be proportional to the velocity raised to its appropriate exponent. The rate of breakdown of a passive film is velocity-sensitive. To review a very detailed and much needed refinement of the information and application of the rotating electrode technique as used for evaluation of the effect of velocity on corrosion, the reader is directed to a recent seminal study by David C. Silverman, "The Rotating Cylinder Electrode for Examining Velocity-Sensitive Corrosion—A Review," *Corrosion*, vol. 60, no. 11, pp. 1003–1023, Nov. 2004.

Environmental Cracking The problem of environmental cracking of metals and their alloys is very important. Of all the failure mechanism tests, the test for stress corrosion cracking (SCC) is the most illusive. Stress corrosion is the acceleration of the rate of corrosion damage by static stress. SCC, the limiting case, is the spontaneous cracking that may result from combined effects of stress and corrosion. It is important to differentiate clearly between **stress corrosion cracking** and **stress accelerated corrosion**. Stress corro-

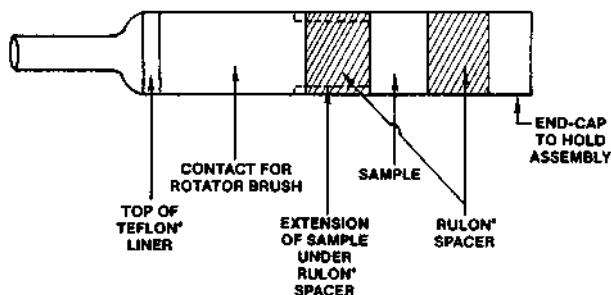


FIG. 25-16 Inner rotating cylinder used in laboratory apparatus of Fig. 25-15.

sion cracking is considered to be limited to cases in which no significant corrosion damage occurs in the absence of a corrosive environment. The material exhibits normal mechanical behavior under the influence of stress; before the development of a stress corrosion crack, there is little deterioration of strength and ductility. Stress corrosion cracking is the case of an interaction between chemical reaction and mechanical forces that results in structural failure that otherwise would not occur. SCC is a type of brittle fracture of a normally ductile material by the interaction between specific environments and mechanical forces, for example, tensile stress. Stress corrosion cracking is an incompletely understood corrosion phenomenon. Much research activity (aimed mostly at mechanisms) plus practical experience has allowed crude empirical guidelines, but these contain a large element of uncertainty. No single chemical, structural, or electrochemical test method has been found to respond with enough consistent reproducibility to known crack-causing environmental/stressed metal systems to justify a high confidence level.

As was cited in the case of immersion testing, most SCC test work is accomplished using mechanical, nonelectrochemical methods. It has been estimated that 90 percent of all SCC testing is handled by one of the following methods: (1) constant strain, (2) constant load, or (3) precracked specimens. Prestressed samples, such as are shown in Fig. 25-17, have been used for laboratory and field SCC testing. The variable observed is "time to failure or visible cracking." Unfortunately, such tests do not provide acceleration of failure.

Since SCC frequently shows a fairly long induction period (months to years), such tests must be conducted for very long periods before reliable conclusions can be drawn.

In the constant-strain method, the specimen is stretched or bent to a fixed position at the start of the test. The most common shape of

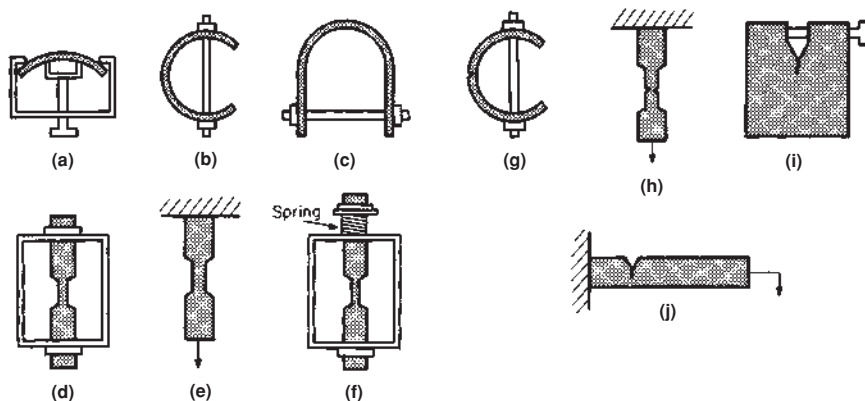


FIG. 25-17 Specimens for stress-corrosion tests. (a) Bent beam. (b) C ring. (c) U bend. (d) Tensile. (e) Tensile. (f) Tensile. (g) Notched C ring. (h) Notched tensile. (i) Precracked, wedge open-loading type. (j) Precracked, cantilever beam. [Chem. Eng., 78, 159 (Sept. 20, 1971).]

the specimens used for constant-strain testing is the U-bend, hair-pin, or horseshoe type. A bolt is placed through holes in the legs of the specimen, and it is loaded by tightening a nut on the bolt. In some cases, the stress may be reduced during the test as a result of creep. In the constant-load test the specimen is supported horizontally at each end and is loaded vertically downward at one or two points and has maximum stress over a substantial length or area of the specimen. The load applied is a predetermined, fixed dead weight. Specimens used in either of these tests may be precracked to assign a stress level or a desired location for fracture to occur or both as is used in fracture mechanics studies. These tensile-stressed specimens are then exposed in situ to the environment of study.

Slow Strain-Rate Test In its present state of development, the results from slow strain-rate tests (SSRT) with electrochemical monitoring are not always completely definitive; but, for a short-term test, they do provide considerable useful SCC information. Work in our laboratory shows that the SSRT with electrochemical monitoring and the U-bend tests are essentially equivalent in sensitivity in finding SCC. The SSRT is more versatile and faster, providing both mechanical and electrochemical feedback during testing.

The SSRT is a test technique where a tension specimen is slowly loaded in a test frame to failure under prescribed test conditions. The normal test extension rates are from 2.54×10^{-7} to 2.54×10^{-10} m/s (10^{-5} to 10^{-8} in/s). Failure times are usually 1 to 10 days. The failure mode will be either SCC or tensile overload, sometimes accelerated by corrosion. An advantage behind the SSRT, compared to constant-strain tests, is that the protective surface film is thought to be ruptured mechanically during the test, thus giving SCC an opportunity to progress. To aid in the selection of the value of the potential at which the metal is most sensitive to SCC that can be applied to accelerate SSRT, potentiodynamic polarization scans are conducted as described previously. It is common for the potential to be monitored during the conduct of the SSRT. The strain rates that generate SCC in various metals are reported in the literature. There are several disadvantages to the SSRT. First, indications of failure are not generally observed until the tension specimen is plastically stressed, sometimes significantly, above the yield strength of the metal. Such high-stressed conditions can be an order of magnitude higher than the intended operating stress conditions. Second, crack initiation must occur fairly rapidly to have sufficient crack growth that can be detected using the SSRT. The occurrence of SCC in metals requiring long initiation times may go undetected.

Modulus Measurements Another SCC test technique is the use of changes of modulus as a measure of the damping capacity of a metal. It is known that a sample of a given test material containing cracks will have a lower effective modulus than does a sample of identical material free of cracks. The technique provides a rapid and reliable evaluation of the susceptibility of a sample material to SCC in a specific environment. The so-called internal friction test concept can also be used to detect and probe nucleation and progress of cracking and the mechanisms controlling it.

The Young's modulus of the specimen is determined by accurately measuring its resonant frequency while driving it in a standing longitudinal wave configuration. A Marx composite piezoelectric oscillator is used to drive the specimen at a resonant frequency. The specimen is designed to permit measurements while undergoing applied stress and while exposed to an environmental test solution. The specimens are three half-wavelengths long; the gripping nodes and solution cup are silver-soldered on at displacement nodes, so they do not interfere with the standing wave. As discussed for SSRT, potentiodynamic polarization scans are conducted to determine the potential that can be applied to accelerate the test procedure. Again, the potential can be monitored during a retest, as is the acoustic emission (AE) as an indicator of nucleation and progress of cracking.

Conjunctive Use of Slow- and Rapid-Scan Polarization The use of the methods discussed in the preceding requires a knowledge of the likely potential range for SCC to occur. Potentiodynamic polarization curves can be used to predict those SCC-sensitive potential ranges. The technique involves conducting both slow- and rapid-scan sweeps in the anodic direction of a range of potentials. Comparison of the two curves will indicate any ranges of potential within which high anodic activity in the film-free condition reduces to insignificant activ-

ity when the time requirements for film formation are met. This should indicate the range of potentials within which SCC is likely. Most of the SCC theories presently in vogue predict these domains of behavior to be between the primary passive potential and the onset of passivity. This technique shortens the search for that SCC potential.

Separated Anode/Cathode Realizing, as noted in the preceding, that localized corrosion is usually active to the surrounding metal surface, a stress specimen with a limited area exposed to the test solution (the anode) is electrically connected to an unstressed specimen (the cathode). A potentiostat, used as a zero-resistance ammeter, is placed between the specimens for monitoring the galvanic current. It is possible to approximately correlate the galvanic current I_g and potential to crack initiation and propagation, and, eventually, catastrophic failure. By this arrangement, the galvanic current I_g is independent of the cathode area. In other words, the potential of the anode follows the corrosion potential of the cathode during the test. The SSRT apparatus discussed previously may be used for tensile loading.

Fracture Mechanics Methods These have proved very useful for defining the minimum stress intensity K_{ISCC} at which stress corrosion cracking of high-strength, low-ductility alloys occurs. They have so far been less successful when applied to high-ductility alloys, which are extensively used in the chemical-process industries.

Work on these and other new techniques continues, and it is hoped that a truly reliable, accelerated test or tests will be defined.

Electrochemical Impedance Spectroscopy (EIS) and AC Impedance* Many direct-current test techniques assess the overall corrosion process occurring at a metal surface, but treat the metal/solution interface as if it were a pure resistor. Problems of accuracy and reproducibility frequently encountered in the application of direct-current methods have led to increasing use of electrochemical impedance spectroscopy (EIS).

Electrode surfaces in electrolytes generally possess a surface charge that is balanced by an ion accumulation in the adjacent solution, thus making the system electrically neutral. The first component is a double layer created by a charge difference between the electrode surface and the adjacent molecular layer in the fluid. Electrode surfaces may behave at any given frequency as a network of resistive and capacitive elements from which an electrical impedance may be measured and analyzed.

The application of an impressed alternating current on a metal specimen can generate information on the state of the surface of the specimen. The corrosion behavior of the surface of an electrode is related to the way in which that surface responds to this electrochemical circuit. The AC impedance technique involves the application of a small sinusoidal voltage across this circuit. The frequency of that alternating signal is varied. The voltage and current response of the system are measured.

The so-called white-noise analysis by the fast Fourier transform technique (FFT) is another viable method. The entire spectrum can be derived from one signal. The impedance components thus generated are plotted on either a Nyquist (real versus imaginary) or Bode (log real versus log frequency plus log phase angle versus log frequency) plot. These data are analyzed by computer; they can be used to determine the polarization resistance and, thus, the corrosion rate if Tafel slopes are known. It is also thought that the technique can be used to monitor corrosion by examining the real resistance at high and low frequency and by assuming the difference is the polarization resistance. This can be done in low- and high-conductivity environments. Systems prone to suffer localized corrosion have been proposed to be analyzed by AC impedance and should aid in determining the optimum scan rate for potentiodynamic scans.

The use of impedance electrochemical techniques to study corrosion mechanisms and to determine corrosion rates is an emerging technology. Electrode impedance measurements have not been widely used, largely because of the sophisticated electrical equipment required to make these measurements. Recent advantages in microelectronics and computers has moved this technique almost overnight from being an academic experimental investigation of the concept

*Excerpted from papers by Oliver W. Siebert, courtesy of NACE International and ASTM.

itself to one of shelf-item commercial hardware and computer software, available to industrial corrosion laboratories.

Other Electrochemical Test Techniques A seminal summary of the present state of electrochemical test techniques can be found in John R. Scully, Chapter 7, "Electrochemical Tests," 2005 *ASTM Manual* 20. Professor Scully performs a praiseworthy job of presenting the theories associated with the mechanisms of corrosion, addressing both the thermodynamics and the kinetics of their electrode reactions. He then follows with a detailed encapsulation of major test methods being used in the academic and industrial research laboratories throughout the world for both basic information as well as to predict the scope and types of metallic corrosion experienced. In addition to some of those methods (already) addressed in *Perry's* above, we have noted several others from his presentation, wherein he has provided a much more thorough review, e.g., concentration polarization effects, frequency modulation methods, electronic noise resistance, methods based upon mixed-potential theory, scratch-repassivation method for local corrosion, and many more. The editors of this section of *Perry's* recommend this timely ASTM review for readers who need a greater treatment of electrochemical corrosion testing than is included in this presentation. In addition, Prof. Scully and his associate at the University of Virginia, Prof. Robert G. Kelly, have coauthored a similar quality presentation, "Methods for Determining Aqueous Corrosion Reaction Rates," in the 2005 *ASM Metals Handbook*, vol. 13.

Use and Limitations of Electrochemical Techniques A major caution must be noted as to the general, indiscriminate use of many electrochemical tests. Corrosion is a surface phenomenon. It must be kept in mind that the only condition present during any type of corrosion test that is a true representation of the real-life circumstance is the so-called open-circuit potential (OCP). The OCP is the electrical circuit that exists on the metal surface during the naturally spontaneous accumulation of the electrical potential that forms on the metal surface when exposed to a liquid environment. Any kind of an electrical current that is *added* to that surface is an artifact that no longer represents the true nature of that corrosion reaction. Is this to say that any form of induced electrical variable makes any corrosion test invalid? Absolutely not. Collectively, we have devised a number of unique electronic instruments that are designated to allow us to test/evaluate the influence of one or more variables present in a given corrosion reaction; for example, the application of anodic protection to control the corrosion of bare steel in concentrated sulfuric acid (see the subsection "Anodic Protection" earlier in Sec. 25). The AP system is designed to introduce an applied electrical current, that is, purposely "making" the steel surface the anode, making it corrode, and then taking advantage of its resultant electrochemical reaction to control that corroding surface into a condition with an extremely low rate of corrosion. We had earlier suggested that those electrochemists very well schooled and experienced in their trade also understand the major limitations of many of these types of techniques in real-life situations. Professor Scully, in the *ASTM Manual* referenced above, included a separate section addressing these limitations. An applicable example is the limitations to the general use of AC and EIS test techniques, for the study of corrosion systems. AC and EIS techniques are applicable for the evaluation of very thin films or deposits that are uniform, constant, and stable—for example, thin-film protective coatings. Sometimes, researchers do not recognize the dynamic nature of some passive films, corrosion products, or deposits from other sources; nor do they even consider the *possibility of a change* in the surface conditions during the course of their experiment. As an example, it is noteworthy that this is a *major* potential problem in the electrochemical evaluation of microbiological corrosion (MIC).

MIC depends on the complex structure of corrosion products and passive films on metal surfaces as well as on the structure of the biofilm. Unfortunately, electrochemical methods have sometimes been used in complex electrolytes, such as microbiological culture media, where the characteristics and properties of passive films and MIC deposits are *quite active* and not fully understood. It must be kept in mind that microbial colonization of passive metals can drastically change their resistance to film breakdown by causing *localized* changes in the type, concentration, and thickness of anions, pH, oxygen gradients, and inhibitor levels at the metal surface during the course of a

normal test; viable single-cell microorganisms divide at an exponential rate. These changes can be expected to result in important modifications in the electrochemical behavior of the metal and, accordingly, in the electrochemical parameter measured in laboratory experiments.

Warnings are noted in the literature to be careful in the *interpretation* of data from electrochemical techniques applied to systems in which complex and often poorly understood effects are derived from surfaces which contain *active* or *viable* organisms, and so forth. Rather, it is even more important to not use such test protocol unless the investigator fully understands both the corrosion mechanism and the test technique being considered—and their interrelationship.

CORROSION TESTING: PLANT TESTS

It is not always practical or convenient to investigate corrosion problems in the laboratory. In many instances, it is difficult to discover just what the conditions of service are and to reproduce them exactly. This is especially true with processes involving changes in the composition and other characteristics of the solutions as the process is carried out, as, for example, in evaporation, distillation, polymerization, sulfonation, or synthesis.

With many natural substances also, the exact nature of the corrosive is uncertain and is subject to changes not readily controlled in the laboratory. In other cases, the corrosiveness of the solution may be influenced greatly by or even may be due principally to a constituent present in such minute proportions that the mass available in the limited volume of corrosive solution that could be used in a laboratory setup would be exhausted by the corrosion reaction early in the test, and consequently the results over a longer period of time would be misleading.

Another difficulty sometimes encountered in laboratory tests is that contamination of the testing solution by corrosion products may change its corrosive nature to an appreciable extent.

In such cases, it is usually preferable to carry out the corrosion-testing program by exposing specimens in operating equipment under **actual conditions of service**. This procedure has the additional advantages that it is possible to test a large number of specimens at the same time and that little technical supervision is required.

In certain cases, it is necessary to choose materials for equipment to be used in a process developed in the laboratory and not yet in operation on a plant scale. Under such circumstances, it is obviously impossible to make plant tests. A good procedure in such cases is to construct a pilot plant, using either the cheapest materials available or some other materials selected on the basis of past experience or of laboratory tests. While the pilot plant is being operated to check on the process itself, specimens can be exposed in the operating equipment as a guide to the choice of materials for the large-scale plant or as a means of confirming the suitability of the materials chosen for the pilot plant.

Test Specimens In carrying out plant tests it is necessary to install the test specimens so that they will not come into contact with other metals and alloys; this avoids having their normal behavior disturbed by galvanic effects. It is also desirable to protect the specimens from possible mechanical damage.

There is no single standard size or shape for corrosion-test coupons. They usually weigh from 10 to 50 g and preferably have a large surface-to-mass ratio. Disks 40 mm (1½ in) in diameter by 3.2 mm (¼ in) thick and similarly dimensioned square and rectangular coupons are the most common. Surface preparation varies with the aim of the test, but machine grinding of surfaces or polishing with a No. 120 grit is common. Samples should not have sheared edges, should be clean (no heat-treatment scale remaining unless this is specifically part of the test), and should be identified by stamping. See Fig. 25-18 for a typical plant test assembly.

The choice of materials from which to make the holder is important. Materials must be durable enough to ensure satisfactory completion of the test. It is good practice to select very resistant materials for the test assembly. Insulating materials used are plastics, porcelain, Teflon, and glass. A phenolic plastic answers most purposes; its principal limitations are unsuitability for use at temperatures over 150°C (300°F) and lack of adequate resistance to concentrated alkalis.

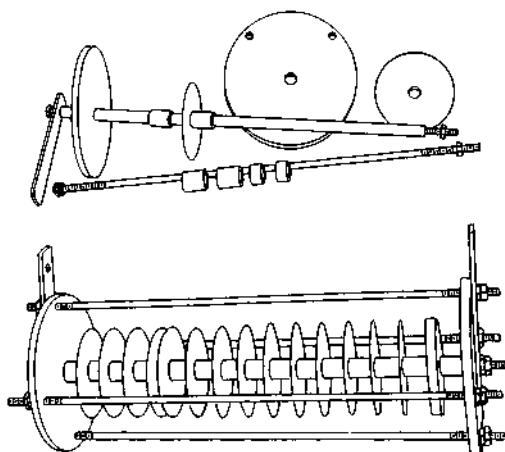


FIG. 25-18 Assembly of a corrosion-test spool and specimens. (Mantell, ed., Engineering Materials Handbook, McGraw-Hill, New York, 1958.)

The method of supporting the specimen holder during the test period is important. The preferred position is with the long axis of the holder horizontal, thus avoiding dripping of corrosion products from one specimen to another. The holder must be located so as to cover the conditions of exposure to be studied. It may have to be submerged, or exposed only to the vapors, or located at liquid level, or holders may be called for at all three locations. Various means have been utilized for supporting the holders in liquids or in vapors. The simplest is to suspend the holder by means of a heavy wire or light metal chain. Holders have been strung between heating coils, clamped to agitator shafts, welded to evaporator tube sheets, and so on. The best method is to use test racks.

In a few special cases, the standard "spool-type" specimen holder is not applicable and a suitable special test method must be devised to apply to the corrosion conditions being studied.

For conducting tests in **pipe lines** of 75-mm (3-in) diameter or larger, a spool holder as shown in Fig. 25-19, which employs the same disk-type specimens used on the standard spool holder, has been used. This frame is so designed that it may be placed in a pipe line in any position without permitting the disk specimens to touch the wall of the pipe. As with the strip-type holder, this assembly does not materially interfere with the fluid through the pipe and permits the study of corrosion effects prevailing in the pipe line.

Another way to study corrosion in pipe lines is to install in the line short sections of pipe of the materials to be tested. These test sections should be insulated from each other and from the rest of the piping

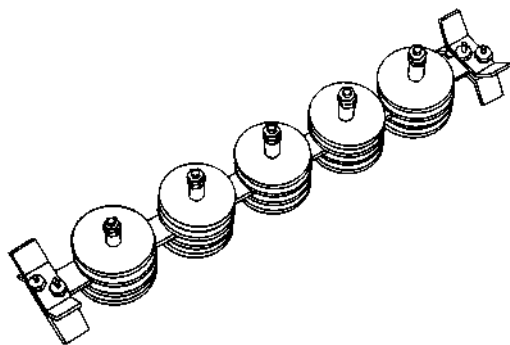


FIG. 25-19 Spool-type specimen holder for use in 3-in-diameter or larger pipe. (Mantell, ed., Engineering Materials Handbook, McGraw-Hill, New York, 1958.)

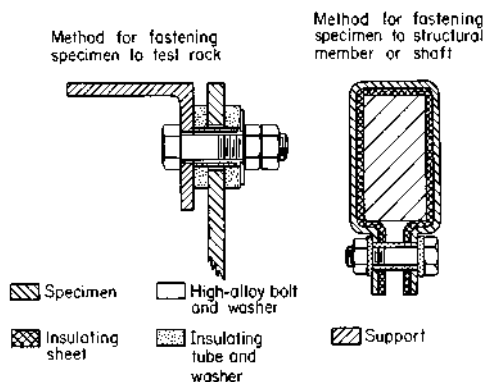


FIG. 25-20 Methods for attaching specimens to test racks and to parts of moving equipment. (Mantell, ed., Engineering Materials Handbook, McGraw-Hill, New York, 1958.)

system by means of nonmetallic couplings. It is also good practice to provide insulating gaskets between the ends of the pipe specimens where they meet inside the couplings. Such joints may be sealed with various types of dope or cement. It is desirable in such cases to paint the outside of the specimens so as to confine corrosion to the inner surface.

It is occasionally desirable to expose corrosion-test specimens in operating equipment without the use of specimen holders of the type described. This can be accomplished by attaching specimens directly to some part of the operating equipment and by providing the necessary insulation against galvanic effects as shown in Fig. 25-20. The suggested method of attaching specimens to racks has been found to be very suitable in connection with the exposure of specimens to corrosion in seawater.

Test Results The methods of cleaning specimens and evaluating results after plant corrosion tests are identical to those described earlier for laboratory tests.

Electrochemical On-Line Corrosion Monitoring* On-line corrosion monitoring is used to evaluate the status of equipment and piping in chemical process industries (CPI) plants. These monitoring methods are based on electrochemical techniques. To use on-line monitoring effectively, the engineer needs to understand the underlying electrochemical test methods to be employed. This section covers many of these test methods and their applications as well as a review of potential problems encountered with such test instruments and how to overcome or avoid these difficulties.

Most Common Types of Probes There are three most common types of corrosion monitoring probes used. Other types of probes are used, but in smaller numbers.

1. **Weight loss probes.** Coupons for measuring weight loss are still the primary type of probe in use. These may be as simple as samples of the process plant materials which have been fitted with electrical connections and readouts to determine intervals for retrieval and weighing, to commercially available coupons of specified material, geometry, stress condition, and other factors, ready to be mounted on specially designed supports at critical points in the process. Coupons can be permanently installed prior to plant start-up or during a shutdown. This type of permanent installation requires a plant shutdown for probe retrieval as well. Shutdown can be avoided by installing the probe in a bypass. Weight loss measurements do not accurately measure the severity of pitting corrosion, including that due to MIC.

2. **Electrical resistance probes.** These probes are the next most common type of corrosion probes after coupons. This type of probe measures changes in the electrical resistance as a thin strip of metal gets

*Excerpted from papers by Oliver W. Siebert, courtesy of NACE International and ASTM.

thinner with ongoing corrosion. As the metal gets thinner, its resistance increases. This technique was developed in the 1950s by Draviniaks and Cataldi and has undergone many improvements since then.

3. Linear polarization resistance probes. LPR probes are more recent in origin, and are steadily gaining in use. These probes work on a principle outlined in an ASTM guide on making polarization resistance measurements, providing instantaneous corrosion rate measurements (G59, "Standard Practice for Conducting Potentiodynamic Polarization Resistance Measurements").

LPR probes measure the electrochemical corrosion mechanism involved in the interaction of the metal with the electrolyte. To measure linear polarization resistance R_p , Ω/cm^2 , the following assumptions must be made:

- The corrosion rate is uniform.
- There is only cathodic and one anodic reaction.
- The corrosion potential is not near the oxidation/reduction potential for either reaction.

When these conditions are met, the current density associated with a small polarization of the metal (less than +10 mV) is directly proportional to the corrosion rate of the metal.

Multiinformational Probes Corrosion probes can provide more information than just corrosion rate. The next three types of probes yield information about the type of corrosion, the kinetics of the corrosion reaction, as well as the local corrosion rate.

Electrochemical impedance spectroscopy, AC probes. EIS, although around since the 1960s, has primarily been a laboratory technique. Commercially available probes and monitoring systems that measure EIS are becoming more widely used, especially in plants that have on-staff corrosion experts to interpret the data or to train plant personnel to do so.

In EIS, a potential is applied across a corroding metal in solution, causing current to flow. The amount of current depends upon the corrosion reaction on the metal surface and the flow of ions in solution. If the potential is applied as a sine wave, it will cause harmonics of the current output. The relationship between the applied potential and current output is the impedance, which is analogous to resistance in a DC circuit.

Since the potential and current are sinusoidal, the impedance has a magnitude and a phase, which can be represented as a vector. A sinusoidal potential or current can be pictured as a rotating vector. For standard AC current, the rotation is at a constant angular velocity of 60 Hz.

The voltage can also be pictured as a rotating vector with its own amplitude and frequency. Both current and potential can be represented as having real (observed) and imaginary (not observed) components.

In making electrochemical impedance measurements, one vector is examined, using the others as the frame of reference. The voltage vector is divided by the current vector, as in Ohm's law. Electrochemical impedance measures the impedance of an electrochemical system and then mathematically models the response using simple circuit elements such as resistors, capacitors, and inductors. In some cases, the circuit elements are used to yield information about the kinetics of the corrosion process.

Polarization probes. Polarization methods other than LPR are also of use in process control and corrosion analysis, but only a few systems are offered commercially. These systems use such polarization techniques as galvanodynamic or potentiodynamic, potentiostatic or galvanostatic, potentiostaircase or galvanostaircase, or cyclic polarization methods. Some systems involving these techniques are, in fact, used regularly in processing plants. These methods are used in situ or in the laboratory to measure corrosion. Polarization probes have been successful in reducing corrosion-related failures in chemical plants.

Polarization probes rely on the relationship of the applied potential to the output current per unit area (current density). The slope of applied potential versus current density, extrapolated through the origin, yields the polarization resistance R_p , which can be related to the corrosion rate.

There are several methods for relating the corrosion current, the applied potential, and the polarization resistance. These methods involve various ways of stepping or ramping either the potential or current. Also, a constant value of potential or current can be applied.

Electrochemical noise monitoring probes. Electrochemical noise monitoring is probably the newest of these methods. The method characterizes the naturally occurring fluctuations in current and potential due to the electrochemical kinetics and the mechanism of

the corroding metal interface. Measurements are taken without perturbing the interface by applying a potential or current to it. In this way, electrochemical processes are not interrupted and the system is measured without being disturbed. Methods including signal processing and mathematical transformation are used to provide information on the reaction kinetics at the surface and the corrosion rate.

This technique, originally discovered in the 1960s, remained a laboratory technique until recently, when some manufacturers began producing commercial devices. There are a few cases where electrochemical noise is being used in process-plant-type environments, and an ASTM committee has been formed to look at standardization of this technique. However, in general it remains a laboratory method with great potential for on-line monitoring. Recent research has demonstrated that electrochemical noise can be used to monitor MIC under laboratory conditions when appropriate probe configurations are used. Tests of these techniques in actual operating environments have been limited.

Indirect Probes Some types of probes do not measure corrosion directly, but yield measurements that also are useful in detecting corrosion. Examples include:

Pressure probes. Pressure monitors or transducers may be of use in corrosion monitoring in environments where buildup of gases such as hydrogen or H_2S may contribute to corrosion.

Gas probes. The hydrogen patch probe allows users to determine the concentration of hydrogen in the system. This is an important measurement since hydrogen can foster corrosion. Detecting production of certain gases may give rise to process changes to eliminate or limit the gaseous effluent and, therefore, lower the possibility of corrosion caused by these gases.

pH probes. Monitoring the pH may also aid in the early detection of corrosion. The acidity or alkalinity of the environment is often one of the controllable parameters in corrosion. Monitoring of the pH can be combined with other corrosion measurements to provide additional data about process conditions and give another level of process control.

Ion probes. Determining the level of ions in solution also helps to control corrosion. An increase in concentration of specific ions can contribute to scale formation, which can lead to a corrosion-related failure. Ion-selective electrode measurements can be included, just as pH measurements can, along with other more typical corrosion measurements. Especially in a complete monitoring system, this can add information about the effect of these ions on the material of interest at the process plant conditions.

Microbially influenced corrosion (MIC) probes. Devices are available which it is claimed can be used to measure the amount of microbial activity in some environments. Most of these "MIC monitoring devices" are limited in their application. The major problem lies in proving that the corrosion being detected is due to MIC. This is an exciting area for development of corrosion probes and monitoring systems.

Use of Corrosion Probes The major use of corrosion monitoring probes is to measure the corrosion rate in the plant or the field. In addition to corrosion-rate measurements, corrosion probes can be used to detect process upsets that may change the corrosion resistance of the equipment of interest. This is usually equally as important a measurement as corrosion rate since a change in the process conditions can lead to dramatic changes in the corrosion rate.

If the upset can be detected and dealt with in short order, the system can be protected. Some of the probes that measure parameters such as pH, ion content, and others are sensitive to process upsets and may give the fastest most complete information about such changes.

Monitoring can also be used to optimize the chemistry and level of corrosion inhibitors used. If too little inhibitor is used, enhanced corrosion can result and failure may follow. If too much is used, costs will increase without providing any additional protection. Optimization of the addition of inhibitor in terms of time, location in the process, and method of addition can also be evaluated through the use of carefully placed probes.

Another area where probes can be used effectively is in monitoring of deposits such as scale. One method of measurement is detecting specific ions that contribute to scale buildup or fouling; another is measuring the actual layers. Scale and fouling often devastate the corrosion resistance of a system, leading to a costly corrosion-related plant shutdown.

A final type of measurement is the detection of localized corrosion, such as pitting or crevice attack. Several corrosion-measuring probes can be used to detect localized corrosion. Some can detect localized corrosion instantaneously and others only its result. These types of corrosion may contribute little to the actual mass loss, but can be devastating to equipment and piping. Detection and measurement of localized corrosion is one of the areas with the greatest potential for the use of some of the newest electrochemically based corrosion monitoring probes.

Corrosion Rate Measurements Determining a corrosion rate from measured parameters (such as mass loss, current, or electrical potential) depends on converting the measurements into a corrosion rate by use of relationships such as Faraday's law.

Information on the process reaction conditions may be important to prolonging the lifetime of process equipment. Techniques such as EIS and potentiodynamic polarization can provide just such information without being tied to a specific corrosion-rate measurement.

This is also the case with methods that yield information on localized corrosion. The overall corrosion rate may be small when localized attack occurs, but failure due to perforation or loss of function may be the consequence of localized attack.

Measuring Corrosion Rate with Coupons Corrosion rate determined with typical coupon tests is an average value, averaged over the entire life of the test. Changes in the conditions under which the coupons are tested are averaged over the time the coupon is exposed. *Uniform attack is assumed to occur.* Converting measured values of loss of mass into an average corrosion rate has been covered extensively by many authors, and standard practices exist for determining corrosion rate.

Corrosion rates may vary during testing. Since the rate obtained from coupon testing is averaged over time, the frequency of sampling is important. Generally, measurements made over longer times are more valid. This is especially true for low corrosion rates, under 1 mil/y, mpy (0.001 in/y). When corrosion rates are this low, longer times should be used.

Factors may throw off these rates—these are outlined in ASTM G31, "Standard Practice for Laboratory Immersion Corrosion Testing of Metals." Coupon-type tests cannot be correlated with changing plant conditions that may dramatically affect process equipment lifetimes. Other methods must be used if more frequent measurements are desired or correlation with plant conditions is necessary.

A plot of mass loss versus time can provide information about changes in the conditions under which the test has been run. One example of such a plot comes from the ASTM Standard G96, "Standard Guide." As mentioned previously, weight loss measurements are appropriate for measurement of localized pitting corrosion, including that caused by MIC.

Heat Flux Tests Removable tube test heat exchangers find an ideal use in the field for monitoring heat flux (corrosion) conditions, NACE TMO286-94 (similar to laboratory test, Fig. 25-6).

The assumption of uniform corrosion is also at the heart of the measurements made by the electrical resistance (ER) probes. Again, ASTM Standard G96 outlines the method for using ER probes in plant equipment. These probes operate on the principle that the electrical resistance of a wire, strip, or tube of metal increases as its cross-sectional area decreases:

$$R = \frac{\rho L}{A}$$

where R = resistance, Ω
 ρ = resistivity, Ω/cm
 L = length, cm
 A = cross-sectional area, cm^2

Usually, in practice, the resistance is measured as a ratio between the actual measuring element and a similar element protected from the corrosive environment (the reference), and is given by R_M/R_R where subscript M is for measured and R is for reference.

Measurements are recorded intermittently or continuously. Changes in the slope of the curve obtained thus yields the corrosion rate.

The initial measurement of electrical resistance must be made after considerable time. Phenomenological information has been determined based on the corrosion rate expected at what period of time to initiate readings of the electrical resistance. Since these values are

based on experiential factors rather than on fundamental (so-called *first*) principles, correlation tables and lists of suggested thicknesses, compositions, and response times for usage of ER-type probes have developed over time, and these have been incorporated into the values read out of monitoring systems using the ER method.

Electrochemical Measurement of Corrosion Rate There is a link between electrochemical parameters and actual corrosion rates. Probes have been specifically designed to yield signals that will provide this information. LPR, ER, and EIS probes can give corrosion rates directly from electrochemical measurements. ASTM G102, "Standard Practice for Calculation of Corrosion Rates and Related Information from Electrochemical Measurements," tells how to obtain corrosion rates directly. Background on the approximations made in making use of the electrochemical measurements has been outlined by several authors.

ASTM G59, "Standard Practice for Conducting Potentiodynamic Polarization Resistance Measurements," provides instructions for the graphical plotting of data (from tests conducted using the above-noted ASTM Standard G103) as the linear potential versus current density, from which the polarization resistance can be found.

Measurements of polarization resistance R_p , given by LPR probes, can lead to measurement of the corrosion rate at a specific instant, since values of R_p are instantaneous.

To obtain the corrosion current I_{corr} from R_p , values for the anodic and cathodic slopes must be known or estimated. ASTM G59 provides an experimental procedure for measuring R_p . A discussion of the factors which may lead to errors in the values for R_p , and cases where R_p technique cannot be used, are covered by Mansfield in "Polarization Resistance Measurements—Today's Status, Electrochemical Techniques for Corrosion Engineers" (NACE International, 1992).

Some data from corrosion-monitoring probes do not measure corrosion rate, but rather give other useful information about the system. For example, suppose conditions change dramatically during a process upset. An experienced corrosion engineer can examine the data and correlate it with the upset conditions. Such analysis can provide insight into the process and help to improve performance and extend equipment lifetime. Changes in simple parameters such as pH, ion content, and temperature may lead to detection of a process upset. Without careful analysis, process upsets can reduce the corrosion lifetime of equipment and even cause a system failure.

Analysis of biological activity is sometimes difficult to correlate to corrosion rates. However, with detection and correlation of microbiological processes, especially those known to be related to corrosion (e.g., respiration rates, amount of acidity being produced) with process conditions, such information may also lead to improvements in the corrosion lifetime of the process equipment.

Evidence of localized corrosion can be obtained from polarization methods such as potentiodynamic polarization, EIS, and electrochemical noise measurements, which are particularly well suited to providing data on localized corrosion. When evidence of localized attack is obtained, the engineer needs to perform a careful analysis of the conditions that may lead to such attack. Correlation with process conditions can provide additional data about the susceptibility of the equipment to localized attack and can potentially help prevent failures due to pitting or crevice corrosion. Since pitting may have a delayed initiation phase, careful consideration of the cause of the localized attack is critical. Laboratory testing and involvement of an experienced corrosion engineer may be needed to understand the initiation of localized corrosion. Theoretical constructs such as Pourbaix diagrams can be useful in interpreting data obtained by on-line monitors.

Combinations of several types of probes can improve the information concerning corrosion of the system. Using only probes that provide corrosion-rate data cannot lead to as complete an analysis of the process and its effect on the equipment as with monitoring probes, which provide additional information.

Other Useful Information Obtained by Probes Both EIS and electrochemical noise probes can be used to determine information about the reactions that affect corrosion. Equivalent circuit analysis, when properly applied by an experienced engineer, can often give insight into the specifics of the corrosion reactions. Information such as corrosion product layer buildup, or inhibitor effectiveness, or coating breakdown can be obtained directly from analysis of the data from EIS or

indirectly from electrochemical noise data. In most cases, this is merely making use of methodology developed in the corrosion laboratory.

Some assumptions must be made to do this, but these assumptions are no different from those made in the laboratory. Recent experience involving in-plant usage of EIS has shown that this technique can be used effectively as a monitoring method and has led to development of several commercial systems.

Making use of the information from monitoring probes, combined with the storage and analysis capabilities of portable computers and microprocessors, seems the best method for understanding corrosion processes. Commercial setups can be assembled from standard probes, cables, readout devices, and storage systems. When these are coupled with analysis by corrosion engineers, the system can lead to a better understanding of in-plant corrosion processes.

Limitations of Probes and Monitoring Systems There are limitations even with the most up-to-date systems. Some of the things which cannot be determined using corrosion probes include:

Specifics on the types and rates of microbiological attack. These must be determined by using other methods such as chemical and microbiological analysis of the solution and materials from the corrosion sites. Consideration must be given to limitations of electrochemical techniques for MIC studies, noted previously under "Corrosion Testing: Laboratory Tests" and subsequent subsections.

Actual lifetime of the plant equipment. Corrosion monitoring provides data, which must then be analyzed with additional input and interpretation. However, only estimates can be made of the lifetime of the equipment of concern. Lifetime predictions are, at best, carefully crafted guesses based on the best available data.

Choices of alternative materials. Corrosion probes are carefully chosen to be as close as possible to the alloy composition, heat treatment, and stress condition of the material that is being monitored. Care must be taken to ensure that the environment at the probe matches the service environment. Choices of other alloys or heat treatments and other conditions must be made by comparison. Laboratory testing or coupon testing in the process stream can be used to examine alternatives to the current material, but the probes and the monitors can only provide information about the conditions which are present during the test exposure and cannot extrapolate beyond those conditions.

Failure analysis. Often, for a corrosion-related failure, data from the probes are examined to look for telltale signs that could have led to detection of the failure. In some cases, evidence can be found that process changes were occurring which led to the failure. This does not mean that the probes should have detected the failure itself. Determination of an imminent corrosion-related failure is not possible, even with the most advanced monitoring system.

The aforementioned limitations are not problems with the probes or the monitoring systems but occur when information is desired that cannot be measured directly or which requires extrapolation. Many of the problems that are encountered with corrosion-monitoring systems and probes are related to the use to which probes are put.

Potential Problems with Probe Usage Understanding the electrochemical principles upon which probes are based helps to eliminate some of the potential problems with probes. However, in some situations, the information desired is not readily available.

For example, consider localized corrosion. Although data from corrosion probes indicate corrosion rate, it is not possible to tell that localized corrosion is the problem. It is certainly not possible to demonstrate the mechanisms responsible for the localized corrosion.

This is an example of measuring the wrong thing. In this case, the probes work adequately, the monitoring system is adequate, as is the monitoring interval, but detection of the type of corrosion cannot be made based on the available data. Different types of probes and testing are required to detect the corrosion problem.

Another problem can arise if the probes and monitors are working properly but the probe is placed improperly (this is very common in natural gas transmission pipelines). Then the probe does not measure the needed conditions and environment. The data obtained by the probe will not tell the whole story and can in fact give very misleading results.

An example of this is in a condenser where the corrosion probe is in a region where the temperature is lower than that at the critical condition of interest. Local scale buildup is another example of this type of situation, as is formation of a crevice at a specific location.

The type of probe, its materials, and method of construction must be carefully considered in designing an effective corrosion-monitoring system. Since different types of probes provide different types of information, it may be necessary to use several types.

Incorrect information can result if the probe is made of the wrong material and is not heat treated in the same way as the process equipment (as well as because of other problems). The probe must be as close as possible to the material from which the equipment of interest is made. Existence of a critical condition, such as weldments or galvanic couples or occluded cells in the equipment of concern, makes the fabrication, placement, and maintenance of the probes and monitoring system of critical importance, if accurate and useful data are to be obtained.

Before electrochemical techniques are used in the evaluation of any situation involving microbes, the test protocol must receive considerable review by personnel quite experienced in *both* electrochemical testing and microbiologically influenced corrosion. It must be demonstrated that the method is capable of detecting and in some cases quantitatively measuring corrosion influenced by microbes.

The data obtained from probes and monitoring systems are most useful when analyzed by a corrosion specialist. Data not taken, analyses not made, or expertise not sought can quickly lead to problems even with the most up-to-date corrosion probes and monitoring system.

ECONOMICS IN MATERIALS SELECTION

In most instances, there will be more than one alternative material which may be considered for a specific application. Calculation of true long-term costs requires estimation of the following:

1. Total cost of fabricated equipment and piping
2. Total installation cost
3. Service life
4. Maintenance costs: amount and timing
5. Time and cost requirements to replace or repair at the end of service life
6. Cost of downtime to replace or repair
7. Cost of inhibitors, extra control facilities, and so on, required to assure achievement of predicted service life
8. Time value of money
9. Factors which impact taxation, such as depreciation and tax rates
10. Inflation rate

Proper economic analysis will allow comparison of alternatives on a sound basis. Detailed calculations are beyond the scope of this section. The reader should review the material in the NACE Publication 3C194, Item No. 24182, "Economics of Corrosion," Sept., 1994.

PROPERTIES OF MATERIALS

MATERIALS STANDARDS AND SPECIFICATIONS

There are obvious benefits to be derived from consensus standards that define the chemistry and properties of specific materials. Such standards allow designers and users of materials to work with confidence that the materials supplied will have the expected minimum

properties. Designers and users can also be confident that comparable materials can be purchased from several suppliers. Producers are confident that materials produced to an accepted standard will find a ready market and therefore can be produced efficiently in large factories.

TABLE 25-2 Unified Alloy Numbering System (UNS)

UNS was established in 1974 by ASTM and SAE to reduce the confusion involved in the labeling of commercial alloys. Metals have been placed into 15 groups, each of which is given a code letter. The specific alloy is identified by a five-digit number following this code letter.

Nonferrous metals and alloys	
A00001–A99999	Aluminum and aluminum alloys
C00001–C99999	Copper and copper alloys
E00001–E99999	Rare-earth and rare-earth-like metals and alloys
L00001–L99999	Low-melting metals and alloys
M00001–M99999	Miscellaneous nonferrous metals and alloys
N00001–N99999	Nickel and nickel alloys
P00001–P99999	Precious metals and alloys
R00001–R99999	Reactive and refractory metals and alloys
Ferrous metals and alloys	
D00001–D99999	Specified-mechanical-properties steels
F00001–F99999	Cast irons and cast steels
G00001–G99999	AISI and SAE carbon and alloy steels
H00001–H99999	AISI H steels
K00001–K99999	Miscellaneous steels and ferrous alloys
S00001–S99999	Heat- and corrosion-resistant (stainless) steels
T00001–T99999	Tool steels

When possible, earlier widely used three- or four-digit alloy numbering systems such as those developed by the Aluminum Association (AA), Copper Development Association (CDA), American Iron and Steel Institute (AISI), etc., have been incorporated by the addition of the appropriate alloy-group code letter plus additional digits. For example:

Alloy description	Former designation		UNS designation
	System	No.	
Aluminum + 1.2% Mn	AA	3003	A93003
Copper, electrolytic tough pitch	CDA	110	C11000
Carbon steel, 0.2% C	AISI	1020	G10200
Stainless steel, 18 Cr, 8 Ni	AISI	304	S30400

Proprietary alloys are assigned numbers by the AA, AISI, CDA, ASTM, and SAE, which maintains master listings at their headquarters. Handbooks describing the system are available. (Cf. ASTM publication DS-56AC.)

SOURCE: ASTM DS-56A. (Courtesy of National Association of Corrosion Engineers.)

While a detailed treatment is beyond the scope of this subsection, a few of the organizations which generate standards of major importance to the chemical-process industries in the United States are listed below. An excellent overview is presented in the *Encyclopedia of Chemical Technology* (5th ed., Wiley, Hoboken, N.J., 2006).

1. **American National Standards Institute (ANSI)**, formerly American Standards Association (ASA). ANSI promulgates the piping codes used in the chemical-process industries.

2. **American Society of Mechanical Engineers (ASME)**. This society generates the Boiler and Pressure Vessel Codes.

3. **American Society for Testing and Materials (ASTM)**. This society generates specifications for most of the materials used in the ANSI Piping Codes and the ASME Boiler and Pressure Vessel Codes.

4. **International Organization for Standardization (ISO)**. This organization is engaged in generating standards for worldwide use. It has 80 member nations.

With only a few noticeable exceptions, metals and their alloys possess a distinct shape. To generate the principal shape, metals are said to be **wrought** or **cast**. Wrought metals receive their shape through deformation steps; cast metals receive their shape through solidification steps. Materials of construction are available in both wrought and cast forms. Each group is discussed separately.

WROUGHT MATERIALS: FERROUS METALS AND ALLOYS

Steel Carbon steel is the most common, least expensive, and most versatile metal used in industry. It has excellent ductility, permitting many cold-forming operations. Steel is also very weldable. The grades of steel most commonly used in the chemical-process industries have tensile strength in the 345- to 485-MPa (50,000 to

TABLE 25-3 Coefficient of Thermal Expansion of Common Alloys*

	UNS	10 ⁻⁶ in./ (in.°F)	10 ⁻⁶ mm/ (mm.°C)	Temperature range, °C
Aluminum alloy AA1100	A91100	13.1	24.	20–100
Aluminum alloy AA5052	A95052	13.2	24.	20–100
Aluminum cast alloy 43	A24430	12.3	22.	20–100
Copper	C11000	9.4	16.9	20–100
Red brass	C23000	10.4	18.7	20–300
Admiralty brass	C44300	11.2	20.	20–300
Muntz Metal	C28000	11.6	21.	20–300
Aluminum bronze D	C61400	9.0	16.2	20–300
Ounce metal	C83600	10.2	18.4	0–100
90-10 copper nickel	C70600	9.5	17.1	20–300
70-30 copper nickel	C71500	9.0	16.2	20–300
Carbon steel, AISI 1020	G10200	6.7	12.1	0–100
Gray cast iron	F10006	6.7	12.1	0–100
4-6 Cr, ½ Mo steel	S50100	7.3	13.1	20–540
Stainless steel, AISI 410	S41000	6.1	11.0	0–100
Stainless steel, AISI 446	S44600	5.8	10.4	0–100
Stainless steel, AISI 304	S30400	9.6	17.3	0–100
Stainless steel, AISI 310	S31000	8.0	14.4	0–100
Stainless steel, AISI HK	J94224	9.4	16.9	20–540
Nickel alloy 200	N02200	7.4	13.3	20–90
Nickel alloy 400	N04400	7.7	13.9	20–90
Nickel alloy 600	N06600	7.4	13.3	20–90
Nickel-molybdenum alloy B-2	N10665	5.6	10.1	20–90
Nickel-molybdenum alloy C-276	N10276	6.3	11.3	20–90
Titanium, commercially pure	R50250	4.8	8.6	0–100
Titanium alloy T1- 6Al-4V	R56400	4.9	8.8	0–100
Magnesium alloy AZ31B	M11311	14.5	26.	20–100
Magnesium alloy AZ91C	M11914	14.5	26.	20–100
Chemical lead		16.4	30.	0–100
50-50 solder	L05500	13.1	24.	0–100
Zinc	Z13001	18.	32.	0–100
Tin	L13002	12.8	23.	0–100
Zirconium	R60702	2.9	5.2	0–100
Molybdenum	R03600	2.7	4.9	20–100
Tantalum	R05200	3.6	6.5	20–100

*Courtesy of National Association of Corrosion Engineers.

70,000 lbf/in²) range, with good ductility. Higher strength levels are achieved by cold work, alloying, and heat treatment.

Carbon steel is easily the most commonly used material in process plants despite its somewhat limited corrosion resistance. It is routinely used for most organic chemicals and neutral or basic aqueous solutions at moderate temperatures. It is also used routinely for the storage of concentrated sulfuric acid and caustic soda [up to 50 percent and 55°C (130°F)]. Because of its availability, low cost, and ease of fabrication, steel is frequently used in services with corrosion rates of 0.13 to 0.5 mm/yr (5 to 20 mils/yr), with added thickness (**corrosion allowance**) to ensure the achievement of desired service life. Product quality requirements must be considered in such cases.

Corrosion under insulation is a major problem with carbon steels and must be taken into consideration in the RBI analysis of any plant to avoid serious failures. Several techniques are available to minimize this attack including painting, spraying with aluminum, and/or wrapping with aluminum foil.

TABLE 25-4 Melting Temperatures of Common Alloys*

	UNS	Melting range	
		°F	°C
Aluminum alloy AA1100	A91100	1190–1215	640–660
Aluminum alloy AA5052	A95052	1125–1200	610–650
Aluminum cast alloy 43	A24430	1065–1170	570–630
Copper	C11000	1980	1083
Red brass	C23000	1810–1880	990–1025
Admiralty brass	C44300	1650–1720	900–935
Muntz Metal	C28000	1650–1660	900–905
Aluminum bronze D	C61400	1910–1940	1045–1060
Ounce metal	C83600	1510–1840	854–1010
Manganese bronze	C86500	1583–1616	862–880
90-10 copper nickel	C70600	2010–2100	1100–1150
70-30 copper nickel	C71500	2140–2260	1170–1240
Carbon steel, AISI 1020	G10200	2760	1520
Gray cast iron	F10006	2100–2200	1150–1200
4-6 Cr, ½ Mo Street	S50100	2700–2800	1480–1540
Stainless steel, AISI 410	S41000	2700–2790	1480–1530
Stainless steel, AISI 446	S44600	2600–2750	1430–1510
Stainless steel, AISI 304	S30400	2550–2650	1400–1450
Stainless steel, AISI 310	S31000	2500–2650	1400–1450
Stainless steel, ACI HK	J94224	2550	1400
Nickel alloy 200	N02200	2615–2635	1440–1450
Nickel alloy 400	N04400	2370–2460	1300–1350
Nickel alloy 600	N06600	2470–2575	1350–1410
Nickel-molybdenum alloy B-2	N10665	2375–2495	1300–1370
Nickel-molybdenum alloy C-276	N10276	2420–2500	1320–1370
Titanium, commercially pure	R50250	3100	1705
Titanium alloy Ti-6Al-4V	R56400	2920–3020	1600–1660
Magnesium alloy AZ 31B	M11311	1120–1170	605–632
Magnesium alloy HK 31A	M13310	1092–1204	589–651
Chemical lead		618	326
50-50 solder	L05500	361–421	183–216
Zinc	Z13001	787	420
Tin	Z13002	450	232
Zirconium	R60702	3380	1860
Molybdenum	R03600	4730	2610
Tantalum	R05200	5425	2996

*Courtesy of National Association of Corrosion Engineers.

Low-Alloy Steels Alloy steels contain one or more alloying agents to improve the mechanical and corrosion-resistant properties over those of carbon steel. A typical low-alloy grade [American Iron and Steel Institute (AISI) 4340] contains 0.40 percent C, 0.70 percent Mn, 1.85 percent Ni, 0.80 percent Cr, and 0.25 percent Mo. Many other alloying agents are used to produce a large number of standard AISI and proprietary grades. Nickel increases toughness and improves low-temperature properties and corrosion resistance. Chromium and silicon improve hardness, abrasion resistance, corrosion resistance, and resistance to oxidation. Molybdenum provides strength at elevated temperatures. The addition of small amounts of alloying materials greatly improves corrosion resistance to atmospheric environments but does not have much effect against liquid corrosives. The alloying elements produce a tight, dense, adherent rust film, but in acid or alkaline solutions corrosion is about equivalent to that of carbon steel. However, the greater strength permits thinner walls in process equipment made from low-alloy steel.

Stainless Steel There are more than 70 standard types of stainless steel and many special alloys. These steels are produced in the wrought form (AISI types) and as cast alloys [Alloy Casting Institute (ACI) types]. Generally, all are iron-based, with 12 to 30 percent chromium, 0 to 22 percent nickel, and minor amounts of carbon, niobium (columbium), copper, molybdenum, selenium, tantalum, and titanium. These alloys are very popular in the process industries. They are heat- and corrosion-resistant, noncontaminating, and easily fabricated into complex shapes.

There are four groups of stainless alloys: (1) martensitic, (2) ferritic, (3) austenitic, and (4) duplex.

The **martensitic alloys** contain 12 to 20 percent chromium with controlled amounts of carbon and other additives. Type 410 is a typical member of this group. These alloys can be hardened by heat treatment, which can increase tensile strength from 550 to 1380 MPa (80,000 to 200,000 lbf/in²). Corrosion resistance is inferior to that of austenitic stainless steels, and martensitic steels are generally used in mildly corrosive environments (atmospheric, freshwater, and organic exposures). In the hardened condition, these materials are very susceptible to hydrogen embrittlement.

TABLE 25-5 Carbon and Low-Alloy Steels*

Steel type	ASTM	UNS	Composition, % ^b	Mechanical properties ^c		
				Yield strength, kip/in ² (MPa)	Tensile strength, kip/in ² (MPa)	Elongation, %
C-Mn	A53B	K03005	0.30 C, 1.20 Mn	35 (241)	60 (415)	
C-Mn	A106B	K03006	0.30 C, 0.29–1.06 Mn, 0.10 min. Si	35 (241)	60 (415)	30
C	A285A	K01700	0.17 C, 0.90 Mn	24 (165)	45–55 (310–380)	30
HSLA	A517F	K11576	0.08–0.22 C, 0.55–1.05 Mn, 0.13–0.37 Si, 0.36–0.79 Cr, 0.67–1.03 Ni, 0.36–0.64 Mo, 0.002–0.006 B, 0.12–0.53 Cu, 0.02–0.09 V	100 (689)	115–135 (795–930)	16
HSLA	A242(1)	K11510	0.15 C, 1.00 Mn, 0.20 min Cu, 0.15 P	42–50 (290–345)	63–70 (435–480)	21
2¼ Cr, 1 Mo	A387(22)	K21590	0.15 C, 0.30–0.60 Mn, 0.5 Si, 2.00–2.50 Cr, 0.90–1.10 Mo	30 (205) ^d 45 (310) ^e	60–85 (415–585) ^d 75–100 (515–690) ^e	18 ^d 18 ^d
4–6 Cr, ½ Mo	A335 (P5)	K41545	0.15 C, 0.30–0.60 Mn, 0.5 Si, 4.00–6.00 Cr, 0.45–0.65 Mo	30 (205)	60 (415)	
9 Cr, 1 Mo	A335 (P9)	K81590	0.15 C, 0.30–0.6 Mn, 0.25–1.00 Si, 8.00–10.00 Cr, 0.90–1.10 Mo	30 (205)	60 (415)	
9 Ni	A333(8), A353(1)	K81340	0.13 C, 0.90 Mn, 0.13–0.32 Si, 8.40–9.60 Ni	75 (515)	100–120 (690–825)	20
	AISI 4130	G41300	0.28–0.33 C, 0.80–1.10 Mn, 0.15–0.3 Si, 0.8–1.10 Cr, 0.15–0.25 Mo	120 (830) ^f	140 (965) ^f	22 ^f
	AISI 4340	G43400	0.38–0.43 C, 0.60–0.80 Mn, 0.15–0.3 Si, 0.70–0.90 Cr, 1.65–2.00 Ni, 0.20–0.30 Mo	125 (860) ^g	148 (1020) ^g	20 ^g

*Courtesy of National Association of Corrosion Engineers. To convert MPa to lbf/in², multiply by 145.04.

^bSingle values are maximum values unless otherwise noted.

^cRoom-temperature properties. Single values are minimum values.

^dClass 1.

^eClass 2.

^f1-in.-diameter bars water-quenched from 1,575°F (860°C) and tempered at 1,200°F (650°C).

^g1-in.-diameter bars oil-quenched from 1,550°F (845°C) and tempered at 1,200°F (650°C).

TABLE 25-6a Typical Physical Properties of Low-Alloy AISI Steels^a

AISI type	Melting temperature, °F	Thermal conductivity, Btu/(h·ft ²) (°F/ft) (212°F)	Coefficient of thermal expansion (0–1,200°F) per °F	Specific heat (68–212°F), Btu/(lb·°F)
13XX		27	7.9×10^{-6b}	0.10–0.11
23XX	2,600–2,620	38.3 ^c	8.0×10^{-6}	0.11–0.12
25XX	2,610–2,620	34.5–38.5 ^c	7.8×10^{-6}	0.11–0.12
40XX		27	8.3×10^{-6b}	0.10–0.11
41XX		24.7 ^d	0.11	
43XX	2,740–2,750	21.7 ^c	8.1×10^{-6}	0.107
46XX		27 ^d	6.3×10^{-6e}	0.10–0.11
48XX	2,750	26 ^f	8.6×10^{-6}	
51XX	2,720–2,760	27–34 ^g	7.4×10^{-6h}	0.10–0.11
61XX		27	8.1×10^{-6b}	0.10–0.11
86, 87XX	2,745–2,755	21.7 ^c	8.2×10^{-6}	0.107
92, 94XX		27	8.1×10^{-6b}	0.10–0.12

XX = nominal percent carbon.

^aDensity for all low-alloy steels is about 0.28 lb/in³.

^b68 to 1200°F.

^c120°F.

^d68°F.

^e0 to 200°F.

^f75°F.

^g32 to 212°F.

^h100 to 518°F.

Ferritic stainless steel contains 15 to 30 percent Cr, with low carbon content (0.1 percent). The higher chromium content improves its corrosive resistance. Type 430 is a typical example. The strength of ferritic stainless can be increased by cold working but not by heat treatment. Fairly ductile ferritic grades can be fabricated by all standard methods. They are fairly easy to machine. Welding is not a problem, although it requires skilled operators. Corrosion resistance is rated good, although ferritic alloys are not good against reducing acids such as HCl. But mildly corrosive solutions and oxidizing media are handled without harm. Type 430 is widely used in nitric acid plants. In addition, it is very resistant to scaling and high-temperature oxidation up to 800°C (1500°F).

Austenitic stainless steels are the most corrosion-resistant of the four groups. These steels contain 16 to 26 percent chromium and 6 to 22 percent nickel. Carbon is typically kept low (0.03 percent maximum) to minimize carbide precipitation. These alloys can be work-hardened, but heat treatment will not cause hardening. Tensile strength in the annealed condition is about 585 MPa (85,000 lbf/in²), but work hardening can increase this to 2000 MPa (300,000 lbf/in²) by creating martensite in the matrix. Austenitic stainless steels are tough and ductile. They can be fabricated by all standard methods. But austenitic grades are not easy to machine; they work-harden and gall. Rigid machines, heavy cuts, and low speeds are essential. Welding, however, is readily performed, although welding heat may cause chromium carbide precipitation, which depletes the alloy of some chromium and lowers its corrosion resistance in some specific environments, notably nitric acid. The carbide precipitation can be eliminated by heat treatment (solution annealing). To avoid precipitation, special stainless steels stabilized with titanium, niobium, or tantalum have been developed (types 321 and 347). The addition of molybdenum to the austenitic alloy (types 316, 316L, 317, and 317L) provides generally better corrosion resistance and improved resistance to pitting. Nitrogen additions have also shown benefit in improving localized corrosion resistance. Nitrogen also increases the strength, and materials today are frequently dual-certified, i.e., 304/304L but with a carbon content equal to a 304L.

In the stainless group, nickel greatly improves corrosion resistance over straight chromium stainless. Even so, the chromium-nickel steels, particularly the 18-8 alloys, perform best under **oxidizing conditions**, since resistance depends on an oxide film on the surface of the alloy. Reducing conditions and chloride ions destroy this film and

TABLE 25-6b Typical Mechanical Properties of Low-Alloy AISI Steels^a

AISI type	Tensile strength, 1,000 lbf/in ²	Yield strength (0.2% offset), 1000 lbf/in ²	Elongation (in 2 in), %	Reduction of area, %	Hardness, Brinell	Impact strength (Izod), ft-lbf
1,330 ^b	122	100	19	52	248	
1,335 ^c	126	105	20	59	262	
1,340 ^c	137	118	19	55	285	
2,317 ^c	107	72	27	71	222	84
2,515 ^c	113	94	25	69	233	85
E2,517 ^c	120	100	22	66	244	80
4,023 ^d	120	85	20	53	255	
4,032 ^e	210	182	11	49	415	
4,042 ^f	235	210	10	42	461	
4,053 ^g	250	223	12	40	495	
4,063 ^h	269	231	8	15	534	
4,130 ⁱ	200	170	16	49	375	25
4,140 ^j	200	170	15	48	385	16
4,150 ^k	230	215	10	40	444	12
4,320 ^l	180	154	15	50	360	32
4,337 ^k	210	140	14	50	435	18
4,340 ^k	220	200	12	48	445	16
4,615 ^d	100	75	18	52	42	
4,620 ^d	130	95	21	65	68	
4,640 ^j	185	160	14	52	390	25
4,815 ^d	150	125	18	58	325	44
4,817 ^d			15	52	355	36
4,820 ^j			13	47	380	28
5,120 ^l	143	114	13	45	302	6
5,130 ^m	189	175	13	51	380	
5,140 ^m	190	170	13	43	375	16
5,150 ^m	224	208	10	40	444	
6,120 ⁿ	125	94	21	56		28
6,145 ^o	176	169	16	52	429	20
6,150 ^o	187	179	13	42	444	13
8,620 ^p	122	98	21	63	245	76
8,630 ^p	162	142	14	54	325	42
8,640 ^p	208	183	13	43	420	18
8,650 ^p	214	194	12	41	423	
8,720 ^p	122	98	21	63	245	76
8,740 ^p	208	183	13	43	420	18
8,750 ^p	214	194	12	41	423	
9,255 ^p	232 ^q	215	9	21	477	6
9,261 ^p	258 ^r	226	10	30	514	12

^aProperties are for materials hardened and tempered as follows: ^bwater-quenched from 1,525°F, tempered at 1,000°F; ^coil-quenched from 1,525°F, tempered at 1,000°F; ^dpseudocarbonized 8 h at 1700°F, oil-quenched, tempered 1 h at 300°F; ^ewater-quenched from 1,525°F, tempered at 600°F; ^foil-quenched from 1,500°F, tempered at 600°F; ^goil-quenched from 1,475°F, tempered at 600°F; ^hoil-quenched from 1,450°F, tempered at 600°F; ⁱwater-quenched from 1,500 to 1,600°F, tempered at 800°F; ^joil-quenched from 1,550°F, tempered at 800°F; ^koil-quenched from 1,525°F, tempered at 800°F; ^lnormalized at 1,650°F, reheated to 1,475°F, oil-quenched, tempered at 800°F; ^mnormalized at 1,625°F, reheated to 1,550°F, water-quenched, tempered at 800°F; ⁿcarburized 10 h at 1,680°F, pot-cooled, oil-quenched from 1,525°F, tempered at 300°F; ^onormalized at 1,600°F, oil-quenched from 1,575°F, tempered at 1,000°F; ^poil-quenched tempered at 800°F; ^qnormalized at 1,650°F, reheated to 1,625°F, quenched in agitated oil, tempered at 800°F; ^rnormalized at 1,600°F, reheated to 1,575°F, quenched in agitated oil, tempered at 800°F.

NOTE: °C = (°F – 32) × 5/9. To convert Btu/(h·ft²·°F) to W/(m²·°C), multiply by 0.8606; to convert Btu/(lb·°F) to kJ/(kg·°C), multiply by 0.2388; to convert lbf/in² to MPa, multiply by 0.006895; and to convert ft-lbf to J, multiply by 0.7375.

bring on rapid attack. Chloride ions tend to cause pitting and crevice corrosion; when combined with high tensile stresses, they can cause stress-corrosion cracking.

A specialty group of alloys has been created with chromium and molybdenum significantly above the standard stainless grades. Many of these grades are proprietary and contain chromium of 20 percent and molybdenum of 6 to 7 percent, and most also contain nitrogen to

25-32 MATERIALS OF CONSTRUCTION

TABLE 25-7 Cast-Iron Alloys

Alloy	ASTM	UNS	Composition, % [†]	Condition	Mechanical properties [‡]			
					Yield strength, kip/in ² (MPa)	Tensile strength, kip/in ² (MPa)	Elongation, %	Hardness, HB
Gray cast iron	A159 (G3000)	F10006	3.1–3.4 C, 0.6–0.9 Mn, 1.9–2.3 Si	As cast		30 (207)		187–241
Malleable cast iron	A602 (M3210)	F20000	2.2–2.9 C, 0.15–1.25 Mn, 0.9–1.90 Si	Annealed	32 (229)	50 (345)	12	130
Gray cast iron	A436(1)	F41000	3.0 C, 1.5–2.5 Cr, 5.5–7.5 Cu, 0.5–1.5 Mn, 13.5–17.5 Ni, 1.0–2.8 Si	As cast		25 (172)		150
Gray cast iron	A436(2)	F41002	3.0 C, 1.5–2.5 Cr, 0.50 Cu, 0.5–1.5 Mn, 18–22 Ni, 1.0–2.8 Si	As cast		25 (172)		145
Gray cast iron	A436(5)	F41006	2.4 C, 0.1 Cr, 0.5 Cu, 0.5–1.5 Mn, 34–36 Ni, 1.0–2.0 Si	As cast		20 (138)		110
Ductile austenitic cast iron	A439(D-2)	F43000	3.0 C, 1.75–2.75 Cr, 0.7–1.25 Mn, 18–22 Ni, 1.5–3.0 Si	As cast	30 (207)	58 (400)		170
Ductile austenitic cast iron	A439 (D-5)	F43006	2.4 C, 0.1 Cr, 1.0 Mn, 34–36 Ni, 1.0–2.8 Si	As cast	30 (207)	55 (379)		155
Silicon cast iron	A518	F47003	0.7–1.1 C, 0.5 Cr, 0.5 Cu, 1.50 Mn, 0.5 Mo, 14.2–14.75 Si	As cast		16 (110)		520

To convert MPa to lbf/in², multiply by 145.04.

[†]Single values are maximum values.

[‡]Typical room-temperature properties.

improve localized corrosion resistance. This family is collectively known as the **6Mo family** and has corrosion resistance intermediate between the standard grades and some of the nickel-based alloys. They also have improved stress corrosion and localized corrosion resistance. Typical of this family are Avesta 254 SMO and Allegheny AL6XN.

Duplex stainless steels were created by adjusting the composition so that approximately equal amounts of the austenitic phase and the ferritic phase are present in the alloy. Three general families of alloys have been developed that contain 18 percent Cr, 22 percent Cr, and 25 percent Cr. As the chromium content increases, so does the general resistance of the alloys. All duplex alloys have better stress-corrosion resistance than the austenitic grades while the general corrosion resistance is not significantly increased. Because the structure is a mixture of ferritic and austenitic grains, the yield strength of this class of materials is significantly higher than that for the austenitic grades. Some of the “super duplex” (25 Cr) alloys have been used in offshore oil platforms because

of their strength and corrosion resistance, and they are finding increasing uses in chemical plants. These alloys suffer from a 474°C (885°F) embrittlement and so have lower maximum-use temperatures than the austenitic alloys. The presence of sigma phase in the microstructure can cause serious corrosion and mechanical problems. ASTM A 923 was developed to assist in determining the presence of sigma phase.

WROUGHT MATERIALS: NONFERROUS METALS AND ALLOYS

Nickel and Nickel Alloys Nickel is available in practically any mill form as well as in castings. It can be machined easily and joined by welding. Generally, oxidizing conditions favor corrosion, while reducing conditions retard attack. Neutral alkaline solutions, seawater, and mild atmospheric conditions do not affect nickel. The metal is widely used for

TABLE 25-8 Standard Wrought Martensitic Stainless Steels

AISI type	UNS	Composition, % [*]					Mechanical properties [†]			
							Yield strength, kip/in ² (MPa) [‡]	Tensile strength, kip/in ² (MPa) [‡]	Elongation, %	Hardness, HB
		Cr	Ni	Mo	C	Other				
403	S40300	11.5–13.0			0.15		40 (276)	75 (517)	35	155
410	S41000	11.5–13.5			0.15		35 (241)	70 (483)	30	150
414	S41400	11.5–13.5	1.25–2.5		0.15		90 (621)	115 (793)	20	235
416	S41600	12–14		0.6	0.15	0.15S§	40 (276)	75 (517)	30	155
416Se	S41623	12–14			0.15	0.15Se§	40 (276)	75 (517)	30	155
420	S42000	12–14			0.15		50 (345)	95 (655)	20	195
420F	S42020	12–14		0.6	0.15§	0.15S§	55 (379)	95 (655)	22	220
422	S42200	11–13	0.5–1.0	0.75–1.25	0.20–0.25	0.15–0.30 V, 0.75–1.25 W	125 (862)	145 (1000)	18	320
431	S43100	15–17	1.25–2.5		0.20		95 (665)	125 (862)	20	260
440A	S44002	16–18		0.75	0.6–0.75		60 (414)	105 (724)	20	210
440B	S44003	16–18.0		0.75	0.75–0.95		62 (427)	107 (738)	18	215
440C	S44004	16–18		0.75	0.95–1.20		65 (448)	110 (758)	14	220
501	S50100	4–6		0.40–0.65	0.10§		30 (207)	70 (483)	28	160
502	S50200	4–6		0.40–0.65	0.10		25 (172)	65 (448)	30	150

^{*}Single values are maximum values unless otherwise noted.

[†]Typical room-temperature properties of annealed plates.

[‡]To convert MPa to lbf/in², multiply by 145.04.

§Minimum.

TABLE 25-9 Standard Wrought Ferritic Stainless Steels

AISI type	Composition, %†								Mechanical properties‡			
	UNS	Cr	C	Mn	Si	P	S	Other	Yield strength, kip/in ² (MPa)*	Tensile strength, kip/in ² (MPa)*	Elongation, %	Hardness, HB
405	S40500	11.5–14.5	0.08	1.0	1.0	0.04	0.03	0.1–0.3 Al	40 (276)	65 (448)	30	150
409	S40900	10.5–11.75	0.08	1.0	1.0	0.045	0.045	(6 × C) Ti§	35 (241)	65 (448)	25	137
429	S42900	14–16	0.12	1.0	1.0	0.04	0.03		40 (276)	70 (483)	30	163
430	S43000	16–18	0.12	1.0	1.0	0.04	0.03		40 (276)	75 (517)	30	160
430F	S43020	16–18	0.12	1.25	1.0	0.06	0.15¶	0.6 Mo	55 (379)	80 (552)	25	170
434	S43400	16–18	0.12	1.0	1.0	0.04	0.03	0.75–1.25 Mo	53 (365)	77 (531)	23	160
436	S43600	16–18	0.12	1.0	1.0	0.04	0.03	0.75–1.25 Mo (5 × C)(Cb + Ta)§	53 (365)	77 (531)	23	160
442	S44200	18–23	0.20	1.0	1.0	0.04	0.03		45 (310)	80 (552)	20	185
446	S44600	23–27	0.20	1.5	1.0	0.04	0.03	0.25N	55 (379)	85 (586)	25	160

*To convert MPa to lbf/in², multiply by 145.04.

†Single values are maximum values unless otherwise noted.

‡Typical temperature properties of annealed plates.

§0.70 maximum.

¶Minimum.

handling alkalis, particularly in concentrating, storing, and shipping high-purity caustic soda. Chlorinated solvents and phenol are often refined and stored in nickel to prevent product discoloration and contamination. A large number of nickel-based alloys are commercially available.

One of the best known of these is **Monel 400**, 67 percent Ni and 30 percent Cu. It is available in all standard forms. This nickel-copper alloy is ductile and tough and can be readily fabricated and joined. Its corrosion resistance is generally superior to that of its components, being more resistant than nickel in reducing environments and more resistant than copper in oxidizing environments. The alloy can be used for relatively dilute sulfuric acid (below 80 percent), although aeration will result in increased corrosion. Monel will handle hydrofluoric acid up to 92 percent and 115°C (235°F) although aeration can produce cracking. Alkalies have little effect on this alloy, but it will not stand up against very highly oxidizing or reducing environments.

Another important alloy is a Ni-Mo alloy, **Hastelloy alloy B-3**. This alloy is 74 percent nickel and 26 percent molybdenum. The alloy is resistant to all hydrochloric acid solutions and other strongly reducing acids. The addition of oxidizers such as ferric or cupric ions or oxygen will accelerate corrosion. Work hardening presents some fabrication difficulties, and machining is somewhat more difficult than for 316L stainless.

The addition of chromium forms a family of Ni-Cr-Mo alloys such as **Hastelloy alloys C-276, C-22, and C-2000**. These alloys contain 16 to 22 percent chromium and 13 to 16 percent molybdenum and are very resistant to a wide variety of chemical environments. They are considered resistant to stress-corrosion cracking and very resistant to localized corrosion in chloride-containing environments. These alloys are resistant to strong oxidizing solutions, such as wet chlorine and hypochlorite solutions. They are among only a few alloys that are completely resistant to seawater. The carbon contents are low enough that weld sensitization is not a problem during fabrication. These alloys are also more difficult to machine than stainless steel, but fabrication is essentially the same.

Replacing some of the nickel with iron produces a family of alloys with intermediate corrosion resistance between stainless steels and the Ni-Cr-Mo alloys. Alloys such as **Incoloy 825** and **Hastelloy G-3 and G-30** are in this family. Incoloy 825 has 40 percent Ni, 21 percent Cr, 3 percent Mo, and 2.25 percent Cu. Hastelloy G-3 contains 44 percent Ni, 22 percent Cr, 6.5 percent Mo, and 0.05 percent C maximum. These alloys have extensive applications in sulfuric acid systems. Because of their increased nickel and molybdenum contents they are more tolerant of chloride-ion contamination than are standard stainless steels. The nickel content decreases the risk of stress-corrosion cracking; molybdenum improves resistance to crevice corrosion and pitting. Many of the nickel-based alloys are proprietary and are covered by the following specifications:

1. Sheet and plate: ASTM B 333, B575
2. Bar: ASTM B 335, B 574
3. Forgings: ASTM B 564

4. Welded pipe and tubing: ASTM B 626, B 619

5. Seamless pipe and tubing: ASTM B 622

6. Fittings: ASTM B 366

All the nickel alloys are readily fabricated and welded. Their design strengths allow use to elevated temperatures and with relatively thin wall thicknesses. The Nickel Institute (<http://www.nickelinstitute.org>) has excellent publications available that detail the properties and fabrication requirements for both stainless steels and nickel-based alloys, as does SSINA (<http://www.ssina.com>) and so do many of the manufacturers of these alloys.

Aluminum and Alloys Aluminum and its alloys are made in practically all the forms in which metals are produced, including castings. Thermal conductivity of aluminum is 60 percent of that of pure copper, and unalloyed aluminum is used in many heat-transfer applications. Its high electrical conductivity makes aluminum popular in electrical applications. Aluminum is one of the most workable of metals, and it is usually joined by inert-gas-shielded arc-welding techniques.

Commercially pure aluminum has a tensile strength of 69 MPa (10,000 lbf/in²), but it can be strengthened by cold working. One limitation of aluminum is that strength declines greatly above 150°C (300°F). When strength is important, 200°C (400°F) is usually considered the highest permissible safe temperature for aluminum. However, aluminum has excellent low-temperature properties; it can be used at -250°C (-420°F). Aluminum has high resistance to atmospheric conditions as well as to industrial fumes and vapors and fresh, brackish, or salt waters. Many mineral acids attack aluminum, although the metal can be used with concentrated nitric acid (above 82 percent) and glacial acetic acid. Aluminum cannot be used with strong caustic solutions.

Note that a number of **aluminum alloys** are available (see Table 25-17). Many have improved mechanical properties over pure aluminum. The wrought heat-treatable aluminum alloys have tensile strengths of 90 to 228 MPa (13,000 to 33,000 lbf/in²) as annealed; when they are fully hardened, strengths can go as high as 572 MPa (83,000 lbf/in²). However, aluminum alloys usually have lower corrosion resistance than the pure metal.

The **Alclad** alloys have been developed to overcome this shortcoming. Alclad consists of a pure aluminum layer metallurgically bonded to a core alloy. The corrosion resistance of aluminum and its alloys tends to be very sensitive to trace contamination. Very small amounts of metallic mercury, heavy-metal ions, or chloride ions can frequently cause rapid failure under conditions which otherwise would be fully acceptable. When alloy steels do not give adequate corrosion protection—particularly from sulfidic attack—steel with an **aluminized surface coating** can be used.

A spray coating of aluminum on a steel is not likely to spall or flake, but the coating is usually not continuous and may leave some areas of the steel unprotected. Hot-dipped “aluminized” steel gives a continuous coating and has proved satisfactory in a number of applications,

particularly when sulfur or hydrogen sulfide is present. It is also used to protect thermal insulation and as weather shields for equipment. The coated steel resists fires better than solid aluminum.

Copper and Alloys Copper and its alloys are widely used in chemical processing, particularly when heat and electrical conductivity are important factors. The thermal conductivity of copper is twice that of aluminum and 90 percent that of silver. A large number of copper alloys are available, including brasses (Cu-Zn), bronzes (Cu-Sn), cupronickels (Cu-Ni), and age-hardenable alloys such as copper beryllium (Cu-Be) and copper nickel tin (Cu-Ni-Sn).

Copper has excellent low-temperature properties and is used at $\sim 200^{\circ}\text{C}$ ($\sim 320^{\circ}\text{F}$). Brazing and soldering are common joining methods for copper, although welding, while difficult, is possible. Generally, copper has high resistance to industrial and marine atmospheres, seawater, alkalis, and solvents. Oxidizing acids rapidly corrode copper. However, the alloys have somewhat different properties from those of commercial copper.

Brasses with up to 15 percent Zn are ductile but difficult to machine. Machinability improves with increasing zinc up to 36 percent Zn. Brasses with less than 20 percent Zn have corrosion resistance equivalent to that of copper but with better tensile strengths. Brasses with 20 to 40 percent Zn have lower corrosion resistance and are subject to dezincification and stress-corrosion cracking, especially when ammonia is present.

Bronzes are somewhat similar to brasses in mechanical properties and to high-zinc brasses in corrosion resistance (except that bronzes are not affected by stress cracking).

Aluminum and silicon bronzes are very popular in the process industries because they combine good strength with corrosion resistance. Copper-beryllium alloys offer the greatest strength and excellent corrosion resistance in seawater and are resistant to stress-corrosion cracking in hydrogen sulfide.

Cupronickels (10 to 30 percent Ni) have become very important as copper alloys. They have the highest corrosion resistance of all copper alloys and find application as heat-exchanger tubing. Resistance to seawater is particularly outstanding. Cu-Ni-Sn spinodal strengthened alloys offer higher strength than the cupronickels along with excellent corrosion resistance. These alloys demonstrate very low weight loss in sulfide-containing environments and negligible weight loss in seawater.

The Copper Development Association has excellent references on copper and its alloys at <http://www.copper.org>.

Lead and Alloys Chemical leads of 99.9 percent purity are used primarily in the chemical industry in environments that form thin, insoluble, and self-repairable protective films, e.g., salts such as sulfates, carbonates, or phosphates. More soluble films such as nitrates, acetates, or chlorides offer little protection. Alloys of antimony, tin, and arsenic offer limited improvement in mechanical properties, but the usefulness of lead is limited primarily because of its poor structural qualities. It has a low melting point and tensile stress as low as 1 MPa (145 lbf/in²).

For information on lead and its uses, contact Lead Development Association International at <http://www.ldaint.org>.

Titanium Titanium has become increasingly important as a construction material. It is strong and of medium weight. Corrosion resistance is very superior in oxidizing and mild reducing media (Ti-Pd alloys Grade 7 and 11 have superior resistance in reducing environments, as does the Ti-Mo-Ni alloy Grade 12). Titanium alloys with lower amounts of Pd and Ru have been developed to reduce the cost (Grades 16 and 26). Titanium is usually not bothered by impingement attack, crevice corrosion, or pitting attack in seawater. Its general resistance to seawater is excellent. Titanium is resistant to nitric acid at all concentrations except with red fuming nitric. The metal also resists ferric chloride, cupric chloride, and other hot chloride solutions. However, there are a number of disadvantages to titanium which have limited its use. Titanium is not easy to form, it has a high springback and tends to gall, and welding must be carried out in an inert atmosphere.

For complete information on titanium, its properties and uses, contact the International Titanium Association at <http://www.ita.org>.

Zirconium Zirconium was originally developed as a construction material for atomic reactors. Reactor-grade zirconium contains very

little hafnium, which would alter zirconium's neutron-absorbing properties. Commercial-grade zirconium, for chemical-process applications, however, contains 2.5 percent maximum hafnium. Zirconium resembles titanium from a fabrication standpoint. All welding must be done under an inert atmosphere. Zirconium has excellent resistance to reducing environments. Oxidizing agents frequently cause accelerated attack. It resists all chlorides except ferric and cupric. Zirconium alloys should *not* be used in concentrations of sulfuric acid above about 70 percent. There are a number of alloys of titanium and zirconium, with mechanical properties superior to those of the pure metals.

Tantalum The physical properties of tantalum are similar to those of mild steel except that tantalum has a higher melting point. Tantalum is ductile and malleable and can be worked into intricate forms. It can be welded by using inert-gas-shielded techniques. The metal is practically inert to many oxidizing and reducing acids (except fuming sulfuric). It is attacked by hot alkalis and hydrofluoric acid. Its cost generally limits use to heating coils, bayonet heaters, coolers, and condensers operating under severe conditions. When economically justified, larger items of equipment (reactors, tanks, etc.) may be fabricated with tantalum liners, either loose (with proper anchoring) or explosion-bonded-clad. Since tantalum linings are usually very thin, very careful attention to design and fabrication details is required.

CAST MATERIALS

Cast Irons Generally, cast iron is not a particularly strong or tough structural material, although it is one of the most economical and is widely used industrially.

Gray cast iron, low in cost and easy to cast into intricate shapes, contains carbon, silicon, manganese, and iron. Carbon (1.7 to 4.5 percent) is present as combined carbon and graphite; combined carbon is dispersed in the matrix as iron carbide (cementite), while free graphite occurs as thin flakes dispersed throughout the body of the metal. Various strengths of gray iron are produced by varying the size, amount, and distribution of graphite. Gray iron has outstanding damping properties, i.e., ability to absorb vibration, as well as wear resistance. However, gray iron is brittle, with poor resistance to impact and shock. Machinability is excellent. With some important exceptions, gray-iron castings generally have corrosion resistance similar to that of carbon steel. They do resist atmospheric corrosion as well as attack by natural or neutral waters and neutral soils. However, dilute acids and acid-salt solutions will attack this material. Gray iron is resistant to concentrated acids (nitric, sulfuric, phosphoric) as well as to some alkaline and caustic solutions. Caustic fusion pots are usually made from gray cast iron with low silicon content; cast-iron valves, pumps, and piping are common in sulfuric acid plants.

White cast iron is brittle and difficult to machine. It is made by controlling the composition and rate of solidification of the molten iron so that all the carbon is present in the combined form. Very abrasion- and wear-resistant, white cast iron is used as liners and for grinding balls, dies, and pump impellers.

Malleable iron is made from white cast iron. It is cast iron with free carbon as dispersed nodules. This arrangement produces a tough, relatively ductile material. Total carbon is about 2.5 percent. Two types are produced: standard and pearlitic (combined carbon plus nodules). Standard malleable iron is easily machined; pearlitic, less so. Both types will withstand bending and cold working without cracking. Large welded areas are not recommended with fusion welding because welds are brittle. Corrosion resistance is about the same as for gray cast iron.

Ductile cast iron includes a group of materials with good strength, toughness, wear resistance, and machinability. This type of cast iron contains combined carbon and dispersed nodules of carbon. Composition is about the same as that of gray iron, with more carbon (3.7 percent) than malleable iron. The spheroidal graphite reduces the notch effect produced by graphite flakes, making the material more ductile. There are a number of grades of ductile iron; some have maximum toughness and machinability, others have maximum resistance to oxidation. Generally, corrosion resistance is similar to that of gray iron. But ductile iron can be used at higher temperatures—up to 590°C (1100°F) and sometimes even higher.

Alloy Cast Irons Cast iron is not usually considered corrosion-resistant, but this condition can be improved by the use of various cast-iron alloys. A number of such materials are commercially available.

High-silicon cast irons have excellent corrosion resistance. Silicon content is 13 to 16 percent. This material is known as Duricon. Adding 4 percent Cr yields a product called Durichlor, which has improved resistance in the presence of oxidizing agents. These alloys are not readily machined or welded. Silicon irons are very resistant to oxidizing and reducing environments, and resistance depends on the formation of a passive film. These irons are widely used in sulfuric acid service, since they are unaffected by sulfuric at all strengths, even up to the boiling point. Because they are very hard, silicon irons are good for combined corrosion-erosion service.

Another group of cast-iron alloys is called **Ni-Resist**. These materials are related to gray cast iron in that they have high carbon contents (3 percent), with fine graphite flakes distributed throughout the structure. Nickel contents range from 13.5 to 36 percent, and some have 6.5 percent Cu. Generally, nickel-containing cast irons have superior toughness and impact resistance compared with gray irons. The nickel-alloy castings can be welded and machined. Corrosion resistance of nickel-containing cast-iron alloys is superior to that of cast irons but less than that of pure nickel. There is little attack from neutral or alkaline solutions. Oxidizing acids such as nitric are highly detrimental. Cold, concentrated sulfuric acid can be handled. Ni-Resist has excellent heat resistance, with some grades serviceable up to 800°C (1500°F). Also, a ductile variety is available as well as a hard variety (Ni-Hard).

Cast stainless alloys are widely used in pumps, valves, and fittings. These casting alloys are designated under the ACI system. All corrosion-resistant alloys have the letter C plus a second letter (A to N) denoting increasing nickel content. Numerals indicate maximum carbon. While a rough comparison can be made between ACI and AISI types, compositions are not identical and analyses cannot be used interchangeably. Foundry techniques require a rebalancing of the wrought chemical compositions. However, corrosion resistance is not greatly affected by these composition changes. Typical members of this group are CF8, similar to type 304 stainless, and CF8M, similar to type CD3MWCuN, which represent the 25 Cr duplex stainless grades and have improved resistance to nitric, sulfuric, and phosphoric acids. Other duplex grades are CD3MN and CE3MN.

In addition to the C grades, there is a series of heat-resistant grades of ACI cast alloys, identified similarly to the corrosion-resistant grades, except that the first letter is H rather than C. Mention should also be made of precipitation-hardening (PH) stainless steels, which can be hardened by heat treatments at moderate temperatures. Very strong and hard at high temperatures, these steels have but moderate corrosion resistance. A typical PH steel, CB7Cu-1, containing 17 percent Cr, 7 percent Ni, and 1.1 percent Al, has high strength, good fatigue properties, and good resistance to wear and cavitation corrosion. A large number of these steels with varying compositions are commercially available. Essentially, they contain chromium and nickel with added alloying agents such as copper, aluminum, beryllium, molybdenum, nitrogen, and phosphorus.

Medium Alloys The group of (mostly) proprietary alloys with somewhat better corrosion resistance than stainless steels is called medium alloys. A popular member of this group is the **20 alloy**, made by a number of companies under various trade names. Durimet 20 is a well-known cast version, containing 0.07 percent C, 29 percent Ni, 20 percent Cr, 2 percent Mo, and 3 percent Cu. The ACI designation of this alloy is CN7M. Worthite is another proprietary 20 alloy with about 24 percent Ni and 20 percent Cr. The 20 alloy was originally developed to fill the need for a material with sulfuric acid resistance superior to that of the stainless steels. Also included are the 6Mo alloys such as CK3MCuN, similar to alloy 254SMO, and CN3MN, similar to alloy AL6XN.

Other members of the medium-alloy group do not have ACI designations but are listed in ASTM specifications (refer to ASTM A 494) such as **Incoloy 825** and **Hastelloy G-3** and **G-30**. Several nickel-based alloys that are normally only wrought have cast equivalents but are not listed in any specifications. Hastelloy G-3 contains 44 percent

Ni, 22 percent Cr, 6.5 percent Mo, and 0.05 percent C maximum while alloy G-30 has 30 percent Cr. These alloys have extensive applications in sulfuric acid systems. Because of their increased nickel and molybdenum contents they are more tolerant of chloride-ion contamination than are standard stainless steels. The nickel content decreases the risk of stress-corrosion cracking; molybdenum improves resistance to crevice corrosion and pitting. **Inconel 600** (80 percent Ni, 16 percent Cr, and 7 percent Fe) should also be mentioned as an intermediate alloy. It contains no molybdenum. The corrosion-resistant grade is recommended for reducing-oxidizing environments, particularly at high temperatures. When heated in air, this alloy resists oxidation up to 1100°C (2000°F). The alloy is outstanding in resisting corrosion by gases when these gases are essentially sulfur-free. The alloy is also used with a number of alkalies.

High Alloys The materials called high alloys all contain relatively large percentages of nickel. **N-12MV** contains 61 percent Ni and 28 percent Mo while **N7-M** has 63 percent Ni and 32 percent Mo. Control of carbon and silicon is better with **N7-M** and is important to receiving quality castings. Work hardening presents some fabrication difficulties, and machining is somewhat more difficult than for type 316 stainless. Conventional welding methods can be used. The alloy has unusually high resistance to all concentrations of hydrochloric acid at all temperatures in the absence of oxidizing agents. Sulfuric acid attack is low for all concentrations at 65°C (150°F), but the rate goes up with temperature. Oxidizing acids and salts rapidly corrode these Ni-Mo cast materials, but alkalies and alkaline solutions cause little damage.

Ni-Cr-Mo castings are covered by **CW-12MW**, which is a nickel-based alloy containing chromium (16 percent), molybdenum (16 percent), and tungsten (3 percent) as major alloying elements. This alloy is a cast modification of Hastelloy alloy C. This alloy is resistant to strong oxidizing chloride solutions, such as wet chlorine and hypochlorite solutions. It is one of the very few alloys which are totally resistant to seawater. **CW-6M** is a similar alloy with improved carbon control. Two improved alloys in this family are **CW-2M** and **CX-2MW**. The lower residuals in **CW-2M** provide for a material with much better thermal stability while the higher chromium of **CX-2MW** provides for better general corrosion resistance than the older grades.

Casting Specifications of Interest Refer to these specifications for the chemical analyses and properties of the various alloys:

1. ASTM A 217/A 217M-04 Standard Specification for Steel Castings, Martensitic Stainless and Alloy, for Pressure-Containing Parts, Suitable for High-Temperature Service (contains CA15)
2. ASTM A 297/A 297M-97(2003) Standard Specification for Steel Castings, Iron-Chromium and Iron-Chromium-Nickel, Heat Resistant, for General Application (contains most of the H grade, e.g., HP, castings)
3. ASTM A 351/A 351M-05 Standard Specification for Castings, Austenitic, Austenitic-Ferritic (Duplex), for Pressure-Containing Parts (contains most of the stainless grades, mostly corrosion, but some high-temperature)
4. ASTM A 494/A 494M-04 Standard Specification for Castings, Nickel and Nickel Alloy (contains many of the corrosion nickel alloy castings)
5. ASTM A 743/A 743M-03 Standard Specification for Castings, Iron-Chromium, Iron-Chromium-Nickel, Corrosion Resistant, for General Application (contains many of the corrosion grades)
6. ASTM A 744/A 744M-00(2004) Standard Specification for Castings, Iron-Chromium-Nickel, Corrosion Resistant, for Severe Service (contains some of the austenitic corrosion grades)
7. ASTM A 747/A 747M-04 Standard Specification for Steel Castings, Stainless, Precipitation Hardening (contains CB7Cu-1 and -2)
8. ASTM A 890/A 890M-99(2003) Standard Specification for Castings, Iron-Chromium-Nickel-Molybdenum Corrosion-Resistant, Duplex (Austenitic/Ferritic) for General Application (contains the major duplex grades)
9. ASTM A 990-05 Standard Specification for Castings, Iron-Nickel-Chromium and Nickel Alloys, Specially Controlled for Pressure Retaining Parts for Corrosive Service [currently contains CW-2M (C-4C) and two others].

TABLE 25-10 Properties of Chemical-Resistant Brick

Property	Red shale	Fireclay	Carbon	Silica
Specific gravity (ASTM C 20)	2.4–2.5	2.2–2.3	1.53	1.8–1.9
Compressive strength, psi (ASTM C 133)	10–30,000	10–30,000	7–10,000	4000–7000
Water absorption, % (ASTM C 20)	1.0–7.0	1.0–7.0	7.0–12.0	6.0–7.0
Apparent porosity, % (ASTM C 20)	4.0–12.0	4.0–12.0	18.0–21.0	12.0–16.0
Coefficient of thermal expansion, per °F (ASTM E 228)	$2.5\text{--}3.0 \times 10^{-6}$	$2.5\text{--}3.0 \times 10^{-6}$	3.5×10^{-6}	$2.2\text{--}2.8 \times 10^{-6}$
Thermal conductivity, Btu/(ft ² ·h·°F) (ASTM C 113)	8–10	8–10	30–40	5–9

INORGANIC NONMETALLICS

Glass and Glassed Steel Glass is an inorganic product of fusion which is cooled to a rigid condition without crystallizing. With unique properties compared with metals, these materials require special considerations in their design and use.

Glass has excellent resistance to all acids except hydrofluoric and hot, concentrated H₃PO₄. It is also subject to attack by hot alkaline solutions. Glass is particularly suitable for piping when transparency is desirable.

The chief drawback of glass is its brittleness, and it is also subject to damage by thermal shock. However, glass armored with epoxy-polyester fiberglass can readily be protected against breakage. On the other hand, glassed steel combines the corrosion resistance of glass with the working strength of steel. Accordingly, **glass linings** are resistant to all concentrations of hydrochloric acid to 120°C (250°F), to dilute concentrations of sulfuric to the boiling point, to concentrated sulfuric to 230°C (450°F), and to all concentrations of nitric acid to the boiling point. Acid-resistant glass with improved alkali resistance (up to 12 pH) is available.

A nucleated crystalline ceramic-metal composite form of glass has superior mechanical properties compared with conventional

glassed steel. Controlled high-temperature firings chemically and physically bond the ceramic to steel, nickel-based alloys, and refractory metals. These materials resist corrosive hydrogen chloride gas, chlorine, or sulfur dioxide at 650°C (1200°F). They resist all acids except HF up to 180°C (350°F). Their impact strength is 18 times that of safety glass; abrasion resistance is superior to that of porcelain enamel. They have 3 to 4 times the thermal-shock resistance of glassed steel.

Porcelain and Stoneware Porcelain and stoneware materials are about as resistant to acids and chemicals as glass, but with greater strength. This is offset by a greater potential for thermal shock. **Porcelain enamels** are used to coat steel, but the enamel has slightly inferior chemical resistance. Some refractory coatings, capable of taking very high temperatures, are also available.

Brick Construction Brick-lined construction can be used for many severely corrosive conditions under which high alloys would fail. Brick linings can be installed over metal, concrete, and fiberglass structures. Acid-resistant bricks are made from carbon, red shale, or acid-resistant refractory materials. Red-shale brick is not used above 175°C (350°F) because of spalling. Acid-resistant refractories can be used up to 870°C (1600°F). See Table 25-10.

TABLE 25-11 Standard Wrought Austenitic Stainless Steels

AISI type	UNS	Composition, % [°]							Mechanical properties [†]			
									Yield strength, kip/in ² (MPa)‡	Tensile strength [†] , kip/in ² (MPa)‡	Elongation, %	Hardness, HB
		Cr	Ni	Mo	C	Si	Mn	Other				
304	S30400	18–20	8–10.5		0.08	1.0	2.0		35 (241)	82 (565)	60	149
304L	S30403	18–20	8–12		0.03	1.0	2.0		33 (228)	79 (545)	60	143
308	S30800	19–21	10–12		0.08	1.0	2.0		30 (207)	85 (586)	55	150
309	S30900	22–24	12–15		0.20	1.0	2.0		40 (276)	95 (655)	45	170
309S	S30908	22–24	12–15		0.08	1.0	2.0		40 (276)	95 (655)	45	170
310	S31000	24–26	19–22		0.25	1.5	2.0		45 (310)	95 (655)	50	170
310S	S31008	24–26	19–22		0.08	1.5	2.0		45 (310)	95 (655)	50	170
316	S31600	16–18	10–14	2.0–3.0	0.08	1.0	2.0		36 (248)	82 (565)	55	149
316L	S31603	16–18	10–14	2.0–3.0	0.03	1.0	2.0		34 (234)	81 (558)	55	146
317	S31700	18–20	11–15	3.0–4.0	0.08	1.0	2.0		40 (276)	85 (586)	50	160
317L	S31703	18–20	11–15	3.0–4.0	0.03	1.0	2.0		35 (241)	85 (586)	55	150
321	S32100	17–19	9–12		0.08	1.0	2.0	(5 × C) Ti§	30 (207)	85 (586)	55	160
347	S34700	17–19	9–13		0.08	1.0	2.0	(10 × C)(Cb + Ta)§ 0.20 Co	35 (241)	90 (621)	50	160

[°]Single values are maximum values unless otherwise noted.

[†]Typical room-temperature properties of solution-annealed plates.

[‡]To convert MPa to lbf/in², multiply by 145.04.

[§]Minimum.

[¶]Minimum except Ta = 0.1 maximum.

TABLE 25-12 Standard Wrought Austenitic/Ferritic Duplex Stainless Steels

Alloy	UNS	Composition, % ^a							Mechanical properties [†]			
		Cr	Ni	Mo	C	Si	Mn	Other	Yield strength, kip/in ² (MPa) [‡]	Tensile strength, kip/in ² (MPa) [‡]	Elongation, %	Hardness, HB
329	S32900	25-30	3-6	1.0-2.0	0.10	1.0	2.0		80 (552)	105 (724)	25	230
3RE60	S31500	18-19	4.25-5.25	2.5-3.0	0.03	1.4-2.0	1.2-2.0	0.05-0.10 N	A789			
2205	S32205	22-23	4.5-6.5	3.0-3.5	0.03	1.0	2.0	0.05-0.20 N	65 (450)	95 (655)	25	293
2304	S32304	21.5-21.5	3.0-5.5	0.5-0.6	0.03	1.0	2.5	0.05-0.25 N	58 (400)	87 (600)	25	290
								0.05-0.6 Cu				
Ferru- lium	S32550	24-27	4.5-6.0	2.9-3.9	0.04	1.0	1.5	0.10-0.25 N	80 (550)	110 (760)	15	302
								1.5-2.5 Cu				
44LN	S31200	24-26	5.5-6.5	1.2-2.0	0.03	1.0	2.0	0.14-0.20 N	65 (450)	100 (690)	25	293
DP3	S31260	24-26	5.5-7.5	2.5-3.0	0.03	1.0	0.75	0.10-0.30 N	70 (485)	100 (690)	25	293
								0.2-0.8 Cu				
2507	S32750	24-26	6.0-8.0	3.0-5.0	0.03	0.8	1.2	0.24-0.32 N	80 (550)	116 (795)	15	310

^aSingle values are maximum values unless otherwise noted.

[†]Typical room-temperature properties of solution-annealed plates.

[‡]To convert MPa to lbf/in², multiply by 145.04.

A number of **cement** materials are used with brick. Standard are polymer resin, silicate, and sulfur-based materials. The most widely used resins are furane, vinyl ester, phenolic, polyester, and epoxies. Carbon-filled furanes and phenolics are good against nonoxidizing acids, salts, and solvents. Silicates and silica-filled resins should not be used in hydrofluoric or fluorosilicic acid applications. Sulfur-based cements are limited to 93°C (200°F), while resins can be used to about 180°C (350°F). Silicate-based cements are available for service temperatures up to 1000°C (1830°F).

Brick porosity, which can be as high as 20 percent, necessitates an intermediate lining of lead, asphalt, rubber, or plastic. This membrane functions as the primary barrier to protect the substrate from corrosion damage. The brick lining provides thermal and mechanical protection for the membrane. The membrane system also allows for the differential thermal expansion between the brick lining and supporting substrate. The design of brick linings exposed to higher operating pressure should take into account chemical expansion in addition to thermal expansion.

Cement and Concrete Concrete is an aggregate of inert reinforcing particles in an amorphous matrix of hardened cement paste. Concrete made of portland cement has limited resistance to acids and bases and will fail mechanically following absorption of crystal-forming solutions such as brines and various organics. Concretes made of corrosion-resistant cements (such as calcium aluminate) or polymer resins can be selected for specific chemical exposures.

Soil Clay is the primary construction material for settling basins and waste treatment evaporation ponds. Since there is no single type of clay even within a given geographic area, the shrinkage, porosity, absorption characteristics, and chemical resistance must be checked for each application. Geotextiles can be incorporated into basin and pond clay construction to improve the performance of the structure.

ORGANIC NONMETALLICS

Plastic polymers are materials made from organic compounds that have been joined to form long-chain, large-molecular-weight molecules that can be easily processed. There are two basic families of plastic polymers. Thermoplastics are a family of polymers that can be repeatedly heated, changed in shape, then cooled and solidified. Typical polymers within this family are polyolefins, polyvinyls, and the fluoropolymers. The second family of polymers is the thermosets. Unlike the thermoplastics, these materials crosslink during initial processing and cannot be reheated and reshaped. Upon reheating thermosets will not melt before they reach their decomposition temperature. Thermosets are typically more rigid than the thermoplastic polymers.

Thermoplastics Thermoplastics are used in a number of ways. They may be freestanding vessels, pipe, and other equipment; as linings in metallic vessels or as dual laminates in combination with thermoset

plastics. The processing of the thermoplastic raw materials has a great degree of influence on the final properties of the thermoplastic part. Processing techniques include extrusion, injection molding, compression molding, blow molding, rotational molding, and powder coating. Thermoplastics can be welded using various techniques. For critical applications it is highly recommended that welders be certified by independent, impartial organizations such as the German Welding Society (DVS) or a similar qualified organization. DVS certification is available in North America for the polymers and welding techniques covered within its guidelines. Visual inspection criteria for thermoplastic welds are available from the American Welding Society (AWS), DVS, and American Society of Mechanical Engineers Bio-processing standard (ASME- BPE). See Table 25-18 for typical thermoplastic properties.

The use of plastic polymers is limited to relatively moderate temperatures and pressures; 450°F (230°C) is considered high for polymers. They are also less resistant to mechanical abuse and have high expansion, low strength (thermoplastics), and only fair resistance to solvents. Plastic polymers are lightweight, are good thermal and electrical insulators, and are easy to fabricate and install. The chemical resistance of the thermoplastics is one of the major attributes of these materials. Virtually all chemical resistance tables and purity data are derived from the screen tests that were designed and for short-term environmental damage. Some long-term environmental data are available in the DVS guidelines. Care must be exercised when using these data as the test results are for individual chemicals and not typically mixtures of chemicals.

Polytetrafluoroethylene (Teflon) (PTFE) is the most corrosion-resistant thermoplastic polymer. This polymer is resistant to practically every known chemical or solvent combination and has the highest useful temperature of commercially available polymers. It retains its properties up to 500°F (260°C). Because of its exceedingly high molecular weight PTFE is processed by sintering. The PTFE resin is compressed into shapes under high pressure at room temperature and then heated to 700°F (371°C) to complete the sintering process.

Fluorinated ethylene propylene (Teflon) (FEP) is a fully fluorinated plastic. This polymer was developed to have a combination of unique properties. It combines the desirable properties of PTFE with advantageous melt processing properties.

Perfluoroalkoxy (Teflon) (PFA) was introduced in 1972 and is a fully fluorinated polymer that is melt-processible with better melt flow and molding properties than the FEP. The PFA has excellent resistance to chemicals. It can withstand acids as well as caustic materials. PFA has better mechanical properties than FEP above 300°F (149°C) and can be used up to 500°F (260°C) for some applications. The low physical strength and high cost of this polymer limit use for some applications.

Polyvinylidene fluoride (Kynar) (PVDF) is probably the most widely used fluorinated polymer for chemical applications. It can be melt-processed using virtually all the melt-processible procedures. PVDF components such as pipe fittings, tubing, sheet, shapes, and

TABLE 25-13 Special Stainless Steels

Alloy	UNS	Composition, % [*]							Mechanical properties [†]			
		Cr	Ni	Mo	C	Mn	Si	Other	Yield strength, kip/ in ² (MPa) [‡]	Tensile strength, kip/ in ² (MPa) [‡]	Elongation, %	Hardness, HB
A-286	S66286	13.5–16	24–27	1.0–1.5	0.08	2.0	1.0	1.90–2.35 Ti, 0.1–0.5 V, 0.001–0.01 B, 8 × C–1.0 Cb	100 (690)	140 (970)	20	
20Cb-3	N08020	19–21	32–38	2.0–3.0	0.07	2.0	1.0	3.0–4.0 Cu	53 (365)	98 (676)	33	185
PH13-8Mo	S13800	12.25–13.25	7.5–8.5	2.0–2.5	0.05	0.2	0.1	0.90–1.35 Al	120 (827)	160 (1100)	17	300
PH14-8Mo	S14800	13.75–15.0	7.75–8.75	2.0–3.0	0.05	1.0	1.0	0.75–1.5 Al, 0.15–45 Cb	55–210 (380–1450)	125–230 (860–1540)	2–25	200–450
15-5PH	S15500	14.0–15.5	3.5–5.5		0.07	1.0	1.0	2.5–4.5 Cu	145 (1000)	160 (1100)	15	320
PH15-7Mo	S15700	14.0–16.0	6.5–7.75	2.0–3.0	0.09	1.0	1.0	0.75–1.5 Al, 0.15–0.45 Cb	55–210 (380–1450)	130–220 (900–1520)	2–35	200–450
17-4PH	S17400	15.5–17.5	3.0–5.0		0.07	1.0	1.0	3.0–5.0 Cu, 0.4 Al	145 (1000)	160 (1100)	15	320
17-7PH	S17700	16.0–18.0	6.5–7.75		0.09	1.0	1.0	0.75–1.5 Al	40 (276)	130 (710)	10	185
Nitronic 60	S21800	16.0–18.0	8.0–9.0		0.10	7.0–9.0	3.5–4.5	0.08–0.18 N	60 (410)	103 (710)	62	210
21-6-9	S21900	18.0–21.0	5.0–7.0		0.08	8.0–10.0	1.0	0.15–0.40 N	68 (470)	112 (770)	44	220
AM350	S35000	16.0–17.0	4.0–5.0	2.5–3.25	0.07–0.11	0.5–1.25	0.5	0.07–0.13 N	60–173 (410–1200)	145–206 (1000–1420)	13.5–40	200–400
AM355	S35500	15.0–16.0	4.0–5.0	2.5–3.25	0.10–0.15	0.5–1.25	0.5		182 (1250)	216 (1490)	19	402–477
Stab. 26-1	S44626	25.0–27.0	0.5	0.75–1.50	0.06	0.75	0.75	7 × (C + Ni)–1.0 Ti, 0.15 Cu	50 (345)	70 (480)	30	165
29-4	S44700	28.0–30.0	0.15	3.5–4.2	0.010	0.3	0.2	0.02 N, 0.15 Cu	70 (480)	90 (620)	25	210
29-4-2	S44800	28.0–30.0	2.0–2.5	3.5–4.2	0.010	0.3	0.2	0.02 N, 8 × C Cb min.	85 (590)	95 (650)	25	230
Custom 450	S45000	14.0–16.0	5.0–7.0	0.5–1.0	0.05	1.0	1.0	1.25–1.75 Cu	117–184 (800–1270)	144–196 (990–1350)	14	270–400
Custom 455	S45500	11.0–12.5	7.5–9.5	0.5	0.05	0.5	0.5	1.5–2.5 Cu, 0.8–1.4 Ti	115–220 (790–1500)	140–230 (970–1600)	10–14	290–460
254 SMO	S31254	19.5–20.5	17.5–18.5	6.0–6.5	0.02	1.0	0.8	0.18–0.22 N	45 (310)	95 (655)	35	223
AL6XN	N08367	20.0–22.0	23.5–25.5	6.0–7.0	0.03	2.0	1.0	0.18–0.25 N	45 (310)	95 (655)	30	241
27-7Mo	S31277	20.5–23.0	26.0–28.0	6.5–8.0	0.02	3.0	0.5	0.3–0.4 N 0.5–1.5 Cu	52 (260)	112 (770)	40	168

*Single values are maximum values unless otherwise noted.

†Typical room-temperature properties.

‡To convert MPa to lbf/in², multiply by 145.04.

TABLE 25-14 Standard Cast Heat-Resistant Stainless Steels

ACI	Equivalent AISI	UNS	Composition, % ^a						Mechanical properties at 1600°F			
									Short term		Stress to rupture in 1000 h	
									Tensile strength, kip/in ² (MPa) [†]	Elonga- tion, %		
			Cr	Ni	C	Mn	Si	Other			kip/in ²	MPa [‡]
HC	446	J92605	26–30	4	0.5	1.0	2.0				1.3	9.0
HD	327	J93005	26–30	4–7	0.5	1.5	2.0				7.0	48 [†]
HE		J93403	26–30	8–11	0.2–0.5	2.0	2.0			23 (159)		
HF	302B	J92603	18–23	9–12	0.2–0.4	2.0	2.0			21 (145)	16	4.4
HH		J93503	24–28	11–14	0.2–0.5	2.0	2.0	0.2 N		18.5 (128)	30	3.8
HH-30	309	J93513	24–28	11–14	0.2–0.5	2.0	2.0	0.2 N		21.5 (148)	18	3.8
HI		J94003	26–30	14–18	0.2–0.5	2.0	2.0			26 (179)	12	4.8
HK	310	J94224	24–28	18–22	0.2–0.6	2.0	2.0			23 (159)	16	6.0
HL		N08604	28–32	18–22	0.2–0.6	2.0	2.0			30 (207)		
HN		J94213	19–23	23–27	0.2–0.5	2.0	2.0			20 (138)	37	7.4
HP		N08705	24–28	33–37	0.35–0.75	2.0	2.5			26 (179)	27	7.5
HT	330	N08002	13–17	33–37	0.35–0.75	2.0	2.5			19 (131)	26	5.8
HU		N08004	17–21	37–41	0.35–0.75	2.0	2.5			20 (138)	20	5.2
HW-50		N08006	10–14	58–62	0.35–0.75	2.0	2.5			19 (131)		4.5
HX		N06006	15–19	64–68	0.35–0.75	2.0	2.5			20.5 (141)	48	4.0

^aSingle values are maximum values; S and P are 0.04 maximum; Mo is 0.5 maximum.

[†]To convert MPa to lbf/in², multiply by 145.04.

[‡]At 1400°F (760°C).

TABLE 25-15 Standard Cast Corrosion-Resistant Stainless Steels

ACI	Equivalent AISI	UNS	Composition, % ^a							Mechanical properties ^b			
										Yield strength, kip/in ² (MPa) ^c	Tensile strength, kip/in ² (MPa) ^c	Elonga- tion, %	Hardness HB
			Cr	Ni	Mo	C	Mn	Si	Other				
CA-15	410	J91150	11.5–14	1.0	0.5	0.15	1.0	1.5	2.5–3.5 Cu	150 (1034) ^d	200 (1379) ^d	7 ^d	390 ^d
CA-15M		J91151	11.5–14	1.0	0.15–1.0	0.15	1.0	1.5		150 (1034) ^d	200 (1379) ^d	7 ^d	390 ^d
CA-6NM		J91540	11.5–14	3.5–4.5	0.4–1.0	0.06	1.0	1.0		100 (690) ^e	120 (827) ^e	4 ^e	269 ^e
CA-40	420	J91153	11.5–14	1.0	0.5	0.20–0.40	1.0	1.5		165 (1138) ^d	220 (1517) ^d	1 ^d	470 ^d
CB-30	431	J91803	18.21	2.0		0.30	1.0	1.5		60 (414) ^f	95 (655) ^f	15 ^f	195 ^f
CC-50	446	J92615	26–30	4.0		0.50	1.0	1.5		65 (448) ^g	97 (669) ^g	18 ^g	210 ^g
CE-30	312	J93423	26–30	8–11		0.30	1.5	2.0		63 (434)	97 (669)	18	190
CB-7Cu ⁻¹	17-4PH	J92180				0.07				165 (1138)		3	418
CB-7Cu ⁻²	15-5PH	J92110	14.0–15.5	4.2–5.5		0.07	0.7	1.0					
CF-3	304L	J92500	17–21	8–12		0.03	1.5	2.0			36 (248)	77 (531)	60
CF-8	304	J92600	18–21	8–11		0.08	1.5	2.0		37 (255)	77 (531)	55	140
CF-3M	316L	J92800	17–21	9–13	2.0–3.0	0.03	1.5	1.5		38 (262)	80 (552)	55	150
CF-8M	316	J92900	18–21	9–12	2.0–3.0	0.08	1.5	2.0		42 (290)	80 (552)	50	160
CF-10M	316H	J92901	18–21	9–12	2.0–3.0	0.12	1.5	1.5		42 (290)	80 (552)	50	160
CG-12	317	J93001	18–21	9–13	3.0–4.0	0.08	1.5	1.5		44 (303)	83 (572)	45	170
CG-3M	317L	J92999	18–21	9–13	3.0–4.0	0.03	1.5	1.5		40 (275)	80 (552)	50	150
CF-8C	347	J92710	18–21	9–12		0.08	1.5	2.0	(8 × C) Cb ^h	38 (262)	77 (531)	39	149
CF-16F	303	J92701	18–21	9–12	1.50	0.16	1.5	2.0		40 (276)	77 (531)	52	150
CH-20	309	J93402	22–26	12–15		0.20	1.5	2.0		50 (345)	88 (607)	38	190
CK-20	310	J94202	23–27	19–22		0.20	1.5	2.0		38 (262)	76 (524)	37	144
CN-7M	Alloy 20	J95150	19–22	27.5–30.5	2.0–3.0	0.07	1.5	1.5	3–4 Cu	32 (221)	69 (476)	48	130
CD-4MCu	2205	J92205	21.0–23.5	4.5–6.5	7.5–3.5	0.03	1.5	1.0	0.1–0.3 N	60 (414)	90 (621)	20	250
		J93372	25–26.5	4.75–6.0	1.75–2.25	0.04	1.0	1.0	2.75–3.25 Cu	82 (565)	108 (745)	25	253

^aSingle values are maximum values. P and S values are 0.04 maximum.

^bTypical room-temperature properties for solution-annealed material unless otherwise noted.

^cTo convert MPa to lbf/in², multiply by 145.04.

^dFor material air-cooled from 1800°F and tempered at 600°F.

^eFor material air-cooled from 1750°F and tempered at 1100 to 1150°F.

^fFor material annealed at 1450°F, furnace-cooled to 1000°F, then air-cooled.

^gAir-cooled from 1900°F.

^h1.0 maximum.

TABLE 25-16 Nickel and Cobalt Alloys

Alloy	UNS	Composition, %°						Mechanical properties†				
		Ni or Co	Cr	Fe	Mo	C	Other	Condition	Yield strength, kip/in² (MPa)‡	Tensile strength, kip/in² (MPa)‡	Elongation, %	Hardness, HB
Corrosion Alloys												
200	N02200	99.		0.4		0.15		Annealed	15–30 (103–207)	55–80 (379–552)	55–40	90–120
201	N02201	99.		0.4		0.02		Annealed	10–25 (69–172)	50–60 (345–414)	60–40	75–102
400	N04400	63–70		1.0–2.5		0.3	28–34 Cu	Annealed	25–50 (172–345)	70–90 (483–621)	60–35	110–149
K-500	N05500	63–70		2.0		0.25	2.3–3.15 Al 0.35–0.85 Ti, 30 Cu	Age-hardened Annealed	85–120 (586–827)	130–165 (896–1138)	35–20	250–315
600	N06600	72.	14–17	6–10		0.15		Annealed	30–50 (207–345)	80–100 (552–690)	55–35	120–170
625	N06625	Bal.	20–23	5	8–10	0.10	3.15–4.15 (Cb + Ta)	Annealed	60–95 (414–655)	120–150 (827–1034)	60–30	145–220
825	N08825	38–46	19.5–23.5	Bal.	2.5–3.5	0.05	1.5–3.0 Cu, 0.6–1.2 Ti	Annealed	35–65 (241–448)	85–105 (586–724)	50–30	120–180
B-3	N10665	65.	1.0–3.0	1.0–3.0	27–32	0.010	Ni + Mo = 94.0–98.0	Annealed	76 (524)	139 (958)	53	210
C-276	N10276	Bal.	14.5–16.5	4–7	15–17	0.010	3.0–4.5 W	Annealed	52 (358)	115 (793)	61	194
C-22	N06022	Bal.	20.0–22.5	2–6	12.5–14.5	0.015	2.5–3.5 W	Annealed	52 (358)	115 (793)	61	194
C-2000	N06200	Bal.	22.0–24.0	3	15.0–17.0	0.010	1.3–1.9 Cu	Annealed	52 (358)	115 (793)	61	194
MAT21	N06210	Bal.	18.0–20.0	1.0	18.0–20.0	0.015	1.5–2.2 Ta	Annealed	52 (358)	115 (793)	61	194
686	N06686	Bal.	19.0–23.0	5.0	15.0–17.0	0.010	3.0–4.4 W, 0.02–0.25 Ti	Annealed	52 (358)	115 (793)	61	194
G-3	N06985	Bal.	21.0–23.5	18.0–21.0	6.0–8.0	0.015	Cb + Ta 0.5	Annealed	52 (358)	115 (793)	61	194
G-35	N06035	Bal.	32.3–34.3	2.0	7.6–9.0	0.05		Annealed	52 (358)	115 (793)	61	194
High-Temperature Alloys												
600	N06600	72.	14–17	6–10		0.15		Annealed	30–50 (207–345)	80–100 (552–690)	55–35	120–170
601	N06601	58–63	21–25	Bal.		0.10	1.0–1.7 Al	Annealed	30–60 (207–414)	80–115 (552–793)	70–40	110–150
625	N06625	Bal.	20–23	5	8–10	0.10	3.15–4.15 (Cb + Ta)	Annealed	60–95 (414–655)	120–150 (827–1034)	60–30	145–220
706	N09706	39–44	14.5–17.5	Bal.		0.06		Solution-treated and aged	161 (1110)	193 (1331)	20.	371.
718	N07718	50–55	17–21	Bal.	2.8–3.3	0.08	4.75–5.5 (Cb + Ta)	Special heat treatment	171 (1180)	196 (1351)	17.	382.
X-750	N07750	70	14–17	5–9		0.08	0.65–1.15 Ti, 0.2–0.8 Al 0.7–1.2 (Cb + Ta)	Special heat treatment	115–142 (793–979)	162–193 (1117–1331)	30–15	300–390
800	N08800	30–35	19–23	Bal.		0.10	2.25–2.75 Ti, 0.4–1.0 Al	Annealed	30–60 (207–414)	75–100 (517–690)	60–30	120–184
800H	N08810	30–35	19–23	Bal.		0.05–0.10	0.15–0.6 Al, 0.15–0.6 Ti	Solution-treated	20–50 (138–345)	65–95 (448–655)	50–30	100–184
801	N08801	30–34	19–22	Bal.		0.10	0.75–1.5 Ti	Stabilized	79.5 (548)	129 (889)	29.5	
803	S35045	32–37	25–29	Bal.		0.06–0.10	0.15–0.6 Al 0.15–0.6 Ti	Annealed				
X	N06002	Bal.	20.5–23	17–20	8–10	0.05–0.15	0.2–1.0 W	Annealed	56 (386)	110 (758)	45	178
HR-120	N08120	35–39	223–27	Bal.	2.5	0.02–0.1		Annealed				
230	N06230	Bal.	20–24	3	1.0–3.0	0.05–0.15	13.15 W, 0.005–0.05 La	Annealed	45 (310)	110 (760)	40	187
617	N06617	44.5	20–24	3	8.0–10.0	0.05–0.15	10–15 Co, 0.8–1.5 Al	Annealed	35 (240)	95 (655)	30	170
HR-160	N12160	Bal.	26–30	3.5	1.0	0.15	27–33 Co, 2.4–3.0 Si	Annealed	35 (240)	90 (670)	40	170
N155R30155		18.2–21	20–22.5	Bal.	2.5–3.5	0.08–0.16	19–21 Ni, 0.75–1.25 Cb 2.0–3.0 W	Annealed	50 (345)	110 (760)	30	192
25 R30605		Bal.	19–21	3		0.38–0.48	9–11 Ni, 14–16 W	Annealed	69 (475)	146 (1005)	50	200
188R30188		Bal.	20–24	3		0.05–0.45	20–24 Ni, 13–16 W 0.03–0.15 La	Annealed	69 (475)	142 (985)	55	200

*Single values are maximum unless otherwise noted.

†Typical room-temperature properties.

‡To convert MPa to lbf/in², multiply by 145.04.

§Single values are minima. Those alloys with N UNS numbers are nickel and R numbers are cobalt alloys.

TABLE 25-17 Aluminum Alloys

AA designation	UNS	Composition, %*						Condition†	Mechanical properties†			
		Cr	Cu	Mg	Mn	Si	Other		Yield strength, kip/in ² (MPa)	Tensile strength, kip/in ² (MPa)	Elongation in 2 in, %	Hardness, HB
Wrought												
1060	A91060						99.6 Al min.	0	4 (28)	10 (69)	43	19
1100	A91100		0.05–0.2				99.0 Al min.	0	5 (34)	13 (90)	45	23
2024	A92024	0.1	3.8–4.9	1.2–1.8	0.3–0.9	0.5		T4	47 (324)	68 (469)	19	120
3003	A93003		0.05–0.2		1.0–1.5	0.6		H14	21 (145)	22 (152)	16	40
5052	A95052	0.15–0.35	0.1	2.2–2.8	0.1			0	13 (90)	2.8 (193)	30	47
5083	A95083	0.05–0.25	0.1	4.0–4.9	0.4–1.0	0.4		0	21 (145)			
5086	A95086	0.05–0.25	0.1	3.5–4.5	0.2–0.7	0.4		0	17 (117)	38 (262)	30	
5154	A95154	0.05–0.35	0.1	3.1–3.9	0.1	0.25		0	17 (117)	35 (241)	27	58
6061	A96061	0.04–0.35	0.15–0.4	0.8–1.2	0.15	0.4–0.8		T6	40 (276)	45 (310)	17	95
6063	A96063	0.1	0.1	0.45–0.9	0.1	0.2–0.6		T6	31 (214)	35 (241)	18	73
7075	A97075	0.18–0.28	1.2–2.0	2.1–2.9	0.3	0.40	5.1–6.1 Zn	T6	73 (503)	63 (572)	11	150
Cast												
242.0	A02420	0.25	3.5–4.5	1.2–1.8	0.35	0.7	1.7–2.3 Ni	S-T571		29 (200)		
295.0	A02950		4.0–5.0	0.03	0.35	0.7–1.5		S-T4		29 (200)	6	
336.0	A03360		0.5–1.5	0.7–1.3	0.35	11–13	2.0–3.0 Ni	P-T551		31 (214)		
B443.0	A24430		0.15	0.05	0.35	4.5–6.0		S-F		17 (117)	3	
514.0	A05140		0.15	3.5–4.5	0.35	0.35		S-F		22 (152)	6	
520.0	A05200		0.25	9.5–10.6	0.15	0.25		S-T4	22 (152)	42 (290)	12	

*Single values are maximum values.

†Typical room-temperature properties.

‡S = sand-cast; P = permanent-mold-cast; other = temper designations.

SOURCE: Aluminum Association. To convert MPa to lb/in², multiply by 145.04.

other configurations are commercially available. PVDF has a use range from –40 to 302°F (150°C). PVDF has a high tensile strength, flex modulus, and heat deflection temperature. It is easily welded, resists permeation, and offers a high-purity smooth polymer surface. This is the polymer of choice for high-purity applications such as semiconductor, bioprocessing, and pharmaceutical industries.

Ethylene chlorotrifluoroethylene (Halar) (ECTFE) has excellent chemical resistance to most chemicals including caustic. ECTFE can be used from –105°F (–76°C) to 302°F (150°C). To obtain good extrusion characteristics, this polymer is usually compounded with a small amount of extrusion aid.

Ethylene trifluoroethylene (Tefzel) (ETFE) has good mechanical properties from cryogenic levels to 350°F (177°C). It has an upper continuous working temperature limit of 300°F (149°C).

Polyethylene (PE) is one of the lowest-cost polymers. There are various types of polyethylene denoted by their molecular weight. This ranges from low-density polyethylene (LDPE) through ultrahigh-molecular-weight (UHMW) polyethylene. Physical properties, processability, and other characteristics of the polyethylene vary greatly with the molecular weight.

Polypropylene (PP) is a crystalline polymer suitable for low-stress applications up to 225°F (105°C). For piping applications this polymer is not recommended above 212°F (100°C). Polypropylene is shielded, pigmented, or stabilized to protect it from uv light. Polypropylene is often a combination of polyethylene and polypropylene which enhances the ductility of the polymer.

Polyvinyl chloride (PVC) has excellent resistance to weak acids and alkaline materials. PVC is commonly utilized for applications that do not require high-temperature resistance or a high-purity resin. Chlorinated PVC (CPVC or PVC-C) represents more than 80 percent of all the PVC used in North America. PVC contains 56.8 percent chlorine by weight in contrast to about 67 percent for CPVC. Both PVC and CPVC are compounded with ingredients such as heat stabilizers, lubricants, fillers, plasticizers, pigments, and processing aids. The actual amount of polymer may range from 93 to 98 percent. The remaining 2 to 7 percent is filler, pigment, stabilizer, lubricant, and plasticizer.

Other commercial thermoplastics include **acrylonitrile butadiene styrene** (ABS), **cellulose acetate butyrate** (CAB), **polycarbonate** (PC), **nylon** (PA), and **acetals**. These resins are frequently used in consumer applications.

Thermosets* There are several generic types of thermosetting resins used for the manufacture of fiberglass-reinforced plastic (FRP composites) equipment. Unlike thermoplastic polymers, thermosetting polymers are hardened by an irreversible crosslinking cure and are almost exclusively used with fiber reinforcement such as glass or carbon fibers in structural applications. It is important to note that because thermoset resins are used with fiber reinforcements, the properties of the resultant laminate are dependent upon the resin and the type, amount, and orientation of reinforcement fibers. To reduce the number of generally used constructions, ASTM and ASME RTP-1 define several standard corrosion-resistant laminate constructions suitable for most equipment. In addition to new construction, thermoset composites are providing practical solutions to the engineer faced with the challenges of restoring structural integrity, increasing load-bearing capabilities, and/or enhancing the strength and stiffness of aging structures. See Table 25-19 for typical thermoset fiber reinforced laminate properties.

The advantages of composites are inherent in their construction. A variety of resin/fiber systems can yield possible solutions for many types of situations. Depending on the product and application, FRP products for civil and mechanical applications can deliver the following benefits:

- Part design (orientation of the fibers) can be optimized for specific loads.

- Reduced structure dead load can increase load ratings.

- Reduced maintenance costs due to resistance from salts and other corrosive agents.

- Engineered system packaging reduces field installation time.

- Faster installation due to lower weight.

- Enhanced durability and fatigue characteristics—FRP does not rust nor is it chloride susceptible.

Myriad FRP products are available for either the repair or the outright replacement of existing structures. In addition to chemical-process pipes and tanks, FRP composite products include structural shapes, bridge systems, grating, handrail ladders, etc.

*Note: Thermosets are also used in non-fiber-reinforced applications such as gel coats and cast polymer.

TABLE 25-18 Typical Thermoplastic Properties

	Units	PP		PVC	CPVC	PVDF		ECTFE	ETFE	FEP	TFE	PFA
		Homopolymer	Copolymer			Homopolymer	Copolymer					
Density	g/cm ³	0.91	0.88–0.91	1.38	1.5	1.75–1.79	1.76–1.79	1.68	1.70	2.12–2.17	2.2–2.3	2.12–2.17
Melting point (crystalline)	°C	160–175	150–175	—	—	160–170	141–160	220–245	270	275	327	310
	°F	320–347	302–347	—	—	320–340	285–320	460	518	527	621	590
Physical Properties												
Break strength; ASTM D 638	kpsi	4.5–6.0	4.0–5.3	6.0–7.5	—	4.5–7.0	3.5–6.0	6.6–7.8	6.5	2.7–3.1	2.0–2.7	4.0–4.5
Modulus flex @ 73°F; ASTM D 790	MPa	1135–1550	345–1035	—	—							
	kpsi	165–225	50–150	—	—	165–325	90–180	180–260	200	80–95	190–235	120
Yield strength; ASTM D 638	kpsi	4.5–5.4	1.6–4.0	—	—	5.0–8.0	2.9–5.5	—	7.1	—	—	—
Thermal Properties												
HDT at 0.46 MPa (66 psi); ASTM D 648	°C	107–121	75–89	57	—	132–150	93–110	90	104	70	221	75
	°F	225–250	167–192	135	—	270–300	200–230	194	220	158	250	166
Linear coefficient of expansion; ASTM D 696	in/(in·°C) × 10 ⁻⁵	10	7–9.5	4.4	3.9	7.2–14.4	14.0	8	6	8–11	10	12
Conductivity; ASTM C 177	W/(m·K)	0.1	0.16	—	—	0.17–0.19	0.16	—	—	—	—	—
	Btu/(ft ² ·h· °F/in)	0.7	1.1	—	—	1.18–1.32	1.11	—	—	—	—	—

TABLE 25-19 Typical Thermoset Fiber-Reinforced Laminate Properties

Property →	Glass fiber, %	Specific gravity	Density, lb/in ³	Tensile strength, 10 ³ psi	Tensile modulus, 10 ⁶ psi	Elongation, %	Flexural strength, 10 ³ psi	Flexural modulus, 10 ⁶ psi	Compress. strength, 10 ³ psi	Impact strength, ft-lb/in of notch	Flam-mability	Specific heat, Btu/(lb·°F)	Thermal coeff. of expansion, 10 ⁻⁶ in/(in·°F)	Heat distortion, °F at 264 psi	Thermal conductivity ^a , Btu/h, ft ² /°F/in	Dielectric strength, V/mil.	Water absorption, % in 24 h	Mold shrinkage, in/in
ASTM test method	D 790	D 792		D 638	D 638	D 638	D 790	D 790	D 695	D 256	UL-94		D 696	D 648	C 177	D 149	D 570	D 955
Polyester preform, low profile	24	1.74	0.063	11.5	1.70	2.5	28.5	1.32	20.0	20.8	°	0.30	14.0	400+	1.3	400	.8	0.000
(Compression) general-purpose	25	1.55	0.056	13.5	1.80	2.5	27.0	1.10	25.0	18.0	°	0.30	14.0	350+	1.5	400	.25	0.001
(Compression) high glass	40	1.70	0.061	21.5	2.25	2.5	38.5	1.50	32.0	23.0	°	0.30	14.0	400+	1.5	400	.4	0.0005
Carbon/epoxy fabric				23.0	2.10	1.0	25.0	2.00					°			N/A		
Polyester SMC LP, low profile	30	1.85	0.067	12.0	1.70	<1.0	26.0	1.60	24.0	16.0	°	0.30	12.0	400+	1.5	400	.2	0.001
(Compression) general-purpose	22	1.78	0.064	10.5	1.70	0.4	21.2	1.40	23.0	12.0	°	0.30	12.0	350+	1.5	400	.2	0.001
(Compression) high glass	50	2.00	0.072	23.0	2.27	1.7	45.0	2.00	32.0	19.4	°	0.30	15.0	500	1.5	375	.5	0.0005
Carbon/vinyl ester		1.50		22.0	6.0	0.4	71.0	4.00					°			N/A	.13	
Polyester BMC (compression)	22	1.82	0.066	6.0	1.75	<0.5	12.8	1.58	20.0	4.3	°	0.30	8.0	500	1.5	200	.5	0.001
Polyester BMC (injection)	22	1.82	0.066	4.9	1.53	<0.5	12.7	1.44	—	2.9	°	0.28	4.0	—	1.5	250	.5	—
Polyester (pultruded)	55	1.69	0.061	30.0	2.50	—	30.0	1.60	30.0	25.0	°	0.31	—	400+	1.50	350	.5	0.003
Polyester (spray-up, lay-up)	30	1.37	0.049	12.5	1.00	1.3	28.0	0.75	22.0	14.0	°	—	—	400+	—	300	.5	0.002
Polyester woven roving (lay-up)	50	1.64	0.059	37.0	2.25	1.6	46.0	2.25	27.0	33.0	V-0	0.23		400+	1.92	—	0.50	0.008
Epoxy (filament-wound)	80	2.08	0.075	80.0	4.00	1.6	100.0	5.00	45.0	45.0	V-0	—		—	—	—	—	—
Polyurethane, milled fibers (RRIM)	13	1.07	0.039	2.8	—	140.0	—	0.05	—	—	V-0	—		—	—		—	—
Polyurethane, glass flake (RRIM)	23	1.17	0.042	4.4	—	38.9	—	0.15	—	2.1							—	—

^aPolyester thermosets can be formulated to meet a wide range of flame, smoke, and toxicity specifications.

Epoxy (Amine-Cured) Bisphenol A-based epoxy resins used for composite fabrication are commonly cured with multifunctional primary amines. For optimum chemical resistance these generally require a heat cure or postcure. The cured resin has good chemical resistance, particularly to basic environments, and can have good temperature resistance.

Epoxy (Anhydride-Cured) Epoxy resins may be crosslinked with various anhydrides by using a tertiary amine accelerator and heat. These cured polymers generally have good chemical resistance especially to acids.

Epoxy Vinyl Ester Vinyl ester resins are typified by a highly aromatic Bisphenol-A structure and exhibit a broad range of chemical resistance, particularly to caustic solutions, and are the most resilient of the corrosion-resistant resins, generally resulting in the toughest laminate. Brominated versions have about the same corrosion resistance as other Bisphenol-A vinyl esters plus they offer a degree of fire retardancy. The novolac vinyl ester resins exhibit high-temperature resistance and a broad range of chemical resistance, particularly to solvents, oxidizing media, and strong acids.

Bisphenol-A Fumarate Polyester These resins exhibit a broad range of chemical resistance, particularly to caustic solutions. Brominated versions have about the same corrosion resistance as other Bisphenol-A vinyl esters plus they offer a degree of fire retardancy.

Chlorendic Acid Polyester These resins have excellent resistance to acids, good resistance to oxidizing acid media and high-temperature resistance, and moderate fire resistance.

Furan Furan resins are thermosetting polymers derived from furfuryl alcohol and Furfural. The cure must be carefully controlled to avoid the formation of blisters and delaminations. To obtain optimum strength and corrosion resistance, furan composites must undergo a postcure schedule at carefully selected temperatures depending upon the laminate thickness. Equipment made with furan resins exhibits excellent resistance to solvents and combinations of acids and solvents. These resins are not for use in strong oxidizing environments.

Isophthalic/Terephthalic Acid Polyester Typically, isophthalic acid is reacted in a two-stage process with glycols such as propylene or neopentyl glycol and then with maleic anhydride. The crosslinked, fully cured isophthalic-based composites exhibit good weathering resistance and good corrosion resistance to acid media.

Dual-Laminate Construction and Linings The availability of some of the thermoplastic resins in sheet form with a bondable surface allows them to be utilized in a fashion similar to metal cladding. The sheets can be bonded in place and welded to form a barrier surface. When they are used in conjunction with thermoset resin technology for the fabrication of vessels, the procedure is referred to as *dual-laminate*. This fabrication technique extends the chemical resistance of the thermoset resin and the structural strength of the thermoplastic, providing a strong vessel that is highly chemically resistant internally and externally yet lightweight and easily fabricated. Standards for the fabrication of dual-laminate structures can be found in the ASME RTP-1 standard. See Table 25-20 for dual-laminate construction and lining properties.

Rubber and Elastomers Rubber and elastomers are widely used as lining materials. To meet the demands of the chemical industry, rubber processors are continually improving their products. A number of synthetic rubbers have been developed, and while none has all the properties of natural rubber, they are superior in one or more ways. The isoprene and polybutadiene synthetic rubbers are duplicates of natural.

The ability to bond natural rubber to itself and to steel makes it ideal for lining tanks. Many of the synthetic elastomers, while more chemically resistant than natural rubber, have very poor bonding characteristics and hence are not well suited for lining tanks.

Natural rubber is resistant to dilute mineral acids, alkalies, and salts, but oxidizing media, oils, and most organic solvents will attack it. **Hard rubber** is made by adding 25 percent or more of sulfur to natural or synthetic rubber and, as such, is both hard and strong. **Chloroprene** or **neoprene rubber** is resistant to attack by ozone, sunlight, oils, gasoline, and aromatic or halogenated solvents but is

easily permeated by water, thus limiting its use as a tank lining. **Styrene rubber** has chemical resistance similar to that of natural. **Nitrile rubber** is known for resistance to oils and solvents. **Butyl rubber's** resistance to dilute mineral acids and alkalies is exceptional; resistance to concentrated acids, except nitric and sulfuric, is good. **Silicone rubbers**, also known as polysiloxanes, have outstanding resistance to high and low temperatures as well as against aliphatic solvents, oils, and greases. **Chlorosulfonated polyethylene**, known as **Hypalon**, has outstanding resistance to ozone and oxidizing agents except fuming nitric and sulfuric acids. Oil resistance is good. **Fluoroelastomers (Viton A, Kel-F, Kalrez)** combine excellent chemical and temperature resistance. **Polyvinyl chloride elastomer (Koro seal)** was developed to overcome some of the limitations of natural and synthetic rubbers. It has excellent resistance to mineral acids and petroleum oils.

The **cis-polybutadiene**, **cis-polyisoprene**, and **ethylene-propylene** rubbers are close duplicates of natural rubber. The newer ethylene-propylene rubbers (EPR) have excellent resistance to heat and oxidation.

Asphalt Asphalt is used as a flexible protective coating, as a brick-lining membrane, and as a chemical-resisting floor covering and road surface. Resistant to acids and bases, asphalt is soluble in organic solvents such as ketones, most chlorinated hydrocarbons, and aromatic hydrocarbons.

Carbon and Graphite The chemical resistance of impervious carbon and graphite depends somewhat on the type of resin impregnated used to make the material impervious. Generally, impervious graphite is completely inert to all but the most severe oxidizing conditions. This property, combined with excellent heat transfer, has made impervious carbon and graphite very popular in heat exchangers, as brick lining, and in pipe and pumps. One limitation of these materials is low tensile strength. Threshold oxidation temperatures are 350°C (660°F) for carbon and 400°C (750°F) for graphite.

Several types of resin impregnates are employed in manufacturing impervious graphite. The standard impregnate is a phenolic resin suitable for service in most acids, salt solutions, and organic compounds. A modified phenolic impregnate is recommended for service in alkalies and oxidizing chemicals. Furan and epoxy thermosetting resins are also used to fill structural voids. The chemical resistance of the impervious graphite is controlled by the resin used. However, no type of impervious graphite is recommended for use in over 60 percent hydrofluoric, over 20 percent nitric, and over 96 percent sulfuric acids and in 100 percent bromine, fluorine, or iodine.

Wood While fairly inert chemically, wood is readily dehydrated by concentrated solutions and hence shrinks badly when subjected to the action of such solutions. It is also slowly hydrolyzed by acids and alkalies, especially when hot. In tank construction, if sufficient shrinkage once takes place to allow crystals to form between the staves, it becomes very difficult to make the tank tight again.

A number of manufacturers offer wood impregnated to resist acids or alkalies or the effects of high temperatures.

Wood, one of people's oldest materials, remains an important (and too often, overlooked) corrosion-resistant material of construction in the chemical-process industry. Wood tanks and wood piping have long met engineers' requirements for dependable service and excellent performance in industrial applications. A very thorough and detailed treatise can be found in a three-part publication, by Oliver W. Siebert, "Wood—Nature's High-Performance Material," NACE, *Materials Performance*, vol. 31, nos. 1-3, January through March 1992.*

Types of wood and their chemical resistance and physical characteristics are reviewed, including examples showing the manufacture of typical tank and pipe construction. In-service case histories are included. While this coverage takes you from the "Forest" to the plants making acetic acid, that is beyond the need for most users; it is hoped that the reader becomes aware that this product family is the *only* MOC for several CPI applications and is a *competitive* choice over some quite exotic materials, e.g., titanium, in others.

*Reference cited courtesy of NACE International.

TABLE 25-20 Dual-Laminate Construction and Lining Properties

Technology	Materials	Fabrication	Design	Size limit	NDT	Repair
Adhesively bonded	PVDF (60, 90, 118 mils) Synthetic and glassbacked ECTFE (60,90)" ETFE (60, 90 mils) glass-backed PTFE (80, 120 mils) " FEP (60, 90 mils) " PFA (90, 110 mils) " MFA (60, 90 mils) "	Contact or thermosetting adhesive Weld rod and cap strip welds Shop or field	Pressure ok Full vacuum up to 120°F Max. temp. 275°F	None	Visual Spark	Possible Testing recommended
Rotolining	ETFE PVDF ECTFE All up to 250 mils (normal thickness: 186 mils)	Rotationally molded No seams No primer Shop only	Pressure ok Limited vacuumability	Max size 8' × 22'	Visual Spark	By hot patching Testing recommended
Spray and baked						
Dispersion	FEP (10 mils), PFA (10–40 mils and up to 80 mils when reinforced or filled) PVDF (20–40 mils and up to 90 mils when reinforced with carbon cloth)	Primer and multicoat conventional spray equipment Each coat baked Shop only	Pressure ok Vacuum ok	12' × 12' × 37'	Visual Spark	By hot patching Testing recommended
Electrostatic spray	ETFE (up to 50 mils), FEP (10 mils), PFA (25–40 mils) ECTFE (35–90 mils), PVDF (35–125 mils)	Primer and multicoat Each coat baked Shop only	Pressure ok Vacuum ok	12' × 12' × 37'	Visual Spark	By hot patching Testing recommended
Loose lining	FEP PFA Modified PTFE All 60, 90, 125, 187 mils	Liner fabricated outside the housing and slipped inside Liner fabricated by hand and machine welding Shop only	No vacuum Pressure ok	Determined by body flange (12') and section height (12')	Visual Spark	Difficult
Dual laminate (fluoropolymer-lined FRP)	Same as adhesively bonded	Fabricate liner first on a mandrel (hand and machine welding) and build FRP laminate over the liner. Use carbon cloth for spark testing Shop and field	Pressure ok (RTP-1 Dual lam. or Section X) Vacuum ok for FRP/fluoropolymer bond. Design FRP for vacuum	~33' diameter max.	Visual Spark AE CRBBD Needs to be defined	Possible Testing recommended

HIGH- AND LOW-TEMPERATURE MATERIALS

LOW-TEMPERATURE METALS

The low-temperature properties of metals have created some unusual problems in fabricating cryogenic equipment.

Most metals lose their ductility and impact strength at low temperatures, although in many cases yield and tensile strengths increase as the temperature goes down.

Materials selection for low-temperature service is a specialized area. In general, it is necessary to select materials and fabrication methods which will provide adequate toughness at all operating conditions. It is frequently necessary to specify Charpy V-notch (or other appropriate) qualification tests to demonstrate adequate toughness of carbon and low-alloy steels at minimum operating temperatures.

Stainless Steels Chromium-nickel steels are suitable for service at temperatures as low as -250°C (-425°F). Type 304 is the most pop-

ular. The original cost of stainless steel may be higher than that of another metal, but ease of fabrication (no heat treatment) and welding, combined with high strength, offsets the higher initial cost. Sensitization or formation of chromium carbides can occur in several stainless steels during welding, and this will affect impact strength. However, tests have shown that impact properties of types 304 and 304L are not greatly affected by sensitization but that the properties of 302 are impaired at -185°C (-300°F).

Nickel Steel Low-carbon 9 percent nickel steel is a ferritic alloy developed for use in cryogenic equipment operating as low as -195°C (-320°F). ASTM specifications A 300 and A 353 cover low-carbon 9 percent nickel steel (A 300 is the basic specification for low-temperature ferritic steels). Refinements in welding and (ASME code-approved) elimination of postweld thermal treatments make 9 percent steel competitive with many low-cost materials used at low temperatures.

TABLE 25-21 Copper Alloys

Alloy	CDA	UNS	Composition, % ^a						Mechanical properties [†]		
			Cu	Zn	Sn	Al	Ni	Other	Yield strength, kip/in ² (MPa)‡	Tensile strength, kip/in ² (MPa)‡	Elongation in 2 in, %
Wrought											
Copper	110	C11000	99.90						10 (69)	32 (221)	55
Commercial bronze	220	C22000	89–91	Rem.					10 (69)	37 (255)	50
Red brass	230	C23000	84–86	Rem.					10 (69)	40 (276)	55
Cartridge brass	260	C26000	68.5–71.5	Rem.					11 (76)	44 (303)	66
Yellow brass	270	C27000	63–68.5	Rem.					14 (97)	46 (317)	65
Muntz metal	280	C28000	59–63	Rem.					21 (145)	54 (372)	52
Admiralty brass	443	C44300	70–73	Rem.	0.9–1.2			0.02–0.1 As	18 (124)	48 (331)	65
Admiralty brass	444	C44400	70–73	Rem.	0.9–1.2			0.02–0.1 Sb	18 (124)	48 (331)	65
Admiralty brass	445	C44500	70–73	Rem.	0.9–1.2			0.02–0.1 P	18 (124)	48 (331)	65
Naval brass	464	C46400	59–62	Rem.	0.5–1.0				25 (172)	58 (400)	50
Phosphor bronze	510	C51000	Rem.	0.3	4.2–5.8			0.03–0.35 P	19 (131)	47 (324)	64
Phosphor bronze	524	C52400	Rem.	0.2	9.0–11.0			0.03–0.35 P	28 (193)	66 (455)	70
Aluminum bronze	613	C61300	86.5–93.8		0.2–0.5	6–8	0.5	3.5 Fe	30 (207)	70 (483)	42
Aluminum bronze D	614	C61400	88.0–92.5	0.2		6–8		1.5–3.5 Fe, 1.0 Mn	33 (228)	76 (524)	45
Nickel-aluminum bronze	630	C63000	78–85	0.3	0.2	9–11	4.0–5.5	2.0–4.0 Fe, 1.5 Mn, 0.25 Si	36 (248)	90 (620)	10
High-silicon bronze	655	C65500	94.8	1.5			0.6	0.8 Fe, 0.5–1.3 Mn, 2.8–3.8 Si	21 (145)	56 (386)	63
Manganese bronze	675	C67500	57–60	Rem.	0.5–1.5	0.25		0.05–0.5 Mn, 0.8–2.0 Fe	30 (207)	65 (448)	33
Aluminum brass	687	C68700	76–79	Rem.		1.8–2.5		0.02–0.1 As	27 (186)	60 (414)	55
90-10 copper nickel	706	C70600	86.5	1.0			9.0–11.0	1.0–1.8 Fe, 1.0 Mn	16 (110)	44 (303)	42
70-30 copper nickel	715	C71500	Rem.	1.0			29–33	0.4–1.0 Fe, 1.0 Mn	20 (138)	54 (372)	45
65-18 nickel silver	752	C75200	63–66.5	Rem.			16.5–19.5	0.25 Fe, 0.5 Mn	25 (172)	56 (386)	45
Copper beryllium	172	C17200	Rem.			0.2	0.6	1.80–2.00 Be	28 (193)	60 (413)	65
Copper nickel tin	729	C72900	Rem.		8.0		15		140 (965)§	165 (1140)§	15
									20 (138)	50 (345)	30
									110 (758)¶	125 (861)¶	10
Cast											
Ounce metal	836	C83600	84–86	4–6	4–6	0.005	1.0	4–6 Pb	17 (117)	37 (255)	30
Manganese bronze	865	C86500	55–65	36–42	1.0	0.5–1.5	1.0	0.4–2.0 Fe, 0.1–1.5 Mn	28 (193)	71 (490)	30
G bronze	905	C90500	86–89	1.0–3.0	9–11	0.005	1.0		22 (152)	45 (310)	25
M bronze	922	C92200	86–90	3.0–5.0	5.5–6.5	0.005		1.0–2.0 Pb	20 (138)	40 (276)	30
Ni-Al-Mn bronze	957	C95700	71			7.0–8.5	1.5–3.0	2.0–4.0 Fe, 11–14 Mn	45 (310)	95 (665)	26
Ni-Al bronze	958	C95800	79			8.5–9.5	4.0–5.0	3.5–4.5 Fe, 0.8–1.5 Mn	38 (262)	95 (655)	25
Copper nickel	964	C96400	65–69				28–32	0.5–1.5 Cb, 0.25–1.5 Fe, 1.5 Mn	37 (255)	68 (469)	28

^aSingle values are maximum values except for Cu, which is minimum.

[†]Typical room-temperature properties of annealed or as-cast material.

[‡]To convert MPa to lbf/in², multiply by 145.04.

[§]Age-hardened condition.

[¶]Spinodal-hardened condition.

Nickel Alloys Alloy C-4 (16 Cr, 16 Mo, Balance Ni) and alloy C-276 (16 Cr, 16 Mo, 3.5 W, 5 Fe, balance Ni) have been used for closure seals on cryogenic gas cylinders because the alloys retain all their ductility down to –327°C (–557°F). The impact strength at liquid nitrogen temperatures is the same as that at room temperature.

Aluminum Aluminum alloys have an unusual ability to maintain strength and shock resistance at temperatures as low as –250°C (–425°F). Good corrosion resistance and relatively low cost make these alloys very popular for low-temperature equipment. For most welded construction the 5000-series aluminum alloys are widely used. These are the aluminum-magnesium and aluminum-magnesium-manganese materials.

Copper and Alloys With few exceptions the tensile strength of copper and its alloys increases quite markedly as the temperature goes

down. However, copper's low structural strength becomes a problem when constructing large-scale equipment. Therefore, alloy must be used. One of the most successful for low temperatures is silicon bronze, which can be used to –195°C (–320°F) with safety.

HIGH-TEMPERATURE MATERIALS*

Metals Successful applications of metals in high-temperature process service depend on an appreciation of certain engineering factors. Among the most important properties are creep, rupture,

*An excellent reference book for the high-temperature corrosion resistance of materials of construction is George Y. Lai, *High-Temperature Corrosion of Engineering Alloys*, ASM International, Metals Park, Ohio, 1990.

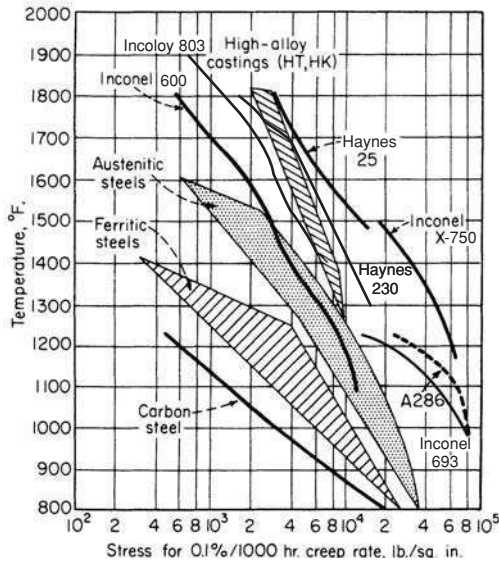


FIG. 25-21 Effect of creep on metals for high-temperature use. $^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times \frac{5}{9}$; to convert lb/in^2 to MPa, multiply by 6.895×10^{-3} . [Chem. Eng., 65, p. 140 (Dec. 15, 1958); modified 2005 using producer data.]

and short-time strengths (see Figs. 25-21 and 25-22). **Creep** relates initially applied stress to rate of plastic flow. **Stress rupture** is another important consideration at high temperatures since it relates stress and time to produce rupture. As the figures show, ferritic alloys are weaker than austenitic compositions, and in both groups molybdenum increases strength. Austenitic castings are much stronger than their wrought counterparts. And higher strengths are available in the nickel- and cobalt-based alloys. Other properties that become important at high temperatures include thermal conductivity,

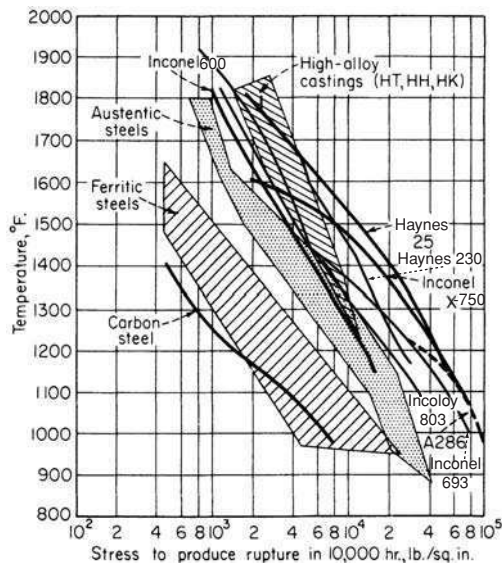


FIG. 25-22 Rupture properties of metals as a function of temperature. $^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times \frac{5}{9}$; to convert lb/in^2 to MPa, multiply by 6.895×10^{-3} . [Chem. Eng., 65, p. 140 (Dec. 15, 1958); modified 2005 using producer data.]

thermal expansion, ductility at temperature, alloy composition, and stability.

Actually, in many cases strength and mechanical properties become of secondary importance in process applications, compared with resistance to the corrosive surroundings. All common heat-resistant alloys form oxides when exposed to hot oxidizing environments. Whether the alloy is resistant depends upon whether the oxide is stable and forms a protective film. Thus, mild steel is seldom used above 480°C (900°F) because of excessive scaling rates. Higher temperatures require chromium. Thus, type 502 steel, with 4 to 6 percent Cr, is acceptable to 620°C (1150°F). A 9 to 12 percent Cr steel will handle 730°C (1350°F). The well-known austenitic stainless steels have excellent oxidation resistance: up to 900°C (1650°F) for 18-8; however, their strength limits their use. Chromium is also important in nickel-based alloys as a very protective chromium-nickel spinel oxide forms at lower temperatures. At temperatures above 1000°C (1832°F) the addition of significant quantities of aluminum or silicon will enhance the oxidation resistance. There is a synergistic effect between silicon or aluminum in the formation of more resistant oxide layers.

Figure 25-23 shows the oxidation resistance of a variety of commercially significant alloys. The penetration shown is a combination of the uniform scaling that occurred during exposure and the internal penetration of oxide into the microstructure. This value of penetration is considered to be conservative but better from a prediction of life basis. The lower curve for alloy 214 shows the beneficial effect of aluminum in reducing oxidation at high temperatures; however, the use of this alloy is limited by its high temperature strength. These curves were generated using a program called ASSET that was developed by Shell Global solutions and is available for purchase. The curves are based on actual data that were generated and then used to calculate a parameter that allows extrapolation to different temperatures. The heavy line is from actual data while the thin line is extrapolated data.

Chromium is the most important material in imparting resistance to sulfidation (formation of sulfidic scales similar to oxide scales). The austenitic alloys are generally used because of their superior mechanical properties and fabrication qualities, despite the fact that nickel in the alloy tends to lessen resistance to sulfidation somewhat.

Aluminum and *silicon* also improve the resistance of alloys to oxidation as well as sulfidation. But use as an alloying agent is limited because the amount required interferes with the workability and

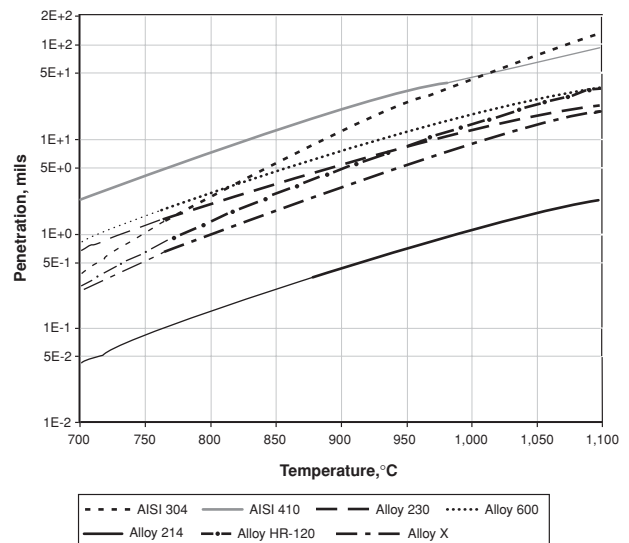


FIG. 25-23 Penetration curves for oxidation in still air of several commercially significant high-temperature alloys.

TABLE 25-22 Miscellaneous Alloys

Alloy	Designation	UNS	Composition, %°	Condition	Mechanical properties†			
					Yield strength, kip/in² (MPa)‡	Tensile strength, kip/in² (MPa)‡	Elongation, %	Hardness, HB
Refractory Alloys								
Niobium (columbium)		R04210	99.6 Cb	Annealed	37 (255)	53 (365)	26	80
Molybdenum		R03600	0.01–0.04 C	Annealed				
Molybdenum, low C		R03650	0.01 C					
Molybdenum alloy		R03630	0.01–0.04 C, 0.40–0.55 Ti, 0.06–0.12 Zn					
Tantalum		R05210	99.8 min. Ta					
Tungsten		R07030	99.9 min. W	Annealed		50 (345)	40	45
Zirconium		R60702	4.5 Hf, 0.2 Fe + Cr, 99.2 Zi + Hf	Annealed	16 (110)	270 (1862)	31	77
Precious Metals and Alloys								
Gold		P00020	99.95 min. Au	Annealed		19 (131)	45	25
Silver		P07015	99.95 min. Ag	Annealed	8 (55)	18 (124)	54	27
Sterling silver		P07931	6.50–7.90 Cu, 92.1–93.5 Ag	Annealed	20 (138)	41 (283)	26	65
Platinum		P04955	99.95 min. Pt	Annealed		18 (124)	38	39
Palladium		P03995	99.80 min. Pd	Annealed		25 (172)	27	38
Lead Alloys								
Chemical lead		L50049	99.9 min. Pb	Rolled	1.9 (13)	2.5 (17)	50	5
Antimonial lead			90 Pb, 10 Sb	Rolled		4.1 (28)	47	13
Tellurium lead			99.85 Pb, 0.04 Te, 0.06 Cu	Rolled	2.2 (15)	3 (21)	45	6
50-50 solder		L05500	50 Pb, 50 Sn, 0.12 max. Sb	Cast		6.8 (47)	50	14
Magnesium Alloys								
Extended Shapes	AZ31B	M11311	2.5–3.5 Al, 0.20 min. Mn, 0.6–1.4 Zn	Annealed	15–18 (103–124)	32 (220)	9–12	56
Cast alloy	AZ91C	M11914	8.1–9.3 Al, 0.13 min. Mn, 0.4–1.0 Zn	As cast	11 (76)	23 (159)		60
Cast alloy	EZ33A	M12330	2.0–3.1 Zn, 0.5–1.0 Zr	Aged	14 (97)	20 (138)	2	50
Wrought alloy	LA141A	M14141	1.0–1.5 Al, 13–15 Li	Stress hard- annealed	24–26 (165–179)	33–34 (228–234)	4	57
Titanium Alloys								
Commercial pure	Gr. 1	R50250	0.20 Fe, 0.18 O	Annealed	35 (241)	48 (331)	30	120
Commercial pure	Gr. 2	R50400	0.30 Fe, 0.25 O	Annealed	50 (345)	63 (434)	28	200
Ti-Pd	Gr. 7	R52400	0.30 Fe, 0.25 O, 0.12–0.25 Pd	Annealed	50 (345)	63 (434)	28	200
Ti-6Al-4V	Gr. 5	R56400	5.5–5.6 Al, 0.40 Fe, 0.20 O, 3.5–4.5 V	Annealed	134 (924)	144 (993)	14	330
Low alloy	Gr. 12	R53400	0.2–0.4 Mo, 0.6–0.9 Ni	Annealed	65 (448)	75 (517)	25	
Ti-LowPd	Gr. 16	R52402	0.30 Fe, 0.25 O, 0.04–0.08 Pd	Annealed	50 (345)	63 (434)	28	200
Ti-Ru	Gr. 26	R52404	0.3 Fe, 0.25 O, 0.08–0.14 Ru	Annealed	50 (345)	63 (434)	28	200

°Single values are maximum values unless otherwise noted.

†Typical room-temperature properties.

‡To convert MPa to lbf/in², multiply by 145.04.

TABLE 25-23 Typical Physical Properties of Surface Coatings for Concrete

	Concrete	Polyester		Epoxy		Urethane*
		Isophthalic	Bisphenol	Polyamide	Amine	
Tensile strength (ASTM C307), lbf/in ²	200–400	1200–2500	1200–2500	600–4000	1200–2500	200–1200
MPa	1.4–2.8	8.3–17	8.3–17	4.0–28	8.3–17	1.4–8.3
Thermal coefficient of expansion (ASTM C531)						
Maximum in/(in·°F)	6.5 × 10 ⁶	20 × 10 ⁶	20 × 10 ⁶	40 × 10 ⁶	40 × 10 ⁶	†
Maximum mm/(mm·°C)	11.7 × 10 ⁶	36 × 10 ⁶	36 × 10 ⁶	72 × 10 ⁶	72 × 10 ⁶	
Compressive strength, (ASTM C579), lbf/in ²	3500	10,000	10,000	4000	6000	†
MPa	24	70	70	28	42	
Abrasion resistance, Taber abraser—weight loss, mg, 1000-g load/1000 cycles		15–27	15–27	15–27	15–27	†
Shrinkage, ASTM C531, %		2–4	2–4	0.25–0.75	0.25–0.75	0–2
Work life, min		15–45	15–45	30–90	30–90	15–60
Traffic limitations, h after application						
	Light	16	16	24	24	24
	Heavy	36	36	48	48	48
	Ready for service	48	48	72	72	72
Adhesion characteristics‡		Poor	Fair	Excellent	Good	Fair
Flexural strength (ASTM C580), lbf/in ²		1500	1500	1000	1500	†
MPa		10	10	7	10	

NOTE: All physical values depend greatly on reinforcing. Values are for ambient temperatures.

*Type of urethane used is one of three: (1) Type II, moisture-cured; (2) Type IV, two-package catalyst; or (3) Type V, two-package polyol. (Ref. ASTM C16.)

†Urethanes not shown because of great differences in physical properties, depending on formulations. Adhesion characteristics should be related by actual test data. Any system which shows concrete failure when tested for surfacing adhesion should be rated excellent with decreasing rating for systems showing failure in cohesion or adhesion below concrete failure.

‡Adhesion to concrete: primers generally are used under polyesters and urethanes to improve adhesion.

SOURCE: NACE RP-03-76, *Monolithic Organic Corrosion Resistant Floor Surfacing*, 1976. Courtesy of National Association of Corrosion Engineers.

high-temperature strength properties. The formation of Ni₃Al precipitates in a nickel alloy increases the strength but reduces the ductility. The development of high aluminum surface layers by spraying, dipping, and cementation is a feasible means of improving the corrosion resistance of low-alloy steels.

Hydrogen Atmospheres Austenitic stainless steels, by virtue of their structure and high chromium contents, are usually resistant to hydrogen atmospheres. Ferritic alloys are more susceptible to hydrogen embrittlement.

TABLE 25-24 Chemical Resistance of Elastomers

Type of rubber	Features
Butadiene styrene	General-purpose; poor resistance to hydrocarbons, oils, and oxidizing agents
Butyl	General-purpose; relatively impermeable to air; poor resistance to hydrocarbons and oils
Chloroprene	Good resistance to aliphatic solvents; poor resistance to aromatic hydrocarbons and many fuels
Chlorosulfonated polyethylene	Excellent resistance to oxidation, chemicals, and heat; poor resistance to aromatic oils and most fuels
cis-Polybutadiene	General-purpose; poor resistance to hydrocarbons, oils, and oxidizing agents
cis-Polyisoprene	General-purpose; poor resistance to hydrocarbons, oils, and oxidizing agents
Ethylene propylene Fluorinated	Excellent resistance to heat and oxidation Excellent resistance to high temperature, oxidizing acids, and oxidation; good resistance to fuels containing up to 30% aromatics
Natural	General-purpose; poor resistance to hydrocarbons, oils, and oxidizing agents
Nitrile (butadiene acrylonitrile)	Excellent resistance to oils, but not resistant to strong oxidizing agents; resistance to oils proportional to acrylonitrile content
Polysulfide	Good resistance to aromatic solvents; unusually high impermeability to gases; poor compression set and poor resistance to oxidizing acids
Silicone	Excellent resistance over unusually wide temperature range [–100 to 260°C (–150 to 500°F)]; fair oil resistance; poor resistance to aromatic oils, fuels, high-pressure steam, and abrasion
Styrene	Synonymous with butadiene-styrene

Halogens (Hot, Dry Cl₂, HCl) Pure nickel and nickel alloys are useful with dry halogen gases. The presence of oxygen with Cl₂ or HCl gas will alter the choice of materials. But even with the best materials, corrosion rates are relatively high at high temperature. There are cases in which equipment for high-temperature halogenation has used platinum-clad nickel-based alloys. These materials have high initial cost but long life. Platinum and gold have excellent resistance to dry HCl even at 1100°C (2000°F).

Refractories Refractories are selected to accomplish four objectives:

1. Resist heat
2. Resist high-temperature chemical attack
3. Resist erosion by gas with fine particles
4. Resist abrasion by gas with large particles

Refractories are available in three general physical forms: solids in the form of brick and monolithic castable ceramics and as ceramic fibers.

The primary method of selection of the type of refractory to be used is by gas velocity:

- <7.5 m/s (25 ft/s): fibers
- 7.5–60 m/s (25–200 ft/s): monolithic castables
- >60 m/s (200 ft/s): brick

Within solids the choice is a tradeoff because, with brick, fine particles in the gas remove the mortar joints and, in the monolithic castables, while there are no joints, the refractory is less dense and less wear-resistant.

Internal Insulation The practice of insulating within the vessel (as opposed to applying insulating materials on the equipment exterior) is accomplished by the use of fiber blankets and lightweight aggregates in ceramic cements. Such construction frequently incorporates a thin, high-alloy shroud (with slip joints to allow for thermal expansion) to protect the ceramic from erosion. In many cases this design is more economical than externally insulated equipment because it allows use of less expensive lower-alloy structural materials.

Refractory Brick Nonmetallic refractory materials are widely used in high-temperature applications in which the service permits the appropriate type of construction. The more important classes are described in the following paragraphs.

Fireclays can be divided into plastic clays and hard flint clays; they may also be classified as to alumina content. **Firebricks** are usually made of a blended mixture of flint clays and plastic clays which is formed, after mixing with water, to the required shape. Some or all of

TABLE 25-25 Properties of Graphite and Silicon Carbide

	Graphite	Impervious graphite	Impervious silicon carbide
Specific gravity	1.4–1.8	1.75	3.10
Tensile strength, lbf/in ² (MPa)	400–1400 (3–10)	2,600 (18)	20,650 (143)
Compressive strength, lbf/in ² (MPa)	2000–6000 (14–42)	10,500 (72)	150,000 (1000)
Flexural strength, lbf/in ² (MPa)	750–3000 (5–21)	4,700 (32)	
Modulus of elasticity ($\times 10^5$), lbf/in ² (MPa)	0.5–1.8 (0.3–12 $\times 10^4$)	2.3 (1.6 $\times 10^4$)	56 (39 $\times 10^4$)
Thermal expansion, in/(in. $^{\circ}$ F $\times 10^{-6}$) [mm/(mm. $^{\circ}$ C)]	0.7–2.1 (1.3–3.8)	2.5 (4.5)	1.80 (3.4)
Thermal conductivity, Btu/[(h·ft ²)($^{\circ}$ F/ft)] [(W/(m·K)]	15–97 (85–350)	85 (480)	60 (340)
Maximum working temperature (inert atmosphere), $^{\circ}$ F ($^{\circ}$ C)	5000 (2800)	350 (180)	4,200 (2300)
Maximum working temperature (oxidizing atmosphere), $^{\circ}$ F ($^{\circ}$ C)	660 (350)	350 (180)	3,000 (1650)

SOURCE: Carborundum Co. Courtesy of National Association of Corrosion Engineers.

the flint clay may be replaced by highly burned or calcined clay, called grog. A large proportion of modern brick production is molded by the dry-press or power-press process, in which the forming is carried out under high pressure and with a low water content. Extruded and hand-molded bricks are still made in large quantities.

The dried bricks are burned in either periodic or tunnel kilns at temperatures ranging between 1,200 and 1,500 $^{\circ}$ C (2,200 and 2,700 $^{\circ}$ F). Tunnel kilns give continuous production and a uniform burning temperature.

Fireclay bricks are used in kilns, malleable-iron furnaces, incinerators, and many portions of metallurgical furnaces. They are resistant to spalling and stand up well under many slag conditions but are not generally suitable for use with high-lime slags or fluid-coal-ash slags or under severe load conditions.

High-alumina bricks are manufactured from raw materials rich in alumina, such as diaspore. They are graded into groups with 50, 60, 70, 80, and 90 percent alumina content. When well fired, these bricks contain a large amount of mullite and less of the glassy phase than is present in firebricks. Corundum is also present in many of these bricks. High-alumina bricks are generally used for unusually severe temperature or load conditions. They are employed extensively in lime kilns and rotary cement kilns, in the ports and regenerators of glass tanks, and for slag resistance in some metallurgical furnaces; their price is higher than that of firebrick.

Silica bricks are manufactured from crushed ganister rock containing about 97 to 98 percent silica. A bond consisting of 2 percent lime is used, and the bricks are fired in periodic kilns at temperatures of 1,500 to 1,540 $^{\circ}$ C (2,700 to 2,800 $^{\circ}$ F) for several days until a stable volume is obtained. They are especially valuable when good strength is required at high temperatures. Superduty silica bricks are finding some use in the steel industry. They have a lowered alumina content and often a lowered porosity.

Silica bricks are used extensively in coke ovens, the roofs and walls of open-hearth furnaces, and the roofs and sidewalls of glass tanks and

as linings of acid electric steel furnaces. Although silica brick is readily spalled (cracked by a temperature change) below red heat, it is very stable if the temperature is kept above this range and for this reason stands up well in regenerative furnaces. Any structure of silica brick should be heated up slowly to the working temperature; a large structure often requires 2 weeks or more.

Magnesite bricks are made from crushed magnesium oxide, which is produced by calcining raw magnesite rock to high temperatures. A rock containing several percent of iron oxide is preferable, as this permits the rock to be fired at a lower temperature than if pure materials were used. Magnesite bricks are generally fired at a comparatively high temperature in periodic or tunnel kilns. A large proportion of magnesite brick made in the United States uses raw material extracted from seawater.

Magnesite bricks are basic and are used whenever it is necessary to resist high-lime slags, as in the basic open-hearth steel furnace. They also find use in furnaces for the lead- and copper-refining industries. The highly pressed unburned bricks find extensive use as linings for cement kilns. Magnesite bricks are not so resistant to spalling as fireclay bricks.

Chrome bricks are manufactured in much the same way as magnesite bricks but are made from natural chromite ore. Commercial ores always contain magnesia and alumina. Unburned hydraulically pressed chrome bricks are also available.

Chrome bricks are very resistant to all types of slag. They are used as separators between acid and basic refractories, also in soaking pits and floors of forging furnaces. The unburned hydraulically pressed bricks now find extensive use in the walls of the open-hearth furnace. Chrome bricks are used in sulfite-recovery furnaces and to some extent in the refining of nonferrous metals. Basic bricks combining various properties of magnesite and chromite are now made in large quantities and have advantages over either material alone for some purposes.

The **insulating firebrick** is a class of brick that consists of a highly porous fire clay or kaolin. Such bricks are light in weight (about one-half to one-sixth of the weight of fireclay), low in thermal conductivity, and yet sufficiently resistant to temperature to be used successfully on the hot side of the furnace wall, thus permitting thin walls of low thermal conductivity and low heat content. The **low heat content** is particularly valuable in saving fuel and time on heating up, allows rapid changes in temperature to be made, and permits rapid cooling. These bricks are made in a variety of ways, such as mixing organic matter with the clay and later burning it out to form pores; or a bubble structure can be incorporated in the clay-water mixture which is later preserved in the fired brick. The insulating firebricks are classified into several groups according to the maximum use limit; the ranges are up to 870, 1,100, 1,260, 1,430, and above 1,540 $^{\circ}$ C (1,600, 2,000, 2,300, 2,600, and above 2,800 $^{\circ}$ F).

Insulating refractories are used mainly in the heat-treating industry for furnaces of the periodic type. They are also used extensively in stress-relieving furnaces, chemical-process furnaces, oil stills or heaters, and the combustion chambers of domestic-oil-burner furnaces. They usually have a life equal to that of the heavy brick that they replace. They are particularly suitable for constructing experimental or

TABLE 25-26 Minimum Temperature without Excessive Scaling in Air (Continuous Service)*

Alloy	$^{\circ}$ F	$^{\circ}$ C
Carbon steel	1050	565
½Mo steel	1050	565
1Cr ½Mo steel	1100	595
2¼Cr 1Mo steel	1150	620
5Cr ½Mo steel	1200	650
9Cr 1Mo steel	1300	705
AISI 410	1300	705
AISI 304	1600	870
AISI 321	1600	870
AISI 347	1600	870
AISI 316	1600	870
AISI 309	2000	1090
AISI 310	2100	1150

*Courtesy of National Association of Corrosion Engineers.

TABLE 25-27 Important Commercial Alloys for High-Temperature Process Service

	Nominal composition, %			
	Cr	Ni	Fe	Other
Ferritic steels				
Carbon steel			Bal.	
2¼ chrome	2¼		Bal.	Mo
Type 502	5		Bal.	Mo
Type 410	12		Bal.	
Type 430	16		Bal.	
Type 446	27		Bal.	
Austenitic steels				
Type 304	18	8	Bal.	
Type 321	18	10	Bal.	Ti
Type 347	18	11	Bal.	Cb
Type 316	18	12	Bal.	Mo
Type 309	24	12	Bal.	
Type 310	25	20	Bal.	
Type 330	15	35	Bal.	
Nickel-base alloys				
Nickel		Bal.		
Incoloy 800	21	32	Bal.	
Hastelloy B		Bal.	6	Mo
Hastelloy C	16	Bal.	6	W, Mo
60-15	15	Bal.	25	
Inconel 600	15	Bal.	7	
80-20	20	Bal.		
Hastelloy X	22	Bal.	19	Co, Mo
Multimet	21	20	Bal.	Co
Rene 41	19	Bal.	5	Co, Mo, Ti
Cast irons				
Ductile iron			Bal.	C, Si, Mg
Ni-Resist, D-2	2	20	Bal.	Si, C
Ni-Resist, D-4	5	30	Bal.	Si, C
Cast stainless (ACI types)				
HC	28	4	Bal.	
HF	21	11	Bal.	
HH	26	12	Bal.	
HK	15	20	Bal.	
HT	15	35	Bal.	
HW	12	Bal.	28	
Superalloys				
Inconel X	15	Bal.	7	Ti, Al, Cb
A 286	15	25	Bal.	Mo, Ti
Stellite 25	20	10	Co-base	W
Stellite 21 (cast)	27.3	2.8	Co-base	Mo
Stellite 31 (cast)	25.2	10.5	Co-base	W

laboratory furnaces because they can be cut or machined readily to any shape. They are not resistant to fluid slag.

There are a number of types of special brick obtainable from individual producers. **High-burned kaolin refractories** are particularly valuable under conditions of severe temperature and heavy load or severe spalling conditions, as in the case of high-temperature oil-fired boiler settings or piers under enameling furnaces. Another brick for the same uses is a high-fired brick of Missouri aluminous clay.

There are on the market a number of bricks made from **electrically fused materials**, such as fused mullite, fused alumina, and fused magnesite. These bricks, although high in cost, are particularly suitable for certain severe conditions.

Bricks of **silicon carbide**, either recrystallized or clay-bonded, have a high thermal conductivity and find use in muffle walls and as a slag-resisting material.

Other types of refractory that find use are forsterite, zirconia, and zircon. Acid-resisting bricks consisting of a dense body like stoneware are used for lining tanks and conduits in the chemical industry. Carbon blocks are used as linings for the crucibles of blast furnaces, very extensively in a number of countries and to a limited extent in the United States. Fusion-cast bricks of mullite or alumina are largely used to line glass tanks.

Ceramic-Fiber Insulating Linings Ceramic fibers are produced by melting the same alumina-silica china (kaolin) clay used in conventional insulating firebrick and blowing air to form glass fibers. The fibers, 50.8 to 101.6 mm (2 to 4 in) long by 3 µm in diameter, are interlaced into a mat blanket with no binders or chopped into shorter fibers and vacuum-formed into blocks, boards, and other shapes. Ceramic-fiber linings, available for the temperature range of 650 to 1,430°C (1,200 to 2,600°F), are more economical than brick in the 650 to 1,230°C (1,200 to 2,250°F) range. Savings come from reduced first costs, lower installation labor, 90 to 95 percent less weight, and a 25 percent reduction in fuel consumption.

Because of the larger surface area (compared with solid-ceramic refractories) the chemical resistance of fibers is relatively poor. Their acid resistance is good, but they have less alkali resistance than solid materials because of the absence of resistant aggregates. Also, because they have less bulk, fibers have lower gas-velocity resistance. Besides the advantage of lower weight, since they will not hold heat, fibers are more quickly cooled and present no thermal-shock structural problem.

Castable Monolithic Refractories Standard portland cement is made of calcium hydroxide. In exposures above 427°C (800°F) the hydroxyl ion is removed from portland (water removed); below 427°C (800°F), water is added. This cyclic exposure results in spalling. Castables are made of calcium aluminate (rather than portland); without the hydroxide they are not subject to that cyclic spalling failure.

Castable refractories are of three types:

1. *Standard.* 40 percent alumina for most applications at moderate temperatures.

2. *Intermediate purity.* 50 to 55 percent alumina. The anorthite (needle-structure) form is more resistant to the action of steam exposure.

3. *Very pure.* 70 to 80 percent alumina for high temperatures. Under reducing conditions the iron in the ceramic is controlling, as it acts as a catalyst and converts the CO to CO₂ plus carbon, which results in spalling. The choice among the three types of castables is generally made by economic considerations and the temperature of the application.

Compared with brick, castables are less dense, but this does not really mean that they are less serviceable, as their cements can hydrate and form gels which can fill the voids in castables. Extra-large voids do indicate less strength regardless of filled voids and dictate a lower allowable gas velocity. If of the same density as a given brick, a castable will result in less permeation.

Normally, castables are 25 percent cements and 75 percent aggregates. The aggregate is the more chemically resistant of the two components. The highest-strength materials have 30 percent cement, but too much cement results in too much shrinkage. The standard insulating refractory, 1:2:4 LHV castable, consists of 1 volume of cement, 2 volumes of expanded clay (Haydite), and 4 volumes of vermiculite.

Castables can be modified by a clay addition to keep the mass intact, thus allowing application by air-pressure gunning (gunite). Depending upon the size and geometry of the equipment, many castable linings must be reinforced; wire and expanded metal are commonly used.

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